

(Project Name)

DPR Environmental Health & Safety Plan

Injury Illness Prevention Plan (IIPP)

Section	Title
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DPR Environmental Health & Safety Program

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DPR Environmental Health & Safety Program

1.0 Safety Policy, Purpose & Scope

The purpose of the DPR Construction Environmental Health & Safety Plan (DPR EHSP or EHSP) is to establish a commitment to creating an Injury Free Environment (IFE) as an integral part of conducting business – contributing to the overall success of the project.

It is the intent of DPR Construction (DPR) to foster and promote the *mission* of an IFE. This calls for the elimination of unsafe acts, unsafe conditions and the elimination of near-miss incidents, which can be accomplished with:

- Collaborative teamwork between the owner, DPR, and subcontractors
- Personal commitment to the success of all projects
- Ownership by the workers
- Accountability

It is the intent of the DPR EHSP to protect and promote the safety and well-being of all personnel by providing a safe and healthy work environment during all phases of construction. The DPR EHSP provides a foundation and a mechanism for protecting people, equipment, materials, and property and for complying with DPR, federal, state and local safety and health regulations and to aid in the development of a tangible safety culture. No employee is permitted to perform any work for which they have not been properly equipped and trained or which they feel is unsafe.

When DPR's EHSP may be in conflict with another employer's Injury & Illness Prevention Program (IIPP), the employer performing the work shall implement the most stringent health and safety requirements. Following the plan will help eliminate at-risk behaviors and unsafe conditions; furthermore, it will facilitate good cooperation and communication between DPR, subcontractors, the public and other agents represented. Everyone is expected and obligated to perform all tasks safely.

By contract, every contractor is obligated to conform to the requirements of the Occupational Safety and Health Act of 1970 (OSHA) and all additions and revisions thereto, as well as other applicable federal, state and local requirements, and the DPR EHSP. The DPR EHSP applies to all work performed on or directly involving the DPR site. The site is the area defined contractually between DPR and the owner.

All subcontractors, tiered subcontractors and all supervisory personnel are responsible for the actions of those they supervise, for maintaining safe and healthy working conditions in their areas of responsibility and for strictly enforcing all authorized safety and health rules and regulations. Everyone is expected to comply with these rules and regulations.

These safety efforts will affect the overall success of the construction project – the mission of an injury free workplace and zero environmental impacts. It is understood that all contractors working on this project will subscribe to the DPR EHSP in addition to their own plan. With the cooperation, dedication and the assistance of everyone, this will ensure a successful and safe job.

To effectively motivate and monitor the safe work practices, all Project Managers, Superintendents, Foremen, Leadmen, Project Engineers, Safety Coordinators and Supervisors are to become familiar with the DPR EHSP, the safety indicators and milestones for the project.

Within the DPR EHSP, contractor is defined as all construction managers, general contractors, contractors, subcontractors, and tiered subcontractors. Further, subcontractor is defined as all contractors and their tiers under contract to DPR.

2.0 Introduction

Safety and environmental compliance issues will be continually developed monitored and reviewed by DPR Construction and subcontractors and therefore are subject to change without notification. Unless otherwise specified, at a minimum, all construction will be conducted in accordance with each subcontractor's safety and health plan and the DPR EHSP, and must fully comply with all federal, state and local environmental and worker protection regulations.

2.1 Reservation of Rights

DPR's Corporate Safety Officer reserves the right to interpret, change, revise or depart from any/all policies and procedures at any time without notice. DPR further reserves the right to promulgate safety standards during the course of construction as may be deemed necessary in the interest of safety, at no additional cost to DPR. Nothing in this safety plan alters an employer's or employee's status or infringes on the rights of any employee or contractor.

Employees remain free to resign their employment at any time, for any reason, without notice and as may be limited by law. DPR retains the right to remove any employee at any time, for any reason and without notice.

The DPR EHSP is one of the construction contract documents. All subcontractors are required to ensure that they and their employees, trade contractors, suppliers, vendors and visitors, while on the job site and in the conduct of contracts, comply with the provisions of the manual. THE PROVISIONS OF THE MANUAL WILL BE STRICTLY ENFORCED. Non-compliance with safety requirements will be treated the same as non-compliance with any contract item. Any non-compliance may result in work stoppage, employee dismissal or further remedies. Willful and/or repeated non-compliance may result in trade contractor's termination of the contract and removal from the project.

3.0 Behavior-Based Safety (BBS) and Injury Free Environment (IFE)

Behavior based safety is designed to create a culture of discipline wherein workers will do the right thing when no one is watching.

3.1 Behavior-Based Safety (BBS)

Behavior based safety starts by enforcing the idea that accidents and injuries don't have to happen, that an Injury Free Environment can exist within construction. In order to promote BBS, we should understand how our individual values affect our personal choices, and how others' priorities influence decisions that workers may make on the job. There are existing concepts that exist within the construction industry that should be addressed in order to ensure the success of a behavior based safety program. Those concepts may fall into one of the following categories.

- Outside Influences:
 - o Exists in each organization, group, trade, and/ or project. Determine what culture you have inherited and how you will influence them to create change
 - o Look for sub-cultures that may exist that promote unacceptable behaviors or short cuts.
 - o Continuously promote the IFE culture:
 - Attitudes are contagious, always display a positive safety attitude
 - Change your language to promote desired culture, watch your words
 - Don't wait for someone else to act, you go first
 - Never stop teaching
 - Plan, Do, Check, Act
- Environment:
 - o Create an environment that promotes the behaviors you want
 - o Use positive re-enforcement
 - o Measure success of the process, not the results

- o Praise desired behavior in public
 - o Coach at-risk behavior in private
 - o New habits take a while to stick, keep coaching
 - o Perception is often seen as reality, always set the example
- Conditions:
 - o Set expectations up front and get commitment
 - o Empower individuals to make positive change, bottom up improvement
 - o Acknowledge those who participate, give feedback often
 - o Do not allow any deviation of expectations, act quickly
 - o Do not re-enforce at-risk behavior by promoting workers who complete work faster by taking short-cuts
 - o Display measurements for all to see in the workspace
 - o Report everything
 - o Share failures and lessons learned

3.2 Injury Free Environment (IFE)

Injury Free Environment (IFE) is not simply a safety program, it is an ideology; a belief that we can work injury free. In order to create an Injury Free Environment, it takes participation from all levels of the organization, including ownership, management, supervision, and individual workers. In order for an Injury Free Environment to exist, the following principles must be in place.

- **Supportive Leadership**
 - o Personally committed to IFE values
 - o Personally care
 - o Positive role model
 - o Provides resources - time, dollars, and manpower
 - o Not afraid to push back if safety or people are compromised
 - o Promotes and acts upon safety and free communications to address safety concerns
- **Personal Commitment To Safety**
 - o Each individual is personally committed
 - o Accepts the responsibility for their safety and their co-workers
 - o Agrees to be held accountable for their actions or lack of actions
 - o Positive role model of IFE behaviors
 - o Individual has knowledge of how to work safely (adequately trained)
- **Free and Open Lines Of Communication**
 - o Employees must become comfortable identifying, correcting & communicating safety concerns to management and each other
 - o Identified safety concerns and injuries are addressed immediately
 - o Feedback is provided on actions taken to address/correct identified safety concerns
 - o Employees are not afraid to report injuries or near misses (emphasis on preventing future incidents)
- **Accountability**
 - o Being responsible for one's own safety, accountability falls on individuals
 - o Failure to deliver will result in someone getting hurt
 - o Everyone working together to monitor safety and to ensure safety is being maintained at the appropriate level

4.0 Health and Safety Communications

4.1 Site Specific IIPP (Site Specific Written Safety Program)

Each Subcontractor and tiered Subcontractor shall develop a written IIPP/safety program specific to the project. The IIPP/Safety Plan should be submitted and reviewed by DPR project management prior to



mobilization. If a Safety Corrective Action Plan (CAP) is required, the CAP should be integrated as part of the site specific requirements.

4.2 Written Hazard Communication Program and Safety Data Sheets (SDSs)

As part of their written Hazard Communication Program, each Subcontractor and tiered Subcontractor shall submit applicable Safety Data Sheets (SDSs), specific to the project to the DPR project team. The SDSs submitted shall include a list of hazardous chemicals utilized within the project, and this list should be updated and maintained by the Subcontractor/tiered Subcontractor for the duration of the project, with any changes submitted to the DPR project team. In addition to SDSs being forwarded to DPR project management during the submittal process, a complete library of SDSs shall be provided for the Right to Understand/Right-To-Know area for the project and prior to any site mobilization. Workers shall be given a means to access information about the chemicals they may be exposed to or come in contact with whether pre-existing or newly introduced during the construction process.

4.3 Site Specific Safety Orientation

Subcontractors are required to communicate the requirements of the DPR EHSP and other DPR applicable documented EHS practices as outlined in this document.

The DPR provided site specific safety orientation may be up to one (1) hour in duration, and should be attended by all personnel working on site. The orientation includes a discussion of site protocol, DPR EHS requirements, evacuation procedures and a description of the logistics of the site. Workers returning to the site after a three month or longer absence should be re-oriented prior to start of work.

On-going safety training (i.e., tool box/tailgate safety meetings, pre-task planning, etc.) shall be conducted by DPR and/or subcontractors prior to the initiation of tasks or activities that may involve new hazards.

4.4 Pre-Planning

4.4.1 Subcontractor Pre-Qualification

Subcontractors should have completed the DPR Pre-Qualification process and been approved. Subcontractors, who have been identified during the DPR Pre-Qualification process as being conditionally approved, may require a Safety Corrective Action Plan (CAP). Where applicable, the CAP should be submitted and approved by DPR Regional Safety Manager or designee, prior to execution of contracts. Once the CAP is approved, the elements of the CAP should be integrated into the Subcontractors' Site IIPP. The CAP should be fully executed on all DPR projects for the duration of their work. CAPs are to remain current and may require updating or amending, should the expiration of their Pre-Qualification approval occur during their contract period.

Where a CAP is required, the CAP will be reviewed during any subsequent incident reviews, instances of non-safety compliance and/or during all Root Cause Analysis discussions. Should any portion of the CAP not be in compliance, corrective measures must be immediately implemented to address gaps.

4.4.2 Pre-Task Planning

Pre-task planning is an identification and communication process with craftsmen, that will aid in the elimination of injuries and minimize both direct and indirect the impacts to the production facility, office buildings, and construction activities. It is important to understand that different construction activities involve different levels of risk to personnel, facility production, and/or construction schedules and will require different levels of controls.

Daily, written, pre-task planning is required for all work activities. All activities should be pre-task planned and reviewed and/or updated as tasks and/or conditions change. Additionally, a Job Hazard Analysis (JHA – reference 3.2.3 in this Section) and /or written plans with greater detail should be prepared for any high hazard activity. These activities include, but are not limited to:

- Chemicals: hazardous & irritant
- Concrete: placement, pre-cast, tilt up, vertical, form work
- Confined Space
- Hoisting/rigging activities: including cranes, derricks, forklifts, straddle buggies, etc.
- Demolition activities & hazardous materials assessment: asbestos, lead, biohazards or other chemicals in the workplace, as well as general demolition hazards assessment
- All framing activities (including drywall)
- Excavation & trenching
- Fall hazards: exposures 6+ feet, overhead work
- Material handling
- Non-routine activities: activities not performed in the last six months
- Public exposure: phased occupancy, partial demolition, traffic control, etc.
- Scaffolding
- Steel erection
- Start Up/Shut Down/System Testing activities: tool hook up, introduction of process chemicals into systems, utility tie ins, lockout/tagout, work on energized equipment

General pre-task planning should including the following:

- Pre-task planning should be led by the supervisor and documented in writing
- Conducted at least daily or as otherwise defined by DPR project management
- All crew members should participate (at the job location) in pre-task planning and should sign the completed plan
- Should include hazards and precautions identified in applicable Job Hazard Analysis (JHA)
- Should be readily available at the work site (posted and/or placed where crew members have knowledge of its location at the work area)

Pre-Task Plans should be reviewed and revised whenever work conditions (or crew membership) change that may affect the ability to safely complete the work. *Refer to the Pre-Planning Section for further guidelines.*

4.4.3 Job Hazard Analysis

Job Hazard Analysis (JHA) is a tool that can be used by project teams and/or subcontractors to preplan high hazard tasks and procedures. Whereas a pre-task plan is task specific, the JHA is intended to be used as a high level planning tool that will serve to breakdown an entire scope of work into tasks and define the potential hazards and controls for each task. The goal is the successful planning and elimination of hazards and/or re-work through such methods as engineering controls, training, site and craft coordination, etc.

Written, detailed JHAs may be developed for high hazard activities on DPR projects. Refer to list of high hazard activities noted in Pre-Task Planning section above. DPR project management reserves the right to require site specific JHAs be prepared by subcontractors and/or their tiered subcontractors prior to start of work. When JHAs are developed and/or required by DPR project management, the following guidelines should be considered:

- Break the job down into individual, basic steps
- Identify potential hazards for each step
- Identify procedural methods and/or control measure to be implemented to address each hazard identified
- Project team members share JHAs w/crew to be performing their work, as a basic guideline to use and refer to during daily pre-task planning

Refer to the Pre-Planning Section 17 for further guidelines.

4.5 Incident/Accident Reporting and Root Cause Analysis

All incidents and accidents shall be immediately reported to DPR project management and fully investigated. Investigation and root cause analysis shall be completed, at a minimum, for any injury requiring medical treatment, near miss incident or unplanned events. Root Cause Analysis is used to identify the primary reason the incident occurred with an action plan developed to prevent recurrence. *Refer to the Incident Reporting Section for further details.*

4.6 Jobsite Safety Audits

Project Management should perform jobsite safety audits. The preferred method for performing jobsite safety audits is utilizing an electronic-based application. Inspecting and correcting items should be an ongoing activity. Documenting the audits is important and should occur on a regular basis. Jobsite Safety audits should be performed and documented, at a minimum, as follows:

- DPR shall ensure that periodic safety audits (inspections) are performed to identify and correct existing and foreseeable hazards.
- DPR Superintendents and all safety personnel should perform documented daily safety audits
- DPR Project Managers, Project Engineers, Foreman should perform documented weekly safety audits
- Subcontractor field supervision should perform documented weekly safety audits

Refer to the Forms Section at the end of the EHSP for audit forms.

4.7 Stretch & Flex

The first of every morning, DPR workers on the project should participate in the Stretch & Flex program. When applicable, additional sessions of exercises should be performed after break and/or lunch. Subcontractor personnel are encouraged to participate. DPR reserves the right to require site wide stretch and flex participation. This program prepares the worker to function both physically and mentally. This program's value is founded on the belief that education and clear information about simple changes in lifestyle are major factors in incident prevention. DPR will use this time to communicate to workers specific safety and/or site information for that day. *Refer to Ergonomics and Manual Material Handling Section for the Stretch & Flex program in its entirety.*

4.8 Substance Abuse Testing

Substance abuse testing, at a minimum, should be conducted for reasonable suspicion and post incident/accident. Other requirements may apply on a per project basis. Refer to the Substance Abuse Program section of this part for additional details.

4.9 Ongoing Communications

Other forms of employer-to-worker communications on safety topics on the project may include, but are not limited to:

- Providing workers with a copy of the Safe Work Practices handbook
- Activity specific safety guidelines listed in the EHSP Appendices
- Safety bulletins
- Brown bag meetings
- Hazard alerts
- Communication boards
- Other verbal and written notices

Communication boards may be utilized to display progress on project sponsored safety indicators and milestones. Weekly safety meetings shall be held, and should be attended by all DPR and subcontractor personnel and should be under the direction of DPR project management. The purpose of these meetings is to discuss the findings of jobsite safety audits and corrective actions, upcoming project activities, etc. This



meeting should also address pertinent safety regulations, any changes in scheduling, engineering, administration controls or PPE that are to be adopted for safety reasons.

Other required meetings may include:

- Pre-Bid & Pre-Construction meetings with subcontractors
- Weekly “tail-gate” safety meetings
- Stretch & Flex and safety briefing
- Craft feedback and appreciation lunches
- And other meetings that may be deemed necessary

4.10 *Safety and Health Training and Information*

In addition to the DPR site specific safety orientation, new workers shall be provided with initial training and/or orientation prior to assignment or when assigned to a new task for which training has not been received. Supervisors are expected to be knowledgeable and informed on hazards and safe work practices in their area of responsibility and to coordinate the disbursement of this information to crews. Training should include general area and specific assignment topics. Documentation of required training should be made available to DPR project management upon request. Training, to include refresher training when required, shall be provided in accordance with Federal/State OSHA guidelines. Refer to the Safety Training Section for additional information on required training. Training may include, but not be limited to:

- Aerial/Boom Lifts
- Confined Space Entry
- CPR/First Aid
- Electrical
- Excavation & Trenching
- Fall Protection
- Fire Watch
- Forklift
- Hazard Communication
- Hazardous Chemicals
- Ladders
- Lasers
- Lockout/Tagout
- Powder Actuated Tools
- Pre-task planning
- Respiratory Protection
- Rigging
- Scaffolding: Use & Erection/Dismantle
- Steel Erection
- Accident investigation training for Foremen and Superintendents
- Protocol refreshers to include, but not limited to:
 - o isolation
 - o protection of tools
 - o HEPA certification
 - o cutting, etc.

4.11 *Required Site Postings*

Federal and state worker protection and labor law regulations require a variety of site postings. Such postings should be provided in a language understandable to site workers and in an area readily accessible to workers. DPR project management should verify that all responsible subcontractors display adequate postings at each site. Such postings shall include, but are not limited to:

- DPR Project Safe Work Practices
- Worker operating instructions (for forklifts, industrial trucks, etc.)
- Work site permits (asbestos, hot work, utility shutdown, trench and excavation, confined spaces, etc.)
- Log of Summary of Occupational Injuries and Illnesses (OSHA 300 Log) as required by law
- Information regarding access to medical and exposure records
- Emergency phone numbers
- Federal Labor Law postings – English & Spanish

5.0 Safety Program Implementation

5.1 Overview

This section contains the requirements for implementing the DPR EHSP. Each subcontractor under the direction of DPR project management is responsible for initiating, maintaining and supervising all safety programs and precautions in connection with all work activities for the safety of their workers. This includes:

- All workers who perform work activities onsite, including any tiered subcontractor personnel
- All who may be affected by the work activities
- All work, materials, and equipment to be incorporated therein, whether in storage on or off site
- Operations which affect other property at the site or adjacent thereto

5.2 Enforcement of the DPR EHSP

Workers should be advised that the provisions of the DPR EHSP, the rules expressed during site specific safety orientation, as well as in the DPR Safe Work Practices Handbook and the use of proper personal protective equipment (PPE) are mandatory and should be enforced by the following:

- Recognition for compliance and injury free environment initiatives should include spot awards, and the use of safety as an evaluation criterion for promotion
- Subcontractors are responsible for any and all citations issued by any and all regulatory agencies, their related expenditures, both direct and indirect, and fines that may be due to DPR
- Progressive discipline shall be enforced for non-compliance with any DPR EHSP or subcontractor's safety and health work rule, refer to the Disciplinary Action section of this part for additional details

All workers should report unsafe conditions and practices and are encouraged to communicate with their supervisor and/or their Safety Coordinator. At any time, workers may anonymously report unsafe conditions and practices. There will be no reprisals or other job discrimination for expressing any concern, comment, suggestion or complaint about a safety related matter.

5.3 Hazard Assessment and Control

Workplace health and safety jobsite audits should be conducted daily. DPR and subcontractor members of project management should participate in this process, as discussed previously in the Jobsite Safety Audits section of this manual. Jobsite safety audits shall be documented with corrective actions implemented, including actions dates and the parties responsible.

Ongoing workplace health and safety jobsite audits should include reviewing new substances, materials, processes, procedures, or equipment introduced into work areas.

Workers have the responsibility to keep their employers informed whenever new substances, materials, processes, procedures or equipment are introduced into work areas that may give rise to new occupational safety and health hazards.

Skill assessments for employees involved in critical jobs that present high risk to other employees should be conducted and training provided and documented as needed to rectify competency deficiencies.

5.4 Hazard Correction

Any identified hazard which poses a significant health or safety risk shall be addressed immediately. Hazards should be corrected according to the following general guidelines:

- **FIRST** - If possible, workers should correct the condition **immediately**. Many safety hazards, such as tripping hazards, are easy to correct. The success of any project requires that everyone participate in the identification and correction of hazards.
- **SECOND** - If workers are not able to take immediate corrective action, the hazardous condition should be rendered safe and reported immediately to supervisory personnel.
- **THIRD** - All supervisory personnel should address all identified hazards in a timely manner. In the event that corrective action is not implemented in a reasonable length of time, the worker is requested to anonymously contact DPR project management.

Documented jobsite safety audits should report on the progress towards abatement and correction of recognized hazards. Documented jobsite safety audits should be filed with DPR project management. Continued noncompliance with the project health and safety requirements may result in progressive disciplinary action, up to and including, removal from the site.

5.5 Recordkeeping and Documentation

All safety and health records and documents, including site specific DPR EHSP & Safety Data Sheets, subcontractor IIPPs/safety plans and SDSs; OSHA logs, training documents, accident investigations and root cause analyses, near-miss reports, pre-task plans and safety related pre-planning documents, equipment inspections, safety related permits, safety related meeting minutes, safety related surveys and logs, etc. shall be maintained by DPR project management. All safety and health records and documents are archived separately within project related archive files, and maintained for a minimum 10 years. Any and all medical exposure records are maintained for 30 years following last date of employment.

Safety reports detailing leading indicators and safety milestones should be reviewed and maintained on a monthly basis, at minimum. These reports should include, but are not limited to, project consistency audits, incident frequency rates (e.g. the number of days without a recordable, first aid or lost time accident), costs per claim, work hours, and employee training records.

6.0 Public Relations

6.1 General Public

We realize that good public relations are essential to the success of a construction project. Mishandling of public relations can damage the reputation of the owner, DPR Construction and subcontractors; and may subject all of us to unduly restrictive regulation by city, county, state or federal agencies. All workers on the project are expected to be professional and courteous in dealing with owners, members of the general public, government inspectors, regulatory agency personnel, and representatives of the media.

Neighbors and other members of the general public may have questions regarding work activities or concerns about perceived problems such as odor, vibration, noise, visual nuisance, or structural support. Everyone on site should actively demonstrate concern with the issues identified, explain what can be expected, and most importantly, direct them to DPR project management or designated spokesperson if questions arise after project work has started. If necessary, a brief, factual response to the issues raised should be provided by a DPR representative. All SDSs for odor producing materials should be provided to the owner upon request.

6.2 Government Inspectors

In the event of a safety and health inspection performed by a government agency, contractors should comply with the following instructions:

- Notify DPR project management immediately upon arrival of the inspector (DPR project management should notify the owner and DPR regional EHS of any inspection or inquiry by a governmental agency before the inspection occurs or prior to a response to an inquiry)
- Choose a single individual to coordinate with the inspector and who will accompany the inspector at all times
- Respectfully verify the identification and authority of the inspector
- Read any warrant, complaint or notice presented
- Determine the nature, scope and purpose of the investigation prior to any physical inspection
- To the extent possible, limit the inspection to the areas identified in the inspection document or to those areas for which the inspector has authority
- Coordinate physical sample gathering, videos or photographs and worker contact with the inspector; a split must be obtained for any sample collected by the inspector and duplicate, as much as possible any videos or photographs, and document any conversations with workers
- Coordinate access to documents and record documents reviewed
- Ask the inspector for a copy of any report produced

Refer to Miscellaneous Section for a copy of the “OSHA / Compliance Inspection Report”. This is a tool which can be utilized by DPR project management to document and capture issues and concerns raised during a compliance inspection.

Additionally:

- Do **not** permit the news media to accompany the inspector
- Do **not** make admissions or volunteer information, and do not admit culpability
- Do **not** voluntarily produce records or documents not clearly required by law to be maintained and produced for review and copying
- Do **not** voluntarily permit the inspector to take any documents or records without specific authority to do so
- Do **not** over-commit the company as to abatement of any particular condition or practice
- Do **not** volunteer to assist the inspector in making the inspection
- Do **not**, under any circumstances, forcibly interfere with an inspector
- Do **not** fail to invoke your “right to remain silent” if a search warrant is presented

For government investigators appearing for purposes other than safety and health inspection, contact DPR project management immediately for instructions.

6.3 Regulatory Agencies

DPR regional EHS should be contacted prior to contacting a public agency regarding EHSP matters. DPR will contact the owner upon notification.

At times, special conditions on projects require subcontractors to deal directly with the representatives of federal, state, and local regulatory agencies. Subcontractors should immediately notify DPR project management of the matters discussed with regulators. No schedules should be agreed to or documents submitted, without first obtaining express written approval from DPR for each request.

6.4 Insurance Broker Risk Management Representative

The insurance broker’s risk management representative may periodically audit site operations to evaluate compliance with the elements of this DPR EHSP and all other federal, state and local regulations. Workers throughout the jobsite should cooperate with this audit as necessary.

7.0 Chemical & Environmental Management

DPR has developed a Chemical and Environmental Management Plan as a part of our EHSP. DPR is committed to reducing the amount of chemicals used and stored on site, the amount of waste generated on



each site and continually increasing the amount of materials recycled. DPR's intent is to protect workers, the public and the environment through coordinated job preplanning and systems review. Under all circumstances, it is the intent of DPR to comply fully with the requirements of EPA, federal, state and local regulations. Subcontractors are contractually required to fully comply with DPR's Chemical and Environmental Protection Plan as well as any site specific environmental, health and safety regulations. *For further information, refer to the Chemical and Environmental Management Section.*

7.1 Community Right to Know & Understand

Community Right to Know and Hazardous Spill reporting requirements shall be followed on all DPR projects. All subcontractors shall provide DPR with all appropriate information to fulfill these requirements. DPR feels strongly that providing accurate information to area emergency response personnel and regulatory agencies will result in significant safety benefits for the facility, the community and responding personnel. *Refer to the Emergency Procedures Section for additional guidelines.*

8.0 Emergency Response

The potential for emergencies exists at any construction site. Emergencies happen quickly, unexpectedly, and require advanced planning to facilitate an immediate and appropriate response. Therefore, a DPR/Subcontractor team is organized and trained on each project. This team's function and purpose is to effectively and efficiently handle the broad range of potential emergencies on each project. *Refer to the Emergency Procedures Section for further guidelines.*

9.0 DPR Crisis Management Plan

The purpose of the DPR Crisis Management Plan, contained within the Emergency Procedures Section, is to provide a systematic approach to managing a crisis without causing a major disruption to normal activities. General procedures, site and area-specific information, as well as notification and communication systems are all outlined in detail for reference and use at the time of the crisis if necessary. Responders should be trained and able to accurately and effectively manage crisis situations. *Refer to the Emergency Procedures Section for additional guidelines.*

10.0 Responsibilities for Safety and Loss Control

10.1 Overview

The primary objective of the DPR EHSP is to coordinate the elimination or reduction of risk associated with the construction of the project. Associated missions are to promote safe work practices/ behaviors, prevent accidents, prevent worker injuries, prevent damage to property, promote maximum efficiency and effect savings by reducing unplanned business interruptions.

Planning for safety starts with project design and continues through purchasing, fabrication and construction in all phases of the project. Practical steps should be taken to maintain an Injury Free Environment. All subcontractors are responsible for preventing accidents and for continual safety and health related training for their workers.

The objective of this section is to establish the concept that safety and health should be addressed throughout the entire project. The prevention of accidents and protection of the public and property are company values and an integral part of our success. All safety issues should receive active support and participation by the entire project team. Active participation by the management of DPR, subcontractors, tiered subcontractors and all workers will make the program effective and successful by coordinating the participants' efforts in performing the following tasks:

- Provide a safe environment in which workers can perform high quality work
- Use job hazard analysis and pre-task safety planning as a tool to reduce injury to persons, the public and property

- Conduct jobsite safety audits to identify and abate unsafe work practices, behaviors and unsafe conditions
- Protect the public and property potentially affected by DPR construction sites
- Educate and train workers through:
 - o site specific safety orientations
 - o safety meetings
 - o task specific safety training, (e.g., hazardous communications, excavation & trenching safety, confined space entry, fall protection, etc.)
 - o mandatory personal protective equipment (PPE) programs
 - o immediate injury reporting and effective record keeping to maintain current accident experience and trends analysis
 - o accident investigation lessons learned

10.2 DPR Regional Leadership

DPR Regional Leadership is responsible for construction management services for the region and for:

- Encouraging, reinforcing and modeling DPR culture, including IFE initiatives
- Participating in the development and assessment of EHS leading indicators
- Reviewing and approving project corrective action plans
- Instituting accountability when action plans and culture are not maintained
- Has the authority to stop any operations that pose a potential threat

10.3 DPR Project Executive

DPR Project Executives are responsible for construction management services for the region and for:

- Encouraging, reinforcing and modeling DPR culture, including IFE initiatives; promoting and supporting IFE culture
- Assist in determining if contract documents and specifications support the project's safety missions and objectives
- Reviewing and approving project corrective action plans; assisting project team with holding subcontractors accountable for implementation
- Instituting accountability when action plans and culture of applicable jobs are not maintained
- Has the authority to stop any operations that pose a potential threat
- Participating in subcontractor selection during the preconstruction prequalification process
- Periodically auditing subcontractor's safety plans for compliance with the DPR EHSP; assisting project team with holding subcontractors accountable for implementation
- Participating in fact finding, root cause analysis and the implementation of corrective actions associated with injury/incident investigations on applicable projects
- Documenting monthly jobsite safety audits
- Participate periodically in jobsite safety meetings
- Supporting DPR EHS personnel and cooperating with all designated personnel in obtaining corrective actions necessary to comply with the DPR EHSP
- Has the authority to stop any operations that pose a potential threat

10.4 DPR Project Manager

The DPR Project Manager is responsible for the construction management services for the project and for:

- Determining if contract documents and specifications support the project's safety missions and objectives
- Participating in subcontractor selection during the preconstruction prequalification process
- Periodically auditing subcontractor's safety plans for compliance with the DPR EHSP
- Participating in subcontractor pre-construction safety (kick-off) meetings

- Monitoring the implementation of job hazard analysis and pre-task planning on their projects
- Being aware of loss control and public protection requirements of the project
- Participating in fact finding, root cause analysis and the implementation of corrective actions associated with injury/incident investigations
- Documenting weekly jobsite safety audits
- Periodically participating in weekly jobsite safety meetings
- Facilitating craft feedback luncheons
- Supporting DPR EHS personnel and cooperating with all designated personnel in obtaining corrective actions necessary to comply with the DPR EHSP
- Has the authority to stop any operations that pose a potential threat
- Promoting and supporting IFE culture

10.5 DPR Superintendent

It is the responsibility of DPR Superintendents to oversee safety on jobsites. Their EHS responsibilities include:

- Overseeing the planning and execution of all work in compliance with the DPR EHSP and contract specifications
- Being aware of loss control and public protection requirements identified in the safety specifications of the contract documents
- Completing daily jobsite safety audits and reviewing completed jobsite safety audits to ensure identified hazards are addressed in a timely manner
- Participating and subcontractor pre-bid, pre-construction safety (kick-off) meetings
- Monitoring and participating in job hazard analysis and pre-task planning
- Requiring supervisors and workers to use personal protective equipment in accordance with the DPR EHSP, subcontractor's safety program, local, state and federal safety regulations
- Participating in fact finding, root cause analysis and the implementation of corrective actions associated with injury/incident investigations
- Ensuring Grab & Go packets and Root Cause Analysis are accurately completed and forwarded to designated individuals
- Participating in and encouraging stretch & flex, weekly tool box/tailgate safety meetings and evaluating their effectiveness
- Taking appropriate action to address identified unsafe conditions and practices and document corrective actions
- Supporting DPR EHS and cooperating with all designated project safety personnel in obtaining corrective actions necessary to comply with the DPR EHSP
- Has the authority to stop any operations that pose a potential threat
- Promoting and supporting IFE culture

10.6 DPR Regional EHS Professional

The DPR Regional EHS Professional (DPR EHS) is considered to be the regional EHS program administrator and has the authority delegated by DPR Corporate EHS to implement and promote safety. Duties of DPR EHS include:

- Has the authority to stop any operations that pose a potential threat, up to and including the removal of personnel and/or subcontractor(s); and termination of DPR workers as necessary
- Help familiarize DPR and subcontractor project personnel with the DPR EHSP. These individuals should be familiar with safety and health hazards to which all workers may be exposed, as well as applicable laws, regulations and safety rules and policies
- Support project management in achieving an injury free environment
- Help assure that all workers are trained in accordance with applicable requirements

- Help assure observation, inspection, recognition, evaluation and abatement of hazards are conducted on a continuing basis
- Help monitor the subcontractor selection process and adherence to established environmental safety and health guidelines
- Assist with the continual development of new methods for abating hazards
- Help assure hazards are addressed in a timely and effective manner
- Perform regular jobsite safety audits of work site activities, share findings, corrective actions and corrective plans with DPR project management
- When necessary, DPR EHS should every resource necessary, including escalation, to ensure the maintenance of an injury free environment
- Administrative duties may include maintaining regional safety files such as safety training documentation, applicable medical certifications, environmental testing and special associated training, investigation reports including near-misses, injury summaries, leading indicator data and other safety and health documentation
- DPR EHS also has the responsibility for overseeing development, implementation and maintenance of the project's safety program which may include, but is not limited to:
 - o encouraging subcontractors to incorporate the requirements of the DPR EHSP into their safety programs and safety orientation if theirs are less protective than those of DPR
 - o expediting corrective actions to address any observed or potential safety exposures to workers
 - o Helping assure that DPR project management and safety coordinators are continuously monitoring DPR and the subcontractor's safety performance and expediting corrective actions
 - o overseeing the development and implementation of emergency response procedures, and helping to assure that DPR and subcontractor's personnel are trained to handle onsite emergencies
 - o assisting with the development of project missions, milestones and indicators

10.7 *DPR MEPF Coordinator*

- Has the authority to stop any operations that pose a potential threat or environmental impact up to and including the stoppage of work and/or removal of any subcontractor(s) as necessary
- Documenting weekly jobsite safety audits
- Help familiarize DPR and MEPF subcontractor project personnel with the DPR EHSP. These individuals should be familiar with safety and health hazards specific to the MEPF trade related tasks to which workers may be exposed, as well as applicable laws, regulations and safety rules and policies
- Support project management in achieving an injury free environment
- Participate in the oversight of the coordination, installation, and testing of MEP systems in compliance with the DPR EHS and contract specifications.
- Provides oversight and coordination for the project's hazardous energy control program in accordance with DPR EHSP requirements.
- Help assure that provisions have been taken to address hazards associated with testing media, methods, and the proximity to other workers and adjacent property.
- Help assure observation, inspection, recognition, evaluation and abatement of hazards specific to the MEPF trades are conducted on a continuing basis
- Assist in the MEPF subcontractor selection process and adherence to established environmental safety and health guidelines when applicable
- Help assure hazards related specifically to MEPF trades are addressed in a timely and effective manner
- Participate in MEPF subcontractor pre-bid, pre-construction (kick-off) meetings
- Periodically monitor and participate in job hazard analysis and pre-task planning
- Actively participate in fact finding, root cause analysis and implementing corrective actions to prevent further occurrences on all MEPF trade related injury/incident investigations
- Attend and participate in jobsite safety meetings

- o Assist the DPR Supervisor/Foreman by participating in daily pre-shift coordination meetings to include review of day's MEPF specific activities and associated hazards
- Assist the project safety coordinator with the weekly toolbox/tailgate and safety meetings as they relate to specific MEPF related hazards
- Help coordinate with owners facility personnel

10.8 DPR Project Engineer

The DPR Project Engineer assists the DPR Project Manager with his/her responsibilities for construction management services for the project. This person should:

- Complete weekly jobsite safety audits
- Participate in subcontractor pre-bid, pre-construction (kick-off) meetings
- Periodically monitor and participate in job hazard analysis and pre-task planning
- Assist with jobsite safety startup, safety orientations and craft feedback luncheons
- Participate in fact finding, root cause analysis and implementing corrective actions to prevent further occurrences on all injury/incident investigations
- Attend and participate in jobsite safety meetings

10.9 DPR Supervisor/Foreman

The DPR Supervisor/Foreman should interface daily with their workers. Therefore, the DPR Supervisor/Foreman should have a major influence on the effectiveness of the subcontractor's safety program and accident experience. Each Supervisor/Foreman's construction safety responsibilities should include:

- Reporting unsafe acts and conditions to the project team and/or safety coordinator for prompt corrective action and stop all life threatening situations immediately upon knowledge
- Training and instructing workers in safe work practices for the tasks they are assigned
- Pre-planning activities to help assure workers are properly trained in applicable safety requirements
- Helping ensure crew participation in pre-task planning and assist with understanding related job hazard analyses
- Participating in daily pre-shift coordination meetings to include review of day's activities and associated hazards
- Helping ensuring availability of and enforce the proper use of jobsite tools and PPE
- Monitoring the work area for unsafe acts and conditions and addressing all identified conditions in a timely manner
- Setting a good example for workers
- Completing weekly jobsite safety audits
- Help ensure all incident reports are properly completed and submitted to appropriate parties
- Participating in fact finding, root cause analysis and the implementation of corrective actions associated with injury/incident investigations
- Promoting and supporting IFE culture
- Leading daily Stretch & Flex and tool box/tailgate safety meetings with the crew to:
 - o encourage participation
 - o discuss observed accident trends and causes
 - o plan construction safety into crew's work activities
 - o take action to correct safety-related concerns

10.10 DPR Project Safety Coordinator (Project Specific)

The DPR Safety Coordinator's primary responsibility is to ensure that observed unsafe acts and unsafe conditions are addressed in a timely manner. This person should:

- Report unsafe acts and conditions to the worker's supervisor, project team and/or subcontractor safety coordinator for prompt corrective action and stop all life threatening situations immediately upon knowledge
- Orient all new DPR workers according to the Project Site-Specific Safety Orientation
- Make daily job site safety audits
- Help facilitate daily safety coordination meetings with subcontractor safety coordinators (as applicable)
- Assist with the weekly toolbox/tailgate and safety meetings and help ensure the appropriate materials are provided, as well as:
 - o review meeting reports for attendance
 - o help implement required training programs for workers
 - o provide necessary information for the obtaining of motor vehicle records for all crane operators on site
- Report to the subcontractor, the names of individuals and their supervisors who are continually observed to violate construction safety requirements
- Report all injuries immediately to DPR project management and regional EHS
- Participate in fact finding, root cause analysis and resolution on all injury/incident investigations, and ensure applicable paperwork is forwarded to appropriate parties
- Promote and support IFE culture
- Keep on file the following:
 - o worker safety orientation logs
 - o the subcontractor's daily safety audit reports
 - o crane documentation
 - o weekly tool box/tailgate safety meeting reports
 - o miscellaneous documentation, i.e., confined space permits, excavation/trenching assessments/logs & inspections, demolition engineering surveys, etc.
 - o updated chemical management plan, including chemical inventory lists and applicable Safety Data Sheets

10.11 DPR Field Office Coordinator (*Project Specific*)

The DPR Field Office Coordinator will assist Project Management with:

- Support and promote an IFE culture
- Compiling and filing of safety related paperwork
- Support with the tracking of submittal of SDSs, subcontractor safety plans, insurance documents and completion of pre-construction meetings
- Assisting with jobsite safety startup and craft feedback luncheons
- Complete period jobsite safety audits

10.12 Subcontractor Responsibilities

It is DPR's expectation that all subcontractors' office and field based personnel should actively participate in creating an injury free work environment on all DPR projects.

The subcontractor has overall responsibility for accident prevention and implementation of this DPR EHSP for all their personnel on site, including their tier-subcontractors, vendors and suppliers. Subcontractor supervision may be required to be on-site and actively managing any tiered subcontractor, vendor or supplier for the duration of their scope of work. Subcontractors may be required to provide full time Safety Engineer(s) when actual or expected manpower reaches 20 employees, the contract value exceeds 2 million dollars, or hazardous activities or behaviors are determined to be present by DPR project management. Where applicable, subcontractor Safety Engineers will be approved in advance by the DPR project management. Additionally, the subcontractor may be required to identify to DPR project management one craft level Safety Coordinator per every 10 employees. When a Safety Coordination has not been established, subcontractor



field supervision will be responsible for the safety of their workers and should be considered the Safety Coordinator for their scope of work. Documented Safety Audits are required - *Reference Pre-Planning Forms*

Safety Engineers/Safety Coordinators may be approved in advance by the DPR project management and/or DPR EHS. Subcontractor Safety Engineers/Safety Coordinators may be required to submit a resume with relevant project safety experience. At a minimum, Safety Engineers/Safety Coordinators will have an experience level of OSHA 30-hour training, as well as competent person designations for applicable scopes of work, authorized by their company.

Subcontractors shall submit a copy of their company's safety program, applicable SDSs, insurance and applicable training verification prior to beginning work.

10.12.1 Subcontractor's Project Manager

The subcontractor's Project Manager is responsible for:

- Planning and monitoring all work performed for compliance with the objectives of the DPR EHSP, subcontractor's safety program, federal, state and local safety and health regulations
- Authorizing immediate correction of any existing construction safety related concerns
- Fully supporting designated safety coordinator and cooperating with all designated project safety personnel in obtaining corrective actions necessary to comply with the DPR EHSP
- Completing weekly safety audits
- Participating in pre-task planning and subcontractor pre-construction (kick-off) meetings
- Participate in fact finding, root cause analysis and resolution on all injury/incident investigations
- When requested, attend special construction safety meetings

10.12.2 Subcontractor Superintendent/Supervisor/Foremen

The same as DPR Superintendent/Supervisor/Foremen, plus:

- Attend weekly subcontractor coordination/safety meetings
- Complete daily reports

10.12.3 Subcontractor's Safety Engineer (Project Specific)

The Subcontractor Safety Engineer's primary responsibility is to ensure that observed unsafe acts and unsafe conditions are addressed in a timely manner. This person should:

- Report all injuries immediately to DPR project management
- Perform continuous safety audits of all their respective trade contractors and their subcontractors' work areas throughout the entire workday and take immediate action to eliminate all unsafe acts and/or conditions. These observations, along with corrective actions taken should be reported to the appropriate member of DPR project management and their own management in a timely manner. These audits will be documented and submitted to DPR project management.
- Serve as technical advisors to their project management team on safety and health planning, training and problem resolution issues. Ensure that prior to the commencement of any work activity; Supervisors/Foremen review each task assignment with affected employees to ensure a comprehensive understanding of the safety requirements and precautions to be followed while performing this work
- Have the right and authority to stop hazardous work activities being performed until necessary corrective actions are taken, regardless of company affiliation
- Ensure that appropriate personal protective equipment is provided and its use enforced
- Enforce their company's safety program and disciplinary procedures
- Accompany DPR's supervisory personnel as directed and perform joint inspections of work areas and activities
- Orient all new subcontractor personnel to the site's safety program prior to work commencement

- Participate in fact finding, root cause analysis and resolution on all injury/incident investigations, and ensure applicable paperwork is forwarded to appropriate parties
- Attend and participate in safety coordination meetings
- Assist with the weekly toolbox/tailgate and safety meetings and help ensure the appropriate materials are provided, as well as:
 - Review meeting reports for attendance
 - Help implement required training programs for workers
 - Provide necessary information for the obtaining of motor vehicle records for all crane operators on site

10.12.4 Subcontractor's Safety Coordinator (Project Specific)

The Subcontractor Safety Coordinator's primary responsibility is to ensure that observed unsafe acts and unsafe conditions are addressed in a timely manner. This person should:

- Report all injuries immediately to DPR project management
- Orient all new subcontractor personnel to the site's safety program prior to work commencement
- Make daily job site safety audits and provide copies of documentation to DPR project management
- Participate in fact finding, root cause analysis and resolution on all injury/incident investigations, and ensure applicable paperwork is forwarded to appropriate parties
- Attend and participate in safety coordination meetings
- Provide appropriate materials for those conducting weekly tool box/tailgate meetings or safety meetings, as well as:
 - o periodically conduct tool box/tailgate meetings
 - o implement required training programs for workers and supervisors
 - o provide necessary information for the obtaining of motor vehicle records for all crane operators on site

10.13 Everyone's Responsibilities

- Work according to good safety practices as posted, instructed and discussed.
- Comply with DPR EHSP and their contractor's safety program, whichever is more stringent
- Use all required safety devices
- Report any unsafe situation or act to supervisor immediately
- Maintain a clean and safe work area
- Come to work alert and free of any impairment that may affect safety
- Promote and support the IFE: agree to be held accountable for their safety, and the safety of others

10.14 Safety Compliance by Owner and Owner's Separate Contractors

The Owner should cause its own employees and those of its separate contractors to comply with this EHSP while on the DPR project site. DPR reserves the right to remove from the Project Site any employee of Owner or Owner's separate contractors who fail after one notification from the contractor to comply with the Contractor's safety plans and policies.

11.0 Safety Recognition Program

The Project Safety Recognition Program should be created to provide encouragement and recognition to workers that take *ownership* and act on the upstream indicators established. This is a multi-leveled program, separate from worker appreciation programs. The Recognition Program is directed at promoting workers and companies to be proactive and interactive in creating an Injury Free Environment. When workers place a value in their work, excellence in safety follows naturally. To help accomplish this goal, appropriate awards and recognition should be developed.

A Recognition Program should focus on elements that drive safety and quality of the project forward as it relates to achieving an Injury Free Environment. Some areas where recognition may occur include:



- Pre-task Planning
- Jobsite Safety Auditing
- Identification and Correction of hazards
- Employees going above and beyond in their safety efforts
- Opportunities for Improvement (project specific upstream indicator development)
- Safety quizzes
- Consistency Survey results
- Etc.

12.0 Project Disciplinary Procedures

DPR reserves the right to discharge "at will" from the project. Workers found performing work in an unsafe manner that could endanger either themselves or other workers will be subject to disciplinary action or termination from the project.

DPR project management will determine the course of action best suited to the circumstances. The steps to be taken at a minimum may include the following:

12.1 Verbal Warning Citation

A documented verbal warning may be issued to the worker. If necessary, the worker may be placed on probation or removed from the project.

12.2 Written Warning Citation

Any written warning should clearly state the safety policy that was violated and corrective actions necessary. A probationary period may be a part of the written warning. Corrective actions may include a training session related to the safety policy violated. This training should be completed within the probationary period, if applicable. Documentation should be maintained.

In the case of probationary action, DPR project management should meet with the worker and their supervision to determine if the worker has achieved the required level of performance necessary to return to unrestricted duty.

12.3 Zero Tolerance

Zero Tolerance items may not follow the typical course of discipline and may result in immediate removal from the project. The following list details several of the zero tolerance items:

- Horseplay
- Drug and alcohol use, possession, or distribution
- Fall protection violations
- Lockout/Tagout violations
- Confined space entry procedure violations
- Harassment or discrimination
- Depending upon the severity of the infraction, we reserve the right to waive the "warning" for knowingly violating a safety procedure

12.4 Termination

DPR reserves the right to remove personnel from the project for any safety related infractions or attitudes. Nothing will relieve the subcontractor of liability and/or obligation under the requirements of state or federal OSHA, other applicable federal, state, and local requirements and the DPR and subcontractor's EHSP, whichever is more stringent.



13.0 Drug, Alcohol, & other Prohibited Articles Corporate Policy

13.1 Scope

This policy applies to all DPR Construction employees and temporary employees. This policy also applies to all employees of subcontractors and tiered subcontractors performing services for DPR Construction. Such employees of subcontractors and tiered subcontractors are considered “employees” within the meaning of this Policy.

13.2 Purpose

It is the policy of DPR Construction not to permit the use, sale or possession of any illegal intoxicating and/or other narcotic substances on company property, within company vehicles, on any jobsite or at any time while receiving compensation from the company. Violating this policy may result in disciplinary action, up to and including immediate suspension or termination.

Note: The ONLY exception to this rule is the limited, responsible use of alcohol outside of work hours at DPR social events in the office.

Therefore, DPR prohibits the use, possession or distribution on its premises, facilities or work places of any of the following:

- Alcoholic beverages
- Intoxicants and narcotics
- Illegal or unauthorized drugs (including marijuana)
- "Look-alike" (simulated) drugs
- Related drug paraphernalia

No employee should report to work and/or operate vehicles or machinery under the influence of any drug, alcoholic beverage, intoxicant or narcotic or other substance (including legally prescribed drugs and medicines) which will in any way adversely affect their working ability, alertness, coordination, response, or pose a direct threat to the safety of others on the job.

Employees who are medically authorized to use drugs or other substances which may impair job performance are responsible to determine from their physician whether or not the substance can impair job performance. When possible, the employee should report the fact of such impairment to DPR project management and provide written authorization from their physician regarding the work that they are able to perform while using such authorized drugs.

DPR has the right, in its discretion, to report to law enforcement officials the use, possession or distribution of any substance named in the first paragraph of this policy to law enforcement officials and to turn over to the custody of law enforcement officials any such substances on company property.

This policy and acknowledgment is not intended to and does not constitute a contract of employment with DPR. Employment with DPR is “at will,” and either the individual or DPR may terminate employment at any time, and for any or no reason. No supervisor or manager has any authority to make any statements or representations that change or conflict with the at-will status of employment with DPR, or that change or conflict with any of the provisions of this policy. This Policy supersedes and revokes all previous practices, procedures, policies, and other statements of the Company, whether written or oral, that modify, supplement, or conflict with the Policy. This Policy may be amended as determined appropriate by DPR.

13.2.1 Smoking / Electronic Cigarettes / Vaporized Cigarettes

Smoking, of any kind, including but not limited to electronic cigarettes should:

- comply with owner requirements,
- should be prohibited Inside any building and in any public or eating area



- is prohibited in the vicinity of hazardous operations or locations and may be further restricted to designated areas only

"No Smoking" signs should be conspicuously placed in hazardous areas and no-smoking rules strictly enforced. If specified smoking areas are provided, suitable ash receptacles should be furnished. Cigarette and cigar butts should only be disposed of into designated ash receptacles.

13.3 *Dissemination of Policy*

All DPR employees should receive a copy of this policy. Subcontractor personnel should review these requirements during site specific safety orientations, and should be provided a copy of this Policy as requested by their employer. Both DPR and subcontractor personnel will be required to sign appropriate acknowledgment and receipt documents.

13.4 *Confidentiality*

Appropriate and reasonable steps should be maintained to ensure the confidentiality of information generated by the implementation and enforcement of this policy and these procedures. The controlling employer is responsible for maintaining the confidentiality of all information generated by the implementation and enforcement of this policy and these procedures for their own workers.

It is the responsibility of subcontractors and tiered subcontractors to present proof that affected workers have completed substance abuse testing; further, upon receipt of testing results, the subcontractor and tiered subcontractors are required to provide DPR with proof that affected workers have passed their substance abuse screening. DPR reserves the right to require proof of negative results prior to that individual being allowed back on the project premises.

13.5 *Right to Search*

To enforce this policy, DPR reserves the right to search any company property and any personal effects for contraband. Personal effects include but are not limited to vehicles, lockers, baggage, lunch boxes, tool boxes, clothing, etc. An individual who has notice of this rule and enters the project premises is deemed to consent to this "right to search" procedure. Searches should be conducted on an as-needed basis as determined after consultation with subcontractor management. A worker representative and/or other witnesses, who may include law enforcement officers, should be present at all searches conducted by DPR.

Consent to inspection under this policy is a condition of employment or continued employment. Any worker who refuses to submit to a search as described in this policy is subject to disciplinary action, up to and including immediate discharge, by the controlling employer. Contractors and/or workers who refuse to submit to a search are subject to removal and denial of future access to the project premises.

Any worker on the project premises in possession of contraband is subject to disciplinary action, up to and including immediate termination from the project with denial of future access to the project premises.

13.6 *Substance Abuse Testing*

Substance abuse testing is applicable under any of the following circumstances:

- *Post-Accident/Post Incident:* Any employee(s) involved in or contributing to an accident or incident may be required to submit to a drug screen. The drug screen should limit Post-Accident/Post-Incident testing to situations in which employee drug use is likely to have contributed to the accident/incident.
- *Reasonable Suspicion Testing:* Should there be a reasonable suspicion (including near miss incidents) that an employee is in violation of this policy, the employee will be required to submit to substance abuse testing to determine the presence or use of drugs and/or alcohol. However, DPR reserves the right to determine whether reasonable suspicion exists.
- *Pre-Placement Testing:* Pre-placement testing may be required as a condition of employment.

13.7 Testing Procedure

Drug and alcohol testing and sample collection should be performed by a certified laboratory which will be expected to comply with scientifically accepted analytical procedures. A Medical Review Officer should oversee drug testing results to ensure validity of the results and preserve employee's rights and confidentiality. Testing procedures should include the following:

- Test samples should be collected under reasonable and sanitary conditions
- Test samples should be labeled, handled, transported and stored in a way reasonably designed to preclude sample contamination, adulteration or misidentification
- At the time of testing, a person should be given the chance to provide any information that may be relevant to the test, including whether the person is taking prescription medication or undergoing any medical treatment that may affect the result of the test. Additionally, individuals should be given the chance to explain a positive drug test result in a confidential manner with a Medical Review Officer.
- An employee's positive test result should be retested for confirmation by use of a different chemical process than was used in the initial screen
- DPR employees and applicants may request a written copy of their drug and/or alcohol test results. These results are confidential, and should be released only to the tested employee or applicant, persons designated by the tested employee in writing; persons designated by the Company to receive and evaluate test results or hear any explanation regarding a positive test result and other persons authorized by law.
- Failure to comply with testing requirements constitutes a refusal to comply with this policy. Should an employee fail to comply, disciplinary action will result, up to and including termination of employment or removal from the project.

13.8 In the Event Problems Arise from the Collection Process

Any individual affected by this policy who believes that his/her specimen was not collected in accordance with established procedures must report any deficiencies to DPR project management.

13.9 Confirmed Positive Result – DPR

If the employee fails the required test, disciplinary action will result. The offending employee will not be allowed to work until a negative test result is provided and confirmation the employee has entered into an approved drug/alcohol rehabilitation program as determined by DPR. Their return to work will begin with a minimum 90-day probationary period in which the employee may be randomly tested. Should an employee test positive during the probationary period or any time thereafter, their employment will be terminated and not eligible for re-hire.

Requirements may vary depending upon state law.

13.10 Confirmed Positive Result – Subcontractor

If subcontractor employee fails the required test, he/she will be immediately removed from the project.

13.11 Financial Obligation of the Control Employer

The cost of time, transportation, and the actual testing of workers will be the responsibility of the controlling employer.

13.12 Training

The company will take the necessary steps to assure that DPR supervisors and management personnel are trained to recognize appropriate objective evidence that indicates that an employee may be under the influence and to administer this policy in a consistent, confidential, and intelligent manner.

13.13 Anti-Retaliation Policy

DPR's post-accident drug testing policy encourages employees to report any work-related illnesses and injuries without fear of reprisal. The intent is that all employees are trained on our Anti-Retaliation Policy during our orientation process. Drug testing is done with the intent to comply with the requirements of a state or federal law or other regulations. Drug testing done in accordance with a state law workers' compensation program, whether mandatory or voluntary, is acceptable.

14.0 Accident, Injury and Incident Management

14.1 Medical Management

Each subcontractor and tiered subcontractor should take a proactive approach to managing injuries and incidents on our projects. Each subcontractor and tiered subcontractor should provide at a minimum, one current CPR/First Aid trained worker per crew. Additionally, the following should be provided:

- A designated medical provider for both business and after hour care should be identified prior to mobilization
- Injured individuals should be accompanied to the designated medical facility, regardless of severity, by their foreman or company designee
- An on-site first aid kit, inspected weekly and contents maintained
- Aggressive return to work program

14.2.1 Accident Reporting

All on-site accidents, injuries or incidents shall be reported to DPR project management and regional EHS immediately. All accidents, injuries or incidents should be thoroughly investigated regardless of severity. Completion of appropriate forms, as defined in the Incident Reporting Section, should be done in a timely manner.

14.2.2 Accident Investigation

All accidents, injuries or incidents should be thoroughly investigated regardless of severity. It is DPR's intent to complete the initial accident investigation as soon as possible with the target time within the first 24 hours. Identification and review process of root causes should also be completed. Corrective actions, persons responsible for corrective actions, and date of completion should be established. Corrective actions should be tracked through completion. Lessons learned should be shared with the project, regionally and globally.

Copies of all accident investigation documentation shall be submitted to the DPR Regional EHS and DPR's Claims Administrator. Injury notification to OSHA should be made as necessary and coordinated with DPR project management and the Corporate EHS Director.

14.3 Recordkeeping

Subcontractors will be required to submit copies of the following documentation to DPR project management for appropriate distribution and/or retention:

- All environmental related reports and documents
- Accident investigation reports
- Pre-task plans
- Equipment inspection reports
- Certificates of insurance
- Training records
- Inspection reports with documentation of abatement methods
- Safety related meeting agendas and minutes
- Subcontractor written safety programs
- Safety data sheets

- Safety plans
- Worker termination notification
- Security reports
- Written safety violations
- Any other safety and health related reports and documents

14.4 DPR Return To Work Program

DPR Construction has developed a program designed to assist workers who are temporarily unable to perform their usual work activities due to a work related illness or injury. This program is called the “Return to Work Program.” This includes a team effort, including injured workers, their attending physician, the insurance carrier and project management.

Studies show that return to work programs are therapeutic and help speed the recovery process. In addition, injured employees stay “in touch” with the work environment and with fellow employees, which helps to facilitate a smooth and speedy transition back to their normal job. This also creates an opportunity for cross training and developing new skills.

If the doctor determines that the employee qualifies for our Return to Work Program, the doctor will complete the appropriate forms indicating the restrictions and conditions for transitional work. We will then attempt to provide an alternative work position until the employee is able to resume full duties. All alternative work is temporary in nature and is designed to facilitate a return to full duties as soon as possible. Alternative work positions may be offered at any project, and/or any shift. Alternative work positions may also be offered on a varied schedule.

This program is not designed as a substitute for reasonable accommodation under any applicable federal and/or state regulations, such as the Americans with Disabilities Act, The Rehabilitation Act 1973 or other applicable laws.

To preserve the ability to meet company needs under changing conditions, we reserve the right to revoke, change or supplement these guidelines at any time with or without written notice. No permanent employment for any term is intended or can be implied by this policy. But, while in effect, all doctor recommended restrictions will be followed. The site management and the claims manager, on a two-week basis or after a doctor’s visit, will review with the worker the availability of continued alternative work.

The policies and procedures in this program are not intended to be contractual commitments and they shall not be construed as such. This policy is not intended as a guarantee of continuity of benefits or rights. *Refer to the Incident Reporting Section for further details.*

15.0 Special Site Conditions

Subcontractors should develop specific safety procedures for the following special site conditions, as applicable. DPR project management should review these procedures with affected subcontractors. The subcontractor will be responsible for ensuring their personnel understand and follow these procedures. Specific safety procedures should be periodically reviewed and updated. Special site conditions include, but are not limited to:

- Environmental or biologically sensitive issues
- Noise sensitive issues
- Vibration sensitive issues
- Contaminated soil and/or water issues
- Owner’s employees/members of the general public in close proximity to site
- Power/gas/sewer/fiber-optic/communications lines in close proximity to site
- Critical operations in close proximity to site
- Hazardous operation in close proximity to site
- Orientation/training/medical screening/security/drug screening requirements

- Unusual working hours
- Traffic/truck haul route considerations
- Manlift(s) and/or personnel hoists
- Tower crane(s)
- Deep excavation(s) of 20 feet or more
- Demolition
- Live electrical tie-ins
- Storm water discharge
- Safety Nets and Helicopter operations (refer to the Elevated Work Section for further guidelines)

16.0 Sanitation and Housekeeping

All contractors, subcontractors, and tiered subcontractors are required to provide, at minimum, basic general requirements for health and sanitation for their workers. These include, but are not limited to:

- Potable water
- Portable toilets
- Flush toilets & hand wash facilities
- Temporary power
- Temporary and task lighting
- First aid kits
- Eye wash stations and/or safety showers
- Portable fire extinguishers
- Adequate covered garbage containers

Additionally, the site, work areas and all premises occupied by the contractor, subcontractor and tiered subcontractor will be maintained in a clean, healthy and sanitary condition. Daily housekeeping is required. Refer to the Sanitation & Housekeeping Section for further guidelines and requirements.

17.0 Project Signs and Barricading

Signs will be placed in positions where they will convey their messages most effectively and placement must therefore be accommodated to enhance design and alignment. Safety signage is important to keep constant visual reminders in front of workers. Refer to the Public Protection Section for further guidelines and requirements.

18.0 Unmanned Aircraft (Drones)

The use of drones on DPR projects has increased exponentially, we have begun to use them for: daily logistics planning, existing conditions documentation, underground utility as-built drawings, change documentation, exterior skin production tracking, as supplements to owner billings, and are beginning to develop map systems for strategic planning purposes. DPR was granted a 333 exemption from the FAA for commercial use of drones in construction under the following guidelines;

- Skycatch and KESPRY drones are approved for flight, additional drones are under review with the FAA.
- Flights must stay below 400 ft.
- The person controlling the drone must have a pilot license
- A second contact needs to be present during the flight (next to the pilot) to act as a spotter
- On-site personnel must be informed of the flight
- The owner must approve the flight

In addition to the bullet points above additional requirements are listed in the details of the 333 exemption which can be found on the toolbox. DPR Project Management should sign off prior to any flight

19.0 Additional Requirements and Resources



Please refer to the various appendices to this EHSP. Many of the items covered independently in each section have wide application to a variety of the tasks and hazards encountered on jobs. Additional resources are available to you should you have further questions including the entire DPR EHSP, DPR Insight, federal and state regulations, articles published by professional organizations, documents provided by manufacturers, and other members of your project team. Our goal is acting on upstream leading indicators in the achievement of an Injury Free Environment.

***INJURY
FREE
ENVIRONMENT

LIVE IT!***



DPR EHSP Acknowledgment Form

It is DPR's expectation that all subcontractors' office and field based personnel should actively participate in creating an injury free work environment on all DPR projects. As such, I recognize that our company has the overall responsibility for accident prevention and implementation of this DPR EHSP for our personnel on this project site, including our vendors and suppliers. I acknowledge that I have been provided with a copy, been informed, and been given the opportunity to ask questions regarding the DPR EHSP.

Project Name:		Job #:	
Subcontractor Company Name:		Date of Receipt:	
Project Manager:			
Signature:		Date:	
Supervisor/Foreman:			
Signature:		Date:	
Safety Professional:			
Signature:		Date:	

Note: Copy filed in project safety files

Section 01

Bloodborne Pathogens

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Bloodborne Pathogens

1.0 Introduction

When a work activity may reasonably result in a worker's exposure to human blood, its' components, or products made from human blood, a plan should be designed and implemented to protect the worker, and others, from unprotected exposure. This (infection control) plan is needed to protect employees who handle, store, use, process, or dispose of blood or other potentially infectious materials. This plan complies with OSHA's requirements and recommendations and includes information regarding personal protective equipment, housekeeping, training, and procedures for reporting exposures.

Some examples of where training may be needed include (but are not limited to): Employees working in laboratories where infectious materials are utilized, hospitals, and those who administer on-site first aid.

Reference: Bloodborne Pathogens CFR 1910.1030. Other applicable Standards, guidelines, and regulations may apply.

Definitions

Blood: Human blood, human blood components, and products made from human blood

Bloodborne Pathogens (BBPs): Pathogenic microorganisms that are present in human blood and can cause disease in humans. These pathogens include, but are not limited to, hepatitis B virus (HBV) and human immunodeficiency virus (HIV).

Other Potentially Infectious Materials: (1) The following human body fluids: semen, vaginal secretions, cerebrospinal fluid, synovial fluid, pleural fluid, pericardial fluid, peritoneal fluid, amniotic fluid, saliva in dental procedures, any bodily fluid that is visibly contaminated with blood, and all body fluids in situations where it is difficult or impossible to differentiate between body fluids; (2) Any unfixed tissue or organ (other than intact skin) from a human (living or dead); and (3) HIV-containing cell or tissue cultures, organ cultures, and HIV- or HBV-containing culture medium or other solutions; and blood, organs, or other tissues from experimental animals infected with HIV or HBV.

2.0 Exposure Control Plan

Every affected subcontractor should be responsible for developing and maintaining an Exposure Control Plan designed to eliminate or minimize employee exposure. This plan should include the subcontractor developing a list of tasks that may or could reasonably be anticipated to involve occupational exposure to blood or other potentially infectious materials.

Further, each affected subcontractor, after identifying hazardous or potentially hazardous tasks should:

- Develop methods to control exposures
- Train applicable employees
- Medically evaluate applicable employees
- Maintain medical records

Each affected contractor should also ensure that a copy of the Exposure Control Plan be accessible to employees and should forward a copy of the plan to DPR project management and DPR prior to mobilization. DPR may review the plan and reserves the right to modify procedures if deemed unsafe or inadequate for preserving worker safety.

The Bloodborne Pathogen Exposure Control Plan should be periodically reviewed for compliance and updated as necessary. This is to occur annually and/or when conditions, tasks, employees involved or procedures change. All affected subcontractors are responsible for auditing their program and ensuring compliance. DPR should be notified of any changes to a subcontractor's Exposure Control Plan.

3.0 Employee Training Program

All attempts should be made to have the owner abate all known hazards prior to work in any hazardous areas; however; this may not be possible in all cases. If a task involves the handling of blood and other potentially infectious materials, employees must be trained to ensure knowledge of how those materials are to be contained, labeled and properly disposed. The necessity for proper housekeeping and personal hygiene techniques, including hand washing, should be emphasized.

Examples where Bloodborne Pathogen training may be needed include, but are not limited to:

- Work in laboratories or research facilities where infectious materials are present
- Demolition or tenant improvement work that may expose workers to bloodborne pathogens
- Those responsible for administering on-site first aid
- Sanitary or sewer work on live systems

Where BBP exposure is a hazard, worker training should take place prior to work assignment and at least annually thereafter. The person conducting the training should be knowledgeable in the subject matter covered by the elements contained in the training program as it relates to the workplace that the training will address. At a minimum, training for those workers covered by the Exposure Control Plan, should include:

- An explanation of the regulation
- Epidemiology and symptoms of bloodborne diseases
- Modes of transmission of bloodborne pathogens
- Review of applicable exposure control plan; access to the plan
- Methods for recognizing tasks and other activities that may involve exposure to blood and other potentially infectious materials
- Use and limitations of methods to prevent or reduce exposure
- Selection of, and types, proper use, location, removal, handling, decontamination and disposal of personal protective equipment
 - o hepatitis B vaccine information (including vaccination provided at no cost)
 - o Information on the appropriate actions to take and persons to contact in an emergency involving blood or other potentially infectious materials
- Post-exposure procedures
- Waste labels and/or color coding required by OSHA

4.0 Waste Container Labeling and Disposal

Each affected contractor is responsible for blood and infectious materials that are properly labeled. All containers used to store or transport blood and other potentially infectious materials should be clearly labeled with warning labels that include the biohazard symbol, contents, hazards involved, and the name and address of the project. Red bags or red containers may be used instead of labeling.

All containers of blood and other potentially infectious materials should be stored and disposed of according to federal/state/local guidelines. Arrangements may be made with a local hospital to receive and dispose of limited quantities of these regulated wastes in cases of first-aid treatment.

5.0 Hazardous Non-Routine Tasks and Nearby Work

Additional training is required when specialized tasks or in areas of work where special hazards exist. This includes work in HIV or HBV research labs and production facilities. This training is to be conducted as per the requirements set forth in part 3.0 of this document. The information should include:

- Specific hazards present
- Controls and protective measures required, including the types of personal protective equipment required

- the use and decontamination of specialized personal protective equipment
- hazards present and any emergency procedures/special precautions required for the task

6.0 Universal Precautions

Universal precautions should be followed on all tasks involving blood or other infectious material. This means that all blood and potentially infectious material are treated as if they contain bloodborne pathogens. Attention should be given to contaminated sharp objects that may penetrate the skin including, but not limited to:

- Needles
- Broken glass
- Exposed ends of wires

Where BBP exposure is likely, work practices and engineering controls should be followed diligently, including, where appropriate:

- Good hygiene practices – no smoking, eating, drinking or handling contact lenses in these areas
- Minimize splashing, spraying, spattering, and generation of droplets of BBP containing substances
- All equipment and environmental and working surfaces shall be cleaned and decontaminated after contact with blood or other potentially infectious materials
- The provision and use of puncture resistant latex or equivalent gloves
- Masks and eye protection
- Resuscitation bags and mouthpieces
- Gowns, aprons or specialized clothing where required by established engineering practices
- Hand-washing facilities with soap and clean water (washing is recommended for a minimum of 30 seconds)
 - o When provision of hand washing facilities is not feasible, the employer should provide either:
 - an appropriate antiseptic hand cleanser in conjunction with clean cloth/paper towels
 - antiseptic towelettes.
 - o Flush mucous membranes with water immediately or as soon as feasible following contact of such body areas with blood or other potentially infectious materials
 - o Use and provision of sharps containers
 - o Other decontamination where required by established engineering practices

Note: When antiseptic hand cleansers or towelettes are used, hands should be washed with soap and running water as soon as feasible.

Decontamination of personal protective items should be conducted by trained personnel following approved procedures. Disposable items should be discarded into red bags or properly labeled containers and delivered for disposal as required elsewhere in this program. Reusable items and any work areas contaminated by blood and other potentially infectious materials should be cleaned and disinfected with an appropriate cleaning solution. When cleaning blood:

- Put on latex or other puncture resistant impermeable protective glove and other PPE as required
- Where splashing or misting is possible, wear eye, nose and mouth protection
- Carefully cover the spill with paper towels or rags
- Gently pour the decontamination solution over the rags or towels until saturated
- Follow manufacturer's directions to ensure that all bloodborne pathogens are killed
- Dispose of all contaminated materials, following approved procedures

7.0 Hepatitis B Vaccination

Hepatitis B vaccinations should be made available to all affected employees within 10 working days of assignment to a task with BBP exposure. These vaccinations will be at no cost to the employee, at a



reasonable time and place, under the supervision of a licensed physician or health care professional and according to the latest recommendations of the U.S. Public Health Service (USPHS). Employees should sign a declination form if they choose not to be vaccinated, but they may later opt to receive the vaccine at no cost. Should booster doses later be recommended by the USPHS, they must be offered to employees at no cost.

8.0 Post-Exposure Evaluation and Follow-up

In the event an employee is exposed or potentially exposed to blood or infectious material, an exposure report should be completed and immediate notification given to DPR. Additionally, a confidential medical evaluation and follow-up should be made immediately available to the exposed employee by the employer. These medical evaluations must be conducted by an accredited clinic/ physician and offered at no cost to the employee. Follow-up procedures should include:

- Reported illnesses
- A confidential medical evaluation documenting the circumstances of exposure
- Testing the exposed employee's blood (if consented by the employee)
- Post-exposure prophylaxis counseling

Note: All diagnoses must remain confidential

9.0 Recordkeeping

Each affected contractor should establish and maintain an accurate medical record of each employee with an occupational exposure to blood and other infectious materials for the duration of employment plus 30 years. Records should be kept in accordance with 29 CFR 1910.1020. These records must remain confidential and may only be disclosed or reported to any party with the expressed consent of the employee. Medical records must be made available to the employee upon request.

Training records should be typed or otherwise legible and include the date of training, agenda, instructor's name, attendance list and signature of attendees. These records are to be maintained for at least 3 years from the training date.

Bloodborne Pathogens Forms

Forms

Bloodborne Pathogens Exposure Awareness Training Verification – English
Bloodborne Pathogens Exposure Awareness Training Verification – Spanish
Hepatitis B Vaccine Fact Sheet – English
Hepatitis B Vaccine Fact Sheet - Spanish



Bloodborne Pathogens Exposure Awareness Training Verification

Archive Document – DO NOT DISCARD

Print Name	Date
Jobsite	Company

All attempts should be made by DPR Construction to have the owner's employees abate all known hazards prior to work in any hazardous areas associated with potential bloodborne pathogen exposure; however, this may not be possible in all cases. Therefore, workers should be trained in hazards associated with activities with the potential of contact with bloodborne pathogens. Bloodborne pathogen activities are considered high hazard work by DPR and therefore require daily, detailed, written pre-task planning prior to start of work. Below is a review of DPR guidelines:

- All construction areas that could have a Bloodborne Pathogen exposure should be cleaned and disinfected before work begins in those areas.
- Workers should not enter areas where blood or bodily fluids are present.
- Workers should not perform clean-up of blood or bodily fluids. Consider all blood and human body fluid as potentially infectious.
- In the event any worker is asked to work in an area that contains Bloodborne Pathogens, training should be provided prior to start of work and Universal Precautions should be used. The list of Universal Precautions includes, but are not limited to:
 - o The provision and use of latex gloves
 - o Masks and eye protection
 - o Resuscitation bags, pocket masks, mouthpieces and/or other ventilation devices
 - o Gowns, aprons or specialized clothing where required by established engineering practices
 - o Hand-washing facilities (washing hands is recommended for a minimum of 30 seconds)
 - o Other decontamination where required by established engineering practices
 - o Use and provision of sharps containers and Biohazard Bags
 - o Good hygiene practices – no smoking, eating, drinking or handling contact lenses in these areas
- Workers who enter healthcare occupied areas may need to conform to gowning, shoe covering and other clothing requirements.
- Remove any PPE before leaving the work area or when the PPE becomes contaminated.
- Potential hazards found in exhaust duct removal and wet system/sanitary drain removal must be considered
- Training must be completed before initial assignment to a job that presents the possibility of exposure to blood and/ or other potentially infectious material (OPIM)
- Training is required annually, or more often, and anytime new or modified tasks are added to the job.

Signature of Employee	Signature of Person Conducting Verification Review
-----------------------	--

A copy of this form should be faxed to the DPR regional office PRIOR TO START OF WORK.



Patógenos Transmitidos Por La Sangre

Verificación de Entrenamiento

Guardar este documento – NO DESTRUIR

Nombre en letra de imprenta

Fecha

Sito de Trabajo

Compañía

Todos los intentos deben ser hechos por DPR Construction para que los trabajadores del proyecto disminuyan todos los riesgos conocidos antes de trabajar en las zonas de riesgo que sean asociados con el potencial de exposición a agentes patógenos transmitidos por la sangre; Aun así, pueda ser que esto no sea posible en todos los casos. Con esto en mente, los trabajadores deben ser entrenados en peligros o riesgos asociados con actividades por el potencial de contacto con patógenos transmitidos por la sangre. Actividades con potencial de exposición a patógenos transmitidos por la sangre son considerados de alto riesgo por DPR y por esta razón se requiere que diariamente se complete un plan de trabajo detallado antes del comienzo del trabajo.

Aquí están los reglamentos de DPR:

- Todas las áreas de trabajo donde podría haber exposición de patógenos transmitidos por la sangre deben ser limpiados efectivamente y desinfectados antes de comenzar el trabajo en esas áreas de trabajo.
- Los trabajadores no deben entrar en las áreas de trabajo donde existe o pueda existir sangre o los fluidos corporales.
- Los trabajadores no deben limpiar sangre o los fluidos corporales. Se debe considerar que toda sangre y fluidos corporales están contaminados y con potencial de infección.
- En el evento de que a algún trabajador se le asigne trabajar en una área que contiene patógenos transmitidos por la sangre, se debe proveer entrenamiento antes del comienzo del trabajo y precauciones universales se deben usar. La lista de precauciones universales incluyen pero no se limita a:
 - Que se provea y el uso de guantes látex
 - Mascaras y protección para los ojos
 - Bolsas de resucitación, mascarar de bolsillo, piezas de boca y/o otros dispositivos de ventilación
 - Batas, mandiles o ropa especial cuando se requiera por prácticas establecidas por ingeniería
 - Estaciones de Lavado de manos (Lavado de manos es recomendado al mínimo por 30 segundos)
 - Cualquier otro tipo de descontaminación que sea requerido por las practicas establecidas por ingeniería
 - El uso y que se provean contenedores (Sharps) y bolsas de riesgo biológico
 - Buenas prácticas higiénicas – no fumar, comer, tomar o manipulación de lentes de contacto en estas áreas
- Los trabajadores que estén en áreas medicas de trabajo pueda que necesiten asumir reglas de ponerse batas, cubiertos de zapato, y otros requerimientos de ropa
- Se debe remover todo equipo de protección personal antes de irse del área de trabajo o cuando el equipo de protección personal sea contaminado
- Peligros potenciales que puedan ser encontrados cuando se remuevan los ductos, y sistemas de agua/sanitarios o drenajes deben ser considerados
- Se debe completar entrenamiento antes de asignar inicialmente a un trabajador a un sitio de trabajo donde se presente la posibilidad de exposición a patógenos transmitidos por la sangre y/o cualquier material que potencialmente sea infectado (OPIM)
- El entrenamiento es requerido anualmente, o más frecuente, y cuando hayan cambios nuevos o tareas modificadas que sean agregadas al trabajo.

Firma de trabajador

Firma de persona haciendo verificación de entrenamiento

Una copia de esta forma se deben mandar por fax a la oficina regional de DPR ANTES DEL COMIENZO DE TRABAJO



Hepatitis B Vaccine Fact Sheet

Archive Document – DO NOT DISCARD

The following information is derived from the Centers for Disease Control and Prevention (CDC) and the National Vaccine Information Center (NVIC).

Hepatitis B Disease

Hepatitis B is primarily an adult disease transmitted through infected body fluids, most frequently infected blood, and is prevalent in high risk populations such as needle using drug users (28%); sexually promiscuous heterosexual (22%) and homosexual (9%) adults; residents and staff of custodial institutions such as prisons; health care workers exposed to blood; persons who require repeated blood transfusions and babies born to infected mothers (2%). Hepatitis B is not common in childhood and is not highly contagious.

Hepatitis B is not a killer disease for most people. Symptoms of hepatitis B include nausea, vomiting, fatigue, low grade fever, pain and swelling in the joints, headache and cough that may occur one to two weeks before the onset of jaundice (yellowing of the skin) and enlargement and tenderness of the liver, which can last for three to four weeks. Fatigue can last up to one year. According to Harrison's Principles of Internal Medicine (1994) 'most patients do not require hospital care', "95% of patients have a favorable course and recover completely" and the case fatality ratio is "very low (approximately 0.1%). Those who recover completely acquire life-long immunity from the disease.

Hepatitis B Vaccine Safety

Public health organizations and the CDC endorse the vaccine as a safe and effective way to prevent disease and death. The Institute of Medicine and NVIC cite a lack of conclusive evidence and adequate unbiased scientific study on the adverse effects or long term safety of the hepatitis B vaccine. During the past decade, there have been many reports in the medical literature that hepatitis B vaccination is causing chronic immune and neurological disease in children and adults.

Common short term reactions to the hepatitis B vaccine include fatigue, weakness, fever, headache and joint pain. One or several of these symptoms were reported in up to 17% of all hepatitis B injections.

Hepatitis B Vaccine Efficacy

All vaccines stimulate only an artificial, temporary immunity, and the length of immunity conferred by the hepatitis B vaccine and the future need for more "booster" doses later in life is still not clear. Current research shows that about "30-50% of people who develop adequate antibody after three doses of vaccine will lose detectable antibody within 7 years".

Vaccine Schedule

The recommended dose schedule for hepatitis B vaccine is a three shot dose including the initial shot, and follow up shots at 1-2 months and 4-6 months. The second and third dose should be separated by at least 2 months.

Hepatitis B Vaccination Acceptance

I understand that due to my reasonable occupational exposure to blood or other potentially infectious materials I may be at risk of acquiring hepatitis B virus (HBV) infection. I have been given the opportunity to be vaccinated with hepatitis B vaccine, at no charge to myself. I understand the above information related to the vaccine and hepatitis B and have been given the opportunity to ask any questions about this to DPR or the health care provider. I further understand that by accepting this vaccine, I expose myself to any and all risks associated with vaccine use. Further, if my employment at DPR Construction is terminated prior to the completion of the full series of vaccinations, I am responsible for ensuring that I follow-up and complete my vaccine schedule. If in the future I choose not to be vaccinated or discontinue the vaccination schedule, I will contact DPR EHS and ask for and complete the necessary Hepatitis B Vaccination Declination form.

Print Name

Date

Signature

Hepatitis B Vaccination Declination

I understand that due to my reasonable occupational exposure to blood or other potentially infectious materials I may be at risk of acquiring hepatitis B virus (HBV) infection. I have been given the opportunity to be vaccinated with hepatitis B vaccine, at no charge to myself. However, I decline hepatitis B vaccination at this time. I understand that by declining this vaccine, I continue to be at risk of acquiring hepatitis B, a serious disease. If in the future I continue to have occupational exposure to blood or other potentially infectious materials and I want to be vaccinated with hepatitis B vaccine, I can receive the vaccination series at no charge to me.

Print Name

Date

Signature



Hoja de datos—vacuna contra la Hepatitis B

Documento de archivo – NO DESECHAR

La siguiente información se obtuvo del Centro para el Control y la Prevención de Enfermedades (Centers for Disease Control and Prevention, o CDC por su sigla en inglés) y del Centro Nacional de Información sobre las Vacunas (National Vaccine Information Center, o NVIC por su sigla en inglés).

La enfermedad conocida como Hepatitis B

La Hepatitis B es primordialmente una enfermedad de adultos transmitida a través de líquidos corporales contaminados, con mayor frecuencia a través de sangre infectada, y prevalece en poblaciones de alto riesgo, tales como los consumidores de drogas por inyección (28%); adultos heterosexuales (22%) y adultos homosexuales (9%) sexualmente promiscuos; residentes y personal de instituciones de custodia tales como prisiones; trabajadores en cuidados de la salud expuestos a sangre; personas que necesitan repetidas transfusiones de sangre y bebés nacidos de madres infectadas (2%). La Hepatitis B no es común en la infancia y no es altamente contagiosa.

Para la mayoría de las personas la Hepatitis B no es una enfermedad mortal. Los síntomas de la Hepatitis B son náuseas, vómitos, fatiga, fiebre leve, dolor e inflamación de las coyunturas, dolor de cabeza y tos, y pueden ocurrir una o dos semanas antes de la manifestación de la ictericia (piel amarilla) y aumento en el tamaño y sensibilidad del hígado, lo que puede durar entre tres a cuatro semanas. La fatiga puede durar hasta un año. Según los Principios de Medicina Interna de Harrison (1944), “la mayoría de los pacientes no requiere cuidados hospitalarios”, “95% de los pacientes tiene una resolución favorable y se recupera totalmente” y la tasa de mortalidad es “muy baja (aproximadamente 0.1%)”. Aquellas personas que se recuperan adquieren una inmunidad vitalicia a la enfermedad.

La seguridad de la vacuna contra la Hepatitis B

Las organizaciones de salud pública y el CDC endosan la vacuna como un método seguro y eficaz de prevenir la enfermedad y la muerte. El Instituto de Medicina y el NVIC indican que no existen pruebas contundentes y suficientes estudios científicos adecuados e imparciales sobre los efectos adversos o sobre la seguridad a largo plazo de la vacuna contra la Hepatitis B. Durante la última década, hubo muchos informes en publicaciones médicas que indican que la vacuna contra la Hepatitis B causa enfermedades neurológicas e inmunológicas crónicas en niños y adultos.

Las reacciones más comunes dentro del corto plazo a la vacuna contra la Hepatitis B incluyen fatiga, debilidad, fiebre, dolor de cabeza o de las coyunturas. Se informó sobre la presencia de uno o varios de estos síntomas en hasta 17% de todas las personas que recibieron inyecciones contra la Hepatitis B.

Eficacia de la vacuna contra la Hepatitis B

Todas las vacunas estimulan exclusivamente una inmunidad artificial y temporal, y aún no tenemos idea clara de la duración de la inmunidad otorgada por la vacuna contra la Hepatitis B y la necesidad futura para aplicar dosis de “refuerzo” en años posteriores. La investigación actual muestra que alrededor de “30-50% de las personas que desarrollan anticuerpos adecuados después de tres dosis de vacuna perderán una cantidad detectable de anticuerpos en 7 años”.

Programa de vacunación

El programa de vacunación recomendado para la vacuna contra la Hepatitis B es una dosis de tres inyecciones, incluyendo la inyección inicial, y dos inyecciones posteriores dentro de 1-2 meses y 4-6 meses. La segunda y la tercera dosis deberán estar separadas por un intervalo de por lo menos 2 meses.

Aceptación de vacunación contra la Hepatitis B

Entiendo que por motivo de mi ocupación es razonable anticipar que podré estar expuesto/a a la sangre y otras materias posiblemente infecciosas, y que puedo correr el riesgo de contraer una infección de virus Hepatitis B (HBV). He tenido la oportunidad de recibir una vacuna contra la Hepatitis B, en forma gratuita. Comprendo la información que antecede sobre la vacuna y sobre la Hepatitis B, y he tenido la oportunidad de presentar todas mis preguntas sobre este tema a DPR o al proveedor de servicios de la salud. Además entiendo que al aceptar esta vacuna me expongo a todos los riesgos asociados con el uso de vacunas. Además, si termina mi empleo en DPR Construction antes de completar toda la serie de vacunas, seré yo responsable de las medidas necesarias para continuar y completar mi programa de vacunación. Si en el futuro decido no ser vacunado/a o no continuar con el programa de vacunación, me comunicaré con el departamento de salud y seguridad (EHS) de DPR y solicitaré y completaré el formulario pertinente para rechazar la vacuna contra la Hepatitis B.

Nombre en letra de imprenta

Fecha

Firma

Rechazo a la vacunación contra la Hepatitis B

Entiendo que por motivo de mi ocupación es razonable anticipar que podré estar expuesto/a a la sangre y otras materias posiblemente infecciosas, y que puedo correr el riesgo de contraer una infección de virus Hepatitis B (HBV). He tenido la oportunidad de recibir una vacuna contra la Hepatitis B, en forma gratuita. Sin embargo, me niego a recibir la vacuna contra la Hepatitis B en este momento. Entiendo que al rechazar esta vacuna, sigo corriendo el riesgo de contraer Hepatitis B, una enfermedad grave. Si en el futuro sigo estando expuesto/a en mis ocupaciones a la sangre y a otras materias posiblemente infecciosas y deseo ser vacunado/a contra la Hepatitis B, puedo recibir la serie de vacunas en forma gratuita.

Nombre en letra de imprenta

Fecha

Firma

Section 02

Chemical & Environmental Management

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Chemical and Environmental Management

1.0 Introduction

This section serves as DPR Construction's Chemical Safety Program, which includes the written Hazard Communication program. All employees in the workplace have the right to know and understand the risks and hazards associated with chemicals they are working with and around; how to recognize signs of over-exposure and how to mitigate these hazards. This plan provides detailed safety guidelines and instructions for receipt, use and storage of chemicals at the workplace for employees and contractors. This plan shall be readily available to all employees.

Reference: CFR 1910.1200 Hazard Communication, CFR 1926.1153 Respirable Crystalline Silica and CCR Section 1532.3 Occupational Exposure to Respirable Crystalline Silica. Other Standards, guidelines and regulations may apply.

Definitions

Action level: A concentration of airborne respirable crystalline silica of 25 $\mu\text{g}/\text{m}^3$, calculated as an 8-hour TWA.

Competent person: (in reference to Respirable Crystalline Silica) An individual who is capable of identifying existing and foreseeable respirable crystalline silica hazards in the workplace and who has authorization to take prompt corrective measures to eliminate or minimize them. The competent person must have the knowledge and ability necessary to fulfill the responsibilities set forth in paragraph (g) of this section

Concrete or Masonry Material: A hard stone-like building material made of clay or made by mixing cement or a combination of cement, sand, gravel, broken stone, or other aggregate with water. Examples include brick, clay brick, concrete block, mortar, natural or manufactured stone, floor, wall, or counter top tile, and terra cotta.

Dust containment device: A device attached to a power tool such as a pouch, bag, plastic container, or similar attachment which is intended to capture dust generated by the power tool. This device is not intended to be a dust reduction system as defined under this section.

Dust reduction system: Technology that utilizes the application of water or local exhaust ventilation to reduce airborne dust generated by the use of powered tools or equipment. Local exhaust ventilation may include vacuum systems, dust collection systems, and dust exhaust systems.

Employee exposure: (in reference Respirable Crystalline Silica) The exposure to airborne respirable crystalline silica that would occur if the employee were not using a respirator.

Exposure: Being exposed to a chemical that is a physical or health hazard, and includes potential (e.g. accidental or possible) exposure. "Subjected" in terms of health hazards includes any route of entry (e.g. inhalation, ingestion, skin contact or absorption)

Globally Harmonized System (GHS): Classification and labeling of chemicals is an internationally agreed-upon system, created by the United Nations

Hazard Statement: A set of standardized phrases about the hazards of chemical substances and mixtures that can be translated into different languages

Hazardous Chemical: Any chemical which is a physical or health hazard

Hazardous Substance: A substance, material or mixture which by reason of being explosive, flammable, poisonous, corrosive, oxidizing, an irritant, or otherwise harmful, is likely to cause injury or illness

Hazardous Waste: A waste that is dangerous or potentially harmful to health or the environment. Hazardous wastes can be liquids, solids, gases, or sludges. They can be discarded commercial products, such as cleaning fluids or pesticides, or the by-products of manufacturing processes.

Health Hazard: A chemical for which there is statistically significant evidence based on at least one study conducted in accordance with established scientific principles that acute or chronic health effects may occur in exposed employees. The term "health hazard" includes chemicals, which are carcinogens, toxic or highly toxic agents, reproductive toxins, irritants, corrosives, sensitizers, hepatotoxins, nephrotoxins, neurotoxins, agents which act on the hematopoietic system, and agents which damage the lungs, skin eyes or mucous membranes.

High-efficiency particulate air [HEPA] filter: A filter that is at least 99.97 percent efficient in removing mono-dispersed particles of 0.3 micrometers in diameter.

Immediate Reporting of a Release or a Threatened Release: An immediate, verbal report must be made if there is knowledge of a release or threatened release of hazardous material (for example, an open-top tank containing a hazardous material is likely to overfill resulting in a release).

Minimum Efficiency Reporting Value (MERV): A standard that rates the overall effectiveness of air filters. The MERV is a single number that is used, along with the air velocity at which the test was performed, to simplify the extensive data generated by the method of testing. MERV is expressed on a 16 point scale and is derived from the PSE for each of the three groups. Higher value MERV rating equates to finer filtration, meaning fewer dust particles and other airborne contaminants can pass through the filter

Notification: A formal written notice.

Objective Data: information, such as air monitoring data from industry-wide surveys or calculations based on the composition of a substance, demonstrating employee exposure to Respirable Crystalline Silica associated with a particular product or material or a specific process, task, or activity. The data must reflect workplace conditions closely resembling or with a higher exposure potential than the processes, types of material, control methods, work practices, and environmental conditions in the employer's current operations.

Permissible Exposure Limit (PEL): The maximum permitted 8-hour time-weighted average concentration of an airborne contaminant.

Physician or other licensed health care professional [PLHCP]: An individual who is legally permitted scope of practice (i.e., license, registration, or certification) allows him or her to independently provide or be delegated the responsibility to provide some or all of the particular health care services required by paragraph (h) of this section.

Pictogram: A pictorial symbol for a word or phrase.

Powered tools or equipment: Tools or equipment for which the motive force that disrupts concrete or masonry materials is provided by a source other than human energy. Powered tools or equipment include those powered by electrical, combustion, hydraulic, chemical, or pneumatic energy.

Precautionary Statement: A set of phrases about the hazards of chemical substances and mixtures that can be translated into different languages.

Respirable crystalline silica: Quartz, cristobalite, and/or tridymite contained in airborne particles that are determined to be respirable by a sampling device designed to meet the characteristics for respirable-particle-size-selective samplers specified in the International Organization for Standardization (ISO) 7708:1995: Air Quality – Particle Size Fraction Definitions for Health-Related Sampling.

Safety Data Sheet: A document that contains information on potential hazards (health, fire, reactivity and environmental) and how to work safely with the chemical product.

Specialist: An American Board Certified Specialist in Pulmonary Disease or an American Board Certified Specialist in Occupational Medicine.

Spill/Release: Any spilling, leaking, pumping, pouring, emitting, emptying, discharging, injecting, escaping, leaching, dumping, or disposing into the environment (air, land or water), unless permitted or authorized by a regulatory agency.

Threatened Release: A condition creating a substantial probability of harm, when the probability and potential extent of harm make it reasonably necessary to take immediate action to prevent, reduce, or mitigate damages to persons, property, or the environment.

Waste: Material that no longer has a prescribed use and is ready for disposal.

2.0 Pre-Planning

Pre-planning of the chemical and environmental processes used on each jobsite is vital to reaching our goal of protecting our workers and the public, as well as reducing our impact on the environment. To facilitate proper planning, DPR provides the following tools:

- Hazard Communication Training Verification: training verification form to assist in documenting hazard communication training for workers.
- Pre-task Planning: work activities involving hazardous materials or chemicals are defined by DPR to be High Hazard and therefore require detailed, written pre-task planning. Steps to protect the worker, the environment and public should be detailed in these pre-task plans. *Refer to the Pre-Planning Section 17 for additional pre-task planning guidelines.*
- Chemical & Environmental Management Checklist: a checklist provided as a tool to assist subcontractors with the appropriate means and methods to developing their site-specific plan.
- Storm Water Protection & Prevention Plan Checklist: a checklist provided as a tool to ensure storm water protection is planned and appropriate for the site.
- Chemical Inventory Form: used for documenting on-site chemicals, their quantities, locations and storage requirements.

3.0 Hazard Communication Program

The Occupational Safety and Health Act (OSHA) defines that employers with workers who are potentially exposed to hazardous chemicals (e.g. fuels, lubricants, foam releases, curing compounds, sealants, paints, coatings, etc.) be advised of the potential hazards and how to guard against those hazards. OSHA has recently adopted the “United Nations Globally Harmonized System of Classification and Labeling of Chemicals” commonly called “GHS”. The GHS initiative has become part of OSHA’s recently updated Hazard Communication Program requirements. It will also provide a more standardized approach to classifying the hazards and conveying the information. All contractors should develop a list of all chemicals used on the project; gather Safety Data Sheets (SDS) for those materials; follow the Globally Harmonized System’s (GHS) labeling system for all materials. Employees must be trained in proper identification of chemicals through labels and SDS, and can identify any potential hazards working with or around a chemical through hazard identifying pictograms and “signal words”. Worker training should be documented. *Refer to Hazard Communication Training Verification the Forms Section at the end of the EHSP.* A hazard evaluation should be completed to determine that all safety precautions have been addressed for use of chemicals prior to performing a task. Use of chemicals can be considered a high hazard activity by DPR and therefore requires daily, detailed, written pre-task planning prior to use. Below is a review of GHS guidelines you should be familiar with prior to using chemicals.

3.1 Safety Data Sheets (SDS)

Every subcontractor will be responsible for the development and maintenance of a list of hazardous chemicals utilized within the project operations and will be further responsible for obtaining and maintaining SDSs for all such hazardous chemicals.

Note: As new labels are noticed on primary containers, efforts should be made to update the chemical list and secure an SDS to replace the MSDS.

One copy of the chemical inventory list and of each of the SDS should be forwarded to DPR project management during the submittal process and prior to any site mobilization. Workers will be allowed access to this information and the specific SDS for chemicals utilized in their work areas. All questions relating to the program should be directed to the contractor's qualified person.

Below is the standardized format all SDS's must follow under the GHS Standard.

- Section 1 - Identification
- Section 2 - Hazard Identification
- Section 3 - Composition Information
- Section 4 - First Aid Measures
- Section 5 - Firefighting Measures
- Section 6 - Accidental Release Measures
- Section 7 - Handling and Storage
- Section 8 - Exposure Controls/PPE
- Section 9 - Physical and Chemical Properties
- Section 10 - Stability and Reactivity
- Section 11 - Toxicological Information
- Section 12 - Ecological Information
- Section 13 - Disposal Considerations
- Section 14 - Transport Information
- Section 15 - Regulatory Information
- Section 16 - Other Information

3.2 Worker Information and Training

All new and present workers should be given information and training regarding the requirements of the Hazard communication standard to include the information contained in the GHS, the hazardous chemicals present in their work place and the physical and health risks of these chemicals. Training should be conducted at the time of their initial assignment, and whenever a new hazardous material is introduced. Training should include the following elements:

- How to determine the hazardous presence or release of a chemical in the workplace
- The physical and health hazards of chemicals in their work area
- Methods to reduce or prevent the exposure to hazardous chemicals
- The location of SDS and written hazard communication program

Refer to the Forms Section at the end of the EHSP for a copy of the Hazard Communication Training Verification form that is available to assist with documentation of training.

3.3 Container Labeling

All chemical containers should be clearly labeled with its contents, a signal word, pictogram(s), hazard statement, precautionary statement, and manufacturer / supplier identification. The label should include at a minimum:

product identifier

signal word

hazard statement

manufacturer/supplier
i.d.

Acetone

Danger!

Highly flammable liquid vapor. Causes severe eye irritation.

Keep away from heat, sparks and flame – No smoking. Take precautionary measures against static discharge. Keep from direct sunlight. Keep container closed when not in use. Store in a cool/low temperature, well-ventilated place away from heat and ignition sources. Use only in a well-ventilated area. Avoid contact with eyes, skin and clothing. Wear appropriate personal protective equipment, avoid direct contact.

IF CONTACT WITH EYES: Flush eyes with water for at least 15 minutes while holding eyelids open.

In case of fire, use water spray, fog or mist. Dry chemicals. Halon. Powder, foam or CO2.

See Safety Data Sheet for further details regarding safe use of this product.

ABC Company, Main Street, Anytown, NJ 00000, Tel: 555 123 4567



pictogram

precautionary
statement

- Product Identifier
 - The name identifying the product that should match the SDS.
- Signal Word
 - The signal word indicates the relative degree of severity of a hazard.
 - Danger: for the more severe hazards
 - Warning: for the less severe hazards
- Pictogram
 - Pictograms are as follows:

Health Hazard  • Carcinogen • Mutagenicity • Reproductive Toxicity • Respiratory Sensitizer • Target Organ Toxicity • Aspiration Toxicity	Flame  • Flammables • Pyrophorics • Self-Heating • Emits Flammable Gas • Self-Reactives • Organic Peroxides	Exclamation Mark  • Irritant (skin and eye) • Skin Sensitizer • Acute Toxicity (harmful) • Narcotic Effects • Respiratory Tract Irritant • Hazardous to Ozone Layer (Non-Mandatory)
Gas Cylinder  • Gases Under Pressure	Corrosion  • Skin Corrosion/ Burns • Eye Damage • Corrosive to Metals	Exploding Bomb  • Explosives • Self-Reactives • Organic Peroxides
Flame Over Circle  • Oxidizers	Environment (Non-Mandatory)  • Aquatic Toxicity	Skull and Crossbones  • Acute Toxicity (fatal or toxic)

- Hazard Statement
 - Hazard statements are standardized and assigned phrases that describe the hazard(s) as determined by hazard classification
- Precautionary Statement
 - Precautionary information supplements the hazard information by briefly providing measures to be taken to minimize or prevent adverse effects from physical, health or environmental hazards
 - There are four types of statements:
 - prevention
 - response in case of accidental spillage or exposure
 - storage
 - disposal
- Manufacturer/ Supplier Identification
 - The name, address and telephone of the manufacturer of the product

Secondary Container:

All secondary containers should be clearly labeled with the equivalent hazard warnings as the original container, and must also be compatible with the chemical properties. *Refer to manufacturers recommendations.*

The NFPA/HMIS hazard rating scale starts with 0, which equals minimal hazard and progresses to 4, which equals a severe hazard. The new GHS hazard rating scale starts with category 1, which is a severe hazard and moves down to category 5, which is a minimal hazard. The two rating systems may confuse chemical users who are new to the GHS.

There are currently no plans to change the existing NFPA and HMIS systems and classifications. These systems will still be used to convey hazardous information to emergency first responders. Employers may choose to label workplace containers with the same label that would be on shipped containers for the chemical under the revised rule. The information supplied on these labels must be consistent with the revised HCS, e.g., no conflicting hazard warnings or pictograms. DPR's best practice is the use of dual-labels.



3.4 Hazardous Non-Routine Tasks and Nearby Work

In the event a worker is assigned to perform or work in an area where a hazardous task, non-routine to their work may be required, the worker should be given the additional information and training related to the hazardous chemicals that may be encountered in the non-routine task. Supervisors, subcontractor's superintendent, or subcontractor safety representative should provide this information and training.

The information should include the:

- Specific chemical hazards of the task

- Controls and protective measures required
- Types of personal protective equipment required
- Proper use of equipment
- Nature of other work being performed in or near the non-routine task
- What emergency procedures are involved with the task

3.5 *Chemicals in Unlabeled Pipes, Vessels and Containers*

Prior to work being conducted on unlabeled pipes, vessels or containers, workers should be informed of the following information:

- Type of chemical in the pipe, vessel or container
- Potential hazards
- Safe precautions which should be taken

3.6 *Audit and Review*

DPR project management should review the Hazardous Communication Program annually, and to revise and update as needed. Additionally, periodic auditing should be performed to ensure that procedures in the use of the hazardous chemicals meet the requirements as set forth in the SDS. Each subcontractor's project management should perform the above responsibilities for all of their materials and procedures.

4.0 *Chemical Management Program*

All contractors, subcontractors and vendors on any DPR project should ensure that all chemicals on the project are handled in such a way as to eliminate or reduce the potential for exposure to workers, the public or the environment.

4.1 *Definitions*

4.2 *Subcontractor Site-Specific Chemical Management Plan*

Each affected subcontractor should develop and submit a written Chemical Management Plan to DPR prior to mobilization. Each plan should include, at minimum, the following components:

- Chemical inventories, to include:
 - name of chemical
 - description
 - quantity
 - method of containment
 - location of storage
- SDS for the entire chemical inventory
- Storage location and quantities of site chemicals
- Emergency, Response and Cleanup Plan

Preventative measures to prevent spills and releases (e.g., secondary containment, inspection schedules, protection, good housekeeping practices etc.)

4.3 *General Guidelines for Chemical Hazard Protection*

These guidelines apply to all substances and chemicals with a potential for exposure, injury or illness to any workers, member of the public or the environment. Products not included within this plan include those that are part of an existing facility or plant operation. In these instances, the owner's chemical management plan should be available and accessible to all workers. Each subcontractor is responsible for ensuring that their workers understand the owner's chemical management plan, hazards present and appropriate controls and preventative measures. They should also ensure that each worker knows where to find more information and will receive appropriate training prior to working around these chemicals. The following general guidelines apply to all work with chemicals on a DPR project:

- Appropriate personal protective equipment should be used as necessary to protect the worker from exposure to chemicals. This includes, but is not limited to respiratory, face, hand/arm and body protection (refer to the SDS for specific requirements on the proper PPE required)
- Direct-reading air monitoring should be performed to evaluate potential respiratory and explosion hazards, as applicable
- Underground pipelines and utilities should be accurately located before drilling or excavating, and "soft dig" potholing should be performed to visually locate all utilities. For more information, refer to the Excavation and Trenching Section 10.
- Spark-ignition equipment should not be used in the vicinity of flammable/combustible materials
- Smoking should be permitted only in designated areas
- Fans should be used as necessary to disperse airborne contaminants at the work site. No sparking or open flame equipment will be permitted inside restricted work areas if there is potential of reaching the Lower Explosive Limit (LEL) of contaminants present at the site.
- Appropriate respiratory protection with applicable filters should be worn if there is a potential for contaminated dust at the site. For more information, refer to the Respiratory Protection Section 19.
- Chemical storage areas should be thoroughly ventilated

4.4 Auditing and Inspection

Subcontractor supervision should inspect their work areas daily for the effectiveness of their chemical management plans and spill and exposure prevention measures. Any unsafe conditions should be immediately corrected. Inspections should be documented and stored on file with DPR.

4.5 Chemical Storage, Handling and Disposal

In addition to the requirements outlined in the Environmental Protection Plan these additional measures should be taken regarding chemical storage, handling and disposal:

- Subcontractors should bring only chemicals and materials anticipated to be used on the project
- Upon being used, all material should be timely removed from site and/or properly disposed of
- Whenever possible, subcontractors should attempt to re-use or recycle as much of their material as possible and plan their work in such a way as to minimize the amount of waste generated
- Containers should be appropriately labeled
- Liquid residues from paints, thinners, solvents, glues or adhesives and cleaning fluids are hazardous wastes and should be treated as such
- Never clean brushes or rinse paint containers into an area having the potential to contact the street, gutters, storm drains, streams or the general construction site
- Rinse water based paints, brushes, etc. to a sanitary sewer or other properly contained system
- When using oil based paints, paint out brushes to the extent possible
- Filter and/or reuse thinners and solvents whenever possible
- Dispose of excess liquids and residues as hazardous waste
- State and federal, or site specific, guidelines may require secondary containment for chemical storage in order to comply with EPA storm water runoff regulations

Refer to Fire Protection Prevention Section 11 for specific guidelines on storage and handling requirements for flammable and/or combustible materials.

4.6 Discovery of Hazardous or Unknown Material

In the event an identified hazardous material or an unidentified material of unknown origin is discovered, the following procedures should be implemented (All unknown materials will be treated as hazardous until identification proves otherwise):

1. Stop work immediately
2. Shutdown all equipment in area (size of area is dependent upon the individual situation)

3. Extinguish ignition sources
4. Evacuate affected area immediately
5. Report incident to DPR project management immediately
6. Protect area and prevent unauthorized personnel from entering. Barricade area to ensure isolation
7. Report incident to appropriate personnel per Emergency Response Plan guidelines. *Refer to Emergency Procedures Section 08 for additional details.*
8. Work should not resume until authorized by DPR project management

4.7 Spill or Release Response

Should hazardous materials be stored on site in quantity, a Spill response plan should be developed. At a minimum, this plan should include:

- Identification of trained responders
- Procedures to be implemented in response to a spill
- Supplies to be used for containment and spill cleanup; these supplies should be based on project needs. The following is a list of potential supplies:
 - soaker booms
 - drip pads
 - absorbents
 - appropriate personal protective equipment
 - DOT approved hazardous waste containers
 - non-sparking shovels, squeegees and other tools needed for proper cleanup and containment
- Area in which containment/spill supplies should be stored
- Provisions for disposal of products and supplies used for containment/spill cleanup

4.8 Release Reporting

DPR requires that all actual and potential spills or releases are reported. DPR divides these reporting requirements into two subcategories; immediate and non-emergency release reporting. All subcontractors are therefore required to **immediately** report any release or threatened release of any hazardous material on the project site. Immediate release reporting is not required if there is a reasonable belief that the release or threatened release poses no significant or potential hazard to human safety and health, property or the environment.

Immediate reporting should occur in accordance with federal, state and local regulations and should include, at a minimum:

- The exact location of the release or threatened release
- The name of the person reporting the release or threatened release
- The hazardous materials involved in the release or threatened release
- An estimate of the quantity of the hazardous materials involved
- If known, the potential hazards presented by the hazardous material involved in the release or threatened release

An immediate verbal report of a release should be made to DPR and the appropriate government agency and when appropriate followed by a written comprehensive report. Written reports should be completed and submitted to the appropriate federal agency when there are releases of extremely hazardous materials or releases of substances listed in CERCLA (Comprehensive Environmental Response Compensation and Liability Act) that exceed the reportable quantities limits.

4.9 Community Right to Know and Understand

A Hazardous Materials Management Plan should be submitted to DPR if the inventory of all hazardous materials in the operation exceeds a specified amount as per CERCLA (e.g., greater than 55 gallons of solvent). Information should include, but not be limited to:

- Emergency procedures, reporting and response
- Storage
- Evacuation procedures for the project, owner facility and public
- Training

5.0 Environmental Protection Plan

DPR is committed to reducing the amount of waste generated on each site and continually increasing the amount of materials recycled. It is DPR's intent to protect workers, the public and the environment through coordinated job preplanning and systems review. Subcontractors are contractually required to fully comply with this Environmental Protection Plan as well as all state, federal, local and site specific rules and regulations.

5.1 Permits and Record Keeping

It is the responsibility of each subcontractor to have obtained the necessary permits in relation to the environment and their work activities. This includes permits for removal, transport and manifesting of hazardous waste, storm water management and other appropriate environmental permits. Further, each subcontractor should appropriately document all activities associated with hazardous waste and abide by all of the requirements for long term record keeping. Copies of these documents should be forwarded to DPR project management for archiving purposes. DPR reserves the right to audit compliance with these regulations and each subcontractor will submit copies of required permits, manifests etc. upon written request.

5.2 Erosion Protection

- Avoid excavation/grading activities during inclement weather when practicable
- Only remove existing vegetation when absolutely necessary
- Where slopes exist and construction activities are immediately planned, plant temporary vegetation as practicable
- Use secured stabilizing materials (e.g., plastic sheeting, filter fabric, straw wattles, etc.) to stabilize exposed soils, topsoil and landscaping materials
- As a last resort, protect downslope drainage courses, streams, catch basins, etc. with hay bales or temporary drainage swales
- Plant permanent vegetation as soon as possible after excavation/grade activities are completed
- Refer to the site-specific Storm Water Pollution Prevention Plan (SWPPP) for site specific requirements

5.3 Control of Runoff

- Diversion dikes/drainage swales should be constructed to channel runoff around the site
- Line swales with inert materials (e.g., grass, roughened pavement, filter fabric etc.) to reduce runoff velocity
- Berms and/or drainage ditches should be used around exposed areas to divert runoff
- Roughen other surfaces by adding rip rap, sand bags, rocks, straw bales, etc. (asphalt or demolition debris should not be used for this purpose)
- Establish sediment traps in catch drains or drain inlets, overflow drain may be appropriate to isolate drain off until sediments can settle out
- Refer to the site-specific Storm Water Pollution Prevention Plan (SWPPP) and state and federal regulations regarding effluent (run off) testing requirements
- In addition, be aware that "run on" carrying pollutants are a cause for concern and should be prevented if possible

5.4 Spill and Leak Prevention

- Designated areas should be established for refueling and maintenance activities
- Spill and leak prevention measures should be incorporated for refueling activities, including, but not limited to drip pans, drop cloths and/or sorbent
- Vehicles and heavy equipment should be properly maintained. Equipment should be inspected daily and if detected, leaks should be addressed immediately.
- When possible, designate one location for equipment parking
- When maintenance is being performed, drip pans, drop cloths, etc. should be used when draining or replacing of fuels, oils or liquids
- Equipment repair areas should be bermed or located in a remote area away from streams and storm drains

5.5 Spill Cleanup

All spill cleanup should follow federal, state, local, and owner regulations and guidelines. These should include at a minimum the following:

- If fuels, fluids or oils are spilled, cleanup should be immediate, automatic and routine
- Sweep up spilled dry materials immediately
- Dry methods should be used to cleanup liquid spills on paved or impermeable surfaces
- Spills on dirt should be cleaned up by digging up the soil and properly treating and/or disposing of contaminated soil
- All spills should be reported to DPR immediately

5.6 Material Storage and Handling

- Store containers of paint, chemicals, solvents and other hazardous materials in designated storage areas with double containment
- Wet or dry building materials with the potential to contaminate runoff should be protected from damage and weather
- Secondary containment without built-in canopy should be stored under cover to prevent water accumulation
- Berm around covered storage areas to prevent contact with runoff
- Chemical storage areas should be properly signed
- Proper grounding practices should be implemented within flammable materials storage areas
- Documentation of material storage areas may be required to be posted on the Storm Water Pollution Prevention Plan (SWPP) site plan

5.7 Material Disposal and Cleanup

- Subcontractors should identify potential sources of waste and develop a plan for managing and minimizing their waste
- Nothing should be placed into storm or sanitary sewer systems without prior approval from DPR and/or the owner
- Removal of wastes from the site should be in compliance with all federal, state and local regulations
- All washout activities (e.g., concrete, plaster, joint/taping compounds, paint etc.) should be completed in their designated washout area
- Concrete activities should be performed so as to contain and/or minimize residue or waste water; all such wastes should be properly disposed
- Appropriate precautions should be taken for specialty concrete applications (e.g., acid washing, dyes, stains etc.)
- Whenever possible, subcontractors should attempt to re-use or recycle materials in such a way as to minimize the amount of waste generated. Materials recycled should include, but are not limited, to:

- Paper
- Cardboard
- Glass
- Plastics
- Wood
- Metal
- Oil
- Concrete
- Drywall
- Asphalt

- Containers used for chemicals, paints, solvents, oils or any product with the potential for environmental contamination should be covered at the end of the day and during rainy weather
- Inspect containers regularly for leaks and repair/replace immediately if leakage is detected
- Dumpsters should not be hosed down on the construction site, dumpster contractors should be contacted for cleaning and all cleaning performed should be off site

6.0 Dust Management and Outdoor Air Quality

Each subcontractor should ensure that measures are taken to eliminate or reduce the amount of dust created and the impact to the environment. Further, measures should be taken to preserve the quality of the air in and around the project. Some methods include but are not limited to:

- Wet cutting and/or coring methods
- Use of smoke eaters, smog-hogs, and vacuum devices
- Negative air/HEPA filtration
- Watering trucks or water buffalos
- Fencing/dust barriers
- Building gravel or paved roadways
- Limiting the use of compressed air to clean
- Water based or non-odorous solvents, paints and chemicals
- Forced fresh air intakes/blowers
- Sweeping compounds

7.0 Control of Employee Exposure to Concrete Dust (CALIFORNIA ONLY – effective until September 23, 2017)

This section applies only to the state of California and remains in effect through June 22, 2017, at which point CFR 1926.1153 Respirable Crystalline Silica **and** CCR Section 1532.3 Occupational Exposure to Respirable Crystalline Silica are fully applicable.

This section applies to the use of powered tools or equipment to cut, grind, core or drill concrete or masonry materials. This section does not apply to:

- Stucco, plastering material, wall cladding, siding, or other similar products
- Downward drilling
- Jack-hammering or chipping when that work is incidental to the scope of work or planned operations of a plumbing or landscaping activity
- Work with powder-actuated tools
- Work incidental to the installation of concrete and masonry materials such as the drilling of holes for plumbing fixtures
- Tile backer board when cut with powered shears or a dust reduction blade having a dust containment device

7.1 Control of Employee Exposures to Airborne Particulates

During operations in which powered tools or equipment are used to cut, grind, core, or drill, concrete or masonry materials, a dust reduction system should be applied to effectively reduce airborne particulates.

Exceptions:

1. A dust-reduction system is not required if the operation, without considering any protection provided by personal protective equipment, does not result in employee exposure exceeding the Permissible Exposure Limits (PEL) for applicable particulates listed in Section 5155 including, but not limited to, crystalline silica, as demonstrated reliably by air sampling data applicable to the specific operation being performed.
2. A dust reduction system is not required for rooftop operations with roofing tile, roofing pavers, or similar materials.
3. During the first 24 hours of an operation undertaken in response to an emergency, a dust reduction system is not required where it can reasonably be demonstrated or foreseen that use of a dust reduction system will materially impair the timely progress of the operation. For the purposes of this exception, "emergency" means an unexpected occurrence requiring immediate action to prevent or mitigate loss of, or damage to, life, health, property, or essential public services. "Emergency" includes, but is not limited to, a fire, flood, earthquake or other soil or geologic movement, structural collapse, damage to a subsurface installation, terrorist act, or sabotage.

7.2 Training

Employee training: An employer whose operations include using powered tools or equipment to cut, grind, core, or drill concrete or masonry materials shall provide training on the following topics to all employees, prior to their assignment to jobs or work areas, where the employer will be conducting these operations:

- The potential health hazards of overexposure to airborne dust generated from concrete and masonry materials, including silicosis, lung cancer, chronic obstructive lung disease (COPD) and decreased lung function
- Methods used by the employer to control employee exposures to airborne dust from concrete and masonry materials, including wet cutting, local exhaust ventilation systems, and isolation of the process from the operator or other employees by means of distance, enclosure, or other method, as applicable
- Proper use and maintenance of dust reduction systems, including the safe handling and disposal of waste materials collected in connection with their use
- The importance of good personal hygiene and housekeeping practices from concrete and masonry materials, when working in proximity to dust, including:
 - Not smoking tobacco products; appropriate methods of cleaning up before eating, and appropriate methods of cleaning clothes
 - Avoid, to the extent practical, activities that would contribute significantly to an employee's exposure to airborne dusts

7.3 Supervisor Training

Prior to engaging in supervision of employees who will be cutting, grinding, drilling, or coring concrete or masonry materials, supervisory employees should be trained on the following topics:

- The information required to be provided by subsection (e)(1) above
- Identification of tasks the employees will perform, which may result in employee exposure to concrete or masonry dust
- Procedures for implementation of the measures used by the employer to reduce the exposure to concrete or masonry dust

Note: Periodic training. The employer should conduct the training required by this section at least annually.

8.0 Respirable Crystalline Silica

This section applies to all occupational exposures to respirable crystalline silica in construction work, except where employee exposure will remain below 25 micrograms per cubic meter of air (25 µg/m³) as an 8-hour time-weighted average (TWA) under any foreseeable conditions. All work with the potential for airborne silica must be pre-planned and a detailed site *and* task specific, written Silica Exposure Control plan (SECP)/ [Job Hazard Analysis (JHA), reference Section 17 Preplanning] must be completed. The SECP/JHA must be submitted to and reviewed by DPR project management prior to start of any work. In addition to including all tasks to be performed, the SECP/JHA must include the company's written exposure control plan (reference sub-section 8.5 below).

Reference: CFR 1926.1153 Respirable Crystalline Silica and CCR Section 1532.3 Occupational Exposure to Respirable Crystalline Silica.

Effective Dates	Implementation Requirements
June 23, 2016	Standard takes effect
September 23, 2017	Contractors must have plan fully implemented (written exposure control plan, air sampling, medical surveillance, employee respiratory testing, employee respiratory training, recordkeeping, wet or vacuum methods)
June 23, 2018	Requirements for methods of sampling analysis

8.1 Specified Control Methods

For each employee engaged in a task identified on Table 1 (see below), the employer shall fully and properly implement the engineering controls, work practices, and respiratory protection specified for the task on Table 1, unless the employer assesses and limits the exposure of the employee to respirable crystalline silica in accordance with 8.2 of this sub-section.

Table 1

Equipment/task	Engineering and work practice control methods	Required respiratory protection and minimum assigned protection factor (APF)	
		≤ 4 hours/shift	> 4 hours/shift
(i) Stationary masonry saws	Use saw equipped with integrated water delivery system that continuously feeds water to the blade Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions	None	None
(ii) Handheld power saws (any blade diameter)	Use saw equipped with integrated water delivery system that continuously feeds water to the blade Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions:		

Equipment/task	Engineering and work practice control methods	Required respiratory protection and minimum assigned protection factor (APF)	
		≤ 4 hours/shift	> 4 hours/shift
(ii) Handheld power saws (any blade diameter) <i>cont'd</i>	-When used outdoors	None	APF 10
	-When used indoors or in an enclosed area	APF 10	APF 10
(iii) Handheld power saws for cutting fiber-cement board (with blade diameter of 8 inches or less)	For tasks performed outdoors only: Use saw equipped with commercially available dust collection system Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions Dust collector must provide the air flow recommended by the tool manufacturer, or greater, and have a filter with 99% or greater efficiency	None	None
(iv) Walk-behind saws	Use saw equipped with integrated water delivery system that continuously feeds water to the blade Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions:		
	-When used outdoors	None	None
	-When used indoors or in an enclosed area	APF 10	APF 10
(v) Drivable saws	For tasks performed outdoors only: Use saw equipped with integrated water delivery system that continuously feeds water to the blade Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions	None	None
(vi) Rig-mounted core saws or drills	Use tool equipped with integrated water delivery system that supplies water to cutting surface Operate and maintain tool in accordance	None	None

Equipment/task	Engineering and work practice control methods	Required respiratory protection and minimum assigned protection factor (APF)	
		≤ 4 hours/shift	> 4 hours/shift
(vi) Rig-mounted core saws or drills <i>cont'd</i>	with manufacturer's instructions to minimize dust emissions		
(vii) Handheld and stand-mounted drills (including impact and rotary hammer drills)	Use drill equipped with commercially available shroud or cowl with dust collection system Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions Dust collector must provide the air flow recommended by the tool manufacturer, or greater, and have a filter with 99% or greater efficiency and a filter-cleaning mechanism Use a HEPA-filtered vacuum when cleaning holes	None	None
(viii) Dowel drilling rigs for concrete	For tasks performed outdoors only: Use shroud around drill bit with a dust collection system. Dust collector must have a filter with 99% or greater efficiency and a filtercleaning mechanism Use a HEPA-filtered vacuum when cleaning holes	APF 10	APF 10
(ix) Vehicle-mounted drilling rigs for rock and concrete	Use dust collection system with close capture hood or shroud around drill bit with a low-flow water spray to wet the dust at the discharge point from the dust collector OR Operate from within an enclosed cab and use water for dust suppression on drill bit	None None	None None

Equipment/task	Engineering and work practice control methods	Required respiratory protection and minimum assigned protection factor (APF)	
		≤ 4 hours/shift	> 4 hours/shift
(x) Jackhammers and handheld powered chipping tools	Use tool with water delivery system that supplies a continuous stream or spray of water at the point of impact:		
	-When used outdoors	None	APF 10
	-When used indoors or in an enclosed area	APF 10	APF 10
	OR		
	Use tool equipped with commercially available shroud and dust collection system		
	Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions		
	Dust collector must provide the air flow recommended by the tool manufacturer, or greater, and have a filter with 99% or greater efficiency and a filter-cleaning mechanism:		
	-When used outdoors	None	APF 10
(xi) Handheld grinders for mortar removal (i.e., tuckpointing)	-When used indoors or in an enclosed area	APF 10	APF 10
	Use grinder equipped with commercially available shroud and dust collection system	APF 10	APF 25
	Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions		
	Dust collector must provide 25 cubic feet per minute (cfm) or greater of airflow per		

Equipment/task	Engineering and work practice control methods	Required respiratory protection and minimum assigned protection factor (APF)	
		≤ 4 hours/shift	> 4 hours/shift
(xi) Handheld grinders for mortar removal (i.e., tuckpointing) <i>cont'd</i>	inch of wheel diameter and have a filter with 99% or greater efficiency and a cyclonic pre-separator or filter-cleaning mechanism		
(xii) Handheld grinders for uses other than mortar removal	For tasks performed outdoors only: Use grinder equipped with integrated water delivery system that continuously feeds water to the grinding surface	None	None
	Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions		
	OR		
	Use grinder equipped with commercially available shroud and dust collection system		
	Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions		
	Dust collector must provide 25 cubic feet per minute (cfm) or greater of airflow per inch of wheel diameter and have a filter with 99% or greater efficiency and a cyclonic pre-separator or filter-cleaning mechanism:		
	-When used outdoors	None	None
	-When used indoors or in an enclosed area	None	APF 10
(xiii) Walk-behind milling machines and floor grinders	Use machine equipped with integrated water delivery system that continuously feeds water to the cutting surface	None	None

Equipment/task	Engineering and work practice control methods	Required respiratory protection and minimum assigned protection factor (APF)	
		≤ 4 hours/shift	> 4 hours/shift
(xiii) Walk-behind milling machines and floor grinders <i>cont'd</i>	Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions OR Use machine equipped with dust collection system recommended by the manufacturer Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions Dust collector must provide the air flow recommended by the manufacturer, or greater, and have a filter with 99% or greater efficiency and a filter-cleaning mechanism When used indoors or in an enclosed area, use a HEPA-filtered vacuum to remove loose dust in between passes	None	None
(xiv) Small drivable milling machines (less than half-lane)	Use a machine equipped with supplemental water sprays designed to suppress dust. Water must be combined with a surfactant Operate and maintain machine to minimize dust emissions	None	None
(xv) Large drivable milling machines (half-lane and larger)	For cuts of any depth on asphalt only: Use machine equipped with exhaust ventilation on drum enclosure and supplemental water sprays designed to suppress dust	None	None

Equipment/task	Engineering and work practice control methods	Required respiratory protection and minimum assigned protection factor (APF)	
		≤ 4 hours/shift	> 4 hours/shift
(xv) Large drivable milling machines (half-lane and larger) <i>cont'd</i>	Operate and maintain machine to minimize dust emissions For cuts of four inches in depth or less on any substrate: Use machine equipped with exhaust ventilation on drum enclosure and supplemental water sprays designed to suppress dust Operate and maintain machine to minimize dust emissions OR Use a machine equipped with supplemental water spray designed to suppress dust. Water must be combined with a surfactant Operate and maintain machine to minimize dust emissions	None None	None None
(xvi) Crushing machines	Use equipment designed to deliver water spray or mist for dust suppression at crusher and other points where dust is generated (e.g., hoppers, conveyers, sieves/sizing or vibrating components, and discharge points) Operate and maintain machine in accordance with manufacturer's instructions to minimize dust emissions Use a ventilated booth that provides fresh, climate-controlled air to the operator, or a remote control station	None	None

Equipment/task	Engineering and work practice control methods	Required respiratory protection and minimum assigned protection factor (APF)	
		≤ 4 hours/shift	> 4 hours/shift
(xvii) Heavy equipment and utility vehicles used to abrade or fracture silica-containing materials (e.g., hoe-ramming, rock ripping) or used during demolition activities involving silica-containing materials	Operate equipment from within an enclosed cab	None	None
	When employees outside of the cab are engaged in the task, apply water and/or dust suppressants as necessary to minimize dust emissions	None	None
(xviii) Heavy equipment and utility vehicles for tasks such as grading and excavating but not including: Demolishing, abrading, or fracturing silica-containing materials	Apply water and/or dust suppressants as necessary to minimize dust emissions	None	None
	OR When the equipment operator is the only employee engaged in the task, operate equipment from within an enclosed cab	None	None

When implementing the control measures specified in Table 1 (above), each employer shall:

- For tasks performed indoors or in enclosed areas, the JHA must detail applicable airflow rates to provide the specific means of exhaust needed to minimize the accumulation of visible airborne dust;
- For tasks performed using wet methods, the JHA must detail applicable water flow rates sufficient to minimize release of visible dust;
- For measures implemented that include an enclosure, ensure that the enclosure:
 - Is maintained as free as practicable from settled dust;
 - Has door seals and closing mechanisms that work properly;
 - Has gaskets and seals that are in good condition and working properly;
 - Is under positive pressure maintained through continuous delivery of fresh air;
 - Has intake air that is filtered through a filter that is 95% efficient in the 0.3-10.0 μm range (e.g., MERV-16 or better); and
 - Has heating and cooling capabilities.
- Where an employee performs more than one task on Table 1 during the course of a shift, and the total duration of all tasks combined is more than four hours, the required respiratory protection for each task is the respiratory protection specified for more than four hours per shift. If the total duration of all tasks on Table 1 combined is less than four hours, the required respiratory protection for each task is the respiratory protection specified for less than four hours per shift

8.2 **Alternative exposure control methods**

Alternative Exposure Control Methods apply for tasks not listed in OSHA's Construction Standard Table 1, or where DPR Construction cannot not fully and properly implement the engineering controls, work practices, and respiratory protection described in Table 1.



First, the employer will assess the exposure of each employee who is or may reasonably be expected to be exposed to Respirable Crystalline Silica at or above the Action Level in accordance with either the Performance Option or the Scheduled Monitoring Option.

Performance Option – the employer will assess the 8-hour TWA exposure for each employee on the basis of any combination of air monitoring data or objective data sufficient to accurately characterize employee exposures to Respirable Crystalline Silica.

Scheduled Monitoring Option:

The employer will perform initial monitoring to assess the 8-hour TWA exposure for each employee on the basis of one or more personal breathing zone air samples that reflect the exposures of employees on each shift, for each job classification, and in each work area. Where several employees perform the same tasks on the same shift and in the same work area, the employer will plan to monitor a representative fraction of these employees. When using representative monitoring, the employer will sample the employee(s) who are expected to have the highest exposure to Respirable Crystalline Silica.

If initial monitoring indicates that employee exposures are below the Action Level, the employer will probably discontinue monitoring for those employees whose exposures are represented by such monitoring.

Where the most recent exposure monitoring indicates that employee exposures are at or above the Action Level but at or below the PEL, the employer will repeat such monitoring within six months of the most recent monitoring.

Where the most recent exposure monitoring indicates that employee exposures are above the PEL, the employer will repeat such monitoring within three months of the most recent monitoring.

Where the most recent (non-initial) exposure monitoring indicates that employee exposures are below the Action Level, the employer will repeat such monitoring within six months of the most recent monitoring until two consecutive measurements, taken seven or more days apart, are below the Action Level, at which time the employer will probably discontinue monitoring for those employees whose exposures are represented by such monitoring, except when a reassessment is required. The employer will reassess exposures whenever a change in the production, process, control equipment, personnel, or work practices may reasonably be expected to result in new or additional exposures at or above the Action Level, or when the employer has any reason to believe that new or additional exposures at or above the Action Level have occurred.

The employer will ensure that all Respirable Crystalline Silica samples taken to satisfy the monitoring requirements of this program and OSHA are collected by a qualified individual (i.e. a Certified Industrial Hygienist) and the samples are evaluated by a qualified laboratory (i.e. accredited to ANSI/ISO/IEC Standard 17025:2005 with respect to Crystalline Silica analyses by a body that is compliant with ISO/IEC Standard 17011:2004 for implementation of quality assessment programs).

Within five working days after completing an exposure assessment, the employer will individually notify each affected employee in writing of the results of that assessment or post the results in an appropriate location accessible to all affected employees.

Whenever an exposure assessment indicates that employee exposure is above the PEL, the employer will describe in the written notification the corrective action being taken to reduce employee exposure to or below the PEL.

Where air monitoring is performed, the employer will provide affected employees or their designated representatives an opportunity to observe any monitoring of employee exposure to Respirable Crystalline Silica. When observation of monitoring requires entry into an area where the use of protective clothing or equipment is required for any workplace hazard, the employer will provide the observer with protective clothing and equipment at no cost and shall ensure that the observer uses such clothing and equipment.

Once air monitoring has been performed, the employer will determine its method of compliance based on the monitoring data and the hierarchy of controls. The employer will use engineering and work practice

controls to reduce and maintain employee exposure to Respirable Crystalline Silica to or below the PEL, unless the employer can demonstrate that such controls are not feasible. Wherever such feasible engineering and work practice controls are not sufficient to reduce employee exposure to or below the PEL, the employer will nonetheless use them to reduce employee exposure to the lowest feasible level and shall supplement them with the use of respiratory protection.

In addition to the requirements of this program, the employer will comply with other programs and OSHA standards (such as 29 CFR 1926.57 [Ventilation]), when applicable where abrasive blasting is conducted using Crystalline Silica-containing blasting agents, or where abrasive blasting is conducted on substrates that contain Crystalline Silica.

8.3 *Respiratory protection*

Where respirator use is required by this section, the employer shall institute and fully implement a written respiratory protection program in accordance with 29 CFR 1910.134. Reference DPR's EHSP Rev 9 Section 19.

8.4 *Housekeeping*

Dry sweeping or dry brushing are prohibited where such activity could contribute to employee exposure to respirable crystalline silica. Wet sweeping, HEPA-filtered vacuuming or other methods that minimize the likelihood of exposure shall be used.

The employer shall not allow compressed air to be used to clean clothing or surfaces where such activity could contribute to employee exposure to respirable crystalline silica unless: The compressed air is used in conjunction with a ventilation system that effectively captures the dust cloud created by the compressed air; or no alternative method is feasible.

8.5 *Written exposure control plan*

A written, site specific silica exposure control plan shall be submitted to the DPR project team, at a minimum, two weeks prior to start of work for project management review. The expectation is that the written plan summarizes steps the company will take to protect their workers and other workers in adjacent areas from exposure to airborne silica. The written plan should include, at a minimum:

- A description of the tasks in the workplace that involve exposure to respirable crystalline silica;
- A description of the housekeeping measures used to limit employee exposure to respirable crystalline silica
- Engineering controls (such as water or ventilation) to be used to limit worker exposure to airborne silica
- Provisions for providing respiratory protection when engineering controls cannot adequately limit exposure to airborne silica
- Provisions to limit worker access to high exposure areas
- Train workers on silica risks and how to limit exposures
- Annual review to verify effectiveness of the plan
- Identify the company's competent person, and schedule for frequent and regular inspections of job sites, materials, and equipment to implement the written exposure control plan

8.6 *Medical Surveillance*

Medical surveillance will be made available for each employee who will be required to use a respirator for 30 or more days per year due to their Respirable Crystalline Silica exposure. Medical surveillance (i.e. medical examinations and procedures) will be performed by a PLHCP and provided at no cost to the employee at a reasonable time and place.

The employer will make available an initial (baseline) medical examination within 30 days after initial assignment, unless the employee has received a medical examination that meets the requirements of the



OSHA Respirable Crystalline Silica Construction Standard within the last three years. The examination shall consist of:

A medical and work history, with emphasis on past, present, and anticipated exposure to Respirable Crystalline Silica, dust, and other agents affecting the respiratory system in addition to any history of respiratory system dysfunction, including signs and symptoms of respiratory disease (e.g., shortness of breath, cough, wheezing), history of tuberculosis, and smoking status and history;

A physical examination with special emphasis on the respiratory system;

A chest X-ray (a single posterior-anterior radiographic projection or radiograph of the chest at full inspiration recorded on either film [no less than 14 x 17 inches and no more than 16 x 17 inches] or digital radiography systems) interpreted and classified according to the International Labor Office (ILO) International Classification of Radiographs of Pneumoconiosis by a NIOSH-certified B Reader;

A pulmonary function test to include forced vital capacity (FVC) and forced expiratory volume in one second (FEV1) and FEV1/FVC ratio, administered by a spirometry technician with a current certificate from a NIOSH-approved spirometry course;

Testing for latent tuberculosis infection; and

Any other tests deemed appropriate by the PLHCP.

The employer will make available medical examinations that include the aforementioned procedures (except testing for latent tuberculosis infection) at least every three years. If recommended by the PLHCP, periodic examinations can be more frequently than every three years.

The employer will ensure that the examining PLHCP has a copy of the OSHA Respirable Crystalline Silica Construction Standard, this program, and the following information:

- A description of the employee's former, current, and anticipated duties as they relate to the employee's occupational exposure to Respirable Crystalline Silica;
- The employee's former, current, and anticipated levels of occupational exposure to Respirable Crystalline Silica;
- A description of any personal protective equipment (PPE) used or to be used by the employee, including when and for how long the employee has used or will use that equipment; and
- Information from records of employment-related medical examinations previously provided to the employee and currently within the control of the employer.

The employer will ensure that the PLHCP explains to the employee the results of the medical examination and provides each employee with a written medical report within 30 days of each medical examination performed. The written report shall contain:

- A statement indicating the results of the medical examination, including any medical condition(s) that would place the employee at increased risk of material impairment to health from exposure to Respirable Crystalline Silica and any medical conditions that require further evaluation or treatment;
- Any recommended limitations on the employee's use of respirators;
- Any recommended limitations on the employee's exposure to Respirable Crystalline Silica; and
- A statement that the employee should be examined by a Specialist if the chest X-ray is classified as 1/0 or higher by the B Reader, or if referral to a Specialist is otherwise deemed appropriate by the PLHCP.

The employer will also obtain a written medical opinion from the PLHCP within 30 days of the medical examination. The written opinion shall contain only the following in order to protect the employee's privacy:

- The date of the examination;
- A statement that the examination has met the requirements of the OSHA Respirable Crystalline Silica Construction Standard; and

- Any recommended limitations on the employee's use of respirators.

If the employee provides written authorization, the written opinion shall also contain either or both of the following:

- Any recommended limitations on the employee's exposure to Respirable Crystalline Silica; and/or
- A statement that the employee should be examined by a Specialist if the chest X-ray is classified as 1/0 or higher by the B Reader, or if referral to a Specialist is otherwise deemed appropriate by the PLHCP.

If the PLHCP's written medical opinion indicates that an employee should be examined by a Specialist, the employer will make available a medical examination by a Specialist within 30 days after receiving the PLHCP's written opinion. The employer will ensure that the examining Specialist is provided with all of the information that the employer is obligated to provide to the PLHCP.

The employer will ensure that the Specialist explains to the employee the results of the medical examination and provides each employee with a written medical report within 30 days of the examination. The written report will contain:

- A statement indicating the results of the medical examination, including any medical condition(s) that would place the employee at increased risk of material impairment to health from exposure to Respirable Crystalline Silica and any medical conditions that require further evaluation or treatment;
- Any recommended limitations on the employee's use of respirators; and
- Any recommended limitations on the employee's exposure to respirable crystalline Silica.

In addition, the employer will obtain a written opinion from the Specialist within 30 days of the medical examination. The written opinion shall contain the following:

- The date of the examination;
- Any recommended limitations on the employee's use of respirators; and
- If the employee provides written authorization, the written opinion shall also contain any recommended limitations on the employee's exposure to Respirable Crystalline Silica.

8.7 Training

- Employee training should cover the information contained within the section. Upon completion, the worker should be able to demonstrate knowledge and understanding of at least the following:
 - The health hazards associated with exposure to respirable crystalline silica; **Cancer, lung effects, immune system effects, and kidney effects**
 - Specific tasks in the workplace that could result in exposure to respirable crystalline silica
 - Specific measures the employer has implemented to protect employees from exposure to respirable crystalline silica, including engineering controls, work practices, and respirators to be used
 - The contents of this section
 - The identity of the competent person
 - The purpose and a description of the medical surveillance program

8.8 Recordkeeping

The employer will make and maintain an accurate record of all exposure measurements taken to assess employee exposure to Respirable Crystalline Silica. This record will include at least the following information:

- The date of measurement for each sample taken;
- The task monitored;
- Sampling and analytical methods used;
- Number, duration, and results of samples taken;
- Identity of the laboratory that performed the analysis;

- Type of personal protective equipment (PPE), such as respirators, worn by the employees monitored; and
- Name, social security number, and job classification of all employees represented by the monitoring, indicating which employees were actually monitored.

The employer will ensure that exposure records are maintained and made available in accordance with 29 CFR 1910.1020. Exposure records will be kept for at least 30 years.

The employer shall make and maintain an accurate record of all objective data relied upon to comply with the requirements of the OSHA Respirable Crystalline Silica Construction Standard. This record shall include at least the following information:

- The Crystalline Silica-containing material in question;
- The source of the objective data;
- The testing protocol and results of testing;
- A description of the process, task, or activity on which the objective data were based; and
- Other data relevant to the process, task, activity, material, or exposures on which the objective data were based.

The employer will ensure that objective data are maintained and made available in accordance with 29 CFR 1910.1020. Objective data records will be kept for at least 30 years.

The employer will make and maintain an accurate record for each employee enrolled in the Medical Surveillance portion of this program. The record shall include the following information about the employee:

- Name and social security number;
- A copy of the PLHCPs' and/or Specialists' written medical opinions; and
- A copy of the information provided to the PLHCPs and Specialists.

The employer will ensure that medical records are maintained and made available in accordance with 29 CFR 1910.1020. Medical records will be kept under lock and key for at least the duration of employment plus 30 years. It is necessary to keep these records for extended periods because Silica-related diseases such as cancer often cannot be detected until several decades after exposure. However, if an employee works for an employer for less than one year, the employer does not have to keep the medical records after employment ends, as long as the employer gives those records to the employee.

9.0 Indoor Air Quality

This section will address ways in which the HVAC equipment and ductwork should be protected from potential collection of dust, odors and other contaminants. *Also refer to the Mold Section of the Infection & Contamination Control Section 14.*

9.1 HVAC Return Side

- During periods of heavy construction (e.g., construction creating excessive amounts of dust or odors) the HVAC equipment should be shut down. In this case the space and hallway outside the space act as a plenum for return air. All vents into the plenum should be sealed throughout construction.
- If the new HVAC equipment is in operation during any construction period, the return side should be covered with an adhering a bag-filter-type media over the return louvers
- Digital photographs should be taken to document and ensure these measures have been implemented appropriately

9.2 HVAC Central Filtration

- When the HVAC equipment is operational during the construction period it should be fitted with filters with a MERV value of 8
- Product cut sheets should be provided for the filters being used

- Digital photographs should be taken to document and ensure these measures have been implemented appropriately

9.3 HVAC Supply Side

- As ductwork installation progresses, the supply side should be sealed off to eliminate dust settling into the ductwork
- Digital photographs should be taken to document and ensure these measures have been implemented appropriately

9.4 HVAC Duct Cleaning

- Ductwork should be cleaned and sealed at the manufacturer's duct fabrication facility prior to shipment to the project
- Ductwork should remain sealed during storage at the manufacture's facility and during shipment to the jobsite
- Ductwork should remain sealed during storage on the jobsite and until it is installed
- Digital photographs should be taken to document and ensure these measures have been implemented appropriately

9.5 Source Control

This section will address ways to prevent potential air quality degradation by controlling pollutants at their source.

9.5.1 Product Substitution

- All adhesives, paints and coatings, carpet systems and composite wood products will comply with the VOC standards as outlined in the LEED Rating System v2.1 Reference Guide, dated May 2003 and submitted under Credit EQ 4
- Documentation to prove these measures have been implemented will be submitted under LEED Credit EQ 4

9.5.2 Modifying Equipment Operation

- When construction equipment is not in use it should be shut down (i.e. generators, forklifts, etc) to eliminate excessive exhaust vapors
- All construction equipment that is operated inside the building should be powered by battery, electricity, pneumatic or other sources so as not to create exhaust vapors within the building. (I.e. saws, nail guns, scissor lifts, etc.)
- Digital photographs and/or video should be taken to document and ensure these measures have been implemented appropriately

9.5.3 Changing Work Practices

- Every effort should be made to use low emitting products in all aspects of the construction process. However, if a product is required for use that is not low emitting it should be used during off hours or a special outside exhaust system should be utilized so as to limit personnel exposure.
- When it is necessary to clean the building by sweeping, a sweeping compound should be utilized to minimize dust
- When it is necessary to clean the building by vacuum, a filter should be placed on the exhaust air
- Digital photographs should be taken to document and ensure these measures have been implemented appropriately

9.5.4 Local Exhaust

Special exhaust systems that vent directly outside the building should be utilized in the event it is deemed necessary to protect personnel from excessive dust, odors and other pollutants.



9.5.5 *Air Cleaning*

If using a special outside exhaust systems is not feasible for a particular construction activity requiring ventilation, a portable air cleaner should be utilized.

9.5.6 *Cover or Seal*

All construction waste materials should be placed directly into construction recycling bins provided on the jobsite.

10.0 *Auditing and Protection*

Documented inspections of the environmental protection methods used should be regularly conducted by DPR and subcontractors. All items observed that are deficient should be corrected immediately.

Chemical and Environmental Management Forms

Forms

- Chemical and Environmental Management Checklist
- Chemical and Environmental Inspection Checklist
- Specifics & Comments – Chemical and Environmental Inspection
- Hazard Communication Training Verification
- Hazard Communication Training Verification – English
- Hazard Communication Training Verification – Spanish
- Respirable Crystalline Silica Awareness Training Verification - English
- Respirable Crystalline Silica Awareness Training Verification – Spanish
- Silica Exposure Control Plan



Chemical and Environmental Management Checklist

To be completed prior to any work involving chemicals or other hazardous substances.

This is an initial review of required procedures but may also be utilized periodically to ensure all systems are maintained.

Project/Project#:		Date:	
Inspector:		Time:	

General	Yes	No	N/A
Has the project been surveyed for hazardous materials and substances?	☐	☐	☐
Have work areas have been protected/isolated from chemical exposure? (rooms decontaminated, lines and tanks removed/purged/capped, systems locked out/tagged out, drains covered etc.)	☐	☐	☐
Has a chemical inventory checklists been complete?	☐	☐	☐
Have SDS been obtained and filed (or readily available) for all hazardous materials on the project, including owner generated hazardous substances?	☐	☐	☐
Have the chemicals to be used been identified and less hazardous alternatives used when practicable? (e.g. water based or less noxious solvents, paint or epoxy)	☐	☐	☐
Has the Right to Know Area been established that is easily accessible to all?	☐	☐	☐
Have all employees completed GHS training?	☐	☐	☐
Have all affected employees received training on site specific chemicals?	☐	☐	☐
Are all employees aware of the hazards posed by site specific chemicals?	☐	☐	☐
Are all chemicals and containers properly labeled with substance, manufacturer, hazard recognition symbols and appropriate warnings?	☐	☐	☐

Storage and Containment	Yes	No	N/A
Have storage areas for all hazardous materials identified?	☐	☐	☐
Are storage areas protected from vehicle and pedestrian traffic, with materials secured?	☐	☐	☐
Are materials double contained, protected from weather and not in close proximity to drains or waterways?	☐	☐	☐
Are all materials stored in appropriate containers?	☐	☐	☐
Have ignition sources been identified and materials stored away from ignition sources?	☐	☐	☐
Have materials of cross reactivity been identified, separated and isolated?	☐	☐	☐
Has appropriate signage been posted?	☐	☐	☐
Are appropriately rated fire extinguishers or other fire systems within required distances?	☐	☐	☐

Usage	Yes	No	N/A
Are appropriate PPE available and utilized during handling, dispensing and use of chemicals?	☐	☐	☐
Is the PPE appropriately cleaned and stored when not in use?	☐	☐	☐
Are hazard specific respirators available and used as necessary for all affected workers, and have workers completed a medical evaluation, been fit tested and trained?	☐	☐	☐
Has adequate ventilation of the space been established and maintained?	☐	☐	☐
When material is not in use, is it protected, covered, lids closed etc.?	☐	☐	☐
Are all secondary containers appropriately labeled?	☐	☐	☐

Disposal	Yes	No	N/A
Are all materials being disposed of per OSHA, EPA, DOT etc. guidelines?	☐	☐	☐
Are all hazardous materials tested and approved by owner prior to removal from site?	☐	☐	☐
Are hazmat kits for clean-up and disposal identified and on site as per requirements?	☐	☐	☐
Are dumpsters covered and protected from the elements (wind, rain, snow etc.)?	☐	☐	☐
Are dumpsters and trash containers in good condition, leak free and regularly inspected?	☐	☐	☐
Are washout areas designated, double contained and material properly disposed of?	☐	☐	☐
Is a material recycling program implemented as practicable?	☐	☐	☐
Is material collected and removed from site in a timely manner?	☐	☐	☐



Chemical and Environmental Inspection Form

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Project Information		Date	
Project/Project#		Week Starting	
Contractor/Subcontractor		Week Ending	
Inspector's Name			
Weather (check all that apply)	Hot ☐	Dry ☐	Clear ☐
	Windy ☐	Cool ☐	Wet ☐
	Overcast ☐	Raining ☐	

Partial List of Safety Items – Identify & Correct All Unsafe Conditions								
A	Inspect at a minimum daily, mark a “+” for items being performed well or in good condition	MON	TUES	WED	THURS	FRI	SAT	SUN
B	For items that need improvement or corrective action, mark a “Δ” and comment on the back any action needed/taken							
1	General Requirements							
	All identified systems decontaminated, locked out/tagged out, capped, purged and otherwise isolated?							
	All affected workers trained for chemicals that they are using?							
	All containers, primary and secondary are appropriately labeled with substance name, manufacturer, hazard recognition symbols and appropriate warnings?							
	SDS available for all chemicals on site?							
	All drains, inlets, waterways etc. blocked, covered, or otherwise protected?							
2	Storage Containment							
	Materials protected from vehicle and pedestrian traffic?							
	Materials double contained, protected from weather and not in close proximity to drains or waterways?							
	Materials stored in appropriate containers and locations?							
	Materials stored away from all ignition sources?							
	Materials or cross reactivity separated?							
	Appropriate signage posted?							
	Appropriately rated fire extinguisher present or other fire system and within required distance?							
3	Usage							
	Appropriate PPE utilized for all material handling, use and disposal?							
	Appropriate respirators used and all workers appropriately trained for use?							
	Adequate ventilation of the space established and being maintained?							
	Material not in use is protected, covered, capped, lids closed etc.?							
	All spills are properly cleaned up?							
4	Disposal							
	Materials disposed of as per OSHA, EPA, DOT etc. guidelines?							
	Washout areas used, maintained and protected?							
	Dumpsters in good condition, covered, maintained and emptied routinely?							
	Hazardous materials are removed from the project as soon as practicable?							



Specifics & Comments – Chemical and Environmental Inspection

Monday

Tuesday

Wednesday

Thursday

Friday

Saturday

Sunday

Hazard Communication Training Verification

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Name:		Date:	
Project/Project#:		Company	

OSHA has adopted the “United Nations Globally Harmonized System of Classification and Labeling of Chemicals” commonly and conveniently called “GHS”. The GHS initiative has become part of OSHA’s recently updated Hazard Communication Program requirements. OSHA has modified the Hazard Communication Standard to adopt the GHS to improve safety and health of workers, as well as bring the US in alignment with the chemical communication standard that’s used in most of the rest of the world. It will also provide a more standardized approach to classifying the hazards and conveying the information.

Workers should be trained in the proper identification of chemicals through labels and Safety Data Sheets, as well as be able to identify any potential hazards working with or around a chemical through hazard identifying pictograms and “signal words”. A hazard evaluation should be completed to determine that all safety precautions have been addressed for use of chemicals prior to performing a task. Use of chemicals can be considered a high hazard activity by DPR and therefore requires daily, detailed, written pre-task planning prior to use. Below is a review of GHS guidelines you should be familiar with prior to using chemicals:

Right to Understand (Formerly Right to Know):

- All workers should be familiar with the chemicals in their work area
- The hazards of the chemicals in the areas where you will be working
- To protect yourself from the hazards
- Location of the right to know area, and be knowledgeable of the chemicals. (*Refer to the SDS as needed*)

How to identify a chemical and its hazards:

Identifying a chemical and the hazards it possesses can be done one of two ways. The first, acquire the SDS. The second, read the label.

Safety Data Sheets (SDS):

The information required on the (SDS) will be standardized in format -- using specific headings in a specified sequence. The standardized 16 sections are as follows:

- Section 1, Product Identification
- Section 2, Hazard Identification
- Section 3, Composition Information
- Section 4, First Aid Measures
- Section 5, Fire Fighting Measures
- Section 6, Accidental Release Measure
- Section 7, Handling & Storage
- Section 8, Exposure Controls and PPE
- Section 9, Physical and Chemical Properties
- Section 10, Stability and Reactivity
- Section 12, Ecological information*
- Section 13, Disposal considerations*
- Section 14, Transport information*
- Section 15, Regulatory information*
- Section 16, Other information, includes the date of preparation or last revision.

*Note: Since other Agencies regulate this information, OSHA will not be enforcing Sections 12 through 15

The hazard category rating system under the GHS system was changed from the Hazardous Materials Identification System (HMIS) / National Fire Protection Agency (NFPA) rating scale. The new GHS ratings range from 1 to 5, category 1 means severe hazard and category 5 means minimal hazard. It is imperative to remember this when reading the SDS to ensure there is no confusion or conflicting information given (Refer to section 16 of the SDS).

Hazard Communication Training Verification

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Marking systems: (each category is rated from 0-4, 0 being the Minimal, and 4 being Extreme)



HMIS/NFPA Hazard Ratings

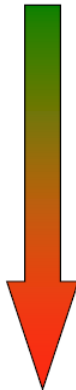
0 = Minimal Hazard

1 = Slight Hazard

2 = Moderate Hazard

3 = Serious Hazard

4 = Severe Hazard



GHS Hazard Categories

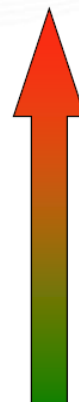
Category 1 = Severe Hazard

Category 2 = Serious Hazard

Category 3 = Moderate Hazard

Category 4 = Slight Hazard

Category 5 = Minimal Hazard



Where will I see those differences?

- The GHS hazard categories are displayed in the chemical's Safety Data Sheet and not typically shown on the label
- Due to the significant change in classifications, it will be imperative to train employees on the change to ensure there is no confusion or conflicting information given

Labels:

Chemical manufacturers and importers will be required to provide a label that includes the following:

- **Signal word**
 - o The signal words used are "**danger**" and "**warning**." "Danger" is used for the more severe hazards, while "warning" is used for less severe hazards. Only one signal word is allowed per package.
- **Pictogram**
 - o Each pictogram consists of a different symbol on a white background within a red square.
 - o There are 9 standardized pictograms used under GHS

Health Hazard  • Carcinogen • Mutagenicity • Reproductive Toxicity • Respiratory Sensitizer • Target Organ Toxicity • Aspiration Toxicity	Flame  • Flammables • Pyrophorics • Self-Heating • Emits Flammable Gas • Self-Flammable • Organic Peroxides	Exclamation Mark  • Irritant (skin and eye) • Skin Sensitizer • Acute Toxicity (harmful) • Narcotic Effects • Respiratory Tract Irritant • Hazardous to Ozone Layer (Non-Mandatory)	Gas Cylinder  • Gases Under Pressure	Corrosion  • Skin Corrosion/ Burns • Eye Damage • Corrosive to Metals	Exploding Bomb  • Explosives • Self-Reactives • Organic Peroxides	Flame Over Circle  • Oxidizers	Environment (Non-Mandatory)  • Aquatic Toxicity	Skull and Crossbones  • Acute Toxicity (fatal or toxic)
--	--	--	--	---	---	---	---	---

- **Hazard statement for each hazard class and category**
 - o a statement assigned to a hazard class and category that describes the nature and degree of a chemical hazard
- **Precautionary statements**
 - o a phrase that describes recommended measures to be taken to minimize or prevent adverse effects resulting from exposure, improper storage or handling of a hazardous chemical
- **Supplier identification**
 - o The name, address and telephone of the manufacturer or supplier of the product should be provided on the label



Hazard Communication Training Verification

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Secondary Containers and labeling:

Employers may choose to label workplace containers within the same GHS label that is used to ship containers, or with label alternatives that meet the requirements for the standard.

This document hereby certifies that I have been instructed on the items listed above and I have had an opportunity to ask questions. I fully understand the DPR requirements for work involving the use of chemicals.

Name/signature of employee

Date

Name/signature of person conducting verification

Date

*A copy of this form should be faxed to the regional safety professional
PRIOR TO START OF WORK.*



Verificación de Entrenamiento en Comunicación de Riesgos

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Nombre:		Fecha:	
Proyecto/# de Proyecto:		Compañía:	

OSHA ha adoptado el "Sistema Globalmente Armonizado de las Naciones Unidas de Clasificación y Etiquetado de Productos Químicos" comúnmente y convenientemente llamado "SGA". La iniciativa GHS se ha convertido en parte de los requisitos del Programa de Comunicación de Riesgos recientemente actualizadas de OSHA. OSHA ha modificado la Norma de Comunicación de Riesgos (HCS) para adoptar el GHS para mejorar la seguridad y la salud de los trabajadores, así como poner a los EE.UU. en alineación con el estándar de comunicación de químicos que se utiliza en la mayor parte del resto del mundo. También proporcionará un enfoque más estandarizado para clasificar los peligros y transmitir la información.

Los trabajadores deben ser entrenados correctamente en la identificación de los productos químicos a través de etiquetas y fichas de datos de seguridad "SDS" (anteriormente conocidos como Ficha de Datos de Seguridad de Materiales, o "SDS"), así como ser capaces de identificar los peligros potenciales de trabajar con o alrededor de una sustancia química a través de identificación de peligro por fotos y "palabras clave". Una evaluación de riesgos debe ser conducida para determinar que todas las medidas de seguridad han sido abordadas para el uso de productos químicos antes de realizar una tarea. El uso de productos químicos se puede considerar una actividad de alto riesgo por DPR y por lo tanto requiere una planificación previa a la tarea (PTP) diaria y detallada por escrito antes de su uso. A continuación se muestra una revisión de las reglas del SGA con las que debe estar familiarizado antes de usar productos químicos:

Derecho a Entender (Anteriormente Derecho a Saber):

- Todos los trabajadores deben estar familiarizados con los productos químicos en su área de trabajo
- Los riesgos de los productos químicos en las zonas donde va a trabajar
- Para protegerse de los peligros
- Localización de la zona del Derecho a Saber, y tener conocimiento de los productos químicos. (*Consulte los SDS, según sea necesario*)

Cómo identificar un producto químico y sus peligros:

La identificación de un producto químico y los peligros que posee se puede hacer de 2 maneras. El primero, simplemente adquiriendo la Hoja de Seguridad que también se conoce como el SDS y formalmente conocido como la Hoja de Datos de Seguridad del Material o SDS localizadas en la zona del Derecho a Saber de los proyectos. El segundo, mediante la lectura de la etiqueta ubicada en el exterior del recipiente de un producto químico.

Fichas de Datos de Seguridad (FDS):

La información requerida en la Hoja de Datos de Seguridad (FDS) se normalizará en formato - con la ayuda de secciones específicas en una secuencia determinada.

Las 16 secciones estandarizadas son las siguientes:

- Sección 1, Identificación del Producto
- Sección 2, Identificación de Riesgos
- Sección 3, Información Sobre la Composición
- Sección 4, Medidas de Primeros Auxilios
- Sección 5, Medidas para Extinción de Incendios
- Sección 6, Medida para Derrame Accidental
- Sección 7, Manipulación y Almacenamiento
- Sección 8, Controles de Exposición y Equipo de Protección Personal (PPE)
- Sección 9, Propiedades Físicas y Químicas
- Sección 10, Estabilidad y Reactividad
- Sección 12, Información Ecológica *

Verificación de Entrenamiento en Comunicación de Riesgos

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Sección 13, Consideraciones al Desechar *

Sección 14, Información Sobre el Transporte *

Sección 15, Información Regulatoria *

Sección 16, Otra Información, que incluye la fecha de formulación o de la última revisión.

**** Nota: Ya que otras Agencias regulan esta información, OSHA no estará enforcing los Artículos 12 al 15**

El sistema de calificación de categoría de peligro bajo el sistema GHS se ha cambiado de la escala de calificación HMIS / NFPA. Las nuevas calificaciones GHS van de 1 a 5, con la categoría 1 significa peligro grave, y una categoría 5 significa un riesgo mínimo. Es indispensable recordar esto cuando lea la SDS para asegurar que no haya confusión o se dé información contradictoria (Consulte la sección 16 de la SDS).

Sistemas de marcado: (cada categoría tiene una clasificación de 0 a 4, siendo 0 el mínimo y 4 siendo Extremo)



Calificación de Riesgos HMIS/NFPA



Categorías de Riesgo del GHS

0 = Riesgo Mínimo

1 = Riesgo Ligero

2 = Riesgo Moderado

3 = Riesgo Alto

4 = Riesgo Grave

Categoría 1 – Riesgo Grave

Categoría 2 = Riesgo Alto

Categoría 3 = Riesgo Moderado

Categoría 4 = Riesgo Ligero

Categoría 5 = Riesgo Mínimo

¿Dónde veré las diferencias?

- Las categorías de riesgos GHS son mostradas en la hoja SDS y no necesariamente en la etiqueta
- Debido al enorme cambio en clasificaciones, es indispensable entrenar a los empleados en el cambio para asegurar que no haya ninguna confusión o se dé información contradictoria.

Etiquetas:

Se requerirán a los fabricantes e importadores de productos químicos que proporcionen una etiqueta que incluya lo siguiente:

- Palabra de Advertencia**
 - Las palabras de señalización utilizados son "peligro" y "advertencia". "Peligro" se utiliza para los peligros más graves, mientras que "advertencia" se utiliza para los peligros menos severos. Sólo una palabra de advertencia es permitido por paquete.
- Pictograma**
 - Cada pictograma consta de un símbolo diferente sobre un fondo blanco en un cuadrado rojo.
 - Hay 9 pictogramas estandarizados utilizados bajo GHS.

Verificación de Entrenamiento en Comunicación de Riesgos

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Peligro para la salud  • Carcinógeno • Mutagenicidad • Toxicidad para la reproducción • Sensibilización respiratoria • Toxicidad específica de órganos diana • Peligro por aspiración	Llama  • Inflamables • Pirofóricos • Calentamiento espontáneo • Desprenden gases inflamables • Reaccionan espontáneamente (autorreactivos) • Peróxidos orgánicos	Signo de exclamación  • Irritante (piel y ojos) • Sensibilizador cutáneo • Toxicidad aguda (daltno) • Efecto narcótico • Irritante de vías respiratorias • Peligros para la capa de ozono (no obligatorio)	Botella de gas  • Gases a presión	Corrosión  • Corrosión o quemaduras cutáneas • Lesión ocular • Corrosivo para los metales	Bomba explotando  • Explosivos • Reaccionan espontáneamente (autorreactivos) • Peróxidos orgánicos	Llama sobre círculo  • Comburentes	Medio ambiente (No obligatorio)  • Toxicidad acuática	Calavera y tibias cruzadas  • Toxicidad aguda (mortal o tóxica)
--	---	---	---	---	--	--	---	---

- **Indicaciones de peligro para cada clase y categoría de peligro**
 - o Una declaración asignada a una clase o categoría de peligro que describa la naturaleza del peligro(s) de una sustancia química, incluyendo, en su caso, el grado de peligro.
- **Declaración de Precaución**
 - o Una frase que describa las medidas recomendadas a tomar para minimizar o prevenir los efectos adversos derivados de la exposición a una sustancia química, o el almacenamiento o manejo inadecuado de una sustancia química peligrosa.
- **Identificación de Proveedor**
 - o El nombre, dirección y teléfono del fabricante o proveedor del producto deberá indicarse en la etiqueta.

Contenedores Secundarios y Etiquetamiento:

Los empleadores pueden optar por etiquetar los recipientes del lugar de trabajo dentro de la misma etiqueta SGA que se utiliza para los contenedores de transporte, o alternativas de etiquetas que cumplen con los requisitos de la norma.

Este documento certifica que me han instruido en los puntos enumerados anteriormente y he tenido la oportunidad de hacer preguntas. Entiendo perfectamente los requisitos que DPR tiene para trabajos que impliquen el uso de productos químicos.

Nombre/Firma de Empleado

Fecha

Nombre/Firma de la persona realizando la verificación

Fecha

Una copia de este formulario debe ser enviado por fax al encargado de seguridad regional
ANTES DE COMENZAR EL TRABAJO.



Respirable Crystalline Silica Awareness Training Verification

Archive Document – DO NOT DISCARD

Name:		Date:	
Project/Project#:		Company	

About two million construction workers are exposed to respirable crystalline silica in over 600,000 workplaces. OSHA estimates that more than 840,000 of these workers are exposed to silica levels that exceed the new permissible exposure limit (PEL). Activities that expose employees to crystalline silica are considered high hazard work by DPR and therefore require daily, detailed, written pre-task planning prior to start of work. Below is a review of DPR guidelines to protect workers from exposure to respirable crystalline silica. Reference CFR 1926.1153 Respirable Crystalline Silica, and DPR EHSP Rev. 9, Section 03 Concrete and Masonry, and Section 19 Respiratory Protection.

Exposure can occur during common construction tasks such as operating masonry saws, grinders, drills, jackhammers, chipping tools; operating vehicle mounted drilling rigs; operating crushing machines; using heavy equipment for demolition or certain other tasks; dry sweeping and dry brushing.

- The health hazards associated with exposure to respirable crystalline silica: Silica dust is hazardous when very small (respirable) particles are inhaled. These respirable dust particles can penetrate deep into the lungs and cause disabling and sometimes fatal lung diseases including:
 - o *Silicosis*
 - o *Lung cancer*
 - o *Other respiratory diseases*
 - o *Kidney disease*

Specific tasks that could result in exposure to respirable crystalline silica when operating the tools below.

Stationary masonry saws	Walk behind saws
Handheld power saws	Drivable saws
Handheld power saws for cutting fiber cement board	Handheld and stand mounted drills including impact and rotary hammer drills
Rig mounted core saws or drills	Handheld grinders for tuck-pointing
Dowel drilling rigs for concrete	Handheld grinders for other uses
Vehicle-mounted drilling rigs	Walk behind floor grinders
Jackhammers and powered chipping tools	Crushing machines
Small drivable milling machines	Large drivable milling machines
Heavy equipment for demolition	Heavy equipment-grading and excavating

- Specific measures to protect from exposure to respirable crystalline silica:
 - o Engineering controls:
 - Substituting silica with a safer option
 - Exhaust ventilation
 - Using containment methods
 - Using wet methods (water)
 - Dust collecting vacuum methods of silica-containing materials
 - o Restrict housekeeping practices that expose workers to silica by:
 - Prohibit dry sweeping or dry brushing
 - Use HEPA-filtered vacuuming
 - Prohibit compressed air to be used to clean clothing or surfaces
 - o Limiting workers' exposure time and requiring workers to shower and change into clean clothes before leaving a worksite.
 - o Wearing proper respiratory protection (refer to DPR EHSP Rev. 9, Concrete and Masonry 03, and Respiratory Protection, Section 19) to keep exposure levels below the OSHA PEL.



Respirable Crystalline Silica Awareness Training Verification

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- For tasks not specifically mentioned in this training, or where engineering controls, work practices, and respiratory protection described in CFR 1926.1153 Table 1 are not fully implemented, alternative exposure control methods will be adhered to.
- Competent Person: DPR Construction has designated a competent person to make frequent and regular inspections of job sites, materials, and equipment to implement the written exposure control plan. Contact DPR's project management for the identity of the competent person. Subcontractors with exposure to silica are also required to have a designated competent person to oversee their activities.
- Medical Surveillance Program: DPR shall make medical surveillance available at no cost to the employee, and at a reasonable time and place, for each employee who will be required under this section to use a respirator for 30 or more days per year.
- DPR's silica exposure plan which is detailed in the DPR EHSP Rev. 9, Section 2 Chemical and Environmental Management is located in the projects "Right to Understand Area" and is available for review. Subcontractors with activities that potentially expose their workers to silica will also be required to have a written silica exposure plan.

Signature of Employee

Signature of Person Conducting Verification Review

A copy of this form should be faxed to the DPR regional office PRIOR TO START OF WORK.



Conocimiento de Sílice Cristalina Respirable

Verificación de Entrenamiento

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Imprimir Nombre

Fecha

Obra - Proyecto

Compañía

Alrededor de dos millones de trabajadores de la construcción son expuestos a la sílice cristalina respirable en más de 600,000 puestos de trabajo. OSHA estima que más de 840,000 de estos trabajadores son expuestos a niveles de sílice que excede el nuevo límite de exposición permisible (PEL). Las actividades que exponen a los trabajadores a la sílice cristalina son consideradas trabajo de alto riesgo por DPR, que por lo tanto requieren diariamente, un escrito detallado de planificación previa de (Pre-Task) antes de empezar a trabajar. A continuación se muestra una revisión de DPR de practicas establecidas para proteger a los trabajadores contra la exposición a sílice cristalina respirable. Referencia CFR 1926.1153 Respirable Crystalline Silica, and DPR EHSP Rev. 9, Section 03 Concrete and Masonry, and Section 19 Respiratory Protection.

La exposición puede ocurrir durante tareas de construcción comunes tales como operar serruchos de concreto, amoladoras, taladros, martillos neumáticos, herramientas de astillado; operar equipos de perforación en plataformas montadas en vehiculo; operar máquinas de trituración; el uso de maquinaria pesada para la demolición; el cepillado y barrido en seco.

- Los riesgos para la salud asociados con la exposición a la sílice cristalina respirable son: El polvo de sílice es peligroso cuando partículas muy pequeñas (se respiran) son inhaladas. Estas partículas de polvo respirables pueden penetrarse profundamente en los pulmones y causar incapacidad y enfermedades pulmonares potencialmente mortales incluyendo:
 - o Silicosis
 - o El Cáncer de Pulmón
 - o Otras enfermedades respiratorias
 - o Enfermedad Renal

Tareas específicas que podrían resultar en la exposición a sílice cristalina respirable al operar las siguientes herramientas:

Serruchos de albañilería estacionarias	Serruchos que se dirigen caminando
Serruchos de mano	Serruchos maniobrables
Serruchos de mano para cortar hojas de fibra de cemento	Taladros de mano e impacto de varios tamaños
Equipo de perforación central de serrucho o taladros	Amoladoras de mano para tuck-pointing
Plataformas de perforación de espigas para concreto	Amoladoras de mano para otros usos
Plataformas de perforación montados en vehículos.	Amoladoras de piso
Martillos neumáticos y herramientas motorizadas de astillado	Máquinas de trituración
Moledoras pequeñas maniobrables	Moledoras grandes maniobrables
Maquinaria pesada para la demolición	Equipo pesado para nivelación y excavación

- Medidas específicas para proteger contra la exposición a sílice cristalina respirable:
 - o Controles de ingeniería:
 - Sustituyendo el sílice con una opción más segura
 - Ventilación de escape
 - Utilizando métodos de contención
 - Utilizando métodos húmedos (agua)
 - Aspiradora con métodos de coleccionar polvos para materiales que contiene sílice



Conocimiento de Sílice Cristalina Respirable

Verificación de Entrenamiento

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- o Restringir prácticas de orden y limpieza que exponen a los trabajadores a la sílice:
 - Prohibir el barrido o cepillado en seco
 - Usar aspiradora con filtro HEPA
 - Prohibir que se utilice aire comprimido para limpiar la ropa o las superficies
- o Limitar el tiempo de exposición de los trabajadores y exigirles a los trabajadores que deben ducharse y cambiarse de ropa antes salir del trabajo.
- o El uso de protección respiratoria adecuada para mantener los niveles de exposición por debajo del PEL de OSHA (consulte DPR EHSP Rev. 9, Concrete and Masonry, and Respiratory Protection, Section).
- Para tareas no mencionadas específicamente en este entrenamiento, o cuando los controles de ingeniería, prácticas de trabajo y de protección respiratoria CFR 1926.1153 descritos en la Tabla 1 no se aplican plenamente; métodos alternativos de control de exposición serán requeridos.
- Persona competente: DPR Construction ha designado a una persona competente para realizar inspecciones frecuentes y periódicas de los lugares de trabajo, materiales, y equipo para aplicar el plan de control de exposición escrita. Contacte a la gerencia de proyectos de DPR para la identidad de la persona competente. Los subcontratistas con exposición a sílice también están obligados a tener una persona competente designada para supervisar sus actividades.
- Programa de Vigilancia Médica: DPR hará disponible un programa de vigilancia médica sin costo alguno para el empleado, y en un lugar y plazo razonable de tiempo, a cada empleado que sea requerido bajo esta sección utilizar un respirador durante 30 o más días al año.
- El plan de exposición a la sílice de DPR es detallado en DPR EHSP Rev. 9, Section 2 Chemical and Environmental Management. Se encuentra en el archivo del proyecto "Area derecho a entender" y está disponible para su revisión. Subcontratistas con actividades que potencialmente exponen a los trabajadores a la sílice también deben tener por escrito un plan de exposición a la sílice.

Firma del Empleado

Firma de la persona que lleva a cabo la verificación

Una copia de este formulario debe enviarse por fax a la oficina regional del DPR ANTES DE COMENZAR EL TRABAJO.



SILICA EXPOSURE CONTROL PLAN

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

Project Information																			
Project/Project#:				Date:															
Contractor/Subcontractor:				Person Developing Plan:															
Competent Person:				Location of Work:															
Task/Activity:																			
Weather (check all that apply)				Hot	☐	Dry	☐	Clear	☐	Windy	☐	Cool	☐	Wet	☐	Overcast	☐	Raining	☐

Exposure Control Methods to be used:

OR

Table 1 (OSHA Specified Control Methods)	Alternative Exposure Control Methods (Documentation must be made available upon request)
☐ Stationary Masonry Saws ☐ Hand held power saws (any blade diameter) ☐ Hand held power saws (cutting fiber cement board ≤ 8" blade diameter) ☐ Walk behind saws ☐ Drivable saws ☐ Rig-mounted core saws or drills ☐ Hand held and stand mounted drills (including impact and rotary hammer) ☐ Dowel drilling rigs for concrete ☐ Vehicle mounted drilling rigs for rock and concrete ☐ Jackhammers and hand held powered chipping tools ☐ Hand held grinders for mortar removal (i.e., tuckpointing) ☐ Walk-behind milling machines and floor grinders ☐ Small drivable milling machines (less than half-lane) ☐ Large drivable milling machines (half-lane and larger) ☐ Crushing machines ☐ Heavy equipment and utility vehicles used to abrade/fracture silica materials or used during demolition of silica containing materials ☐ Heavy equipment and utility vehicles for tasks such as grading and excavating but not including demolishing, abrading or fracturing silica-containing materials	☐ Performance Option (historical air testing data, manufacturer's guidelines, air monitoring from industry wide surveys, etc.) OR ☐ Scheduled Monitoring Option (in-house testing data)



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Respiratory Protection				Engineering & Work Practice Controls					
Will respirators be used?		☐	Yes	☐	No				
If yes, what type									
☐	APF 10	☐	APF 25 (tuckpointing only)						
Duration of use per worker				☐					
☐	≤ 4 hours per shift		☐	> 4 hours per shift					
Restricted Area Access Methods									
Describe methods to be used, to prevent access to work areas to workers not directly engaged in work tasks described in this control plan (signage, barricading, barriers, containment, etc.)				☐	Floor sweep				
				☐	Wet sweep				
				☐	HEPA vac				
				Disposal Methods					
				Describe disposal methods to be used, and how dust will be controlled (disposal details, e.g. when, where to, how, etc.)					
Training (verify the following was completed for all workers on crew)				Medical Surveillance					
Silica training (to include at minimum)				Will any worker on the crew have the potential to wear a respirator more than 30 days in a year?		☐	Yes	☐	No
☐	Health hazards			If YES, ensure each worker has completed a baseline medical exam that includes, at minimum:					
☐	Tasks that could result in exposure			☐	Medical/work history				
☐	Specific measures to protect employees from exposure			☐	Physical examination				
☐	Identity of competent person performing work (on site during work)			☐	Chest x-ray				
☐	Purpose of medical surveillance program			☐	Pulmonary function test				
				☐	TB test				
				☐	Other physician indicated tests				
				☐	Periodic examinations (minimum every three years)				
				If NO, ensure if respirators will be worn, the company has a written respiratory protection program and medical evaluation, fit testing and training is completed.					



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CRITICAL PROCEDURES: Where work involves any of the following hazards, applicable critical procedures must be incorporated into the plan				
☐	Demolition			
☐	Facility/owner specific requirements (work plans, permits, etc.)			
☐	Sub-surface installations (rebar, post-tension cables, utilities, etc.)			
☐	Abrasive blast (sandblasting)			
☐	Confined Space Entry			
☐	Activities in traffic areas			
☐	Work at heights (6 feet or above)			
☐	Mobile and heavy equipment activity (excavators, dump trucks, scissor lifts)			
☐	Lockout/Tagout			
☐	Pile driving/shoring			
☐	Other:			
Manufacturer's Tool Requirements Checklist:				
Identify tools to be used		Yes	No	N/A
Are the manufacturer's instructions onsite, and have they been reviewed by the employees engaged in the task?		☐	☐	☐
Is there an adequate supply of water for dust suppression?		☐	☐	☐
Is the spray nozzle working properly to supply water at the point of dust generation?		☐	☐	☐
Is water applied at the flow rate specified by the manufacturer?		☐	☐	☐
Where required by Table 1, is the tool equipped with an integrated water delivery system?		☐	☐	☐
Is the tool equipped with a commercially available shroud and dust collection system?		☐	☐	☐
Is the shroud, or cowl installed in accordance with the manufacturer's instructions?		☐	☐	☐
Does the manufacturer provide an air filter of 99% or greater efficiency, and have a filter cleaning mechanism?		☐	☐	☐
When using handheld grinder and a vacuum system; does the dust collector provide 25 cfm of airflow per inch of diameter and have a filter with a 99% or greater efficiency, and a cyclonic pre-separator or filter cleaning mechanism.		☐	☐	☐
Is the hose connecting the tool to the vacuum intact without kinks, or bends?		☐	☐	☐
Are the filters on the vacuum cleaned, or changed per the manufacturer's instructions?		☐	☐	☐
Are the dust collection bags/filters being disposed of properly?		☐	☐	☐
Have the tools, appliances, and accessories been inspected for damage?		☐	☐	☐
Are tools stored in a secure, dry place where they are inaccessible to unauthorized users?		☐	☐	☐
<i>*Note: Always unplug the supply cord from the power outlet when the appliance is not in use, before cleaning and maintenance, and before changing the filter or accessories. Never handle or pull tools by their supply cords. Do not run over supply cords or hoses with equipment. If tools or appliances malfunction or stop working, red tag the item and remove from service until they are repaired by a qualified repair person.</i>				
Manufacturer's Checklist Completed By:		Date:		



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Task Specific Exposure Control Plan: (Complete in detail for all work to be performed for the task, ensure all hazards and controls are identified.)		
Sequence of Basic Job Steps (Order in which the work will be carried out and brief details of how tasks will be performed)	Potential Hazards (Examples: worker exposure to dust, electrical exposures, water intrusion, noise, vibration, etc.)	Safety Controls to Reduce or Eliminate Hazard (Describe the precautions that will be taken)
EXAMPLE: Inspect drill prior to use	EXAMPLE: Damaged cord, missing components, broken/malfunctioning parts	EXAMPLE: Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions and follow other safety instructions



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Task Specific Exposure Control Plan: (Complete in detail for all work to be performed for the task, ensure all hazards and controls are identified.)

Sequence of Basic Job Steps (Order in which the work will be carried out and brief details of how tasks will be performed)	Potential Hazards (Examples: worker exposure to dust, electrical exposures, water intrusion, noise, vibration, etc.)	Safety Controls to Reduce or Eliminate Hazard (Describe the precautions that will be taken)

Task Specific Exposure Control Plan: (Complete in detail for all work to be performed for the task, ensure all hazards and controls are identified.)		
Sequence of Basic Job Steps <i>(Order in which the work will be carried out and brief details of how tasks will be performed)</i>	Potential Hazards <i>(Examples: worker exposure to dust, electrical exposures, water intrusion, noise, vibration, etc.)</i>	Safety Controls to Reduce or Eliminate Hazard <i>(Describe the precautions that will be taken)</i>



SILICA EXPOSURE CONTROL PLAN

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PREPARED BY (COMPETENT PERSON):		POSITION:		DATE:	
CREW MEMBERS WHO HAVE REVIEWED AND UNDERSTAND THE WORK PLAN:		SIGNED:			
APPROVED BY (SITE SUPERINTENDENT):		SIGNED:		DATE:	

Section 03

Concrete and Masonry

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Concrete and Masonry Construction

1.0 Introduction

This section is intended to address high risk activities associated with concrete and masonry work and to provide a set of best practices associated with those high-risk activities. It should be viewed as supplemental and inseparable from the other sections contained within the DPR EHSP.

Reference: CFR 1926.700–705 – Subpart Q – Concrete and Masonry Construction and ANSI/ASSE A10.9 – 2013 Safety Requirements for Concrete and Masonry Work. Other standards, guidelines and regulations may apply. Note: any work which could potentially expose workers to respirable crystalline silica should reference Section 02 – Chemical and Environmental Management, subsection 7.0 Respirable Crystalline Silica in

Definitions

Concrete Bucket: A cylindrical shaped steel container designed to transport concrete, typically lifted by a crane for placement.

Impalement Protection: An adequate means of protecting employees from impalement hazards. There are various means, each with limitations to the amount of protection afforded.

Precast Concrete: Refers to elements of a structure, made of concrete, which are brought from elsewhere and placed/hung in a finish condition. May include deck planks, columns, beams, tilt walls, and decorative panels. These can also be constructed onsite known as tilt up panels.

Post-Tension (PT): Elevated decks with beams and slab on grade with beams designed such that cast-in-place cables are stressed after the concrete has reached a desired strength to provide the final design characteristics.

Post Tension Jack: A hydraulic ram device designed to pull tension on previously cast-in-place cables. They operate under extreme pressure.

Reshoring: Devices installed after formwork is removed, intended to support partially cured concrete and construction loads.

2.0 Pre-Planning

The Concrete phase sets the initial tone of ground-up projects. Sequence, logistics, and pace should be in line with the overall project plan and schedule, and should be facilitated by continuous communication with the project team and affected subcontractors. Construction activities in this section are defined as high hazard activities and therefore require a detailed Job Hazard Analysis (JHA), and daily written pre-task planning by all crews. Refer to the Pre-Planning Section 17 for additional JHA and pre-task planning guidelines and the Pre-Task Planning Worksheet. A detailed JHA should be submitted to DPR Project Management and DPR EHS two weeks prior to erecting tilt-up panels.

2.1 Site Specific Concrete Construction Plan

An appropriate level of planning, conducted in a timely manner, which addresses safety risks, is a key element to a successful project. Items that should be addressed in this plan are, at a minimum:

- Sequencing plan
- Logistics, lay down, traffic flow.
- Equipment to be used
- Fall protection systems to be used, during:
 - Vertical formwork
 - Concrete placement

- Form stripping
- Leading edge work
- Falsework and/or shoring
- Rigging plan
- Silica exposure control plan
- PPE expectations
- Weather conditions to be expected
- Training required for all of the above

2.2 Pre-task Planning

Pre-task planning is covered in detail in Section 17 – Pre-Planning. Daily task specific, and crew specific, pre-task plans should be completed and reviewed prior to the start of work activities. PTPs should address individual steps, existing conditions, adjacent work, other trades, as well as the specifics listed in Section 17.

3.0 Formwork and Shoring Operations

The term “concrete formwork” denotes an almost endless array of materials, systems, configurations, equipment, locations, and elevations. The expectation is that those responsible for executing the work are, or become familiar with the systems to be used and verify those workers are trained to install and build it safely. Formwork should be addressed in the Site Specific Concrete Construction Plan which must include safety provisions.

General guidelines:

- Reference other sections of the EHSP, including PPE, Rigging, Motorized Equipment, Safety Training, etc.
- Fall protection is required at 6 feet above a lower level. Reference Section 07 – Elevated Work. Consideration should be made for equipment needed prior to worker’s exposure to elevated work.
- All protruding reinforcing steel, onto and into which employees could fall, shall be guarded to eliminate the hazard of impalement.
- Wire mesh shall be secured to prevent recoiling or placed in such a way that the tension of wire is facing down.
- Formwork shall be designed, fabricated, erected, supported, braced and maintained so that it will be capable of supporting without failure all vertical and lateral loads that may reasonably be anticipated to be applied to the formwork.
- Pre-engineered wall systems should be constructed per manufacturer’s instructions and published documentation.
- Drawings or plans of jack layout, formwork, shoring, working decks and scaffolding systems must be available at the job site
- Concrete placement loads and pour rates should be considered and monitored.
- The riding of concrete buckets for any purpose is prohibited
- Considerations for rebar attachment and support should be addressed.
- A qualified rigger shall inspect the rigging prior to each lift.
- Formwork rigging should be planned to minimize the amount of climbing needed to detach forms from the cranes and other lifting equipment.
- Erected shoring equipment shall be inspected immediately prior to, during, and immediately after concrete placement of concrete.

3.1 Formwork Removal (Stripping Operations)

Only trained, qualified workers should perform stripping operations. A teamwork approach utilizing the buddy system is preferable. Job Hazard Analysis should be prepared and submitted to the DPR project team prior

to start of work. Daily, detailed pre-task planning is required for duration of work. At a minimum, the following elements should be incorporated into the Job Hazard Analysis:

- Determine the specific strength at which point concrete will support its weight and superimposed loads. Such determination must be based on compliance with one of the following:
 - The plans and specifications stipulate conditions for removal of forms and shores, and such conditions have been followed.
 - The concrete has been properly tested with an appropriate American Society for Testing and Materials (ASTM) standard test method designed to indicate the concrete compressive strength, and the test results indicate that the concrete has gained sufficient strength to support its weight and superimposed loads.
- Develop and communicate a form removal plan and sequence.
- Review reshoring plan to verify location and installation.
- Designate debris pile locations, a methodology for removal, which should include provisions for sharp object removal.
- Stripped lumber and materials intended for re-use shall have all nails removed or bent over and all wire removed.
- Employees or others at lower levels shall be protected from falling materials by approved warning signs, barricades or overhead protection.
- Verify workers are familiar with their task at hand, including any required training for fall protection, equipment operation, respiratory protection, etc.

4.0 Rebar and Post Tension Cable Operations

This subsection is intended to address recognized safety hazards associated with concrete reinforcing installation and those hazards that may remain after the work is complete. The most common hazard when working with rebar is impalement, however other potential hazards exist. Post tension cables are put under extreme pressure, which is an invisible threat that must be addressed.

4.1 Rebar

- Appropriate impalement protection should be installed as soon as practicable. Both horizontal and vertical impalement hazards should be addressed.
 - Guard all exposed ends that extend up to 6 feet above grade or other work surfaces.
- Examples of protection include, but are not limited to:
 - Rebar caps.
 - Job-built wooden troughs, should be so constructed to protect varying heights of adjacent projections and secure from dislodgement. In California see 1712(c)(4)(A) for design requirements
 - Job-built troughs may be used when employees are working at heights not greater than 6 feet above grade or other working surfaces.
 - Scaffold erected over projections, where higher elevated work is expected.

Note: Hook and 90-bars are not considered adequate impalement protection.

- Impalement protection should remain in place as long as practicable, prior to subsequent operations.
- A plan for “guying” or otherwise securing vertical rebar should be in place.
- Adequate manpower should be provided for placing rebar. A single person carrying lengths of rebar is not considered adequate.
- When climbing vertical rebar, verify the wire ties can support your weight.
- Proper fall arrest systems are required when accessing vertical rebar positions. Positioning devices alone are not an acceptable form of fall protection, they must be used in conjunction with a shock absorbing or retractable lanyard.

- It should be determined that shoring has sufficient strength to support the weight of the rebar when loaded on shoring or falsework.

4.2 Post Tension (PT) Activities

Placing, tensioning, and working around completed post tension cables can present unforeseen hazards. During placement and tensioning of cables, unnecessary personnel should be kept at a distance. Cables sitting in a bundle can unravel similar to a tensioned spring and consume a person. During the tensioning process, a multitude of hazards exist, including concrete failure. Once completed, a finished building constructed with post tension cables requires special consideration.

4.2.1 Delivery, loading, and PT placing cables

- Although PT cables are tough, they are subject to damage. Care should be taken when unloading. Proper equipment should be used with trained personnel.
- It should be determined that shoring has sufficient strength to support the weight of the cables when loaded on the deck.
- Unnecessary personal should be kept out of areas where cables are being placed (i.e. pulled from their coil into a straight length).
- Care should be taken after cable placement so as not to introduce an unwanted curve in the cable. This can occur when repositioning inserts. An unwanted curve in the cable can cause devastating results when stressing.
- If possible, it is prudent to mark or paint the deck at the low points of the cable to facilitate locating them from the bottom during subsequent construction.
- Prior to drilling, saw cutting, chipping, shooting, or making attachments to a post-tensioned concrete deck, the location of the PT cables must be verified.

4.2.2 Tensioning PT cables

Tensioning PT cables poses a hazard all along the entire cable length, not just at the ends.

- Signs and barriers shall be erected to limit access into the stressing area only to personnel engaged in stressing or de-tensioning.
- No one should be permitted to stand behind, in line with, or directly above the stressing equipment or the full length of the tendons including the fixed end anchorage.
- A PT cable broken during tensioning can rebound and break out up through the finished floor.
- Tensioning equipment should have current calibration certificates.
- Workers should be trained in the proper use of this equipment. DPR reserves the right to request training records at any time.
- A plan as to how to access the end of the PT cables should be in place. Leaning over the edge of the deck is not an adequate plan.
- An adequate work platform is preferred.
- Equipment should be tethered or otherwise secured from falling.
- Engineered drawings should be consulted to verify sequence of tensioning.
- Gauges should be removed when transporting equipment.

4.2.3 Working around/on existing post tensioned concrete

When working in existing buildings, it should be verified whether PT cables exist. If it is found that they do exist, cables must be located prior to penetrating the concrete deck. Any work that has the potential of penetrating (to any degree, i.e., core drilling, roto-hammering, chipping, shot pins, sawing, etc.) to the PT concrete must be pre-planned and coordinated with the DPR project team (DPR PT cable contact). The Post Tensioned Concrete Deck Work Permit must be reviewed with the DPR PT cable contact prior to start of work.

5.0 Concrete Placement and Finishing Operations

Mechanical Concrete Placement Systems (pump trucks, placing booms, etc.) are covered under Motorized and Site Equipment Section 15. Refer to Section 15 for safe working guidelines. Other means of concrete transport exist, such as wheel barrow, truck chute, concrete pump truck, and concrete bucket. Each means of concrete placement presents a different set of hazards.

Over exposure of the skin to wet concrete can cause dermal burning, sometimes referred to as concrete poisoning. This condition is painful; however, it may not be noticed until intensive medical attention is already required.

All workers exposed to wet concrete should take steps to protect eyes and skin. Reference Section 22 Sanitation and Housekeeping (Hand wash requirements) and reference the applicable SDS.

5.1 Concrete Buckets

- Where concrete buckets are to be used, a detailed JHA should be completed as part of the pre-lift crane plan. The plan should include where and how buckets will be loaded, flown, and discharged. Methods of placing concrete in vertical formwork will be given special attention in this planning.
- All buckets must be operated and maintained in accordance with the manufacturer's guidelines, and modifications must be done only with written approval of the manufacturer.
- Buckets should be inspected daily prior to use. The following items should be inspected at a minimum:
 - Inspect all bolts for abrasion, damage or wear and are properly tightened.
 - Inspect all welds for damage, abrasion, cracks and wear.
 - Inspect the lifting bail for cracks, distortion, abrasion, excessive wear, or any other unsafe condition, especially at lifting and pivot points.
- Buckets equipped with hydraulic or pneumatically operated gates must have positive safety latches or similar safety devices to prevent premature or accidental dumping.
- Buckets with manually operated gates must be self-closing.
- A Bucket lift plan should be completed in accordance with this section and should comply with DPR's Crane and Rigging policies. The plan should include where and how buckets will be loaded, flown, and discharged. Methods of placing concrete in vertical formwork will be given special attention in this planning.
- Riding buckets is prohibited.
- Buckets are not to be used to transport tools or material other than concrete.
- Do not route elevated concrete buckets over employees.

5.2 Wheel Barrow Use

- Do not overfill wheel barrows with concrete.
- An adequate path of travel, free of obstructions should be provided.
- Appropriate width planks, fastened to prevent separation, should be provided when crossing uneven terrain.
- Additional "Stretch & Flex" may be necessary after breaks.

5.3 Concrete Power Trowels (Finishing Machines)

- Walk Behind trowels will be equipped with a control switch that will turn off the engine should the operator lose contact with or control of the machine.
- Trowels must be rigged and moved in accordance with manufacturer specifications.
- The Site Specific Concrete Construction Plan should address where machines will be staged.
- An individual pre-task plan should be implemented for transport of finishing machines.

5.4 Concrete Hand Tools

- Bull float handles, which could come in contact with energized electrical conductors, shall be constructed of nonconductive material.

6.0 Precast Concrete

Reference: CFR 1926.704 and for projects in California CA OSHA 1715

A precast erection plan should be submitted to the DPR project team, a minimum of two weeks prior to start of the work, for project management review. Reference Cranes Section 05. Other sections may apply. The expectation is that this final plan submission summarizes previous planning exercises that include the entire team.

A precast erection plan should include, at a minimum:

- Logistics, transportation, and storage of precast elements.
- Crane requirements, including insurance.
 - Considerations for FAA (this element may take 30 – 45 days)
- Lifting insert/rigging specifications and design
- Erection sequence
- Temporary bracing
- Permanent connections
- Welding procedures
- Barricade plan and safe working distances for other subcontractors
- Fall protection plan
- No employee should be permitted under precast concrete members being lifted or tilted into position except those employees required for the erection of those members.

7.0 Tilt-Up Construction

Reference: CFR 1926.704, CCR 1715, TCA Safety Guidelines for Tilt-Up Concrete Construction, Tilt-Up Concrete Association, June 2013

Tilt-up construction and precast concrete operations are very similar. The differences between the two forms of construction is that precast construction building components usually are formed and casted offsite and then trucked to the construction site to be erected. Whereas with tilt-up construction workers onsite form and cast wall panels which are tilted into place with a crane. The panels are temporarily braced while the process of permanently securing the panels to the concrete footing below and the adjoining panels at top is completed.

7.1 Planning

Planning begins with an evaluation of the site. A large, flat, open site is ideal, but not common. Therefore, evaluate the site regarding slab layout and the movement of materials and equipment around the slab. If the site is tight around the building, consider sequencing the pouring of slabs and panels from within the building. Engineering is a critical phase of Tilt-Up projects. Concrete erection stresses, lifting point attachments and locations in tilt-up concrete panels shall be designed for expected loads, including impact. Check local requirements regarding engineer's qualifications.

7.2 Prior to Construction

- Check the floor slab design for adequate strength to support the designed brace loads, and crane if the crane will be on the floor slab
- Obtain approved shop drawings for each panel showing all pertinent information
- Obtain a properly designed and detailed Tilt-Up package stamped by a registered professional engineer

- Obtain a bracing manual with braces designed for the proper wind loads
- Develop a panel casting and erection sequence
- Select a crew experienced in handling Tilt-Up panel construction
- Select a crane with a capacity capable of lifting the heaviest panel plus the weight of the rigging gear. (Note: Crane selection will not only be based upon weight but also how far the crane must reach and how far the crane may have to travel with a panel.)
- Obtain all crane certification and inspection documentation as well as certification of insurance. Refer to DPR EHSP Section 05 for crane requirements
- Verify the bond breaker compatibility with all curing or sealing compounds, paint and sealers that may have been or will be used on the floor slab or panels

7.3 *During the Lift (Tilt)*

- Maintain a clean working area with all debris and obstacles removed
- Do not lift panels when wind conditions would produce unsafe conditions
- Keep personnel not involved with the panel lifting procedure clear of the lifting area
- Maintain fully extended outriggers and use cribbing to spread the outrigger loading.
- Inspect all rigging gear prior to loading the inserts for proper alignment and free of snags
- Verify the rigging configuration matches what are shown in the erection manual
- Check to be sure that the rigging will not trap braces once the panel is in its final position
- Be alert for panels that may be stuck to the casting surface. Loads to the lifting inserts may be twice that designed for causing possible insert failure
- Carefully release the panel off the casting surface using pry bars and wedges. Keep hands and feet clear of panel and braces
- Be alert to all obstacles in the path of the crane and the crew, especially if you must walk a panel. Never allow personnel to place themselves:
 - Under a panel while it is being tilted
 - On the blind side of the panel while the crane is traveling with it
 - Between the crane and the panel
 - Between the panel being lifted and an adjacent panel
- Never allow workers to reach their hands under a panel to adjust a shim or a bearing pad
- Take extra precautions when lifting panels with special shapes or special rigging
- Do not use any damaged or bent braces, lifting hardware or bolts
- Make certain that any strong backs shown on the erection details are included on the panels
- Address fall protection requirements if aerial lift equipment or ladders are used during panel erection

7.4 *After the Lift (Tilt)*

- Be alert when plumbing panels to their final upright position. Be sure that the panel being plumbed does not strike another previously erected panel
- Never release the crane load if the bracing does not appear adequate
- All bracing should be installed on all erected panels at the end of the workday
- At the beginning and end of the workday, check all brace connections to ensure that they are tight and have not worked loose throughout the night or day
- Check all panel alignments each work day
- If at all possible, grout under all erected panels prior to the end of the workday
- Do not remove any braces until all the structural connections are completed and the lateral resistive system is in place and completed. The structural engineer of record can help you determine if it is safe to remove any or all of the panel braces.
- Be careful when backfilling the pour strip so that you do not exert excessive pressure on the tilt-up panel

8.0 Masonry Construction

Masonry walls, also known as CMU, or concrete masonry units are common place. Not all masonry walls are equal. In structural applications, masonry walls are solid grouted with concrete, and include rebar reinforcing. Some are intended to be free standing, others are not. Masonry walls that are not intended to be free standing pose a potential over turn threat until tied into the structure, even if fully grouted and cured.

Reference: CFR 1926.706(b). **All masonry walls 8 feet in height and over shall be adequately braced to prevent overturning and collapse.** The bracing shall remain in place until the wall is adequately tied in to the structure. Unless the wall is designed to be freestanding.

8.1 *Building a wall*

- A limited access zone should be established whenever a masonry wall is being constructed. This zone shall meet the following:
 - Established for all standing masonry walls.
 - Be located on the non-scaffold side of the wall and run the entire length of the wall
 - Shall be equal to the height of the wall plus 4 feet.
 - Marked off for access to only those involved in the construction of the wall
 - Remain in place until the wall is adequately supported or structurally tied-in to prevent overturn or collapse.
- Masonry saws shall be guarded over the blade by a semicircular enclosure. Refer to DPR EHSP Rev. 9 Section 24, subsection 2.5 for additional requirements
- A face shield and safety glasses are required when workers cut block.
- Proper respiratory protection is required. Wet methods are preferred. Refer to DPR EHSP Rev. 9 Section 02, sub-sections 7 and 8.
- Mortar Mixing Silos
 - Mixing silos will be set-up, operated and maintained in accordance with manufacturer's guidelines.
 - Activities involving mixing silos must have manufacturers safe work practices and guidelines readily available.
 - Steps should be taken to protect adjacent work areas from the hazards associated with the mortar mixing operation. These hazards include silica dust, heavy equipment and overhead lifting hazards.

Concrete & Masonry Forms

Forms

Post Tensioned Concrete Deck Work Permit



Post Tensioned Concrete Deck Work Permit

Archive Document – DO NOT DISCARD

Audited by: _____

Date: _____

Project Name: _____ Project Number: _____ Date: _____

Company: _____ Location of Work (Be Specific) _____

Answer the following when evaluating your work (circle YES, NO or N/A):				Check PPE Required	
Have you reviewed your work plan with the DPR project team member who is the PT cable contact?	YES	NO	N/A	☐	Hard Hat / Safety Glasses
Do the structural drawings give guidelines for penetrating the post tension deck?	YES	NO	N/A	☐	Fall Protection / Rescue Plan
If there are no guidelines given in the drawings have you contacted the structural engineer about the planned penetration?	YES	NO	N/A	☐	Hand / Arm PPE
Have you reviewed the X-ray and layout of hidden elements with the X-ray technician?	YES	NO	N/A	☐	Hearing PPE
Has the DPR PT contact reviewed the X-ray and layout of hidden elements?	YES	NO	N/A	☐	Foot PPE
Have you done a final walk through of the work area with the DPR PT contact?	YES	NO	N/A	☐	Respirator
Has the DPR PT contact approved your proposed work?	YES	NO	N/A		
Any safety item identified in the above question list must be addressed on the table below.					

Tasks: _____

What is the quality item you will focus on today? _____

Steps for Work	Tools	Hazards	Steps Taken to Address Hazards

* If you need more space, attach an additional PRE-TASK PLAN SHEET

Work Planner:		DPR PT Contact:		Safety Coordinator:	
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Crew Members – (additional crew members sign on back or separate page if necessary):

Section 04

Confined Space

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Confined Space Entry

1.0 Introduction

The Confined Space Entry plan is provided to protect authorized employees that will enter confined spaces and may therefore be exposed to hazardous atmospheres, engulfment in materials, conditions which may trap or asphyxiate due to converging or sloping walls, or contains any other physical or health hazards.

This section covers OSHA standards requirements for practices and procedures to protect employees engaged in construction activities at a worksite with one or more confined spaces. Examples of locations where confined spaces may occur include, but are not limited to, the following: Bins; boilers; pits (such as elevator, escalator, pump, valve or other equipment); manholes (such as sewer, storm drain, electrical, communication, or other utility); tanks (such as fuel, chemical, water, or other liquid, solid or gas); incinerators; scrubbers; concrete pier columns; sewers; transformer vaults; heating, ventilation, and air-conditioning (HVAC) ducts; storm drains; water mains; precast concrete and other pre-formed manhole units; drilled shafts; enclosed beams; vessels; digesters; lift stations; cesspools; silos; air receivers; sludge gates; air preheaters; step up transformers; turbines; chillers; bag houses; and/or mixers/reactors.

Exceptions to this section include: Excavations, Underground Construction, Caissons, Cofferdams and Compressed Air, Diving.

Reference: Confined Spaces Standard for Construction Subpart AA, CFR 1926.1200 – 1926.1213 other applicable standards, guidelines, and regulations may apply.

Definitions

Acceptable entry conditions: Conditions that must exist in a permit space, before an employee may enter that space, to ensure that employees can safely enter into, and safely work within, the space.

Attendant: An individual stationed outside one or more permit spaces who assesses the status of authorized entrants and who must perform the duties specified in § 1926.1209.

Authorized entrant: An employee who is authorized by the entry supervisor to enter a permit space.

Barrier: A physical obstruction that blocks or limits access.

Blanking or blinding: The absolute closure of a pipe, line, or duct by the fastening of a solid plate (such as a spectacle blind or a skillet blind) that completely covers the bore and that is capable of withstanding the maximum pressure of the pipe, line, or duct with no leakage beyond the plate.

Competent person: One who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has the authorization to take prompt corrective measures to eliminate them.

Confined space: A space that:

1. Is large enough and so configured that an employee can bodily enter it;
2. Has limited or restricted means for entry and exit; and
3. Is not designed for continuous employee occupancy.

Controlling Contractor: The employer that has overall responsibility for construction at the worksite.

Early-warning system: The method used to alert authorized entrants and attendants that an engulfment hazard may be developing, i.e. alarms activated by remote sensors; lookouts with equipment for immediately communicating with the authorized entrants and attendants.

Engulfment: The surrounding and effective capture of a person by a liquid or finely divided (flowable) solid substance that can be aspirated to cause death by filling or plugging the respiratory system or that can exert enough force on the body to cause death by strangulation, constriction, crushing, or suffocation.

Entry: The action by which any part of a person passes through an opening into a permit-required confined space. Entry includes ensuing work activities in that space and is considered to have occurred as soon as any part of the entrant's body breaks the plane of an opening into the space, whether or not such action is intentional or any work activities are actually performed in the space.

Entry employer: Any employer who decides that an employee it directs will enter a permit space.

Entry permit (permit): The written or printed document that is provided by the employer who designated the space a permit space to allow and control entry into a permit space

Entry supervisor: The qualified person (such as the employer, foreman, or crew chief) responsible for determining if acceptable entry conditions are present at a permit space where entry is planned, for authorizing entry and overseeing entry operations, and for terminating entry as required by this standard. Note: an entry supervisor also may serve as an attendant or as an authorized entrant, as long as that person is trained and equipped for each role he or she fills. Also, the duties of entry supervisor may be passed from one individual to another during the course of an entry operation.

Hazardous atmosphere: An atmosphere that may expose employees to the risk of death, incapacitation, impairment of ability to self-rescue (that is, escape unaided from a permit space), injury, or acute illness from one or more of the following causes:

- Flammable gas, vapor, or mist in excess of 10 percent of its lower flammable limit (LFL)
- Airborne combustible dust at a concentration that meets or exceeds its LFL
- Atmospheric oxygen concentration below 19.5 percent or above 23.5 percent
- Atmospheric concentration of any substance or material above published PELs (Toxins)

Host employer: The employer that owns or manages the property where the construction work is taking place.

Immediately dangerous to life or health (IDLH): Any condition that would interfere with an individual's ability to escape unaided from a permit space and that poses a threat to life or that would cause irreversible adverse health effects.

Inerting: Displacing the atmosphere in a permit space by a noncombustible gas (such as nitrogen) to such an extent that the resulting atmosphere is noncombustible.

Isolate or isolation: The process by which employees in a confined space are completely protected against the release of energy and material into the space and/or contact with a physical hazard

Limited or restricted means for entry or exit: A condition that has a potential to impede an employee's movement into or out of a confined space. Such conditions include, but are not limited to, trip hazards, poor illumination, slippery floors, inclining surfaces and ladders.

Lockout Tagout: The placement of a lock and a tag on an energy isolating device, ensuring that the energy isolating device and the equipment being controlled cannot be operated until the lockout device is removed.

Lower flammable limit or lower explosive limit means: The minimum concentration of a substance in air needed for an ignition source to cause a flame or explosion.

Monitor or monitoring: The process used to identify and evaluate the hazards after an authorized entrant enters the space. This is a process of checking for changes that is performed in a periodic or continuous manner after the completion of the initial testing or evaluation of that space.



Non-entry rescue: Occurs when a rescue service, usually the attendant, retrieves employees in a permit space without entering the permit space.

Non-permit confined space: Means a confined space that meets the definition of a confined space but does not meet the requirements for a permit-required confined space, as defined in this subpart.

Permit-required confined space (permit space) means: A confined space that has one or more of the following characteristics:

- Contains or has a potential to contain a hazardous atmosphere;
- Contains a material that has the potential for engulfing an entrant;
- Has an internal configuration such that an entrant could be trapped or asphyxiated by inwardly converging walls or by a floor which slopes downward and tapers to a smaller cross section; or
- Contains any other recognized serious safety or health hazard.

Physical hazard: An existing or potential hazard that can cause death or serious physical damage; e.g., mechanical, electrical, hydraulic and pneumatic energy; radiation; temperature extremes; engulfment; noise; and inwardly converging surfaces.

Qualified person: One who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training, and experience, has successfully demonstrated his ability to solve or resolve problems relating to the subject matter, the work, or the project.

Rescue: To retrieve and provide medical assistance to, one or more employees who are in a permit space.

Rescue service: Personnel designated to rescue employees from permit spaces.

Retrieval system: The equipment (including a retrieval line, chest or full body harness, wristlets or anklets, if appropriate, and a lifting device or anchor) used for non-entry rescue of persons from permit spaces.

Ventilate or ventilation: Controlling a hazardous atmosphere using continuous forced-air mechanical systems.

2.0 Application

This policy applies to all contractor operations. It is designed to protect workers from the hazards of entry into permit-required and non-permit-required confined spaces. Every effort shall be made to identify and classify potential confined spaces through communication between host employer (owner), DPR project management and the entry employer. All areas that meet the definition of a confined space should be treated as permit-required confined spaces unless testing and documentation are performed and finalized.

When a competent person reclassifies a space to a non-permit required confined space, the reclassification must be approved by DPR project management and DPR EHS.

DPR EHS or designee should be notified, a minimum of 24 hours prior to the intent to perform confined space entry work. The DPR project team should review the subcontractor's confined space entry plan (JHA), including a written site rescue plan, prior to start of work. In addition to a confined space entry permit, a detailed, written pre-task plan outlining the work to be performed must be completed prior to each entry. The complete confined space entry package must be submitted for approval a minimum of 24 hours prior to the planned entry. The complete confined space entry package includes, at a minimum:

- Written entry plan
- Space specific JHA (job hazard analysis)
- Permit Space Entry Communication & Coordinator document
- Safety Data Sheets, if applicable
- Rescue plan
- Confined space entry permit
- Pre-task plan

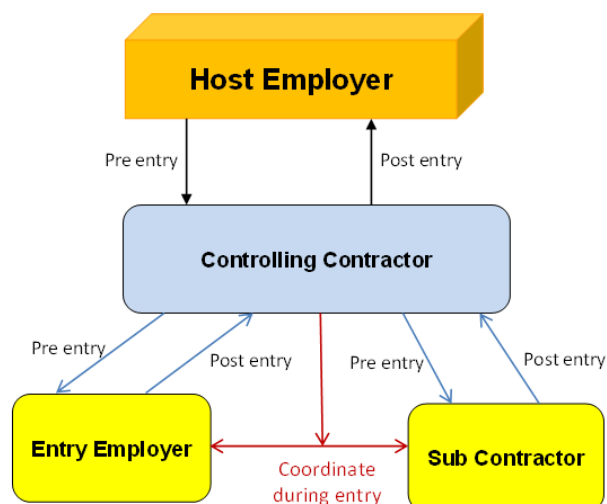
- Verification of training for all crew members and
- Any other pertinent documentation

The complete package must be available at the location of the confined space entry work being performed, for the duration of work. The complete package, completed copies provided to DPR project management upon completion of work. *Refer to Pre-Planning Section 17 for further guidelines on pre-planning.*

3.0 Responsibilities of Employers under the Confined Spaces Standard

The following table summarizes the responsibilities of the various categories of employer. Refer to the Permit Space Entry Communication & Coordinator document in the forms section to facilitate planning and communication responsibilities for all parties.

Category of Employer	Employer Responsibilities
All Employers	• Identify all confined spaces in which their workers may work and determine whether any are permit spaces. If its workers are supposed to enter permit spaces, the employer is an “entry employer.” • Employers who are not “entry employers” must make sure their workers stay out of any permit spaces present on the site, unless the workers are
Entry Employers	• Protect workers against permit space hazards by complying with the standard. • Inform controlling contractor of the program followed and hazards encountered in permit spaces. • Keep any unauthorized persons from entering the permit space, and remove any unauthorized persons who do enter the space.
Controlling Contractors	• Share information it has about permit space hazards with entry employers and other employers whose activities may create hazards in the permit space. • Coordinate entry operations when there is more than one entry employer. • Coordinate operations when permit space entry occurs during other activities at the site that might create a hazard in the space.
Host Employers	• Share information it has about permit space hazards with the controlling contractor.



4.0 Characteristics of Confined Spaces

If the workplace contains confined spaces, they must be categorized as to the hazards and entries must be planned to protect workers from those hazards. The protection that is required for each space depends upon the type and severity of the hazard present. The following tables lists the three categories of spaces for which different levels of protection are specified:

Characteristics of Space	Protective Actions Required
Permit required spaces that do not qualify for one of the following two exceptions.	- If the employer's workers will enter the space, develop and follow a written permit- required confined space program, or permit space program. The permit space program specifies, among other things, how the employer will (1) regulate worker entry into permit spaces; and (2) control permit space hazards. Subsection 4.1 below describes the features that a permit space program must contain. - All employers must inform their workers about the locations and dangers of each permit space (e.g., post signs), and take additional steps to ensure that workers do not enter permit spaces if they are not authorized to do so.
Exception 1: Spaces that contain only physical (non- atmospheric) hazards.	If the physical hazards are eliminated or isolated so that they no longer present a hazard, the space may be reclassified as a non-permit space, with no further precautions required. Reference subsection 4.2.
Exception 2: Spaces containing an atmospheric hazard that can be controlled by continuous forced air ventilation.	As long as the atmospheric hazard is controlled by continuous forced air ventilation and any physical hazards are eliminated or isolated, the alternate procedures listed in Subsection 4.3 may be used instead of full permit space procedures, although the space is still classified as a permit space.

5.0 Types of Confined Space Entries

If an employer's workers will enter a permit space, the employer must develop a written permit space program, and entry is only permitted under the terms of the program. The written program must be submitted and reviewed by DPR project management prior to entry.

Note: *should the contractor wish to implement procedures for attendants overseeing multiple spaces, these procedures must be agreed upon by all parties prior to start of any entry.*

5.1 Permit Required Confined Space Entry

The permit space program must establish a system for preparing, using, and canceling entry permits and control entry into permit spaces. The program must include, at a minimum:

- Implement measures to prevent unauthorized entry;
- Identify and evaluate permit space hazards before allowing employee entry;
- Provide for the atmospheric conditions in the permit space to be tested before entry operations and for the space to be monitored during entry;
- Require appropriate testing for the following atmospheric hazards in this sequence:
 - oxygen

- combustible gases or vapors
 - toxic gases or vapors
- Establish and implement the means, procedures and practices to eliminate or control hazards necessary for safe permit space entry operations;
- Identify employee job duties;
- Provide and maintain (at no cost to the employee), PPE and any other equipment necessary for safe entry, and require employees to use the equipment properly;
- Ensure that an attendant is stationed outside the permit space during entry operations;
- Coordinate entry operations (in consultation with the controlling contractor) when employees of more than one employer are working in the permit space; and
- Define the procedures to be followed during an emergency to include the process for summoning rescue and emergency services and preventing unauthorized personnel from attempting rescue.
- The employer must review the program and correct any deficiencies when any of the following occur:
 - injury or near-miss during entry
 - unauthorized entry
 - detection of a new hazard or
 - a condition prohibited by the permit, or
 - an employee complaint about the program's effectiveness.
- The employer must annually review the program using the cancelled permits, and revise the program as necessary to ensure that employees are protected.
- The permit must be posted at the entrance to the space or be otherwise made available to authorized entrants and their authorized representatives at the time of entry.

5.1.1 *Canceling Entry Permits*

The employer must make sure that the entry supervisor cancels entry permits when an assignment is completed or when new conditions exist. Once a permit is cancelled, entry under it is no longer permitted. New conditions must be noted on the canceled permit and used in revising the permit space program. The employer must keep all canceled entry permits for at least one year.

5.1.2 *Suspending Entry Permits*

An entry supervisor may suspend an entry permit instead of cancelling it if a temporary condition has occurred in or near the space that, once corrected, is not expected to reoccur. The permit may be reinstated and entry may occur under the permit if the entry supervisor has determined that the conditions in the space match the allowable conditions listed on the permit.

5.2 *Reclassifying a Permit Space as a Non-Permit Space*

A permit space that contains only physical hazards may be reclassified as a non-permit space if:

- the physical hazards are eliminated or isolated without entering the space; or
- the physical hazards are eliminated or isolated by entering the space using permit space procedures.
- Physical hazards include all hazards that are not atmospheric hazards, including:
 - explosives (other than explosive atmospheres)
 - mechanical
 - electrical
 - hydraulic and pneumatic energy
 - radiation
 - temperature extremes
 - engulfment
 - noise
 - inwardly converging surfaces

- chemicals that can cause death or serious physical harm through skin or eye contact (rather than through inhalation)
- Whenever a permit space is reclassified as a non-permit space, the entry employer must document the basis for determining that all hazards in the permit space have been eliminated through a certification (refer to the confined space entry permit in the forms section). The certification must be made available to each worker entering the space or to that worker's authorized representative.
- If any hazards arise within a permit space that has been reclassified as a non-permit space, each worker in the space must exit the space. The entry employer must then reevaluate the space and determine how to proceed:
 - If the new hazards are physical hazards that can be eliminated or isolated, the employer can reclassify the space as a non-permit space after that has been accomplished. If the new hazard is an atmospheric hazard that can be made safe for entry by continuous forced air ventilation, the employer can proceed in accordance with the procedures outlined in subsection 5.3 below.
 - In all other situations, further entry can only be made pursuant to a full permit space program

5.3 *Alternate Procedures for Certain Permit Spaces*

Unlike a space that contains only physical hazards, a space containing an atmospheric hazard cannot be reclassified as a non-permit space. However, if the atmospheric hazard in a permit space can be controlled by forced air ventilation, the employer may use less stringent procedures instead of full permit space procedures when workers enter the space. The alternate procedures may be used if the employer can show that:

- All physical hazards are eliminated or isolated;
- The only hazard is an actual or potential hazardous atmosphere that can be and is made safe for entry using continuous forced air ventilation; and
- In the event the ventilation system stops working, entrants can exit the space safely.

5.3.1 *Alternative Procedures*

An employer who uses the alternate procedures must document the reasons, and the supporting data, for concluding that these criteria are met. The documentation must be made available to each worker who enters the space or to that worker's authorized representative. Alternate procedures are as follows:

- Entry and Exit:
 - Any conditions making it unsafe to remove an entrance cover must be eliminated before the cover is removed
 - When entrance covers are removed, the opening must be immediately guarded by a railing, temporary cover, or other temporary barrier that will prevent an accidental fall through the opening and that will protect each employee working in the space from foreign objects entering the space
 - Before a worker enters the space, the internal atmosphere must be tested, with calibrated direct-reading instruments, for oxygen content, for flammable gases and vapors, and for potential toxic air contaminants, in that order. The testing must demonstrate that the atmosphere in the space is not hazardous. Any worker who enters the space, or that worker's authorized representative, must be provided an opportunity to observe the pre-entry testing
 - The employer must verify that the space is safe for entry and that the pre-entry measures discussed above have been taken, through a written certification that contains the date, the location of the space, and the signature of the person providing the certification. The certification must be made before entry and must be made available to each worker entering the space or to that worker's authorized representative
 - The employer must provide workers with a safe method of entering and exiting the space. Any hoisting system that is used must either:
 - Be designed and manufactured for hoisting personnel or
 - Be approved for personnel hoisting by a registered professional engineer prior to use

- Ventilation Requirements:
 - Continuous forced air ventilation must be used. (Exhaust ventilation is not an acceptable substitute for forced air ventilation)
 - If the forced air ventilation stops for any reason, the entrant(s) must immediately leave the space.
 - The forced air ventilation must be directed to ventilate the immediate areas where each entrant is or will be located within the space and must continue until all employees have left the space.
 - The air supply for the forced air ventilation must be from a clean source and must not increase the hazards in the space
- During Entry:
 - The atmosphere within the space must be monitored to ensure that the forced air ventilation is preventing the accumulation of a hazardous atmosphere. Monitoring must be conducted continuously unless the entry employer can demonstrate that equipment for continuous monitoring is not commercially available or that periodic monitoring is sufficient to ensure that the atmosphere remains non- hazardous. Employers must provide any worker who enters the space, or that worker's authorized representative, with an opportunity to observe the testing required by this paragraph
 - If a hazard is detected during entry:
 - Employers must make sure each worker leaves the space immediately
 - The space must be evaluated to determine how the hazard developed
 - The employer must implement measures to protect workers from the hazard before any subsequent entry takes place

6.0 Comparison of Requirements for Permit-Required Spaces and Alternate Procedure Spaces

The following table compares the requirements for spaces where permits are required and for spaces where alternate procedures are permitted.

Precautions Required	Permit- Required Space	Alternate Procedures Space
Written Permit Space Program Meeting (Reference OSHA 1926.1204-1206)	Yes	No
Worker Training Meeting (Reference OSHA 1926.1207)	Yes	Yes
Duties of Entrants, Authorized Attendants and Entry Supervisors (Reference OSHA 1926.1208, .1209, .1210)	Yes	No
Rescue and Emergency Services (Reference OSHA 1926.1211)	Yes	No
Information exchange between controlling contractor and entry employer(s) (Reference OSHA 1926.1203(h))	Yes	Yes
Permit Space Warning Signs and Entry Restrictions (Reference OSHA 1926.1203(b) and (c))	Yes	Yes

7.0 Roles of Entrants, Attendants and Entry Supervisors

Three categories of workers have specific duties whenever work is performed in a permit space: authorized entrants; attendants, and entry supervisors. The entry employer is responsible for ensuring that these workers carry out those duties.

Role	Responsibilities
Authorized Entrant	• Know the hazards of the space • Use appropriate PPE • Stay in communication with attendants • Exit from the space as soon as possible when: ○ Ordered by the attendant or entry supervisor ○ When warning signs and symptoms of exposure are recognized ○ A prohibited condition exists ○ An automatic alarm is activated • Alerts the attendant when a prohibited condition exists or when warning signs or symptoms of exposure exist
Attendant	• Remain outside the permit space during entry operations unless relieved by another authorized attendant • Perform non-entry rescues when specified by the employer's rescue procedure • Know existing and potential hazards, including information on the mode of exposure, and signs or symptoms, and consequences, and possible behavioral effects of exposure; • Maintain communication with and keep an accurate account of those workers entering the permit space; • Assess conditions inside and outside the space and order evacuation of the permit space when: ○ A prohibited condition exists ○ A worker shows behavioral effects of hazard exposure ○ A situation exists outside the confined space that could endanger entrants ○ The attendant cannot effectively and safely perform required duties. • Summon rescue and other services during an emergency; • Ensure that unauthorized people stay away from permit spaces or exit immediately if they have entered the permit space; • Inform authorized entrants and the entry supervisor if any unauthorized person enters the permit space; and • Perform no other duties that interfere with the attendant's primary duties.
Entry Supervisor	• Know space hazards including information on the mode of exposure, signs or symptoms and consequences; • Verify that specified entry conditions are satisfied, including permits, tests, procedures and equipment before allowing entry; • Terminate entry and cancel or suspend permits when entry operations are completed or if a condition that is not allowed under the permit arises; • Verify that rescue services are available and that the means for summoning them are operable; • Take appropriate measures to remove unauthorized entrants; and • Ensure that entry operations remain consistent with the entry permit and that acceptable entry conditions are maintained

8.0 Worker Training

Before any worker enters a confined space, the entry employer must train authorized entrants, attendants, entry supervisors, and other employees with duties under the standard to understand the hazards in the confined space and the methods used to protect against those hazards.

If a worker is not authorized to perform entry rescues, the training must include the dangers of attempting such rescues. After the training, the employer must ensure that the workers have the understanding, knowledge and skills necessary to safely perform their assigned duties. Training must be documented and made available upon request to DPR project management. The required training must be provided:

- In both a language and vocabulary that the worker can understand;
- Before the worker is first assigned duties and before there is a change in the worker's assigned duties;
- Whenever there is a change in permit space entry operations that presents a new hazard about which the worker has not previously been trained;
- Whenever the worker's actions show inadequacies in the worker's knowledge or use of entry procedures.
- Rescue personnel must also be trained to perform their duties.

9.0 Rescue and Emergency Services

Relying on 911 emergency services for confined space rescue should not be the sole remedy due to actual response times, specialized training required, or other emergency obligations. Rescue equipment, sufficient for the number of entrants, must be maintained in a ready state. Permit required confined space rescue plans fall into two basic categories:

- Non-entry Rescue – allows rescue personnel to perform rescue without entering the permitted space
 - Non-entry rescue plans require at a minimum:
 - At least one standby person at the site, trained and immediately available to perform rescue and emergency services, other than the Attendant.
 - A clear method of retrieval, including a list of equipment and PPE to be utilized
 - A mechanical device must be available to retrieve someone from vertical type permit spaces more than five feet deep.
 - When non-entry rescue is selected, the entry employer must also confirm, before entry begins, that emergency assistance would be available if the non-entry rescue fails.
- Entry – requires rescue personnel to “Enter” the permitted space
 - Entry into the permitted space must meet the following requirements, at a minimum:
 - Trained personnel, proficient with PPE and the actual equipment to be used.
 - Each member of the rescue service should be trained to perform the assigned rescue duties
 - Each member of the rescue service should also receive the training required of authorized entrant(s)
 - Each member of the rescue service should practice making permit space rescues at least once every 12 months, by means of simulated rescue operations in which they remove dummies, manikins or actual persons from the actual permit spaces or from representative permit spaces
 - Each member of the rescue service should be trained and hold current certification in basic first aid and cardiopulmonary resuscitation (CPR), and AED
 - Rescue services should be selected such that they have the capability to reach the victim(s) within a time frame that is appropriate for the permit space hazards
 - Representative permit spaces should, with respect to opening size, configuration and accessibility, simulate the types of permit spaces from which rescue is to be performed
 - Each member of the rescue service must be informed of the hazards they may confront when called on to perform rescue

- The rescue service must be provided with access to all permit spaces from which rescue may be necessary so that the rescue service can develop appropriate rescue plans and practice rescue operations

Confined Space Forms

Forms

- Confined Space Entry Permit – English
- Confined Space Entry Permit – Spanish
- Air Monitoring Log
- Confined Space Entry Sign In / Sign Out Log
- Controlled Access Non-Permit Confined Space Entry Sign In / Sign Out Log
- Permit Space Entry Communication and Coordination



Confined Space Entry Permit

Archive Document – DO NOT DISCARD

General Information

Project/Project#:		Date:	
Project Address:			
Entry Employer:		Entry Start Time*:	
Location on Site:		Entry End Time*:	
Purpose of Entry:			
Description of Activity:			
Communication Method:			
Type of Entry:	☐ Permit Required	☐ Alternate Methods Permit**	☐ Non-Permit**

* Permit is valid for one shift only. The type of permit, regardless of type is valid for only ONE shift, **Document Methods for Reclassification on 2nd page

Emergency Contacts

Host Employer Contact	Name:	Contact:
DPR Management/EHS	Name:	Contact:
Entry Competent Person	Name:	Contact:
Fire Department	Name:	Contact:
Rescue/Emergency Services	Name:	Contact:
Nearest Medical Facility:	Address:	

Entry and Exit Log

Authorized Entrant (Print Name)	Time		Time		Time		Time		Time		Time	
	In	Out	In	Out	In	Out	In	Out	In	Out	In	Out

Air Monitoring Log

Gas Monitor (Model #):										Calibration Date:									
Date	Time	Reading and Location														Tester Name			
		Top					Middle					Bottom							
		O2	LEL	CO	H2S		O2	LEL	CO	H2S		O2	LEL	CO	H2S				

Chemical Action Levels (For Additional Chemicals Refer to Federal/State Regulations)

Oxygen (O2)	<19.5% or >23.5%	Carbon Monoxide (CO)	50 ppm (Fed) 25ppm (CA)
Lower Explosive Limit (LEL)	Any % over 10	Hydrogen Sulfide (H2S)	10 ppm



Confined Space Entry Permit

Archive Document – DO NOT DISCARD

Hazards of the Confined Space							
Hazard Description	Yes	No	N/A	Hazard Description	Yes	No	N/A
Lack of Oxygen	☐	☐	☐	Temperature	☐	☐	☐
Combustible Gases**	☐	☐	☐	Engulfment	☐	☐	☐
Combustible Vapors	☐	☐	☐	Entrapment	☐	☐	☐
Combustible Dusts	☐	☐	☐	Pressure Systems	☐	☐	☐
Toxic Gases	☐	☐	☐	Welding/Burning/Open Flame	☐	☐	☐
Toxic Vapors	☐	☐	☐	Dehydration/Heat Exhaustion	☐	☐	☐
Chemical Contact	☐	☐	☐	Falls	☐	☐	☐
Electrical Hazards	☐	☐	☐	Unknown or Spilled Liquids	☐	☐	☐
Mechanical Exposure	☐	☐	☐	Other:	☐	☐	☐

Methods for Abating Hazards					
☐	Purge/Flush & Vent	☐	Buddy System	☐	Goggles/Faceshield
☐	Mechanical Ventilation	☐	Evacuation Alarm	☐	Gloves
☐	Lockout/Tagout	☐	Lifelines & Escape Harnesses	☐	Hearing Protection
☐	Inerting	☐	Emergency Egress Procedure	☐	Protective Clothing
☐	Blanking, Blocking, Bleeding	☐	Tripod Escape Unit	☐	Hot Work Permit
☐	External Barricades	☐	Fire Extinguishers	☐	Other:
☐	Signage	☐	Respirator	☐	
☐	Shields/Barriers	☐	Supplied Air/SCBA	☐	
☐	Lighting	☐	Fall Protection	☐	
☐	Resuscitator	☐	Safety Glasses	☐	

**Competent Person's Methods for Reclassifying Space(s) to Non-Permit Required	
☐	I.D. permit required hazard(s):
☐	Hazard(s) have been eliminated or isolated through:
☐	Verify that work will create no additional hazards:
☐	Competent person verifies no atmospheric hazard is present (document on air monitoring Log) initial:
☐	Competent person verifies there no potential for atmospheric hazard to develop initial::
Compete Person Signature: _____	

NOTE: if ventilation is required to control the atmosphere and make it safe, the space CANNOT be downgraded to a non-permit space; instead consider the alternate entry procedures

Required Signatures						
Description	Name (print)	Signature	Training Completed		Date	
DPR Management/EHS			☐	Yes	☐	No
Host Employer			☐	Yes	☐	No
Entry Supervisor			☐	Yes	☐	No
Qualified Attendant			☐	Yes	☐	No
Qualified Entrant 1			☐	Yes	☐	No
Qualified Entrant 2			☐	Yes	☐	No
Qualified Entrant 3			☐	Yes	☐	No
Competent Person			☐	Yes	☐	No

Permit Closeout			
Entry Stop Time:		Company:	
Entry Supervisor:		Entry Supervisor Signature:	



Air Monitoring Log

Archive Document – DO NOT DISCARD

General Information			
Project		Date	
Contractor		Monitor	
Gas Meter		Serial Number	
Fresh air calibration	☐ Yes ☐ No	Next calibration due	Days
Area Monitored			

Time	Location	O2	LEL	CO	H2S



Permit Space Entry Communication and Coordination

Project/Project #:		Date:	
Entry Supervisor		Company:	
DPR PM		DPR Super:	
Host Employer:		Entry Employer	
Controlling Contractor		Entry Employer	
DPR EHS		Entry Employer	

Host Employer Entry Information				
Comments		Yes	No	N/A
Has the host employer provided the locations of each known permit space? <i>List each known permit space in comments column and on additional sheet</i>		☐	☐	☐
Has the host employer provided the hazards or potential hazards in each space or the reason it is a permit space? <i>Use addition sheets if necessary</i>		☐	☐	☐
Has the host employer provided any precautions that the host employer or any previous controlling contractor or entry employer implemented for the protection of employees in the permit space. <i>Use addition sheets if necessary</i>		☐	☐	☐

Controlling Contractor Pre Entry				
Comments		Yes	No	N/A
Has the controlling contractor obtained the host employer's information about the permit space hazards and previous entry operations?		☐	☐	☐
Has the controlling contractor provided the information to each entity entering a permit space and any other entity at the worksite whose activities could foreseeably result in a hazard in the permit space? <i>Review below:</i>		☐	☐	☐
<i>The information received from the host employer?</i>		☐	☐	☐
<i>Any additional information the controlling contractor has about the host employers permit spaces?</i>		☐	☐	☐
<i>The precautions that the host employer, controlling contractor, or other entry employers implemented for the protection of employees in the permit spaces?</i>		☐	☐	☐

Entry Employer Pre Entry				
Comments		Yes	No	N/A
Has the entry employer obtained all of the controlling contractor's information regarding permit space hazards and entry operations?		☐	☐	☐
Has the entry employer informed the controlling contractor of the permit space program that the entry employer will follow, including any hazards		☐	☐	☐



Permit Space Entry Communication and Coordination

likely to be confronted or created in each permit space?				
--	--	--	--	--

Controlling Contractor/Entry Employer Coordination				
	Comments	Yes	No	N/A
Has the controlling contractor and entry employer(s) coordinated entry operations when more than one entity performs permit space entry at the same time? <i>List additional entities</i>		☐	☐	☐
Has the controlling contractor and entry employer(s) coordinated entry operations when permit space entry is performed at the same time that any activities that could foreseeably result in a hazard in the permit space are performed? <i>List activities</i>		☐	☐	☐

Post Entry Operations				
	Comments	Yes	No	N/A
Has the controlling contractor debriefed each entity that entered a permit space regarding the permit space program followed and any hazards confronted or created in the permit space(s) during entry operations? <i>Use additional sheets if necessary</i>		☐	☐	☐
Has the entry employer informed the controlling contractor in a timely manner of the permit space program followed and of any hazards confronted or created in the permit space(s) during entry operations? <i>List Hazards</i>		☐	☐	☐
Has the controlling contractor apprised the host employer of the information exchanged with the entry entities pursuant to 1926.1293(h)? Note to paragraph (h). <i>Unless a host employer or controlling contractor has or will have employees in a confined space, it is not required to enter any confined space to collect the information specified in this paragraph (h).</i>		☐	☐	☐
If there is no controlling contractor present at the worksite, the requirements for, and role of, controlling contractors in this section must be fulfilled by the host employer or other employer who arranges to have employees of another employer perform work that involves permit space entry. Notify DPR Project Management!		☐	☐	☐

Note: DPR and the entry employer should retain each canceled entry permit for at least 1 year to facilitate the review of the permit-required confined space program required by § 1926.1204(n). Any problems encountered during an entry operation should be noted on the pertinent permit so that appropriate revisions to the permit space program can be made

Missing Information	Responsible Party	Due Date	Date Completed



Permit Space Entry Communication and Coordination

Section 05

Cranes

(Hydraulic, Conventional, Tower & Helicopters)

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Cranes

(Includes hoists and rigging devices)

1.0 Introduction

Many types of cranes, hoists, and rigging devices are used on DPR Construction workplaces for lifting and moving material. DPR Construction's plan is to maintain a safe workplace for its employees and contractors; therefore, it cannot be overemphasized that only qualified and licensed individuals should operate and direct these devices. The safety rules and guidance in this section apply to all operations at DPR Construction workplaces that involve the use of cranes and hoists installed in, set-up in, or attached to buildings and to all DPR Construction employees and contractor personnel who use such devices.

DPR reserves the right to require independent third party evaluation, re-certification of cranes re-training or personnel.

Reference: Cranes & Derricks in Construction CFR 1926.1400 (Subpart CC); D.O.T. 49 CFR Part 391. California Code of Regulations, Title 8, Section 1610. ANSI/ASME B30. Other applicable standards, guidelines, and regulations may apply.

Definitions

Assembly/Disassembly Director (A/D Director): An individual who meets this subpart's requirements for an A/D director, irrespective of the person's formal job title or whether the person is non-management or management personnel

Assembly/Disassembly: The assembly and/or disassembly of equipment covered under this standard. With regard to tower cranes, "erecting and climbing" replaces the term "assembly," and "dismantling" replaces the term "disassembly". Regardless of whether the crane is initially erected to its full height or is climbed in stages, the process of increasing the height of the crane is an erection process.

Blocking (also referred to as "cribbing"): Wood or other material used to support equipment or a component and distribute loads to the ground. It is typically used to support lattice boom sections during assembly/disassembly and under outrigger and stabilizer floats.

Certified operator: With respect to certification from an accredited testing organization, an operator must be certified for the type and capacity of crane he or she is going to operate. Each accredited testing organization develops its own categories for crane type and capacity.

Conventional crane: A device for lifting and moving heavy weights in suspension.

Crane Operator Evaluation: OSHA requirement that employer or third party vendor must conduct an evaluation of each operator to ensure he/she is qualified by a demonstration of (i) the skills and knowledge necessary to operate the equipment safely, and (ii) the ability to recognize and avert risks associated with the operation. This must be conducted by someone who has the knowledge, training and experience to assess an operators abilities.

Directly under load: Any part or all of an employee is directly beneath the load

Dismantling: Includes partial dismantling (such as dismantling to shorten a boom or substitute a different component)

Encroachment: Where any part of the crane, load line or load (including rigging and lifting accessories) breaches a minimum clearance distance that this subpart requires to be maintained from a power line.

Helicopter: A type of aircraft that derives both lift and propulsion from one or more sets of horizontally revolving overhead rotors



Hydraulic crane: A crane that relies on electric or diesel-powered motors, hydraulic cranes include an internal hydraulic system that allows the crane to lift/hoist heavier loads

Lift Director: When required, a Lift Director is an individual who directly oversees the work being performed by a crane, crane operator and associated rigging crew

Proximity Alarm: A device that provides a warning of proximity to a power line and that has been listed, labeled, or accepted by a Nationally Recognized Testing Laboratory in accordance with 29 CFR 1910.7

Qualified Evaluator (third party): An entity that, due to its independence and expertise, has demonstrated that it is competent in accurately assessing whether individuals meet the Qualification Requirements in this subpart for a signal person

Qualified Person: A person who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training and experience, successfully demonstrated the ability to solve/resolve problems relating to the subject matter, the work, or the project

Qualified Person, Attendant or Operator (CA): In CA, a person designated by the employer who by reason of training, experience or instruction has demonstrated the ability to safely perform all assigned duties and, when required, is properly licensed in accordance with federal, state, or local laws and regulations

Qualified Rigger: A person who meets the criteria for a qualified person. Employers must determine whether a person is qualified to perform specific rigging tasks. Each qualified rigger may have different credentials or experience. A qualified rigger is a person that:

- possesses a recognized degree, certificate, or professional standing, or
- has extensive knowledge, training, and experience, and
- can successfully demonstrate the ability to solve problems related to rigging loads

Signal Person: A signal person is required when:

- The point of operation is not in full view of the operator
- The operator's view is obstructed in the direction the equipment is traveling
- Either the operator or the person handling the load determines that a signal person is needed because of site-specific safety concerns

Only a qualified person should be allowed to give signals. The employer of the signal person should confirm that each signal person is qualified to signal. This requirement should be met by using either a third party qualified evaluator or an employer qualified evaluator (non-portable)

Special Hazard Warnings: Warnings of site-specific hazards (i.e., proximity of power lines)

2.0 Pre-Planning

Work involving crane and lifting activities are defined as High Hazard by DPR and therefore requires a detailed, written pre-task plan in addition to the crane pre-lift checklist. Refer to the Pre- Planning Section for additional pre-task planning guidelines.

2.1 Advanced Notice

DPR EHS and the project team must be notified, prior to on-site arrival of any crane. Cranes should not be permitted on site until all documentation requirements are met.

2.1.1 Mobile Cranes

At a minimum, notification to DPR for conventional or hydraulic and/or assist cranes should be made one (1) week prior to the first lift. For critical lifts, a minimum of two (2) weeks' notice is required. Special consideration should be made for insurance requirements, which could require up to a two-week lead time.

2.1.2 Tower Cranes

Considerations should be made when planning installation of a tower including, but not limited to:

- Logistics plan for site routing of support/assist vehicles and emergency access
- Design and installation of foundation
- Availability of crane with desired configuration
- Availability of crane operator and backup operators
- Boom encroachment to adjacent structures or multiple cranes
- Counterweight encroachment on adjacent load line
- State and local permits or special requirements
- FAA considerations
- Air Rights
- Capacity and Reach
- Assembly
- Provisions for increasing (raising) height of crane
- Disassembly of crane and removal of base and trucking routes

2.2 Site Plan & Crane Assembly/Disassembly

Coordination of crane arrival and set up should be conducted with the project team and all other affected parties. Items to consider include, but are not limited to:

- Crane selection
- Site logistics
- Coordination with owners/ users schedule crane activities
- Crane set up/break down and lifting location(s)
- Duration crane will be in (each) location
- Location of utilities and services

2.3 Insurance Requirements

A certificate of insurance should fulfill DPR insurance requirements (reference contract Attachment A for specific requirements).

- The requirements include, but are not limited to:
 - Insurance must be valid (i.e., not expired)
 - Insurance must include general liability
 - As needed, insurance must include excess liability insurance
 - If the equipment or material to be lifted is in excess of \$50K in value, then additional riggers or hook liability insurance will be needed
 - DPR, the owner and subcontractor (if applicable) must be named as an additional insured

2.4 Crane Certification & Repair Logs

Documented daily, monthly and annual inspection records and any other records pertaining to crane inspections should be kept with the crane. If during any safety inspection, the operator cannot produce the required crane inspection, work should be stopped as soon as the current load is off the hook and inspected immediately. Copies of inspection records must be submitted to DPR on a weekly basis (originals will be kept on unit).

Repair and/or maintenance logs should be kept with the crane and maintained. Should any systems considered critical to safe operation be found deficient, all repairs must be made prior to start of work. Copies of repair and/or maintenance logs should be provided to DPR upon request.

3.0 Pre-Mobilization Documentation

Documents required prior to the first crane pick on site include the following:

- Annual Certification
- Contractor's Certificate of Insurance
- Operator's certification
- Documentation of Crane Operators Evaluation
- Copy of load charts pertinent to boom configuration to be used
- Written verification from crane owner that maintenance has been performed pursuant to manufacturer's recommendations
- Written verification of qualified rigger(s) training
- Written verification of qualified signaler(s) training
- Written verification of competent/qualified assembly/disassembly director
- Written verification of assembly/disassembly procedures
- Where required, written verification of qualified lift supervisor

4.0 Qualifications/Certifications & Responsibilities

Below is a list of qualifications/ certifications & responsibilities of essential personnel to all crane activity. Additional responsibilities may be required pursuant to local, state, and/ or federal agencies including manufacturer's requirements.

4.1 Crane Operator

4.1.1 Crane Operator Certifications

Operator certification must comply with all local, state and federal requirements including ANSI/ASME B30 Standards. Operators must be certified by an accredited crane operator testing organization. Operators must maintain in their possession, documentation of their certification for the type and configuration of the crane being operated. The employer or third party vendor must conduct an evaluation of each operator to ensure he/she is qualified by a demonstration of (i) the skills and knowledge necessary to operate the equipment safely, and (ii) the ability to recognize and avert risks associated with the operation. This must be conducted by someone who has the knowledge, training and experience to assess an operators abilities. This should include Name of operator and Evaluator; date of evaluation; make model and configuration of crane(s) and be signed by the evaluator.

Current copies of the following must be provided:

- Operators certification Card
- Current Medical Examiner's Certificate
- Documentation of Crane Operators Evaluation

4.1.2 Crane Operator Responsibilities

The operator is responsible for the following:

- Having overall responsibility and authority for ensuring loads are safely lifted, moved and landed
- Following all manufacturers' recommendations, limitations, specifications and procedures
- The proper placement of the crane in relationship to the load to be handled and the landing area so as to obtain the best rated lift capacity
- The ground and footings are stable for all lifts
- That sufficient cribbing and supporting materials are used and verifying the manufacturer's specifications for adequate support and degree of level are met
- The installation and maintenance of crane swing radius protection and exclusion work zones control lines are properly installed and maintained

- Completing daily/monthly inspections; tagging out of service equipment that is defective; confirming all adjustments and repairs are made prior to operating equipment
- Not engaging in any practice or activity that diverts attention while engaged in operating
- Verifying the crane is operated within its rated capacities; and ensuring:
 - The weight of all auxiliary handling devices such as hoist blocks, headache balls, hooks and rigging are considered by the operator as part of the total load
 - Additionally, the weight of all items added to the load at the site must be determined and added to the total weight
- Not leaving the controls while the load is suspended
- Ensuring, at a minimum, the manufacturer's guidelines are followed for securing the equipment in the event of local storm warnings
- Ensures tag lines are used, and sufficient number are attached to the load, to control loads
 - Must obey the stop signal, regardless of who gives the signal
 - Have the authority to stop any lift deemed unsafe

4.2 Rigger Qualification

All rigging must be performed by a qualified rigger. Documentation of rigger's qualification should be on the employer's company letterhead or documentation provided by an evaluating third party and submitted to the DPR project team prior to start of work. Copies of these qualifications will be kept on file at the jobsite.

4.2.1 Qualified Rigger Responsibilities

The person designated as the qualified rigger must have the ability to properly rig the load for a particular job, including inspection of all rigging to be used. It does not mean that a rigger must be qualified to do every type of rigging job. Each load that requires rigging has unique properties that can range from the simple to the complex. Riggers must ensure all loads are rigged with a "tag-line" and have the ability to stop any lift deemed unsafe.

Once it has been verified that a person is qualified to rig, that person needs to be outfitted, at the expense of their employer, with a differentiating colored reflective vest, a whistle or air horn, and a "X" on the top of their hardhat made of a reflective tape or sticker. Whistle or air horn is to be blown during picks to make other workers aware that a load is being lifted. Alternative means should be reviewed and signed off by EHS.

4.3 Signal Person

4.3.1 Signal Person Qualifications

All signaling must be performed by a qualified signal person. Verification of signal person(s) qualifications should be submitted on the employer's company letterhead with qualified evaluator identified **OR** documentation from third party. Hand signal charts must be either posted on the crane or conspicuously posted in the vicinity of the hoisting operations.

4.3.2 Signal Person Responsibilities

The signal person is responsible for:

- Knowing and understanding the type of signals used
- Understanding the operation and limitations of the equipment, including the crane dynamics involved in swinging, raising, lowering and stopping loads and in boom deflection from hoisting loads
- Having passed an oral or written test and a practical test
- Having the authority to stop any lift deemed unsafe

Once it has been verified that a person is qualified to signal, that person needs to be outfitted, at the expense of their employer, with a differentiating colored reflective vest, a whistle or air horn, and an "X" on the top of

their hardhat made of a reflective tape or sticker. Whistle or air horn is to be sounded during picks to make other workers aware that a load is being lifted. A building “map” should be distributed to the Crane Operator as well as to all signalers so that clear verbal direction can be given as to where the load is heading once it is rigged and lifted. Alternative means should be reviewed and signed off by EHS.

4.4 Assembly/Disassembly (A/D) Director

4.4.1 A/D Director Qualifications

The A/D director must meet the requirement, of **both** a competent and a qualified person. Verification of the A/D director’s qualification should be submitted on the worker’s company letterhead with qualified evaluator identified **OR** documentation from third party qualified evaluator. When assembling or disassembling equipment (or attachments), the AD must comply with all:

- Manufacturers assembly and disassembly procedures
- Crane Contractor’s assembly and disassembly procedures

4.5. A/D Director Responsibilities

The assembly/disassembly Director (A/D Director) is responsible for the following:

- Ensuring all crew members involved in assembly and/or disassembly understand the task they are assigned to perform, hazards of their assigned tasks and all hazards associated.
- Confirming operator is aware of each crew member’s location, if out of view of the operator
- Ensuring no one works under the boom, jib or other components when the potential for falling objects exists, or has a plan for minimizing exposure
- That the rated capacity of load limits is not exceeded during assembly or disassembly activities
- The following specific hazards are identified and addressed, including, at a minimum:
 - Adequate site and ground bearing conditions
 - Adequate load blocking materials are properly located to maintain stability
 - Verification of assist crane (if used) loads
 - Suitable boom and jib pick points are utilized
 - Center of gravity of load identified and maintained
 - Understanding the potential of stored energy present during assembly/disassembly, and ensuring all crew members understand as well
- Has the authority to stop any lift deemed unsafe

4.6 Lift Director Responsibilities

A qualified lift director (who may be the operator) is responsible, for the following:

- Being present at the jobsite during lifting operations
- Stopping crane operations if alerted to unsafe conditions affecting operations
- Ensuring preparation of the area needs to support crane operations has been completed prior to start of crane operations
- The necessary traffic controls are in place to restrict unauthorized access to the crane’s work area
- That personnel involved in crane operations understand their responsibilities, assigned duties and associated hazards
- Addressing safety concerns raised by the operator or other personnel
- Appointing qualified signal & rigging personnel and ensuring the operator has that information
- That hand, voice or audible signals are agreed upon by all parties prior to lifting
- Ensuring crane operations in the vicinity of power lines are performed in accordance with this EHSP and federal, and state OSHA guidelines
- Ensuring precautions are implemented when hazards associated with special lifting operations (i.e., multiple crane lifts, lifting personnel, pick and carry operations, etc.) are present

- Informing the crane operator of the weight of loads to be lifted as well as the lifting, moving and placing locations for those loads; and obtaining the crane operator's verification that those weights do not exceed the crane's rated capacity
- That the load is properly rigged and balanced prior to lifting
- Has the authority to stop any lift deemed unsafe

5.0 Inspection

Daily and annual inspections must be performed in accordance with the manufacturer's recommendations and as mandated by OSHA standards. Annual inspections shall be performed by an independent third party. Annual inspections must be received and reviewed by DPR prior to the arrival of equipment. Daily, documented inspections (pursuant to manufacturer's requirements) must be performed by a competent person prior to lifting each day maintained on the unit.

Additionally, cranes must be inspected after set up and prior to initial lift, before each shift, and after any malfunction or incident.

6.0 Ground Conditions

DPR will ensure firm, drained and graded ground conditions are provided in the mutually agreed upon lift location(s). Location(s) of any underground hazards beneath crane set up locations must be identified and communicated.

7.0 Crane Selection and Set Up

The operator is responsible for ensuring the crane being operated has sufficient capacity and follows manufacturer's recommendations for intended lifts. The operator has final responsibility and control over crane operations. The operator should comply with the manufacturer's specifications and limitations applicable to the operation of any and all crane and lifting devices and derricks. Where manufacturer's specifications are not available, the limitations assigned to the equipment should be based on the determinations of a qualified engineer competent in this field. Such determinations should be appropriately documented and recorded and available at the job site. ***All cranes will be equipped with an integrated anti-two block system.***

Once the appropriate crane has been selected, the crane supplier is responsible for obtaining the proper permits to transport the equipment to the site.

7.7 Working Around Power Lines

Overhead line contact still remains one of the leading causes of accidents/fatalities on construction sites. OSHA outlines three options of programmatic requirements to prevent contact. An assessment of the work area is required which must include one of the three following options:

- Option 1: Dennergizing and grounding the power lines in the work area: It must be presumed that all lines are energized until the line is deenergized and visibly grounded at the worksite. Deenergization alone is never acceptable.
- Option 2: 20 foot clearance: An assessment must be performed to ensure no part of the crane loadline or load (including all rigging accessories) can get closer than 20 feet to a power line. If the assessment shows the potential for any part of the crane to come within 20 feet of the power lines, additional actions are required, including but not limited to:
 - A planning meeting with minimum specific agenda items to be covered
 - Non-conductive tag lines (meeting ASTM standards)
 - Erection and maintenance of an elevated warning line, barricade or line of signs in view of the operator. If the operator is unable to see elevated warning line, dedicated spotter must be used who can:
 - Reposition to continually assess clearance distance

- Have direct and continuous communication with the crane operator to ensure clearance distances are maintained
- Additionally, one of the following must also be implemented:
 - A device that automatically limits the crane's range of movement to prevent encroachment
 - Proximity alarm set to give the operator sufficient warning to prevent encroachment
 - A device that automatically warns the operator when to stop movement, set to give the operator sufficient warning to prevent encroachment
 - An insulating link or device, installed at a point between the end of the load line (or below) and the load. *Only devices listed by an OSHA Nationally Recognized Testing Laboratory are allowed.*
- Option 3: If crane work has the potential for encroaching boundaries listed in Table A of the standard (and listed below), no work may be performed until a detailed work plan is developed, reviewed and approved by the DPR project team. The plan must meet all requirements under 29 CFR 1926.1400 (Subpart CC)

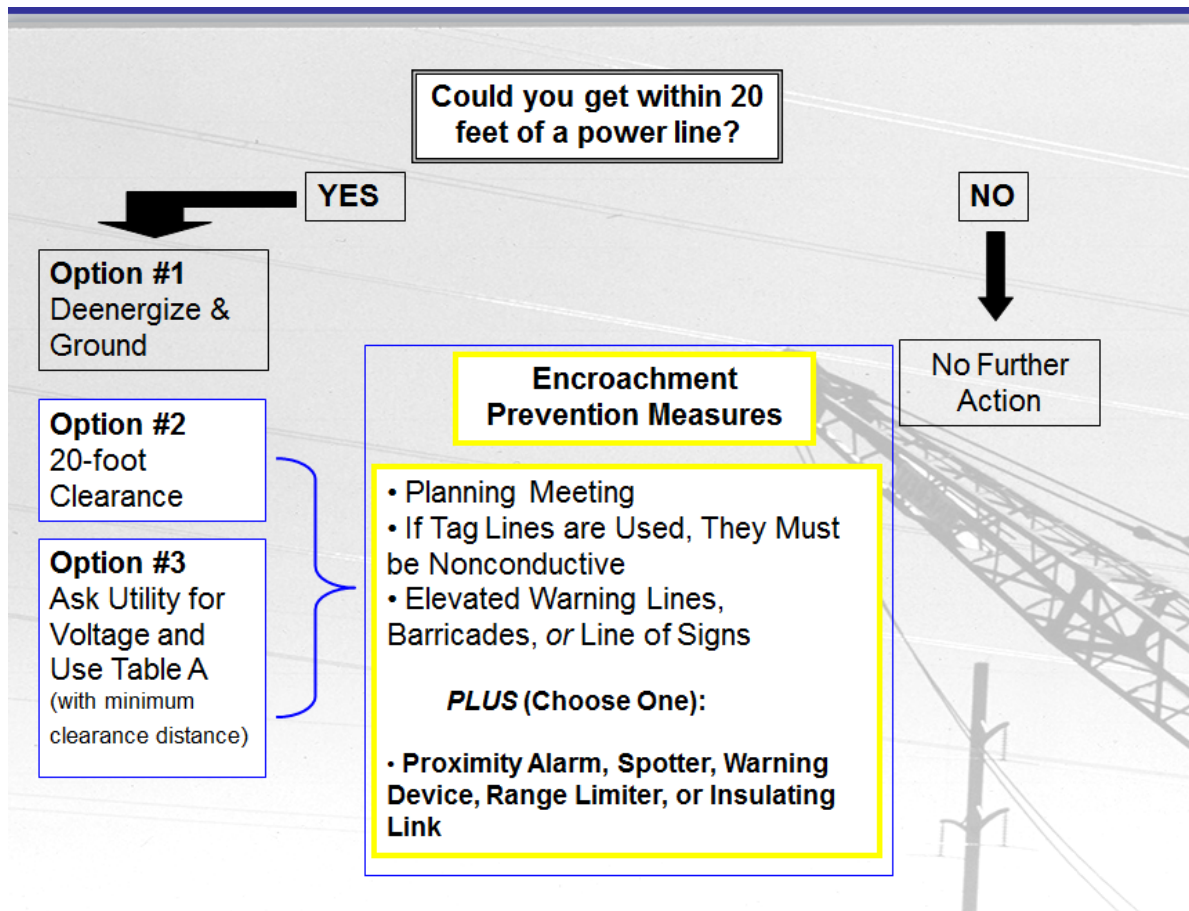


Table A – Minimum Clearance Distances

Voltage (nominal, kV, alternating current)	Minimum clearance distance (feet)
up to 50	10
over 50 to 200	15
over 200 to 350	20
over 350 to 500	25
over 500 to 750	35
over 750 to 1000	45
over 1000	(as established by the power line owner/operator or registered professional engineer who is a qualified person with respect to electrical power transmission and distribution)

Intentionally Working Closer Than Table A Zone

Must Show:

- Staying Outside Zone is Infeasible
- Infeasible to Deenergize and Ground



All of the following are required:

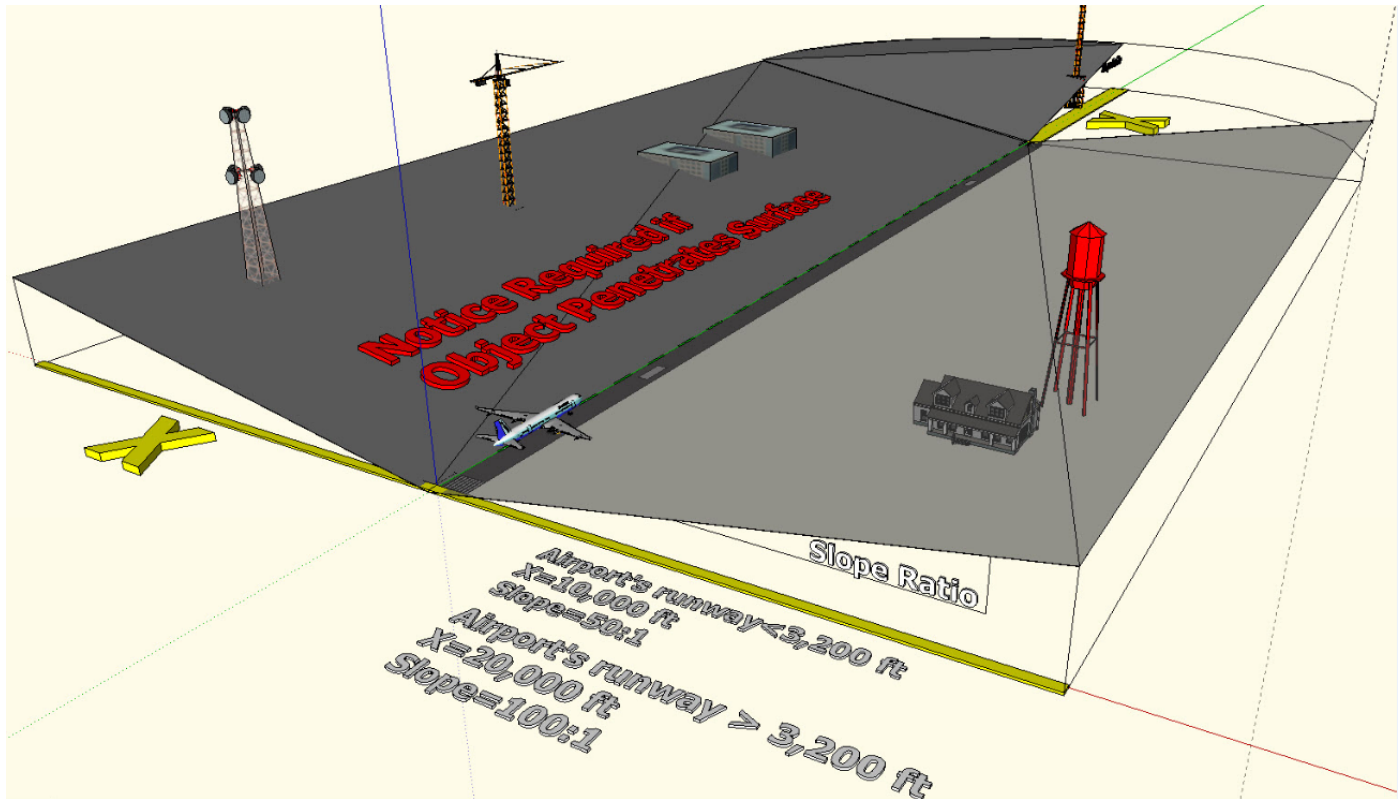
1. Power Line Owner – **Sets Minimum Approach Distance**
2. Planning Meeting – Procedures
3. Dedicated Spotter
4. Elevated Warning Line or Barricade
5. Insulating Link/Device
6. Nonconductive Rigging
7. Range Limiter (if Equipped)
8. Nonconductive Tag Line (if used)
9. Barricades - 10 feet from Equipment
10. Limit Access to Essential Employees
11. Ground Crane
12. Deactivate Automatic Re-energizer

8.0 FAA Notification

Construction equipment or other temporary structures such as tower cranes, derricks, mobile cranes are examples of structures that require notice to the FAA when the following conditions apply:

- When they are greater than 200 feet above the ground level at their location
- When they are near an airport or heliport:
 - within 20,000 feet of a major airport
 - within 10,000 feet of a minor airport
 - within 5,000 feet of a heliport
- When they are located on an airport or heliport
- When requested by the FAA
- Check local regulations for overhead clearances

Notice to the FAA should be submitted at least 30 days before the date the proposed construction or alteration is to begin. Refer to <http://jan.ucc.nau.edu/~cvm/latlongdist.html> for further information and requirements.



9.0 Critical Lift

A critical lift plan should be submitted a minimum of two (2) weeks in advance of the anticipated lift to DPR project management for review. Critical lifts are defined as any lift that:

- Any pick which puts personnel, equipment or structures at risk
- Gross load weight exceeds 75% of crane's rated capacity

Note: *At no time should the load exceed 80% of the crane's rated capacity.*

- Any lift which requires unusual rigging or procedures
- Any lift where the crane is supported on a structure
- Crane suspended personnel platform lift
- Tandem picks
- Net weight of load exceeds 25 tons
- Replacement time for damaged load that exceeds four months
- Any lift with energized power lines within 20'

9.1 Critical Lift Procedures

In addition to completing the Pre-Lift Checklist and Crane Pre-task Plan (refer to the Forms Section at the end of the EHSP), the following requirements apply:

- All lift activities should be conducted under the direction of a lift director. The lift director should have no other duties other than directing the lift, and should be on-site during all lifting activities
- Coordination meeting, to include a pre-job walk, should be conducted with DPR project management/EHS no more than one (1) week prior to lift. The meeting should review the contractor's written lift plan to address at a minimum:
 - Public protection
 - Crane set up
 - Dry-run procedures

- Load conditions
- Rigging configuration
- FAA - Airports
- Significant site condition changes
- Boom configuration
- “Call off” conditions
- Maximum wind allowed
- Ground conditions
- Overhead obstructions
- Assembly/Disassembly areas
- Form of communications to be used during the lift
- Exclusion Zones
- Weather / Wind

10.0 Operating Procedures

It is DPR’s intent that all cranes be operated by a certified operator. It is the responsibility of the subcontractor to provide an operator meeting these requirements, and to ensure safe operating practices are followed at all times.

- The operator should not engage in any practice that may divert attention while engaged in crane operations, cell phones should only be used for signaling, operate the crane if physically or mentally impaired, taking prescription drugs that may affect judgment and should not respond to any signal which is unclear or is given by anyone other than appointed signaler (*Exception*: the operator should respond to an emergency stop signal given by anyone)
- The operator should not permit trainees to make trial lifts
- The operator should perform the trial lift to determine lift stability, crane function and safety
- The operator has final responsibility and control over crane operations
- Whenever there is any doubt as to safety, the operator has the authority to stop and refuse to handle loads until safety concerns have been addressed
- The operator should be familiar with the crane and its care, the operator's manual and load charts
- The operator should be responsible for notifying their lift director of any needed adjustments or repairs, and for logging their findings in the crane log
- Upon request, the operator should demonstrate their ability to determine total load weight and its relationship to the crane load charts
- No crane shall be loaded beyond its rated capacity (over 75% would be considered a critical lift), except for test purposes under controlled circumstances
- Swing radius should be barricaded prior to any lifting activities
- When loads to be handled are limited by structural competence rather than by stability, the operator and/or lift director should determine concurrently that the weight of the load is known within (+)/(-) 5 percent *before* the load is lifted
- All loads should be rigged by a qualified rigger
- All loads must be controlled by a tagline(s)
- Safety latches on hooks shall not be deactivated or made inoperable except:
 - When equivalent protection is provided in a site-specific erection plan, and accepted by DPR project management
 - Open hooks are used only for the unloading of and staging of material
- The operator should not suddenly accelerate or decelerate a moving load and should maintain the control of the load at all times
- At no time should any part of the crane or any part of the load contact any obstructions
- The operator must not swing loads over personnel
- Only essential employees may be in the fall zone during lifting activities

- The operator must avoid side-loading of booms
- Lifts should be limited to freely suspended loads
- Cranes should not be used to drag loads sideways
- Fall protection is required at or above six feet
- Lift operations should not begin until all safety devices are functional; if safety devices are not operable, equipment shall be taken out of service until repairs are made and tested
- At no time should the suspended load be lifted/flown outside the limits of construction

11.0 Crane-Suspended Work Platforms

The use of crane-suspended work platforms is prohibited. Crane or derrick suspended personnel platforms may not be used unless the erection, use and dismantling of conventional means of accessing the worksite is not possible or would be more hazardous; lifting of personnel is considered a critical lift. In the event that all other conventional means are not possible, a request for the use of crane-suspended work platforms should be submitted in writing to DPR EHS and/or DPR Corporate EHS, a minimum of two (2) weeks prior to the scheduled work. The plans should be submitted and approved prior to use. The following items, at a minimum, should be included:

- Statement as to why conventional means are not feasible and a list of the options considered infeasible
- Full compliance with the requirements set out in 29 CFR Section 1926.1501
- Description of the crane to be used and the manufacturer's instructions and requirements in the use of the crane to lift, lower and/or suspend personnel on a work platform
- Drawing of the work platform, suspension bridge and other components with computations used in the design
- Documented emergency plan in the event of a crane failure
- Communication method
- Certified third party inspection of the basket to be used and associated components
- An integrated anti-two-block device on the crane at all times
- All persons on the lift must have a 100% fall protection system in place at all times during the lift

Should the use of crane suspended work platforms be approved, the contractor should complete the following:

- Crane Pre-Lift Checklist
- Crane Pre-Task Plan
- Crane Personnel Platform Safety Checklist
- Documentation that all individuals who are to be lifted have been trained in personnel platform lifting protocols per OSHA, state and/or local requirements (and owner requirements, where applicable)

Also, the jobsite must perform a proof test prior to hoisting employees. The platform and rigging should be proof tested to 125% of the platform's posted rated capacity. Inspect the crane, rigging and the personnel platform after the test. *Refer to Fed/OSHA regulations for additional procedures to be followed when using a personnel platform lift.*

12.0 Tower Cranes

Tower crane operations are potentially hazardous. This requires coordination between estimating, preconstruction, DPR on-site project team and affected subcontractors. In addition to tower crane specific planning, all documentation required by sub-section 3.0 Pre Mobilization should be completed prior to operation. (California requires permit)

Tower cranes should be operated and inspected in accordance with manufacturer's recommendations and limitations. Only certified operators, trained for the specific piece of equipment, are permitted to operate a

tower crane. *Refer to the Tower Crane Pre-Planning Checklist in the Forms Section at the end of the EHSP to guide the project team through the planning process.*

12.1 Tower Crane Security Guidelines (public protection)

- Access to tower cranes should be carefully planned to prevent unauthorized access to the crane structure and controls
- The following list of security and access considerations should be considered as minimum precautions. High profile urban settings should be assessed with the Owner to see if they require any additional security measures above DPR's standard policy. Local jurisdictional requirements must also be evaluated. The stricter requirement should be followed.
 - Barricade around the perimeter of the crane base to prevent unauthorized access. The barricade can be plywood, chain link fence or other material. The barricade should be at least a minimum of 8 feet high and include a lockable gate/door.
 - Where tiebacks occur, the perimeter of the structure shall be sealed off to prevent access to the crane via the tie-backs
 - Operators' access bridges shall be secured with a self-closing and self-locking door
 - Tower cranes inside the building footprint shall be sealed off horizontally above the roof level and/or highest level of the building prior to jacking the crane
 - The tower crane operator is to remove the key to the crane after each shift
 - The DPR project team shall keep a spare set of keys for the barricade gate(s) and the tower crane in the project office in case of an emergency
 - Inspections of crane security measures and maintenance should be conducted on a frequent basis
 - The tower crane operator is responsible for the security of the tower crane. Tower crane operators should report any deficiencies to security measures.
 - Tower crane bases built into the recess of a building or below grade, where water may accumulate, should have a pre-installed sump pump system with adequate discharge capacity.
- Crane Operator Hours
 - Tower crane operator hours should not exceed 12 hours in the seat per shift and that an 8-hour rest period should follow each shift. Tower crane operators should not work more than 60 hours in any given week and never more than 6 days in a row.

13.0 Steel Erection Activities

Refer to Steel Erection Section for specific guidelines on Steel Erection activities.

14.0 Helicopters

Helicopter hoisting operations are not permitted unless all other methods have been considered and determined to be infeasible. If the DPR Project Team determines that a helicopter lift is the only viable option, a written plan must be developed by the DPR Project Team and submitted at least-six (6) weeks in advance of the requested lift date. The plan must be submitted to DPR's Regional Safety Manager, Business Unit Leader and appropriate Management Committee Member for review and approval. Once the plan is approved at the Regional Level, it must be forwarded to DPR's Corporate President for review and approval.

The written plan must include, but not limited to:

- List all other lifting methods considered and include a detailed explanation/justification for why no other method is feasible.
- Explain what is being lifted, its weight, the helicopter being used and its load capacity (which will vary based on fuel level), helicopter refueling location, and emergency plans
- Rigging and lifting plans
- Insurance requirements, including coverages, limits, and evidence of compliance

- FAA Permit that includes flight plan (take off/landing locations, path of travel and staging area)
- Pilot and ground crew qualifications and current licenses
- Helicopter Work Permit completed and all required signatures obtained
- Detailed Helicopter Lift Flight Checklist completed (refer to Helicopter Lift Checklist in the Forms Section at the end of the EHSP)
- To the extent subcontractors are involved in the lift, each subcontractor should develop its own written plan for DPR review that addresses the applicable items above and any other important aspects of the lift.
- Fire watch communication system
- Any hazardous substances, which may be included in load

15.0 Weather Related Impacts

It is the responsibility of the subcontractor to ensure all lifting activities are performed safely within the parameters and limitations specified in crane manufacturer's operating manual. Special consideration should be given to adverse weather conditions. DPR project management reserves the right to cease any activity should weather conditions change affecting work performance; i.e., windspeed, wind gusts, lightning, high humidity, temperature extremes, or any other severe weather condition etc.

Cranes Forms

Forms

- Crane Subcontractor Pre-Planning Package
- Crane Pre-Lift Checklist
- Crane & Material Lifting Pre-Task Plan
- Crane Hand Signals
- Instructions for Completing FAA Form 7460-1
- Tower Crane Pre-Planning Checklist
- Helicopter Work Permit
- Helicopter Lift Checklist



Crane Subcontractor Pre-Planning Package

Date: _____

Dear Subcontractor:

The following package is provided to you to assist you and your workers in maintaining an Injury Free Environment. As you know, crane operations are very hazardous and there are many steps necessary prior to a crane arriving on the project. Though crane accidents rarely occur, when they do there are usually serious consequences such as loss of life, property or otherwise. Please review and complete the following documentation. These should be filled out as completely as possible and submitted to DPR at least one (1) week prior to first lift and two (2) weeks for critical lifts.

Required documentation:

- 1) Please complete the Pre-Lift Checklist as completely as possible.
- 2) Provide copies of annual crane certification.
 - a. Certification must match the crane that shows up on site.
 - b. Certification must be current.
 - c. Crane must have passed certification; including verification of completed repairs identified at time of certification.

Note: cranes are required to be equipped with an integrated anti-two block systems.
- 3) Provide copies of insurance (*acceptance of crane insurance certificate is required prior to the first lift*).
 - a. Insurance must include general liability.
 - b. As needed, insurance must include excess liability insurance.
 - c. If a equipment or material to be lifted is in excess of \$50K in value, then additional riggers or hook liability insurance will be needed.
 - d. DPR and the owner must be named as an additional insured.
 - e. Insurance must be valid (i.e., not expired).
- 4) Current copy of operator's certification
- 5) Identify the Assembly/Disassembly director (if required)
- 6) Manufacturer's procedures for assembly and disassembly **or** Crane Contractor's procedures for assembly and disassembly.
- 7) Copy of load charts pertinent to specific boom configuration to be used.
- 8) Written verification from crane owner that maintenance has been performed pursuant to manufacturer's recommendations.
- 9) Written verification, of competent - qualified assembly/disassembly director, when required
- 10) Written verification of qualified rigger(s) training
- 11) Written verification of signaler(s) training
- 12) Written verification of qualified lift supervisor, when required
- 13) Complete a pre-task/pre-lift plan, detailing likely sequencing and what safety precautions will be implemented to address any potential hazards. A pre-task plan checklist and worksheet have been provided. Safety items to consider:
 - a. Fall Protection
 - b. Public & Worker Exposure
 - c. Weather
 - d. Communication
 - e. Powerline Proximity
 - e. Guarding
 - f. Crane Set Up
 - g. Rigging
 - h. Control of Load / Pinch Points

Additionally:

- 1) A safety meeting will be held prior the lift with the crew to review all of the above listed items.
- 2) A daily inspection (as required by OSHA) must be performed on all cranes. Typically these inspection logs will already be with the cranes. A sample inspection form is provided.
- 3) **For steel erection activities, refer to Steel Erection Section 23 for specific guidelines on hoisting and lifting.**

If you have any questions regarding any information contained within, please contact the DPR Project Team. Thank you for your assistance in helping us to create an Injury Free Environment.

DPR Construction



Crane Pre-Lift Checklist

Archive Document – DO NOT DISCARD

Contact DPR project management and EHS at least one (1) week prior to a lift and at least two (2) weeks prior to a critical lift

Project/Project#: _____ Date: _____
Contractor Requesting Crane: _____ Date Crane will be Lifting: _____
Contractor Providing Crane: _____ Date of Crane Arrival: _____
Contractor Supervisor: _____ Critical Lift: Yes / No (see section 10.0)

Documentation

☐ Valid annual crane certification with documentation of all repairs completed	
☐ Identify Assembly/Disassembly Director (if applicable)	
☐ Manufacturer's or crane contractor's assembly and disassembly procedures (if applicable)	
☐ Written verification from crane owner that maintenance has been performed pursuant to manufacturer's recommendations.	
☐ Verify operator's certification, written verification of competency testing specific to the make/model of crane.	
☐ Verification of qualified rigger(s) training	☐ Verification of signaler(s) training
☐ Subcontractor to submit copies of load charts pertinent to specific boom configuration to be used.	
☐ Obtain a copy of the crane contractor's certificate of insurance with additional insured endorsement, Verify limits are within terms and conditions per DPR contract.	
☐ Verification of meeting FAA requirements (as required).	
☐ Complete detailed, written pre-task planning with operator, oiler, and crew and foreman involvement. Contractor to submit a written plan one (1) week prior to crane arrival two (2) weeks for critical lifts to include:	
☐ An Erection Plan for steel erection activities.	☐ Start & finish times for all lifts
☐ A detailed list of protective measures for workers, the environment, & the public	☐ A detailed flail protection plan
☐ Crane set-up locations w/swing path, travel plan & loading positions.	☐ Communication plan
☐ Documented cut sheets on all units to be lifted, including weights & dimensions and the anticipated load calculations	☐ Hoisting/Rigging Plan (single pick preferred, maximum 3 members w/approved multi-rigging assembly(prior approval required)

Pre-Mobilization

☐ DPR Superintendent & requesting crane contractor perform site walk. Review at minimum:	
☐ Integrated anti-two block system	
☐ Condition of asphalt, concrete, soil, underground hazards, etc. in crane operation location(s)	
☐ Location of overhead power lines	
Rated line voltage (verified by owner): (kv)	Max.working radius of crane: (ft)
☐ Lines Energized ☐ No (contact owner STOP until voltages are verified)	☐ Yes (distance greater than 20' or use of alternative device required)
☐ Distance to neighboring buildings, roadways & walkways	
Person designated to observe line clearance for operation	
☐ Crane contractor to provide:	☐ Building height
☐ Gross capacity of crane to be used	☐ Sling(s) type
☐ Net capacity of crane: (Max load weight+blocks+rigging=total weight	
☐ Max. radius of lift	☐ Sling(s) angle
☐ Maximum boom length	☐ Sling(s) capacity
☐ Minimum boom angle	☐ Accessory (i.e., shackles) capacity

Mobilization

☐ Operator and crew complete site safety orientation	☐ Weather conditions & related impacts	
☐ Determine operator's qualifications and length of experience with specific make and model of crane.	☐ Maximum wind speed per manufacturer verified	
☐ Documentation of Crane Assembly Safety Meeting & Boom Inspection.	☐ Evacuation plan	
☐ A copy of the manufacturers Operating Manual is on the crane.	☐ Rigging prodcedures	
☐ A minimum of a 5 BC portable fire extinguisher with current maintenance tag is on the crane	☐ Verify & review means of communication	
☐ Conduct a safety meeting, covering at least	☐ Safety precautions working around crane	
☐ Review Pre-task Plan	☐ Public protection	
☐ Outriggers must be fully extended, pads level, on solid bearing & pinned. Retracting outriggers without a signaler is prohibited	☐ Barricading in place to protect the swing of crane and for traffic control	
☐ Rigging in good condition & additional rigging available.	PRINT	SIGNATURE
☐ Ensure these are in place		
☐ Pre-task Plan (review with all lift participants)		DPR Superintendent
☐ PPE (hard hat, safety glasses, gloves, vests, etc)		Requesting Contractor Supervisor
☐ Fall Protection (harnesses, lanyards, etc)		
☐ Taglines & air horns		Lift director/Crane Operator
☐ There is no work under the load or pinch points		
Signaler		Qualified Person for Critical Lift



Crane & Material Lifting Pre-Task Plan

Archive Document – DO NOT DISCARD

Project/Project# _____ Date _____
Company# _____ Specific Location of Work _____

Answer the following when evaluating your work YES, NO or N/A:							Are any of the following Required?		
1.	Have you walked your work area to identify live systems, overhead power lines, utilities, underground conditions, public exposures, lighting, etc?	☐	YES	☐	NO	☐	N/A	PPE	
2.	Are there any hazards created by any other workers in your area or does your work create hazards for others?	☐	YES	☐	NO	☐	N/A	☐	Hard Hat
3.	Have all rigging, equipment and tools been inspected prior to use?	☐	YES	☐	NO	☐	N/A	☐	Fall Protection
4.	Are materials, tools and manpower adequate to perform the job safely?	☐	YES	☐	NO	☐	N/A	☐	Eye / Face PPE
5.	Have weights of all loads been verified?	☐	YES	☐	NO	☐	N/A	☐	Hand / Arm PPE
6.	Is this a critical lift? (i.e. load exceeds 75% of crane capacity, or puts personnel, equipment or structures at risk, etc.)	☐	YES	☐	NO	☐	N/A	☐	Hearing PPE
7.	Does this task require any additional permits or procedures? (FAA Notification, etc.)	☐	YES	☐	NO	☐	N/A	☐	Foot PPE
8.	Have you addressed any barricading, warning system or signage requirements appropriate to the task?	☐	YES	☐	NO	☐	N/A	LOCATE	
9.	Have you established a communication system for signaling?	☐	YES	☐	NO	☐	N/A	☐	Respirator
10.	Are outriggers fully extended with adequate pads?	☐	YES	☐	NO	☐	N/A	☐	Emergency Telephones
11.	Are all riggers & signalers qualified?	☐	YES	☐	NO	☐	N/A	☐	Fire Extinguisher
12.	Will this task require any lifting, bending or twisting? Have you completed Stretch & Flex today?	☐	YES	☐	NO	☐	N/A	☐	Emergency Exit Routes
13.	Are enough people assigned to safely complete the task? (lifting, spotting, guiding w/taglines etc.)	☐	YES	☐	NO	☐	N/A	☐	First Aid Equipment
14.	Is there a new hire, or new team member on the project who need support?	☐	YES	☐	NO	☐	N/A	☐	Other
15.	Do you have an injury to report or were you injured the prior working day?	☐	YES	☐	NO	☐	N/A	☐	Other

Any safety item identified in the above question list must be addressed on the back page.

Steps for Work	Tools	Hazards	Steps Taken to Address Hazards

*If you need more space, attach another PRE-TASK PLAN SHEET

Lift Supervisor		Foreman		Safety Coordinator	
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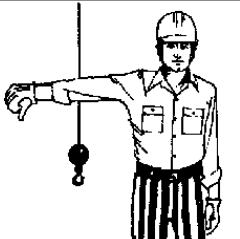
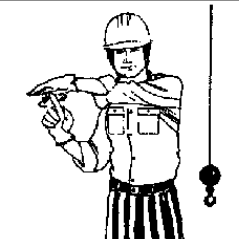
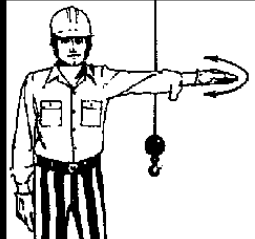
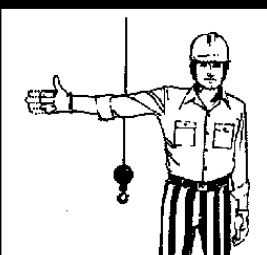
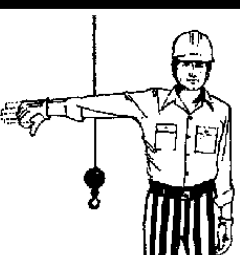
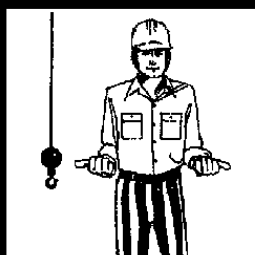
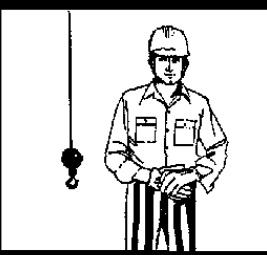
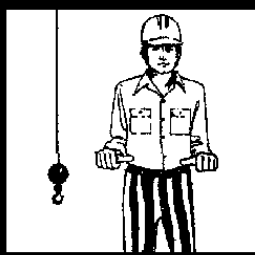
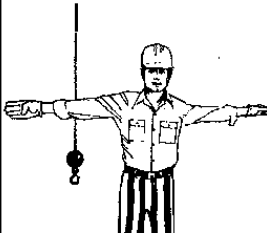
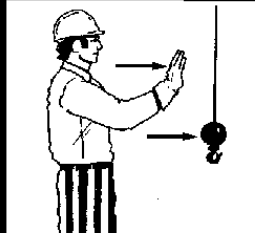
Crew Members – list additional crew members on back or a separate page if necessary:



Crane Lift Plan

LIFT PLAN – Please draw a map to detail set up points, sequencing of picks, protection etc.

Crane Hand Signals

			
HOIST	LOWER	USE MAIN HOIST	USE WHIPLINE
			
RAISE BOOM	LOWER BOOM	HOIST SLOWLY	STOP
			
RAISE THE BOOM AND LOWER THE LOAD	LOWER THE BOOM AND RAISE THE LOAD	SWING	EXTEND BOOM
			
DOG EVERYTHING	TRAVEL	TRAVEL - ONE TRACK	RETRACT BOOM
			
EMERGENCY STOP	BRING UP ARM (TELESCOPIC ARM)	EXTEND THE ARM (TELESCOPIC ARM)	RUN



INSTRUCTIONS FOR COMPLETING FAA FORM 7460-1

PLEASE TYPE or PRINT

ITEM #1. Please include the name, address and phone number of a personal contact point as well as the company name.

ITEM #2. Please include the name, address and phone number of a personal contact point as well as the company name.

ITEM #3. New Construction would be a structure that has not yet been built.

Alteration is a change to an existing structure such as the addition of a side mounted antenna, a change to the marking and lighting, a change to power and/or frequency, or a change to the height. The nature of the alteration shall be included in ITEM #21 "Complete Description of Proposal".

Existing would be a correction to the latitude and/or longitude, a correction to the height, or if filing on an existing structure which has never been studied by the FAA. The reason for the notice shall be included in ITEM #21 "Complete Description of Proposal".

ITEM #4. If Permanent, so indicate. If Temporary, such as a crane or drilling derrick, enter the estimated length of time the temporary structure will be up.

ITEM #5. Enter the date that construction is expected to start and the date that construction should be completed.

ITEM #6. Please indicate the type of structure. DO NOT LEAVE BLANK.

ITEM #7. In the event that obstruction marking and lighting is required, please indicate type desired. If no preference, check "other" and indicate "no preference" DO NOT LEAVE BLANK. NOTE: High Intensity lighting shall be used only for structures over 500' AGL. In the absence of high intensity lighting for structures over 500' AGL, marking is also required.

ITEM #8. If this is an existing tower that has been registered with the FCC, enter the FCC Antenna Structure Registration number here.

ITEM #9 and #10. Latitude and longitude must be geographic coordinates, accurate to within the nearest second or to the nearest hundredth of a second if known. Latitude and longitude derived solely from a hand-held GPS instrument is NOT acceptable. A hand-held GPS is only accurate to within 100 meters (328 feet) 95 percent of the time. This data, when plotted, should match the site depiction submitted under ITEM #20.

ITEM #11. NAD 83 is preferred; however, latitude and longitude may be submitted in NAD 27. Also, in some geographic areas where NAD 27 and NAD 83 are not available other datums may be used. It is important to know which datum is used. DO NOT LEAVE BLANK.

ITEM #12. Enter the name of the nearest city and state to the site. If the structure is or will be in a city, enter the name of that city and state.

ITEM #13. Enter the full name of the nearest public-use (not private-use) airport or heliport or military airport or heliport to the site.

ITEM #14. Enter the distance from the airport or heliport listed in #13 to the structure.

ITEM #15. Enter the direction from the airport or heliport listed in #13 to the structure.

ITEM #16. Enter the site elevation above mean sea level and expressed in whole feet rounded to the nearest foot (e.g. 17'3" rounds to 17', 17'6" rounds to 18"). This data should match the ground contour elevations for site depiction submitted under ITEM #20.

ITEM #17. Enter the total structure height above ground level in whole feet rounded to the next highest foot (e.g. 17'3" rounds to 18"). The total structure height shall include anything mounted on top of the structure, such as antennas, obstruction lights, lightning rods, etc.

ITEM #18. Enter the overall height above mean sea level and expressed in whole feet. This will be the total of ITEM #16 + ITEM #17.

ITEM #19. If an FAA aeronautical study was previously conducted, enter the previous study number.

ITEM #20. Enter the relationship of the structure to roads, airports, prominent terrain, existing structures, etc. Attach an 8-1/2" x 11" non-reduced copy of the appropriate 7.5 minute U.S. Geological Survey (USGS) Quadrangle Map MARKED WITH A PRECISE INDICATION OF THE SITE LOCATION. To obtain maps, contact USGC at 1-800-435-7627 or via internet at "<http://mapping.usgs.gov>". If available, attach a copy of a documented site survey with the surveyor's certification stating the amount of vertical and horizontal accuracy in feet.

ITEM #21.

- For transmitting stations, include maximum effective radiated power (ERP) and all frequencies.
- For antennas, include the type of antenna and center of radiation (Attach the antenna pattern, if available).
- For microwave, include azimuth relative to true north.
- For overhead wires or transmission lines, include size and configuration of wires and their supporting structures (Attach depiction).
- For each pole/support, include coordinates, site elevation, and structure height above ground level or water.
- For buildings, include site orientation, coordinates of each corner, dimensions, and construction materials.
- For alterations, explain the alteration thoroughly.
- For existing structures, thoroughly explain the reason for notifying the FAA (e.g. corrections, no record or previous study, etc.).

Filing this information with the FAA does not relieve the sponsor of this construction or alteration from complying with any other federal, state or local rules or regulations. If you are not sure what other rules or regulations apply to your proposal, contact local/state aviations and zoning authorities.

Paperwork Reduction Work Act Statement: This information is collected to evaluate the effect of proposed construction or alteration on air navigation and is not confidential. Providing this information is mandatory for anyone proposing construction or alteration that meets or exceeds the criteria contained in 14 CFR, part 77. We estimate that the burden of this collection is an average 19 minutes per response. An agency may not conduct or sponsor, and a person is not required to respond to a collection of information unless it displays a currently valid OMB control number. The OMB control number for this collection is 2120-0001. Comments concerning the accuracy of this burden and suggestions for reducing the burden should be directed to the FAA at: 800 Independence Ave. SW, Washington, DC 20591, Attn: Information Collection Clearance Officer, ABA-20



Tower Crane Pre-Planning Checklist

Project/Project# _____ Date: _____

Your Name: _____

Tower Crane Company _____ Erection Crane Company _____

Pad/Foundation Design Engineer _____ Third Party Consultant _____

Tower Crane Buy Out	
☐	Allow 3-6 months planning time
☐	Include at least 1 -2 weeks on schedule for crane pad installation
☐	Is the use of a tower crane essential? (use of a conventional crane is preferred when conditions allow). How will a tower crane affect the rest of the job? Ground cranes will usually require multiple staging / set-up areas along with acceptable ground conditions and swing protection.
☐	Determine type of tower crane that will be needed (free standing, guyed/braced, internal/self climbing), based on: reach, height of building, capacity, logistics of site, location of adjacent structures, allowable room for lay down, power lines, etc. (Note: in CA, if power lines are on site, then auto shutoff "zone protection" is required on the crane)
	• Configuration • Jib length • Height under hook • Maximum capacity at tip: (also max capacity needed for special items such as the buildings, generator, etc.)
☐	Consider "air space" issues, i.e., adjacent building owners, etc. not wanting the crane over their buildings/property – what will happen in the event of over-sail? If necessary, contact adjacent owners and work out issues.
☐	FAA requirements: will they apply? If so, consider having a temporary beacon available during installation phases (i.e., if crane power is not complete, temporary beacon will need to burn throughout the night until power is connected). If FAA requirements apply, allow sufficient time for application and approval.
☐	QC Tech – Estimating should plan on 4-5 visits by the Tech for each job, plan on two days per visit per job (Does a service contract make sense? Note: even with a service contract, the chance of extra charges is high).
☐	Crane company will need to review and sign off on pad design.
☐	Pad design & installation: allow 8-10 weeks planning time prior to construction, figure 1-2 weeks for pad installation.
☐	Determine if independent third party is needed for pad design, or if it will be bought out through the crane company.
☐	Figure a "per day" charge during crane pad installation for the QC Service Tech and the installation crew. There may be rolling fees incorporated if changes must be made or re-work to pad must be performed.
☐	Any/all RFIs should be answered and changes documented by the design engineer and approved by the crane company.
☐	Plan for sufficient concrete set/cure time for pad per design (figure 30 days, unless design states differently).
☐	Get power to crane pad ahead of time, or additional charges for generator may result.
☐	Representative from crane erection company should walk site 1-2 weeks prior to verify underground utilities, vaults, etc. it's our responsibility to have foundations prep'd for crane's bearing pressure, etc. Crane vendor to verify what ground bearing capacity is need for his crane
☐	Ensure sufficient area is available at the end of the job to safely dismantle and stage trucks and crane parts.
Site Preparation	
☐	Approved design by crane company.
☐	Installation is vital – zero tolerance allowed from design (any rework required will result in Ensure lower section of crane with embed anchors, anchor stools are on site (delivery should be scheduled for prior to start) (a week before preferable)additional \$\$'s for QC Tech's, installation crew, etc.)
☐	Base section of crane should be installed prior to concrete pour
☐	Allow sufficient set time for concrete, per design (figure 30 days, unless design states differently).
☐	Power must be on-line (may require separate/individual transformer, need a disconnect box at the crane base -- 480 up to 300amp service).
☐	Determine logistics of deliveries and/or off-loading of parts/components, i.e., flag personnel and/or spotters available, staging of tower sections, laydown for jib, street permits, street closures, properly prepped roadways, reroute non-crane deliveries to a different area, install perimeter/controlled access zone to keep non-authorized personnel out of area (best case scenario – only those involved in crane set up on site that day), etc. (clear area as much as possible, move forms, material, knock down dirt piles, fill in holes, make ground for jib pre-assemble as flat and un-obstructed as possible.)

Tower Crane Pre-Planning Checklist

Installation	
☐	Will the installation crew be provided through the crane company, or by an independent company?
☐	What company will provide the erection crane? "ANY" and "ALL" correspondence with regards to the crane should have a paper trail, no matter how trivial, back anything discussed with an email (trust no-one's word).
☐	What's the history of the installation crew & crane company working together? Have they prepared a method statement, sequence of events. What ever they promise, allow an extra day before crane will be ready for operation. Ensure erection plan is submitted and reviewed by DPR project team prior to start.
☐	Verify rigging is in good shape (do not allow nylon rigging w/o softeners – steel chokers/rigging preferred), and verify rigging crews training/competency. Qualified riggers, signal person, assembly/disassembly director and operators required. Insist that rigging be in excellent condition and not to exceed 75% of capacity. Verify cylinder breaks of concrete pad have been completed.

Inspection	
☐	Thorough inspection while parts are on the ground, prior to install. Best to have inspector present during erection.
☐	Once installed, third party inspection should be performed: Start @ power box (clearly marked, and i.d. disconnect), verify foundations are in good shape and per design (foundation matches crane it was configured/designed for), bolts, structure (visual cracks), slew rings, counter jib, correct # counterweights and in correct slots, check wire rope, pins, crane operation/functions, check gear box for leaks, verify i.d.'d maintenance items are corrected immediately, not put off till end of project (extra charges can add up at the end of the project if maintenance is put off). Verify function of limit switches, mark load limit switches to prevent, verify jam nut. Test all weights and limits.
☐	Every 5-6 weeks, have a more thorough, extensive inspection, i.e., trolley out, check lacing, bolts, pins, welds, hoist rope dead end (verify socket/becket is secure and wire is not frayed) Designate someone to work with the operator for these types of inspections. Have the inspector explain items of importance (there is no formal training on inspections for operators). Verify all tower bolts are re-torqued after 40 hours of operation. Verify 3 rd party inspection is completed after any parts changes (limit switch, load line, new motor, etc), prior to crane being put back into use.

Use	
☐	Daily pre-use inspection required (inspected per manufacturer's requirements, i.e., weekly checks, monthly checks, etc.) Check for operators pencil whipping the inspection.
☐	Verify operator's past experience, don't rely just on CCO card.
☐	Trained signalman/rigger Anyone who will work the crane, i.e. your sub-contractors should have training.

Dismantling	
☐	3-4 weeks prior, have crane company submit a plan for dismantling, to include
☐	Will the dismantling crew be provided through the crane company, or by an independent company?
☐	What company will provide the dismantling crane?
☐	Verify rigging is in good shape (do not allow nylon rigging w/o softeners – steel chokers/rigging preferred), and verify rigging crews training/competency. Qualified riggers, signal person, assembly/disassembly director and operators required.



Helicopter Work Permit

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General Information

Project/Project #:		Date(s) of proposed pick:	
Location of Activity:		Start Date/Time:	
Contractor Name:		End Date/Time:	
Contractor Contact #:			
Description of Activity:			
Compelling Reason:			
☐	Crane of sufficient reach/capacity unavailable		
☐	Severe Economic Impact (shutting down major operational systems, disrupting R&D prototypes etc)		
☐	Helicopter lift is the safest/most logically practical way of performing the activity		
☐	Other:		
Detailed explanation of compelling reason:			

Specific Information

Has a detailed, step by step pre-pick plan and work procedure been created? If yes, attach to Helicopter Permit. If no, complete detailed pre-pick plan before proceeding.	☐	Yes	☐	No
Has the owner been notified and all potential impacts to operations reviewed?	☐	Yes	☐	No
Has insurance requirements been reviewed and sufficient insurance obtained to cover all operations?	☐	Yes	☐	No
Has the FAA guidelines been reviewed and been found feasible for all activities?	☐	Yes	☐	No
Does the project schedule accommodate no other work activities during the time of the scheduled Helicopter pick?	☐	Yes	☐	No

Owner Representative Signature

By signing above, you verify that the above work is infeasible and have been adequately informed as to the risks and benefits using helicopters for moving/picking materials. Further, you verify that the work is authorized as per the procedures and precautions outlined in the detailed, written helicopter pick plan.

Regional Manager Signature

By signing above, you verify that the above work is infeasible and have been adequately informed as to the risks and benefits using helicopters for moving/picking materials. Further, you verify that the work is authorized as per the procedures and precautions outlined in the detailed, written helicopter pick plan.

Regional EHS Manager Signature

Corporate EHS Manager Signature

The attached Helicopter Work Permit and associated backup documentation has been reviewed and approved for the lift in question.



Helicopter Lift Checklist

Archive Document – DO NOT DISCARD

Project/Project#: _____ Date: _____
Contractor Requesting Helicopter Lift: _____ Date of Lift: _____
Contractor Providing Helicopter Lift: _____ Location of Lift: _____

Documentation

- ☐ DPR Safety Manager must be notified when the work has been awarded and it has been determined that lift by helicopter is the safest and most logistically practical way of performing the activity. This notification should happen, at a minimum, 2 months in advance of the activity.
- ☐ All DPR pre-task documentation must be completed
- ☐ All personnel involved with the Helicopter hoisting operation must review the OSHA regulation for helicopter operations. (29 CFR 1926.551 – Subpart N)
- ☐ All personnel shall be properly equipped with all required personal protective equipment. This shall include but will not be limited to: Hardhats with chinstraps, foam gasket eye protection, hearing protection and gloves.
- ☐ The Helicopter service contractor shall conduct a safety briefing, each day, on all applicable FAA guidelines prior to any lift being performed.

Site Preparations

The wind or rotor wash generated by the Helicopter lift operation will be strong. The following preparation and safety precautions must be completed before any lift operations may begin:

- ☐ All loose materials of any type must be tied down or adequately weighted to prevent any materials from becoming airborne. This must be completed at the pick-site and at the set-site.
- ☐ No other trades may work on the project site while the helicopter lifting activities are being performed without prior approval of EHS
- ☐ Only personnel critical to the process will be allowed on site during these activities
 - ☐ All entries to the site will be blocked and manned to assure that no one enters the site once the activities begin
- ☐ Ensure that pick-site has been adequately sized to accommodate the operation. A minimum area of 200' x 200' will be required.
- ☐ Ensure that all items to be picked are properly spaced. A minimum of 20' between items will be required to avoid a potential caught between incident.
- ☐ Only personnel rigging the equipment are allowed at the pick site
- ☐ Only personnel setting the equipment are allowed at the setting site
- ☐ Ensure that all items are oriented as they will be set. This will ensure a proper set time and keep the Helicopter from hovering due to equipment being improperly oriented and/or rigged.
- ☐ Federal regulations prohibit anyone not connected to the rigging or setting of equipment from being under the flight path of the helicopter and in areas where items will be set
- ☐ If the pick and/or setting area is sandy or dusty, the areas should be adequately wet down before and during the lift operations
- ☐ Notify all local law enforcement, EMS Services and Fire Departments have been notified of the Helicopter operations and ensure direct contact with these services on the day(s) of the lifting operations.
- ☐ Notify the local FAA Officials of the helicopter operations
 - ☐ Ensure that all permits and approvals from the FAA have been obtained prior to lifting operations
- ☐ Control all vehicles, workmen and pedestrian traffic at the job site. No access will be permitted on the site by anyone not directly connected to the lifting operations being performed.

Preparing Items:

- ☐ Remove all crating and packaging material from the items and dispose of prior to lift operations commencing
- ☐ Ensure that items or equipment are clear of any overhead obstructions (power lines, phone lines, pipes, roof overhangs)
- ☐ Develop a lift schedule
- ☐ When items or equipment must be "precision" set, the use of guides are recommended

Helicopter Arrival

- ☐ Confirm the ETA for the service provider.
- ☐ Service provider must review the pick-site and set-site for conformance to OSHA and FAA regulations. **WE WILL NOT BEGIN LIFTING ACTIVITIES UNTIL THE SERVICE PROVIDER HAS ISSUED THE ALL CLEAR AND READY TO LIFT.**
- ☐ The service provider must hold a safety briefing with those responsible for attaching, detaching and directing the load into place
- ☐ Service provider must conduct a briefing related to the manner in which the helicopter work will be performed. Lifting may only proceed once a mutual agreement pertaining to the work has been reached. Failure to cooperate with the service provider will result in:
 - ☐ Stoppage of activities until an agreement is reached or,
 - ☐ Cancellation and leave the project site
 - ☐ Everyone associated with the lifting activities must have a copy of the lift schedule

DPR Superintendent/EHS

Date

Contractor Supervisor

Date

Helicopter Operator

Date

Section 06

Demolition

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Demolition

1.0 Introduction

Employees performing demolition work are exposed to many hazardous conditions and materials. The general public, other contractors' employees and property may also be at risk to these hazardous conditions.

All DPR Construction employees and contractors should follow the requirements of this plan during all demolition projects. This plan does not establish specific demolition protocol for your jobsite; these are guidelines. DPR requires that specific demolition plans be established in writing and reviewed with the superintendent supervision/foremen, safety, the crew and all affected contractors prior to demolition commencement. This plan outlines control measures DPR Construction and contractors may implement as part of their existing program to plan for a successful and safe demolition project.

Reference: Demolition CFR 1926 Subpart T. Other applicable Standards, guidelines, and regulations may apply.

Definitions

Asbestos-containing material (ACM): Any material containing more than one percent asbestos

Class I asbestos work: Activities involving the removal of TSI and surfacing ACM and PACM

Class II asbestos work: Activities involving the removal of ACM which is not thermal system insulation or surfacing material. This includes, but is not limited to, the removal of asbestos-containing wallboard, floor tile and sheeting, roofing and siding shingles, and construction mastics.

Class III asbestos work: Repair and maintenance operations, where "ACM", including TSI and surfacing ACM and PACM, is likely to be disturbed

Class IV asbestos work: Maintenance and custodial activities during which employees contact but do not disturb ACM or PACM and activities to clean up dust, waste and debris resulting from Class I, II, and III activities

Concrete or masonry material: A hard stone-like building material made of clay or made by mixing cement or a combination of cement, sand, gravel, broken stone, or other aggregate with water. Examples include brick, clay brick, concrete block, mortar, natural or manufactured stone, floor, wall, or counter top tile, and terra cotta.

Critical barrier: One or more layers of plastic sealed over all openings into a work area or any other similarly placed physical barrier sufficient to prevent airborne asbestos in a work area from migrating to an adjacent area

Decontamination area: An enclosed area adjacent and connected to the regulated area and consisting of an equipment room, shower area, and clean room, which is used for the decontamination of workers, materials, and equipment that are contaminated with asbestos

Dust containment device: A device attached to a power tool such as a pouch, bag, plastic container, or similar attachment which is intended to capture dust generated by the power tool. This device is not intended to be a dust reduction system as defined under this section.

Dust reduction system: Technology that utilizes the application of water or local exhaust ventilation to reduce airborne dust generated by the use of powered tools or equipment. Local exhaust ventilation may include vacuum systems, dust collection systems, and dust exhaust systems.

Glovebag: Not more than a 60 x 60 inch impervious plastic bag-like enclosure affixed around an asbestos-containing material, with glove-like appendages through which material and tools may be handled.

HEPA: HEPA stands for high-efficiency particulate air. A HEPA filter is a type of mechanical air filter; it works by forcing air through a fine mesh that traps harmful particles such as molds, dusts, particulates, etc.

Negative initial exposure assessment: A demonstration by the employer that employee exposure during an operation is expected to be consistently below the PELs.

Powered tools or equipment: Tools or equipment for which the motive force that disrupts concrete or masonry materials is provided by a source other than human energy. Powered tools or equipment include those powered by electrical, combustion, hydraulic, chemical, or pneumatic energy.

Presumed Asbestos Containing Material (PACM): Thermal system insulation and surfacing material found in buildings constructed no later than 1981

Regulated area: An area established by the employer to demarcate areas where Class I, II, and III asbestos work is conducted, and any adjoining area where debris and waste from such asbestos work accumulate; and a work area within which airborne concentrations of asbestos, exceed or there is a reasonable possibility they may exceed the permissible exposure limit

Thermal System Insulation (TSI): ACM applied to pipes, fittings, boilers, breeching, tanks, ducts or other structural components to prevent heat loss or gain

2.0 Pre-Demolition Activities

Planning for demolition is as important as performing the actual work. Therefore, a competent person experienced in all phases of the demolition work to be performed should perform all planning. When preplanning demolition work (complete or partial), consider the following:

- The condition of the structure
- Any items surrounding the structure (e.g., above, next to, below, underground etc.)
- The presence of any hazardous materials (e.g., asbestos, lead, PCB'S, molds, stockpiled chemicals or wastes from prior operations etc.)
- The methods of bringing down the structure
- Dust control and/or mitigation plan (reference Section 2 Chemical and Environmental Management if the potential for employee exposure to Respirable Crystalline Silica exists).
- The routing of construction debris out of the building
- The storage and removal of construction debris
- Any necessary equipment
- The measures to be taken to perform the work safely
- Condition of adjacent structures
- Public protection
- Location of and/or removal or disconnection of utilities

2.1 Pre-Task Planning

As a high hazard activity, demolition will require daily detailed and written pre-task planning per each work crew. Refer to the Pre-Planning Section 17 for additional guidelines and the Pre-task Planning worksheet.

2.2 Hazardous Materials and Engineering Survey

Prior to conducting renovations, demolitions or maintenance, a hazardous materials survey, conducted by an independent third party must be conducted. The completed hazardous building materials survey, with reports and clearance tests should be readily available at the job site.

An engineering survey should be conducted prior to the start of demolition. Only a competent person (qualified person- CA only) should perform an engineering survey of the project. The purpose of the engineering survey is to determine the condition of the framing, floors, ceilings and walls. Adjacent structures should also be surveyed to eliminate the potential for adversely impacting them during demolition. A copy of the engineering survey should be retained with the DPR project management team. Ensure a survey of hazardous materials is completed, e.g., asbestos, lead, mold, etc. A sample demolition survey can be found at the end of this section.

If the structure to be demolished has been damaged by fire, flood, explosion or some other cause, appropriate measures, including bracing and shoring of walls and floors, shall be taken to protect workers, the public and any adjacent structures. Also, any type of hazardous chemical, gas, explosives, flammable material or other similarly dangerous substances identified will be properly protected, stored or removed from the site prior to demolition. If the nature of any substance cannot be determined, it must be analyzed by a proper authority prior to beginning work.

2.3 Utility Identification and Protection

Prior to any demolition work, all existing utility systems should be identified and protected. This includes site underground and overhead utilities, as well as interior utilities (overhead, under floor or slab, in or on the other side of walls etc.).

Special precautions should be taken with any concrete demolition or saw cutting operations, post tensioning cables and/or any affected utilities should be identified and protected prior to these activities commencing. When planning the demolition phase, a copy of the “as-builts” and civil drawings should be obtained illustrating the premises, property boundaries, building, etc. to aid in identifying common utilities such as electrical, gas, water, sewage, fire protection system, and telecommunication and security systems.

Prior to coring and/or cutting activities, the following tasks, at a minimum, should be performed:

- Verify as-builts
- Consult with facilities personnel
- Perform utility locating
- Perform potholing to visually locate applicable utilities
- *In addition, one of the two below should be used*
 - Ground Penetrating Radar (GPR)
 - X-ray

Note: Where excavating will be required, Section 10 Excavation & Trenching requirements must be followed, including the use of soft-dig for all utility locating.

Areas of demolition should be clearly marked with paint or some other standardized method established by the project team and those areas to be demolished clearly communicated to all workers. Marking on walls with lines of paint and instructions delineating start and stop areas or an ‘X’ or the instructions “Demo” or “Save” are examples of suitable marking systems.

3.0 Demolition Activities

3.1 Worker Protection

3.1.1 Access and Egress

Only those stairways, passages, and ladders designated as a means of access and egress should be used. Worker entrances to multi-story buildings being demolished should be completely protected by sidewalk sheds or canopies. In addition:

- At least one stairway, passageway, or ladder should be provided as a means of access and egress at all times
- All stairs, passageways, and ladders should be periodically inspected and kept clear of obstructions
- Means of access and egress should be maintained in a clean and safe condition
- Stairways and passageways that are under demolition should be barricaded and posted with appropriate signage
- Ensure that alternative means of egress are clearly identifiable
- Ensure proper walking/working surfaces are available
- Ladders and scaffolds should be clear of all demolished material

3.1.2 Signs and Lighting

- Signs should be used to identify hazards that workers or the public might otherwise overlook
- Standardized signs are more easily recognized than others
- Areas of demolition should be clearly identified (signage) and properly barricaded
- Include exit/evacuation routes with standard signage

Sufficient light is vital to safe working conditions. Artificial lighting should be provided if there is any question about the ability of employees to work safely under natural lighting conditions. All stairwells and paths of egress and other work areas should be lit with a minimum of 5 candle watt power.

3.1.3 Personal Protective Equipment (PPE)

Hazards associated with the work to be performed should be evaluated prior to start of work and the appropriate PPE provided to address these hazards. Potential hazards include, but are not limited to exposure to dust, chemicals, sharp objects, hazardous chemicals, falling objects, electricity etc. Required PPE consists of, hard hats, safety glasses and substantial leather work boots that reach above the ankle must be worn at all times. Additionally, PPE that should be considered include, but are not limited to:

- Face shield or goggles
- Toe caps/metatarsal guards will be required when performing jack hammering operations or other operations requiring toe protection
- Safety harnesses and retractable lanyards
- Long sleeve shirts or Kevlar sleeves
- Gloves (appropriate for activity or task):
 - leather gloves for wood, general use, etc
 - rubber or vinyl coated gloves for work in wet areas
 - chemical resistant gloves for work involving hazardous chemicals
 - Kevlar gloves for metal stud, glass, sheet metal, duct, etc.
- Kevlar or cut resistant gauntlets
- Hearing protection must also be worn when circumstances warrant/hazardous conditions exist
- Respirators and additional PPE may be required depending upon the hazards present (refer to Respiratory Section 19)
- During concrete demolition, respirators must be worn unless environmental surveying is completed ensuring silica exposure will not exceed PEL
- High Visibility Apparel

Additionally:

- Work pants and shirts should not have cuffs or open pockets during welding or cutting operations as they can trap hot slag
- Jewelry, loose fitting clothing or anything that could become entangled, should not be worn during demolition activities. Long hair should be contained.

3.2 Tools and Equipment

Refer to the Tools & Equipment Section 24 for specific guidelines.

3.3 Core Drilling/Sawcutting

The subcontractor's competent person should be on the job site at all times to monitor coring/cutting. Daily, detailed, written pre-task plans are required, and should be reviewed and signed off by the crew (this plan must include provisions for utility locating, per subsection 2.3 above). When coring/cutting activities are being performed above occupied areas, work must be coordinated with the DPR project team. Coring/cutting operators should be equipped with appropriately rated rubber gloves and boots to insulate the operator in case of electrical contact. All coring/cutting requires full cleanup and water containment as work is being performed, up to and including one spill kit per floor. Additionally, a plan for controlling water should be developed and implemented

for all work. Coring/cutting materials must be removed from the work area and disposed of properly at the end of each shift.

3.4 Electrical Demolition

In addition to completing a detailed demolition plan, a competent person experienced in hazardous energy must be present and actively supervising all demolition work. Prior to the start of any demolition work where the worker may come into contact with energized systems, the Owner's Facilities Department, DPR Project Management and DPR EHS and electrical subcontractor will review the project and identify electrical hazards. Steps should be taken to eliminate all identified and reasonably anticipated electrical hazards. Such steps include:

- Verifying that the circuitry at each piece of equipment has been shut-off, completely de-energized and that the circuitry has been locked out and tagged out. Tags must be labeled with the person's name, location of work, contact # and the date clearly written. *Refer to Hazardous Energy Section 12 for additional guidelines.*
- For all electrical panels, boxes, fixtures, receptacles, conduit, etc. to be removed, the systems and circuits will be protected and all wires should be pulled by the electrical subcontractor. Only after the wires have been pulled from the system should any other demolition of these systems begin.
- All systems that cannot have the wires pulled will be protected from accidental removal, contact or energizing by lock-out/tag-out measures, cover plates or other protective measures, with flagging and/or markings indicating a live system.
- Each junction box cover should be removed and the circuit tested with a volt tester. If the circuit cannot be locked out and tagged-out, it should be clearly marked for identification and each employee informed of the hazard. These areas should be treated differently in as much as the employee must be supervised during the demo of these potentially deadly conditions.
- In the event workers are performing demolition work around live electricity, all live circuits will be clearly identified with flagging and/or markings.
- All safe systems for demolition will have wires pulled and will also be appropriately flagged or marked in such a manner as to not be confused with live systems.
- Written, detailed pre-task planning must be completed by the project management team (DPR and Subcontractor), including the affected workers, prior to the demolition of any electrical systems where wires are present.

3.5 Debris Removal

Establishing methods for safe removal of all debris is critical in maintaining a safe work environment. Debris removal should be part of the written demolition plan. At a minimum, the following guidelines should be incorporated in the plan, as applicable.

3.5.1 Floor Openings

When they are available, existing floor openings, such as elevator and ventilation shafts, should be used for debris removal. If these openings are not available, a qualified engineer should be consulted before cutting floor openings or chutes may be used. Further:

- Any openings cut in a floor for the removal of debris should not be larger than 25% of the aggregate floor area, unless lateral supports are left intact or additional supports are installed (engineering survey should identify post tensioning cables and other hazards)
- Floors weakened or otherwise made unsafe should be shored to carry the original or intended floor load, whichever is greater
- Each entrance to each level with one or more floor openings should be posted with appropriate WARNING signage
- All floor openings which are not in use should be covered with a material capable of withstanding at least 400 lbs or twice the maximum intended load imposed or protected by a guardrail, midrail and toeboard. Covers must be secured and clearly labeled "HOLE; OPENING; OPENING-DO NOT REMOVE, etc."

3.5.2 Chutes

- Chutes should be designed and constructed to eliminate failure due to the impact of materials or debris; this includes the possible use of paddles, baffles and to design the chute to change direction a minimum of every 120 feet
- All chutes should be adequately secured
- Any chute opening, into which workers dump debris, shall be protected by a substantial guardrail approximately 42 inches above the floor or other surface on which the workers stand to dump the material. Any space between the chute and the edge of openings in the floors through which it passes shall be solidly covered over
- Areas into which debris is dropped outside the exterior wall should be adequately protected to prevent persons from entering those areas
- A substantial gate should be installed in each chute, at or near the discharge end and should be closed when not in use
- A competent person should be assigned to control the operation of the gate and to control vehicular traffic near the chute
- Under no circumstance should anyone enter a chute
- Use an enclosed chute whenever materials are dropped 20 feet or more to any exterior point of a building
- When debris is dropped through floor holes without a chute, the area where the material is dropped must be enclosed with a substantial barricade at least 42 inches high and not less than 6 feet back from the projected edges of the opening above
- Post warning signs at each level
- Disposal chutes positioned $>45^\circ$ from the horizontal must be entirely enclosed

3.6 Fire Protection

The potential for serious fires at demolition sites is great due to the accumulation of materials and debris and the presence of many possible sources of ignition. Early planning of fire prevention measures and providing protection equipment are vitally important and should be part of the demolition plan. This subpart deals with general fire protection methods to be used during demolition activities. *More information may be found in the Fire Protection and Prevention Section 11.* The following items should be established, at a minimum, when performing demolition activities:

- A *written* fire plan must be incorporated into the demolition plan; once accepted by DPR project management, the plan should be reviewed with all crew members.
- The written fire plan should include possible ignition sources, key responding personnel, portable fire extinguisher placement, evacuation plan and corrective/abatement measures being taken
- Readily maintain accessible fire-fighting equipment at all times
- Periodic inspection and maintenance of all fire-fighting equipment; defective equipment must be removed immediately from service
- Maintain access from the street to fire hydrants and to outside connections for standpipes, sprinklers or other fire extinguishing equipment
- An ample number of fully charged portable fire extinguishers (suitable for the fire hazards involved) should be provided at convenient, conspicuous, accessible locations near work areas
- Existing standpipes should be maintained in conformity with the progress of demolition in such a manner that they are ready for fire department use
- Roadways between and around combustible storage piles should be at least 20' wide and maintained free of rubbish, equipment and other materials
- In cold weather demo operations, building heat should be maintained to permit sprinklers, hoses, and extinguishers to be maintained on lower floors, or within enclosed areas, without danger of freezing
- Temporary heating equipment when used should be monitored and maintained by a competent person

- An alarm system (e.g., siren, telephone, radio, etc.) should be established in such a way that workers on the site and the local fire department can be alerted in case of an emergency
- Paths of egress should always be maintained as per those detailed in the Public Protection Section 18

3.7 Explosives

The transportation, handling, storage and use of explosives are subject to numerous federal, state and local regulations. These regulations, in addition to manufacturer's recommended procedures, should be consulted and followed through each stage of the blasting operation.

3.7.1 Planning

Prior to the blasting of any structure or portion thereof, a complete written survey should be made by a qualified person of all underground utilities and adjacent structures in addition to the target. When there is the possibility of extensive vibration, seismic or vibration tests must be taken to determine proper safety limits to prevent inadvertent damage.

3.7.2 General Guidelines

- Only trained, qualified and authorized employees are permitted to handle and use explosives
- Smoking, firearms, matches, lighters and other potential sources of ignition or heat sources are prohibited during the handling, transportation and use of explosives
- All explosives should be kept in a locked magazine and inventoried daily
- When blasting is done in confined areas appropriate safeguards, such as blast mats and barriers, should be taken to protect other structures, personnel and materials
- Every reasonable precaution, including but not limited to, visual and audible warning signals, signage and physical barriers should be used to ensure employee safety
- All mobile radio transmitters less than 1000 feet from blasting operations should be turned off and de-energized

3.7.3 Qualified Persons

A blaster is a person who is competent in using and detonating explosives. A blaster must be qualified by reason of training, knowledge and/or experience in the field of transporting, storing, handling and use of explosives. In addition, the blaster should have a working knowledge of state and local regulations pertaining to explosives. Blasters will be required to furnish satisfactory evidence of competency in handling explosives and safely performing the type of blasting requires. Copies should be retained with the job files.

3.7.4 Transportation of Explosives

Only a competent, trained individual who carries all appropriate licenses as it relates to transportation of explosives should be permitted to operate or be in charge of a vehicle containing explosives. This individual should ensure all local, state and federal regulations governing the transportation of explosives are met. Additional guidelines for the transportation of explosives are as follows:

- No passengers are permitted in the vehicle
- The vehicle used must be strong enough to carry the load without difficulty and be in good mechanical condition
- Have tight floors, and any spark producing agent should be covered with wood or other non-sparking material
- The vehicle should be clearly and appropriately marked or placarded on all four sides with the word "Explosives" in red lettering on a white background
- Explosives, blasting agents and blasting supplies should not be transported with other materials
- Blasting caps should not be transported in the same vehicle as explosives (unless allowed by local, state or federal regulations)

- The entire load should be protected from the elements and resistant to fire
- No motor vehicle transporting explosives should be left unattended at any time

3.7.5 Storage of Explosives

Explosives and related materials should be stored and locked in approved facilities at all times. Other requirements include:

- Blasting caps, electrical blasting caps, detonating primers and primed cartridges should not be stored in the same magazine with other explosives or blasting agents
- Smoking or other open flames are not to be permitted within 50 feet of any storage magazine

3.7.6 Loading of Explosives

Procedures for the safe and efficient loading of explosives should be established prior to any loading to begin. These procedures should include, but are not limited to, the following guidelines:

- Tamping should only be done with non-sparking tools such as wood or plastic tamping rods without any exposed metal parts
- Violent tamping is not permitted and primers should never be tamped
- Only holes to be fired during the immediate round of blasting are to be loaded
- Drilling of holes should only occur after all prior charges and holes have been inspected and any unexploded charges fired
- No explosives or blasting agents are to be left unattended at the blast site
- Warning signs indicating the blast area should be posted and maintained at all approaches to the blast area
- No loaded holes should be left unattended or unprotected
- Loading of holes should be conducted in a manner so that employees and other crews are separated and protected from each other as much as practicable

3.7.7 Blasting of Explosives

The following general guidelines apply to the blasting of explosives:

- A code of blasting signals should be posted in a conspicuous location and all affected employees made familiar with the code. Signals should be equivalent to the following:
 - *warning signal* – a 1 minute series of long blasts 5 minutes in length prior to the blast signal
 - *blast signal* – a series of short blasts 1 minute prior to the shot
 - *all clear signal* – a prolonged blast following the inspection of the blast area
- Flag personnel should be stationed on highways and roadways which pass through the blasting area to stop traffic during blasting operations
- Prior to firing the shot, the blaster should make certain that all surplus explosives are safely stowed, all employees, vehicles, and equipment are a safe distance away and all protective measures are in place

3.7.8 Post-Blast Inspection

- Immediately after the blast, the firing line should be disconnected from the blasting machine and short circuited
- Where power switches are used, they should be locked open or in the off position
- Sufficient time should be allowed for smoke, fumes and dust to leave the blast area before returning to inspect the blast area and surrounding rubble
- Inspections of the blast area for unexploded charges and the safety of structures prior to signaling the all clear warning
- All unexploded charges will be addressed and the hazard removed before permitting those employees unrelated to the abatement and inspection operation to return to the blast area
- Only employees necessary to do this work should be permitted in the blast area

4.0 Hazardous Materials

4.1 Introduction

The purpose of this section is to establish a program for the safe identification, abatement and disposal of hazardous materials. This program should be conducted in accordance with 29 CFR 1926 Subpart Z guidelines and all applicable OSHA standards (including CALOSHA). This section applies to all work where an employee has the potential to be occupationally exposed to the hazardous materials listed below. Disturbance, demolition and/or removal of any hazardous material should be performed by qualified personnel under the guidance of third party oversight.

It is DPR's intent that all handling, removal and disposal of hazardous materials be performed by third party agents. ***The best case scenario is the total removal and clearance by the owner/client prior to DPR mobilizing on the project.*** Ultimately, the responsibility for the safe abatement of asbestos falls upon the owner/client and DPR may guide and consult the owner/client in the methods, safe removal and requirements. When DPR takes on the responsibility for managing the abatement process, an independent third party contractor should be hired to create and oversee the process and ensure, at a minimum:

- frequent and regular inspections of jobsite materials and equipment are conducted
- procedures are followed per the contractor's written plan
- setup & maintenance of regulated areas
- integrity of containment or enclosure is maintained
- entry to and exit from enclosure is properly set up and maintained
- ensure employee exposure monitoring is maintained
- work is performed only within containment area(s) or glovebag(s), as applicable
- proper use of protective clothing and respirators
- hygiene facilities and decontamination procedures are properly set up and maintained
- engineering controls are functioning properly and proper work procedures are being utilized
- signage and notification requirements are met
- clearance sampling is conducted at end of abatement

4.2 Asbestos

The presence of asbestos should be determined by a third party performing a hazardous materials survey. As a general guide, assume suspect material contains asbestos until objective tests have determined otherwise. Material typically found on high heat equipment or used in fireproofing may be presumed to contain Asbestos. Locations for asbestos containing material may include insulation for pipes, furnaces, reactors, boilers, sprayed materials on walls for fire protection or sound deadening, and in certain types of floor, wall, ceiling, old fire doors and roofing materials. Demolition or removal of asbestos should be performed by asbestos certified personnel at all times.

It is DPR's intent that all handling, removal and disposal of asbestos be performed by third party agents. Ultimately, the responsibility for the safe abatement of asbestos falls upon the owner/client and DPR may guide and consult the owner/client in the methods, safe removal and requirements. The best case scenario is the total removal and clearance by the owner/client ***prior*** to DPR mobilizing on the project. When DPR takes on the responsibility for managing the abatement process, an independent third party contractor should be hired to create and oversee the process and ensure, at a minimum:

- frequent and regular inspections of jobsite materials and equipment are conducted
- procedures are followed per the contractor's written plan
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- ensure employee exposure monitoring is maintained

- work is performed only within containment area(s) or glovebag(s), as applicable
- proper use of protective clothing and respirators
- hygiene facilities and decontamination procedures are properly set up and maintained
- engineering controls are functioning properly and proper work procedures are being utilized
- signage and notification requirements are met
- clearance sampling is conducted at end of abatement

4.2.1 Permissible Exposure Limits

The sampling, testing and analysis of asbestos should be conducted, at minimum, in accordance with 29 CFR 1926.1101 guidelines. At no times should employees be exposed to an airborne concentration of asbestos in excess of 0.1 fibers per cubic centimeter of air as an eight-hour time weighted average (TWA).

4.2.2 General Requirements

- In the event that ACM's are found and the establishment of a regulated area is required, workers should be notified of the nature of the work being performed, location of ACM's or PACM's, measures taken to ensure that workers are not exposed and where to obtain more information
- Barriers should be erected for containment when airborne concentrations exceed or when there is a reasonable possibility that airborne concentrations may exceed the PEL
- The integrity of the barrier systems should be regularly inspected and documented

4.2.3 Regulated Areas

- The regulated area should be clearly marked and should sufficiently limit access
- Appropriate signage should be in place to demarcate the regulated areas
- Access to regulated areas should be limited to authorized personnel only
- Entrants should wear appropriate respirators as required
- Workers should not eat, drink, smoke or chew tobacco or gum while engaged in these activities. Further, workers should wash their hands and face prior to eating, smoking etc.

4.2.4 Exposure Assessments and Monitoring

When required, monitoring should be performed in accordance with federal, local and state requirements. The following guidelines apply to exposure assessments and monitoring in the workplace:

- Monitoring should accurately determine the airborne concentrations
- The assessment should occur before or at the initiation of activities; the assessment should be performed and completed in a timely manner to ensure that all appropriate controls can be implemented unless a 'negative initial exposure assessment' exists
- Assessments should attempt to reproduce conditions most likely to be above the excursion limit (1.0 fiber/cc of air over a 30-minute sampling time)
- A Negative Exposure Assessment can be utilized in lieu of monitoring if it meets the following criteria:
 - objective data shows that the material containing asbestos or activity involving the material cannot release airborne fibers in concentrations exceeding the TWA given the work conditions having the greatest potential to release fibers, or
 - the employer has properly monitored prior asbestos activities for the PEL and excursion limit within 12 months and has demonstrated that the activities are within a high degree of certainty of not exposing workers; the current activity must closely resemble the current activities in all areas, including but not limited to, the level of employee training, type of material, type of activities, workplace conditions etc., or
 - the results of initial exposure monitoring for the current activities are representative for all future activities and they have demonstrated levels below the PEL and Excursion Limit.

- If a Negative Exposure Assessment cannot be utilized then periodic monitoring may be required. Periodic monitoring should meet the following requirements:
 - class I and class II operations should be performed daily under condition representative of the exposure
 - all other operations outside of class I and class II work should conduct periodic monitoring at sufficient intervals to document the validity of the exposure assessment
 - when all affected employees are equipped with supplied air respirators in the supply demand mode or other positive pressure respirator, air monitoring will not be required; the employees must meet all applicable requirements for control methods for this to be allowable
- When conditions change substantially additional monitoring may be required
- When results show exposures consistently and substantially below the PEL or Excursion Limit then monitoring may be discontinued
- All affected workers should be informed of the results of the monitoring and their exposures

4.2.5 Control Measures

General Provisions

- vacuum cleaners equipped with HEPA filters to collect all debris and dust containing ACM and PACM should be used
- wet methods should be used as practicable to control exposure during handling, mixing, removal, cutting, cleanup etc.
- all wastes and debris contaminated with asbestos should be cleaned up and disposed of in leak tight containers as soon as practicable
- local exhaust ventilation should be equipped with HEPA filter dust collection systems
- enclosures or isolation of practices should be used as practicable
- high speed abrasive wheels or saws should not be used without appropriate controls in place
- compressed air should not be used for cleaning or removal operations
- dry sweeping, shoveling or other dry clean-up should be avoided
- all affected individuals should be appropriately trained

Class I Requirements

- critical barrier systems should be used
- HVAC systems should be isolated using a double thickness of a minimum 6 mil plastic sheeting
- impermeable drop cloths should be placed below all removal operations
- contaminated air should be ventilated away from the breathing zone
- when using Negative Pressure Enclosures (NPE) the following requirements apply:
 - at least 4 air changes per hour should be maintained
 - a minimum of -0.02 column inches of water pressure differential should be maintained
 - a negative pressure should be maintained throughout the period of use of the NPE
 - the NPE should be inspected for leaks prior to each shift
 - electrical circuits within the NPE should be inactivated unless otherwise protected
- When using a glove bag system they should meet all applicable federal, state and local requirements including but not limited to the following:
 - glovebag systems should only be used on straight runs of piping
 - glovebags should be durable, seamless and designed for that purpose
 - glovebags should completely enclose the pipe
 - glovebags are single use only and should be disposed of appropriately
- Water spray systems should meet the following requirements:
 - asbestos should be wetted down using a fine spray

- wetting should not be used when temperatures are below freezing (32° f - 0° c); however, the following protocols must also be implemented:
- removed asbestos should remain wet during all phases of removal and disposal

Class II Requirements

- critical barrier systems should be used
- HVAC systems should be isolated
- impermeable drop cloths should be placed below all removal operations
- resilient sheeting should be cut while wetting the snip point and wetting during delamination
- ripping-up or ripping-out of sheeting should not be performed
- all scraping, cutting or removal should be done using wet methods
- dry sweeping should not be performed
- mechanical chipping should not be used unless additional precautions are taken
- all material should be removed intact as practicable

Class III Requirements

- work should be done using wet methods
- work should be done using local exhaust ventilation as practicable
- when particulates are being generated (i.e. drilling, cutting, sanding, etc) drop cloths should be used and placed below the removal activity in addition to additional protective measures such as barriers, glovebags or other similar means of protection

Class IV Requirements

- vacuum cleaners equipped with HEPA filters to collect all debris and dust containing ACM and PACM should be used
- wet methods should be used as practicable to control exposure during handling, mixing, removal, cutting, cleanup etc.
- all wastes and debris contaminated with asbestos should be cleaned up and disposed of in leak tight containers as soon as practicable

4.2.6 Worker Protection

- Workers should be provided with Tyvek coveralls, gloves, head coverings and booties if the asbestos fiber concentration exceeds the PEL or Excursion limit
- Light, disposable protective clothing should be used; this clothing should never be reused, worn in public areas or taken off the site
- Decontamination areas, clean rooms, and showers should be provided for exposures above the PEL
- Small scale, short duration operations, such as pipe repair, valve replacement, etc. may be exempt from these regulations
- Respiratory protection should be used under the following conditions at a minimum:
 - class I work
 - class II work when ACM is not removed in a substantially intact state
 - class II and III work that is not done using wet methods
 - class II and III work that has not had a negative exposure assessment
 - class III work when TSI or surfacing ACM or PACM is being disturbed
 - class IV work that is performed in regulated areas where respiratory protection is required
 - all other operations where the TWA or excursion limits are exceeded

Refer to the Respiratory Protection Section 19 for the proper selection and usage of respirators.

4.2.7 Recordkeeping

All documentation related to monitoring, test results, clearance of work areas, training, medical surveillance etc. should be maintained for the duration and made available to affected employees and government agencies when requested. These documents are part of the archive and should be filed accordingly.

4.3 Polychlorinated Biphenyls (PCBs)

PCBs are most commonly found on construction sites in transformers and capacitors which are a part of the structure's electrical system. PCB's may also be found within older high output Fluorescent Fixture lamp ballasts. (Fluorescent tubes contain Mercury and must be disposed of per EPA guidelines) The presence of PCBs should be determined in the engineering survey. Check the name plates of each piece of equipment for the insulating fluid. If the label does not indicate what insulating fluid has been used then the equipment should be presumed to contain PCB's, unless otherwise indicated. Some of the more common PCB containing chemicals are as follows:

Askarel	Pryoclor	Hyvol
Aroclor	Santotherm	No-Flamol
Pydraul	Puralene	Saf-T-Kuhl
Therminol	Pyranol	Aroclor
Diachlor	Inerteen	Clorinol
Dykanol	Asbestol	Clorphen
Elemex	Chlorextol	Eucarel

Whenever it is determined that PCBs are present, the equipment should be exempted from the demolition contract, making its removal and disposal the owner's sole responsibility. The PCBs should be safely contained and removed prior to any other work resuming with these materials and DPR should not be listed as the name shown on any disposal manifest. Further, all workers should be notified of the potential hazard. Whenever handling these materials, appropriate gloves, such as neoprene, polyvinyl alcohol, FEP Teflon and Viton fluorocarbon rubber, should be worn. Other gloves such as latex do not prevent exposure.

4.4 Lead

This section applies to all construction activities where an employee may be occupationally exposed to lead. These activities include but are not limited to abatement activities involving lead paints, coatings, sheeting or wall components. It also applies to all sandblasting, cutting and welding of lead coated steel and the handling of lead containing materials, including dust.

4.4.1 Lead-Work Pre-Job Notification (Refer to Cal/OSHA 1532.1, 1926.62)

The employer shall provide written notification to the nearest California Division District Office.

4.4.2 Permissible Exposure Limits

At no times should employees be exposed to lead of concentrations greater than 50 micrograms per cubic meter (50 $\mu\text{g}/\text{m}^3$) averaged over an 8-hour period (TWA). The Action Level for lead is 30 micrograms per cubic meter (30 $\mu\text{g}/\text{m}^3$) of air per 8-hour period.

4.4.3 Exposure Assessment and Monitoring

The presence of lead should be determined in the initial engineering survey. If lead has been determined to be present on the jobsite, that lead should be deemed in excess of the Action Level unless proven otherwise in writing from an authorized and/or licensed entity. Where objective data demonstrates that a particular product or material and activity or operation involving lead cannot result in an exposure to lead at or above the action level, this data may be used in lieu of initial monitoring provided it meets all the other requirements of such an assessment. Depending on the levels present, air monitoring of airborne lead may also be necessary.

4.4.4 Control Measures

Steps to be taken if the lead is in excess of the PEL or Action Level, or is presumed to be in excess include:

- Appropriate respiratory protection – at minimum a $\frac{1}{2}$ mask air purifying or supplied air/in demand respirator with HEPA filter (99.97% effective)
- Appropriate personal protective clothing – coveralls or similar full body clothing, gloves, hats, disposable booties, and eye/face protection. PPE should meet Level C requirements
- Change areas where workers can remove contaminated clothing and change into clean street clothes prior to leaving the work area
- Hand washing facilities
- Biological monitoring, to consist of blood sampling and analysis for lead and zinc levels
- All affected workers should be informed of the results of the monitoring, testing and their exposures
- Applicable training in lead hazards, routes of entry, safety procedures and precautions etc as per 40 CFR 1926.62
- Appropriate signage should be placed in all work areas where lead exposure may exist

4.4.5 Worker Protection

- All surfaces should be maintained as free as practicable of accumulations of lead
- Clean-up of floors should be done by vacuuming or other methods to reduce the likelihood of lead becoming airborne
- Vacuums should be equipped with HEPA filters
- Shoveling, dry sweeping or compressed air should not be used when working with lead
- Good sanitary practices should be employed including hand washing prior to smoking, eating, drinking etc.
- Contaminated clothing should be removed prior to leaving the change area to prevent cross contamination
- Shower areas may also be required
- Eating areas should be separate from work areas
- Personal vehicles should be parked a safe distance from the site to ensure that contaminants are not collected and transported with dust particles
- Pre-employment physicals, including determination of baseline blood-lead levels and the ability to work while wearing proper respiratory protection should occur
- Proper handling procedures, safe work practices and decontamination practices should also be implemented and followed

4.4.6 Recordkeeping

All documentation related to monitoring, test results; training, medical surveillance etc should be maintained for the duration of employment plus 30 years, and made available to affected employees when requested. These documents are part of the archive and should be filed accordingly.

4.5 Dust

Dust control measures are extremely important during all demolition activities to minimize exposure to hazardous materials, silica, bacteria, fungal, viral agents, etc. Reference Chemical and Environmental Management Section 2 if the potential for employee exposure to Respirable Crystalline Silica exists. Additional references for dust control are included in *Infection Control Section 14 and Respiratory Protection Section 19*. In addition to concrete dust, general dust control measures should be implemented. These measure include, but are not limited to:

- Housekeeping must be a priority with debris being removed continuously and in a timely manner
- Sweeping activities should always be performed using floor sweep or other dust controlling agents. Use vacuums with HEPA filters, wet sweeping or use wet sweeping instead of dry sweeping whenever possible

- Negative air pressure machines with HEPA filters should be utilized; these units may need to be operated continuously and their use should be coordinated with the client/owner
- The use of HEPA vacuums for general cleanup is recommended
- “Tacky Mats” should be installed at all exists from the construction area
- Dust collection systems available on many types of dust-generating equipment should be used when available and practicable
- Use local exhaust ventilation to prevent dust from being released into the air
- Watering to minimize dust raised by vehicles, concrete demolition or earth moving etc.
- Use good work practices to minimize exposures and to prevent nearby workers from being exposed, e.g., remove dust from equipment with a water hose rather than compressed air
- Do not eat, drink or use tobacco products in dusty areas
- Wash hands and face before eating, drinking or smoking outside dusty areas
- Park cars where they will not be contaminated with dust
- Change into disposable Tyvek clothes at the site; these clothes should remain on-site at all times until disposed
- Respirators should not be used as a primary means of preventing or minimizing exposures to airborne contaminants; instead effective source controls such as substitution, automation, enclosed systems, local exhaust systems, wet methods, and good work practices should be used

Demolition Forms

Forms

Demolition Pre-Planning Checklist

Site Demolition Utility Decommissioning Schedule



Demolition Pre-Planning Checklist

Archive Document – DO NOT DISCARD

To be completed prior to any demolition work. This is an initial review of required procedures but may also be utilized during the demolition process to ensure all systems are maintained.

Project/Project#:		Date:	
Company:		Time:	
Competent/Qualified Person:			
Signature:			

Note: if any of the below items are marked “no”, an updated Site Specific Demolition Plan must be developed to address each outstanding item.

Engineering Survey/Site Specific Demolition Plan	Yes	No	N/A
Has engineering survey been completed by a competent person? (Note: the contractor performing the demolition is required to provide the engineering survey)	☐	☐	☐
Has the engineer survey addressed the following?			
Structural layout	☐	☐	☐
Condition of framing	☐	☐	☐
Floors	☐	☐	☐
Walls	☐	☐	☐
The possibility of unplanned collapse of any portion of the structure(s)	☐	☐	☐
The existence of other hazards	☐	☐	☐
Has the results of the engineering survey been communicated to all affected individuals?	☐	☐	☐
Does the submitted Demolition Plan adequately cover the hazards that may be encountered during the course of demolition operations?	☐	☐	☐
Is a plan in place for photographic or video documentation of the existing area, as well as work in progress and surrounding areas?	☐	☐	☐

Utilities & Hazardous Materials Preparation	Yes	No	N/A
Have the following been decommissioned?			
Building automation / control	☐	☐	☐
Chilled water supply and return	☐	☐	☐
Compressed air (including HVAC units)	☐	☐	☐
Condensed return	☐	☐	☐
Domestic water	☐	☐	☐
Electrical (including exhaust fans, switchgear, transformers, control panels)	☐	☐	☐
Fire protection	☐	☐	☐
Fuel lines	☐	☐	☐
Heating water supply and return (including pumps, boilers, and chillers)	☐	☐	☐
Natural gas	☐	☐	☐
Propane / butane / LPG	☐	☐	☐
Sanitary sewer	☐	☐	☐
Steam	☐	☐	☐
Storm sewer	☐	☐	☐
Voice / data	☐	☐	☐
Life safety alarms	☐	☐	☐
Fire alarm	☐	☐	☐
Did the owner conduct a survey for hazardous materials and provide a copy? Has it been verified by an independent Licensed Environmental Firm?	☐	☐	☐
Has hazardous materials training been completed for all affected workers?	☐	☐	☐
Are all permits, notices and certificates authorizing demolition work in place?	☐	☐	☐
Has an approved disposal facility been identified? Is there a protocol for disposal established with the owner of the materials?	☐	☐	☐
Did a qualified person review the specifications referring to environmental considerations? Was this review communicated to applicable personnel?	☐	☐	☐



Demolition Pre-Planning Checklist

Archive Document – DO NOT DISCARD

To be completed prior to any demolition work. This is an initial review of required procedures but may also be utilized during the demolition process to ensure all systems are maintained.

Preparatory Operations	Yes	No	N/A
Site specific Demolition Plan complete and all affected individuals trained?	☐	☐	☐
Has the demolition area been properly marked for demolition?	☐	☐	☐
Are there any party walls that need to be addressed?	☐	☐	☐
Are wall ties required?	☐	☐	☐
Are there any structural hazards present?	☐	☐	☐
Will any shoring of walls or floors be required?	☐	☐	☐
Any protection of adjacent structures or property needed?	☐	☐	☐
Any existing damage of adjacent structures or property? If so, document with photos.	☐	☐	☐
Is there a physical separation between the structure that will remain and structures that are being removed?	☐	☐	☐
Any tanks or vaults on the property currently or in the past?	☐	☐	☐
All tanks have been tested and decontaminated?	☐	☐	☐
Walls or floors are secure prior to workers beginning work in areas?	☐	☐	☐
All service lines shut off, capped or otherwise controlled?	☐	☐	☐
All utilities that must be maintained are clearly marked and protected?	☐	☐	☐
Hazardous chemicals or other dangerous substances (lead, asbestos, mold etc.) identified and abated?	☐	☐	☐
Employees and public protected from glass fragments?	☐	☐	☐
Employees protected from wall openings?	☐	☐	☐
Protection in place and maintained around chute openings and/or material drops?	☐	☐	☐
All floor openings that are not used as material drops are adequately covered & secured?	☐	☐	☐
Demolition of exterior walls and floors is sequenced from top down?	☐	☐	☐
Sidewalk sheds, canopies, catch platforms and other protective measures in place?	☐	☐	☐

Stairs, Passageways and Ladders	Yes	No	N/A
All secondary access ways closed and blocked off (except access & egress points) and adequate signage posted?	☐	☐	☐
All stairs, passageways, ladders etc. inspected and maintained?	☐	☐	☐
Appropriate signage placed for exiting, debris removal pathways etc.?	☐	☐	☐
At least one stairway, passageway or ladder maintained at all times?	☐	☐	☐
All stairwells and passageways have adequate lighting?	☐	☐	☐

Chutes	Yes	No	N/A
Material not dropped to exterior without adequate protection?	☐	☐	☐
Chutes used whenever material dropped from a height of 20' or greater?	☐	☐	☐
Material chutes >45 degrees from horizontal are fully enclosed?	☐	☐	☐
A substantial gate installed around chute discharge and controlled by competent person?	☐	☐	☐
Discharge end of chute closed when not in use?	☐	☐	☐
Drop area protected by adequate barricades and signage?	☐	☐	☐
Chute opening protected by substantial guardrail?	☐	☐	☐
A toeboard or stop log at least 4" thick and 6" high installed for mechanical removal?	☐	☐	☐
Chutes designed and constructed to eliminate failure?	☐	☐	☐

Storage and Fire Protection	Yes	No	N/A
Storage of debris and demolition materials does not exceed allowable floor loads?	☐	☐	☐
Fire equipment readily accessible at all times during demolition, including access roads?	☐	☐	☐
Fire equipment regularly inspected and maintained?	☐	☐	☐
All flammable and combustible material properly stored?	☐	☐	☐
Debris removal performed in a timely manner?	☐	☐	☐
Stairways, passageways and egress paths kept free of debris?	☐	☐	☐
Fire watch posted as necessary?	☐	☐	☐

Floor Openings	Yes	No	N/A
Lateral supports in place for holes larger than 25% of aggregate floor area?	☐	☐	☐
Shoring provided for weakened or otherwise unsafe floors?	☐	☐	☐
Openings cut in floor extend the full span of the arch between supports?	☐	☐	☐
Debris and material within 20' of floor opening removed?	☐	☐	☐
Floor is capable of withstanding the load of equipment, debris and materials?	☐	☐	☐

Mechanical Demolition	Yes	No	N/A
Employee access to areas where balling and clamming operations performed restricted?	☐	☐	☐
When pulling over walls or portions of walls, all steel members affected cut free?	☐	☐	☐
A competent person performs a continuing inspection during demolition process?	☐	☐	☐
Demolition ball, crane boom and load line meet necessary requirements?	☐	☐	☐
Equipment is inspected daily before use?	☐	☐	☐
Adequate venting or clearing of equipment exhaust being taken?	☐	☐	☐
Rigging is regularly inspected and all riggers adequately trained?	☐	☐	☐

Worker Protection	Yes	No	N/A
Affected workers provided with respirators and appropriate fit testing as needed?	☐	☐	☐
Dust control measures implemented?	☐	☐	☐
Workers use goggles, Kevlar sleeves, gloves, toe caps, face shields etc. as appropriate?	☐	☐	☐
Affected workers provided with adequate hearing protective devices?	☐	☐	☐
Affected workers provided with adequate fall arrest devices?	☐	☐	☐

Tools & Equipment	Yes	No	N/A
Tools inspected prior to use?	☐	☐	☐
Proper tools used for the job? e.g. crowbars, pry bars, cat's paw, sledge hammers etc.	☐	☐	☐
Jackhammers, compressors and other pneumatic tools properly maintained/inspected?	☐	☐	☐
Concrete cutting operations are adequately protected and cleaned up?	☐	☐	☐
Welding and cutting equipment in good condition and properly stored and maintained?	☐	☐	☐
Locator service used prior to cutting any floors, walls or other area with potential utilities?	☐	☐	☐
Hoses inspected, no leaks present, adequately secured with whip-checks, pins etc?	☐	☐	☐
Cords inspected and protected prior to use?	☐	☐	☐

Hazardous Energy	Yes	No	N/A
All reasonably foreseeable energy sources identified and protected (i.e., lockout/tag-out, drained, isolated, blinded etc.)?	☐	☐	☐
Utilities potholed and/or exposed?	☐	☐	☐
Wires pulled from conduit whenever practicable?	☐	☐	☐
Electrical demolition has all energy sources clearly identified and marked?	☐	☐	☐

Soft Demolition	Yes	No	N/A
Are as-builts and blueprints available for the area to be demoed and have they been reviewed by the applicable work groups? Do they appear accurate, or does additional investigation need to be completed?	☐	☐	☐



Have hazardous building materials survey been completed, with reports and clearance tests available? Have all findings been communicated to workers?	☐	☐	☐
Soft Demolition	Yes	No	N/A
Is the project evacuation plan current and communicated to all workers, with exit signage posted along designated routes, and designated emergency gather area established?	☐	☐	☐
Are all power and data wires safe'd off in all demolition areas (i.e., walls, ceiling systems, floor monument, etc.)?	☐	☐	☐
Are lockout/tagout procedures in place, with signage installed and non-authorized personnel prevented access at breaker panels, electrical rooms, etc.	☐	☐	☐
Have fire sprinkler shut off valves and drain locations been verified and signage posted?	☐	☐	☐
Have waste streams been identified and recycle program been established?	☐	☐	☐
Have separate dumpsters been identified for applicable recycle waste streams?	☐	☐	☐
Have waste dumpster locations been established and clear path of travel verified?	☐	☐	☐
Are there any types of construction equipment, tools and/or materials prohibited on the project or facility?	☐	☐	☐
Are all designated work areas clear and ready for demolition to start?	☐	☐	☐
Are there any items designated for re-use? Has a storage area been i.d.'d?	☐	☐	☐
Has a job specific JHA been developed for the scope of work to be performed incorporating the above information, and has this JHA been shared with demolition crew members?	☐	☐	☐



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Section 07

Elevated Work

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Elevated Work

(Includes Aerial and Scissor Lifts; Ladders; Scaffolds; Stilts; Floor and Wall Openings; Stairways; Hoists and Elevators, etc.)

1.0 Introduction

The purpose of this plan is to establish guidelines for the safe operation and inspection of elevated work surfaces/platforms, and the duty to have fall protection to protect employees from falls when working from these surfaces. This plan includes, but is not limited to the use of aerial and scissor lifts and various types of scaffolding and ladders.

Reference: Stairways and Ladders CFR 1926.1055-1060; Scaffolding CFR 1926.451, Subpart L; Aerial Lifts CFR 1926.453; Mobile Scaffolds (including Scissor Lifts) CFR 1926.452; Fall Protection CFR 1926 Subpart M; Cal/OSHA §3212. Floor Openings, Floor Holes, Skylights and Roofs. Other applicable Standards, guidelines, and regulations may apply.

2.0 Pre-Task Planning

All work activities that expose worker(s) to fall hazards of 6 feet or more, work on/around scaffolding, as well as overhead work requiring the worker to be 6 feet or more above the work platform are activities defined by DPR to be High Hazard and therefore require detailed, written pre-task planning. *Refer to the Pre-Planning Section 17 for additional pre-task planning guidelines.*

3.0 Duty to Have Fall Protection

It is the responsibility of the employer to ensure workers are protected from the hazard of falls whenever work is being completed at heights of six feet (6') or greater measured from the work platform to the bottom of the sole of the foot. All workers potentially exposed to fall hazards must be trained in recognition of those hazards as well as any applicable fall protective systems. The six-foot rule, applies, but is not limited to the following conditions:

- Ladders (refer to Ladder sub-section of this Section)
- Unprotected sides and edges
- Leading edges
- Hoist areas
- Holes
- Formwork and reinforcing steel
- Ramps, runways, and other walkways
- Excavation and trenching
- Dangerous or large pieces of equipment
- Overhand bricklaying and related work
- All roof work
- Precast and tilt-up concrete erection
- Wall openings
- Walking & working surfaces
- Floor openings
- Skylights
- Scaffolding erection/dismantle
- Wood framing
- Steel erection (refer to Steel Erection Section 23 for specific guidelines on connecting and decking activities)
- Any additional fall protection circumstance that may be deemed necessary by DPR

Note: It is the intent of DPR that proper fall protection measures be utilized at all times where a fall hazard of 6' or greater is present. DPR may require the use of an Elevated Work Permit located in the Forms Section at the end of the EHSP, in conjunction with a written, detailed pre-task plan.

4.0 Fall Protection Systems

Where potential fall hazards of 6 feet or more exists, a suitable fall protection system must be provided to protect the worker. Examples of suitable systems include the following:

- Guardrail Systems
- Safety Net Systems
- Personal Fall Arrest Systems
- Positioning Device Systems
- Warning Line Systems (Roofing Only)
- Controlled Access Zone
- Safety Monitoring System
- Covers
- Protection from Falling Objects
- Fall Protection Plan

5.0 Guardrail Systems

A standard guardrail should consist of a top rail, intermediate/mid-rail, toeboard and posts. Components of a guard rail system may consist of 2" X 4" stock, 3/8-inch wire rope with three clamps per OSHA 1926.251. Guard rail components must meet the following minimum requirements:

5.1.1 Top Rail

- 42 inches + 3 inches (CA between 42" and 45") from the upper surface of the rail to the floor, platform, or ramp level
- Secured to withstand a 200-pound, horizontal force with maximum of 3" deflection
- Have a smooth surface throughout its length
- Wire rope top rails must be flagged with high visibility flagging at 6' intervals

5.1.2 Midrail

- 21 inches, or halfway between the top rail and the floor, runway, platform, or ramp
- Secured to withstand a 150-pound, horizontal force with maximum 3" deflection
- The ends of the rail should not overhang the terminal posts except when it does not constitute a projection hazard

5.1.3 Toeboards

- Should have a 4-inch nominal, minimum height
- Securely fastened in place
- No more than 1/4 inch clearance above the floor level
- Secured to withstand a 50-pounds of force

5.1.4 Posts

- Wooden railing posts (verticals) should be made of at least 2-inch by 4-inch stock or its equivalent
- Be spaced so as not to exceed 8 feet on center

5.1.5 Access/Egress Points

- When guardrail systems are used around holes which are used as points of access (such as ladder ways), they should be provided with a gate, or be so offset that a person cannot walk directly into the hole.

6.0 Safety Net Systems

Safety Nets are not to be used unless all other methods have been considered and determined to be infeasible, and approval has been granted in writing from the DPR Corporate EHS Director. Should safety nets be utilized, they will comply with CFR 1926.502 requirements. Below are guidelines for Safety Nets:

- Safety nets should be provided by the subcontractor or tiered subcontractor when work places are more than 25 feet above the ground or other surfaces where the use of ladders, scaffolds, catch platforms, temporary floors, safety lines or safety harnesses are impractical
- When safety net protection is required, operations should not be undertaken until the net is in place and has been thoroughly tested
- Safety nets should extend 8 feet beyond the edge of the work surfaces where workers are exposed and should be installed as close under the work surface as practical; in no case should the safety net be more than 25 feet below the work surface
- Nets should be hung with sufficient clearance to prevent the user's contact with surfaces or structures below; clearances should be determined by impact load testing
- The mesh size of the nets should not exceed 6 inches by 6 inches
- Edge ropes should have a minimum breaking strength of 5,000 pounds
- Forged steel safety hooks or shackles should be used to fasten the net to its supports. Connections between net panels should develop the full strength of the net.
- All nets should meet the requirements of ANSI A10.11-1989

7.0 Personal Fall Arrest Systems

Personal fall arrest systems are designed to control the fall of a worker and minimize the injury once a worker has fallen. Personal fall arrest systems consist of the following components:

- Training
- Full body harness (body wear)
- Shock absorbing lanyard or retractable (connecting device)
- Tie off point (anchorage)
- Rescue Plan

7.1 Specific Requirements

- The use of body belts is not acceptable for fall protection on any DPR project.
- All fall protection equipment must be inspected prior to each use. Documentation of inspections are required per State and Federal OSHA requirements
- Safety harness is the only acceptable means of personal fall arrest system permitted on any DPR project
- Positioning device systems may be used only in conjunction with a suitable personal fall arrest system
- Self-Retracting Lifelines are the preferred method for personal fall arrest
- Each subcontractor and tiered subcontractor is responsible for providing and requiring the use of safety harnesses, lifelines and lanyards when workers are exposed to a fall of 6 feet or greater
- All subcontractors must provide all fall protection equipment at their cost when fall protection is required
- All lanyards must be equipped with double action locking snap hooks
- Length of the potential fall must be calculated prior to selection and use of shock absorbing lanyards used for fall protection

- Shock absorbing lanyards are not to be used in combination with a retractable lanyard
- Fall arrest equipment, positioning devices systems and/or fall restraint systems **MUST** be immediately removed from service when evidence of wear is detected or in the event of a fall or in-service loading to the system
- All safety harnesses, lifelines and lanyards must have a nominal breaking strength of 5,000 lbs
- The anchorage (tie off point) for fall arrest systems must be capable of withstanding a minimum 5,000 lbs tensile strength per worker attached
- Anchorage used for attachment of personal fall arrest equipment should be secured above the point of operation whenever possible
- Fall arrest systems should be selected with consideration to fall distances so that no worker will strike any lower level surface or object, should a fall occur
- All fall protection equipment should be inspected daily/monthly and before each use, with documentation made available upon request that it is in proper working order
- Where perimeter cables or cable systems are to be used as part of a personal fall arrest system, they should be designed to meet all applicable requirements for fall protection (e.g., 5000 lbs per person to be attached)
- Where perimeter or cable systems are used as fall arrest anchorage, systems should be installed following manufacturer's guidelines or stamped, engineered drawings of perimeter or cable systems are required
- Systems should be inspected and maintained on a regular basis

7.2 Rescue Plans

Each employer should provide for prompt rescue of employees in the event of a fall or should assure that employees are able to rescue themselves. Specific plans for rescue of workers should be developed and rehearsed prior to initiating work requiring the use of a personal fall arrest system. Rescue plans and the basic work plan should be submitted to the DPR Project Management and DPR EHS for review and comment.

8.0 Positioning Device Systems

A positioning device system is a fall harness system rigged to allow a worker to be supported on an elevated, vertical surface (i.e., wall or formwork) and work with both hands free, while leaning. This system must be worn in conjunction with a personal fall arrest system.

The following guidelines apply:

- Positioning devices should not be rigged such that a worker cannot free fall more than two (2) feet
- Anchorage points should be capable of withstanding twice the impact of a worker's fall or 3,000 lbs, whichever is greater

9.0 Personal Fall Restraint Systems

A fall restraint system is a fall protection system that prevents the user from falling any distance. The system is comprised of a body harness, along with an anchorage, connectors and other necessary equipment. The other components typically include a lanyard, and may also include a lifeline and other devices. When a fall restraint system is used, the employer must ensure that the lanyard and anchor are arranged so that the employee is not potentially exposed to falling any distance. Where fall restraint systems are used, the following applies:

- Anchorage points should be capable of supporting 4 times the maximum intended load and be secured to a substantial structural member or securely rigged lines
- Restraint protection should be rigged to allow the movement of employees only as far as the sides of the working level or working area

10.0 Warning Line Systems (Roofing Only)

DPR does not recognize the use of warning line systems as an effective means of ensuring the safety of persons at elevated heights; hence, **the use of a warning line system is only allowed when all other means have been demonstrated to be infeasible.** A written fall protection plan must be developed to address the actual work to be performed. The written plan must be submitted and reviewed by DPR project management and DPR EHS prior to start of work. A warning line system is allowed only for roofing activities on low sloped roofs.

Should a roofing activity warning line system be necessary, the following guidelines will apply:

- The warning line should extend along all side of roof work:
 - At least 10' from the edge when mechanical equipment is in use
 - At least 6' from the edge for all other activities
 - Points of access and loading zone areas should be connected to the work area with an access path formed by two warning lines
- Roofing activity warning lines should consist of ropes, wires, chains, and supporting stanchions as follows:
 - Each line should be flagged or otherwise clearly marked at not more than 6-foot intervals with high-visibility material
 - Each line should be rigged and supported in such a way that its lowest point (including sag) is not less than 34 inches from the walking/working surface and its highest point is not more than 39 inches from the walking/working surface
 - Each line should have a minimum tensile strength of 500 lbs.
 - Stanchions must be capable of withstanding, without tipping, a minimum of 16 lbs. of applied force

Note: Mechanical equipment should not be used or stored in areas where workers are unprotected from falling from warning line systems, guard rails system or personal fall arrest system.

11.0 Controlled Access Zones

DPR does not recognize the use of controlled access zones as an effective means of ensuring the safety of persons at elevated heights; hence, **the use of a controlled access zone is only allowed when all other means have been demonstrated to be infeasible.** A written fall protection plan must be developed to address the actual work to be performed. The written plan must be submitted and reviewed by DPR project management and DPR EHS prior to start of work. A controlled access zone is an area in which certain work may take place without the use of guardrail systems, personal fall arrest systems or safety net systems and access to the zone is controlled.

Should a controlled access zone be necessary, the following guidelines will apply:

- The controlled access zone will be defined by a control line or by any other means that restricts access
- The controlled access zone must be clearly identified using appropriate signage, detailing name of company, contact person and contact number
- Control lines will be erected not less than 6 feet nor more than 25 feet from the unprotected or leading edge
- The control line should extend along the entire length of the unprotected or leading edge and should be approximately parallel to the unprotected or leading edge
- The control line should be connected on each side to a standard railing or wall, or securely anchored on each end, and should enclose all workers performing work in the designated area
- Control lines should consist of ropes, wires, tapes, or equivalent materials, and supporting stanchions as follows:
 - Each line should be flagged or otherwise clearly marked at not more than 6-foot intervals with high-visibility material

- Each line should be rigged and supported in such a way that its lowest point (including sag) is not less than 39 inches from the walking/working surface and its highest point is not more than 45 inches from the walking/working surface
- Each line should have a minimum tensile strength of 200 lbs.

12.0 Safety Monitoring Systems

DPR does not recognize the use of safety monitors as an effective means of ensuring the safety of persons at elevated heights; hence, **the use of a safety monitor is only allowed when all other means have been demonstrated to be infeasible.** A written fall protection plan must be developed to address the actual work to be performed. The written plan must be submitted and reviewed by DPR project management prior to start of work.

The employer should designate a competent person to monitor the safety of employees; this person will be clearly identified from other workers. Additionally, the competent person will:

- Be in the immediate work area
- Be trained and competent to recognize fall hazards
- Warn the employee when it appears that the employee is unaware of a fall hazard or is acting in an unsafe manner
- Be within visual sighting distance of the employee at all times and will always be in immediate communication with the employees being monitored
- Have no other responsibilities that will take the monitor's attention from the monitoring function
- The safety monitor should be on the same walking/working surface (same roof level)

13.0 Covers

All floor openings, should be protected by a cover or suitable guard rail system. Where there is a danger of workers or materials falling through floor, roof, perimeter edges or wall openings, such openings should be:

- Covers should be secured in place to prevent accidental removal or displacement
- Covers should be legibly labeled and color coded (i.e., HOLE; OPENING; OPENING-DO NOT REMOVE, etc.)
- Covers should be designed by a qualified person
- Capable of supporting at least 400 lbs. or twice the maximum intended load

14.0 Skylights

Any employee approaching within 6 feet of any skylight shall be protected from falling through the skylight or skylight opening by any one of the following methods:

- Skylight screens installed above the skylight
- Skylight screens installed below the skylight
- Guardrails meeting the requirements of subsection 5.0
- The use of a personal fall protection system
- Covers
- Skylight nets

15.0 Protection from Falling Objects

Where potential hazards from falling objects exist, suitable systems must be provided to protect workers. Materials, equipment, debris, etc. with the potential to fall from an upper level should be stored or staged a minimum of 6 feet from any edge, unless secured from movement. Examples of suitable falling object protection systems include the following:

- Covers
- Toeboards



- Canopies
- Screens

16.0 Removal of Floor or Wall Openings or Guard Rail Systems

The opening of guardrails or the removal of floor or wall protective covers or systems requires written, detailed pre-task planning, to include:

- Alternate means of fall protection required to safely perform the work
- Means and methods to protect other workers in the vicinity of the fall exposure
- Those who remove of guardrails or the removal of floor or wall protective covers or systems are responsible for replacement in compliance with this section

17.0 Fall Protection Plan

DPR does not recognize the need to create a site specific fall protection plan that cannot incorporate conventional fall protection systems. **Fall protection plans and other monitoring systems are not allowed unless all other means have been demonstrated to be infeasible.** Should a company wish to utilize a fall protection plan, a detailed, written fall protection plan must be developed to address the actual work to be performed. This plan must clearly outline the reason conventional fall protection systems would not be feasible. **The written plan must be submitted and reviewed by DPR EHS and DPR project management prior to mobilization.**

18.0 Fall Protection Training

Subcontractors and all tier subcontractors must provide as a minimum, by a competent person, the following training. Documentation of training must be forwarded to DPR upon request. Training must include, at a minimum:

- The nature of the fall hazards in the work area
- The correct procedure for erecting, maintaining, disassembling and inspecting the fall protection systems to be used (the installation of personal fall protection systems cannot in themselves create a fall hazard exposure to the worker installing the system)
- The use and operations of guardrail systems, personal fall arrest systems, safety net systems, warning line systems, safety monitoring systems (refer to sub-section 12.0 on Safety Monitoring of this Section); controlled access zones and any other methods of protection to be used
- The role of each worker in the safety monitoring system (refer to sub-section 12.0 on Safety Monitoring of this Section) when this system is approved for use
- The limitations on the use of mechanical equipment during the performance of roofing work on low-sloped roofs
- The correct procedures for the handling and storage of equipment and materials and the erection of overhead protection
- The role of workers in fall protection plans

19.0 Material Hoists, Personnel Hoists and Elevators

Temporary material hoists, personnel hoists and elevators should be designed, constructed, installed, and maintained in compliance with the manufacturer's instructions and the provisions of applicable statutes and regulations of governing authorities. The design of the hoist or elevator, and support systems, should be approved by a qualified structural engineer, with calculations for capacity and load dynamics available to the project team. Hoist and elevator personnel, by verifiable documentation and training and experience, will be familiar with the function, safety, and maintenance of the equipment.

No hoists or elevators should be used for the movement of materials and personnel until the devices have been inspected and certified by a licensed third party inspector qualified to approve the equipment.

19.1 Operation

- All operators must be trained by an authorized trainer with documentation of the training maintained on the project.
- The hoist/elevator must be inspected by trained personnel, with the documentation readily available, prior to use.
- Operations must be performed within the specifications of the manufacturers recommendations.
- Work should be coordinated to ensure that at least one hoist remains in operation at all times
- At end of day, hoists should be brought down to lowest level and locked to prevent unauthorized operation.
- During maintenance/repair, the operator should vacate hoist cab for the duration of the work to be performed.
- During maintenance/repair the hoist gate should be locked and “out of service” sign installed.

19.2 High Hazard Operations:

Any work performed in or near (within 5’ the path of travel of the hoist) should be considered high hazard work. Other examples of high hazard work include:

- Erection
- Maintenance
- Dismantling
- Jumping

The following elements should be included in the in the management level Job Hazard Analysis (JHA) and as applicable, in the crew’s daily, documented pre-task planning:

- Service interruption/Shutdown requests should be submitted to the DPR project team and DPR EHS
- Shutdowns should be identified ahead of time and communicated during the previous week’s “look ahead”/ coordination meetings
- Written work plan created by work crew (PTP, JHA, MOP, etc.) and reviewed by DPR project team
- Employees are trained, understand work procedure, and maintain documentation of the training
- After maintenance/repair, the requesting contractor is only given custody of the hoist by DPR project team or DPR EHS
- Contractors given custody of the hoist(s) must implement their Lockout/ Tagout procedures, after review by DPR project management

19.3 Emergency Response:

- Fire life safety measures should be in place during the operation of hoists and elevators.
- Fire extinguisher(s) should be made available inside of the cab.
- Hazardous materials and flammable materials shall not be transported in the cab.
- An emergency action plan should be in place, and communicated to job site personnel.
- The operator should be made aware of emergency response protocol, and will have means to evacuate the cab safely during an emergency.

Additional requirement will be found in Section 11 - Hazardous Energy regarding the control of Hazardous Energy (lockout/Tagout)

20.0 Crane-Suspended Work Platforms

The use of crane-suspended work platforms is prohibited. Crane or derrick suspended personnel platforms may not be used unless the erection, use and dismantling of conventional means of reaching the worksite is not possible or would be more hazardous; lifting of personnel is considered a critical lift. In the event that all other conventional means are not possible, a request for the use of crane-suspended work platforms should be submitted in writing to DPR EHS and/or DPR Corporate EHS, a minimum of three (3) weeks prior to the

scheduled work. The plans should be submitted and approved prior to use. The following items, at a minimum, should be included:

- Statement as to why conventional means are not feasible and a list of the options considered infeasible
- Full compliance with the requirements set out in 29 CFR Section 1926.1501
- Description of the crane to be used and the manufacturer's instructions and requirements in the use of the crane to lift, lower and/or suspend personnel on a work platform
- Drawing of the work platform, suspension bridge and other components with computations used in the design
- Documented emergency plan in the event of a crane failure
- Certified third party inspection of the basket to be used and associated components
- An integrated anti-two-block device on the crane at all times
- All persons on the lift must have a 100% fall protection system in place at all times during the lift

Should the use of crane suspended work platforms be approved, the contractor should complete the following:

- Crane Pre-Lift Checklist
- Crane Pre-Task Plan
- Crane Personnel Platform Safety Checklist
- Documentation that all individuals who are to be lifted have been trained in personnel platform lifting protocols per OSHA, state and/or local requirements (and owner requirements, where applicable).

Also, the jobsite must perform a proof test prior to hoisting employees. The platform and rigging should be proof tested to 125% of the platform's posted rated capacity. Inspect the crane, rigging and the personnel platform after the test. *Refer to Fed/OSHA regulations for additional procedures to be followed when using a personnel platform lift.*

21.0 Scaffolding

Makeshift platforms are never to be used. Each task should be carefully planned to assure that appropriate scaffolding is used where required, up to and including non-standard and engineered scaffolding. At a minimum, all scaffolding will conform to the applicable OSHA regulations.

21.1 General Scaffold Requirements

Scaffolds, including two-point suspension swinging scaffolds, shall be designed, built and inspected by a competent person. Scaffolds should be inspected by a competent person prior to start of work/shift each day (*a scaffolding inspection checklist is included in the Forms Section at the end of the EHSP*). At no time may any person, other than the competent person, alter or modify a scaffold for any purpose. Employees working on suspended scaffolds should be protected by personal fall arrest system securely attached to an, approved anchorage point above the level of the work platform. Personal fall arrest systems must be independent from the lines securing the scaffolding. Ladders, boxes, blocks or other items may not be used on a scaffold platform to gain height. Additional requirements include:

- Scaffolds should be provided with all of the proper structural components and be capable of supporting 4 times the maximum intended load, and should be erected on sound, rigid footing capable of carrying the maximum intended load without settling or displacement
- Load limits should be verified and communicated for areas designated for material loading/storage
- Guardrails and toeboards should be installed on all open sides and ends of platforms more than 6 feet above ground or floor
- Planking should be Scaffold Grade or equivalent, as recognized by approved grading rules for the species of wood used; overlap scaffold planking by a minimum
- Each platform on all working levels should be fully planked or decked

- Scaffold planks should extend over end-supports not less than 6 inches nor more than 12 inches
- Scaffold planks should be secured from displacement
- Scaffolding and accessories with defective parts should be immediately replaced or repaired
- Scaffolds should be secured as per local, state and federal regulations
- The front edge of all scaffold platforms should not be more than 14 inches (16 inches CA) from the face of the work (18" for plastering and lathing operations), unless guardrail systems and/or personal fall arrest systems are utilized
- Scaffolding erection crews will be trained by a competent person
- Fall protection is required for employees erecting or dismantling scaffolds, unless determined by the designated competent person to be infeasible or will create a greater hazard; additionally:
 - a written fall protection plan should be developed with consideration to the sequencing of scaffold erection and dismantling and to address the actual work to be performed
 - the written plan must be submitted and reviewed by DPR project management and DPR EHS prior to start of work
- Training should be provided to all individuals working from a scaffold
- For scaffolds greater than 24 feet in height access to the scaffold will consist of:
 - internal access systems or
 - stair tower
 - vertical access / egress ladders should be placed and coordinated with project teams

21.2 Mobile Rolling Scaffolds

The use of mobile rolling scaffolds will be **in accordance with manufacturer's requirements** and local, state and federal regulations. In addition, the following guidelines will apply:

- The rolling scaffold must be inspected and a scaffold inspection form completed prior to use each day
- Rolling scaffolds should be used on solid, flat floor surfaces
- The maximum load limit of the scaffold should be verified prior to use
- Safety rails are required on platforms above 6 feet; if safety rails are not feasible, some other type of fall protection has been identified and will be used
- Platforms must be fully planked with no gaps greater than 1 inch
- Mobile rolling scaffolds, regardless of height, should have their wheels appropriately locked or otherwise secured from incidental movement while work activities are being performed; i.e., all four caster wheels must be equipped with suitable locking devices which should be locked at all times while work is being performed from the scaffold
- Under no circumstances should an employee "self-propel" a mobile scaffold while remaining on it
- All components of the scaffold must be compatible and securely fastened to prevent uplift
- Riding a mobile scaffold may be permitted when ALL conditions below are fully met:
 - the ratio of height to minimum base dimension does not exceed 2:1
 - the floor surface is completely free of pits, holes, obstructions, etc.
 - the floor surface is smooth, flat and within 3° of level
- When outriggers are required (stacking of two sections), they must be installed on both sides of the scaffold, or per manufacturer's requirements
- Stacking of more than 2 sections of rolling scaffold requires additional training

21.3 Mobile Rolling Scaffolds (California only)

Riding on a Self-Propelled Scaffold – One employee may ride on and move a rolling scaffold while on the platform without assistance from others below provided the following conditions are met:

- All of the provisions in subsection (20.2) of this Section should be met
- The scaffold platform should not be more than 4 feet above the floor level;

- The working platform should be no less than 20 inches in width with a maximum 1 inch space between platform planks;
- Wheels or casters of rolling scaffolds should be provided with an effective locking device that is used in accordance with subsection (20.2) of this section; or rolling scaffolds should be provided with an effective device that is used to prevent movement of the scaffold when workers are climbing or working on the scaffold; and,
- The use of power systems such as motor vehicles, add-on motors, or battery powered equipment to propel a rolling scaffold is prohibited
- Training. Employees who ride on rolling scaffolds and employees that assist in moving employees riding on a rolling scaffold should be trained in accordance with the requirements of this Section and with the requirements of the Construction Safety Orders, Section 1509 to recognize the hazards associated with riding on a rolling scaffold

22.0 Aerial Lifts

- Lifts should be inspected each day prior to use to verify they are in safe working condition (*refer to Scissor/Boom Lift Inspection form located in the Forms Section at the end of the EHSP or use manufacturer's inspection guidelines*)
- Only authorized persons should operate an aerial lift, and must be trained on the equipment they will be operating
- Always stand on the floor of the basket; do not sit or climb on the edge of the basket or use planks, ladders, or other devices for a work position
- A personal fall arrest/restraint system should be attached to a designated anchor point on the boom or basket when working from an aerial lift; tying off to an adjacent pole, structure or equipment is not permitted
- Boom and basket load limits specified by the manufacture should not be exceeded
- The brakes should be locked and when outriggers are used, they should be positioned on pads or a solid surface; wheel chocks must be used before using an aerial lift on an incline, provided they can be safely installed
- An aerial lift truck should not be moved when the boom is elevated with personnel in the basket
- Aerial lifts should have both platform (upper) and lower controls, and:
 - upper controls should be in or beside the platform within easy reach of the operator
 - lower controls should provide for overriding the upper controls
 - controls should be plainly marked as to their function
 - lower level controls should not be operated unless permission has been obtained from the employee in the lift, except in case of emergency
- Lifts must be thoroughly inspected to determine if they require two hands or a hand and a foot to operate; any lift that does not meet these conditions must immediately be removed from service and either returned, replaced, or modified to meet this requirement
- A spotter may be needed when there is a potential for operator injury or potential for damage due to physical contact with facility systems or structures, in congested areas; spotters may also be needed when there is a potential for damage to sensitive facility systems or structures
- It will be the responsibility of the subcontractor to ensure all lift activities are safely performed within the parameters and limitations specified in manufacturer's operating manual. Special consideration should be given to adverse weather conditions. DPR project management reserves the right to cease any activity should weather conditions change to affect work performance

23.0 Scissor Lifts

- Scissor lifts should be inspected each day prior to use to determine that they are in safe working condition (*refer to Scissor/Boom Lift Inspection form located in the Forms Section at the end of the EHSP or use manufacturer's inspection guidelines*)

- Only authorized persons should operate a scissor lift, and must be trained on the equipment they will be operating
- Lifts will be operated in accordance with manufacturer's recommendations
- Lift controls should have a double action in order to engage lift or motion controls

Note: A spotter may be needed when there is a potential for operator injury due to physical contact with facility systems or structures and in congested areas. Spotters may also be needed when there is a potential for damage to sensitive facility systems or structures.

24.0 Wood Framing

DPR Construction's Environmental Health and Safety Plan (EHSP) is intended to protect and promote the safety and well-being of all personnel by providing a safe and healthy work environment during all phases of construction. The guidelines discussed below are provided in addition to the EHSP and are intended to work in conjunction with the larger EHSP document.

Wood Framing – Floor Joist Installation and Decking:

- It is the responsibility of the employer to ensure workers are protected from the hazard of falls whenever work is being completed at heights of six feet (6') or greater measured from the work platform to the bottom of the sole of the foot. All workers exposed to fall hazards must be trained in recognition of those hazards as well as any applicable fall protective systems.
- Walking the top plates of framed walls is strictly prohibited unless the worker(s) engaged in this activity are 100% fall protected as required by OSHA regulations, any and all applicable harness / lanyard manufacturer requirements and the DPR EHSP. It is the responsibility of the employer to provide any required personal fall arrest systems and to ensure that any PFAS's used are properly worn and maintained.
- DPR Construction recommends that all wood framing activities that encourage workers to walk on the top plate of wood framed walls, or require workers to lay decking on top of joists, be conducted in accordance with the guidelines provided by OSHA in the "OSHA Guidance Document: Fall Protection in Residential Construction". In this document, OSHA explains and illustrates several methods of executing wood framing work that are feasible and that have been proven to be safe and effective.

25.0 Stairways

- During construction, stairs should be provided on all structures of two or more floors or more than 24 feet in height (California only)
- Permanent steel stairways having hollow pan type treads and landings that are to be used prior to concrete placement should have the pans completely filled to the level of the nosing
- Temporary stairs should have a landing not less than 30 inches wide, in the direction of travel, for every 12 feet of vertical rise; additionally, landings with wooden treads for temporary service should be the full width of the opening
- Stairway placement should follow, as soon as practical, placement of the upper floor deck
- Safe means of access; i.e., stairways, ramps, ladders, etc. should be provided at all personnel points of access where there is a change in elevation of 19" or more (18" in CA where the use of such is in frequently traveled areas)
- All parts of stairways should be free of hazardous projections
- Debris and other loose material should not be allowed to accumulate on stairways
- The construction of stair railings should be similar to that of standard railing:
 - vertical height, however, should not be more than 37 inches or less than 36 inches (no more than 38" or less than 34" – Cal/OSHA) from the top rail to the surface of the tread in line with the face of the riser at the forward edge of the riser

- all handrails should be provided with a clearance of approximately 3 inches between the handrail and any other surfaces or objects

Riser height and tread width should be uniform throughout any flight of stairs

- Flights of stairs having four or more risers or rise more than 30 inches, whichever is less, should be equipped with standard stair railings or handrails as specified below:
 - stairways less than 44 inches wide with one side open must have at least one stair railing on the open side
 - stairways less than 44 inches wide having both sides open must have one stair railing on each side
 - stairways more than 44 inches wide but less than 88 inches wide must have one handrail on each enclosed side and one stair railing on each open side

25.1 Stairway & Ladder Training

Training must be provided for all employees, as necessary, using ladders and stairways. Training must enable each employee to recognize hazards related to ladders and stairways and procedures to be followed to minimize such hazards. Retraining shall be provided as necessary to assure employee competency.

26.0 Ladders

It is the responsibility of the subcontractor to demonstrate to DPR, through means of hazard recognition and safe work practices, that the use of a ladder is the safest means available to perform a particular task. Particular attention should be taken around leading edges and other floor and wall openings where additional fall protection measures may be required. All employees should be properly trained in ladder safety. Documentation of training must be available for review at DPR's request.

26.1 General Ladder Requirements

The following guidelines apply to work on all types of ladders:

- Ladders should be inspected for visible defects prior to use and after any incident that could affect their safe use
- Ladders should have legible safety warning labels and weight rating labels, or be removed from service until replaced
- Ladders should be used in accordance with manufacturer's recommendations
- Ladders should be maintained free of oil, grease and other slipping hazards
- Ladders should not be loaded beyond the maximum intended load for which they were designed nor beyond manufacturer's rated capacity
- Ladders placed in areas such as passage-ways, doorways, or driveways:
 - where they can be displaced by workplace activities or traffic, the ladder should be secured to prevent accidental movement, or
 - a barricade/ signage should be used to keep traffic or activities away from the ladder
- Ladders should not be moved, shifted(walked) or extended while in use
- Only type 1, 1A and 1AA fiberglass portable ladders will be permitted on site; wooden and metal ladders are not permitted

26.2 Step Ladders

If it is determined that the use of a step ladder is the only feasible method available to perform work at elevations greater than six feet and any of the below conditions cannot be met, then alternative means of access (i.e., work platform, scissor lift, scaffold, etc.) are required. For work being performed from a step ladder (A frame) fall protection will not be required at elevations of six feet or greater provided all the conditions listed below are met:

- All manufacturer's recommendations are followed
- All work is performed on a level surface, fully opened

- All work is performed with worker facing the ladder
- Each employee's body stays between vertical rails (i.e., belt buckle rule)
- A grab-hold is available in the event the worker was to lose balance (i.e., uni-strut or structural member) OR a coworker stabilizes the ladder for duration of work.

**Note: At no time may utilities be used as a grab-hold*

26.3 Extension Ladders

Without regard to cost, all employees should be provided with the safest elevated work platform available. Extension ladders should be used to gain access from one elevation to another. In the event an employee must use an extension ladder to perform work at elevations of six feet or more, additional means of fall protection should be required. This additional fall protection includes, but is not limited to:

- When guardrail systems are used around holes which are used as points of access (such as ladder ways), they should be provided with a gate, or be so offset that a person cannot walk directly into the hole
- Ladders should be used at an angle where the horizontal distance from the top support to the foot of the ladder is approximately $\frac{1}{4}$ of the working length of the ladder
- The side rails should extend 36 inches above the landing; when this is not practical, grab rails should be installed
- Ladders should be secured to prevent accidental movement
- The top must be placed with 2 rails supported equally unless the ladder is equipped with a single support attachment
- The worker must face the ladder while ascending or descending and while conducting work activities
- Worker must use at least one hand to grasp the ladder when moving up or down the ladder (i.e., three points of contact)
- Worker must not carry any object or load that could cause the loss of balance or cause a fall
- Rule of thumb: The employee's belt buckle stays between the vertical rails of the ladder

26.4 Manufactured Ladders

Manufactured ladders on the project should comply with the ANSI requirements. Only type 1, 1A or 1AA fiberglass portable ladders will be permitted on site. Wooden and metal ladders are not permitted. All ladders should be used in the manner and for the purposes for which they were designed and constructed.

- Ladders should be inspected prior to use and those with broken or missing rungs, broken or split side rails, or otherwise damaged, should not be used
- All portable ladders should be equipped with non-skid safety feet and should be placed on a stable base
- The access areas at the top and bottom of ladders in use should be kept clear and free of debris
- All ladders in use should be tied, blocked or otherwise secured to prevent accidental displacement
- The top two rungs/steps of any ladder should not be used as a point to work from at any time
- All work performed on any ladder must be conducted facing in towards the center point

26.5 Job-Made Ladders

Job-made ladders should be constructed for intended use. Job-made ladders should be fabricated and used in compliance with OSHA, state or other regulations. *For specific guidelines, refer to the Job Made Ladder Guidelines located in the Forms Section at the end of the EHSP.*

26.6 Stairway & Ladder Training

Training must be provided for all employees, as necessary, using ladders and stairways. Training must enable each employee to recognize hazards related to ladders and stairways and procedures to be followed to minimize such hazards. Retraining shall be provided as necessary to assure employee competency.

27.0 Stilts

DPR considers the use of stilts to be a high hazard activity and reserves the right to prohibit the use of stilts on any project. Prior to start of work, a written, detailed Job Hazard Analysis must be submitted to DPR project management. Upon review, the determination will be made to either to allow or prohibit the use of stilts on the project. Should the use of stilts be allowed, a written, detailed pre-task plan must be completed by crew prior to start of work each day. DPR reserves the right to terminate the use of stilts on a project, based on compliance by subcontractors and their tiered subcontractors with requirements of this section. On projects where the use of stilts is permitted, **(not allowed in California)** the following guidelines will apply:

- Surfaces on which stilts are used should be flat and free of pits, holes and obstructions, such as debris, as well as other slip, trip and/or fall hazards
- The use of personal stilts will not be allowed
- Stilts should be properly maintained and inspected prior to each use
- Any alteration of the original equipment shall be approved by the manufacturer
- Only a stable, elevated area should be used for attachment, adjustment and removal of stilts
- Overall height of stilts should not exceed 36 inches
- Areas where stilts are to be utilized should be clearly designated and identified, including appropriate signage and/or barricades; these areas should be treated as an area of controlled access whenever possible
- Only personnel trained in the above requirements for stilts and their safe use will be permitted to utilize this equipment (*Reference the Stilts Training Verification Form located in the Forms Section at the end of the EHSP*)
- A task specific pre-task plan is available upon request, refer to DPR Toolbox

Elevated Work Forms

Forms

Elevated Work Permit – English
Elevated Work Permit – Spanish
Fall Protection Training Verification – English
Fall Protection Training Verification – Spanish
Scaffold Use Agreement
Scaffold Safety Checklist – English
Scaffold Safety Checklist – Spanish
Rolling Towers/Baker/Perry Scaffolds – English / Spanish
Scaffold User Training Verification – English
Scaffold User Training Verification – Spanish
Suspended Scaffold User Training Verification - English
Suspended Scaffold User Training Verification – Spanish
Scissor Lift/Boom Lift Inspection – English
Scissor Lift/Boom Lift Inspection – Spanish
Scissor-Boom Lift Operator Training Verification – English
Scissor-Boom Lift Operator Training Verification – Spanish
Ladder Training Verification – English
Ladder Training Verification – Spanish
Job-Made Ladder Guidelines – English
Job-Made Ladder Inspection
Job-Made Ladder Components
Job-Made Ladder Guidelines – Spanish
Stilts Training Verification – English
Stilts Training Verification – Spanish
Stilts Pre-Task Plan – English
Stilts Pre-Task Plan – Spanish



Elevated Work Permit

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General Information

Project/Project#:		Date:	
Contractor Performing Work:		Permit Start Date:	
Nature of Work:		Permit End Date:	
Name of Person(s) Performing Elevated Work:			
Competent Person Overseeing Elevated Work:			

This Elevated Work Permit may be required for all temporary operations involving work where the worker may be exposed to a fall distance of 6' or greater. It includes, but is not limited to:

Leading edge work	Work on unprotected and/or non-walkable surfaces
Work through hand or guard rails	Work around floor, roof or wall openings
Receiving material at loading gates where the cable gate or guardrail is down	Work on a deck without a Controlled Decking Zone plan in place
Any other elevated work	

Fall Prevention and Protection Checklist

Complete the following checklist for all steps taken / needed to protect the work area.

Area Preparation		Fall Protection	
☐	All tripping hazards and other housekeeping issues addressed	☐	Appropriate floor protection/covers have been installed where applicable and they are of suitable strength to bear 2 times the maximum intended load
☐	Area has been barricaded to prevent other trades from entering the area	☐	Floor protection/covers are secured to prevent accidental displacement
☐	Other trades in the area have been notified of all work in the area	☐	Ensure that guardrail system is in place where applicable
☐	Workers are trained on all equipment used and equipment is inspected	☐	All workers have personal fall arrest devices, lanyards, etc.
Falling Object Protection		☐	A suitable anchorage point is identified and utilized
☐	Toeboards are in place as applicable	☐	Ensure ladders are secured and used appropriately
☐	Area below the work has been safely barricaded to protect against falling objects, equipment, etc.	☐	Wheels on scaffolds are locked and guardrails on scaffolds are used
☐	Tool lanyards are used as applicable	☐	All personnel within 6' of the leading edge to have appropriate protection
☐	Only tools needed for the work are used near the leading edge	☐	All affected personnel have received fall protection training
☐	Material will be secured and/or placed in appropriate containment	Other Precautions	

Contractor Supervisor Signature		Date	
By signing above, you verify that the above location has been examined and that the above precautions have been taken			
DPR Supervisor Signature		Date	
By signing above, you verify that the above location has been examined and that the above precautions have been taken			

Post Work Sign-off

Person Performing Elevated Work Signature		Date	
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By signing above, you verify that the above location has been examined and that the above precautions have been taken



Permiso para trabajar en alturas

Documento de archivo – NO DESECHAR

Información general

Proyecto/Proyecto#:		Fecha:	
Contratista que hace el trabajo:		Fecha de inicio del permiso:	
Naturaleza del trabajo:		Fecha de vencimiento	
Nombre de la(s) persona(s) que realiza(n) el trabajo en alturas:			
Persona Competente Supervisando Trabajo Elevado:			

Esta autorización para trabajo en alturas puede ser exigida para toda operación temporal donde el trabajador pueda estar expuesto a caídas de distancias de 6 o más pies. Sin limitarse a éstas, incluye lo siguiente:

Trabajo en bordes extremos	Trabajo en superficies sin protección o intransitables
Trabajo a través de pasamanos o barandas de protección	Trabajo alrededor de aperturas en pisos, techos o paredes
Recepción de material en zonas de carga sin puerta de cables ni baranda de protección	Trabajo en pisos sin contar con un Plan para el Control de las Zonas del Piso
Cualquier otro trabajo en Alturas	

Lista de verificación–prevención y protección contra caídas

Complete la siguiente lista para verificar el cumplimiento de toda actividad necesaria para proteger la zona de trabajo.

Preparación de la zona		Protección contra caídas	
☐	Eliminar los peligros de tropiezos y hacer orden y limpieza.	☐	Según corresponda, se han instalado protectores o cubiertas para el piso, con suficiente resistencia para soportar el doble del peso de la carga anticipada.
☐	Colocar barreras en la zona para impedir que otros oficios entren a la zona.	☐	Los protectores o cubiertas para el piso están sujetas para impedir su desplazamiento accidental.
☐	Los demás oficios en la zona han sido informados de todos los trabajos en la zona.	☐	Según corresponda, tomar medidas para instalar el sistema de barandas de protección.
☐	Los trabajadores han sido entrenados en todo el equipo usado, y el equipo ha sido inspeccionado.	☐	Verificar que todos los trabajadores tengan sus sistemas personales de detención de caídas, acolladores, etc.
Protección contra objetos que caen		☐	Verificar que se ha identificado y se usa un punto conveniente de anclaje.
☐	Según corresponda, se han colocado tablas de tope para los pies.	☐	Verificar que las escaleras portátiles estén sujetas y usadas en forma apropiada.
☐	Se han colocado barricadas de seguridad en la zona que está debajo de los trabajos, para protegerla contra objetos, equipos, etc. que caen.	☐	Verificar que las ruedas de los andamios tengan traba de seguridad y que las barandas de seguridad estén en uso.
☐	Según corresponda, se usan acolladores para las herramientas.	☐	Verificar que todo el personal ubicado a 6 pies de distancia del borde exterior tenga protección adecuada.
☐	Sólo aquellas herramientas necesarias para el trabajo se encuentran cerca del borde exterior.	☐	Verificar que todo el personal afectado haya recibido capacitación en protección contra caídas.
☐	El material estará sujeto y/o colocado en algún contenedor adecuado.	Otras precauciones:	

Firma del supervisor del contratista	Fecha
Al firmar arriba, usted confirma que el lugar indicado arriba ha sido examinado y que se han tomado las precauciones que se indican	
Firma del supervisor de DPR	Fecha

Visto bueno posterior al trabajo

Al firmar arriba, usted confirma que el trabajo indicado arriba ha sido autorizado conforme a los procedimientos y precauciones marcados arriba.

Persona que hace el trabajo en alturas:	Date
---	------

Al firmar arriba, usted confirma que las zonas abandonadas al final del turno/trabajo han sido protegidas apropiadamente contra caídas y objetos que caen.



Fall Protection Training Verification

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Print Name

Date

Jobsite

Company

A suitable fall protection system must be utilized whenever a potential fall hazard exists **of 6 feet or more**. Failure to comply with DPR fall protection requirements constitutes disciplinary actions, up to and including immediate termination.

Harness & Lanyards

☐	Personally owned fall protection equipment may not be used. DPR supplies all fall protection equipment for DPR employees, and it remains the property of DPR Construction
☐	Personal fall arrest systems consist of: full body harness, shock absorbing or retractable lanyards w/locking snap hooks and an anchorage point of at least 5,000 lbs per person attached .
☐	Body belts are not allowed under any circumstance.
☐	The anchorage point for fall arrest equipment should be secured above shoulder height whenever possible and positioned to ensure the worker will not strike any lower level surface or object should a fall occur. Ensure anchorage attachments are adequate – do not wrap lanyard on itself, do not attach two lanyards together, no tie off to sprinklers, utilities, duct work, conduit, scaffolding, etc.
☐	The proper inspection of fall protection equipment has been demonstrated. All fall protection equipment I use must be inspected daily and/or prior to use. If defects are found in the equipment during inspection, it is to be removed from service immediately. The inspection process includes checking, at a minimum, the following items: • frayed edges, broken fibers, pulled stitches, cuts, heat or chemical damage on harness or lanyards • oxidation or distortion of any metal connection devices, cracks, breaks, and rough or sharp edges on d-rings, • unusual wear patterns on any area of harness or lanyard
☐	The proper method for putting on and wearing the harness has been demonstrated.
☐	Fall protection equipment must be stored in a protected area that is clean, dry and free of exposure to chemicals or damage.
☐	Any equipment exposed to a fall must be taken out of service immediately and given to DPR management.
☐	Rescue plan, i.e. how will the individual be rescued, should an event occur, needs to be considered prior to the setup.

Safety Monitoring System

☐	DPR does not recognize the use of safety monitors as an effective means of ensuring the safety of persons at elevated heights. The use of a safety monitor is only allowed when all other means have been demonstrated to be infeasible.
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Covers

☐	Floor covers should be capable of supporting the maximum intended load.
☐	Floor covers should be marked with "HOLE" and secured at all times.
☐	Floor openings larger than 2 inches should be covered

Guardrail Systems

☐	A top rail should be 42 inches ($\pm 3"$) and capable of withstand a 200 lb load. If wire rope is used it should be flagged with high visibility material at 6 foot intervals
☐	A midrail should be 21 inches ($\pm 3"$) and capable of withstanding a 150 lb load.
☐	A toeboard should be 4 inches in height, securely fastened and have no more than $\frac{1}{4}$ inch gap between the floor level and toeboard. It must be capable of withstanding a 50 lb load

This document hereby certifies that I have been instructed on the items listed above and I had an opportunity to ask questions. I fully understand the DPR fall protection requirements.

Signature of employee	Date
-----------------------	------

Signature of person conducting verification review	Date
--	------

A copy of this form must be faxed to your Regional Safety Professional PRIOR TO START OF WORK.



Verificación de protección de caídas

Documento archivos - NO SE DESCARTEN POR FAVOR

Nombre	Fecha
Proyecto/Proyecto#	Compania

Se debe utilizar un sistema de protección de caídas adecuado siempre que exista el peligro potencial de que se produzca una caída desde **6 pies o más** de altura. De no cumplirse los requisitos de protección de caídas de DPR se tomarán medidas disciplinarias, que pueden incluso llegar al cese inmediato en el trabajo.

Arnés y cabos de sujeción	
☐	No se pueden usar equipos de protección de caídas propios. DPR suministra todos los equipos de protección de caídas, y permanecen en propiedad de DPR Construction
☐	Los sistemas personales de prevención de caídas constan de lo siguiente: arnés para el cuerpo completo, cabos de sujeción replegables o que absorben los impactos con ganchos de traba de resorte y un punto de anclaje de al menos 5.000 lb por persona sujeta .
☐	No se permiten cinturones para el cuerpo en ninguna circunstancia.
☐	El punto de anclaje para los equipos de prevención de caídas debe fijarse por encima de la altura del hombro siempre que sea posible y colocarse de forma que el trabajador no se golpee contra ninguna superficie u objeto de un nivel inferior en caso de que se produzca una caída. Asegúrese de que las sujeciones de anclaje sean adecuadas – no se enrolle el cabo de sujeción sobre sí mismo, no ate los cabos de sujeción entre sí, no los ate a aspersores, conducciones de servicio, tuberías, conductos, andamios, etc
☐	Ha tenido lugar la inspección apropiada de los equipos de protección de caídas. Todos los equipos de protección de caídas que use deben ser inspeccionados a diario y antes de usarse. Si se encuentran defectos en los equipos durante la inspección, se deben retirar del servicio inmediatamente. El proceso de inspección incluye la comprobación, como mínimo, de los componentes siguientes:
	• bordes deshilachados, fibras rotas, puntadas en mal estado, cortes, daños por calor o productos químicos en el arnés o cabos de sujeción • oxidación o deformación de cualquier dispositivo de conexión de metal, rajaduras, roturas y bordes ásperos o afilados en los anillos en D, • formas de desgaste inusuales en cualquier área del arnés o del cabo de sujeción
☐	Se ha demostrado el método apropiado de ponerse y llevar puesto el arnés.
☐	Los equipos de protección de caídas deben guardarse en un área protegida que esté limpia, seca y no expuesta a los productos químicos o daños.
☐	Cualquier equipo expuesto a una caída debe retirarse inmediatamente del servicio y entregarse a la gerencia de DPR.
☐	Un plan de rescate: necesita estar listo antes que comienza la tarea- como van a recatar el empleado
Sistema de supervisión de seguridad	
☐	DPR no reconoce el uso de monitores de servicio como medio eficaz de garantizar la seguridad de las personas a alturas elevadas. El uso de un monitor de seguridad solamente se permite cuando se haya demostrado que los demás medios son inviables.
Cubiertas	
☐	El larguero superior debe medir 42 pulgadas (± 3 ") y soportar una carga de 200 lb. Si se usa cable metálico debe señalarse con material de alta visibilidad a intervalos de 6 pies
☐	Las cubiertas de piso deben estar marcadas con la palabra "HOLE" (agujero) y estar fijas en todo momento.
☐	Se deben cubrir las aberturas de los pisos que sean mayores que 2 pulgadas
Sistemas de barandillas	
☐	La tabla de pie debe medir 4 pulgadas de alto, estar bien sujeta y no dejar un espacio de más de $\frac{1}{4}$ pulgada entre el nivel del piso y la tabla de pie. Debe soportar una carga de 50 lb.

Este documento certifica que he recibido instrucciones sobre los puntos mencionados arriba y que he tenido la oportunidad de hacer preguntas. Entiendo completamente los requisitos de empleo de los andamios de DPR.

Firma del empleado	Date
Firma del revisor de verificación	Date

Se debe enviar por fax una copia de este formulario a DPR regional office ANTES DE EMPEZAR A TRABAJAR.



SCAFFOLD USE AGREEMENT

This Scaffold Use Agreement ("Agreement") is made and entered into this _____ day, of 20_____, by and between DPR Construction, a General Partnership, ("DPR"), and its subcontractor, _____ (referred to as "User"). User desires to have access to and use certain scaffolding which has been erected at the following project:

[insert name and address] _____ ("Project")
and DPR agrees to make available to User such scaffolding subject to the provisions in the Agreement set forth below:

- Use of Scaffolding.** User certifies that the User's employees, listed on Attachment 1 to this Agreement, have been properly trained in utilizing such scaffolding in a safe and responsible manner, at a minimum in strict accordance with OSHA 1926 Subpart L or its state law equivalent. User further certifies that it shall not allow any other personnel to use the scaffolding, unless their names have been submitted to DPR, in writing, twenty-four (24) hours in advance of their first use of the scaffolding with certification that such personnel have been properly trained in accordance with this Section and Attachment 1 shall be updated accordingly by User. User acknowledges and agrees that it will have no ownership interest in the scaffolding.
- Acknowledgement of Scaffolding Condition.** User hereby acknowledges that _____ [Insert Competent Person's Name], User's Employee, is a Competent Person pursuant to OSHA regulations or its state equivalent, and has inspected the scaffolding and determined that it is in a safe and satisfactory condition for use by User's employees.
- Inspection and Repair.** Prior to each use of the scaffolding, the User's designated Competent Person shall examine/inspect the scaffolding and take all such action and make minor modifications and repairs as shall be reasonably necessary to assure the scaffolding is in a "useable and safe condition". This inspection is to be done daily utilizing the attached inspection form, a completed copy of which is to be provided to DPR after each inspection. "Useable and safe condition" means that User has assured that the scaffolding is in compliance with all applicable laws and regulations including but not limited to, subpart "L" of the OSHA standard for construction (29 CFR 1926). If the scaffolding is determined to be unsafe or not useable, User and its employees shall refrain from using it until all such defects and deficiencies have been corrected. User hereby releases and absolutely and forever discharges DPR's client on the Project ("Owner"), DPR, its joint venture and joint venture partners, if any, including their respective owners, officers, agents, employees, members, partners, parents, affiliates and subsidiaries, and all other entities which DPR is obligated to indemnify, defend or hold harmless pursuant to the provisions of DPR's agreement with Owner (the "Indemnitees") from any and all claims, damages, liabilities, obligations, costs, expenses and actions of every kind and nature whatsoever, whether now known or unknown, arising from or in connection with its use of the scaffolding, including its design, capacity, description, manufacture, condition, quality, durability, suitability, merchantability and delivery, to the fullest extent permitted by law. User shall promptly report to DPR any problems with the scaffolding that it discovers. The User is accepting the scaffolding "as is" and "where is". Notwithstanding for foregoing, User shall not make any permanent or material modifications to the structural components of the scaffolding without the express written approval of DPR, and shall return it to DPR in the same condition in which it was received by User.
- Loss or Damage.** Should loss or damage occur to the scaffolding while User is granted use thereof under this Agreement, User shall, at its sole cost and expense, replace and repair the scaffolding sufficient to restore the same to a condition at least as good as at the beginning of the term of this Agreement.
- Indemnification and Insurance.** User hereby agrees to defend, indemnify, and hold harmless Owner, DPR, and all other Indemnitees, from any and all claims of any nature made by User or third parties, including User's employees, arising out of User's use of the scaffolding pursuant to the indemnification obligations set forth in the Subcontract between User and DPR, including all documents incorporated by reference into the Subcontract. User also agrees to insure its obligations under this Agreement pursuant to the insurance requirements and obligations set forth in the Subcontract, Attachment A. The Subcontract is incorporated by reference and made part of this Agreement. PRIOR TO ANY USE OF SCAFFOLDING UNDER THIS AGREEMENT, USER SHALL SUBMIT EVIDENCE THAT IT FULLY COMPLIES WITH THE DPR INSURANCE REQUIREMENTS SET FORTH IN THE SUBCONTRACT.
- Use of Scaffolding by Others.** User shall not allow any persons other than its own trained employees, certified in accordance with Sections 1 and 2 above, to use the scaffolding without written authorization from DPR. If User or its employees become aware of persons trespassing upon the scaffold, the User shall immediately notify DPR of such trespass, identifying the parties involved if possible.
- The provisions in this Agreement are in addition to any terms and conditions contained in a DPR Subcontract or other contract documents and nothing in Agreement shall affect the obligations of User to DPR under the terms and conditions of its Subcontract. In the event of a conflict between this Agreement and the DPR Subcontract or other contract documents, the strictest provision imposing the greatest requirement on the User shall control.
- Miscellaneous.** This Agreement shall be interpreted in accordance with the laws of the State in which the Project and scaffolding are located. This Agreement may be modified or amended only by a written instrument duly executed by both parties hereto. This Agreement is binding on the parties and their respective representatives, successors and assigns. No party may assign this Agreement or any rights hereunder without the written consent of the other parties; any unconsented assignment shall be void. User represents that its signatory is duly authorized by User to bind User to this Agreement. This Agreement may be executed in counterparts, each of which shall be deemed an original. Any copy, whether electronic or photocopy, of this Agreement showing the true signature of the parties shall be valid for all purposes as an original executed document.

IN WITNESS, WHEREOF, the parties have executed this Agreement as of the date set forth above.

FOR USER [Insert Name of Subcontractor]

For DPR Construction, a General Partnership

(Signature of User's Authorized Representative)

(Signature of DPR's Authorized Representative)

(Signature of User's Scaffolding Competent Person)



Scaffold Safety Checklist

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Project/Project#:		Date:	
Competent Person:		Company:	

	Item	Mon		Tues		Wed		Thurs		Fri		Sat		Sun	
		Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
1.	Are Sills properly placed and adequately sized?														
2.	Have screw jacks been used to level and plumb scaffold instead of unsafe objects such as concrete blocks?														
3.	Are base plates and/or screw jacks in firm contact with sills and frames?														
4.	Are all scaffold legs braced with braces properly attached?														
5.	Is guard railing in place on all open sides and ends above the 6-foot level?														
6.	Have ladders been provided as a means of access to the scaffold?														
7.	Have free standing towers been guyed or tied so as not to exceed the 4 to 1 base height ratio IAW Subpart "L" of the OSHA standards 29 CFR 1926.45 (c) (1) (ii).														
8.	Are working level platforms fully planked between guard rails with no split planking used?														
9.	Does planking have minimum 12" overlap extended beyond supports and cleated at ends?														
10.	Are toe-boards installed properly?														
11.	Has a tie off analysis been performed, (list details on back)														
12.	Are safety harnesses available for use when needed?														
13.	Have all employees working on scaffold been informed of and trained in safe working practices while working on the scaffold?														
14.	Are out riggers properly installed at 90 degree angles perpendicular to the bldg.?														
15.	Have scaffold components been properly inspected for damage and compatibility?														
16.	The load limit for this scaffold is _____ lbs per square foot. Has this information been communicated to workers utilizing the scaffold?														



Lista de seguridad diaria de andamios

Documento archivos - NO SE DESCARTEN POR FAVOR

Proyecto/Proyecto#:	Fecha:
Persona competente:	Compañía:

		Lun	Mar	Mie'	Jue	Vie	Sab	Sun	
Componente									
1.	¿Están bien colocadas las soleras y son del tamaño adecuado?	Sí	No	Sí	No	Sí	No	Sí	No
2.	¿Se han usado gatos de tornillo para nivelar y aplomar el andamio en vez de objetos inseguros tales como bloques de hormigón?								
3.	¿Están las placas de base y los gatos de tornillo en contacto firme con las soleras y armazones?								
4.	¿Están reforzadas todas las patas de los andamios con riostras bien sujetas?								
5.	¿Están colocadas las barandillas en todos los lados abiertos y extremos ubicados a más de 6 pies de altura?								
6.	¿Se han proporcionado escaleras de mano como medio de acceso al andamio?								
7.	¿Se han amarrado o anclado las torres autónomas de modo que no se exceda la razón 4 a 1 de la altura de la base IAW Subparte “L” de las normas OSHA 29 CFR 1926.45 (c) (1) (ii).								
8.	¿Están completamente cubiertas por tablonces las plataformas horizontales de trabajo entre las barandillas sin usar tablonces divididos?								
9.	¿Tienen los tablonces un solape mínimo de 12” más allá de los soportes y tienen listones en los extremos?								
10.	¿Están instaladas las tablas de pie?								
11.	¿Se ha realizado un análisis de anclajes? (indique los detalles en el reverso)								
12.	¿Se dispone de arneses de seguridad para usarlos cuando sea necesario?								
13.	¿Se ha informado a todos los empleados que trabajan en el andamio y se les ha capacitado en las prácticas de trabajo seguras mientras trabajan en el andamio?								
14.	¿Están bien instaladas las vigas voladizas perpendiculares al edificio?								
15.	¿Se han inspeccionado los componentes de los andamios para ver si están dañados y son compatibles?								
16.	¿El límite del esquiaffo es _____ libras por piel cuadrado. Esta information ha sido transferida a todos los trabajadores del andamios?								



Rolling Towers/Baker/Perry Scaffolds – English/Spanish

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Competent Person:		Company:	

	Mon	Tues	Wed	Thurs	Fri	Sat	Sun	
Item	Yes	No	Yes	No	Yes	No	Yes	No
1. Are outriggers (if required) properly installed on both sides of rolling towers?								
2. Are platforms fully planked with no gaps greater than 1 inch?								
3. Are wheel brakes operable, and have employees been instructed to set brakes while in use?								
4. Are safety rails installed at the 6-foot level, or some other type of fall protection?								
5. Have employees been properly instructed in the safety procedures for using rolling tower, Baker or Perry Scaffolds?								
6. Have caster or wheel stems been pinned or otherwise secured to prevent them from coming separated from the scaffold legs?								
7. The load limit for this scaffold is _____ lbs.								

Proyecto/Proyecto#:		Compañía:	
Inspector:		Fecha:	

	Lun	Mar	Mié	Jue	Vie	Sáb	Dom
Componente							
1.	¿Están bien instaladas las vigas voladizas (si son necesarias) a ambos lados de las torres móviles?						
2.	¿Están completamente cubiertas las plataformas con tablonos sin espacios de separación de más de 1 pulgada?						
3.	¿Funcionan los frenos de las ruedas, y se ha indicado a los empleados que conecten los frenos mientras trabajan?						
4.	¿Se han instalado barandillas de seguridad, o algún otro tipo de protección contra caídas, al nivel de 6 pies?						
5.	¿Se ha indicado bien a los empleados cuáles son los procedimientos de seguridad para usar torres móviles y andamios Baker o Perry?						
6.	¿Se han instalado pasadores en los vástagos de las roldanas o ruedas o se han fijado para impedir que se separen de las patas de los andamios?						
7.	El peso limitado del andamio es _____ libras						



Scaffold User Training Verification

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Company:			

Scaffolding must be designed, built & inspected only by a competent person. This verification is for scaffold users only.

Frame Scaffold	
☐	At no time may I alter or modify a scaffold. That may only be done by the competent person
☐	A scaffold inspection checklist must be completed every day prior to using the scaffold by a competent person
☐	Ladders, boxes, blocks or other items may not be used on a scaffold platform to gain height
☐	Guardrails and toeboards should be installed on all open sides and ends of platforms more than 6 feet above ground or floor. If these are missing or damaged, I will immediately notify my foreman.
☐	If toeboards are not in place on the scaffold, the area below the scaffold must be barricaded prior to use
☐	Each level of the platform I am working on must be fully planked or decked
☐	Scaffold planks should be overlapped 6-12 inches or should be cleated or otherwise secured from movement. The overlap should only occur over frames/end supports
☐	If I find any damaged or defective scaffolding parts, I will immediately notify my foreman
☐	I will verify the maximum intended load for the scaffolding I will be working on prior to use . This means the maximum total weight of all people, equipment and materials that can be allowed on the scaffold. The maximum intended load should be noted on my pre-task plan prior to start of work each day.
☐	Cross braces may not be used to climb up or down a scaffold
☐	Ladders used for access to the scaffold should be secured
☐	My pre-task plan will identify any electrical or fall hazards in the work area
☐	The plan for safely loading materials to the scaffold and where materials will be stored will be discussed with my foreman and noted on the pre-task plan
☐	If guardrails must be removed to load materials to the scaffold, a harness and lanyard secured to an anchorage point of at least 5,000 lbs must be used

Rolling Towers/Baker/Perry Scaffolds	
☐	Perry/Baker/Rolling scaffolds should be used on solid, flat floor surfaces.
☐	The scaffold must be inspected and a scaffold inspection form completed prior to use each day
☐	If any part is missing, defective or incorrectly assembled, the scaffold must be corrected prior to use
☐	The maximum load limit of the scaffold should be verified prior to use
☐	My pre-task plan will identify any electrical or fall hazards in the work area
☐	Safety rails are required on platforms above 6 feet, if safety rails are not feasible, some other type of fall protection has been identified and will be used
☐	Platforms must be fully planked with no gaps greater than 1 inch
☐	Wheel brakes must be operable and must be set on all four wheels while in use
☐	The caster or wheel stems must be pinned or otherwise secured to prevent them from separating from the scaffold legs
☐	When outriggers are required (stacking of two or three sections), they must be installed on both sides of the scaffold.
☐	Under no circumstances may an employee "self-propel" a mobile scaffold while remaining on the platform. Partners may move the lift with their co-worker on the platform if the rider is notified prior to movement and the floor is level w/out holes or depressions.

This document hereby certifies that I have been instructed on the items listed above and I had an opportunity to ask questions. I fully understand the DPR scaffold use requirements.

Employee Print/Sign	Date
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Verification Reviewer Print/Sign	Date
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A copy of this form must be faxed to the DPR regional office PRIOR TO START OF WORK.



Verificación de la capacitación de usuarios de los andamios

Documento archivos - NO SE DESCARTEN POR FAVOR

Proyecto/Proyecto#:		Fecha:	
Compania:			

Solamente una persona competente debe diseñar, fabricar e inspeccionar los andamios. Esta verificación es para usuarios de andamios solamente.

Andamios de armazón	
☐	En ningún momento debo alterar o modificar un andamio. Esto sólo puede hacerlo una persona competente
☐	Se debe completar una lista de comprobación de inspección de andamios todos los días antes de que el andamio sea utilizado por una persona competente
☐	No se pueden usar escaleras de mano, cajas, bloques ni otros artículos en la plataforma de un andamio para ganar altura
☐	Deben instalarse barandillas y tableros de pie en todos los lados abiertos y extremos de las plataformas a más de 6 pies por encima del terreno o piso. Si faltan o están dañados, notificaré inmediatamente a mi capataz.
☐	Si los tableros de pie no están colocados en el andamio, el área debajo del andamio debe estar cerrada por barricadas antes de usarse.
☐	Cada nivel de plataforma en el que trabajo debe estar completamente cubierto por tablonos o cubiertas.
☐	Los tablonos del andamio deben superponerse 6-12 pulgadas o deben tener tablillas clavadas o estar fijos para evitar su movimiento. La superposición debe tener lugar solamente sobre armazones/soportes de extremo
☐	Si encuentro piezas de andamio dañadas o defectuosas, lo notificaré inmediatamente a mi capataz
☐	Verificaré la máxima carga prevista para el andamio en el que vaya a trabajar antes de usarlo . Esto significa el peso total máximo de todas las personas, equipos y materiales que pueden permitirse en el andamio. La máxima carga prevista debe anotarse en mi plan de actividades anteriores a la tarea antes de empezar el trabajo de cada día.
☐	No se pueden usar las riostras para subirse o bajarse de un andamio
☐	Se deben fijar las escaleras de mano usadas para acceder al andamio
☐	Mi plan de actividades anteriores a la tarea identificará cualquier peligro eléctrico o de caída en el área de trabajo
☐	Hablaré con el capataz del plan para cargar materiales con seguridad en el andamio y sobre el lugar en que se guardarán los materiales y se anotará en el plan de actividades anteriores a la tarea
☐	Si se deben quitar las barandillas para cargar materiales en el andamio, se debe usar un arnés y un cabo de sujeción sujeto a un punto de anclaje de al menos 5,000 lb

Andamios de torres rodantes/Baker/Perry	
☐	Los andamios rodantes/Perry/Baker deben usarse sobre superficies sólidas y planas
☐	El andamio debe inspeccionarse y se debe completar un formulario de inspección de andamios antes de usarlo cada día
☐	Si falta cualquier pieza, es defectuosa o está mal montada, el andamio debe corregirse antes de usarse
☐	El límite máximo de carga del andamio debe verificarse antes de usarse
☐	Mi plan de actividades anteriores a la tarea identificará cualquier peligro eléctrico o de caída en el área de trabajo
☐	Se necesitan largueros de seguridad en las plataformas por encima de 6 pies. Si no son viables los largueros de seguridad, se habrá identificado otro tipo de protección contra caídas y se utilizará.
☐	Las plataformas deben estar completamente cubiertas de tablonos sin distancias de separación mayores que 1 pulgada
☐	Los frenos de las ruedas tienen que funcionar y deben fijarse en las cuatro ruedas mientras se usen
☐	Los vástagos de las ruedas deben sujetarse por medio de pasadores o fijarse para impedir que se separen de las patas del andamio
☐	Cuando se necesiten estabilizadores (apilamiento de dos o tres secciones), se deben instalar a ambos lados del andamio
☐	En ninguna circunstancia puede "autopropulsarse" un empleado sobre un andamio móvil mientras esté sobre la plataforma. Puede mover el elevador con su compañero de trabajo en la plataforma si se lo notifica al mismo antes del movimiento y el piso está horizontal sin agujeros o concavidades.

Este documento certifica que he recibido instrucciones sobre los puntos indicados arriba y que he tenido la oportunidad de hacer preguntas. Entiendo completamente los requisitos de empleo de los andamios de DPR.

Firma del empleado	Fecha
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Firma del revisor de verificación	Fecha
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Se debe enviar por fax una copia de este formulario DPR regional office ANTES DE EMPEZAR A TRABAJAR.



Suspended Scaffold User Training Verification

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Company:			

Scaffolding should be designed by a qualified person; and built and inspected only by a competent person. This verification is for suspended (swing stage) scaffold **users** only.

Capacity:

- The weight of the scaffold and its occupants (including tools & equipment) should be supported by both the structure to which it is attached and by each of the scaffold components that make up the anchorage system.
- Suspended scaffold and its components should be capable of supporting at least 6 times the maximum intended load or two times the stall load, whichever is greater. The stall load is that load at which the scaffold will stall or power will automatically shut off.
- The rated capacity of the scaffold must be verified prior to use.

Outrigger beams:

- Outrigger beams should be restrained to prevent movement and stabilized.
- Outrigger beams should be placed perpendicular to their bearing support, i.e., face of building or structure.
- Should be provided with stop bolts or shackles at both ends, securely fastened together, installed with all bearing supports perpendicular to the beam center line and attached to the scaffold rope by shackle or clevis placed directly over the stirrup.

Counterweights:

- Those used to balance adjustable suspension scaffolds must be capable of resisting:
 - o at least 4 times the tipping moment imposed when operating at the **rated load** of the hoist, or:
 - o a minimum of 1 ½ times the tipping moment imposed by the scaffold when it is operating at the **stall load, whichever is greater.**
- Only items specifically designed as counterweights may be used, i.e., masonry units, buckets of sand, etc. cannot be used.
- Should be secured by mechanical means to the outrigger beams to prevent accident displacement.
- Should not be removed from outrigger beams until the unit is dismantled.

Suspension ropes:

- Should be designed for use with brake and hoist mechanisms.
- Repaired wire rope should not be used; should not be spliced or joined together except as authorized by manufacturer.
- Load ends should be equipped with proper sized thimbles and secured by eye splicing.
- Ropes should be inspected prior to use, and replaced if defects are noted.

Hoists:

- The stall load should not exceed three times the rated load.
- When winding drum hoists are used, there should be a minimum of four wraps around the drum when the scaffold is at the lowest point of travel.
- Gasoline powered hoists may not be used.
- Gears and brakes must be enclosed.
- In addition to the normal operating brake, a secondary automatic braking device must be utilized.

Fall Protection

- Both a guard rail system and personal fall arrest should be utilized.
- Personnel should be trained in fall protection, to include equipment being used for the particular set up.
- Personal fall arrest systems should have a fixed anchorage, independent of the scaffold and be protected from abrasion or sharp edges.



Suspended Scaffold User Training Verification

Archive Document – DO NOT DISCARD

- Guard rails should be installed on all open sides & ends, capable of withstanding 200 lbs force (top rail) and between 38 and 45 inches in height. Midrails should be capable of withstanding 150 lbs of force.

Falling Object Protection

- When the danger of tools, material or other equipment falling exists the following should apply:
 - o The area below should be barricaded and workers not permitted to enter; or,
 - o Toeboards installed; or,
 - o Objects secured; or,
 - o Canopies or other devices such as catch platforms should be used.

Platform

- Platforms should be fully decked and secured, at least 18 inches wide, and no longer in length than approved by the manufacturer.
- Platforms should be at least 18 inches wide and no greater than 36 inches wide and no longer in length than approved by the manufacturer.
- The distance of the platform should be no more than 14 inches (18 inches for plastering or stucco) from the face of the structure being worked on, or guardrails must be installed on the front side.
- Toeboards should be installed.

Inspection and Use

- Scaffolding should be inspected by a competent person prior to use.
- Debris should not be permitted to accumulate on the platform.
- Suspended scaffolding should be secured to prevent swaying, as determined necessary by the competent person.
- Scaffolding should not be used during extreme weather (maximum 25 mph winds).
- Scaffolding should be loaded and used in accordance with manufacturer's specifications.
- Any potential electrical hazards should be evaluated and controlled prior to using scaffolding. Any such hazards should be noted on the pre-task plan.
- All portable electric equipment or tools used on suspended scaffolding must be protected by GFCIs.
- Only trained personnel should use suspended scaffolding.

This document hereby certifies that I have been instructed on the items listed above and I had an opportunity to ask questions. I fully understand the DPR scaffold use requirements.

Name/signature of employee

Date

Name/signature of person conducting verification review

Date

A copy of this form should be faxed to the regional safety professional PRIOR TO START OF WORK.

Verificación de la capacitación para usuarios de andamios de suspensión

Documento para archivo – NO DESECHAR

Proyecto/Proyecto#:		Fecha:	
Compania:		Fecha de contratación:	

Los andamios tienen que ser diseñados por una persona calificada. Solamente las personas competentes pueden construir e inspeccionar andamios. Esta verificación se refiere solamente a **usuarios** de andamios suspendidos (guindola).

Capacidad

- El peso del andamio y lo que soporta (personas, herramientas y equipos) debe estar apoyado por la estructura a la cual está conectado y por cada uno de los componentes del andamio que integran el sistema de anclaje.
- El andamio suspendido y sus componentes tienen que ser capaces de soportar por lo menos seis (6) veces la carga máxima diseñada o dos (2) veces la carga de detención, según cuál sea mayor. La carga de detención es la que provoca el paro automático del andamio o el corte automático de la energía eléctrica.
- Es necesario verificar la capacidad nominal del andamio antes de su uso.

Vigas voladizas:

- Las vigas voladizas se deben fijar y estabilizar para evitar su movimiento.
- Hay que colocar las vigas voladizas en posición perpendicular a su soporte, es decir, el frente del edificio o de la estructura.
- Ambos extremos deben tener pernos o grilletes, creando una unión firme, instalados con todos los soportes de carga colocados en posición perpendicular a la línea central de la viga y conectados a la cuerda del andamio mediante un grillete o abrazadera colocados directamente sobre el estribo.

Contrapesos:

- Los contrapesos usados para equilibrar los andamios de suspensión ajustables tienen que ser capaces de resistir:
 - o por lo menos 4 veces el momento de volcado impuesto cuando funciona el andamio con la carga nominal del montacargas, o:
 - o por lo menos 1.5 veces el momento de volcado impuesto cuando el andamio funciona con la carga de paro, cualquiera que sea mayor.
- Se pueden usar solamente artículos específicamente diseñados como contrapesos. Es decir, no se pueden usar ladrillos, cubos de arena, etc.
- Se deben asegurar mediante medios mecánicos a las vigas voladizas para prevenir el desplazamiento de accidentes.
- No deben quitarse de las vigas voladizas hasta el momento de dismantelar la unidad.

Cuerdas de suspensión:

- Tienen que estar diseñadas para usarse con mecanismos de freno y guinche.
- No deben usarse cuerdas de cable reparadas; los cables no deben tener empalmes ni uniones excepto los que autorice el fabricante.
- Los extremos de carga deben estar equipados con casquillos del tamaño adecuado, fijados por empalme de ojetera.
- Hay que inspeccionar las cuerdas antes de su uso y reemplazarlas si se observan defectos.

Montacargas:

- La carga de paro no debe ser superior a tres veces la carga nominal.
- Cuando se usan montacargas con tambor de enrollamiento, el tambor debe tener por lo menos cuatro vueltas cuando el andamio está en el punto más bajo de su carrera.
- No deben usarse montacargas a gasolina.
- Los engranajes y frenos deben estar encerrados dentro de una estructura.
- Además del freno de funcionamiento normal, debe usarse un dispositivo de freno automático secundario.

Protección contra caídas

- Debe usarse un sistema de barandas de seguridad y un sistema personal para impedir caídas.



Verificación de la capacitación para usuarios de andamios de suspensión

Documento para archivo – NO DESECHAR

- Es necesario capacitar al personal en la protección contra caídas, incluyendo el equipo usado para cada instalación.
- Los sistemas personales para impedir caídas deben tener un anclaje fijo, independiente del andamio y protegido contra los bordes filosos o superficies abrasivas.
- Hay que instalar barandas de protección en todos los lados y extremos abiertos, que sean capaces de resistir una fuerza de 200 libras (baranda de arriba) y estar a una altura de 38 pulgadas a 45 pulgadas. Las barandas intermedias deben ser capaces de resistir una fuerza de 150 libras.

Protección contra objetos que caen

- Cuando hay peligro porque podrían caer herramientas, materiales o equipos, debe aplicarse lo siguiente:
 - o La zona que está debajo debe estar cercada y prohibirse la entrada a los trabajadores; o
 - o Deben instalarse tablas de capellada; o
 - o Los objetos deben estar fijados; o
 - o Deben usarse toldos o algún otro dispositivo como plataformas de contención.

Plataformas

- Es necesario entablar completamente las plataformas, con tablas bien fijadas y de por lo menos 18 pulgadas de ancho, y de una longitud que no sea mayor a la aprobada por el fabricante.
- Las plataformas deben tener un ancho de por lo menos 18 pulgadas y no más de 36 pulgadas, y su longitud no debe exceder las recomendaciones del fabricante.
- La plataforma debe estar a una distancia no mayor de 14 pulgadas (18 pulgadas si es para yeso o estuco) de la superficie de trabajo, o hay que instalar barandas de seguridad en el frente.
- Es necesario instalar tablas de capellada.

Inspección y uso

- Una persona competente debe inspeccionar los andamios antes de su uso.
- No debe permitirse la acumulación de desperdicios sobre la plataforma.
- Los andamios de suspensión tienen que estar anclados para evitar el vaivén, en la medida necesaria que determinará la persona competente.
- No se deben usar los andamios cuando hay mal tiempo (no usar si el viento es de más de 25 millas por hora).
- La carga y el uso de los andamios deben ajustarse a las especificaciones del fabricante.
- Debe evaluarse todo potencial peligro por electricidad y controlarse antes de usar los andamios. Dichos peligros deben anotarse en el plan previo al trabajo.
- Todos los equipos o herramientas eléctricos portátiles usados en los andamios de suspensión tienen que estar protegidos con interruptores de circuito con conexión a tierra (GFCI).
- Solamente los trabajadores capacitados deben usar los andamios de suspensión.

Este documento certifica que me han instruido con respecto a los puntos antedichos y que he tenido la oportunidad de hacer preguntas. Entiendo completamente los requisitos de DPR para usar andamios.

Firma del empleado

Fecha

Firma de la persona encargada de la verificación

Fecha

Enviar por fax una copia de este formulario a oficina de regional ANTES DE INICIAR EL TRABAJO.



Scissor Lift/Boom Lift Inspection

Archive Document – DO NOT DISCARD

Project/Project #:		Date:	
Company Name:		Contact Number:	
Type of Lift		Model or Equipment Number:	

Date		Mon	Tues	Wed	Thurs	Fri	Sat	Sun
<i>Initials of person performing inspection:</i>								
Has the operator been instructed in the safe operation of this type of lift?		Y/N	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N
Inspection Item & Description Pass Fail Status		P/F	P/F	P/F	P/F	P/F	P/F	P/F
1.	The manufacturer's operations manual is stored on unit.							
2.	Safety decals are in place and readable.							
3.	All guardrails are sound and in place, including basket chains.							
4.	Work platform & extension slides are clean, dry, & clear of debris.							
5.	Work platform extension slides in and out freely with safety locking pins in place to lock setting on models with extension platforms.							
6.	Inspect for defects such as cracked welds, fuel leaks, hydraulic leaks, damaged control cables or wire harness, etc.							
7.	Tires and wheels are in good condition, with adequate air pressure if pneumatic. Check water level on batteries.							
8.	Emergency lowering function operates							
9.	Lower operating controls successfully over-ride the upper controls. Check hydraulic fluid level.							
10.	Control panel is clean & all buttons/switches are clearly visible (no fire proofing, paint over spray, etc.)							
11.	All switch & mechanical guards are in good condition and properly installed.							
12.	Both upper and lower controls are adequately protected from inadvertent operation							
13.	Operating and emergency controls are in proper working condition -- EMO button or Emergency Stop Device.							
14.	Upper drive control interlock mechanism is functional (i.e., foot pedal, spring lock, or two hand controls).							
15.	All Safety Indicator lights work.							
16.	Drive controls function properly & accurately labeled (up, down, right, left, forward, back).							
17.	Motion alarms are functional.							
18.	Braking devices are operating properly.							

Survey work area for potential hazardous operating conditions prior to use of lift. Ensure all the hazards identified are addressed in pre task planning with sufficient strategies to mitigate the hazards and or risks.

Workplace Assessment	Are Conditions Present ☒						
Floor conditions: Drop offs, holes, uneven surfaces, sloped floors, etc.	☐	☐	☐	☐	☐	☐	☐
Housekeeping: debris, floor obstructions, cords, construction material/ supplies, etc.	☐	☐	☐	☐	☐	☐	☐
Hazardous Energy: Electrical power cables or panels, chemical lines, gas lines, drain lines, utilities, etc.	☐	☐	☐	☐	☐	☐	☐
Overhead obstructions: tight working conditions, adjacent structures, pipe racks, ceiling grids, etc.	☐	☐	☐	☐	☐	☐	☐



Scissor Lift/Boom Lift Inspection

Archive Document – DO NOT DISCARD

Proyecto/Proyecto Numero:		Fecha:	
Compania:		Nombre:	
Tipo de plataformas de trabajo		Modelo o numero del equipo	

Fecha		Lun	Mar	Mie	Jue	Vie	Sab	Dom
Nombre del operador o del Inspector:(Persona haciendo la inspeccion)								
¿Ha sido entrenando el operador a utilizar el equipo seguarmente?		Si/No	Si/No	Si/No	Si/No	Si/No	Si/No	Si/No
Inspection Item & Description Pass Fail Status		P/NP	P/NP	P/NP	P/NP	P/NP	P/NP	P/NP
1.	El manual de operación del fabricante esta en la unidad y en el idioma de el operador.							
2.	Letreros de seguridad estan en su lugar y se pueden leer bien.							
3.	Todos los resguardos de seguridad y la cadena de proteccion estan en buenas condiciones							
4.	La plataforma de trabajo y la extension de la plataforma esta limpia, seca, y no tiene polvo.							
5.	La plataforma y la extension de la plataforma se desliza libremente y tiene todos los pins de seguridad en su lugar.							
6.	Inspeccionar por defectos como, soldadores quebradas o dañados, fugas de aceite, fugas de gasolina, o cables dañados, etc.							
7.	Ruedas y llantas estan en buena condicion, con la adecuada presion de aire, si son de aire.							
8.	¿Valvula de emergencia funciona apropiadamente?							
9.	¿Los controles de abajo funcionan igual que los controles de arriba?							
10.	Todos los controles y botones estan limpios y claramente visibles (no pintura, polvo etc..)							
11.	Todos resguardos estan en buena condicion y debidamente instalados.							
12.	Los controles de arriba y los controles de abajo estan debidamente protegidos de una operacion inadecuada.							
13.	¿Controles de operacion y emergencia estan en buena condicion, y el boton de emergencia funciona?							
14.	Controles de operacion y boton de emergencia funcionan (e.g. el pedal, boton rojo, y los dos controles;)							
15.	Todas las luces de seguridad funcionan adecuadamente							
16.	Todos los controles de operacion funcionan adecuadamente y se pueden leer claramente (arriba, abajo, derecha, izquierda, para enfrente y para atras.)							
17.	La alarma de movimiento funciona adecuadamente.							
18.	Los frenos funcionan adecuadamente.							

Inspeccionar area de trabajo para eliminar peligros potenciales antes de su uso.

Asegurese de que todos los peligros sean identificados en su PRE-TASK PLAN para eliminar los peligros potenciales.

INSPECCION DEL AREA DE TRABAJO	☑ Si encuentra una condicion insegura marque una						
Condiciones del piso: peligro de caidas, hoyos, etc..	☐	☐	☐	☐	☐	☐	☐
La limpieza: polvo, obstrucciones en el piso, extensiones electricas, material de construccion, etc..	☐	☐	☐	☐	☐	☐	☐
Energía peligrosa: Cables de electricidad o cajas de electricidad, lineas quimicas, lineas de gas, etc..	☐	☐	☐	☐	☐	☐	☐
Obstrucciones en el techo: Condiciones de trabajo muy cerradas, pipas, techo, parrilas, etc..	☐	☐	☐	☐	☐	☐	☐



Scissor-Boom Lift Operator Training Verification

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Employee Name:		Company Name:	
Testing Date:		Field Lead:	

Equipment			
Manufacturer(s):			
Model(s):		Capacity:	

Check boxes that are successfully completed for the type of equipment applicable.

Item to be Verified	Scissor Lift	Boom Lift
Pre Use Inspection: Daily, documented DPR inspection required prior to use.	☐	☐
▪ Use DPR inspection form to walk employee through proper process of inspection (see back side of sheet).	☐	☐
▪ Inspection document must remain on vehicle for duration of shift (once the form is completed, turn in to DPR Project Management)	☐	☐
3-point contact getting into Aerial Operational Platform – maintain three points of contact while accessing platform.	☐	☐
▪ Ensure gates/chain is in place	☐	☐
Work Area Survey		
▪ Perform assessment walk of work area, check of area of operation and surrounding work area for potential hazards	☐	☐
Fall Protection Verification		
▪ Demonstrate proper inspection of harness & lanyard	☐	☐
▪ Hand rails are fall protection on scissor lifts unless feet leave floor of working platform.	☐	☐
▪ Harness & lanyard required immediately upon entering working platform on boom lifts.	☐	☐
▪ Anchorage points (structural w/scissor lift, manufacturer's provided anchorage on boom lifts)	☐	☐
Start Up & Maneuvering		
▪ Control panel – legible, ensure operation functions are understood	☐	☐
▪ Backup alarm – verify operational if equipped	☐	☐
▪ Test brake or traction system (to make sure it will hold)	☐	☐
▪ Operator turns body and head in direction of travel for best visibility.	☐	☐
▪ Looking around for surface problems, avoid those problems	☐	☐
▪ Adjust speed to conditions of work area	☐	☐
▪ Raise and lower platform	☐	☐
▪ Extend and retract platform	☐	☐
▪ Traverse Left and Right	☐	☐
▪ Turn unit slowly and at a proper speed for conditions	☐	☐
▪ Maneuver around obstacles – similar to those that will be encountered in the work area.	☐	☐
▪ Arms and legs are kept inside operations compartment at all times	☐	☐
Parking		
▪ Assure machine is lowered all the way and is stopped on as level surface as possible	☐	☐
▪ Engine or electrical stop switch is in the off position	☐	☐
▪ Building or area access/egress is not blocked.	☐	☐

Name/signature of employee

Date

Name/signature of person conducting verification review

Date

Ensure all sections are completed and form is signed by both employee & instructor. Fax to Regional Safety Professional prior to start of work.



Verificación de entranamiento para operadores de (plataforma en forma de tijeras)

Documento archivos - NO SE DESCARTEN POR FAVOR

Proyecto/Proyecto#:		Fecha:	
Nombre del empleado:		Nombre de la compañía:	
Fecha de prueba:		Líder del área:	

Equipo

Fabricante(s):			
Modelo(s):		Capacidad:	

Marca todos los cuadros que sean satisfactoriamente completados para todo tipo de equipo que se aplicó.

Artículos para verificar	Scissor Lift	Boom Lift
Pre-inspección: Documentado diariamente, DPR requiere inspección antes de usar.	☐	☐
▪ Use la forma de inspección de DPR para llevar al empleado de punto a punto por el proceso adecuado de inspección.	☐	☐
▪ El documento de inspección debe mantenerse en el vehículo por la duración de cada turno (una vez que complete la forma entreguela a la oficina de DPR).	☐	☐
▪ 3 puntos de contacto cuando entre a la plataforma operación aérea – Debe mantener 3 puntos de contacto al entrar en esta plataforma.	☐	☐
▪ Asegúrese que las puertas y las cadenas estén en su lugar.	☐	☐
Examina el área de trabajo		
▪ Evaluar el área de trabajo, examinar el área de operación y sus alrededores para descubrir los peligros potenciales.	☐	☐
Verificación para protección contra caídas.		
▪ Demostrar la inspección apropiada de harness y cable para amarrarte (lanyard).	☐	☐
▪ El barandal en el (scissor lift) es la protección contra caída con la excepción de remover los pies de la plataforma.	☐	☐
▪ Se requiere inmediatamente el arnés y el (lanyard) al subir a la plataforma del boomlift.	☐	☐
▪ Los puntos de anclaje para el scissor lift bienen en la estructura, para el boomlift el anclaje es proporcionado por fabricante.	☐	☐
Empezar y Maniobrar		
▪ Tablero de control – legible, asegurarse de que todas las instrucciones se entiendan.	☐	☐
▪ Alarma de reversa, verificar el funcionamiento.	☐	☐
▪ Examinar las brechas o sistema de tracción (asegúrese de que funcionan).	☐	☐
▪ El operador debe voltear su cuerpo y cabeza en la dirección de que va manejando para su mayor visibilidad.	☐	☐
▪ Revisar la superficie por algún peligro, evitar esos peligros.	☐	☐
▪ Ajustar la velocidad a las condiciones de su área de trabajo.	☐	☐
▪ Subir y bajar la plataforma.	☐	☐
▪ Extender y meter la plataforma.	☐	☐
▪ Dar vuelta a la izquierda y a la derecha.	☐	☐
▪ De vuelta a la unidad lentamente y a la velocidad apropiada a sus condiciones.	☐	☐
▪ Maniobrar alrededor de obstáculos – similares a los que existan en la área de trabajo.	☐	☐
▪ Brazos y piernas deben mantenerse dentro del compartimiento todo el tiempo.	☐	☐
Estacionamiento		
▪ Asegúrese que la máquina esté completamente hasta abajo y sobre una superficie anivelada.	☐	☐
▪ Asegúrese que el motor o la llave de electricidad estén apagados.	☐	☐
▪ No se debe bloquear el acceso o entradas al edificio.	☐	☐

Nombre/Firma del empleado

Fecha

Nombre/Firma del Instructor

Fecha

Ensure all sections are completed and form is signed by both employee & instructor. Fax to Regional Safety Professional prior to start of work



Ladder Training Verification

Archive Document – DO NOT DISCARD

Name:		Date:	
Project/Project#:		Company	

Workers should be trained in ladder safety prior to using a ladder. A hazard evaluation should be completed to determine a ladder is the safest means available to perform the task. Use of ladders is considered a high hazard activity by DPR and therefore requires daily, detailed, written pre-task planning prior to use. Below is a review of DPR guidelines:

General Ladder Guidelines

- Ladders should be used in accordance with manufacturer's recommendations
- Ladders should be maintained free of oil, grease and other slipping hazards
- Ladders should not be loaded beyond the maximum intended load for which they were designed nor beyond manufacturer's rated capacity
- For ladders placed in areas such as passage-ways, doorways, or driveways where they can be displaced by workplace activities or traffic:
 - o They should be secured to prevent accidental movement, or
 - o a barricade should be used to keep traffic or activities away from the ladder
- Ladders should not be moved, shifted or extended while in use
- Ladders should be inspected by a competent person for visible defects on a periodic basis and after any incident that could affect their safe use
- Only type 1, 1A and 1AA fiberglass ladders should be used; wooden and metal ladders should not be on DPR projects.
- Ladders should be inspected daily before use and those with broken or missing rungs, broken or split side rails, or otherwise damaged, should not be used
- All portable ladders should be equipped with non-skid safety feet and should be placed on a stable base
- The access areas at the top and bottom of ladders in use should be kept clear
- All ladders in use should be tied, blocked or otherwise secured to prevent accidental displacement
- The top two rungs/steps of any ladder should not be used as a point to work from at any time
- All work performed on any ladder should be conducted facing in towards the center point (i.e. rule of thumb -- the employee's belt buckle stays between the vertical rails of the ladder)

Step Ladders

- For work being performed from a step ladder (A frame) fall protection will not be required at elevations of six feet or greater provided all the conditions listed below are met. ***If it is determined that the use of a step ladder is the only feasible method available to perform work at elevations greater than six feet and any of the below conditions cannot be met then additional means of fall protection will be required.***
- All work is performed for an A-frame ladder on a level surface, fully opened
- All work is performed with worker facing the ladder
- Belt buckle rule applies (i.e., employee's body is stays between vertical rails)
- A grab-hold is available in the event the worker was to lose balance (e.g., uni-strut or structural member) **OR** a co-worker holds the ladder to stabilize, for the duration of work **note: at no time may utilities be used as a grab-hold**
- All other manufacturer's recommendations are followed

Note: At elevations above six feet where the above requirements cannot be fully met, one of the following fall protection measures must be implemented:

- o Harness and retractable or shock absorbing lanyard are utilized; consideration should be made to ensure the anchorage point is capable of holding a minimum of 5,000 lbs and the fall distance has been considered to ensure the worker will not contact surfaces should the retractable/lanyard be engaged
- o Use alternative work platform, i.e., scissor lift, scaffold, etc.

Extension Ladder

- Ladders should be used at an angle where the horizontal distance from the top support to the foot of the ladder is approximately $\frac{1}{4}$ of the working length of the ladder
- The side rails should extend 36 inches above the landing; when this is not practical, grab rails should be installed
- Ladders should be secured to prevent accidental movement
- The top should be placed with 2 rails supported equally unless the ladder is equipped with a single support attachment
- The worker should face the ladder while ascending or descending and while conducting work activities



Ladder Training Verification

Archive Document – DO NOT DISCARD

- Worker should use at least one hand to grasp the ladder when moving up or down the ladder
- Worker should not carry any object or load that could cause the loss of balance or cause a fall; rule of thumb -- the employee's belt buckle stays between the vertical rails of the ladder

Ladder Inspection

- All ladders should be inspected prior to use, ladders found with defects should be tagged and removed from service until repaired
- The following items should be checked when inspecting any ladder:
 - o missing or loose steps or rungs
 - o damaged or worn non-slip feet
 - o loose nails, screws, bolts or nuts
 - o loose or faulty spreaders, locks, and other metal parts in poor repair
 - o rot, decay or warped rails in wooden ladders
 - o cracks and exposed fiberglass
 - o cracked, split, worn or broken rails, braces, steps or rungs
 - o sharp edges on rails and rungs
 - o rough or splintered surfaces
 - o corrosion, rust, oxidization and excessive wear, especially on treads
 - o twisted or distorted rails; check ladders for distortion by sighting along the rails
 - o missing identification labels
- Additionally, the following should be inspected on extension ladders:
 - o loose, broken or missing extension locks
 - o defective locks that do not seat properly when ladder is extended
 - o sufficient lubrication of working parts
 - o defective or missing cords, chains and ropes
 - o missing or defective pads or sleeves

This document hereby certifies that I have been instructed on the items listed above and I have had an opportunity to ask questions. I fully understand the DPR requirements for work in involving the use of ladders.

Name/signature of employee

Date

Name/signature of person conducting verification review

Date

A copy of this form should be faxed to the regional safety professional PRIOR TO START OF WORK.



Verificación de Capacitación sobre el Uso de Escaleras de Mano

Documento de archivo – NO DESECHAR

Nombre:		Fecha:	
Obra:		Compania	

Se debe capacitar a los trabajadores sobre la seguridad del uso de las escaleras de mano. Se debe completar una evaluación de los peligros para determinar si una escalera de mano es el medio más seguro disponible para realizar la tarea. DPR considera que el uso de las escaleras de mano es una actividad de alto riesgo y por lo tanto requiere una planificación diaria, detallada y por escrito antes de su empleo. A continuación repasamos las normas de DPR:

Normas generales para el uso de escaleras de mano

- Las escaleras deben usarse según las recomendaciones del fabricante
- En las escaleras no debe haber aceite, grasa ni otros peligros de resbaladura
- La carga de las escaleras no debe superar la carga máxima prevista para las que se diseñaron ni tampoco la capacidad nominal del fabricante
- En el caso de escaleras de mano colocadas en áreas como pasillos, puertas o vías de acceso donde puedan ser desplazadas por actividades del lugar de trabajo o tráfico:
 - o Deben fijarse para impedir el movimiento por accidente, o
 - o Se debe usar una barrera para alejar el tráfico o las actividades de la escalera
- Las escaleras no deben moverse, desplazarse o extenderse durante su empleo
- Las escaleras deben ser inspeccionadas periódicamente por una persona competente para ver si tienen defectos visibles y después de cualquier incidente que pudiera afectar su uso seguro
- Solamente se deben usar escaleras de fibra de vidrio de tipo 1, 1A y 1AA; no se deben usar escaleras de madera y metal en los proyectos de DPR.
- Las escaleras deben inspeccionarse diariamente antes de su uso y no se deben usar las que tengan peldaños rotos o que falten, largueros rotos o hendidos, o que estén dañadas de otra forma
- Todas las escaleras portátiles deben estar equipadas con zapatas de seguridad antideslizantes y deben colocarse sobre una base estable
- Se deben mantener despejadas las áreas de acceso de las partes superior e inferior de las escaleras
- Todas las escaleras en uso deben amarrarse, bloquearse o de lo contrario fijarse para impedir que se desplacen por accidente
- No se deben usar en ningún momento los dos peldaños/escalones superiores de cualquier escalera como punto desde el que se trabaje
- Todo el trabajo realizado en cualquier escalera debe hacerse haciendo frente hacia el punto central (por ejemplo, regla general – la hebilla del cinturón del empleado debe estar entre los largueros verticales de la escalera)

Escaleras con escalones

- Para trabajos que se realicen en una escalera de escalones, no se requiere un sistema de protección contra caídas (armazón) a elevaciones de seis pies o más siempre que se cumplan todas las condiciones indicadas a continuación. **Si se determina que el uso de una escalera de escalones es el único método viable disponible para realizar trabajos a elevaciones mayores que seis pies y si no se puede cumplir con alguna de las condiciones siguientes, entonces se requieren medios adicionales de protección contra caídas.**
 - o No se deben usar escaleras de tijera como escaleras rectas
 - o La escalera debe abrirse completamente con los brazos de soporte trabados en la posición extendida
 - o La escalera debe colocarse sobre una superficie horizontal
 - o El empleado debe hacer frente a la escalera en todo momento
 - o Se debe mantener un contacto de tres puntos en todo momento al subir y bajar por la escalera
 - o El empleado debe permanecer centrado en la escalera; regla general – la hebilla del cinturón del empleado debe estar entre los largueros verticales de la escalera
 - o No se deben usar los dos últimos escalones de la escalera como punto de apoyo en ningún momento
 - o Se debe disponer de un asidero en el caso de que el trabajador vaya a perder el equilibrio (por ejemplo, uni-strut o miembro estructural) o además de los requisitos anteriores un compañero de trabajo puede sujetar la escalera para estabilizarla

Nota: A elevaciones superiores a seis pies donde no se puedan cumplir completamente los requisitos anteriores, tal vez se deban aplicar las siguientes medidas de protección contra caídas:

- o Se utilizan arneses y cabos retráctiles o contra impactos; se debe asegurar que el punto de anclaje pueda soportar un mínimo de 5,000 lb y que se haya determinado la distancia de la caída para cerciorarse de que el trabajador no vaya a hacer contacto con superficies si se utiliza el cabo retráctil
- o Uso de una plataforma de trabajo alternativa, por ejemplo, elevador de tijera, andamio, etc.



Verificación de Capacitación sobre el Uso de Escaleras de Mano

Documento de archivo – NO DESECHAR

Escalera extensible

- Se deben usar las escaleras en ángulo cuando la distancia horizontal del soporte superior a la base de la escalera sea aproximadamente $\frac{1}{4}$ de la longitud de trabajo de la escalera
- Los largueros deben extenderse 36 pulgadas por encima de la superficie donde se vaya a subir; cuando esto no sea práctico, se deben instalar barandillas
- Las escaleras deben fijarse para impedir el movimiento por accidente
- La parte superior debe colocarse con 2 largueros soportados por igual a menos que la escalera esté equipada con una conexión de un solo soporte
- El trabajador debe hacer frente a la escalera al subir o bajar por la misma y durante las actividades laborales
- El trabajador debe usar al menos una mano para agarrarse a la escalera al subir o bajar por la misma
- El trabajador no debe llevar ningún objeto ni carga que pueda causar la pérdida de equilibrio o una caída; regla general – la hebilla del cinturón debe estar entre los largueros verticales de la escalera

Inspección de las escaleras

- Se deben inspeccionar las escaleras antes de usarlas, y las que tengan defectos deben etiquetarse y retirarse del servicio hasta repararse
- Se deben comprobar los componentes siguientes al inspeccionar cualquier escalera:
 - o Escalones o peldaños que falten o estén sueltos
 - o Zapatas antideslizantes dañadas o desgastadas
 - o Clavos, tornillos, pernos o tuercas sueltos
 - o Separadores y trabas sueltos o defectuosos, y otras partes de metal en mal estado
 - o Largueros podridos, deteriorados o deformados en las escaleras de madera
 - o Rajaduras y fibra de vidrio expuestos
 - o Largueros, tirantes, escalones o peldaños rajados, hendididos, desgastados o rotos
 - o Bordes afilados en los largueros y peldaños
 - o Superficies ásperas o astilladas
 - o Corrosión, herrumbre, óxido y desgaste excesivo, especialmente en la superficie de los escalones
 - o Largueros torcidos o deformados; compruebe las escaleras para ver si están deformadas mirando a lo largo de los largueros
 - o Faltan etiquetas de identificación
- Además, se debe inspeccionar lo siguiente en las escaleras extensibles:
 - o Trabas de extensión sueltas, rotas o que faltan
 - o Trabas defectuosas que no se asientan debidamente cuando se extiende la escalera
 - o Lubricación suficiente de las piezas de trabajo
 - o Cordones, cadenas y cuerdas defectuosos o que faltan
 - o Almohadillas o manguitos que faltan o sean defectuosos

Este documento certifica que he recibido instrucción sobre los asuntos mencionados arriba y he tenido la oportunidad de hacer preguntas. Entiendo completamente los requisitos de DPR para trabajos en los que haya que utilizar escaleras de mano.

Nombre/Firma del empleado

Fecha

Nombre/Firma de la persona que realiza la revisión de verificación

Fecha

Se debe enviar por fax este formulario a regional safety profesional. ANTES DE EMPEZAR EL TRABAJO.

Job-Made Ladder Guidelines

The following guidelines are offered to assist in the construction of job-made wooden ladders:

- Job-made ladders should be constructed for intended use. If a ladder is to provide the only means of access or exit from a working area for 25 or more employees, or simultaneous two-way traffic is expected, a double cleat ladder should be installed. *NOTE: Cleats for job-made ladders are defined as crosspieces used by a person in ascending or descending a ladder. Cleats are also known as steps or rungs.*
- Side rails, when of wood, should be the equivalent of dressed Douglas fir "selected lumber", free from sharp edges and splinters, and should not have knots, except for an occasional one less than 1/2-inch in diameter that appears only on the wide face and is at least 1/2-inch back from either edge.
- If splicing of side rails is necessary to attain the required length, the splice should develop the full strength of a continuous side rail of the same length.
- Cleats of wood should be clear, straight-grained and absolutely free from knots of any size that appear in the narrow face. Knots appearing in the wide faces of cleats should not exceed a diameter of 1/4-inch. Cleats should be uniformly spaced within 1/4-inch tolerance, and not farther apart than 12 inches measured from the tops of cleats. The clear space in the plane of the cleats between the top of any cleat and an obstruction above should be at least 4 1/2 inches.
- Double cleat ladders should not exceed 24 feet in length.
- Single cleat ladders should not exceed 30 feet in length between supports (base and top landing). If ladders are to connect different landings, or if the length required exceeds this maximum length, two or more separate ladders should be used, offset with a platform between each ladder. Guardrails and toeboards should be erected on the exposed sides of the platforms.
- The width of single cleat ladders should be at least 15 inches, but not more than 20 inches, between rails at the top.
- Side rails should be parallel or flared top to bottom by not more than one-quarter of an inch for each 2 feet of length.
- 2-inch by 4-inch lumber should be used for side rails of single cleat ladders up to 16 feet long; 3-inch by 6-inch lumber should be used for single cleat ladders from 16 to 30 feet in length.
- 2-inch by 4-inch lumber should be used for side and middle rails of double cleat ladders up to 12 feet in length; 2-inch by 6-inch lumber for double cleat ladders from 12 to 24 feet in length.
- Cleats should be inset into the edges of the side rails one-half inch, or filler blocks should be used on the rails between the cleats. The cleats should be secured to each rail with three 10d common wire nails or other fasteners of equivalent strength. Cleats should be uniformly spaced, 12 inches top-to-top. Double-head nails should not be used for ladder construction.
- Every portable ladder should be of such material, size, and construction that it will safely carry the load to be placed thereon. Ladders which are customarily used for special purposes, such as those used for access to and additional support for overhead platforms, ladder brackets, and ladder scaffolding, should be of such design, material, and construction that they will support all normal loads which may be imposed upon them.



Job-Made Ladder Inspection

Name:		Date:	
Project/Project#:		Company	

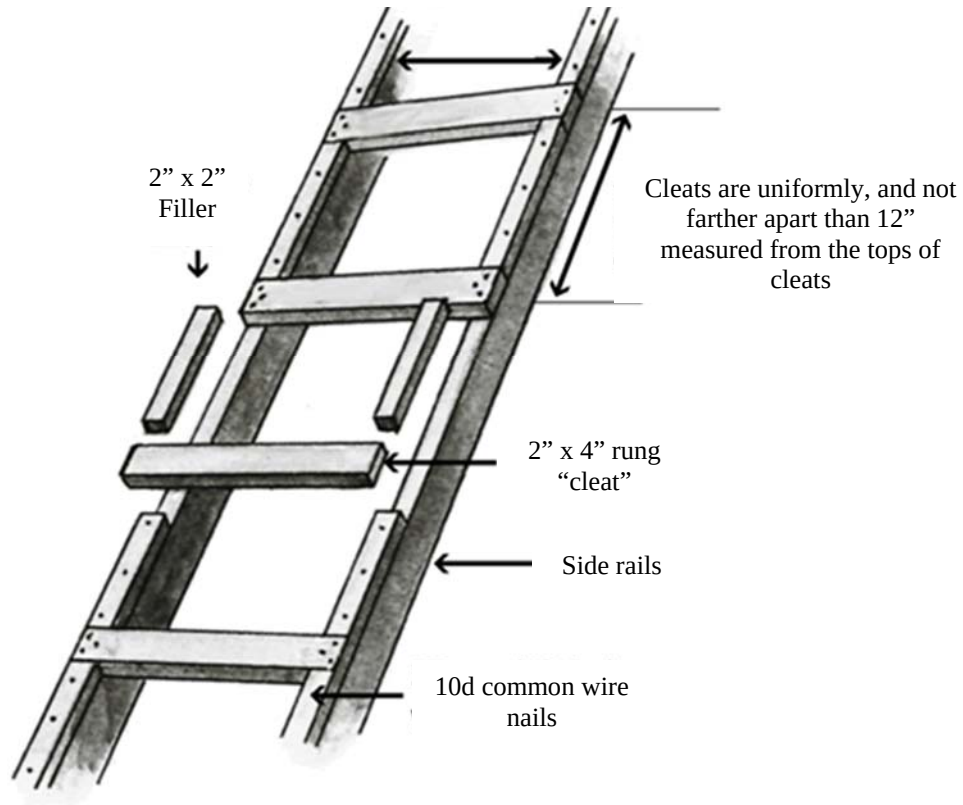
Job-made ladders should be constructed for intended use. If a ladder is to provide the only means of access or exit from a working area for 25 or more employees, or simultaneous two-way traffic is expected, a double cleat ladder should be installed. **NOTE: Cleats for job-made ladders are defined as crosspieces used by a person in ascending or descending a ladder. Cleats are also known as steps or rungs.**

DATE OF INSPECTION								
Initials of competent person performing inspection:								
Has the inspector been trained in the safe use of ladders?		Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N
Inspection Item & Description (Yes/No/Not Applicable)		Y/N/NA	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N
1	Is 2"x 4" lumber used for side rails of single cleat ladders up to 16' long or 3"x 6" lumber used for single cleat ladders from 16' to 30' in length? NOTE: Single cleat ladders should not exceed 30' feet in length between supports (base and top landing).							
2	Is 2"x 4" lumber used for side and middle rails of double cleat ladders up to 12' in length or 2"x 6" lumber for double cleat ladders from 12' to 24' in length? NOTE: Double cleat ladders should not exceed 24' feet in length between supports (base and top landing).							
3	Side rails, when constructed of wood, are the equivalent of dressed Douglas fir "selected lumber"?							
4	Side rails are free of knots, except for an occasional one less than 1/2" in diameter that appears only on the wide face and is at least 1/2" back from either edge?							
5	If splicing of side rails is necessary to attain the required length, the splice is developed to the full strength of a continuous side rail of the same length?							
6	Side rails are parallel or not flared from top to bottom by more than 1/4" of an inch for each 2' of length, and are constructed not less than 16" or more than 20" apart?							
7	Cleats, when made of wood are clear, straight-grained and absolutely free from knots of any size that appear in the narrow face?							
8	The wide faces of the cleats are free of knots exceeding a diameter of 1/4"?							
9	Cleats are uniformly spaced within 1/4" tolerance, and not farther apart than 12" measured from the tops of cleats?							
10	Cleats are inset into the edges of the side rails 1/2", or 2"x 2" filler blocks are used on the rails between the cleats?							
11	Cleats and fillers are secured to each rail with three 10d common wire nails or other fasteners of equivalent strength? Note: Double headed nails are not permitted.							
12	Cleats are free of any slippery substances?							
13	Are side rails and cleats free from sharp edges, splinters, or protruding objects?							
14	Ladder is secured or fastened from movement at top and bottom of ladder?							
15	All debris, floor obstructions, cords, construction material/supplies, etc. have been removed from base of ladder area?							
16	A guardrail is in place at top of landing that is so offset that a person cannot walk directly into the hole, or protects employees from leading edges?							
17	Rails extend at least 36" but not more than 42" above top landing?							

Note: Any items answered "no" below should be immediately addressed, with corrective measures tracked to completion.

Job-Made Ladder Components

Side rails are parallel, and are constructed not less than 16" or more than 20" apart



Job-Made Ladders

Job-made ladders should be constructed for intended use. Job-made ladders should be fabricated and used in compliance with OSHA, state or other regulations.

Normas para escaleras portátiles para tareas específicas

Se ofrecen las siguientes normas para ayudar en la fabricación de escaleras portátiles de madera para tareas específicas:

- Las escaleras portátiles para tareas específicas deben estar fabricadas conforme al uso final que se les destine. Si la escalera tiene como fin ofrecer la única forma de acceso y de egreso de una zona de trabajo para 25 o más empleados, o si se anticipa un tránsito simultáneo en ambas direcciones, entonces se deberá instalar una escalera portátil de doble peldaño. *NOTA: Los peldaños para escaleras para tareas específicas se definen como traveseros usados por la persona al subir o bajar la escalera. Se les denomina también listones o gradas.*
- Los rieles laterales, cuando son de madera, deben ser de madera equivalente al pino Douglas cepillado, “madera seleccionada”, libre de astillas, bordes afilados y nudos, con excepción de algunos pocos nudos que tengan menos de media pulgada de diámetro sobre la faz ancha y ubicados a por lo menos media pulgada de distancia de cualquiera de los bordes.
- Si fuera necesario empalmar los rieles laterales para lograr el largo deseado, el empalme debe tener toda la resistencia de un riel lateral del mismo largo que no tenga empalmes.
- Los peldaños de madera deben ser lisos, de fibra recta y totalmente libres de nudos de todo tamaño que aparezcan sobre la faz angosta. Los nudos que aparecen sobre la faz ancha del peldaño no podrán exceder un diámetro de $\frac{1}{4}$ de pulgada. Los peldaños deben estar colocados a distancia uniforme, con tolerancia de $\frac{1}{4}$ de pulgada, y a una distancia que no supere las 12 pulgadas, medida a partir de la parte superior del peldaño. El espacio vacío en el plano de los peldaños entre la parte superior de cualquier peldaño y una obstrucción arriba deberá medir por lo menos $4 \frac{1}{2}$ pulgadas.
- Las escaleras portátiles de doble peldaño no deben tener un largo superior a los 24 pies.
- Las escaleras portátiles de peldaño simple no deben tener un largo superior a los 30 pies entre sus puntos de apoyo (base y descansillo final). Si tienen como fin conectar diferentes plataformas, o si el largo exigido excede este largo máximo, se deben usar dos o más escaleras individuales, separadas por una plataforma entre cada escalera. Sobre los costados expuestos de las plataformas, deben instalarse barandas de protección y tablas de tope para los pies.
- El ancho de la escalera de peldaño simple debe ser de por lo menos 15 pulgadas, pero no más de 20 pulgadas entre los rieles en su parte superior.
- Los rieles laterales deben ser paralelos o acampanados, desde arriba hacia la base, en no más de $\frac{1}{4}$ de pulgada por cada 2 pies de largo.
- Deben usarse tablas de 2 por 4 pulgadas para los rieles laterales de escaleras de peldaño simple de hasta 16 pies de largo; deben usarse tablas de 3 por 6 pulgadas para las escaleras de peldaño simple de 16 a 30 pies de largo.
- Deben usarse tablas de 2 por 4 pulgadas para los rieles laterales y centrales de las escaleras portátiles de doble peldaño de hasta 12 pies de largo; deben usarse tablas de 2 por 6 pulgadas para las escaleras de doble peldaño de 12 a 24 pies de largo.
- Los peldaños deben estar insertados en media pulgada en los bordes de los rieles laterales, o deben usarse bloques de relleno en los rieles entre los peldaños. Los peldaños deben estar sujetos a cada riel con 3 clavos ordinarios 10d de alambre, u otro tipo de sujetador de resistencia equivalente. Los peldaños deben estar separados por distancias uniformes, 12 pulgadas de una superficie superior a la otra. Para la construcción de escaleras portátiles no deben usarse los clavos de doble cabeza.

Cada escalera portátil deberá ser del material, tamaño y construcción necesario para poder resistir sin peligro la carga que se le imponga. Las escaleras que comúnmente se usan con fines especiales (tales como aquellas que se usan para el acceso y apoyo adicional a plataformas superiores, las agarraderas de escalera y las escaleras para andamios) deben ser de un diseño, material y construcción que les permita resistir todas las cargas normales que se les impongan.



Stilts Training Verification

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Name:		Company:	

Only personnel trained in the safe use of stilts will be permitted to utilize this equipment. DPR considers the use of stilts to be a high hazard activity. Therefore, written, detailed pre-task planning is required prior to start of work.

General Requirements

☐	Stilts are not permitted in California.
☐	Surfaces on which stilts are used must be flat and free of pits, holes and obstructions.
☐	Debris, as well as any other slip, trip and/or fall hazard must be removed prior to start of work.
☐	Debris, as well as any other slip, trip and/or fall hazard must be removed prior to start of work.
☐	Stilts user must walk the work area without stilts to inspect for debris and any other hazards prior to putting on stilts and entering the work area.
☐	The use of personal stilts will not be allowed.
☐	Stilts must be properly maintained and inspected prior to each use.
☐	No alteration of the original equipment is allowed.
☐	Only a stable, elevated area will be used for attachment, adjustment and removal of stilts.
☐	Overall height of stilts should not exceed 36 inches.
☐	Areas where stilts are to be utilized must be clearly designated and identified, including appropriate signage and/or barricades. These areas should be treated as an area of controlled access whenever possible.
☐	Stilts must be used in accordance with manufacturer's recommendations.

Equipment Inspection

☐	All bolts must be tight, with no cracking or missing parts on calf brackets.
☐	Calf straps must be in good shape, with no frays or damaged areas. Buckles must be in good working order.
☐	Screws on calf support clamps must be tight, with no damage or dents to the extension arms.
☐	Buckles and straps on foot straps must be in good condition.
☐	Manufacturer provided screws on foot pedals must be bolted in the bottom and tight. Heel plate on foot pedal must be in place and adjusted for the boot size of wearer.
☐	Bolts and rivets on heel plate brackets must be tight and in place.
☐	The spring must be in one piece and without damage. The tension must be adjusted to wearer's personal preference.
☐	Wing nuts on height adjustment system must be in place and tight. Bushing must be tight with no cracks or missing surfaces.
☐	The overall height of the stilts must be adjusted to no higher than 36", measured from floor to top of foot pedal.
☐	The base plate must be properly attached to the height adjustment system.
☐	Base plate bolts and rivets must be in place and tight.
☐	The rubber on the bottom of the base plate must be properly attached and secure. The rubber must be clean of any build-up.
☐	The bottom of the rubber pad must be slip resistant.

This document hereby certifies that I have been instructed on the items listed above and I have had an opportunity to ask questions. I fully understand the DPR requirements for the use of stilts.

Name/signature of employee	Date
----------------------------	------

Name/signature of person conducting verification review	Date
---	------

A copy of this form must be faxed to the DPR regional office PRIOR TO START OF WORK.



Verificación de la capacitación en el uso de zancos

Documento de archivo – NO DESECHAR

Proyecto/Proyecto#:		Fecha:	
Nombre:		Compañía:	

Las únicas personas autorizadas para utilizar este equipo serán aquellas entrenadas en el uso adecuado de los zancos. DPR considera el uso de zancos como una actividad de alto riesgo. Por lo tanto, antes del comienzo del trabajo exigimos una planificación previa detallada y por escrito de las tareas.

Requisitos generales	
☐	El uso de zancos no está permitido en California.
☐	Las superficies donde se usarán los zancos deben ser planas y estar libres de agujeros, hoyos y obstrucciones.
☐	Antes de comenzar el trabajo, deben eliminarse los escombros y todas las posibles causas de resbalones, tropiezos y caídas.
☐	Antes de ponerse los zancos y entrar a la zona de trabajo, la persona que utilizará los zancos deberá caminar sin zancos por la zona de trabajo, para verificar la presencia de escombros y de cualquier otro peligro.
☐	No se permite el uso de zancos de propiedad personal.
☐	Los zancos deberán ser inspeccionados y adecuadamente mantenidos antes de cada uso.
☐	No se permite ninguna alteración del equipo original.
☐	Para colocarse, ajustarse y quitarse los zancos, sólo se usarán lugares estables y elevados.
☐	La altura total de los zancos no deberá sobrepasar las 36 pulgadas.
☐	Las zonas donde se usarán los zancos deberán estar claramente marcadas e identificadas, incluyendo el uso de carteles y/o de barricadas. Cuando sea posible, estas zonas deberán ser consideradas como zonas de acceso controlado.
☐	Los zancos deben usarse conforme a las recomendaciones del fabricante.
☐	El uso de zancos no está permitido en California.

Inspección del equipo	
☐	Todos los pernos deben estar apretados, sin grietas ni piezas faltantes en las abrazaderas de las pantorrillas.
☐	Las correas de las pantorrillas deben estar en buen estado, sin desgastes o partes dañadas. Las hebillas deben estar en buen estado de funcionamiento.
☐	Los tornillos en los sujetadores del soporte de las pantorrillas deben estar apretados, sin daños ni abolladuras a los brazos extensibles.
☐	Las hebillas y las correas para los pies deben estar en buen estado.
☐	Los tornillos del fabricante en los pedales de pie deben estar atornillados firmemente en la parte de abajo. La chapa para el talón del pedal del pie debe estar en su lugar y ajustada conforme al tamaño de la bota del usuario.
☐	Los pernos y remaches en las abrazaderas de las chapas para el talón deben estar apretados y en su lugar.
☐	El resorte debe estar en una sola pieza y sin daños. La tensión debe estar ajustada según el gusto personal del usuario.
☐	Las tuercas de mariposa para el sistema de ajuste a la altura deben estar en su lugar y apretadas. Los cojinetes deben estar apretados, sin grietas ni superficies faltantes.
☐	La altura total de los zancos debe estar ajustada de forma que no sobrepase las 36 pulgadas, medidas desde el piso hasta la parte superior del pedal del pie.
☐	La chapa de base debe estar conectada adecuadamente al sistema de ajuste a la altura.
☐	Los pernos y remaches de la chapa de base deben estar en su lugar y apretados.
☐	El caucho en la parte de abajo de la chapa de base debe estar bien conectado y sujetado. El caucho debe estar libre de toda acumulación.
☐	La parte de abajo del cojinete de caucho debe ser antideslizante.

This document hereby certifies that I have been instructed on the items listed above and I have had an opportunity to ask questions. I fully understand the DPR requirements for the use of stilts.

Nombre/Firma del empleado

Fecha

Nombre/Firma de la persona que lleva a cabo la verificación

Fecha

A copy of this form must be faxed to the DPR regional office PRIOR TO START OF WORK.



Stilts Pre-Task Plan

Archive Document – DO NOT DISCARD

Audited by: _____

Date: _____

Date	Job Name	Company
Specific Location of Work		

Answer the following when evaluating your work YES, NO or N/A:		Are any of the following Required?						
1.	Is the stilts work area provided with adequate lighting to allow the review of all walking surfaces?	☐	YES	☐	NO	☐	N/A	PPE
2.	Are there any hazards created by any other workers or material that can be removed from stilts work area?	☐	YES	☐	NO	☐	N/A	☐ Hard Hat
3.	Have you walked your work area to address housekeeping, slip/trip issues (e.g. pipe cut offs, holes, debris, etc.)?	☐	YES	☐	NO	☐	N/A	☐ Fall Protection
4.	How many workers will be working on stilts? ____ How many workers to pick up/ clean?	☐	YES	☐	NO	☐	N/A	☐ Eye / Face PPE
5.	Do you have a system to ensure cords/ hoses are off the walking surface in the stilt work area?	☐	YES	☐	NO	☐	N/A	☐ Hand / Arm PPE
6.	Are you working around live systems, energized equipment or overhead obstructions	☐	YES	☐	NO	☐	N/A	☐ Hearing PPE
7.	Have you installed a hard barricade and/or signage to keep other trades out of stilt work area?	☐	YES	☐	NO	☐	N/A	☐ Foot PPE
8.	Do you need to review SDS's to proceed with this work?	☐	YES	☐	NO	☐	N/A	LOCATE:
9.	Have you performed an inspection of the stilts and corrected all necessary repairs?	☐	YES	☐	NO	☐	N/A	☐ Respirator
10.	Have you been provided a stable area to sit while attaching, adjusting, or removing stilts?	☐	YES	☐	NO	☐	N/A	☐ Emergency Telephones
11.	Are workers who walk stilts trained and experienced?	☐	YES	☐	NO	☐	N/A	☐ Fire Extinguisher
12.	Have each worker's stilts been adjusted for the individual's comfort (e.g. spring/foot pads)?	☐	YES	☐	NO	☐	N/A	☐ Emergency Exit Routes
13.	Have you completed Stretch & Flex today? Will this task require any lifting, bending or twisting?	☐	YES	☐	NO	☐	N/A	☐ First Aid Equipment
14.	Is there a new hire, or new team member on the project who will need support?	☐	YES	☐	NO	☐	N/A	☐ Other
15.	Do you have an injury to report or were you injured the prior working day?	☐	YES	☐	NO	☐	N/A	☐ Other
Any safety item identified in the above question list must be addressed on the back page.								

Tasks, Tools & Potential Hazards

Steps for Work	Tools	Hazards	Steps Taken to Address Hazards

*If you need more space, attach another PRE-TASK PLAN SHEET

Work Planner:	Foreman	Safety Coordinator
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Crew Members – list additional crew members on back or a separate page if necessary:

What is the quality item you will focus on today?



Planificación previa a las tareas zancos

Documento de archivo – NO DESECHAR

Revisado por: _____

Fecha: _____

Fecha		Nombre del proyecto:		Compañía:	
Ubicación específica del trabajo					

Responda a lo siguiente cuando evalúe su trabajo (haga un círculo alrededor de SÍ, NO o N/A):		¿Se necesitan los siguientes?						
1.	¿Es el área de trabajo zancos disponer de iluminación adecuada para permitir la revisión de todas las superficies para caminar?	☐	Si	☐	NO	☐	N/A	Equipo de protección personal:
2.	¿Existen los peligros creados por cualquier otro trabajador o ha material que se puede quitar de área de trabajo zancos?	☐	Si	☐	NO	☐	N/A	☐ Casco
3.	¿Han caminado su área de trabajo para revisar la limpieza, riesgos de resbalar o tropezar (piezas de tubería, agujeros, basura, etc.)	☐	Si	☐	NO	☐	N/A	☐ PPE de ojos/cara
4.	¿Cuántos trabajadores estarán trabajando sobre zancos _____ Cuántos trabajadores estarán trabajando sobre pilotes _____	☐	Si	☐	NO	☐	N/A	☐ PPE de manos/brazos
5.	¿Tiene usted un sistema para asegurarse que los cables o mangueras están fuera del área de tránsito en el área de trabajo zancos?	☐	Si	☐	NO	☐	N/A	☐ PPE auditiva
6.	¿Está trabajando alrededor a los sistemas vivos, equipos eléctricos o obstrucciones por lo alto?	☐	Si	☐	NO	☐	N/A	☐ Respirador
7.	¿Ha instalado una barricada duro, o señalización para mantener otros equipos fuera del área de zancos?	☐	Si	☐	NO	☐	N/A	☐ Casco
8.	¿Necesita revisar las SDS antes de proceder con este trabajo?	☐	Si	☐	NO	☐	N/A	Ubique:
9.	¿Ha realizado una inspección de los zancos y corregir todas las reparaciones necesarias?	☐	Si	☐	NO	☐	N/A	☐ Teléfonos de emergencia
10.	¿Ha sido un área estable para sentarse para poner, ajuste o quitar los zancos?	☐	Si	☐	NO	☐	N/A	☐ Extinguidor de incendio
11.	¿Son los trabajadores que utilizan zancos capacitados y experiencia?	☐	Si	☐	NO	☐	N/A	☐ Rutas hacia salida de emergencia
12.	¿Tienen los zancos han ajustado para la comodidad de la persona? (muelles/ cojines del pie)	☐	Si	☐	NO	☐	N/A	☐ Ducha/lavado de ojos
13.	¿Deberá levantar peso, agacharse o torcerse durante esta tarea? ¿Hizo hoy sus ejercicios de estiramiento y flexión?	☐	Si	☐	NO	☐	N/A	☐ Equipo de primeros auxilios
14.	¿Hay algún empleado nuevo, o un nuevo miembro del equipo que esté trabajando en el proyecto y que necesite ayuda?	☐	Si	☐	NO	☐	N/A	☐ Otros
15.	¿Tiene una lesión que deba informar o se lesionó el día laboral anterior?	☐	Si	☐	NO	☐	N/A	☐

Todo punto relativo a la seguridad identificado en la lista de arriba debe ser abordado en la tabla siguiente.

Tarea - *Si necesita más espacio, adjunte otra hoja de planificación previa a las tareas

Pasos del trabajo	Herramienta	Peligros (potencial o imminente)	Medidas tomadas para resolver los peligros

Planificador del trabajo:		Nombre del capataz		Coordinador seguridad	de	
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Cuadrilla - Si necesita un diagrama del trabajo, adjúntelo en una hoja separada.

En que punto referente a calidad vas a estar enfocado hoy?	
--	--

Section 08

Emergency Procedures

Sub-Section	Topic	Page #
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2.0	Crisis Management Team Responsibilities	2
3.0	Emergency Procedures	4
4.0	Active Shooter Emergency	10
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6.0	Emergency Response Plan	13

Emergency Procedures

1.0 Introduction

It is our goal to provide and maintain an Incident Free Environment (IFE). Although we do everything we can to provide an IFE, the possibility of a crisis occurring still exists. It is an expectation that these procedures be made familiar to every employee on the project through New Employee Orientation, jobsite postings, Safety Kickoff/Pre-construction Meetings, and reinforced at site staff and contractor meetings as well as a frequent topic at weekly safety meetings.

This plan provides the actions and assignments for various types of emergencies. The plan includes evacuation procedures and designated assembly areas for which all site employees will gather depending on the type of emergency posted throughout the site. Specific actions are assigned to various project team members. A list of employees who are trained in First Aid / CPR/ AED should be provided to DPR upon request.

Local emergency response provider(s) should be contacted to ensure that they know of any new projects, their locations and scope of work. It may be learned at this time that the site may actually fall under the jurisdiction of multiple response companies which may need to be coordinated prior to need. It is also an opportunity for the project team to meet the local responders who may also express an interest in utilizing the site for training exercises, which may be mutually beneficial.

Reference: Emergency Action Plans CFR 1910.38. Other applicable Standards, guidelines, and local regulations may apply.

2.0 Crisis Management Team Responsibilities

2.1 *Crisis Team Leader*

The crisis team leader is the center point for all emergency communications and coordination. This person should direct the emergency response measures and ensure that all team member positions are assigned to the appropriate people and that these people are aware of their responsibilities.

2.2 *Spokesperson*

The project spokesperson is the designated person who communicates with the media and whom also may be responsible for communication to the owner. The project spokesperson should be media-trained and work closely with the corporate marketing and communications team to develop messaging for any communication with the media. A spokesperson should not be a firsthand witness to the incident, should be able to remain objective and fact-based when speaking about what occurred. Please reach out to pr@dpr.com for assistance with media training, messaging and selecting a spokesperson.

2.3 *Administrator*

The project team administrator provides support to the crisis team, i.e., screening phone calls, making travel arrangements, providing clerical support and assisting families in the event of an injury or fatality. This person maintains a log of all emergency related inquiries received at the jobsite and ensures calls are returned by appropriate personnel in a timely fashion.

2.4 *Project Management Team*

DPR project management should be knowledgeable of emergency response procedures and be prepared to assist those responding in the event of an emergency. DPR project management is primarily responsible for ensuring that everyone understands the Emergency Response Plan, that they are trained properly and that all appropriate measures are being taken. Further, the DPR project management team is responsible for the project site and should be the party responsible for assessing and investigating the situation and notifying all workers when the hazardous situation has cleared and it is safe to return to work. DPR project management



should also ensure at least one DPR supervisor is First Aid/CPR/AED trained and their certification(s) is/are current.

2.5 Subcontractor Supervision

During an emergency, subcontractor supervision should ensure worker safety and take all necessary steps to protect the site and assist in all emergency procedures deemed necessary by the crisis team leader. They further should provide notification to DPR project management that all personnel are accounted for. No subcontractor should talk to any media personnel and all media should be politely referred to the DPR spokesperson or project management team. Subcontractor should also ensure at least one supervisor is First Aid/CPR/AED trained and their certification(s) is/are current.

2.6 All Workers

All workers are expected to participate in and follow all emergency response procedures contained within this section. Workers must respond immediately to instructions from supervision, DPR incident command and from emergency response agency personnel. No worker should talk to any media personnel and all media should be politely referred to the DPR spokesperson or project management team.

If workers witness an accident or incident, they should (in order):

- Immediately call out for first aid or other needed assistance
- Obtain access to radio or telephone and contact the appropriate emergency response contacts
- Protect the worker or situation and secure the area from further harm
- Verify DPR and their crew supervisor is immediately advised of the situation
- Document observations and details of the incident on a Witness Statement Form provided by DPR

Role	Name	Contact Numbers
Crisis Team Leader		
<i>Backup Crisis Team Leader</i>		
Spokesperson		
<i>Backup Spokesperson</i>		
Team Administrator		
<i>Backup Team Administrator</i>		
Regional EHS		

Note: Refer to Emergency Response Plan for project specific contact information.

3.0 Emergency Procedures

All communications during an emergency should be on a specified radio station, word of mouth, telephone or other prearranged form of communication. All personnel should be informed of an evacuation via a specified radio station, word of mouth, telephone, air horn, whistle or other prearranged form of communication. **During this time, all communication and traffic over the radios should stop.**

All affected personnel should be required to observe the following procedures in the event of an evacuation:

- Stop all work activities
- If possible, shut down all equipment, energy sources and ignition sources
- All employees should proceed, if safe to do so, to their designated emergency assembly area at a prearranged specified location using designated evacuation routes
- Upon reaching the designated emergency assembly area, all workers are to check in with their supervisor
- Supervisors should report to DPR project management after taking a head count and report any unaccounted for employees
- All workers are to take all necessary steps to keep non-essential personnel away from the evacuated area
- All workers are to cooperate fully with emergency services personnel
- No one is to re-enter secured areas until the all clear is given by the authorities (Fire Department, etc.) and DPR project management

3.1 Naturally Occurring Emergency Situations

Many areas offer weather warnings or watches. In these areas consult your local weather watch service or reports for current updates and needed actions. The following storm conditions may require additional emergency procedures and serve as a guide for actions. Project teams should consult with local authoritative Jurisdictions. When any of these storm conditions may affect a project, the following procedures should be observed.

3.2 Severe Lightning Storm

- Work activities most at risk:
 - Explosives Handling
 - Roofing/Roof Top Activities
 - Steel Erection/Rodbusters
 - Landscaping
 - Scaffolding Work
 - Heavy Equipment Operation
 - Cranes/Drill Rigs/Pump Trucks
 - Plumbing & Pipe Fitting
 - Any work activity outside of an enclosed or grounded structure.
- How to Monitor Lightning Producing Storm Systems
 - <http://lightningmaps.org> – provides real time lightning strike data for specified area
- When to Seek Shelter
 - Minimum of 10 Miles from Lightning Producing Storm Cells
 - Lightning is unpredictable and can strike up to 10 miles outside the heaviest rainfall
 - Construction equipment should be shut down and all personnel moved to a safe location until the storm has passed
 - Remain in shelter a minimum of 30 minutes after last sound of thunder.
- Where to Seek Shelter
 - Enclosed Grounded Buildings
 - Hard-topped metal vehicles with rolled up windows
 - On roadways, when view is severely obstructed during storm, vehicles should be pulled off the road and the 4-way flashers activated until the storm has passed

3.3 Earthquake

Should the situation arise where there is structural collapse or the threat of imminent collapse due to an earthquake, the following general procedures should be followed:

- All people to evacuate the project as per normal emergency procedures
- The area of the earthquake should be secured
- The nearest supervisors should be notified
- Lower booms on all crawler and mobile cranes; cranes not capable of lower booms are permitted to weathervane per manufacturer's recommendations, ensure booms will not contact power lines, structures, etc.
- People should be kept out of the area except for those rendering emergency aid
- All the area utilities should be turned off as quickly as possible providing it is safe to do so

3.4 Landslide

- Find shelter immediately
- Take all necessary action to preserve life and prevent injury, including leaving the premises

3.5 Floods

Employees working in areas where flood conditions could occur should be prepared to take the appropriate actions when a weather warning or potential flood warning is issued:

- If possible, post a watch (a signaling or warning device should be provided)
- Preparations should be made to leave on very short notice
- Only the required equipment or tools should be carried
- In the event of evacuation, all equipment, gas valves, etc. are to be shut down
- Only dry electrical equipment should be touched
- Avoid traveling through flooded areas, particularly at night when it is difficult to determine hidden dangers
- Employees should be aware of designated high ground safety areas, evacuation routes and the location of emergency equipment

3.6 Hurricanes

In the event of a hurricane threat, a three-stage procedure shall be adopted as outlined below. An alert will be issued by the Hurricane Coordinator when a threat has been identified. The Project Team should follow the varying stages of threat assessment and ensure the applicable responsibilities are completed.

3.6.1 Stage One

This alert will be issued when a hurricane is located in the Gulf of Mexico or approaching our Atlantic coastline. In general, this alert establishes the need to identify locations for securing objects, equipment, materials, and construction debris. It also serves as a notice to all supervisors to review the hurricane preparedness plan.

Upon issuance, all departments will:

- Monitor the storm with on-line weather reports
- Track the storm utilizing maps and radio broadcasts
- Organize unused materials and adequately dispose of trash and debris

3.6.1.1 Jobsite Procedures

- Review the jobsite and establish priority preparations for potential storm impact
- Review construction fencing and ensure it is "storm ready". Check all posts, anchors, and fasteners for security. Review wire ties and strapping for adequate anchorage.

- Pour all possible foundations
- Pour all open forms to secure in-place formwork
- Direct the formwork subcontractor to secure and organize all loose forms
- Set up an emergency contact with a water intrusion contractor company
- Secure all jobsite signage
- Remove all unused scaffolding from the jobsite. If removal is not possible, all scaffold boards must be disassembled and secured on the ground. Follow the recommendations of the manufacturer and the scaffolding engineer for Hurricane preparedness procedures. Bundle materials together as needed.
- Secure all loose lumber with heavy objects, strapping, and/or equipment
- Secure all ladders, small tools, and equipment
- Review the site for proper tie down of all miscellaneous tools and materials
- Review the progress of all subcontractors' preparations. Assist subcontracting teams where necessary to ensure preparations are adequate. Direct subcontractors as needed to ensure all potential hazards are handled appropriately.
- Allow cranes to weather vane or lower booms to rest on the ground. Follow the tower crane manufacturer's recommendations for hurricane preparedness procedures. Remove any signs on the crane booms.

3.6.2 Stage Two

In this threat stage, steady movement of the hurricane toward the project has been forecasted. Upon issuance, the project team will ensure the below noted items are completed. In general, the items noted below consist of activities which secure equipment and materials that may not be practically moved offsite. This procedure may be terminated if the project is no longer threatened by the projected path of the storm.

- Fill all gasoline and diesel tanks
- Fill and/or store all storage tanks
- Locate and retain portable generating equipment
- Secure all pumps/hoses and store in an easily accessible location. Make preparations for water removal and remediation.
- Make arrangements for removal of equipment and materials from all areas
- Ensure all dumpsters and recycling bins are either emptied or covered/secured
- Verify all drains and inlets are unclogged and check for the use of the retention ponds
- Have a tire repair kit and small 12 volt air pump
- Temp dry critical areas of the building (electrical and pump rooms)

3.6.3 Stage Three

- During this stage, an imminent threat by the storm is expected and the project area is under storm warning. The activities associated with this phase finalize the preparation process for an impending storm. If the storm changes course and is no longer expected to impact the project, this process may be terminated.
- Re-check anchors and security means on all temporary buildings, offices, or storage containers. Verify all subcontractor trailers and storage containers are adequately safeguarded.
- Remove any scrim from site fencing. Review fence anchorage and ensure all gates are appropriately locked once all personnel are offsite. May need added support at entry gates.
- Complete a final clean-up of all construction and landscape areas.
- Group all trash bins and portable toilets together and secure them to stable objects with adequate strapping.
- Fill all portable water kegs with potable water and place them inside gang boxes.

- Welding machines, air compressors and other light-weight mobile equipment are to be grouped together and secured to stable objects. All leads, hoses and torches on trailers are to be secured together with their associated trailers parked and sufficiently anchored.
- All scaffolding is to be taken to the scaffold storage area and secured from movement. Group, secure and anchor down scaffolding parts as recommended by the scaffold manufacturing company and the engineer of the scaffold erection team.
- Wooden pallets, forms and wooden decks are to be secured or moved to an indoor facility.
- All crane and equipment booms are to be lowered to a safe position and made “hurricane ready”; as recommended by the equipment manufacturer.
- All light-weight equipment is to be parked and secured with the brakes set.
- Earth moving equipment should be left on high ground where applicable. Pick-up trucks, flatbeds, and buses should be secured with brakes set. Equipment keys stored on site should be accessible.
- All material hoppers should be tied down and secured.
- All gang boxes must be grouped together, secured, and anchored.
- Anchor all tanks, drums and vessels that may be affected by high winds and water.
- Ensure all materials stored in buildings are not lying directly on floor. Utilize dunnage, as needed, to elevate materials off of an exposed surface.
- Ensure that all underground piping is protected against flotation.
- Onsite vehicles should be parked on high ground with brakes set.
- Iceboxes should be secured with a full supply of ice on hand.
- Turn off all electrical power at main switches.
- Secure all exterior doors and sandbag, as needed.
- Turn off main power to the project, if necessary.
- Security coverage will be directed by the project senior superintendent and/or owner.
- Walk the project site, with the owner, for final review and preparation acceptance prior to evacuation.
- Video tape or take photos of the condition of the project.
- Provide names and phone #'s of ride-out crew to the owner's representatives. The ride-out crew will consist of two or more people. Make sure their contact information is noted in the below contact lists.
- Have a safe neutral meeting location for your ride-out crew, have hand held radios if possible (phones may be offline).

3.6.3.1 *Jobsite Trailer*

- Box up important files and documentation for removal
- Remove computers and other mobile electronic equipment
- Elevate items off the floor to minimize chances of water damage
- Wrap immobile and susceptible equipment/materials in plastic and keep them elevated from the floor
- Tape windows with duct tape – may need to cover windows with plywood
- Take pictures of existing conditions of the office trailer prior to evacuating
- Back-up electronic information, as needed, on portable hard drives and store in a safe, dry location

3.6.3.2 *Personnel*

- Identify shelter locations and evacuation routes
- Purchase and store plenty of bottled water and canned food
- Fill all vehicles with gasoline/fuel
- Stock-up on batteries, flashlights, candles, matches, insect repellent, sunscreen, and first aid supplies
- Make sure cell phones are fully charged. Store your cell phone in zip lock bags. To preserve battery life, keep your electronic devices in the off position until they are needed.

- Identify the safest part of your home to be used as a shelter from high winds, flying objects, and broken glass. Avoid rooms with windows.
- Anticipate that the worst winds will come from the northeast to east
- Don't forget to properly prepare for and shelter your pets
- Do not drive through water-covered roads. Of note, running water a few inches deep can sweep a car off the road. A cubic yard of water weights 1,648.8 lbs. The average car weighs the same as two cubic yards of water. Drive with caution.
- Notify a friend or relative (located outside of the hurricane's protected path) of your intentions and establish them as the single point of contact in the event of communication difficulties after the storm
- Clear any debris off of roof and gutter areas that can clog a roof drain. (Small items will add up)

3.6.3.3 Post Hurricane Assessment

- Project management, in concert with the superintendents and the facility managers, shall assess the situation and establish a definitive schedule to resume full operations.
- The DPR project managers, superintendents and each contractor will assign personnel to ascertain the degree of damage to their work. This assessment, accompanied by the estimate for repairs and work recommencement will be reported to the hurricane coordinator, project managers and superintendents as soon as practical.
- The DPR team shall document all damage from the storm and expenses associated with clean-up and preparation. This information will be submitted to project management and the Hurricane Coordinator for use in preparing an "after action" report that will be submitted to the insurance carrier.

Note: *Notify employees through media and emergency lines if normal operations cannot resume immediately following the storm.*

Note: *Refer to the Hurricane Preparedness Plan checklist in the Forms Section at the end of the EHSP.*

3.7 Blizzard

Employees working in remote areas where weather warnings are not applicable to the area should adopt the following procedures:

- Awareness of prevailing weather conditions
- Dress to suit conditions; e.g. heavy-duty winter protection or survival equipment
- If possible, employees should work in pairs and conduct ongoing buddy checks for frostbite
- Pedestrian travel routes should be known. Safety ropes should be erected as a guide
- Employees should not travel alone unnecessarily. Survival equipment should be carried

3.8 Tornado

- Seek shelter below grade (staying inside structures is recommended but exterior doors and windows are to be avoided)
- Operations of mobile crane to be suspended-boom should be laid down if time permits or the load line hooked to either the house or the structure at some low point; the equipment should be left and refuge taken in a shelter
- Should shelter not be available, employees should proceed to low ground or a ditch and lie down with the head protected

3.9 Other than Naturally Occurring Emergency Situations

3.9.1 Fire

Prudent safety practices indicate that "if you attempt to put out a fire in its incipient stages (the first 30 seconds), you stand a good chance of being successful". During the first 30 seconds, anyone discovering or becoming involved in a fire should follow emergency assistance procedures.



In the event a fire occurs on the project, supervisors should respond to the area and follow these procedures as applicable:

- The fire is to be evaluated with regards to controlling it
- Necessary evacuation steps should be taken
- Security measures should be established as necessary to keep non-essential people out and to safeguard records and equipment
- Turn off (do not unplug) all spark producing devices, i.e., electric motors, portable heaters, etc.
- Shut off all gases, if possible, remove high pressure cylinders from the area

When directed to evacuate a building or facility under threat of fire, employees should observe the following points:

- Permanent elevators are not to be used unless directed
- Lights to be left on, doors and windows closed but not locked
- Employees should stay as low as possible and try to keep out of the smoke (possibly toxic)
- Should clothes catch fire, the best thing to do is to drop and roll

3.9.2 Bomb Threats

DPR project management should be notified immediately of any bomb threat. As best as possible, preserve the evidence for law enforcement officials. All attempts should be made to determine the source of the threat and for more information consult the Bomb Threat Information Sheet in the Forms Section at the end of the EHSP. All threats are to be taken seriously.

3.9.3 Structure or Equipment Damage

Should a situation arise where a structure has collapsed or is in danger of collapse or equipment has been involved in an accident, the following general procedures apply:

- The normal emergency assistance procedures should be followed
- The area where the incident has occurred should be secured
- DPR and the nearest supervisors are to be notified
- People are to be kept out of the area except for those rendering medical assistance
- If necessary and safe to do so, area utilities are to be turned off as quickly as possible
- Attempts to clean up or repair should not be made until DPR Project Management has given clearance

3.9.4 Hazardous Substance Release or Spill

Immediately upon a release or a spill, steps should be taken to implement procedures for containment, control and cleanup of the spill. These procedures should comprise the following basic steps:

- Ensure the safety of all individuals in the area and evacuate the area as necessary
- The area is to be secured
- If possible, identify the hazardous material released
- If the material can be identified, use the Safety Data Sheets (SDS) for detailed procedures
- If the release is an airborne vapor spill, gas or a large uncontrollable spill of liquid, the local County Health Department or 911 dispatch should be notified immediately; these authorities will mobilize assistance and commence public evacuation in the immediate vicinity if necessary
- In most cases, clean-up procedures should start as soon as possible to prevent further spread of the substance into flowing water, ground water or sewer systems
- Consult the SDS and the Chemical & Environmental Management Section for more information

For small spills or spills of vehicle fluids, follow the guidelines and procedures set forth in the SDS for proper clean up and disposal.

3.9.5 **Loss of Utilities**

In the event of **loss of water**, the following general procedures apply:

- Discontinue any current work procedures that require the use of water for safe work practices (e.g., asbestos abatement, core drilling, etc.)
- Discontinue the use of all water

In the event of **loss of electricity**, the following general procedures apply:

- Discontinue any current work procedures that require the use of electricity to operate equipment used as an engineering control for safe work practices
- Turn off any temporary equipment in the work area (e.g., negative air units, lighting, etc.)
- If work is to be continued, ensure adequate access/egress and task lighting is provided
- Do not attempt to exit a building in the dark if there are no egress lights – wait for emergency personnel to escort you from the building
- If an area blackout exists, contact the power company to obtain the latest estimates of service resumption and nature of all possible dangers at the site

3.10 **Security Issues**

Should **public demonstrations** occur, the following general procedures apply:

- Work on the jobsite can continue where not encumbered by the public demonstration; however, work in the immediate area should be stopped and all workers evacuated
- Notify police and fully cooperate with their instructions
- Allow only authorized personnel to enter the jobsite
- Revoke all visitor passes and escort all visitors from the jobsite
- If necessary, utilize police or flag personnel to stop or detour traffic
- Refer to the Public Protection Section and DPR Toolbox for additional information regarding union activities

Should other **in-progress emergency security issues** occur, e.g., assault, theft, fights, attempted suicides and/or hostile persons, contact the necessary authorities and let them handle the situation. Be prepared to give details of the incident to responding agencies, to include:

- Direction of travel of person(s) involved if leaving the scene
- Description of person(s) involved
- Description of incident
- Description of property damaged or stolen

4.0 **Active Shooter Emergency**

Active shooter incidents are often unpredictable and evolve quickly. In the midst of the chaos, anyone can play an integral role in mitigating the impacts of an active shooter incident.

(<https://www.dhs.gov/active-shooter-preparedness>)

4.1 **Option One – Run**

- Have an escape route and plan in place
- Leave your belongings behind
- Evacuate regardless of whether others agree to follow
- Help others escape, if possible
- Do not attempt to move the wounded
- Prevent others from entering an area where the active shooter may be
- Keep your hands visible
- Call 911 when you are safe. Provide the following information:

- Location of the active shooter
- Number of shooters
- Physical description of shooters
- Number and type of weapons
- Number of potential victims

4.2 Option Two – Hide

- Hid in an area out of the shooter's view
- Lock door or block entry to your hiding place
- Silence your cell phone (including the vibrate mode) and remain quiet.

4.3 Option Three – Fight

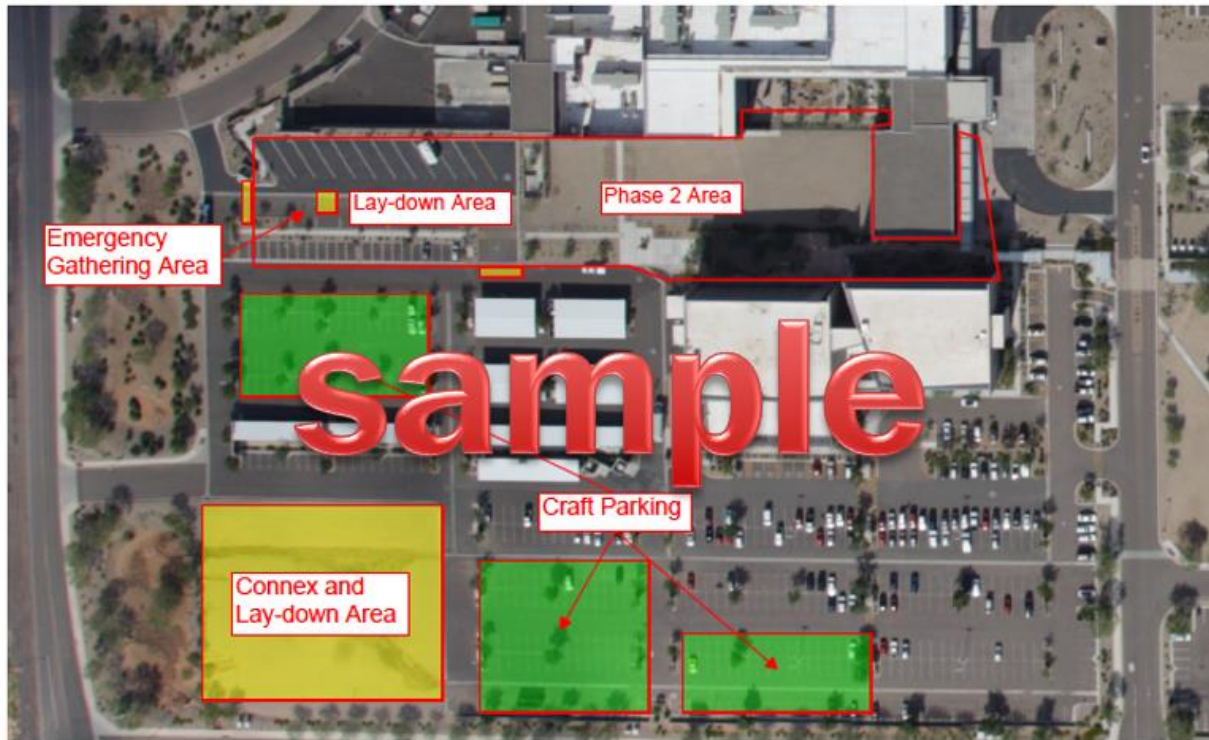
- Fight as a last resort and only when your life is in imminent danger
- Attempt to incapacitate the shooter
- Act with as much physical aggression as possible
- Improvise weapons or throw items at the active shooter
- Commit to your actions – your life depends upon it!!!

4.3 When Law Enforcement Arrives

- Remain calm and follow instructions
- Drop items in your hands (e.g., bags, jackets, etc.)
- Raise hands and spread fingers
- Keep hands visible at all times
- Avoid quick movements towards officers
- Avoid pointing, screaming or yelling
- Do not ask questions or otherwise distract officers when evacuating
- The first officers to arrive on scene will not stop to help the injured.
- Rescue teams will arrive after initial officers, they will treat and remove the injured.
- Once you have reached a safe location, you will remain or be held in the safe location by law enforcement until the situation is under control, and all witnesses have been identified and questioned.
- Do not leave the safe area until law enforcement have instructed you to do so.

5.0 Project Site Plan

A Site Plan indicating access gates, streets, telephones, firefighting equipment, first aid equipment, evacuation routes and emergency gathering area(s) should be posted and/or available for review at the project. This plan should be updated periodically and as site conditions change. The following is a copy of the site plan:



6.0 Emergency Response Plan

The Emergency Response Plan should be completed and posted at the job site trailer. Obtaining this document should be coordinated with your regional safety manager and should be printed on 11x17 and posted throughout project (this is a two-page document).



(project name)
(project address)
(project phone number)

EMERGENCY RESPONSE PLAN

Jobsite Personnel	Name & Cell Numbers	Emergency Numbers	Contact Numbers
Project Manager		911	
Project Superintendent		Police Department	
Project Engineer		Fire Department	
Project FOC		Water Service	
Regional Safety Manager		Power Service	
Foreman		Sewer / Storm Drain	
DPR Site Office		Natural Gas Service	
DPR Regional Office		Other Utilities	
Weather		Traffic Signal	
OSHA		Road Closures	
Disaster Services		811 Locate	
Environmental Consultant		Water Intrusion Restoration	
Hazard Material		Indoor Air Quality	

In the event of an emergency required site evacuation or assistance from an outside emergency services provider, the following procedure should be followed:

Medical Emergencies Requiring Outside Assistance

First Person on Scene - Ask DPR office to call (insert call number here)

Call (insert call number here)

- Dial (insert call number here) from a land line telephone in the office, if possible. (If the emergency call is from a cellular telephone, also dial 911).
- Describe emergency and inform dispatcher to send the emergency response vehicles to:
THE DPR CONSTRUCTION JOBSITE AT (enter cross roads or general direction). ENTRANCE TO THE SITE IS OFF (enter directions for jobsite access).
- Inform dispatcher the emergency vehicle will be met by an escort vehicle: **(such as a white DPR pickup truck, or DPR personnel)**

Escorting Emergency Vehicles

- DPR will escort the emergency vehicle from the Jobsite entrance to the incident scene.
- Often a fire truck will arrive separately from an ambulance. The DPR escort will return to the Jobsite entrance to escort the second vehicle when it arrives.

At the accident scene

- Responsibilities at the scene are assigned in the chart below.
- All Site Safety Coordinators report to the scene for direction by DPR.
- Secure the scene until after injured person is transported and investigation is complete.

The following is a summary of personnel assigned responsibilities in the event of an emergency:

	Primary	Secondary
Call 911 and field incoming calls		
Direct emergency response		
Direct evacuation and secure area		

Emergency Procedures Forms

Forms

Bomb Threat Information Sheet
Severe Weather Preparedness Plan Checklist
Emergency Response Plan Site Posting

Bomb Threat Information Sheet

Project/Project#:		Date:	
-------------------	--	-------	--

QUESTIONS TO ASK:

(The goal is to keep the caller talking, to ensure sufficient time for authorities to be notified and respond.)

When is the bomb going to explode?	
Where is it right now?	
What does it look like?	
What kind of bomb is it?	
What will cause it to explode?	
Did you place the bomb?	
Why?	
What is your name?	
Do you have a nickname?	
Are you on Facebook?	
Are you on Twitter?	
What is your address?	
Have you worked here in the past?	
Are you doing this alone?	

(Note if the caller is familiar with specifics of the jobsite)

Exact wording of the threat: (Ask the caller to repeat the message, if necessary)

--

Sex of Caller:	Male / Female	Race:		Approx. Age:	
Time of Call:	am/pm	Date:		Length of Call:	

Caller's Voice:

☐ Calm	☐ Slow	☐ Crying	☐ Nasal	☐ Deep	☐ Read Message
☐ Angry	☐ Rapid	☐ Normal	☐ Stutter	☐ Ragged	☐ Clearing Throat
☐ Excited	☐ Soft	☐ Distinct	☐ Lisp	☐ Accent	☐ Deep Breathing
☐ Foul	☐ Loud	☐ Slurred	☐ Raspy	☐ Familiar	☐ Cracking Voice
☐ Laughter	☐ Taped	☐ Disguised	☐ Irrational	☐ Educated	☐ Incoherent

If voice is familiar, whom did it sound like?	
---	--

Background Noises:

☐ Street Sounds	☐ PA System	☐ Factory Machines	☐ Local	☐ Other
☐ Pots & Pans	☐ House Noises	☐ Animal Noises	☐ Long Distance	
☐ Voices	☐ Motor	☐ Clear	☐ Phone Booth	
☐ Music	☐ Office	☐ Static		

Comments:

--



Severe Weather Preparedness Plan Checklist

Project/Project#:		Date:	
Project Address:			

Severe Weather	Shelter Location	Site Preparation Prevention Steps
Tornado		
Hurricane		72 hrs out, 48 hrs out, 24hrs out, site evacuation
Earthquake		
High Wind Warnings		Example- Lower crane booms, secure all loose material, walk rooftops, band porta-johns
Other (Flooding, extreme temperatures)		

72 hours out:

- Review roles and responsibilities
- Identify essential personnel – who leaves last/ who comes back first
- Notify workers on site of the coming conditions, process to prepare and their role
- Notice for workers to leave for evacuation/ stay and prepare home for storm
- Securing generators and additional equipment to keep facility running/ back up and running
- Securing additional resources (workers, services, etc.)
- Securing material to protect property (i.e., plywood)
- Begin fortifying most venerable areas of site
- Discontinue any work that could increase or compromise facility (trenching, securing any steel erection activities, etc.)
- Created an inventory of chemicals that cannot be removes from site and location
- Review evacuation routes

48 hours out:

- Reducing fuel sources or stockpiled supplies that can be compromised by storm
- Secure material and equipment on site (lumber, port-o-lets)
- Shut down project, complete fortification of project/site
- Place deposits for any long lead/ hard to get materials/ equipment



Severe Weather Preparedness Plan Checklist

- Pre-prepare equipment for storm(i.e. weathervane crane)
- Review evacuation routes

24 hours

- Final walk to ensure site is storm ready
- Final deposits for any material, equipment or services not secured
- All essential personnel off site and briefed on communication strategy and bringing the site “back online”
- Review evacuation routes

Before a Hurricane

To prepare for a hurricane, you should take the following measures:

- To begin preparing, you should [build an emergency kit](#) and [make a family communications plan](#).
- Know your surroundings
- Learn the elevation level of your property and whether the land is flood-prone. This will help you know how your property will be affected when storm surge or tidal flooding are forecasted.
- Identify levees and dams in your area and determine whether they pose a hazard to you.
- Learn community hurricane evacuation routes and how to find higher ground. Determine where you would go and how you would get there if you needed to evacuate.
- Make plans to secure your property:
- Cover all of your home’s windows. Permanent storm shutters offer the best protection for windows. A second option is to board up windows with 5/8” marine plywood, cut to fit and ready to install. Tape does not prevent windows from breaking.
- Install straps or additional clips to securely fasten your roof to the frame structure. This will reduce roof damage.
- Be sure trees and shrubs around your home are well trimmed so they are more wind resistant
- Clear loose and clogged rain gutters and downspouts
- Reinforce your garage doors; if wind enters a garage it can cause dangerous and expensive structural damage
- Plan to bring in all outdoor furniture, decorations, garbage cans and anything else that is not tied down
- Determine how and where to secure your boat
- Install a generator for emergencies
- If in a high-rise building, be prepared to take shelter on or below the 10th floor
- Consider building a safe room

Hurricanes cause heavy rains that can cause extensive flood damage in coastal and inland areas. Everyone is at risk and should consider flood insurance protection. Flood insurance is the only way to financially protect your property or business from flood damage. To learn more about your flooding risk and how to protect yourself and your business, visit the Federal Insurance and Mitigation Administration (NFIP) Web site, www.floodsmart.gov or call 1-800-427-2419.



(project name)
(project address)
(project phone number)

EMERGENCY RESPONSE PLAN

Jobsite Personnel	Name & Cell Numbers	Emergency Numbers	Contact Numbers
Project Manager		911	
Project Superintendent		Police Department	
Project Engineer		Fire Department	
Project FOC		Water Service	
Regional Safety Manager		Power Service	
Foreman		Sewer / Storm Drain	
DPR Site Office		Natural Gas Service	
DPR Regional Office		Other Utilities	
Weather		Traffic Signal	
OSHA		Road Closures	
Disaster Services		811 Locate	
Environmental Consultant		Private Utility Locating	
Hazardous Materials		Water Intrusion Restoration	
Corporate Mktg/Communication	415.816.8993 or pr@dpr.com	Indoor Air Quality	

In the event of an emergency required site evacuation or assistance from an outside emergency services provider, the following procedure should be followed:

Medical Emergencies Requiring Outside Assistance

First Person on Scene - Ask DPR office to call (insert call number here)

Call (insert call number here)

- Dial (insert call number here) from a land line telephone in the office, if possible. (If the emergency call is from a cellular telephone, also dial 911).
- Describe emergency and inform dispatcher to send the emergency response vehicles to:
THE DPR CONSTRUCTION JOBSITE AT (enter cross roads or general direction). ENTRANCE TO THE SITE IS OFF (enter directions for jobsite access).
- Inform dispatcher the emergency vehicle will be met by an escort vehicle: (such as a white DPR pickup truck, or DPR personnel)

Escorting Emergency Vehicles

- DPR will escort the emergency vehicle from the Jobsite entrance to the incident scene.
- Often a fire truck will arrive separately from an ambulance. The DPR escort will return to the Jobsite entrance to escort the second vehicle when it arrives.

At the accident scene

- Responsibilities at the scene are assigned in the chart below.
- All Site Safety Coordinators report to the scene for direction by DPR.
- Secure the scene until after injured person is transported and investigation is complete.

The following is a summary of personnel assigned responsibilities in the event of an emergency:

	Primary	Secondary
Call 911 and field incoming calls		
Direct emergency response		
Direct evacuation and secure area		
Crowd Control		
Escort emergency service vehicle to scene		
Escort Victim to medical facility		
Subcontractor Supervision		
Initiate investigation process		
Take photos of scene		
Notify DPR EHS		
Notify DPR Corporate Safety Director		
Notify Owner		
Notify Corporate Marketing/Communications	Yumi @ 415.816.8993 or pr@dpr.com	

Emergencies Requiring Site Evacuation

In the event of a site evacuation event such as a fire, explosion, earthquake, severe weather, etc., all personnel should assemble (enter location, i.e., outside the DPR office trailers or designated assembly location) and report to their foreman for a head count. All foremen should report any unaccounted personnel to (insert name of superintendent here). All personnel are to remain in the assembly area to receive specific instructions. Individuals will be allowed to leave the assembly area (to go home or return to work as directed) only when instructed by D.P.R.

Notes:

- All phone and radio communications during an emergency should be limited to that which is absolutely required for immediate emergency service response. All other phone and radio traffic should cease.
- No one is authorized to re-enter a secured area until cleared by DPR (or enter other designated authority).

Non-Emergency Medical Injuries

The following procedure should be followed for ALL non-emergency medical injuries regardless if the injured person is DPR or a Subcontractor employee.

- DPR or subcontractor representative will escort the injured person to the (insert clinic information & address here). Directions to the clinic are outlined on the map below (insert map posting below).
- Inform the clinic immediately that this case is for a project which has a light duty / return to work policy.
- Escort individuals are listed below in priority order:
 -
 -



(project name)
(project address)
(project phone number)

EMERGENCY RESPONSE PLAN

4. Investigation of the injury should begin immediately and be completed when the injured party returns to the jobsite. (insert superintendent name here) should initiate the investigation. Start Grab & Go.

Active Shooter Emergencies

When an “active shooter” is in your vicinity, you must be prepared to deal with the situation both physically and mentally. You have three options:

RUN OPTION 1 (preferred)	HIDE Option 2	FIGHT OPTION 3 (last resort)
- Have an escape route and plan in mind - Leave your belongings behind - Evacuate regardless of whether others agree to follow - Help others escape, if possible - Do not attempt to move the wounded - Prevent others from entering an area where the active shooter may be - Keep your hands visible - Call 911 when you are safe. Provide the following information: - Location of the active shooter - Number of shooters - Physical description of shooters - Number and type of weapons - Number of potential victims	- Hid in an area out of the shooter’s view - Lock door or block entry to your hiding place - Silence your cell phone (including the vibrate mode) and remain quiet.	- Fight as a last resort and only when your life is in imminent danger - Attempt to incapacitate the shooter - Act with as much physical aggression as possible - Improvise weapons or throw items at the active shooter - Commit to your actions . . . your life depends upon it

When law enforcement arrives:

- Remain calm and follow instructions
- Drop items in your hands (e.g., bags, jackets, etc)
- Raise hands and spread fingers
- Keep hands visible at all times
- Avoid quick movements towers officers
- Avoid pointing, screaming or yelling
- Do not ask questions or otherwise distract officers when evacuating

Other items to remember:

- The first officers to arrive on scene will not stop to help the injured.
- Rescue teams will arrive after initial officers, they will treat and remove the injured.
- Once you have reached a safe location, you will remain or be held in the safe location by law enforcement until the situation is under control, and all witnesses have been identified and questioned.
- Do not leave the safe area until law enforcement have instructed you to do so.

Insert map to normal hours’ medical provider

Insert map to off- hours medical provider

Section 09

Ergonomics

Manual Material Handling

Mechanical Handling Aids

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Ergonomics & Manual Material Handling

1.0 Introduction

DPR's ergonomics and material handling section provides general guidelines for the elimination and reduction of the risk factors that contribute to musculoskeletal injuries. Ergonomics is the science of designing and matching the physical demands of workplace to the capabilities and limitations of the worker. The goal of ergonomics is to allow work to be done without undue stress to our labor force. Making ergonomic improvements to the workplace does not have to be expensive or complicated. Significant changes can be made if problems are approached with common sense, good planning and sequencing of work methods, proper handling aids, and knowledge of basic ergonomic principles. Subcontractors need to identify potential ergonomic risks associated with the scope of their work in the bid phase so they can develop and price their work plans to control musculoskeletal risks.

Some of the most common overexertion injuries in construction are the result of specified or unspecified work methods that push the human body beyond its natural limits. Workers often lift, stoop, kneel, twist, grip, stretch, reach overhead, or work in other awkward positions to do a job are at risk of developing a work-related musculoskeletal disorder (WMSD). These can include back problems, knee problems, carpal tunnel syndrome, tendinitis, rotator cuff tears, sprains, and strains.

To aid in the prevention of these injuries work plans should consider possible risk factors and use the knowledge gained from past projects to communicate the many simple and inexpensive ways to make construction tasks easier, more comfortable, and better suited to the needs of the human body.

DPR believes that sprains and strains are not just the nature of the business. There are tools, methods and materials available that can make work less risky and increase productivity.

The following sub-section examines general material handling guidelines, construction ergonomics and a pre-work stretching program to assist workers.

2.0 Material storage

2.1 General Requirements

- All materials stored in tiers shall be stacked, racked, blocked, interlocked, or otherwise secured to prevent sliding, falling or collapse
- Maximum safe load limits of floors within buildings and structures, in pounds per square foot, shall be conspicuously posted in all storage areas, except for floor or slab on grade. Maximum safe loads shall not be exceeded
- Aisles and passageways shall be kept clear to provide for the free and safe movement of material handling equipment or employees. Such areas shall be kept in good repair
- When a difference in working levels exist, means such as ramps, blocking, or grading shall be used to ensure the safe movement of vehicles between the two levels

2.2 Material storage

- Material stored inside buildings under construction shall not be placed within 6 feet of any hoistway or inside floor openings, nor within 10 feet of an exterior wall which does not extend above the top of the material stored
- Each employee required to work on stored material in silos, hoppers, tanks, and similar storage areas shall be equipped with personal fall arrest equipment
- Non-compatible materials shall be segregated in storage
- Bagged materials shall be stacked by stepping back the layers and cross-keying the bags at least every 10 bags high

- Materials shall not be stored on scaffolds or runways in excess of supplies needed for immediate operations
- Brick stacks shall not be more than 7 feet in height. When a loose brick stack reaches a height of 4 feet, it shall be tapered back 2 inches in every foot of height above the 4-foot level.
- When masonry blocks are stacked higher than 6 feet, the stack shall be tapered back one-half block per tier above the 6-foot level
- Lumber:
 - Used lumber shall have all nails withdrawn before stacking
 - Lumber shall be stacked on level and solidly supported sills
 - Lumber shall be so stacked as to be stable and self-supporting
 - Lumber piles shall not exceed 20 feet in height
 - Lumber to be handled manually shall not be stacked more than 16 feet high
- Structural steel, poles, pipe, bar stock, and other cylindrical materials, unless racked, shall be stacked and blocked so as to prevent spreading or tilting

2.3 Housekeeping

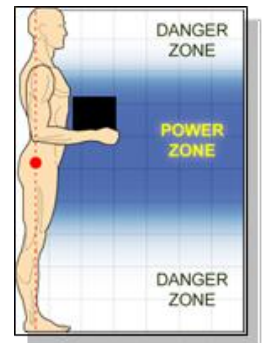
Storage areas and work areas shall be kept free from accumulation of materials that constitute hazards from tripping, fire, explosion, or pest harborage. Vegetation control will be exercised when necessary.

3.0 Material Handling

All workers should address the efficient organization, handling and storing of materials in their daily work plans. Further, each subcontractor should examine their operations to ensure that they provide a continuous flow of materials, parts, tools and assemblies through the workplace, and ensure that materials are available when needed. More importantly, they should also ensure the timely removal of excess materials and debris. The following general guidelines apply to a diverse set of operations, from manual material handling to storage and disposal. *Additional guidelines for material handling and equipment can be found in the Cranes Section 05 and Ergonomics, Material Handling and Mechanical Handling Aids Section 09 and Motorized & Site Equipment Section 15.*

3.1 Manual Handling Risk Factor Reduction – Key Concepts

- Eliminate the manual task if possible
- Reduce the distance the materials must be moved
- Keep all handling/movements in mid-range (between knuckles and shoulders)
- Convert lift/lowers to push/pulls
- Keep weights and forces to a minimum
- Keep all handling/movements close to the body
- Reduce twisting and bending motions, move feet
- Correction – doing it twice, handling it twice doubles worker exposure to risk



3.2 Manual Material Handling – Reducing Project Waste

- Overproduction – Providing more material than is needed before it is needed
- Motion – Unnecessary repeated worker movement (time and energy)
- Material movement – unnecessary transport of materials, parts or tools. (Increases likelihood of injury)
- Waiting (delays) – any non-work time waiting for transport of materials, inspecting damaged materials, parts or tools
- Inventory – maintaining excess inventory of materials, parts, or tools where they won't be needed
- Processing (over processing) – doing more work than is necessary, or making work more complex than is necessary

3.3 *Ergonomic Planning – Trade Specific*

Moving materials is a leading cause of stress injury in the construction industry. Moving it once, to the correct location, will help reduce the hazards associated with material movement.

The actions below are recognized as best practices for ergonomic methods to reduce injuries while performing the following tasks.

Concrete

- When moving concrete forms, place them directly onto pallets or into containers which can then be moved mechanically
- Plan for slip form vs. fixed forms for concrete
- Consider the use of flying forms
- Pump or meter water through a hose to a concrete mixer, instead of handling individual buckets.
- Pump concrete to the point of use
- Equip ready mix trucks with conveyors so the concrete can be mixed and conveyed where it is needed
- Use a concrete pump instead of crane and bucket
- Before handling used lumber, best practice is to remove nails in lieu of bending nails
- Provide stripping/re-shore crew with a banding machine to secure stacks of materials for hoisting
- Self-propelled buggies and barrows are preferred over manual push buggies for moving concrete
- Proper ramps should be used to bridge slight elevation differences to permit rolling carts and hand trucks
- Lifting lugs on pumps and other equipment are to be used in accordance manufactures recommendations

Electrical

- Use the same cart for both the transport of spools of wire or cable, and to support the spools during unwinding.
- Purchase assembled components that are mechanically installed rather than field assembled.
- Lifting lugs on generators are to be used in accordance manufactures recommendations.

Interior Finish

- Special lifting devices for lifting and positing dry wall panels onto ceilings.
- Large caster drywall carts with maximum load limit marks

Masonry/Brick Layers

- Relocate masonry saw as work proceeds to keep it close to where it is needed most
- Provide concrete debris box as close to cutting operations when practical, to prevent re-handling of material
- Keep a stock of blocks at masonry saw to reduce re-handling
- Provide a storage platform for mortar cement bags at the same height at which they will need to be loaded into the mixer
- Stack mortar bags on pallets, not on the ground. Stacks of bags or blocks should not be higher than 6 feet
- Mortar pans and boards are on raised support stands for ease of access at the point of use
- Move mortar from the mixer to the scaffold in a mortar box
- Keeping a variety of handling aids available to fit different task, i.e., the brick tongs, block carts, hand trucks, etc.

Mechanical Contracting

- "Batch" plumbing materials; unload directly onto floor where work will be performed
- Fabricate pipe headers and assemblies in shop on wheeled lifting frames which can be trucked to the job and hoisted to the needed floor and rolled to the work area
- Store pipes in movable pipe storage racks that are capable of being hoisted
- Stack pipe only on solid, level sills and block to prevent rolling
- Purchase assembled components that are mechanically installed rather than field assembled
- Have the sprinkler contractors cut and thread at the fabrication facility or shop
- Use "hoistable" duct work trailers that can be lifted mechanically to individual floors
- Special inexpensive handle-like devices are available for handling sheets of drywall and sheet glass
- Pre-fab as much as possible in a controlled "shop" environment before delivery to the job

Forming - Rebar Installers

- Order materials for delivery so that they will be readily available in the sequence in which they are to be used
- Pre-plan adequate lay-down, shake-out and pre-assembly areas

Roofers

- On large roofing job, plan for material hoisting operation to be done at as many points as possible
- Place dumpsters close to area being cleaned
- A mechanized material hoist may be much more likely to be used for lifting material onto a roof than a manual rope and pulley system

Wrecking/Disposal

- Ensure disposal chutes empty directly into dumpsters
- Evaluate dumpster change-out needs during weekly planning meetings
- Place scrap bins around the site that can be removed by forklift truck or crane
- Locate the trash chutes and dumpsters as far away as possible from material landings, delivery areas and hoists

HVAC

- Pre-fab as much as possible in a controlled "shop" environment before delivery to the job.
- Unload material from delivery trucks directly onto dollies, power trucks which can then be easily moved or hoisted around the job site
- Do not put construction material, scrap or spoil near pedestrian or public areas, as it will need to be moved later

Delivery and Storage

- Order materials for delivery so that they will be readily available in the sequence in which they are to be used
- Pre-plan adequate lay-down, shake-out and pre-assembly areas
- Keep all storage out from under power lines
- Provide site maps for delivery drivers and have them unload at designated point
- Have suppliers place materials directly into the use area
- Unload material from delivery trucks directly onto dollies, power trucks which can then be easily moved or hoisted around the job site
- Place dumpsters close to area being cleaned
- Require all sub-contractors to consult with the site supervisor before storing any material
- Do not put construction material, scrap or spoil near pedestrian or public areas, as it will need to be moved later

- Place barricades around the planned storage site to "reserve" it and to make sure it is available when materials are delivered
- Pre-plan with suppliers where materials are to be delivered. Ask for delivery trucks supplied with elevating equipment if necessary to deliver materials to elevated areas.
- Stack lumber on dunnage to allow forklift positioning.
- Storage sites should be level for ease of movement and handling.

4.0 Manual Material Handling

- All material should be properly stored and at the appropriate height to minimize struck by and crushed by injuries
- In general, if more than two people are required to move equipment, tools, etc., mechanical means of movement should be considered
- When lifting material, it should be lifted in such a way as to minimize the amount of bending, twisting, turning, and jerking motions
- Material and equipment should be delivered and stored as close to the work area as practicable
- All workers should examine their workplaces to detect any unsafe conditions prior to handling materials; such conditions include, but are not limited to:
 - slip and trip hazards
 - uneven and rough surfaces
 - poor lighting
 - overhead material drops
 - holes
 - equipment
 - tools
 - stowed material
 - limited access
- Workers should never attempt to lift material that is too large or too heavy alone; alternatives include:
 - using mechanical methods such as carts, pallet jacks, dollies, wheeled racks, lift gates etc.
 - using equipment such as forklifts, cranes, conveyers etc.
 - breaking the material down into smaller and lighter pieces
 - seeking assistance from other workers
- Proper PPE should be used when handling material; examples are as follows:
 - puncture or cut resistant gloves and arm/hand protection for objects with sharp or rough edges
 - steel toed shoes, with good tread, when working with heavy objects
 - rubberized gloves for slippery or low friction material
 - for more information, refer to the Personal Protective Equipment Section 16
- When unloading and off-loading material and equipment from trucks only essential workers should be in the area and even then, all workers should stay clear of the load to protect against objects that may have shifted during transport, rolling structural steel, material displacement, etc. Areas around truck should be flagged with danger tape to ensure no one passes behind the backside of the load. Drivers should maintain eye contact with operator and maintain distance from material being unloaded.

5.0 Storage

- All stored materials, tools and equipment should be secured or otherwise protected to prevent accidental displacement; further:
 - all items such as ladders, frames or other materials stored against walls should be secured
 - rolling carts and tool boxes should have their wheels blocked or locked

- materials, tools and equipment should not be stored on top of tool vans, carts or other overhead storage areas unless it is secured
- boxed materials should be banded or held in place using cross-ties or shrink plastic fiber
- pipe, bar, conduit, all-thread, and other rolling objects should be stored in racks
- safe procedures should be established for the unbundling and unloading of any materials delivered with special care taken for objects that can roll, wrack or shift
- structural steel should be stacked in such a way as to prevent rolling or shifting
- Aisles and passageways should be kept clear to provide for the free and safe movement of material handling equipment or employees
- No material should be brought to the project that cannot be readily utilized and installed
- At no time should stocks of material exceed that needed for more than two (2) weeks of activities without prior approval from DPR project management

6.0 Disposal

- All workers are required to clean up their work areas continually, but at minimum, daily
- Storage areas should be kept free from debris, accumulated materials and non-essential equipment
- All debris, scrap material and general rubbish should be removed from the work area as soon as practicable
- All workers should place their waste materials into dumpsters, buggies, carts, barrels (with wheels or rated lifting eye) or other trash receptacles first to avoid handling of material twice; this will require ensuring these disposal containers are located near to work and emptied regularly. Trash receptacles that must be manually dumped should be smaller in size to minimize weight and force.
- Each subcontractor is responsible for the timely disposal and removal of debris from the project

6.1 Training

Training on safe material handling should be conducted for all employees engaged in these activities. Training should include the following topics:

- General principles of ergonomics
- Recognition of hazards and injuries
- Safe lifting practices
- Proper use of equipment
- Procedures for reporting hazardous conditions
- Methods and procedures for early reporting of injuries
- Health risks related to improper lifting and material handling

7.0 Construction Ergonomics

7.1 Introduction

The management tools, jobsite tips and checklists that follow in the Forms Section at the end of the EHSP are identified to help minimize the damaging effects to the construction industry from musculoskeletal disorders. The information includes examples of best practices currently being used in the industry. The background information on risk factors for musculoskeletal disorders is provided for informational purposes only and to provide a structure for the management of the tools, jobsite tips and check lists. This information is provided to enable interventions for existing musculoskeletal disorders and to assist DPR, subcontractors and vendors in identifying potential musculoskeletal disorders.

7.2 Risk Factors

Risk factors for musculoskeletal disorders of the hands, wrist, neck and shoulder include, but are not limited to:

- Force or muscle effort
- Awkward body postures
- Repetitive work
- Contact pressure
- Vibration
- Temperature

7.2.1 Force or Muscle Effort

Force is the amount of effort it takes to do an activity or work. Lifting, pushing, pulling, and gripping a tool are examples of activities that require exerting force or muscle effort. Keeping the body in one position for a period of time (e.g., doing overhead work) also requires muscle effort; the more force that is exerted, the greater the stress on the body.

- Ensure the right equipment for the job is used
- Ensure tools are sharp, well maintained and fully charged
- In drilling operations, pre-drilling holes can ease the force of screwing into heavy gauge metal studs
- When possible use a power grip (full-hand grip) versus a pinch grip that only uses the fingers
- Two hands should be utilized whenever practicable
- Ensure tools are not forced beyond their capacity, which should reduce stress transferred to the body
- Ensure handles of tools are clean, dry or have a non-slip surface
- Ensure that tools have a handle or grip that is not too large or small for the hand
- Ensure that the wrist and hand are kept in a neutral position as much as possible when using tools; minimize bending of the wrist
- Ensure gloves fit and are of the appropriate type; gloves that are too tight or too loose or made of certain materials (some chemical-resistant gloves) may cause the user to grip tools more tightly than without
- Keep loads close to the body when carrying or lifting
- Reduce the size or weight of the item to be lifted when possible
- Combine "light" loads to discourage manual handling and encourage use of mechanical equipment. Order bulk size vs bags.
- Get help to lift heavy or awkward materials or objects
- Store material at the level so it is not on or near floor level requiring deep bends at floor level
- Push loads versus pulling loads and ensure good body position when pushing
- Ensure the walking surface is suitable, e.g., good traction, material can easily be rolled or moved over it, free from slip or trip hazards, ideal slope, etc.
- When holding a posture for a prolonged period of time, ensure that we change positions periodically or take a rest; padded mats, new shoes or orthopedic inserts may help minimize some of the stress

7.2.2 Awkward Body Postures

Generally, the most comfortable working postures are when the shoulders are down and relaxed, the arms are close to the sides, elbows are bent and wrists and hand straight, like the posture used to shake hands with someone. This is called a neutral posture or position. It takes more muscle effort to work at eye height than to work at elbow height. When the working posture is out of the neutral position, there is increased stress on the body and greater muscle effort is required.

- When possible, work with the arms in the neutral position
- Minimize the amount of overhead and over the shoulder height work
- Avoid reaching and awkward body positions
- Position the tool, equipment or work surface in a way that places the worker in a more neutral working position
- Use work stands and saw horses to keep from working on knees

- Position work between knuckle and shoulder height
- Keep work close to the body
- Ensure ladders face the work surface or ladder platforms are utilized to minimize twisting
- Pre-fabrication or assembling materials may help minimize awkward body positioning
- Minimize the amount of bending, twisting and lifting that is performed – especially of heavier loads
- Good housekeeping practices and proper storage of materials should be maintained at all times

7.2.3 Repetitive Work

Nailing a deck, screwing drywall, tying rebar, and making terminations are examples of repetitive tasks. When possible, minimize the amount of repetitive type of work being performed. Some general guidelines include:

- Change tasks or positions as much as possible
- Ensure different muscle groups are used when changing tasks
- Ensure adequate breaks are taken
- Job rotation
- Use mechanical means to minimize the amount of repetitive work being done such as nut driver, wire stripper, pneumatic deck crimper, nail gun etc.
- Engineer or redesign the task when possible, especially if the task is of long duration

7.2.4 Contact Stress (Tools and Sharp Objects)

Tools and materials (wood, metal) can press against the soft part of the palm of the hand or other soft tissues of the body. Continued compression of soft tissues should be avoided. The following general guidelines should be used to minimize this risk factor include:

- Use appropriate PPE to cushion, e.g., knee pads, padded gloves, padded tool handles, mats etc.
- Ensure that the proper tool is used for the task
- Ensure proper footwear to minimize foot contact stress, especially for tasks involving walking on rebar, standing on ladders etc.
- Walkways or scaffolds may be used to minimize these risk factors for rebar and ladders respectively
- When possible, the use of platform ladders should be used in lieu of standard A-frame ladders
- Reduce the weight of tool belts to only the tools needed, wear suspenders or remove the tool belt altogether

7.2.5 Vibration from Hand Tools

Rotary hammers, chainsaws, grinders, jackhammers and even screw guns all cause vibrations to travel through the hand. To minimize the risk factors associated with vibration the following general guidelines can be used:

- Anti-vibration gloves and padding should be used as needed
- Anti-vibration seats in heavy machinery
- Job rotation
- Ensure the right tool for the task
- Ensure tools are clean, dry and have easily gripped handles
- Ensure tools and equipment are well maintained, sharp and are not forced beyond capacity
- Ensure two hands are used on tools as appropriate

7.2.6 Temperature

Cold can increase the risk as muscles are tenser and less circulation is present. Heat increases the risk of dehydration, heat cramps, heat stress, heat exhaustion and heat stroke. Also, the following guidelines can be used to minimize factors associated with temperatures:

- Ensure Stretch & Flex is performed as needed
- Gloves and protective clothing should be worn as indicated

- Ensure working environment is of a suitable and safe working temperature
- Provide ventilation as needed
- Ensure appropriate breaks are taken and workers remain well hydrated
- Use appropriate gloves to reduce slippage of hands, tools and equipment in hot conditions
- Sufficient water should be provided to provide workers throughout their shift; approximately one quart of water per hour per worker
- Frequent short breaks for re-hydration should be encouraged
- Access to shade should be provided with either ventilation or cooling capabilities; workers appearing to suffer from heat should be encouraged to use shaded areas
- Supervision should be trained in recognition and control of hazards associated with heat or cold temperature stresses

7.3 *Arms, Hands and Shoulders*

The type of grip used to hold a tool makes a difference in the amount of effort expended. A pinch grip that uses only the fingers to hold an item requires more muscle effort than a power grip. A power grip (full-hand grip) which uses the larger muscles of the arm has about four times the strength of a pinch grip. A pinch grip can cause fatigue and injury.

A slippery handle or a handle that is too large or too small for the hand is hard to hold. These factors cause the user to grip the handle more tightly or to apply more force.

The position of hands and arms in relation to the body also affects the amount of force that needs to be exerted. If using a tool causes the wrist to bend downward, backward, or to one side or the other causes more force to be used. The more often work is done with a bent wrist, the greater the risk of developing a musculoskeletal disorder of the hand, wrist, elbow and even the shoulder.

Gloves that are too tight or too loose or made of certain materials (some chemical-resistant gloves) also cause the user to grip tools more tightly than without. However, it is important to remember that gloves are an important part of Personal Protective Equipment (PPE) and should be worn when required.

7.4 *Back and Neck*

There are a number of common risk factors associated with injuries to the back. They include:

- Lifting
- Pushing, pulling, tugging
- Jerking motions while lifting or twisting
- Twisting, reaching, sideways bending, unequal lifting
- Working in a single position
- Whole body vibration

7.4.1 *Lifting*

Lifting is a forceful movement requiring energy and muscle effort. It stresses muscles, tendons and ligaments and increases forces on the spine. This is true even if proper lifting techniques are used. By bending from the waist and lifting at the same time, forces on the spine are increased even more. Forces are also increased if lifting is done at the same time as twisting and even more by adding jerking motions.

Forces also increase in the spine when you reach, hold your breath or hold the object away from the body as you lift. Using one arm to lift an object or carrying an object on one shoulder or one hip also places extra (and uneven) stress on the spine. Factors that affect the stress of a lift on the back and other parts of the body include:

- **Size and weight of the object.** If the object is too big or bulky you won't be able to do a squat lift (bent legs, straight back). Bending from the waist to lift an object, the stress on the lower back is three times greater than lifting with a straight back.

7.4.2 Pushing, Pulling, Tugging and Sliding

Pushing, pulling, tugging and sliding objects require muscle force or effort which puts strain on the lower back. This also stresses the muscles, tendons and joints of shoulders, arms, upper back and legs. The amount of force these movements require depends upon:

- The amount of force used to pull or push the object; pushing uses less force and allows the use of upper body weight
- Body posture while pushing or pulling; awkward postures force the body to exert more force to move the object
- Condition of the walking surface; high friction between the object and the surface, for example moving an object over gravel requires more force to move the object
- Slippery surfaces reduce the amount of traction, leading to the possibility of slips and falls
- Slope of the work surface. The greater the slope, the harder it is to push the item
- Weight of the object; the greater the weight, the more stress on the shoulder, back, and legs
- Position of the object; pushing or pulling an object above the shoulder or below the waist is more stressful

7.4.3 Twisting, Reaching, Jerking, Sideways Bending and Unequal Lifting

- Any amount of twisting, reaching or bending while lifting causes more stress on the back
- Reaching upward usually causes the back to arch. This increases the forces on the lower spine; it also puts stress on the upper back, shoulders and arms
- Forward reaches that are longer than the length of the arm and require bending or stretching put stress on the lower back and on the legs
- Carrying an object on one shoulder, arm, hand or hip puts more stress on one side of the spine; ensure sides are alternated as objects are carried
- The more reaching, jerking, twisting, sideways bending and unequal lifting/carrying being performed, the greater the risk of back injury

7.4.4 Working in a Single Position (Static Posture)

Working in one position for an extended period of time may cause muscle fatigue. Bending over, leaning forward, and/or working with arms above shoulder height for a period of time may stress the lower back, neck and shoulders.

If low back pain, constant standing may be painful. Putting inserts in your boots and standing on padding rather than hard ground or concrete may help. Also, ensure that appropriate and supportive footwear are being worn. Boots with a moderate heel vs. flat sole help promote good posture. Worn-out boots and soles should be replaced. Boots should also be laced up to fully support the ankle, lower leg, and prevent falls. It's important to move around, change positions and rest your muscles as often as possible or improving the work area (e.g., placing walking and working platforms over rebar mats, steps over edge forms, platforms on ladders, etc.). Another important factor in standing is abdominal strength. By increasing the strength of our stomach muscles we can help ward off potential back pain.

7.4.5 Whole Body Vibration

Heavy equipment operators are the chief workers exposed to whole body vibration. Ensure that anti-vibration seats are utilized and that equipment is properly maintained at all times.

7.5 Other Risk Factors

Other risk factors contribute to the development of musculoskeletal disorders, including:

- Moving/carrying materials combined with trip/fall from floor surfaces and obstacles in the work area - housekeeping practices
- Height of the work

- Working beyond the body's capacity
- Lifting techniques
- Tool belts

7.5.1 Floor Surfaces and Obstacles – Housekeeping Practices

Good housekeeping practices are important to reducing musculoskeletal disorders:

- Mud, wet, oily surfaces and poor housekeeping are serious slip, trip and fall hazards
- Materials, debris, wires, power cords and tools on the ground are common causes of slips and trips
- Reaching over, under or around an object or barrier increases the stresses on the lower back
- Debris causes us to change how we work; position the body in such a way to work around the debris

7.5.2 Height of the Work

Work areas and tasks should be designed for the person who is completing the task.

- Tall people may need to bend over or stoop to do certain jobs; this increases the stress to the lower spine and fatigues the muscles
- Shorter people may need to reach up to do a job; bending backwards while reaching up increases the pressure on the lower spines
- Working with raised arms can also cause fatigue and shoulder, neck and elbow problems

7.5.3 Working Beyond Capacity

- Strong muscles and fitness do not necessarily protect the spine from injury; lifting affects spinal discs no matter how fit or strong the individual
- Workers should not lift beyond individual limitations; working beyond limitations puts the worker at risk for injury
- Ensure adequate breaks are taken and adequate personnel are utilized for all tasks

7.5.4 Lifting Techniques

- No matter the lifting technique used, the size and shape of the object should be small enough to allow the worker to hold it against the body to lift
- Heavy or bulky objects should be lifted with assistance or with alternate methods
- Freestyle lifting (back bent, legs almost straight) puts more stress on the back but requires less energy than squat lifting and should not be used
- You should lift comfortably with or without a straight back
- Avoid jerking motions while lifting
- Lifts should be performed with a minimum of twisting, move your feet
- Repetitive lifting should be evaluated and alternative methods used whenever possible to prevent fatigue

7.5.5 Tool Belts

- Add suspenders to tool belts can help redistribute the weight, but requires adjustment to get the proper shoulder-waist-hip distribution
- Reduce the amount of weight being carried in the tool belt (only tools to be used) to prevent shoulder discomfort
- Ensure that the weight is evenly distributed from side to side

8.0 Pallet Trucks



A Pallet truck may also be known as Pallet Jack, Pallet Mule or a manually powered fork truck. The operator physically pushes the forks into the opening on the pallet and "jacks" the ratcheting lift mechanism so as to raise the pallet off the floor. Once lifted, the load can be manually moved from place to place. Pallet trucks may be motorized or manual. Riding of pallet trucks is prohibited. The following general guidelines should be followed, at a minimum:

- Only trained and authorized persons should operate a pallet jack
- Manufacturer warnings and instructions should be understood and followed
- Inspections should be performed each day prior to operation
- Do not stand under or allow personnel under the machine when the load is raised
- Keep hands and fingers away from folding legs and other potential pinch points
- Body parts should be kept away from the load, lift linkage and forks, as well as under the load and/or unit
- Proper, neutral body position should be maintained during operation, i.e., the worker stays square to the load and utilizes both hands while pushing. Do not pull pallet jacks.
- The load on the pallet jack should not be stacked too high where the worker pushing the jack cannot see over the top of the load.
- Pallet jacks should be used on smooth concrete floors only
- Pallet jacks should not be utilized on ramps or grades, unless specifically designed for the purpose
- Watch for overhead obstructions when handling tall loads
- Park the pallet jack in a safe, level place with the handle up and the forks lowered

9.0 Material Lifts (Duct Lift)



A material lift may also be known as a Duct Lift and are often used during the installation of ductwork and related equipment. The use of all material handling equipment should be in accordance with Federal OSHA, DPR and manufacturer's operating instructions. All material handling are activities defined by DPR to be High

Hazard and therefore require detailed, written pre-task planning. Only trained and authorized personnel should be permitted to operate material lifts.

Inspections should be performed each day prior to operation and documentation should be maintained on the unit. Refer to the *Equipment Pre-Operation Checklist* in the Forms Section at the end of the EHSP. Be sure that the operator manual is complete and in the storage container located on the machine. At a minimum, the following guidelines apply:

- Before operating and material lift read, and understand all instructions contained in the manufacturers operator's manual
- Body parts should be kept away from the load and cable, as well as under the load and/or lift
- Do not stand under or allow personnel under the lift when the load is raised
- Keep hands and fingers away from folding legs and other potential pinch points
- Decals particularly rated capacity should be legible and in place
- Inspections should include checking all components or areas for damage, modifications and improperly installed or missing parts:
 - Winch and related components
 - Base components
 - Legs
 - Stabilizers and latch plates (if equipped)
 - Mast columns
 - Exterior plastic shim for safety brake (if equipped)
 - Cable hold-down bar
 - Cable anchor
 - Cable and pulley. Check cable for kinks, frays or abrasions
 - Wheels and casters
 - Load handling attachments (if equipped)
 - Nuts, bolts and other fasteners
 - Check the entire machine for dents, corrosion, oxidation, cracks in welds or structural components
- Use of the lift as a personnel lifting platform or climbing on the mast or load handling attachments is prohibited
- Rated load capacity of the lift should never be exceeded
- Loads should not be raised unless the stabilizers and legs have been fully lowered and the load is centered and locked
- Casters should be in full contact with a firm, level surface
- Load handling attachments should be properly secured to the lift
- Blocks cannot be used to level the lift
- Lifts should not be moved with a raised load, except for minor positioning
- Lifts should not be operated in strong or gusty wind
- Ladders or scaffolding should not be leaned against any part of the lift
- Lifts should not be used on a moving or mobile surface
- Travel paths should be maintained clear of debris and uneven surfaces while rolling a material lift
- Prior to use, check the work area for drop-offs, holes, bumps, unstable or slippery surfaces or other possible hazardous conditions

These machines are not electrically insulated and will not prevent protection from contact with or proximity to electrical current. Maintain safe distances away from electrical power lines in accordance with the following chart.

VOLTAGE RANGE (Phase to Phase)	MINIMUM SAFE APPROACH DISTANCE	
	(Feet)	(Meters)
0 to 300V	AVOID CONTACT	
Over 300V to 50KV	10	3.05
Over 50KV to 200KV	15	4.60
Over 200KV to 350KV	20	6.10
Over 350KV to 500KV	25	7.62
Over 500KV to 750KV	35	10.67

10.0 Chain Falls, Come-Along Devices



Chain falls and come-along devices are hoisting devices consisting of a chain suspended from or laid over a fixed, raised structure such as a beam, used to lift heavy objects. Only manufacturer's designed components should be used, alterations or supplemental devices are prohibited. These general guidelines should be followed, at a minimum:

- Verify the capacity of the anchorage point, and ensure the maximum load will not exceed beam or anchorage capacity
- Inspect the device for damage prior to use, remove from service if found to be defective; i.e. check the cable, chain or strap for fraying, splitting, etc.
- Don't exceed the rated load capacity. Follow manufacturer's recommendations for connecting to anchor point
- Don't use a come-a-long in lieu of blocking raised equipment
- Never straddle the tool and keep your body out of harm's way
- Be certain that all parts of the 'system' can take the load; verify anchor points

11.0 Drywall Carts



A drywall cart is designed to move drywall in a vertical position through narrow spaces like doorways and aisles. The cart, at a maximum, should hold no more than 10 drywall sheets; it should be pushed and steered by two people. Only carts specifically designed to move drywall should be used on the project. A PTP or JHA should be developed for moving rock carts and drywall stocking of projects. These general guidelines should be followed, at a minimum:

- A maximum of 10 sheets should be loaded or moved using a drywall cart
- Moving carts requires two persons, minimum
- Ensure the travel path is clear prior to cart movement
- Persons moving drywall should be trained on proper use of the cart. The training, at a minimum should include:
 - Push the carts rather than the board
 - Move slowly and allow enough time for turning and stopping the cart
 - Uneven surfaces should be avoided so the cart will not tip over

12.0 Stretch & Flex

12.1 Introduction

Through implementation of the Stretch & Flex program, DPR hopes to eliminate the number of and reduce the severity of work place related injuries. In addition, we also hope to enhance the overall health and welfare of each one of our employees so they can enjoy life to its fullest both at work and home.

12.2 Benefits

There are many benefits to the Stretch & Flex program. Following are some of the major ones we have identified. (For detailed explanations of benefits, refer to sub-section 5.0 of this Section).

12.2.1 Physical Benefits

- improved physical performance
- increased flexibility
- increased relaxation of the muscle
- muscles are more efficient and have greater endurance
- decreased muscle tension
- greater range of motion
- improved coordination of our muscles and joints
- improves circulation to the muscles (e.g., warming it up)
- increased strength of the muscle
- improved posture
- reduces degeneration of the joints (e.g., lubricating joints)
- improved nerve function

12.2.2 Psychological Benefits

- personal gratification of increased health
- increased sense of well-being
- increased enjoyment
- increased team morale
- increased circulation stimulates the brain
- prepares the mind for work activities
- relaxes the mind prior to activity
- establishes a healthy routine to each day

12.2.3 Miscellaneous Benefits

- reduces the incidence and severity of workplace injuries
- allows time to discuss the day's events (e.g., pre-task plan)
- allows time to conduct a mini-safety meeting
- everyone benefits financially
- improve leadership and communication skills

12.3 Guidelines

Warming up and stretching is intended to reduce injuries and increase health...not the other way around. Therefore, in order to get the most benefit out of the Stretch & Flex program certain stretching guidelines must be followed.

- Always spend 2-4 minutes warming up prior to beginning stretches. "Warming up" provides a number of benefits, while preparing the body and mind for more strenuous exercises. Warm ups increase both core body and individual muscle's temperature. By increasing the core and muscle temperature, muscles will stay loose and more flexible.
- Before beginning any stretch, we should put our body into a neutral position
- This is standing up straight and relaxed, with our feet shoulder width apart, shoulders, neck and arms relaxed, shoulders and head back and stomach slightly tensed
- At all times during stretching we should be breathing fully and deeply
- Also, breath slowly and regularly; AVOID HOLDING YOUR BREATH WHILE STRETCHING
- When performing stretches, we never want to force the stretch or aggravate previous injuries
- If you have a muscle or joint which has been recently injured, do the stretches lightly at first
- If the stretch significantly increases the pain in the muscle or joint, then stop the stretch and allow your body to heal; consult a doctor if needed
- Stretches should be taken to a comfortable tension:
 - once we reach the point of comfortable tension, let your muscles relax, breath and hold the stretch
 - DO NOT BOUNCE and DO NOT STRAIN
 - stretching should NOT be painful, therefore if you are experiencing pain, back off a little on your tension
- To get the maximum out of your stretches, they should be held for 15-30 seconds at a time:
 - any time less than this provides no real benefits
 - after 30 seconds we are not adding any further benefit to the muscle
 - a little warm-up prior to stretching may increase your benefit
- It is important to do all of the stretches at your own pace and according to what your body tells you
 - just because everyone can bend over and touch the ground with their hands does not mean you have to force or bounce yourself into that position too
 - everyone's body is different – IT IS NOT A COMPETITION
- To maximize the stretches, you may need to change the stretch slightly; trying slightly different positions or a different type of stretch may be needed to target the area you need stretched the most

12.4 *The Stretches*

12.4.1 *The Program*

The Stretch & Flex program is designed to be both efficient and effective. It is organized to help everyone get the maximum benefit from the program and the stretches can be performed in a minimal amount of time. In addition to the warmup, there are 8 categories of stretches, each with several options that you may use to provide variety, as well as freedom to choose alternative stretches, that may focus on an area of need.

After the warm up, the object is to perform at least one stretch from each category each day. Each job site can incorporate as many or as few stretches into their program as desired. Ideally all of these stretches should be performed at least once per day. For improved health we recommend you perform these stretches additionally on your own time either at home, during lunch or after work. The 8 categories of stretches are:

- Neck
- Shoulder
- Arm and Forearm
- Wrist
- Upper Body and Chest
- Lower Back
- Leg and Thigh
- Calf

Warm Up

Core Body & Muscle Warm up

Stand erect, facing forward with the back straight and arms relaxed at the sides.

Begin walking in place.

Swing arms back and forth in opposition to the legs, keep arm swing motion at the sides of the body do not swing the arms across the front of the body.

Maintain slow to moderate pace; adjust the pace as necessary.

Maintain normal, steady breathing.

Perform for 1-2 minutes.

Switch to marching in place.

With each step, lift leg high enough that the thigh is parallel with the ground.

Swing arms back and forth in opposition to the legs, keep arm swing motion at the sides of the body.

Maintain slow to moderate pace; adjust the pace as necessary

Maintain normal, steady breathing.

Perform for 1-2 minutes, then begin the stretching exercises.

TIPS:

- Do not wing the arms across the front of the body, keep them swinging at the sides of the body.
- Always keep your head in neutral midline as much as possible. (i.e. don't drop your head forward or back when tilting your neck sideways)
- To increase the stretch, use your hands to VERY GENTLY pull your head/neck in the direction of stretch.

Neck Stretches

General Neck Stretch

Slowly lower the head/neck sideways, pulling the left ear toward the left shoulder.

Hold for 15-30 seconds.

Repeat on the right side, to the front, to the back and by turning as far as you can to both the left and right.



TIPS:

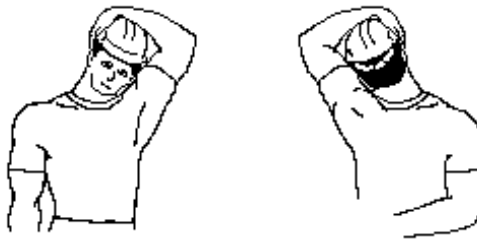
- Do not lift up your shoulder to touch your ear to it. Just move your head.
- Always keep your head in neutral midline as much as possible. (i.e. don't drop your head forward or back when tilting your neck sideways)
- To increase the stretch, use your hands to VERY GENTLY pull your head/neck in the direction of stretch.

Ear-Hole Stretch

Place the middle finger of your left hand in the right ear (over your head).

Now gently pull your head towards your left shoulder, keeping your head facing straight forward and midline.

Hold for 15-30 seconds.



TIPS:

- Do not pull too hard as you may strain a muscle – this stretch should be done GENTLY.
- Push your right shoulder down (when stretching to the left) to increase the stretch.
- Keep your nose facing forward and your head upright.

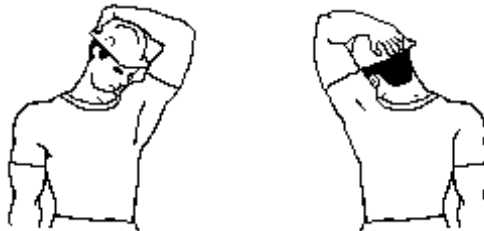
“Smelly Armpit” Stretch

Turn your head as far as you can to the left and then from that position try putting your nose into your left armpit. (Do not breathe too deeply here!)

Using your left hand reach behind your head and GENTLY pull your head down into your armpit, in the direction your nose is pointing.

Hold for 15-30 seconds.

Let go and move your head slightly towards the right and repeat. Eventually you will end up with your nose in the right armpit (changing hands you pull with as you cross the middle of your chest).



TIPS:

- Do not pull down too hard – you may strain a muscle. This stretch should be done GENTLY.
- Push/pull down your opposite shoulder to increase the stretch.
- Make sure you pull in the same direction your nose is pointing to get the best stretch possible.
- Relax your shoulders and neck.

Shoulder Stretches

Shoulder Rotations

Start with arms straight out on both sides with palms facing down and start moving SLOWLY in small circles in a forward direction. Gradually increase the size of the circles until you have hit the max.

Start over except this time with palms up, starting with small circles going in a backwards direction and gradually increasing.



TIPS:

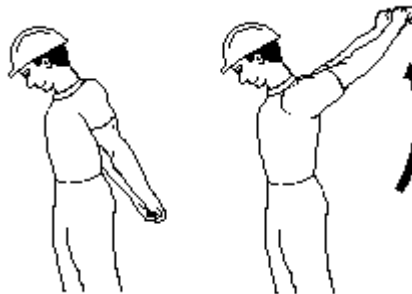
- Do not do this stretch quickly. You get a better stretch if you go slowly. This slow speed also helps prevent injury.
- Stand up straight with shoulders and head back to increase the benefit.

Shoulder Extensions

Clasp your hands behind your back and bend your knees slightly.

Slowly start to raise your arms up as you slightly bend forward at the waist.

Once your arms have gone as high as they can go, try to keep them there as you slowly try to stand up straight.



TIPS:

- Relax your chest and shoulders to maximize the stretch.
- Push your chest up and forward to increase the stretch.
- If you have shoulder problems (i.e. dislocations/chronic pain) perform this stretch with caution.

Shoulder Rolls

Stand straight up with shoulders and head back, arms to your sides.

Slowly rotate both shoulders forward 5 times, in as wide of circles as possible.

Repeat with 5 circles in the opposite direction.



TIPS:

- To increase the stretch, stand as tall and straight as possible.
- By pushing your chest forward and shoulders back you can further maximize the stretch.
- Do not let your head fall forward.

Arm and Forearm Stretches

Triceps Stretch

Reach overhead and place your fingers as far as you can down the center of your back.

With your other hand, gently push the elbow backwards and up.

Hold for 15-30 seconds and repeat on the other side.



TIPS:

- To change the focus of the stretch, you can change the direction you push (a little to the left or right) or the position of your arm. (i.e. Moving your fingers from your spine to directly behind your shoulder)
- To maximize the stretch, you can try thrusting your chest/armpit forward as you push back on your elbow.

Wrist Stretches

Prayer Stretch

Place your palms together like you are praying.

Keeping the palms together raise your elbows and/or lower your hands.

Hold for 15-30 seconds.



TIPS:

- You can turn your fingertips toward your body, away from your body or point them down to focus the stretch on the areas you need most.
- You do not need to press your hands together too tight. Try to relax your arms and wrists as much as possible.

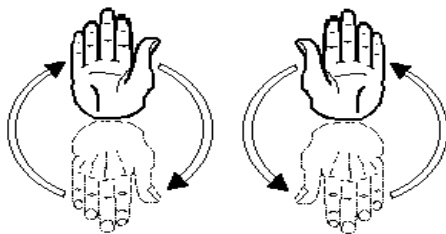
Wrist Rotations

Put both arms straight out in front of you with palms facing down.

SLOWLY rotate your wrists in wide circles (as far as you can go).

Do 5 circles of each wrist and then change direction.

Do 5 slow circles in the opposite direction.



TIPS:

- If you keep your elbows straight, you will get a better stretch.
- By doing the stretch slowly, you get a better stretch and avoid possible wrist injury.

Upper Body and Chest Stretches

Upper Back Stretch

Bend the right elbow and lift it to the same height as your shoulder.

Pull your right elbow across the front of your body as far as you can (with your left hand), while relaxing your right shoulder and back.

Turn your head in the opposite direction you pull and hold for 15-30 seconds.

Repeat on the opposite side.



TIPS:

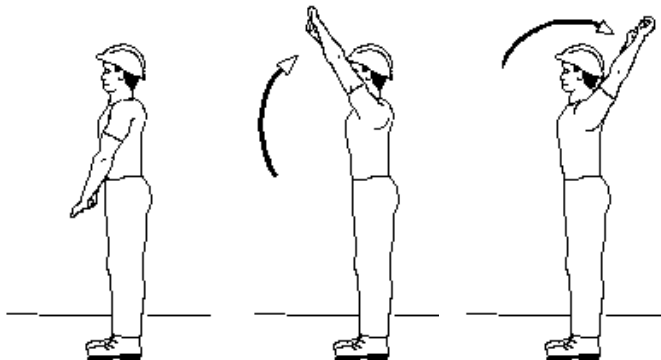
- Keep the elbow high – especially if you have a large chest or arms.
- Make sure that you relax your back and shoulder as much as possible when performing this stretch.
- To maximize the stretch, you get (to target the tighter muscles), change the height of your elbow and/or the direction you pull.

Overhead Stretch

Interlock your hands or thumbs in front of your body.

Slowly raise your arms as high as you can, making sure you do not arch your back.

Reach as high and as far back as you can and hold for 15-30 seconds.



TIPS:

- Make sure you are standing straight up, with head up and chest out.
- You may need to unlock your hands/thumbs to get a better stretch.
- Envision reaching back as high and far as you can.

Lateral Stretch

Put your right hand beside your right hip.

With your left hand (arm bent) reach up over your head as far as you can. Bend your upper body and head to the right as well. Reach down with your right hand.

Hold for 15-30 seconds and repeat on the opposite side.



TIPS:

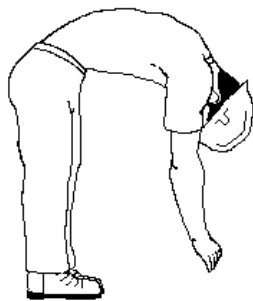
- Ensure that you bend with your upper body and not your lower body.
- Do not twist or bend forward or backward.
- Envision lifting your armpit as high as you can to help with the stretch.
- Keep both feet on the ground firmly.

Lower Back Stretches

Back Stretch

Keeping the knees bent slightly, SLOWLY bend over and try to touch your toes.

Tuck the chin as well and hold for 15-30 seconds.



TIPS:

- Do not bounce in this stretch. Deep breathing can be used to help further the stretch (on the exhale).
- To maximize the stretch, try pushing your bellybutton straight up/back.
- This stretch should be done slowly to avoid injury. Use your legs to crawl up with your hands, if necessary, to help you get back up slowly.

Lateral Rotation Stretch

Grasp your left hip with your right hand and raise your left arm to shoulder height, with the palm facing up.

Turn and reach as far back as you can to your left. Pull with your right hand to help twist.

Hold for 15-30 seconds and repeat on the opposite side.



TIPS:

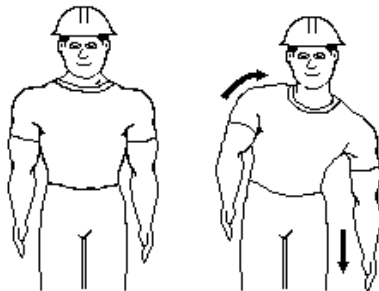
- To avoid injury you must do this stretch slowly and return to the starting position slowly. No whipping or throwing our body into position.
- Try pushing your left hip forward (if turning to the left) to maximize the stretch.
- Do not force or bounce this stretch.

Side bends

Stand straight with the arms to the side.

With your right hand slide down the side of your right leg as far as you can and bend your lower back to the right side.

Hold for 15-30 seconds and repeat on the opposite side.



TIPS:

- Do not lean forward or backward.
- Do not twist your body.
- Push your waist in the opposite way you are stretching to maximize the stretch.

Legs and Thigh

Hamstring Stretch

Assume a stride position with the right leg forward. Put your hands above your right knee and shift most of your weight to your left leg.

Keeping your back straight and head upright, slowly bend your left knee and bend your body over your right leg (keeping your right leg straight).

Hold for 15-30 seconds and then use your arms to stand up straight and repeat on other leg.



TIPS:

- To focus the stretch on your tightest muscles you can point your toe in or out.
- Always keep your back straight and head up.
- You can lift your toes up off the ground to get a greater stretch as well.

Quadriceps Stretch

Reach and grab your left ankle with your left hand and pull up. Keeping your knee pointed downward, pull up on your ankle and thrust your left hip forward.

Hold for 15-30 seconds and repeat on opposite side.



TIPS:

- Changing where your ankle is can help focus the stretch on the tight muscles. Pull it in towards the middle or away to the side to change the area of stretch.
- Pushing your hip forward and standing up straight will help increase the stretch.
- Use something for balance (a coworker, wall etc.).

Runner's Stretch

Assume a stride position with the hands on the front knee or hips.

Keeping your back and head straight, go into a squat with your front leg, using your back leg for balance. Push your hips forward.

Hold for 15-30 seconds.

Use your arms to help push yourself back up to a standing position.



TIPS:

- Pushing your hips forward and down will help with the stretch.
- Use your arms to straighten out your upper body to increase the stretch.
- Keep your front knee midline to avoid injury.

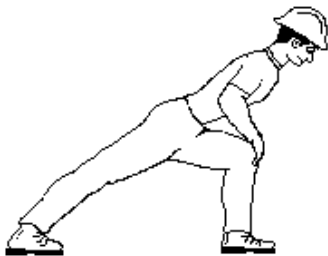
Calf Stretches

Calf Stretch

Assume a stride position with the left leg forward. Keep the right leg straight with the heel on the ground.

Bend your left knee and place your hands on top. Lean forward and push down on your back heel.

Hold for 15-30 seconds and then repeat on the opposite leg.



TIPS:

- Pushing the hips forward and knee back can help maximize the stretch.
- Lengthening your stride can also focus the stretch.
- Point your toe straight, in or out to change the area of the stretch on your calf.

Soleus Stretch

Assume a stride position with the left leg forward. Keep the right leg *slightly bent* with the heel on the ground.

Bend your left knee and place your hands on top and lean forward and push down on your right heel.

Hold for 15-30 seconds and then repeat on the opposite leg.



TIPS:

- To increase this stretch, push your knee down towards the floor while keeping your heel on the ground.

Crossed-Leg Stretch

Stand with one leg crossed over the other.

Slowly bend forward and try to touch your hands to the ground.

Keep your knees as straight as possible.

Hold for 15-30 seconds and then change legs.



TIPS:

- Push your front leg against your back leg to increase the stretch.
- Change the position your toe points to maximize the stretch.

12.5 *Benefits of Stretch & Flex in Detail*

12.5.1 *Physical Benefits*

The main benefit of stretching is to **reduce the tension on the muscle** by slowly and gradually lengthening the muscle. This decrease in tension results **in relaxation of the muscle** and **improved flexibility**.

By increasing your flexibility, you become more limber and as a result the joints are able to move through **a greater range of motion**. By decreasing the tension of the muscles, tendons and ligaments the joints are able to move more freely and with a greater degree of freedom. Further, by decreasing the tension on our muscles or relaxing them, we greatly reduce the amount of energy required by the muscles overall to do their job and therefore **prevent muscle fatigue**. To illustrate this point, picture someone with very tight triceps. To overcome the pull of the tight triceps (the antagonist), the biceps muscle (the agonist) has to work harder to go through a normal flex of the arm. Therefore, the tight muscle in this case causes the other muscles to get tired earlier than they normally would. Along this same train of thought, a more relaxed antagonistic muscle **improves the strength** of the agonistic muscle. Using the same example above, the tight triceps is exerting a force against the action of the biceps. As a result, the biceps is unable to lift as much weight. If the triceps were relaxed, the biceps could lift a greater amount, and therefore would be a much stronger muscle. All of these benefits mentioned combine to improve a person's overall **physical performance**.

By allowing a muscle to relax, the muscle is not in constant tension or prolonged contraction. A muscle in constant contraction needs more energy to accomplish activities as mentioned above. A side effect of a muscle being in constant tension is that the circulation to the muscle is choked off. Fresh blood and nutrients to the muscle are reduced and toxins and by-products from the muscle cannot be flushed from the area. This lack of circulation further puts the muscle at risk. Toxins and lack of proper nutrition to the muscles promote inflammation, shortening, soreness and tightening of the muscle. Therefore, tight muscles promote further tightening. Stretching **helps promote circulation** by causing the muscles to relax and hence taking the clamp off the blood vessels that travel through the muscle. Improved circulation also helps to warm up the muscle by improving the flow of blood to the area. A warmer muscle is more flexible, much like hot metal is easier to bend than cold metal.

Like the blood vessels, a tight muscle may clamp on a nerve, resulting in numbness, weaker muscles and slight pain or soreness. Stretching will **improve the function of the nerves**.

Decreasing the tension in the muscle may help to **prevent injury** caused by over-stretching or over-exerting the muscle. If the muscle is relaxed, then a sudden force applied to a muscle may not exceed its normal ability to stretch and therefore not tear. Another way that stretching helps prevent injury is that it can help restore balance between the muscles. Most people have unbalanced muscles, a stronger arm or leg and tasks during the day which stress different muscles. Stretching helps **to balance out the muscles** and can also **help with posture**. Postural problems are most often due to muscular imbalances in addition to poor habits and the effects of gravity. Having poor posture puts a greater strain on the body overall. By re-balancing and re-aligning the soft tissue structures we can help prevent injury, reduce muscle fatigue and improve the physical performance of the body. There have also been some studies that show stretching may help reduce the muscle soreness found after heavy exercise or activity.

Some other physical benefits of stretching that result from the joint going through a greater range of motion, is that the **wear and tear on a joint will be decreased** and there will be **improved coordination** of the muscles and joints. Moving the joints increases the production of synovial fluid (fluid within the joint for lubrication). Synovial fluid is both nutritious and a lubricant. Therefore, increased fluid will maintain the healthy condition of our joints, preventing degeneration or arthritis. Furthermore, if our joints can move more freely, we will have better coordination of our actions.

12.5.2 Psychological Benefits

It has been proven that people who actively pursue a healthy lifestyle are **happier** and obviously are **healthier**. With that, comes a **personal satisfaction** and gratification of being healthy. All of this leads to a greater enjoyment of work and life in general.

A secondary benefit of the Stretch & Flex program is that by allowing time for every crew member to get together and talk each morning, stronger friendships are gained and there is an **increase in team morale**.

Though not strictly a psychological benefit, engaging in an active stretching program improves the circulation to the muscles as well as the rest of the body. Exercise (stretching) promotes the release of hormones in the body, which give us energy and make us more alert. The **mind is stimulated** and you are **more clear-headed** and **alert**, thus making you more ready to face the hazardous environment. The time also allows the **mind to relax** from the commute or stresses of the morning and focus on the task at hand, easing you into the day.

By engaging in Stretch & Flex each morning, everyone maintains **a healthy routine** to every day. Research has proven that establishing a set routine to each day promotes a balance in hormones and patterns throughout the body. This balance further adds to your health and mental functioning. We all are aware of how getting thrown off our daily cycle makes us fatigued and messes up our body and mind (e.g. jet lag, shift work or oversleeping).

12.5.3 Miscellaneous Benefits

Research has proven that an active morning stretch program **reduces the incidence and severity of workplace injuries**. This benefit is obvious. Not only does everyone go home healthy each day, which is the primary goal, but everyone benefits by having healthy, happy and productive employees. Everybody indirectly benefits financially and DPR can maintain their costs for handling claims, medical bills, administration, insurance etc. to a minimum. By spending money in prevention, **money is saved in the long term**.

The Stretch & Flex can also be used as a forum each morning to discuss the day's events or conduct **morning spot safety meetings**. By identifying particular hazards each morning that a worker may face and the method of work (Pre-Task Planning), workers are prepared to handle risks and hazards more readily and can take significant steps toward prevention. Lastly, the Stretch & Flex program promotes different individuals leading the stretch each day. This **improves leadership skills** in addition to **improving the communication between the workers**. By everyone actively taking part in this program, the benefits increase tenfold.

Ejercicios de Estiramiento y Flexibilidad

Pautas

La finalidad del calentamiento y estiramiento es reducir las lesiones y mejorar la salud, y no al revés. Por lo tanto, para aprovechar al máximo el programa de estiramiento y flexibilidad (stretch & flex) se deben seguir ciertas pautas.

- Siempre tome de 2 a 4 minutos en calentar antes de comenzar los ejercicios de estiramiento. “Calentar” lo provee de muchos beneficios, mientras prepara la mente y el cuerpo para los ejercicios más extenuantes e intensos.
- Antes de empezar a estirarse, debemos de poner nuestro cuerpo en una posición neutral
- Esto significa estar de pie, derecho y relajado, con nuestros pies separados a la altura de los hombros, con el cuello, los hombros y los brazos relajados, los hombros y la cabeza hacia atrás y teniendo el abdomen ligeramente tenso
- Durante todo el estiramiento, debemos respirar profundamente
- Respire despacio y con regularidad. **NO CONTenga EL ALIENTO MIENTRAS SE ESTIRE**
- Cuando haga los ejercicios de estiramiento, no exagere el movimiento ni se empeore molestias anteriores
- Si ha sufrido una lesión en un músculo o articulación recientemente, empiece con un estiramiento ligero
- Si al estirarse aumenta mucho el dolor de ese músculo o articulación, pare de estirarse y permita que su cuerpo sane; consulte a un médico de necesitarlo
- Los estiramientos deben hacerse hasta lograr una tensión que no sea molesta:
 - Una vez que llegue a una tensión cómoda, deje que sus músculos se relajen, respire y mantenga el estiramiento
 - **NO HAGA REBOTES Y NO SE ESTIRE EN FORMA EXAGERADA**
 - El estiramiento **NO** debe de ser doloroso, por lo tanto, si siente dolor, reduzca un poco la tensión del estiramiento
- Para aprovechar el estiramiento al máximo, mantenga la posición de 15 a 30 segundos:
 - Un estiramiento de menos tiempo no lo beneficia
 - Después de 30 segundos, el músculo no se sigue beneficiando
 - Un pequeño calentamiento antes de estirarse puede aumentar el beneficio
- Es importante estirarse a un ritmo al que a usted le sea cómodo y de acuerdo con lo que le diga su cuerpo
 - No porque sus compañeros puedan doblarse y tocar el piso con las manos quiere decir que usted tenga que forzarse para lograr esa misma posición
 - Cada uno tiene un cuerpo diferente - **NO ES UNA COMPETENCIA.**
 - Para aprovechar el estiramiento al máximo, tal vez tenga que cambiar un poco el estiramiento. Si cambia un poco de posición o se estira en forma diferente tal vez se enfoque mejor en la zona que más necesite estirar.

Los Estiramientos

El Programa

El programa de estiramiento (stretch & flex) está diseñado para ser eficiente y efectivo. Está organizado para ayudar a todos a obtener los máximos beneficios de este programa y los estiramientos pueden ser realizados en un corto tiempo. Además del calentamiento, hay 8 categorías de estiramientos, cada uno de ellas con varias opciones que usted puede usar para proveerle con variedad, así como también con la libertad de escoger entre diferentes estiramientos, que pueden enfocarse en un área que necesite atención.

Después del calentamiento, el objetivo es de realizar al menos un estiramiento de cada categoría diariamente. Cada obra de trabajo puede incorporar tantos estiramientos como lo desee. Idealmente, todos

estos estiramientos tienen que ser realizados por lo menos una vez al día. Para mejorar su salud le recomendamos que haga estos estiramientos en su propio tiempo cuando se encuentre ya sea en casa, en la hora de la comida o incluso después del trabajo. Las ocho categorías de estiramiento son:

- Cuello
- Hombros
- Brazo y antebrazo
- Muñeca
- Cuerpo superior y pecho
- Espalda baja
- Pierna y muslo
- Pantorrilla

Calentamiento

Área del Torso y Calentamiento de los Músculos

Párese derecho, volteando hacia enfrente con la espalda derecho y con los brazos en posición de descanso hacia los lados.

Empiece a caminar en su mismo lugar.

Mueva los brazos de atrás hacia adelante en dirección opuesta a las piernas, continúe moviendo los brazos por un lado del cuerpo y no enfrente.

Mantenga un ritmo entre lento y moderadamente rápido; haga un ajuste al ritmo de ser necesario.

Respire de manera normal y controlada, a un ritmo parejo.

Haga esto de uno a dos minutos.

Ahora, marche en su lugar.

Con cada paso, levante la pierna suficientemente alto de modo que su muslo quede paralelo a la tierra.

Mueva los brazos de atrás hacia adelante en dirección opuesta a las piernas, continúe moviendo los brazos por un lado del cuerpo y no enfrente.

Mantenga un ritmo entre lento y moderadamente rápido; haga un ajuste al ritmo de ser necesario.

Respire de manera normal y controlada, a un ritmo parejo.

Ejecutar un trabajo durante 1-2 minutos, luego comenzar los ejercicios de estiramiento.

CONSEJOS:

- No mueva los brazos de un lado hacia a otro por enfrente del cuerpo, por lo contrario, hágalo a los lados del cuerpo.

Estiramientos para el cuello

Estiramiento general para el cuello

Lentamente doble el cuello hacia un lado, acercando la oreja izquierda hacia el hombro izquierdo.

Mantenga la posición durante 15 a 30 segundos.

Repita hacia el lado derecho, hacia adelante, hacia atrás, y gire la cabeza lo más que pueda hacia la izquierda y hacia la derecha.



CONSEJOS:

No levante el hombro para acercar la oreja. Simplemente deje caer la cabeza.

Siempre mantenga la cabeza en la línea media neutral lo más posible. Es decir, cuando incline la cabeza hacia un lado, no la deje caer hacia adelante ni hacia atrás.

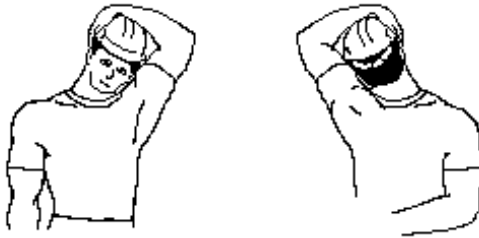
Para ampliar el estiramiento, use sus manos para halar MUY SUAVEMENTE la cabeza en el sentido del estiramiento.

Estiramiento desde el oído

Coloque el dedo del medio de su mano izquierda dentro de su oreja derecha (pasando el brazo por encima de la cabeza).

Lentamente tire la cabeza hacia su hombro izquierdo, mirando siempre directamente hacia adelante y sin inclinar la cabeza hacia adelante o hacia atrás.

Mantenga la posición durante 15 a 30 segundos.



CONSEJOS:

- No tire demasiado fuerte porque podría dañar un músculo. Este estiramiento debe hacerse en forma MUY suave.
- Para ampliar el estiramiento, empuje el hombro derecho hacia abajo (cuando se estira hacia la izquierda).
- Mantenga la nariz apuntando hacia adelante y la cabeza erguida.

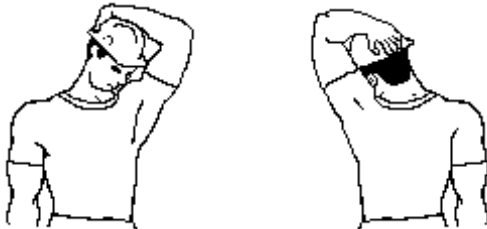
Estiramiento de “la axila sucia”

Gire la cabeza hacia la izquierda, lo más que pueda, y desde esa posición intente acercar la nariz a la axila izquierda. (¡No es necesario respirar profundamente!)

Con la mano izquierda agarre su cabeza y MUY SUAVEMENTE hale su cabeza hacia abajo, hacia la axila, en la dirección de su nariz.

Mantenga la posición durante 15 a 30 segundos.

Suelte la cabeza y muévela ligeramente hacia la derecha, y repita. Eventualmente terminará con la nariz apuntando hacia la axila derecha (cambie de manos cuando cruce la línea media del tórax).



CONSEJOS:

- No tire demasiado fuerte porque podría dañar un músculo. Este estiramiento debe hacerse en forma MUY suave.
- Empuje el hombro opuesto hacia abajo para ampliar el estiramiento.
- Para estirarse de la mejor forma posible, estire la cabeza en la misma dirección donde está apuntando con la nariz.
- Relaje los hombros y el cuello.

Estiramientos para los hombros

Rotaciones de hombros

Ponga los brazos en cruz, con las palmas hacia abajo. LENTAMENTE empiece a hacer círculos pequeños, hacia adelante. Vaya aumentando gradualmente el tamaño de los círculos hasta llegar al máximo.

Repita el movimiento pero esta vez con las palmas hacia arriba. Empiece con círculos pequeños, hacia atrás, y aumente gradualmente el tamaño de los círculos.



CONSEJOS:

- No haga este movimiento rápidamente. El estiramiento es mejor si lo hace lentamente. La lentitud también evita las lesiones.

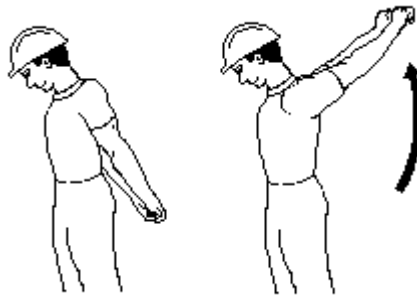
Párese bien derecho, con los hombros y la cabeza hacia atrás, para aumentar el beneficio.

Extensión de hombros

Agárrese las manos detrás de la espalda y doble un poco las rodillas.

Lentamente empiece a levantar los brazos hacia arriba, inclinándose un poco hacia adelante desde la cintura.

Cuando ha levantado los brazos lo más que pueda, trate de mantenerlos en esa posición a medida que intenta lentamente enderezar el cuerpo.



CONSEJOS:

- Relaje el pecho y los hombros para ampliar el estiramiento.
- Empuje el pecho hacia arriba y hacia adelante para ampliar el estiramiento.
- Si tiene problemas con los hombros, como dislocaciones o dolor crónico, haga este estiramiento con cuidado.

Círculos con los hombros

Párese derecho, con los hombros y la cabeza hacia atrás y los brazos a los lados.

Lentamente haga rotaciones hacia adelante con los dos hombros, cinco veces, haciendo el círculo más amplio que pueda.

Repita en el sentido opuesto, haciendo 5 círculos.



CONSEJOS:

- Para ampliar el estiramiento, manténgase bien erguido.
- Puede ampliar el movimiento aún más si empuja el pecho hacia adelante y los hombros hacia atrás.
- No incline la cabeza hacia adelante.

Estiramientos para los brazos y antebrazos

Estiramientos del triceps

Levante un brazo y coloque los dedos en el centro de la espalda, lo más abajo que pueda.

Con la otra mano, suavemente empuje el codo hacia atrás y hacia arriba.

Mantenga la posición durante 15 a 30 segundos y repita el movimiento en el otro lado.



CONSEJOS:

- Para cambiar el enfoque del estiramiento, puede empujar en otra dirección (un poco hacia la izquierda o hacia la derecha) o cambiar la posición del brazo (es decir, en vez de tocar su espina dorsal, colocar los dedos detrás del hombro)
- Para ampliar el estiramiento, trate de empujar el pecho y la axila hacia adelante a medida que empuja el codo hacia atrás.

Estiramientos para la muñeca

Estiramiento de oración

Junte las palmas como si estuviera rezando.

Con las palmas juntas, levante los codos y/o baje las manos.

Mantenga la posición durante 15 a 30 segundos.



CONSEJOS:

- Puede girar las manos para que los dedos apunten hacia su cuerpo, o hacia afuera, o hacia abajo, para enfocar el estiramiento en las zonas que más necesitan trabajo.
- No es necesario juntar las manos con mucha fuerza. Trate de relajar sus brazos y muñecas lo más posible.

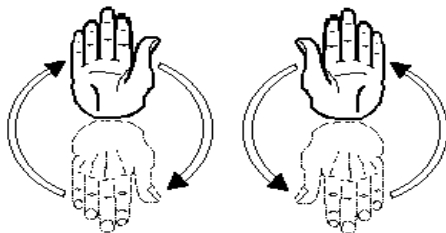
Círculos con las muñecas

Estire los brazos hacia adelante, con las palmas hacia abajo.

LENTAMENTE haga círculos amplios con las manos (lo más que pueda).

Haga 5 círculos con cada mano y luego cambie de dirección.

Haga 5 círculos lentamente en sentido opuesto.



CONSEJOS:

- Si mantiene los brazos estirados, sin doblar los codos, el estiramiento es mejor.
- Si se estira lentamente, el estiramiento es mejor y evita lesionarse las muñecas.

Estiramientos del tronco y tórax

Estiramiento de la parte de arriba de la espalda

Doble el codo derecho y elévelo hasta que esté a la altura del hombro.

Con la mano izquierda, hale el codo derecho para que cruce el frente del tórax, lo más posible, manteniendo relajados el hombro derecho y la espalda.

Gire la cabeza en sentido opuesto y mantenga la posición durante 15 a 30 segundos.

Repita al otro lado.



CONSEJOS:

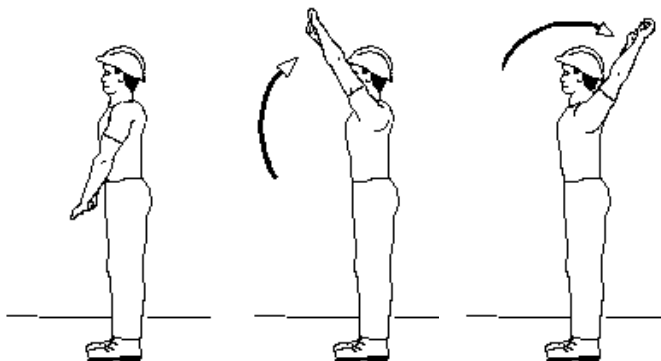
- Mantenga el codo bien arriba, especialmente si tiene el tórax amplio y los brazos gruesos.
- Relaje la espalda y el hombro lo más posible cuando se estire.
- Para ampliar el estiramiento y enfocarse en los músculos más tensos, cambie la elevación del codo o la dirección en que hala el codo.

Estiramiento con las manos arriba

Enlace los dedos o los pulgares frente al cuerpo.

Lentamente levante los brazos lo más que pueda, sin arquear la espalda.

Estírese hacia arriba y hacia atrás, lo más que pueda, y mantenga la posición durante 15 a 30 segundos.



CONSEJOS:

- Manténgase bien erguido, con la cabeza erguida y el pecho hacia afuera.
- Es posible que tenga que desenlazar las manos o los pulgares para estirarse mejor.
- Trate de imaginarse que está alcanzando algo bien arriba y bien atrás de su cuerpo.

Estiramiento lateral

Coloque la mano derecha al lado de la cadera derecha.

Doble el brazo izquierdo y con la mano izquierda estírese hacia el lado derecho, lo más que pueda. Inclíne el tórax y la cabeza hacia la derecha también. Con la mano derecha, estire hacia abajo.

Mantenga la posición durante 15 a 30 segundos y repita al otro lado.



CONSEJOS:

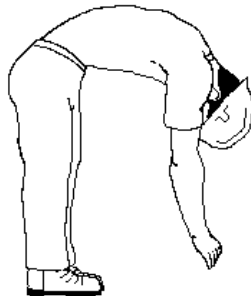
- Haga la flexión con el tórax y no con la parte inferior del cuerpo.
- No haga torsiones con el cuerpo ni se incline hacia adelante ni hacia atrás.
- Para estirarse mejor, imagínese que está levantando la axila hacia arriba, lo más posible.
- Mantenga los dos pies bien plantados en el suelo.

Estiramientos de la espalda lumbar

Estiramiento de la espalda

Doble las rodillas un poco y LENTAMENTE agáchese, intentando tocarse los pies.

Acerque el mentón al cuello y mantenga la posición durante 15 a 30 segundos.



CONSEJOS:

- No haga rebotes mientras se estira. Puede respirar profundamente para ampliar el estiramiento durante la exhalación.
- Para ampliar el estiramiento, trate de empujar el ombligo hacia atrás y hacia arriba.
- Haga este estiramiento lentamente para no lesionarse. Para enderezarse lentamente, puede apoyar las manos en las piernas y lentamente ir subiéndolas.

Estiramiento de rotación lateral

Con la mano derecha agarre la cadera izquierda y levante el brazo izquierdo hasta la altura del hombro, con la palma hacia arriba.

Gire hacia la izquierda y estírese hacia atrás lo más que pueda. Hale con la mano derecha para poder girar mejor.

Mantenga la posición durante 15 a 30 segundos y repita al otro lado.



CONSEJOS:

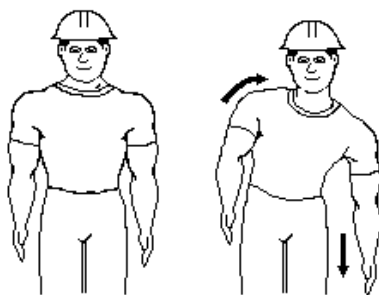
- Para no lesionarse, haga este estiramiento lentamente y vuelva lentamente a la posición inicial. No adopte la posición en forma brusca.
- Trate de empujar la cadera izquierda hacia adelante si está girando hacia la izquierda, para ampliar el estiramiento.
- No haga rebotes ni fuerce este estiramiento.

Flexiones hacia los lados

Párese en posición erguida con los brazos a los lados.

Deslice la mano derecha hacia abajo, a lo largo de la pierna derecha, lo más que pueda, e incline la columna hacia la derecha.

Mantenga la posición durante 15 a 30 segundos y repita al otro lado.



CONSEJOS:

- No se incline hacia adelante ni hacia atrás.
- No tuerza el cuerpo.
- Con la cintura, empuje hacia el lado opuesto al estiramiento, para ampliar el estiramiento.

Piernas y muslos

Estiramiento de los aductores de la rodilla

Tome un paso grande con el pie derecho. Coloque las manos más arriba de la rodilla derecha y apoye su peso en la pierna izquierda.

Mantenga la espalda derecha y la cabeza erguida. Lentamente doble la rodilla izquierda e incline el cuerpo sobre la pierna derecha, que se mantiene sin doblar.

Mantenga la posición durante 15 a 30 segundos y luego ayúdese con los brazos para volver a enderezarse. Repita con la otra pierna.



CONSEJOS:

- Para enfocar el estiramiento en los músculos más tensos, puede apuntar con los dedos del pie hacia adentro o hacia afuera.
- Siempre mantenga la espalda derecha y la cabeza erguida.
- Para ampliar el estiramiento, puede levantar los dedos del pie hacia arriba.

Estiramiento del cuadriceps

Agarre el tobillo izquierdo con la mano izquierda y tire hacia arriba. Con la rodilla apuntando hacia el suelo, tire el tobillo hacia arriba y empuje la cadera izquierda hacia adelante.

Mantenga la posición durante 15 a 30 segundos y repita al otro lado.



CONSEJOS:

- Si cambia la posición del tobillo puede enfocar el estiramiento en los músculos tensos. Si tira el tobillo hacia un lado, sea hacia adentro o hacia afuera, puede cambiar la zona de estiramiento.
- Si empuja la cadera hacia adelante y se mantiene bien erguido, aumenta el estiramiento.
- Use algo para equilibrarse (la pared, un compañero, etc.).

Estiramiento para corredores

Dé un paso grande hacia adelante, con las manos sobre las caderas o sobre la rodilla doblada frente al cuerpo.

Mantenga la espalda derecha y la cabeza erguida. Doblando la rodilla de la pierna que está adelante, baje el cuerpo, manteniendo el equilibrio con la ayuda de la pierna que está atrás. Empuje las caderas hacia adelante.

Mantenga la posición durante 15 a 30 segundos.

Empuje con los brazos para volver a ponerse de pie.



CONSEJOS:

- Si empuja las caderas hacia adelante y hacia abajo, puede ampliar el estiramiento.
- Use los brazos para enderezar el tórax y así ampliar el estiramiento.
- Mantenga la rodilla de adelante en su línea media para no lesionarse.

Estiramientos de la pantorrilla

Estiramientos de la pantorrilla

Dé un paso grande hacia adelante con el pie izquierdo. Mantenga la pierna derecha sin doblar, con el talón en el suelo.

Doble la rodilla izquierda y coloque las manos sobre la rodilla. Inclínese hacia adelante y empuje el talón derecho hacia el suelo.

Mantenga la posición durante 15 a 30 segundos y repita con la otra pierna.



CONSEJOS:

- Si empuja las caderas hacia adelante y la rodilla hacia atrás, puede ampliar el estiramiento.
- Según el largo del paso, puede enfocar el estiramiento en distintas zonas.
- Apunte con el dedo gordo hacia adelante, hacia adentro o hacia afuera para cambiar la zona de estiramiento de la pantorrilla.

Estiramiento del soleo

Dé un paso grande hacia adelante con el pie izquierdo. La pierna derecha se mantiene un poco doblada, con el talón en el suelo.

Doble la rodilla izquierda, coloque las manos sobre la rodilla, inclínese hacia adelante y empuje el talón derecho hacia el suelo.

Mantenga la posición durante 15 a 30 segundos y repita con la otra pierna.



CONSEJOS:

- Para ampliar el estiramiento, empuje la rodilla hacia el suelo a la vez que mantiene el talón en el suelo.

Estiramiento con las piernas cruzadas

Estando de pie, cruce las piernas.

Lentamente inclínese hacia adelante y trate de tocar el suelo con las manos.

Mantenga las rodillas tan derechas como le sea posible.

Mantenga la posición durante 15 a 30 segundos, luego cambie las piernas de posición y repita el movimiento.



CONSEJOS:

- Empuje la pierna de adelante contra la pierna de atrás, para ampliar el estiramiento.
- Cambie la posición de la punta de los pies para ampliar el estiramiento.

Ergonomics & Manual Material Handling Forms

Forms

Ergonomics Pre-Planning Checklist
Ergonomic Training Verification – English
Ergonomic Training Verification – Spanish
Material Lift (Duct Lift) Pre-Use Inspection



Ergonomics Pre-Planning Checklist

Hazard Assessment

Project/Project #:		Date:	
Inspector:		Company:	
Foreman:		Superintendent:	

Material Handling				
	Comments	Yes	No	N/A
What heavy materials or equipment >20 lbs. is being handled on site? (blocks, drywall, lumber, rebar, buckets, forms, etc.)		☐	☐	☐
Do any workers need to lift > 50 lbs. one time without help?		☐	☐	☐
Do any workers need to lift > 20 lbs. repeatedly? ○ How can this be prevented?		☐	☐	☐
Are there handles for materials that must be carried? ○ If yes, are handles easy to use and comfortable?		☐	☐	☐
Are workers encouraged to get help to lift heavy material? ○ Is help readily available?		☐	☐	☐
Are there carts, dollies, etc. readily available to help move material? ○ If yes, are carts being used? ○ If no, why not? ○ If no, is the site clear enough to permit cart use?		☐	☐	☐
Are materials delivered as close as practicable to where they will be used? ○ If no, how could this change?		☐	☐	☐
Are materials stored in walkways or in working areas?		☐	☐	☐
Are materials stored at floor or ground level? ○ If yes, must workers bend to lift materials? ○ Could the material be stored at waist height?		☐	☐	☐
What jobs require workers to lift overhead? ○ How could this lifting be avoided?		☐	☐	☐
What tasks require workers to reach far to pick up or lift materials? ○ Could the material be moved closer?		☐	☐	☐

Training				
	Comments	Yes	No	N/A
Has the schedule of work and sequencing plan been integrated with a weather projection plan to protect as-built components and various stored building materials?		☐	☐	☐
Are tools maintained, kept sharp and in good condition?		☐	☐	☐
Which tools are heavy (> 20 lbs.)?		☐	☐	☐
Which tools vibrate excessively?		☐	☐	☐
Which tools have cold or very warm handles?		☐	☐	☐

Ergonomics Pre-Planning Checklist

Hazard Assessment

Which tools must be used in an awkward or difficult position?		☐	☐	☐
Which tools have poor handle design?		☐	☐	☐
Are gloves used? ○ <i>Do the gloves make it harder or easier to grip tools?</i>		☐	☐	☐
Have workers been trained in ergonomics?		☐	☐	☐
Have supervisors or safety personnel been trained in ergonomics? (material handling, proper lifting, body position, work procedures, prevention of risks, etc.) ○ <i>What are some of the obstacles for training?</i> ○ <i>What steps can be taken to ensure proper training?</i>		☐	☐	☐
Looking at the past injuries, is there any specific training that may help? ○ <i>What trends do the injury logs suggest?</i>		☐	☐	☐

Environment				
	Comments	Yes	No	N/A
Are working and walking surfaces ○ <i>Clean and dry?</i> ○ <i>Unobstructed?</i> ○ <i>Even?</i>		☐	☐	☐
Do workers have enough protection from heat, cold, rain, wind and sun?		☐	☐	☐
Are work areas well lit? ○ <i>Is there enough light to adequately perform the work?</i> ○ <i>Do lights get in the way, limit vision or produce heat?</i> ○ <i>Are there areas where glare is a problem?</i> ○ <i>What can be done to change these things?</i>		☐	☐	☐
Are supervisors or workers under production pressure? ○ <i>Is the project ahead or behind schedule?</i> ○ <i>Can job rotation or rest breaks be used?</i> ○ <i>Can reduced hours, increased help be utilized?</i> ○ <i>What other changes can be done? Safety people, meetings, etc.?</i>		☐	☐	☐
Are working and walking surfaces ○ <i>Clean and dry?</i> ○ <i>Unobstructed?</i> ○ <i>Even?</i>		☐	☐	☐

Work Positioning/Practices				
	Comments	Yes	No	N/A
What trades or people must do the same motion > one hour per day? ○ <i>What are the motions and how could they be changed?</i> ○ <i>Can job rotation or rest breaks be utilized?</i>		☐	☐	☐
What trades must work overhead for > one hour per day? ○ <i>Can scaffolds, platforms, or other equipment reduce overhead postures?</i>		☐	☐	☐

Ergonomics Pre-Planning Checklist

Hazard Assessment

What trades must work at floor level or on their knees > one hour per day? - Are knee pads or cushions available and used? - Can equipment be used to reduce kneeling or bending?		☐	☐	☐
What tasks require workers to stay in one position for a long period of time? Can job rotation or rest breaks be utilized?		☐	☐	☐
What tasks require twisting and turning? How can the twisting and bending be reduced?		☐	☐	☐
What tasks require a lot of bending? How can the bending be reduced?		☐	☐	☐
What tasks require workers to stand all day, especially on concrete floors? - Can anti-fatigue matting or insoles be used? - Does the worker's boots show excessive heel or sole wear? If so, they may need to obtain new boots - Is it possible to use something to allow workers to rest?		☐	☐	☐
What tasks require sitting for extended periods of time? - Are the seats well-designed and comfortable? - Has the worker removed all objects from back pockets? - In heavy equipment, do workers lean forward to work? - Is there heavy vibration through the seat? - What can be done to change these things?		☐	☐	☐
Are hazards abated or problems fixed quickly when reported? If not, why not?		☐	☐	☐

General				
	Comments	Yes	No	N/A
Do any workers currently have any signs and symptoms of injury? If yes, how can further injury be prevented?		☐	☐	☐
Do workers feel comfortable reporting injuries or problems? If no, how can this be changed?		☐	☐	☐
Do workers often appear exhausted at the end of the day?		☐	☐	☐
Do the workers complain a lot about conditions, work, etc.?		☐	☐	☐
Are the workers eating well during the day?		☐	☐	☐
Are the workers drinking enough fluids during the day?		☐	☐	☐
Are any of the workers under excessive stress?		☐	☐	☐
Are any of the workers currently taking any medications? (ensure worker confidentiality is maintained)		☐	☐	☐



Ergonomics Pre-Planning Checklist

Hazard Assessment

All above issue are indicators or signs of problems, if you answered yes to any of the above, what can be done to help or changes these things?		☐	☐	☐
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Corrective Actions	Responsible Party	Due Date	Date Completed



Ergonomic Training Verification

Archive Document – DO NOT DISCARD

Project/Project #

Date

Print Name

Company

Material Handling

Manual Handling

- All material should be properly stored and at the appropriate height
- In general, if more than two people are required to move equipment, tools, etc., mechanical means of movement should be considered
- Material and equipment should be stored as close to the work area as practicable
- All workers should examine their workplaces to detect any unsafe conditions prior to handling materials; such conditions include, but are not limited to: (slip and trip hazards, uneven surfaces, holes, equipment, tools, stowed material, limited access)
- Using mechanical methods such as carts, pallet jacks, dollies, wheeled racks, lift gates, forklifts, etc.
- Proper PPE should be used when handling material; i.e. puncture or cut resistant gloves and arm/hand protection
- Loading/unloading: only essential personnel should be in the area -- all workers should stay clear of the load to protect against objects that may have shifted, rolling structural steel, material displacement, etc.

Storage

- All stored materials, tools and equipment should be secured or otherwise protected to prevent accidental displacement
- Safe procedures should be established for the unbundling and unloading of any materials delivered with special care taken for objects that can roll

Disposal

- All workers are required to clean up their work areas at minimum daily
- Ensure that disposal containers are located near to work and emptied regularly

Risk Factors

Force or muscle effort

- Ensure the right equipment for the job is used, are sharp, well maintained and fully charged
- When possible use a power grip (full-hand grip) versus a pinch grip that only uses the fingers, two hands should be utilized whenever practicable
- Ensure that tools have a handle or grip that is not too large or small for the hand
- Ensure that the wrist and hand are kept in as neutral a position as possible
- Push loads versus pulling loads and ensure good body position when pushing
- Ensure that positions are change periodically or take a rest; padded mats, etc.

Awkward Body Postures

- Work with the arms in the neutral position, minimize the amount of time spent over-the-shoulder height work
- Avoid reaching and awkward body positions
- Position the tool, equipment or work surface in a way that places the worker in a more neutral working position
- Ensure ladders face the work surface or ladder platforms are utilized whenever possible to minimize twisting
- Pre-fabing or assembling materials may help minimize awkward body positioning

Repetitive Work

- Change tasks or positions as much as possible, ensure different muscle groups are used
- Use mechanical means to minimize the amount of repetitive work being done such as nut driver, wire stripper, pneumatic deck crimper, nail gun etc.

Contact Stress (Tools and Sharp Objects)

- Use appropriate PPE for cushioning, e.g., knee pads, padded gloves, padded tool handles, mats etc.
- Ensure that the proper tool is used for the task
- Ensure the condition of footwear being worn is adequate to minimize foot contact stress, especially for tasks involving walking on rebar, standing on ladders etc.; walkways or scaffolds may be used to minimize these risk factors for rebar and ladders respectively
- When possible, the use of platform ladders should be used in lieu of standard A-frame ladders
- Reduce the weight of tool belts to only the tools needed, wear suspenders or remove the tool belt altogether when possible

Vibration from Hand Tools

- Anti-vibration gloves and padding should be used as needed



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- Anti-vibration seats should be provided in heavy machinery
- Utilize job rotation when possible to prevent extended exposure to vibration
- Ensure the right tool for the task is being used
- Ensure tools are clean, dry and have easily gripped handles
- Ensure tools and equipment are well maintained, sharp and are not forced beyond their capacity
- Ensure two hands are used on tools as appropriate

Temperature

- Ensure Stretch & Flex is performed; in the morning prior to work and as needed throughout the day if shift length is extended
- Gloves and protective clothing should be worn as indicated
- Ensure working environment is of a suitable and safe working temperature
- Provide ventilation as needed
- Ensure appropriate breaks are taken and workers remain well hydrated
- Use appropriate gloves to reduce slippage of hands, tools and equipment in hot conditions
- Sufficient water should be provided to provide workers throughout their shift; approximately one quart of water per hour per worker
- Frequent short breaks for re-hydration should be encouraged
- Access to shade should be provided with either ventilation or cooling capabilities; workers appearing to suffer from heat should be encouraged to use shaded areas
- Supervision should be trained in recognition and control of hazards associated with heat or cold temperature stresses

Lifting

- No matter the lifting technique used, the size and shape of the object should be small enough to allow the worker to holding it against the body to lift
- Heavy or bulky objects should be lifted with assistance or with alternate methods
- Freestyle lifting (back bent, legs almost straight) should not be used
- Lifts should be performed with a minimum of twisting
- Repetitive lifting should be evaluated and alternative methods used whenever possible to prevent fatigue

Back belts

- Long term use of back supports should not be allowed as muscle atrophy may develop
- If back supports are used, workers should be trained in proper technique, appropriate use, benefits and risks of improper use

This document hereby certifies that I have been instructed on the items listed above and I had an opportunity to ask questions. I fully understand the DPR ergonomic principles.

Employee Signature

Date

Person Conducting Verification Review Signature

Date

A copy of this form should be faxed to the DPR regional office PRIOR TO START OF WORK

Proyecto/Proyecto #

Fecha

Nombre

Compañía

Manejo de materiales

Manejo manual

- Guardar todos los materiales debidamente y a la altura apropiada
- Por lo general, si se necesitan más de dos personas para trasladar un equipo, una herramienta, etc., considere usar un medio mecánico para hacerlo
- Almacenar los materiales y equipos lo más cerca al lugar de trabajo que sea prácticamente posible
- Cada trabajador debe examinar su lugar de trabajo antes de manejar materiales, para asegurar que no existan condiciones peligrosas, que incluyen las siguientes, entre otras: peligro de resbalarse o tropezarse, superficies dispares o con hoyos, equipos, herramientas, materiales almacenados, acceso limitado
- Usar métodos mecánicos tales como carritos, elevadores de paletas, carretillas “dolly”, bastidores con ruedas, rejas que se levantan, grúas horquilla (forklift), etc.
- Usar equipo de protección personal adecuado para el material. Por ejemplo, guantes resistentes a los objetos filosos, protección en los brazos y manos
- Durante la carga y descarga, solamente deben estar presentes los empleados esenciales. Todos los trabajadores deben alejarse de la carga para evitar el peligro de objetos que se han desplazado, acero estructural rodante, desplazamiento de materiales, etc.

Almacenamiento

- Fijar o proteger de alguna forma todos los materiales, herramientas y equipos almacenados para que no se desplacen accidentalmente
- Establecer procedimientos seguros para desembalar y descargar materiales entregados en la obra, especialmente para objetos capaces de rodar

Desechos

- Cada trabajador debe limpiar y ordenar su lugar de trabajo, por lo menos una vez al día
- Colocar basureros cerca del lugar de trabajo, que deben vaciarse con regularidad

Factores de riesgo

Hacer fuerzas o esfuerzo muscular

- Usar el equipo correcto para cada tarea. Verificar que los equipos estén afilados, bien mantenidos y completamente cargados
- En lo posible, agarrar los objetos con la mano entera y no con los dedos. Cuando sea práctico, usar las dos manos
- Usar herramientas con mango o cacha que no sea demasiado grande ni demasiado pequeña para la mano
- Dentro de lo posible, mantener la muñeca y la mano en posición neutra
- Empujar las cargas en vez de tirarlas o jalarlas. Mantener el cuerpo en posición correcta al empujar una carga
- Cambiar de posición regularmente, o tomarse un descanso. Usar una colchoneta para descansar, etc.

Posiciones retorcidas o difíciles

- Trabajar con los brazos en posición neutra. Reducir al mínimo el trabajo con los brazos estirados encima de los hombros
- Evitar estirarse para alcanzar los objetos. No colocar el cuerpo en posiciones difíciles
- Colocar las herramientas, equipos o superficie de trabajo en lugares que aseguren al trabajador una posición más neutra durante el trabajo
- Colocar las escaleras de frente a la superficie de trabajo o utilizar plataformas cuando sea posible, para no torcer tanto el cuerpo
- Procurar el ensamble de los materiales de antemano o usar técnicas de prefabricación, para reducir la cantidad de trabajo con el cuerpo en posiciones difíciles

Trabajo repetitivo

- Cambiar de tarea o de posición lo más posible. Usar distintos grupos de músculo
- Usar medios mecánicos para reducir la cantidad de trabajo repetitivo. Usar medios como destornillador portapuntas, pelacables, plegador neumático, martillo neumático, etc.

Estrés por contacto (con herramientas y objetos filosos)

- Usar equipo de protección personal apropiado para acolchar las superficies de contacto. Por ejemplo, rodilleras, guantes acolchados, manillas especiales para herramientas, colchonetas, etc.
- Verificar que la herramienta sea la que corresponda para la tarea
- El calzado debe ser adecuado para reducir el estrés por contacto con los pies, especialmente para tareas donde hay que caminar sobre barras de refuerzo, estar de pie sobre escaleras, etc. Se pueden usar pasarelas para reducir el riesgo de caminar sobre barras de acero, y andamios para evitar las escaleras.

- Cuando sea posible, usar escaleras de plataforma en vez de las escaleras de tijera
- Reducir el peso de las herramientas cargadas en el cinturón al usar solamente las herramientas necesarias. Usar suspensores o, en lo posible, no usar el cinturón con herramientas

Vibración por herramientas de mano

- Usar guantes anti-vibración y acolchados cuando sea necesario
- Usar asientos anti-vibración para las maquinarias pesadas
- Rotar el personal cuando sea posible, para evitar una prolongada exposición a la vibración
- Usar herramientas que correspondan a cada tarea
- Usar herramientas limpias, secas y de manillas fáciles de agarrar
- Usar herramientas y equipos bien mantenidos y afilados. No forzarlos durante su funcionamiento
- Usar las dos manos en las herramientas que lo exijan

Temperatura

- Hacer ejercicios de estiramiento y flexibilidad en la mañana antes del trabajo y según sea necesario durante el día si el turno se prolonga
- Usar guantes y vestimenta de protección cuando sea necesario
- Trabajar en un ambiente que esté a una temperatura adecuada y no peligrosa
- Ventilar según sea necesario
- Tomar los descansos apropiados y asegurar que los trabajadores consuman suficiente líquido
- Usar guantes apropiados para evitar que se resbalen las manos, herramientas y equipos cuando hace calor
- Proporcionar suficiente agua para beber durante todo el turno de los trabajadores. Cada trabajador necesita aproximadamente un litro ("1 quart") de agua por hora
- Facilitar descansos breves y frecuentes para que los trabajadores se hidrate
- Proporcionar lugares con sombra, con ventilación o aire acondicionado. Si un trabajador parece estar demasiado acalorado, hay que pedirle que se ponga a la sombra
- Capacitar a los supervisores para que sepan reconocer y controlar los peligros del calor o del frío

Levantar objetos

- Cualquiera que sea la técnica para levantar un objeto, o el tamaño y la forma del objeto, debe ser lo suficientemente pequeño como para que el trabajador pueda sostenerlo contra el cuerpo cuando lo levanta
- Cuando el objeto es pesado o voluminoso, hay que levantarlo con ayuda de otro o mediante algún otro método
- No levantar objetos agachándose hacia adelante casi sin doblar las rodillas
- Tratar de no torcer el cuerpo al levantar un objeto
- Si hay que levantar objetos repetidamente, la tarea debe evaluarse y dentro de lo posible usar otro método, para evitar el cansancio

Cinturones para apoyar la espalda

- No permitir el uso prolongado de cinturones que apoyan la espalda porque se pueden atrofiar los músculos.
- Si el trabajador usa un cinturón para apoyar la espalda, debe estar capacitado en su uso apropiado, las técnicas apropiadas, los beneficios y los riesgos de no usarlo apropiadamente.

Este documento certifica que me han capacitado en los temas listados anteriormente y que me han dado la oportunidad de hacer preguntas. Entiendo completamente los principios ergonómicos de DPR/HARD.

Firma del empleado

Fecha

Firma de la persona que efectúa la revisión de verificación

Fecha

ANTES DE EMPEZAR EL TRABAJO se debe enviar una copia de este formulario por favor



Material Lift (Duct Lift) Pre-Use Inspection

Project/Project#:		Date:	
Model:		Equipment#:	
Company:		Hour Meter:	

Date							
-------------	--	--	--	--	--	--	--

Initials of person performing inspection:							
--	--	--	--	--	--	--	--

Has the operator been instructed in the safe operation of this type of lift and are they certified?	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N
---	------------	------------	------------	------------	------------	------------	------------

Inspection Item & Description Pass Fail Status

Inspection Item & Description Pass Fail Status	P/F	P/F	P/F	P/F	P/F	P/F	P/F
---	-----	-----	-----	-----	-----	-----	-----

Pre-operation Inspection

1	Fork Assembly Structure						
2	Mast Columns						
3	Base Components						
4	Legs						
5	Stabilizers & latch plates						
6	Legible Warning Placards						
7	Operator's Manual						
8	Safety Brake						
9	Carriage Hold Down Bar						
10	Cable Anchor						
11	Cable and Pulleys						
12	Wheels and Casters						
13	Load Handling Attachments						
14	Nuts, Bolts and Fasteners						
15	Dents or Damage						
16	Corrosion or Oxidation						

Workplace Inspection

1	Drop-Offs or Holes						
2	Floor Obstructions/Debris						
3	Overhead Obstructions and High Voltage						
4	Inadequate Surfaces to Support Load Forces						
5	Wind and Weather Conditions						

**** If any item(s) are found to be missing or defective, the equipment is to be repaired/serviced prior to use**

To: _____ Date: _____

Lift Taken Out of Service ☐ Yes ☐ No

Repairs Made

By:		Date:	
-----	--	-------	--

Section 10

Excavation & Trenching

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Excavations & Trenching

1.0 Introduction

This plan outlines procedures and guidelines for the protection of employees working in and around excavations and trenches. Compliance is needed to ensure employee protection when working in or around excavations. Other plans in this manual such as Confined Space, Chemical Safety, Hazardous Energies, Respiratory Protection, and any other safety plan or procedures deemed essential for employee protection should be used in conjunction with this plan.

Reference: Excavations CFR 1926.650, Subpart P; Excavation Equipment Manufacturer Safety Procedures. Other applicable Standards, guidelines, and regulations may apply.

Definitions

811: The universal federally designated number to be used to **Call Before Digging**. Below is a list current states and their local nomenclature:

Blue Stake (AZ)	Montana 811/ Montana One Call Center (Flathead and Lincoln Counties) (MT)
USA (No CA)	North Carolina 811 (NC)
Dig Alert (So CA)	Diggers Hotline of Nebraska (NE)
CBYD (CT)	New Jersey One Call (NJ)
District One Call (DC)	New Mexico One Call (NM)
Sunshine State One Call (FL)	Oregon Utility Notification Center (OR)
Georgia 811 (GA)	Dig Safe System (RI)
Iowa One Call (IA)	South Carolina 811 (SC)
LA One Call (LA)	Tennessee 811 (TN)
Dig safely New York(North of the 5 Boroughs)/ New York 811 New York (5 Boroughs and Long Island)	Texas 811/ Lone Star Notification Center (TX)
Dig Safe (ME)	Blues Stakes of Utah (UT)
Miss Utility of Maryland (west of Chesapeake Bay (MD)	Virginia 811 (VA)
Miss Utility of Delmarva (east of Chesapeake Bay (MD)	Dig Safe System (VT)
Dig Safe System (MA)	Utility Notification Center (WA)
Miss Utility of West Virginia (WV)	

Cemented soil: Soil in which the particles of the earth are held together by a chemical agent, such as calcium carbonate. A hand-size sample cannot be crushed into powder or individual soil particles by finger pressure alone.

Cohesive soil: Soil that is clay (fine grained soil) or a soil with a high clay content, which has cohesive strength. Cohesive soil does not crumble, can be excavated with vertical side slopes and is plastic when moist. Cohesive soil is hard to break up when dry and exhibits significant cohesion when submerged. Cohesive soils include clayey silt, sandy clay, silty clay, clay and organic clay.

Competent person: One who is capable of identifying existing and predictable hazards in the surroundings, or working conditions which are hazardous, unsanitary, or dangerous to workers, and who has authorization to take prompt corrective elimination measures

Dry soil: Soil that does not exhibit visible signs of moisture content

Excavation: Any man-made cut, cavity, trench or depression in an earth surface, formed by earth removal. Regardless of depth, provisions of this section should be followed.

Fissured soil: A soil material that has a tendency to break along definite planes of fracture with little resistance, or a material that exhibits open cracks, such as tension cracks, in an exposed surface

Granular soil: Gravel, sand or silt (coarse-grained soil) with little or no clay content. Granular soil has no cohesive strength. Some moist granular soils exhibit apparent cohesion. Granular soil cannot be molded when moist and crumbles easily when dry.

Hazardous atmosphere: An atmosphere which by reason of being explosive, flammable, poisonous, corrosive, oxidizing, irritating, oxygen deficient, toxic; or otherwise harmful, deadly, injurious; or has the potential to cause illness

High priority subsurface installations: High pressure natural gas pipelines with normal operating pressures greater than 415 kPA gauge (60 p.s.i.g), petroleum pipelines, pressurized sewage pipelines, conductors or cables that have a potential to ground of 60,000 volts or more, or hazardous materials pipelines that are potentially hazardous to employees, or the public, if damaged (CA only)

Horizontal Directional Drilling or "HDD": A type of trenchless excavation that uses guidable boring equipment to excavate in an essentially horizontal plane without disturbing or with minimal disturbance to the ground surface

Layered soil: Two or more distinctly different soil or rock types arranged in layers. Micaceous seams or weakened planes in rock or shale are considered layered.

Moist soil: A condition in which a soil looks and feels damp. Moist cohesive soil can easily be shaped into a ball and rolled into small diameter threads before crumbling. Moist granular soil that contains some cohesive material will exhibit signs of cohesion between particles.

Saturated soil: Soil in which voids are filled with water. Saturation does not require flow. Saturation, or near saturation, is necessary for the proper use of instruments such as a pocket penetrometer or shear vane.

Stable rock: A natural solid mineral matter that can be excavated with vertical sides and remain intact while exposed

Submerged soil: Soil that is underwater or free seeping

Trench (trench excavation): A narrow excavation (in relation to its length) made below the surface of the ground. In general, the depth is greater than the width

Unconfined compressive strength: The load per unit area at which a soil will fail in compression. Predominantly determined by laboratory testing or estimated in the field using a pocket penetrometer or by thumb penetration tests

Wet soil: Soil that contains significantly more moisture than moist soil but in such a range of values that cohesive material will slump or begin to flow when vibrated. Granular material that would exhibit cohesive properties when moist will lose those cohesive properties when wet.

2.0 Pre-Task Planning

Work activities involving excavation & trenching are defined by DPR to be High Hazard and therefore require detailed, written pre-task planning. *Refer to the Pre- Planning Section 17 for additional pre-task planning guidelines.*

3.0 Job Planning

Before a contractor is on site, the following items should be obtained in writing:

- Permit for excavation/trenching activities (CA – Excavation Notification Form as applicable)
- Excavation and trenching plan
- Name(s) of competent person(s)
- Soils analysis report

3.1 Excavation/Trenching Plan

The excavation/trenching plan provides an overall scope of the work, identifies potential hazards and methods for abatement. The excavation/trenching plan is in addition to written, detailed pre-task planning. A competent person should complete this plan prior to the start of any excavation activities. The subcontractor's plan should be reviewed with DPR project management prior to start of work. At a minimum, this plan should include:

- **Pre-Planning Excavation & Trenching Checklist** should be completed by the competent person prior to the start of work. In addition, the foremen of all subcontractors which may be adversely impacted should review and sign off, and then the checklist should be submitted to DPR project management for review and approval. The Checklist will be audited periodically.
- **Excavation & Trenching Inspection Form** is the documentation required to comply with state and federal regulations. The competent person should, at a minimum, complete this form daily. Copies should be forwarded to DPR project management.
- **The Soils Analysis Report** should be completed when the soil type(s) have been determined. A separate analysis should be performed if the excavation/trench is stretched over a distance where soil types change.

3.2 Tools for Excavation & Trenching Management

Information pertaining to excavation and trenching should be maintained throughout the project in a manner readily accessible by members of the DPR and subcontractor project teams. This should include:

- **Utilities Location Map** should be maintained throughout the duration of the project. The map should clearly identify location and type of existing utilities and should be updated as additional utilities are installed. The map should be maintained in a location visible to affected parties.
- **Utility Location Tracking Board** should be maintained while excavating activities are in progress to visually track the status of utility locators/location(s) being used on the project.

3.3 Potential Hazards to Consider

- Surface encumbrances are safe guarded
- Underground utilities (sub-surface installations-CA only)
- Access and egress
- Vehicular traffic
- Falling loads
- Warning systems/mobile equipment
- Hazardous atmospheres (less than 19.5% or greater than 23.5% oxygen)
- Emergency rescue procedures
- Water accumulation

- Stability of adjacent structures
- Loose rock or soil
- Inspections
- Fall protection

3.4 Contingency Planning

On existing or critical facilities, contingency planning should be completed prior to beginning excavation activities. At a minimum, the plan should address potential impacts involving utilities in and around existing facilities with the potential to impact normal business operations or impact worker or public health or safety. The following should be considered when developing the contingency plan:

- Review site plans and as-builts, where available
- Review plans and potential impacts with applicable utility company(ies)
- Check for redundancy in the facility's systems; identify any system that is not redundant and develop a work-around plan
- Identify any backup systems
- Meet with facility personnel to understand what systems are critical
- Verify as-builts and plans for accuracy
- Develop the contingency plan with the facility owner and utility provider's input; share with owner and subcontractors
- Incorporate contingency plan into the site emergency response plan

4.0 Competent Person

DPR reserves to right to request and verify credentials for all competent persons. Responsibilities of the competent person include, but are not limited to:

- Perform daily, documented inspection of protection equipment, trench conditions and adjacent areas:
 - inspections should be made prior to the start of work and as needed throughout the shift
 - inspections should be made after any adverse weather event or other hazard increasing occurrence
- Remove employees from hazardous conditions and make all changes necessary to ensure their safety
- Categorize soil conditions and conduct visual and manual tests as the trench work progresses
- Determine the appropriate protective system to be used
- Obtain appropriate permits and supply necessary notices when needed (e.g., Cal/OSHA Activity Notification form for Holders of Annual Permits, Confined Space Permit, etc.)
- All on-site records in relation to all excavation and trenching activities should be maintained and copies provided to DPR
- A competent person, provided by the subcontractor, should be named for all excavation activities
- The competent person should be on the job site at all times to monitor soil conditions and inspect protective systems when personnel are working in or around any excavation of three (3') feet or greater

5.0 Utilities (Sub-Surface installations)

When planning for excavation or trenching, a copy of underground "as-builts" and civil drawings should be obtained illustrating the premises, property boundaries, building, etc. to aid in identifying

common utilities such as electrical, gas, water, sewage, fire system, telecom and security systems. Areas to be excavated should be marked with white paint, flagging or similar marking. Only qualified persons should perform subsurface installation locating activities. In addition, you should contact 811, or the actual utility company to identify existing utilities and obtain its approval or services. Many larger firms and campuses have a records department that can provide invaluable information on site layout.

CA Only: When the excavation is proposed within 10 feet of a high priority subsurface installation, the excavator should be notified by the facility owner/operator of the existence of the high priority subsurface installation. Additionally, an on-site meeting should be held between the excavator and the utility operator before the legal excavation start. *Reference and follow 1541(b)(1)(B)*

In addition to “as-builts”, **811** or similar utility locator and private locator services should be contacted to locate all existing utilities both public and private. Prior to excavating, confirmation of current **811** should be verified. After locating a utility, said utility should be marked with a color coded marking paint to standards established by the American Public Works Association. The marks should remain visible for the duration of work, and **811** or similar maintained current.

Color	Utility
Red	Electric Power Lines, Cables, Conduit & Lighting Cables
Yellow	Gas, Oil, Steam, Petroleum or Gaseous Materials
Orange	Communication, Cable TV, Alarm or Signal Lines, Cables or Conduit
Blue	Water, Irrigation and Slurry Lines
Green	Sanitary Sewer System
Pink	Temporary Survey Markings
White	Proposed Excavation
Purple	Reclaimed Water

After identifying all utilities, their location should be positively marked using colored paint or other similar marking system. Using the above information, all utilities should be “pot holed” to determine the exact location of the utility in the ground prior to the use of any mechanical or heavy equipment.

“Pot holing” will be defined as exploratory excavation to expose utilities. Pot holing should be performed around each utility until the utility is completely exposed. The preferred method of pot holing is “soft dig”, i.e., the use of air, water or vacuum systems in lieu of hand digging.

Where there is the potential for workers to be exposed, i.e., existing facilities or where the potential to impact public health and/or site safety exists, “soft dig” potholing is required. In any event where the “soft dig” method is not utilized, a compelling reason for not using this method must be submitted in writing, reviewed and the compelling reason approved by the DPR project team, prior to start of work.



For the purposes of this section, hand tools will be defined as hand shovels only. Picks or other similar tools, in addition to pneumatic or electric impact type tools with spade attachments should not be used for hand digging. Critical and hazardous utilities should always be 'pot holed'. Pot holing should occur at a minimum of every 25 feet for the following high risk activities:

- Voltage in excess of 480V
- Fiber optics
- Gas, chemical or fuel lines
- Mission critical (activities with the potential for non-scheduled facility shutdown or interruption)

Other utility hazards that should be considered when developing an excavation and trenching plan:

- Redundancies, back-feeds and other service abnormalities
- Older facility with unknown alterations, repairs, installations etc.
- Air handling systems that may allow the flow of debris, dust and other contaminants into occupied spaces
- Electrical panel diagrams that are not accurate
- Utilities buried in footings, equipment pads or in/under slabs
- Lack of back-flow devices that may allow cross-contamination between services
- Service valves and disconnects to be located and marked even if there is no immediate plan or need to shut them off
- Bio or chemical hazards created from sewer and drain lines being removed or modified

6.0 General Requirements

All workers involved in excavation activities should be trained to recognize all hazards associated with the work being performed. (*Reference Excavation Training Verification Form in the Forms Section at the end of the EHSP*).

- It is the responsibility of the employer to ensure no person enters an excavation until the procedures within this section have been fulfilled
- Adequate precautions should be taken to protect employees working in excavations against the hazards posed by water accumulation
- Where the stability of adjoining buildings, walls or other structures is endangered by excavation operations, support systems such as shoring, bracing or underpinning should be provided to ensure the stability of such structures and for the protection of employees
- Sidewalks, pavement, and supported structures should not be undermined unless a support system such as shoring is provided to protect employees from the possible collapse of such structures
- Employees should be protected from excavated or other materials and equipment that could pose a hazard by falling or rolling into excavations
- Protection should be provided by placing and keeping such materials and equipment a minimum of two (2') feet from the edge of excavations
- A stairway, ladder, or ramp should be used as a means of access or egress in trench excavations that are four (4') feet or more in depth, additionally:
 - ladder(s), stairways(s) or ramp(s) should be spaced so that no employee in the trench excavation is more than 25 feet from a means of safe egress
 - when ladder(s) are employed, the top of the ladder should extend a minimum of three (3') feet above the ground and be properly secured

- When excavations are exposed to vehicular traffic, each employee should wear appropriate garments made of high visibility and/or reflective materials
- Equipment operators should be trained in the safe operation of the equipment they are using, training documentation should be made available to DPR upon request.
- Operators should inspect equipment daily, prior to use, inspection should be documented.
- Employees should not be permitted underneath loads handled by lifting or digging equipment.
- Employees should be required to stand away from any vehicle being loaded or unloaded to avoid being struck by any spillage or falling material.
- Whenever oxygen deficiency or a hazardous atmosphere exists or could reasonably exist, the area should be continuously monitored to assure that employees are adequately protected; in such excavations, a confined space permit should be obtained and procedures followed (*refer to the Confined Space Section 04 for additional guidelines*).
- Where oxygen levels are below 19.5% or above 23.5%, the area must be continuously ventilated until the oxygen levels fall within normal levels.
- Where a flammable gas condition exists, the area should be ventilated until the flammable gas concentration is below 10% of the lower explosive limit (LEL).
- Emergency rescue equipment and an attendant should be provided where hazardous atmospheres exist or have the potential to develop during the scope of work. Workers in the trench cannot be designated attendants.
- In work areas where the exact location of underground electrical power over 50 volts is unknown, workers using hand digging equipment with the potential for contacting live lines should be provided with insulated protective gloves and/or insulated/shielded materials.
- Where personnel/equipment are required/permitted to cross over any excavation walkways/bridges with standards guardrails should be provided.
- Where workers are expected to work within 6' of an excavation that is 6 feet or more in depth, provisions for fall protection must be implemented.

7.0 Soil Types

Soil classifications should be obtained by an independent third party specializing in soil testing and classification. The following general descriptions are for preliminary evaluation and general understanding.

7.1 Type A Soil

Cohesive soils with an unconfined, compressive strength of 1.5 tons per square foot (tsf) (144 kPa) or greater. Examples of cohesive soils are: clay, silty clay, sandy clay, clay loam and, in some cases, silty clay loam and sandy clay loam. Cemented soils such as caliche and hardpan are also considered Type A. However, *no* soil is Type A if:

- The soil is fissured
- The soil is subject to vibration from heavy traffic, pile driving or similar effects
- The soil has been previously disturbed
- The soil is part of a sloped, layered system where layers dip into the excavation on a slope of $\frac{3}{4}$ vertical to one horizontal (3/4:1) or greater
- The material is subject to other factors that would require it to be classified as a less stable material

7.2 **Type B Soil**

Cohesive soils with an unconfined compressive strength greater than 0.5 tsf (48 kPa) but less than 1.5 tsf (144 kPa).

- Granular cohesionless soils including:
 - angular gravel (similar to crushed rock)
 - silt, silt loam, sandy loam and, in some cases, silty clay loam and sandy clay loam
 - previously disturbed soils except those which would otherwise be classed as Type C soil
- Soils that meet the unconfined compressive strength or cementation requirements for Type A, but are fissured or subject to vibration
- Dry rock that is not stable
- Material that is part of a sloped, layered system where layers dip into the excavation on a slope less steep than one horizontal to one vertical (1:1), but only if the material would otherwise be classified as Type B

7.3 **Type C Soil**

Cohesive soil with an unconfined compressive strength of 0.5 tsf (48 kPa) or less.

- Granular soils including gravel, sand, and loamy sand
- Submerged soil or soil from which water is freely seeping
- Submerged rock that is not stable
- Material in a sloped, layered system where the layers dip into the excavation on a slope of one and a half horizontal to one vertical (1.5:1) or steeper

7.4 **Stable Rock**

Natural solid mineral material that can be excavated with vertical sides and should remain intact while exposed.

8.0 **Methods for Determining Soil Types**

Only competent persons should make the determination of soil types and perform the following tests.

8.1 **Manual Tests**

An informal manual analysis of soil samples can be conducted to determine quantitative as well as qualitative properties of soil and to provide more information to classify soil.

- **Plasticity.** Mold a moist or wet sample of soil into a ball and attempt to roll it into threads as thin as 1/8-inch in diameter. Cohesive material can be successfully rolled into threads without crumbling. For example, if at least a two-inch length of 1/8-inch thread can be held on one end without tearing, the soil is cohesive.
- **Dry strength.** If the soil is dry and crumbles on its own or with moderate pressure into individual grains or fine powder, it is granular (any combination of gravel, sand or silt). If the soil is dry and falls into clumps which break up into smaller clumps, but the smaller clumps can only be broken up with difficulty, it may be clay in any combination with gravel, sand or silt. If the dry soil breaks into clumps which do not break up into smaller clumps and which can only be further broken with difficulty, and there is no visual indication the soil is fissured, the soil may be considered unfissured.
- **Thumb penetration.** The thumb penetration test can be used to estimate the unconfined compressive strength of cohesive soils. The thumb can readily indent Type A soils with an

unconfined compressive strength of 1.5 tsf only with very great effort. Type C soils with an unconfined compressive strength of 0.5 tsf can be easily penetrated several inches by the thumb, and can be molded by light finger pressure. This test should be conducted on an undisturbed soil sample, such as a large clump of spoil, as soon as practicable after excavation to minimize the effects of exposure.

8.2 Visual Tests

Visual analysis is conducted to determine qualitative information regarding the excavation site in general, the soil adjacent to the excavation, the soil forming the sides of the open excavation and the soil taken as samples from excavated material. The following guidelines apply:

- Observe samples of soil that are excavated and soil in the sides of the excavation
 - estimate the range of particle sizes and the relative amounts of the particle sizes
 - soil that is primarily composed of fine-grained material is cohesive material
 - soil composed primarily of coarse-grained sand or gravel is granular material
- Observe soil as it is excavated
 - soil that remains in clumps when excavated is cohesive
 - soil that breaks up easily and does not stay in clumps is granular
- Observe the side of the opened excavation and the surface area adjacent to the excavation
 - crack-like openings such as tension cracks could indicate fissured material; if chunks of soil spall off a vertical side, the soil could be fissured
 - small spalls are evidence of moving ground and are indications of potentially hazardous situations
- Observe the area adjacent to the excavation and the excavation itself for evidence of existing utility and other underground structures and to identify previously disturbed soil
- Observe the opened side of the excavation to identify layered systems
 - examine layered systems to identify if layers slope toward the excavation
 - estimate the degree of slope of the layers
- Observe the area adjacent to the excavation and the sides of the opened excavation for evidence of surface water, water seeping from the sides of the excavation or the location of the level of the water table
- Observe the area adjacent to the excavation and the area within the excavation for sources of vibration that may affect the stability of the excavation face

9.0 Protective Systems

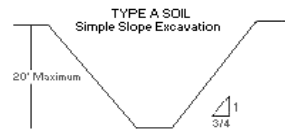
All excavations should be protected from any cave in potential especially when the public or employees and equipment are working in near trenching activities. A competent person shall inspect all trenching activities, and at no time should workers be exposed to any hazard associated with excavations and trenching. *DPR reserves the right, at no additional cost to the owner, to require any subcontractor, at any time, to provide protective systems at four (4') feet in excavations where the potential for cave-in exists. This may be protection over and above that deemed necessary by the subcontractor's competent person.*

- The competent person should determine the need for the use of protective systems sufficient to protect workers from all potential hazards; at a minimum, the use of protective systems is required for all excavations five (5') feet or greater.
- The competent person must determine, based upon soil types, conditions and stability, if protective systems will be required for excavations less than 5 feet in depth.

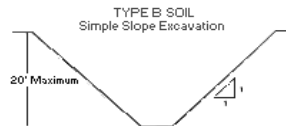
- When sloping, benching or protective systems are used for excavations less than 20 feet in depth, refer to diagram below. Or, refer to federal, state or local requirements (refer to Fed OSHA CFR 1926.652).
- Protective systems for excavations 20 feet and greater should be designed by a registered professional engineer, with documentation maintained onsite.
- Whenever support systems, shield systems or other protective systems are being used, a copy of the manufacturer's specifications, recommendations and limitations sheet should be in written form and maintained at the job site.

Sloping

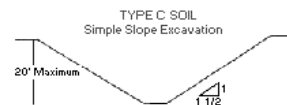
Type A - 3/4:1



Type B - 1:1

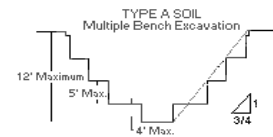


Type C - 1 1/2 :1

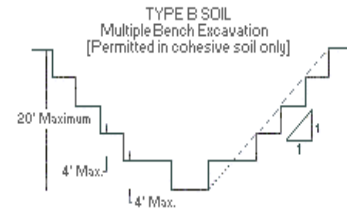


Benching

Type A



Type B



Type C – N/A type C soil cannot be benched

Excavation & Trenching Forms

Forms

Excavation & Trenching Pre-Planning Checklist
Trenching and Excavation Inspection Form – English
Trenching and Excavation Inspection Form – Spanish
Excavation & Trenching Awareness Training Verification – English
Excavation & Trenching Awareness Training Verification – Spanish
Heavy Equipment Training Verification Form - English
Heavy Equipment Training Verification Form – Spanish
Heavy Equipment Inspection – English / Spanish
Guardrail Safety Checklist



Excavation & Trenching Pre-Planning Checklist

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To be completed prior to any excavation or trenching work 811 Locate must be verified and current, pre-task plan must be completed and reviewed with DPR & subcontractor supervision prior to start of work. Periodic audits may be conducted by DPR

Project/Project#:		Company:	
Competent Person:		Date:	
Competent Person Signature:		Start Time:	
		End Time:	
Excavation Depth/Width:		Soil Type:	
Protective System(s) to be used:		Locator Service Number:	
Locator Effective Date:		Locator Expiration Date:	

DPR Auditor:		Date	
ACTIVITIES REQUIRED PRIOR TO START:		DPR Mon Auditor:	
☐	811 Locate complete and current	DPR Tues Auditor:	
☐	Private Locator complete	DPR Wed Auditor:	
☐	Soft Dig for all existing utilities	DPR Thurs Auditor:	
☐	Detailed Pre-task plan reviewed w/crew	DPR Fri Auditor:	
☐	Pre-Planning checklist complete	DPR Sat Auditor:	
☐	Verify competent person credentials	DPR Sun Auditor:	

General Site Conditions	Yes	No	N/A
Excavation, adjacent areas, and protective systems inspected by the competent person daily prior to start of work and as conditions change.	☐	☐	☐
Competent person has the authority to remove employees from the excavation immediately and stop work.	☐	☐	☐
Surface encumbrances removed or supported.	☐	☐	☐
Employees protected from loose rock/soil that could pose a hazard by falling or rolling into the excavation.	☐	☐	☐
Spoils, materials, and equipment set back as least 2' from the edge of the excavation.	☐	☐	☐
Barriers provided at all remotely located excavations, wells, pits, shafts, etc.	☐	☐	☐
Walkways and bridges over excavations 6' or greater in depth are equipped with standard guardrails and toeboards.	☐	☐	☐
Warning vests or other highly visible clothing provided and worn by all employees exposed to vehicular traffic.	☐	☐	☐
Employees are required to stand away from vehicles/equipment being loaded or unloaded.	☐	☐	☐
Employees are prohibited from going or working under suspended loads.	☐	☐	☐
Employees are prohibited from working on the faces of sloped or benched excavations above others.	☐	☐	☐
Walkways or bridges with standard guardrails are provided where personnel or equipment are required or permitted to cross over any exaction 6 feet or more in depth.	☐	☐	☐
Fall protection provisions are implemented where workers are expected to work within 6' of an excavation that is 6 feet or more in depth.	☐	☐	☐

Utilities	Yes	No	N/A
Utility company or private locator service contacted and/or utilities located. (Call may be necessary to private locator service when on private property.)	☐	☐	☐
Locations of utilities marked and markings maintained visible for duration of work.	☐	☐	☐
Underground installations protected, supported or removed when excavation is open.	☐	☐	☐
Critical and hazardous utilities should be potholed at minimum every 25' horizontally.	☐	☐	☐
At least a 24" radius should be air, water or vacuum dug, i.e., "pot-holed" around each utility to expose it. Mechanical means should not be used for pot holing.	☐	☐	☐
On existing or critical facilities, contingency plans in case of utility disruption have been developed and shared with owner/ subcontractors and integrated into site emergency response plan.	☐	☐	☐
Map of existing utilities maintained current.	☐	☐	☐

Means of Access & Egress	Yes	No	N/A
Unobstructed lateral travel to access point no $\geq 25'$ in excavations $\pm 4'$ in depth.	☐	☐	☐
Ladders used in excavations secured and extended 3' above the edge of the trench.	☐	☐	☐
Structural ramps used by employees designed by a competent person.	☐	☐	☐
Structural ramps used for equipment designed by a registered professional engineer.	☐	☐	☐
Ramps constructed of materials of uniform thickness, cleated together on the bottom and equipped with a no-slip surface.	☐	☐	☐
Employees are protected from cave-ins when entering or exiting excavations.	☐	☐	☐

Hazardous Atmospheres	Yes	No	N/A
Atmosphere within the excavation is tested where there is a reasonable possibility of an oxygen deficiency, combustible, or other harmful contaminant posing a hazard.	☐	☐	☐
Adequate precautions taken to protect employees from exposure to an atmosphere containing less than 19.5 % or more than 23.5% oxygen and/or other hazardous atmosphere.	☐	☐	☐
Ventilation provided to prevent employee exposure to an atmosphere containing flammable gas in excess of 10% of the lower explosive limit of the gas.	☐	☐	☐
Testing conducted often to ensure that the atmosphere remains safe.	☐	☐	☐
Emergency equipment, such as breathing apparatus, safety harness and lifeline and/or basket stretcher readily available where hazardous atmospheres could or do exist.	☐	☐	☐
Employee are proficient in the use of PPE and rescue equipment	☐	☐	☐
Safety harness and lifeline used and individually attended for each individual entering bell bottom or other deep confined excavations.	☐	☐	☐

Wet Conditions	Yes	No	N/A
Precautions taken to protect employees from the accumulation of water.	☐	☐	☐
Water removal equipment and conditions monitored by the competent person.	☐	☐	☐
Surface water or runoff diverted or controlled to prevent accumulation in the excavation.	☐	☐	☐
Inspections made after any adverse weather event or other hazard increasing occurrence.	☐	☐	☐

Protective and Support Systems	Yes	No	N/A
Materials and/or equipment for support systems selected based on soil analysis, depth, width and expected loads.	☐	☐	☐
Materials and equipment used for protective systems inspected and in good condition.	☐	☐	☐
Materials and equipment not in good condition have been tagged and removed from service.	☐	☐	☐
Damaged materials and equipment used for protective systems have been inspected by a Registered Professional Engineer after repairs and before being placed back into service.	☐	☐	☐
Protective systems installed without exposing employees to the hazards of cave-ins, collapse or threat of being struck by materials or equipment.	☐	☐	☐
Members of support system securely fastened to prevent failure.	☐	☐	☐
Support systems provided to ensure stability of adjacent structures, buildings, roadways, sidewalks, walls, etc.	☐	☐	☐
Excavations below the level of the base or footing supported and approved by a Registered Professional Engineer.	☐	☐	☐
Excavations 20 feet or more in depth have protective systems designed by a Registered Professional Engineer.	☐	☐	☐
Removal of support systems progresses from the bottom and members are released slowly as to note any indication of possible failure.	☐	☐	☐
Backfilling progresses with removal of the support system.	☐	☐	☐
Excavate material to a level no greater than 2' below the bottom of the support system and only if the system is designed to support the loads calculated for the full depth.	☐	☐	☐
Shield system placed to prevent lateral movement.	☐	☐	☐
Employees are prohibited from remaining in shield system during vertical movement.	☐	☐	☐

Foreman:		Foreman:	
Foreman:		Foreman:	
Foreman:		Foreman:	
Foreman:		Foreman:	
Foreman:		Foreman:	

Comments:



Trenching and Excavation Inspection Form

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Project Information		General Information		Date	
Project/Project#		Purpose of Excavation		Week Starting	
Contractor/Subcontractor		Location		Week Ending	
Competent Person Name:		Greatest Depth			
Competent Person Signature:		Approximate Size (sf)			
Soil Classification		(Stable Rock, A, B or C)			
Mon	Tues	Wed	Thurs	Fri	Sat
					Sun
Partial List of Safety Items - Identify & Correct All Unsafe Conditions					
A Inspect at a minimum daily, mark a "+" for items being performed well or in good condition					
B For items that need improvement or corrective action, mark a "Δ" and comment on the back any action needed/taken					
1 Trench Protection					
Delineation, barricades and signage in place, trench plates in place and not displaced					
Benching or sloping done when appropriate (as determined by competent person, but required 5 feet or more in depth)					
Trench shield (box), trench plates and/or shoring in place, in good condition and secure					
Rebar caps or other protection on all impalement hazards					
Fall protection provided for trenches greater than 6' in depth					
Other					
2 General Site Conditions					
Surface encumbrances or impediments removed or supported					
Spoils and other materials are set back at least 2' from edge of excavation					
Sources of vibration (equipment, traffic etc.) are not too close to excavation and workers					
Employees not working under suspended loads, around equipment, around vehicles being loaded or in actively excavated holes					
Employees are wearing garments of high visibility reflective material when working around moving equipment					
Other					
3 Access/Egress					
Ladders provided for all trenches greater than 4' in depth, proper ladder used and ladder secured and extends 36" out of hole.					
Walkways and bridges over excavations (required for trenches >6') are necessary width, have toe boards, guardrails etc. in place					
Unobstructed lateral access to a means of egress no less than 25' of travel distance					
Other					
4 Utilities					
All utilities have been precisely marked, potholed and/or protected					
Sewer and gas lines are exposed, supported and/or protected					
Critical and hazardous utilities are potholed at a minimum of every 25' horizontally - pot holes dug (air, water or vacuum systems preferred) in 24" radius around utilities. No mechanical equipment should be used for pot holing.					
5 Trench Conditions					
Employees protected from water accumulation and/or surface runoff					
Trench inspected for potential cave ins, sloughing, cracking etc.					
Other					
6 Hazardous Atmospheres					
Atmosphere tested where oxygen deficiency, explosion, combustion or other harmful contaminants possible					
Adequate precautions taken to protect atmosphere, emergency equipment available and employees trained					
Other					



Specifics & Comments of Excavation and Trenching Inspection
Ensure all open items are closed/addressed in a timely manner.

Monday	
Tuesday	
Wednesday	
Thursday	
Friday	
Saturday	
Sunday	



Formulario de inspección de zanjas y excavaciones

Documento de archivo – NO DESECHAR

Información del proyecto		Información general				Fecha	
Proyecto/Proyecto#:		Propósito de la excavación		Fecha al comienzo de la semana			
Contratista/Subcontratista		Localidad		Fecha al final de la semana			
Nombre de persona competente		Máxima profundidad					
Persona competente firma		Tamaño aproximado (pies cuadrados)					
Clasificación de tierra		(piedra estable, A, B, or C)					
Lunes	Martes	Miércoles	Jueves	Viernes	Sábado	Domingo	
Lista parcial de artículos de seguridad—Identificar y corregir toda situación peligrosa							
A	Inspeccionar como mínimo a diario; marque una "4" para los artículos que se hacen bien o que están en buenas condiciones.						
B	Marque una "A" en los artículos que necesitan mejorarse o corregirse, y comente al dorso sobre las medidas tomadas o necesarias.						
1	Protección de zanjas						
	Delineación, barricadas y avisos colocados, cubiertas de zanjas colocadas y sin desplazar.						
	Benching o inclinándose hecho cuando son apropiados (según lo determinado por la persona competente, pero requerido 5 pies o más profundizados)						
	Caja de zanja, colocación de cubiertas de zanjas o muros de contención, aseguradas y en buenas condiciones.						
	Cubiertas para hierros y otra protección contra todo peligro de empalamiento.						
	Protección contra caídas en zanjas de profundidades mayores de 6 pies.						
	Otros.						
2	Condiciones generales del sitio						
	Los escollos o impedimentos en la superficie han sido eliminados o asegurados.						
	Los despojos y otros materiales han sido corridos hacia atrás por lo menos 2 pies del borde de la excavación.						
	Las causantes de vibración (equipos, tráfico, etc.) no están demasiado cerca de la excavación y de los trabajadores.						
	Los empleados no trabajan bajo cargas suspendidas, cerca de equipo o de vehículos en proceso de carga, o dentro de pozos en excavación activa.						
	Los empleados usan vestimentas de material reflexivo de alta visibilidad al trabajar cerca del equipo en movimiento.						
	Otros						
3	Entrada/salida						
	Toda zanja de más de 4 pies de profundidad está provista de escalera portátil; se usa escalera adecuada y sujeta, que se extienda en 36 pulgadas por fuera de la zanja.						
	Las pasarelas y puentes sobre las excavaciones (exigidas para zanjas de más de 6 pies) deben ser del ancho necesario, contar con tablas de tope para los pies, barandas de protección, etc.						
	Acceso lateral sin obstrucciones a una salida, a un recorrido de no menos de 25 pies de distancia.						
	Otros						
4	Servicios públicos						
	La ubicación de todos los suministros de servicios públicos ha sido marcada con precisión, protegida con canaleta o de otra forma.						
	Se exponen, se apoyan y/o se protegen las líneas de la alcantarilla y de gas.						
	Los servicios públicos indispensables y peligrosos han sido protegidos como mínimo con canaletas horizontales cada 25 pies; canaletas cavadas a mano en un radio de 24 pulgadas alrededor de los servicios.						
5	Condición de la zanja						
	Los empleados están protegidos de la acumulación de agua y agua proveniente de la superficie o del desgaste.						
	Inspección de la zanja por posibles derrumbes, desprendimientos, grietas, etc.						
	Otros						
6	Atmósferas peligrosas						
	Pruebas de la atmósfera cuando existe la posibilidad de falta de oxígeno, explosión, combustión o contaminantes dañinos						
	Precauciones adecuadas para proteger la atmósfera, disponibilidad de equipo de emergencia, capacitación de empleados						
	Otros						



Datos específicos y comentarios sobre la inspección de zanjas y excavaciones
Asegúrese que todo asunto pendiente haya sido tratado y solucionado puntualmente.

Lunes	
Martes	
Miércoles	
Jueves	
Viernes	
Sábado	
Domingo	



Excavation & Trenching Awareness Training Verification

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Project/Project#:		Date:	
Name:		Company:	

Workers in and around excavations and trenches should be aware of the potential hazards associated with this work. Below is a review of DPR guidelines:

Competent Person

- A competent person, provided by the subcontractor, should be named for all excavation activities
- The competent person should be on the job site at all times to monitor soil conditions and inspect protective systems when personnel are working in or around any excavation of three (3') feet or greater
- The competent person should perform daily, documented inspections of protection equipment, trench conditions and adjacent areas:
- inspections should be made prior to the start of work and as needed throughout the shift
- inspections should be made after every rainstorm or other hazard increasing occurrence
- The competent person should have the authority to remove employees from hazardous conditions and make all changes necessary to ensure their safety (e.g., unstable sides or edges, water accumulation, etc.)

Utilities & Underground Installations

- Common utilities such as electrical, gas, water, sewage, fire system, telecom and security systems should be clearly identified and marked prior to start of work
- The location should be positively marked using colored paint or other easily distinguishable marking system
- Critical and hazardous utilities should always be 'pot holed' (i.e., using air, water or vacuum systems in lieu of hand digging in at least a 24" radius around each utility until the utility is exposed). Pot holing should occur at a minimum of every 25 feet for high voltage (+480V), fiber optics, gas, chemical or fuel lines and mission critical activities. Mechanical equipment should not be used for pot holing.

Access & Egress

- A stairway, ladder, or ramp should be used as a means of access or egress in trench excavations that are four (4') feet or more in depth, additionally:
 - ladder(s), stairways(s) or ramp(s) should be spaced so that no employee in the trench excavation is more than 25 feet from a means of safe egress
 - when ladder(s) are employed, the top of the ladder should extend a minimum of three (3') feet above the ground and be properly secured
 - appropriate ladders should be used

Exposure to Vehicular Traffic

- When excavations are exposed to vehicular traffic, each employee should wear appropriate garments made of high visibility and/or reflective materials
- Hard barricades and other protective devices should be installed to protect workers and operators of vehicles from exposure to the trench

Exposure to Falling Loads

- Employees should not be permitted underneath loads handled by lifting or digging equipment
- Employees should stand away from any vehicle being loaded or unloaded to avoid being struck by any spillage or falling material

Hazardous Atmospheres

- Whenever oxygen deficiency or a hazardous atmosphere exists or could reasonably exist, the area should be continuously monitored to assure that employees are adequately protected; in such excavations, a confined space permit should be obtained (refer to the Confined Space Section 04 for additional guidelines)



- Where oxygen levels are below 19.5% or above 23.5%, the area must be continuously ventilated until the oxygen levels fall within normal levels
- Where a flammable gas condition exists, the area should be ventilated until the flammable gas concentration is below 10% of the lower explosive limit (LEL)

Stability of Adjacent Structures

- Where the stability of adjoining buildings, walls or other structures is endangered by excavation operations, support systems such as shoring, bracing or underpinning should be provided to ensure the stability of such structures and for the protection of employees
- Sidewalks, pavement, and supported structures should not be undermined unless a support system such as shoring is provided to protect employees from the possible collapse of such structures

Personnel Protection & Protective Systems

- All surface encumbrances which could create a hazard to workers in the excavation should be removed or supported as necessary to safeguard workers. Employees should be protected from excavated or other materials and equipment that could pose a hazard by falling or rolling into excavations
- Protection should be provided by placing and keeping such materials and equipment a minimum of two (2') feet from the edge of excavations
- Protective systems should be designed by a registered professional engineer for excavations greater than 20 feet deep
- Employees in excavations should be protected from cave-ins by an adequate protective system, which should be routinely inspected by a competent person
- The competent person should determine the need for the use of protective systems sufficient to protect workers from all potential hazards; at a minimum, the use of protective systems is required for all excavations five (5') feet or greater, unless:
 - the excavation is made entirely within stable rock, or
 - the competent person has determined there is no potential of cave-in
- Depending upon soil types, conditions and stability, protective systems may be required for excavations less than 5 feet in depth
- Whenever support systems, shield systems or other protective systems are being used, a copy of the manufacturer's specifications, recommendations and limitations sheet should be in written form and maintained at the job site

This document hereby certifies that I have been instructed on the items listed above and I have had an opportunity to ask questions. I fully understand the DPR requirements for work in and around excavations and trenches.

Employee's Signature

Signature of person conducting verification review

A copy of this form should be faxed to the DPR regional office PRIOR TO START OF WORK.



Verificación de la capacitación en zanjas y excavaciones

Documento de archivo – NO DESECHAR

Proyecto/Proyecto#:		Fecha:	
Nombre:		Compañía:	

Los trabajadores que estén en el lugar o las proximidades de zanjas y excavaciones deben conocer los posibles peligros de este trabajo. A continuación encontrará un repaso de las guías generales de DPR:

Persona competente

- Se debe nombrar a una persona competente, provista por el subcontratista, para todas las actividades de excavación.
- La persona competente debe estar en el lugar de la obra en todo momento para vigilar las condiciones del suelo e inspeccionar los sistemas de protección cuando el personal trabaja en el lugar o las proximidades de una excavación de tres (3) pies o más.
- La persona competente debe realizar inspecciones documentadas en forma diaria de los equipos de protección, las condiciones de las zanjas y las áreas circundantes:
- las inspecciones deben realizarse antes de comenzar el trabajo y según sea necesario durante el turno
- las inspecciones deben realizarse después de cada tormenta de lluvia u otro evento que aumente la posibilidad de peligro
- La persona competente debe tener autoridad para alejar a los empleados de las condiciones peligrosas y hacer todos los cambios que sean necesarios para procurar su seguridad (por ejemplo, lados o bordes inestables, acumulación de agua, etc.)

Servicios públicos e instalaciones subterráneas

- Se deben identificar y marcar claramente los servicios públicos comunes como la electricidad, gas, agua, alcantarillado, sistema contra incendios, sistemas de seguridad y telecomunicaciones antes de comenzar el trabajo.
- El lugar debe marcarse claramente usando pintura de color u otro sistema de identificación fácilmente distinguible.
- Los servicios públicos indispensables y peligrosos deben siempre estar protegidos por canaletas (es decir, excavado manual en al menos un radio de 24 pulgadas alrededor de cada servicio público hasta que el mismo quede expuesto). La protección con canaletas debe colocarse cada 25 pies como mínimo para alto voltaje (+480V), fibra óptica, tuberías de gas, productos químicos o combustible y actividades de misión crítica.

Entrada y salida

- Se debe usar una escalera, escalera portátil, o rampa como medio de entrada y salida en las excavaciones de zanjas que tengan cuatro (4) pies o más de profundidad, además:
 - la(s) escalera(s) portátil(es), la(s) escalera(s) o las rampa(s) deben estar espaciadas de manera que ningún empleado en la excavación de zanjas se encuentre a más de 25 pies de distancia de un medio de salida seguro.
 - cuando se use una o más escaleras portátiles, la parte superior de la escalera debe extenderse como mínimo tres (3) pies por encima del suelo y estar sujeta firmemente.
 - se deben usar escaleras portátiles adecuadas.

Exposición al tráfico vehicular

- Cuando las excavaciones están expuestas al tráfico vehicular, cada empleado deberá usar la vestimenta adecuada confeccionada con materiales de alta visibilidad y/o reflexivos.
- Se deben instalar barricadas resistentes y otros dispositivos de protección para proteger a los trabajadores y operadores de vehículos de la exposición a la zanja.

Exposición a cargas suspendidas

- No se debe permitir a los empleados que permanezcan bajo cargas manejadas por equipos elevadores o de excavación.
- Los empleados deben alejarse de cualquier vehículo que esté en el proceso de carga o descarga para evitar que les caiga algún derrame o material suspendido.

Atmósferas peligrosas

- Siempre que exista o que pudiera razonablemente existir una deficiencia de oxígeno o atmósfera peligrosa, se debe supervisar el área constantemente para verificar que los empleados estén protegidos adecuadamente. Para dichas
- excavaciones se debe obtener un permiso para espacios confinados (consulte guías generales adicionales en el Anexo Espacios Confinados).
- Cuando el nivel de oxígeno sea inferior a 19,5% o superior a 23,5%, se debe ventilar el área constantemente hasta que el nivel de oxígeno esté dentro del rango normal.



- Ante la presencia de gas inflamable, se debe ventilar el área hasta que la concentración del gas inflamable sea inferior a 10% del límite explosivo inferior (LEI).

Estabilidad de las estructuras adyacentes

- Cuando la estabilidad de los edificios, paredes u otras estructuras colindantes esté en peligro por las operaciones de excavación, se deben proveer sistemas de soporte como muros de contención, apuntalamiento o socalzado para asegurar la estabilidad de dichas estructuras y para protección de los empleados.
- No se deben socavar las aceras, el pavimento y las estructuras de sostén, salvo que se provea un sistema de soporte como los muros de contención para proteger a los empleados contra el posible derrumbe de dichas estructuras.

Protección del personal y sistemas de protección

- Se deben eliminar o sujetar, según sea necesario, todos los escollos de la superficie que puedan crear un peligro para los trabajadores en la excavación, para protección de los trabajadores. Los empleados deben protegerse contra los materiales excavados u otros materiales o equipos que pudieran representar un peligro de caerse o rodar hacia las excavaciones.
- Como medida de protección se deben colocar o guardar dichos materiales o equipos a un mínimo de dos (2) pies del borde las excavaciones.
- Para excavaciones de más de 20 pies de profundidad, los sistemas de protección deben ser diseñados por un ingeniero profesional matriculado.
- Los empleados en excavaciones deben protegerse contra los derrumbes mediante un sistema protector adecuado, el cual debe ser inspeccionado periódicamente por una persona competente.
- La persona competente deberá determinar la necesidad de usar sistemas protectores suficientes para proteger a los trabajadores contra todos los peligros posibles. Como mínimo, se requiere el uso de sistemas protectores para todas las excavaciones de cinco (5) pies o más, salvo que:
 - la excavación esté hecha por completo dentro de roca estable, o
 - la persona competente haya determinado que no hay peligro de derrumbe
- Dependiendo del tipo de suelo, las condiciones y la estabilidad, se podrán requerir sistemas protectores para excavaciones de menos de 5 pies de profundidad.
- Cuando se usen sistemas de soporte, u otros sistemas protectores, se debe colocar en el lugar de la obra una copia de la hoja de especificaciones, recomendaciones y limitaciones del fabricante.

Este documento por el presente certifica que he recibido instrucción sobre los puntos listados anteriormente y que he tenido oportunidad de formular preguntas. Entiendo en forma cabal los requisitos de DPR para trabajar en el lugar o las proximidades de excavaciones o zanjas.

Firma del empleado

Firma de la persona que realiza la verificación del repaso

Se debe enviar por fax una copia de este formulario a la oficina regional de DPR ANTES DE COMENZAR EL TRABAJO.



Heavy Equipment Training Verification Form

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Name:		Company:	

Equipment:			
Manufacturer:			
Model(s):		Capacity:	

Pre-planning

- Operators should be trained in the safe operation of the equipment they will be utilizing, and understand the equipment's limitations.
- Utilities have been located and marked prior to start of work; critical systems/utilities have been pot holed (potholing requirements per the DPR EHSP have been reviewed and understood).
- Where applicable, the operator should reference blue prints and as-builts pertinent to the work area.
- Operator's manual is readily available on the equipment, and the operator has reviewed and understands the requirements outlined within.

Inspection

- Perform equipment inspection per operator's manual. At a minimum, ensure the following are inspected:
 - Ensure cab is free of loose materials and debris.
 - Ensure operator's manual is in the cab and in legible condition.
 - Check tires for cuts, bulges, chunking and proper inflation
 - Check for proper level of fluids and ensure no leaks are present
 - Check that belts, air pressure, air filter, etc. are in place and in good working condition,
 - All fittings should be adequately greased
 - Look for broken welds, cracks in the paint/metal or obvious damage
 - Inspect roll over protection for tightness of bolts/quality of welds, seat belt is in place and operational

Operation

- When mounting or dismounting, maintain three points of contact with the machine at all times.
- Use proper start procedures, consult the operator's manual for equipment specific requirements.
- Perform equipment inspection, *using the attached equipment inspection document*.
- Allow the equipment to warm up prior to operating the machine
- If applicable, the slope angle should be kept to a minimum.
- Imposed loads and how they affect adjacent excavations and / or other structures should be clearly understood
- If applicable, set stabilizers and raise the machine slightly off the ground, ensure the machine is level.
- Make sure ALL personnel and other equipment are clear of the work area, and be alert to other people & equipment
- Match ground speed with conditions of terrain
- Allow no riders
- If applicable, keep the bucket low for stability and visibility

Employee's Signature

Signature of person conducting verification review

A copy of this form should be faxed to the DPR regional office PRIOR TO START OF WORK.



Formulario de Verificación de Equipos Pesados

Documento para archivo – NO DESECHAR

Project/Proyecto#:		Fecha:	
Nombre:		Compañía:	

Equipo:			
Fabricante(s):			
Modelo(s):		Capacidad(es):	

Planificación previa

- Los operadores deben recibir capacitación sobre el funcionamiento seguro del equipo que utilizarán, y entender las limitaciones de los equipos.
- Se deben ubicar y marcar los servicios públicos antes de comenzar el trabajo y hacer los pozos necesarios en los sistemas/servicios públicos de primera necesidad (se deben revisar y entender los requisitos para hacer pozos de acuerdo con la política EHSP de DPR).
- El operador debe consultar los planos y los dibujos pertenecientes al área de trabajo, si corresponde.
- El manual del operador debe estar disponible en el equipo, El operador ha revisado y entiende los requisitos que se encuentran en la información presentada

Inspección

- Inspeccione el equipo según el manual del operador. Debe inspeccionar, como mínimo, lo siguiente:
 - Verifique que no haya materiales sueltos ni escombros en la cabina.
 - Verifique que el manual del operador se encuentre en la cabina y que se pueda leer bien.
 - Revise los neumáticos para ver si están combados, si tienen cortes, si emiten ruidos y si tienen el nivel de inflado apropiado.
 - Revise que el nivel de fluidos sea el correcto y que no haya ninguna pérdida.
 - Verifique que los cinturones de seguridad, la presión de aire, el filtro de aire, etc., estén en su lugar correcto y en buenas condiciones.
 - Todos los accesorios deben estar adecuadamente engrasados.
 - Verifique que no hayan soldaduras rotas, grietas en la pintura/metal o daños visibles.
 - Para inspeccionar la protección contra vuelcos verifique que los pernos estén ajustados correctamente, revise la calidad de las soldaduras y que los cinturones de seguridad estén en su lugar y en buen funcionamiento.

Operación

- Cuando deba subirse o bajarse de una máquina, mantenga tres puntos de contacto con la máquina en todo momento.
- Inicie la puesta en marcha del equipo de la manera correcta, consultando los requisitos específicos de los equipos en el manual del operador.
- Realice una inspección de los equipos Usando el document para inspeccionar equipos que esta anexo
- Deje que el equipo se caliente antes de operar la máquina.
- El ángulo de inclinación debe mantenerse al mínimo, si corresponde.
- Fije los estabilizadores y eleve la máquina suavemente del piso; verifique que la máquina se encuentre nivelada, si corresponde.
- Verifique que el área de trabajo se encuentre TOTALMENTE despejada de personas y equipos y esté alerta frente a las demás personas y equipos.
- Fije la velocidad de acuerdo con las condiciones del terreno.
- No acepte pasajeros en la máquina.
- Mantenga la cubeta baja, para mayor estabilidad y visibilidad, si corresponde.

Firma del empleado

Firma de la persona que realiza la verificación del repaso

Enviar por fax una copia de este formulario a la oficina regional de DPR ANTES DE INICIAR EL TRABAJO



Heavy Equipment Inspection / Inspeccion de Equipo

Archive Document - DO NOT DISCARD

Project/Project#/Proyecto/Proyecto#:				Last Oil Change / Untimo Cambio de Acieta:		
Operator Name / Operador:				Foreman / Miodomo:		
Company / Compania:				Hours to Next Oil Change / horas de cliente cambio de acieta		
Model / Modelo:				Equipment Number / Equipo de numero:		

Date/ Fecha:						
Initials of person performing inspection / Iniciales de persona realizar inspecciones:						
Inspection Item / Description (Ok/R/A)		Ok/R/A	Ok/R/A	Ok/R/A	Ok/R/A	Ok/R/A
1	Current Hours / Horas Corriente					
2	Operator's Manual / Manual de Operacion					
3	Engine Oil or Leaks / Aceite de Motor o Escape					
4	Transmission Oil / Aceite de Transmission					
5	Hydraulic Oil / Aceite de Hidraulico					
6	Coolant Leaks / Escape de Refrigerante					
7	Fire Extinguisher / Extinguisher de Fuego					
8	Horn / Cuerno de Vehiculo					
9	Backup Alarm / Alarma Secundaria					
10	Seatbelt or ROPs / Cinturon de Seguridad					
11	Glass & Wipers / Ventanta & Limpiaparabrisas					
12	Lights & Blinkers / Luces & Indicador					
13	Steering & Brakes / Volante & Freno					
14	Tires & Rims / Llantia y Rines					
15	Panels & Covers / Paneles y Cubiertas					
16	Cab Clean / Limpieza de Cabina					
17	Exhaust Leaks / Escape de Gases					
18	Belts / Cinto					
19	Pivot Pins / Clavos de Pivote					
20	Bucket condition / Condicion de Bote					

** If any item(s) are found to be missing or defective, the equipment is to be repaired/serviced prior to use.

Defects Reported		Date	Equipment taken out of service		Yes	No
To			☐	☐	☐	☐
Legend/Legendario		Date	Repair needed/ Reparacion necesitado		A - Added/ Agregar	
☐ OK - checked okay/ esta bien la mequira			☐ R			



Guardrail Safety Checklist

Archive Document – DO NOT DISCARD

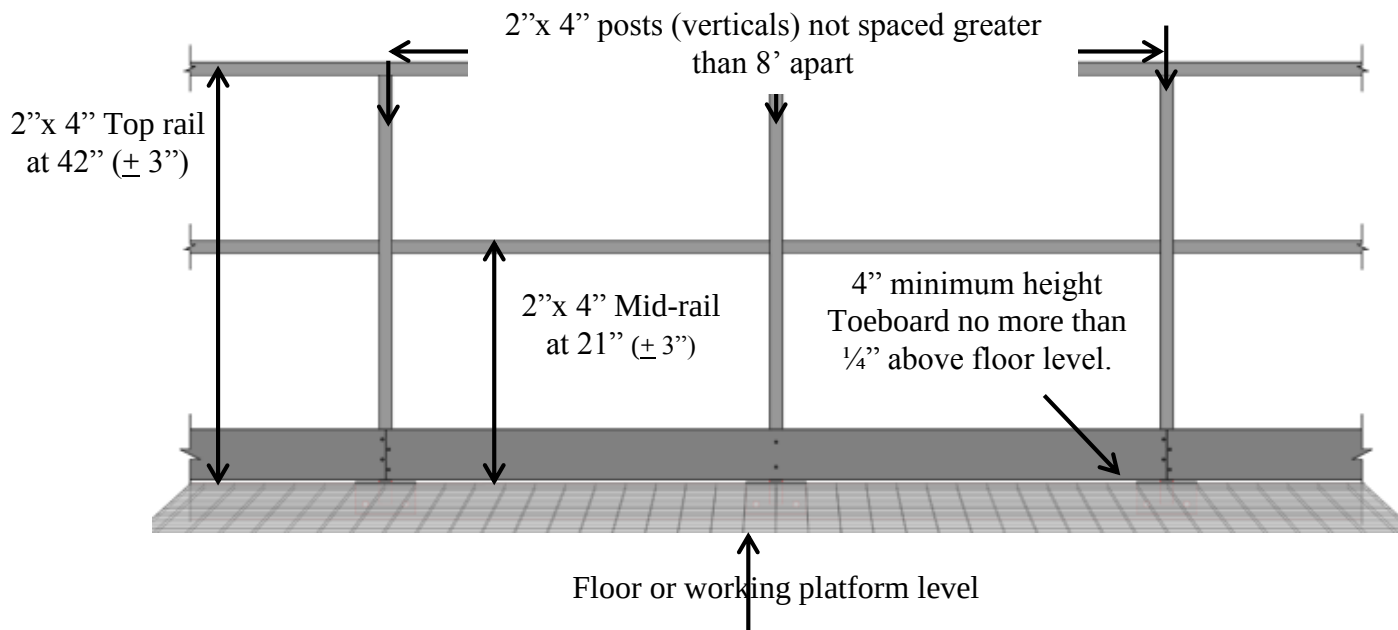
Project/Project#:	Date:
Competent Person:	Company:

		Mon		Tues		Wed		Thurs		Fri		Sat		Sun	
	Item	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
1.	A top rail is installed at 42" (1.1 m) ($\pm$ 3") above the walking/working level?														
2.	Top rail is capable of withstanding a 200lb load of force applied within 2" (5.1 cm) of the top edge, in any direction and at any point along the edge, and without causing the top edge of the guardrail to deflect downward to a height less than 39" (1.0 m) above the walking/working level?														
3.	A midrail is installed at 21" (53 cm) ($\pm$ 3") above the walking/working level?														
4.	Midrail is capable of withstanding a 150lb load of force applied in any direction at any point along the midrail or other member?														
5.	A 4" nominal, minimum height toeboard is installed?														
6.	Toeboard is no more than 1/4 inch clearance above the floor level, and is secured to withstand a 50lb load of force?														
7.	Wooden railing posts (verticals) should be made of at least 2" x 4" stock or its equivalent?														
8.	If the top rail, midrail, or toeboard must be "spliced", the splice has been made with the integrity to withstand the appropriate load for the applicable rail?														
9.	Posts (verticals) are spaced so as not to exceed 8 feet on center?														
10.	Top rails and midrails are installed not to create a projection hazard by overhanging the vertical posts?														
11.	Toeboards are installed not to create a trip hazard by overhanging the vertical posts?														
12.	Guardrail system has been maintained so that it does not have rough or jagged surfaces that would cause punctures, lacerations, or snagged clothing?														
13.	Wire rope top rails are flagged with high visibility flagging at 6' intervals?														
14.	Screens and mesh, when used, shall extend from the top rail to the walking/working level and along the entire opening between top rail supports?														
15.	When guardrail systems are used around holes which are used as points of access (such as ladders), they are provided with a gate, or be so offset that a person cannot walk directly into the hole?														

Note: Any items answered "no" below should be immediately addressed, with corrective measures tracked to completion.

Guardrail Safety Checklist

Archive Document – DO NOT DISCARD



Guardrail Systems

A standard guardrail should consist of a top rail, intermediate/mid-rail, toeboard and posts. Components of a guard rail system may consist of 2" X 4" stock, 3/8-inch wire rope with three clamps per OSHA 1926.251.

Section 11

Fire Protection & Prevention

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Fire Protection & Prevention

1.0 Introduction

The DPR Construction Fire Protection and Prevention Plan have been developed to work in conjunction with company emergency plans and other safety programs. This includes reviewing all new building construction and renovations to ensure compliance with applicable local, state, and national fire and life safety standards. Fire prevention measures reduce the incidence of fires by eliminating opportunities for ignition of flammable materials.

Reference: Portable Fire Extinguishers CFR 1910.157; Portable Fire Extinguishers NFPA Standard #10; Fire Test Standard-ANSI/UL 711, Standard for Rating and Fire Testing of Extinguishers; Fire Protection and Prevention CFR 1926.150 – Subpart F. Other applicable Standards, guidelines, and regulations may apply.

2.0 Fire Protection

2.1 General Requirements

It is DPR's intent that the installation and maintenance of fire protection systems and the abatement of any identified fire hazard should be completed as soon as practicable. Further, all fire-fighting equipment should be inspected periodically and located in a conspicuous area easily accessed by all employees.

2.2 Fire Fighting Equipment

Portable fire extinguishers rated not less than 2A should be provided at a minimum of every 3,000 square feet of building under construction or any occupied space.

- Maximum travel distance from any point to the nearest extinguisher should not exceed 100 feet (75 feet in CA).
- At least one 2A portable fire extinguisher should be located adjacent to the landing of each stairway.
- Extinguishers should be visually inspected daily.
- Extinguisher should be inspected monthly with documentation provided in accordance with project requirements; an annual inspection tag or log must be maintained.
- Fire extinguishers should be readily available within 10 feet of any hot work; fire extinguishers provided for building protection are not adequate to fulfill this requirement.
- An unobstructed path and a clear area of at least 36 inches should be provided around all fire extinguishers or firefighting equipment for access at all times.
- All fire extinguishers or firefighting equipment should conspicuously marked.
- A portable multi-purpose dry chemical fire extinguisher with a minimum of 5BC rating or higher should be available at the operator's station of all cranes, derricks, hoists, material lifting devices, equipment and elevators.

Additional requirements include:

- Fire extinguisher training should be provided for any worker expected to fight a fire.
- Using a fire extinguisher only to escape from a building or to fight a very small (incipient stage) fire; do not attempt to fight large fires with a fire extinguisher.
- All construction work vehicles and equipment should be equipped with an appropriately rated fire extinguisher.

2.3 Fixed Fire Fighting Equipment

2.3.1 Sprinkler Protection

If automatic sprinkler protection is to be provided, the installation should be placed in service as soon as practicable and should be in compliance with NFPA 13 "Standard for Installation of Sprinkler Systems". If sprinklers are required for life safety, the building should not be occupied until the installation has been

completed and tested. If sprinkler protection is being regularly turned off and on to connect newly completed segments, the control valves should be checked at the end of each work period to determine that protection is in service.

2.3.2 Standpipes and Risers

In all buildings where standpipes are required, they should be maintained in such a manner that they are always ready for use. The standpipes should be provided with conspicuously marked Siamese connections on the outside of the building at street level, with at least one standard hose outlet at each floor. The installation should be in compliance with NFPA 14, "Standard for Installation of Standpipes". Standpipes should be carried up with each floor and capped at the top. Temporary standpipes should remain in service until the permanent standpipe is completely installed.

2.4 Crisis Management Plan

It is our goal to maintain the safest projects in the industry. Although we do everything we can to provide an Incident Free Environment, the possibility of a crisis occurring on one of our projects still exists. The Crisis Management Plan is designed to help project teams react to crisis situations, should one arise.

It is expected that these emergency response procedures be made familiar to every worker on the project through the Site Specific Safety Orientation, jobsite postings, Pre-construction Safety Meetings and reinforced at the DPR Construction Staff Meetings, Subcontractor Meetings and as a topic for Weekly Safety Meetings. *For further information on the site specific crisis management plan please refer to the Emergency Procedures Section 08.*

2.5 Fire Fighting Access

- A pre-planning meeting should be completed with local fire department officials to discuss the scope and potential hazards of the project and to plan access routes for fire department apparatus
 - o this meeting should occur prior to actual construction.
 - o it is recommended that periodic site visits be completed with the fire department throughout the course of construction.
- Every building should be accessible to fire department with free unobstructed access to fire fighting apparatus via access roads with all-weather driving surfaces of not less than 20 feet of unobstructed width, able to withstand the live loads of fire department apparatus and having a minimum of 13'6" of vertical clearance
 - o the required width of access roadways should not be obstructed in any manner, including parking of vehicles.
 - o "No Parking" signs or other appropriate notice prohibiting obstruction should be posted in these areas.
- Access for fire-fighting equipment should be provided to the immediate job site from the start of the project and through completion.
- Where a bridge is required to be used as access, it should be constructed and maintained using design live loading sufficient to carry the imposed loads of the firefighting apparatus.
- Where hoists and elevators provide the only efficient means of transporting hose and other cumbersome firefighting equipment to upper floors, they should be available to the fire department whenever necessary.

2.6 Fire Emergency Notification

A system for notification of employees and the local fire department in the event of an emergency should be established. Examples include standard alarm systems, sirens, bell, air horns etc. *Refer to the Emergency Procedures Section 08 for further information.*

3.0 Fire Prevention

3.1 Ignition Hazards

3.1.1 Smoking

Smoking, of any kind, is prohibited in the vicinity of hazardous operations or locations and may be further restricted to designated areas only. “No Smoking” signs should be conspicuously placed in hazardous areas and no-smoking rules strictly enforced. If specified smoking areas are provided, suitable ash receptacles should be furnished. Cigarette and cigar butts should only be disposed of in designated ash receptacles.

3.1.2 Temporary Electrical Equipment

All electric installations and repairs should be performed by qualified personnel and must comply with applicable provisions of the National Electric Code. The type and size of conductors should be adequate for the expected load. Proper fuses and circuit breakers should be provided. Conductors should not rest on floors and should not be installed so as to be subject to physical damage. In areas containing flammable vapors or dusts, only electrical equipment specifically approved for use in such hazardous locations should be used. Ground fault circuit protection should be provided in accordance with the National Electric Code requirements.

3.1.3 Equipment Maintenance

Proper maintenance of equipment should be completed, particularly on gasoline or diesel powered equipment. Maintenance checks should be made on fuel tanks and lines, carburetors, hydraulic reservoirs and lines, and exhaust systems. Fixed equipment should be clear of combustible materials. Equipment should be regularly inspected and maintained by qualified personnel and complete records maintained. Particular attention should be given to fueling methods. Mobile equipment should be fueled outside the building in a manner minimizing potential spillage and contamination. Employees involved in fueling operations should not operate equipment unless their clothing and PPE is free of fuels, oils or grease. *Refer to the Chemical and Environmental Management Section 02 for specific requirements on equipment fueling and maintenance procedures.*

3.1.4 Welding, Cutting and Brazing

The following requirements will assist with welding, cutting and brazing fire protection and general safety requirements during these operations:

- Combustible materials of any kind are to be removed from the work area; when it is impractical to remove combustibles, a flame-retardant tarpaulin should be used to protect combustibles.
- “Hot Work Permits” are required with all welding, cutting and brazing operations, *refer to the Hot Work Permit in the Forms Section of the EHSP.*
- No welding, cutting or heating should be done where the application of flammable paints or the presence of other flammable compounds or heavy dust concentrations creates a fire hazard Note: See SDS for all materials.
- All torches should be equipped with anti-flashback devices.
- Appropriately rated fire extinguishers are required at all times in accordance with federal state and local regulations.
- Arc welding and cutting operations should be shielded by non-combustible or flameproof shields to protect the public and workers from direct arc rays.
- When electrode holders will be left unattended, electrodes should be removed and holder should be placed or protected so they cannot make electrical contact.
- All arc welding and cutting cables should be completely insulated.
- There should be no tape, repairs, cuts or splices within 10 feet of electrode holder. Defective cables should be repaired or replaced.

- Some state programs may allow splices to cables, provided insulation is equal to the original cable; some state programs may have more stringent requirements for cables, check your specific requirements.
- Fuel gas and oxygen hoses should be easily distinguishable and not interchangeable.
- Hoses should be inspected at the beginning of each shift and repaired or replaced if defective.
- General mechanical or local exhaust ventilation or appropriate respirators should be provided as required when welding, cutting or heating hazardous materials, or in confined spaces.
- Approved tinted eye protection should be worn when welding or when in areas where welding is being performed.

3.1.5 Woodworking

The woodworking area preferably should be set up in a remote area of the project. Dust accumulation should be minimized by good housekeeping procedures; and, if necessary, dust collecting apparatus should be installed for power equipment. Dust, scraps and refuse should be removed regularly and properly disposed. Smoking, of any kind, is prohibited in woodworking areas.

3.1.6 Temporary Heating Devices

The use of temporary heaters fueled by liquefied petroleum gas (LPG), liquid-fuel oil, natural gas or other fuels should meet the following general guidelines:

- Fresh air should be present in sufficient quantity to maintain the safety of workers.
- Storage of propane/LPG is prohibited inside buildings and on scaffolds.
- Gas heaters should be equipped with an automatic flame loss device which will shut off gas supply if the flame or pilot is extinguished.
- Heaters should meet manufacturer's clearance requirements from all combustible materials; at a minimum, heaters should have a clearance of at least 12 inches for circulating type heaters and 36 inches for radiant type heaters.

3.2 Fire Watch

Hot work operations that may create a fire hazard will require appropriate fire watch personnel for the duration of the task and for a minimum of ½ hour following the conclusion of these tasks; owner or project specific requirements may vary. Fire watch personnel should be trained and provided with adequate fire protection devices. Fire extinguishers provided for building protection are not adequate to fulfill this requirement. Where there exists a possibility of sparks falling to another level or location, fire watch personnel should be posted in a manner to adequately protect these areas. *Refer to the Forms Section at the end of the EHSP for Fire Watch training verification forms at the end of the EHSP.*

3.3 Temporary Structures

Temporary structures located inside a building should be of non-combustible construction or protected with automatic sprinklers. Temporary structures located outside should be separated from each other by at least 10 feet and not clustered together.

3.4 Storage

Flammable and combustible materials such as forms, scaffolding, shoring, plastics, trim, wall and roofing materials etc. should meet the following general guidelines:

- Good housekeeping practices should be maintained on a daily basis.
- Surplus materials should be removed from the site as soon as possible.
- Flammable and combustible materials should be separated from each other and stored in a remote location away from construction activities and other hazards.
- Flammable and combustible materials should be brought into the building in minimal quantities, and at a minimum, meet federal, state and local requirements.

- Stairs, landings, stairwells and egress paths should not be used for storage of equipment, tools and/or materials.
- At least 36 inches should be maintained between the top level of the stored material and the sprinkler deflectors/dispersers and around fire door openings.
- Appropriate clearance should be maintained around electrical equipment, lights and heating units to allow access and prevent ignition of flammable and combustible materials.
- For outdoor storage, weeds, grass and other vegetation should be kept down and a regular procedure provided for periodic cleanup.

3.5 *Blasting*

Highly specialized controls are necessary whenever explosives of any kind are to be used. All recognized precautions relating to qualifications of personnel, storage locations, containers, transportation and handling should be strictly followed. *Refer to the Demolition Section 06 for further information.*

4.0 Flammable and Combustible Liquids

Flammable and combustible liquids such as fuels, solvents, paints and adhesives etc. should meet the following general guidelines:

4.1 *Storage and Handling*

Flammable and combustible liquids should be stored in appropriate locations and manner.

- Fuels should be stored and handled in accordance with NFPA 30, which includes secondary containment; large capacity storage of fuel must have prior approval from DPR project management.
- Flammable liquids must be stored overnight in a secondary containment system.
- Locate storage areas 20 feet at a minimum from any building and keep free from weeds, debris and other combustible materials.
- Keep flammable liquids in closed containers when not in use.
- Provide adequate ventilation
- Storage areas should be graded to divert possible spills away from buildings or other exposures, or surround storage areas with a curb or dike.
- Storage areas should be marked with appropriate signage.
- Smoking, of any kind, is prohibited around fuel storage areas and while handling.
- "NO SMOKING" signs should be conspicuously posted around service, refueling and storage areas.
- Only approved containers and portable tanks should be used for the storage and handling of flammable and combustible liquids.
- All equipment using liquefied petroleum gases should be fueled in accordance with NFPA 58 *Liquefied Petroleum Gases*.
- An adequate number of appropriately rated fire extinguishers should be conveniently and conspicuously located in all storage and service areas.
- No more than 25 gallons of flammable or combustible liquids should be stored in a room outside of an approved storage cabinet.
- When oxygen and fuel gas tanks are not used for a period of greater than 24 hours, they should be separated by at least 20 feet or by a half hour rated firewall.
- Flammable chemicals or liquids should not be used within 35 feet of areas where open flames are present or Hot Work operations are in progress.

4.2 *Fuel Storage Tanks*

It is DPR's intent to reduce the quantity of chemicals stored on site. The preferred method for providing large quantities fuel and fueling of equipment is by "just-in-time" fueling services. Where this option is not feasible, total quantities of fuels stored will be limited to 550 gallons per project location. Where on-site fuel storage is necessary, prior approval from DPR project management must be obtained. Should approval for a fuel storage tank be granted, the following guidelines apply:

- All federal, state and local requirements for handling, storage and disposal should be followed.
- The fuel storage tank location should be in an isolated area, away from building, laydown areas, public traffic, etc.
- Site specific refueling procedures incorporating the requirements of this section should be developed, communicated to affected workers and strictly followed.
- Usage logs should be maintained.
- Fuel tanks should be installed per manufacturer's specifications, as well as including, at a minimum:
 - o Double containment
 - o Berms
 - o Signage
 - o Fire protection
 - o Canopy or appropriate weather protection

5.0 Fire Protection during Demolition

Specific procedures for fire prevention during demolition should be outlined within the site specific demolition plan. All potential sources of ignition should be evaluated, and any corrective measures identified should be taken. This plan will also include assignments of key personnel and define the emergency procedures. *For more information, refer to the Demolition Section 06.*

6.0 Inspections

A regular documented inspection should be conducted by a qualified person to determine the condition of fire protection equipment and to identify fire hazards. Corrective actions should be identified and addressed in a timely manner.

7.0 Fire Investigations

DPR should be notified immediately of any fires. All fires should be promptly investigated and appropriate corrective actions implemented.

8.0 Hot Work Permits

Hot work is defined to be any operation that involves an open flame or produces heat and/or sparks. Hot work permits should be obtained prior to any hot work activities. The hot work permit should be posted or readily available at the location of the hot work. **Permits should have all signatures in place prior to initiating hot work and prior to leaving the site.** *Refer to the Forms Section at the end of the EHSP for a copy of the Hot Work Permit and hot work signage.*

DPR project management will determine the necessity of a permit based on the activities being performed. Hot work permits are issued each day by a member of the DPR project management, and are only valid for the duration of the date specified on the permit, up to a maximum of one week. Upon completion of the work, hot work permits will be signed and submitted to DPR.

DPR hot work procedures call for a walk of the work area to be conducted prior to issuance of the Hot Work Permit. Items to be evaluated during this walk include:

- Are flammable or combustibles within 35 feet clear or protected?
- Is a fire extinguisher available within 10 feet of hot work to be performed? Is this extinguisher in addition to extinguishers provided for building protection?
- Is the fire protection system on-line in the work area?
- Is the equipment in good working order?
- Adequate number and type of fire extinguishers provided?
- Emergency procedures clearly defined and communicated?
- Is proper ventilation provided for the potential of fumes or smoke?
- Are there fresh air/return air intakes in close proximity to work area that should be covered/protected?



- Is sufficient public protection provided? Welder's screens and/or Welding blankets provided and in good condition?
- Does the work involve a confined space?
- Is a fire watch required?
- Has the fire watch been properly trained?

Fire Protection & Prevention Forms

Forms

Fire Watch Training Verification Form – English
Fire Watch Training Verification Form – Spanish
Hot Work Permit – English
Hot Work Permit – Spanish
Hot Work Warning – English / Spanish
Oxy-Acetylene Cutting & Welding Torch Inspection Checklist



Fire Watch Training Verification Form

Archive Document – DO NOT DISCARD

Name:		Date:	
Project/Project#:		Company:	

Workers should be trained in hazards associated with hot work activities. Hot Work is considered a high hazard activity by DPR and therefore requires daily, detailed, written pre-task planning prior to use; additionally, a hot work permit should be completed and signed off by DPR project management prior to start of work. Below is a review of DPR guidelines:

Fire Watch Guidelines:

- Fire watch personnel should be trained on usage of the fire extinguisher, i.e., P-A-S-S:
 - P: Pull the pin
 - A: Aim at the base
 - S: Squeeze the handle
 - S: Sweep side to side
- Fire watch personnel should have a fully charged extinguisher, appropriate for hazards in the work area, and the extinguisher should be within 10 feet of the work area at all times.
- Combustible and flammable materials should be removed or protected within 35 feet of hot work.
- Fire watch personnel should have no other duties assign while the hot work is in progress.
- Fire watch personnel should be able to communicate with workers performing hot work.
- Fire watch personnel are authorized to stop work if necessary and restore safe conditions within the hot work area.
- Fire watch personnel should remain in the hot work area for at least 30 minutes after completion of the hot work.
- Fire watch personnel, if unable to extinguish fire in the areas exposed to the hot work, activate the alarm and/or warning system, and evacuate the building.

This document verifies that I have been instructed on the items listed above and I have had an opportunity to ask questions. I fully understand the DPR requirements for Fire Watch/Hot Work activities.

Employee's Signature

Signature of person conducting verification review

A copy of this form should be faxed to the DPR regional office PRIOR TO START OF WORK.



Formulario de verificación de vigilancia contra incendios

Archivar documento – NO DESECHAR

Nombre:		Fecha:	
Proyecto/Proyecto#:		Compania:	

Los trabajadores deben recibir capacitación sobre los riesgos que entraña el trabajo realizado a altas temperaturas. DPR considera este tipo de trabajo una actividad de alto riesgo y, por este motivo, requiere una planificación diaria, detallada y por escrito antes de su inicio. Además, antes del inicio de las tareas es necesario obtener un permiso para trabajar a temperaturas elevadas firmado por la gerencia de proyectos de DPR. A continuación se presenta un resumen de las directrices de DPR:

Directrices de vigilancia contra incendios:

- El personal de vigilancia contra incendios debe recibir capacitación sobre el uso de extintores de incendios, por ejemplo, los cuatro pasos de utilización:
 1. Tirar del pasador
 2. Apuntar a la base
 3. Apretar el mango
 4. Hacer un barrido de lado a lado
- El personal de vigilancia contra incendios debe tener un extintor totalmente lleno, adecuado para los riesgos potenciales del área de trabajo, y éste debe estar ubicado en todo momento a una distancia máxima de diez pies del área de trabajo.
- Todo material combustible e inflamable debe retirarse o colocarse en una zona protegida situada a unos 35 pies de distancia de la zona donde se trabaje a temperaturas elevadas.
- El personal de vigilancia contra incendios no debe tener asignada ninguna otra tarea mientras se esté realizando algún trabajo a temperaturas elevadas.
- El personal de vigilancia contra incendios debe poder comunicarse con los trabajadores que están realizando el trabajo a temperaturas elevadas.
- El personal de vigilancia contra incendios está autorizado para detener el trabajo si fuera necesario para restablecer las condiciones de seguridad dentro del área de trabajo a temperaturas elevadas.
- El personal de vigilancia contra incendios debe permanecer en el área de trabajo a temperaturas elevadas durante al menos 30 minutos después de la finalización del trabajo a temperaturas elevadas.
- El personal de vigilancia contra incendios, si no es capaz de extinguir el incendio en las zonas expuestas al trabajo a temperaturas elevadas, debe activar la alarma o advertir sistema y evacuar el edificio.

Este documento verifica que he recibido instrucción en lo referente a los elementos enumerados anteriormente y he tenido la oportunidad de hacer preguntas. Entiendo por completo los requisitos de DPR relacionados con la vigilancia contra incendios y las actividades de trabajo a temperaturas elevadas.

Firma del empleado

Firma de la persona que realiza la verificación del
repasso

ANTES DE COMENZAR EL TRABAJO es necesario enviar por fax una copia de este formulario a la oficina regional de DPR.



Hot Work Permit

Permiso Para Trabajos Peligrosos
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This Hot Work Permit is required for all operations involving open flames or producing heat and/or sparks. Permit must be posted in the work area with appropriate signage. Hot Work includes, but is not limited to: Brazing – Cutting – Grinding – Soldering – Thawing Pipe – Torch Applied Roofing – Welding – Chop Saw Use

Project/Project#:		Date:	
Contractor Performing Work:		Permit Start Date*:	
Nature of Work:		Expiration Date**:	
Name of Person(s) Performing Hot Work:			

***Occupied Facilities - Maximum duration one shift**

****Non-Occupied – Maximum duration one week**

Fire Prevention and Protection Checklist			
Complete the following checklist for all steps taken / needed to protect the work area.			
Fire Systems		Work on enclosed equipment or spaces	
☐	Available sprinklers, hose streams, and extinguishers are in service / operable	☐	Enclosed equipment or space cleaned of all combustibles
☐	Suitable fire extinguisher or fire prevention system provided specific for hot work is within 10 feet of work	☐	Containers purged of flammable liquids/vapors
☐	Sprinklers, standpipe systems, etc. are protected from hot work, as applicable	☐	Work does not create hazardous gas or potential for buildup (argon, welding fumes etc.)
Area Preparation		Hot Work on Walls, Ceilings or Upper Decks/Floors	
☐	Flammable liquids, dust, lint and oily deposits removed/cleaned	☐	Construction material is noncombustible
☐	Welding screens installed	☐	Construction material is covered with noncombustible covering or insulation
☐	Explosive atmosphere in area eliminated	☐	Combustibles on other side of walls moved away.
☐	No flammable chemicals or liquids within 35 feet of Hot Work	☐	All areas and floors below have been examined for combustibles
☐	Floors are swept clean and combustible floors are wet down, covered with damp sand, fire protective tarp or other protection	Fire Watch/Hot Work Area Monitoring	
☐	Remove all combustibles where possible. If not possible, ensure all combustibles are protected with fire-resistive sheets or tarps or metal shields	☐	Fire watch will be provided during and for 30 minutes after work, including any coffee or lunch breaks
☐	All wall and floor openings covered to prevent falling sparks	☐	Fire watch is supplied with suitable extinguishers, charged small hose and is trained in the use of this equipment
☐	Fire-resistive tarps suspended beneath work	☐	Fire watch is in communication with hot work activities
☐	Work coordinated with other trades in area	☐	Separate fire watch may be needed for other floors, behind walls or other areas
☐	Hot Work equipment is in good repair	☐	Fire monitor has no other responsibilities other than fire watch
☐	Attached checklist completed for oxy-acetylene tanks	Personal Protective Equipment	
Other Precautions Taken		☐	Proper eye / face protection – hood, tinted glasses, etc.
☐		☐	Proper shroud, gloves, gauntlets etc. used
☐		☐	Proper protective clothing worn

Contractor Supervisor Signature _____

By signing above, you verify that the above location has been examined and that the above precautions have been taken.

DPR Supervisor Signature _____

By signing above, you verify that the above location has been examined and that the above precautions have been taken.

Fire Prevention and Protection Checklist

Person Performing Hot Work _____

By signing above, you verify that the above location has been examined and that the above precautions have been taken.

Fire Watch Monitor _____



Hot Work Permit

Permiso Para Trabajos Peligrosos

Archive Document – DO NOT DISCARD

Se exige este Permiso para Trabajos Peligrosos para toda operación con llamas al descubierto o donde se produce calor y/o chispas. El Permiso debe ser colocado en la zona de trabajo junto con los carteles del caso. Sin limitarse a éstos, el Trabajo Peligroso incluye lo siguiente: Soldadura amarilla Descongelado de tuberías – Cortes Techo instalado a soplete – Pulido Soldadura pesada - Soldadura blanda Uso de sierra

Proyecto/Proyecto#:		Fecha:	
Contratista que hace el trabajo:		Fecha de inicio del permiso*:	
Naturaleza del trabajo:		Fecha de vencimiento**:	
Nombre de la persona que hace el trabajo peligroso:			

Instalaciones Ocupados – la duracion maxima de un turno** *No-Ocupado – Maximo una semana de duracion**

Lista de verificación-prevención y protección contra incendios			
Complete la siguiente lista para verificar cumplimiento de toda medida tomada o necesaria para proteger la zona de trabajo.			
Sistemas contra incendios		Trabajo en espacio o equipo confinado	
☐	Los rociadores, mangueras y extinguidores disponibles están en servicio y operables	☐	Eliminación de todo combustible del equipo o espacio confinado.
☐	Un sistema apropiado de extinguidores o de prevención de incendios está a una distancia de 10 pies del trabajo.	☐	Purgado de todo líquido o vapor inflamable de los contenedores
☐	Aspersores (sprinklers), la pipa de soporte, etc. estan protegidas de trabajo caliente, como aplique	☐	Asegurar que el trabajo no produce gases peligrosos ni la posibilidad de acumulación de gases (argón, vapores de la soldadura, etc.)
Preparación de la zona		Trabajo peligroso en paredes, cielos rasos o pisos altos	
☐	Limpieza o eliminación de líquidos inflamables, polvo, pelusas y depósitos aceitosos	☐	El material de construcción no es combustible
☐	Instalación de pantallas para soldar	☐	El material de construcción está cubierto por un aislamiento o cobertura no combustible
☐	Eliminación de gases explosivos de la zona	☐	Se han retirado los combustibles del otro lado de la pared.
☐	Eliminación de sustancias químicas o líquidos inflamables 35 pies a la redonda de la zona del trabajo peligroso	☐	Se ha verificado la presencia de combustibles en todas las zonas y pisos inferiores
☐	Limpieza de pisos con escoba. Mojar los pisos combustibles, cubrirlos con arena húmeda y colocar una lona protectora contra incendios u otro tipo de protección	☐	
☐	Donde sea posible, eliminar todo combustible. Si esto no es posible, asegurar que todos los combustibles estén protegidos con telas o lonas resistentes a los incendios o pantallas metálicas	Vigilancia contra incendios/zona de trabajo peligroso	
☐	Cubrir toda las aperturas en paredes o pisos para impedir que caigan chispas	☐	Se mantendrá vigilancia contra incendios durante el trabajo y durante los 30 minutos que sigan, incluyendo descansos y horas de almuerzo
☐	Colgar lonas resistentes a los incendios debajo del trabajo	☐	El guardián contra incendios contará con extinguidores apropiados, una manguera pequeña cargada y estará entrenado en el uso de este equipo
☐	Coordinar el trabajo con otros oficios en la zona.	☐	El guardián contra incendios está en comunicación con las actividades peligrosas.
☐	El equipo para el trabajo peligroso se encuentra en buenas condiciones	☐	Es posible que se necesite vigilancia separada contra incendios en otros pisos, detrás de paredes o en otras zonas
☐	Attached checklist completed for oxy-acetylene tanks	☐	El guardián contra incendios tiene como única responsabilidad la vigilancia contra incendios
Otras medidas de precaución tomadas		Equipo de protección personal (PPE)	
		☐	Protección adecuada de ojos y cara – capucha, gafas oscuras, etc.
		☐	Uso de la pantalla protectora adecuada, guantes, guantes industriales, etc.
		☐	Uso de vestimenta protectora adecuada

Firma del supervisor del contratista

Mediante su firma, usted confirma que la localidad indicada arriba ha sido examinada y que las precauciones que se indican arriba han sido tomadas.

Firma del supervisor de DPR

Mediante su firma, usted confirma que la localidad indicada arriba ha sido examinada y que las precauciones que se indican arriba han sido tomadas.

Visto bueno—Vigilancia contra incendios

Persona que realiza el trabajo peligroso

Mediante su firma, usted confirma que la localidad indicada arriba ha sido examinada y que las precauciones que se indican arriba han sido tomadas.

Guardián contra incendios

Mediante la firma de arriba, de comprobar que el area de trabajo Y todas las zonas adyacentes fueron monitoreados durante 30 minutos siguientes el trabajo caliente y se encontraron a salvo.



WARNING!

¡ADVERTENCIA!

HOT WORK IN PROGRESS
TRABAJO PELIGROSO EN CURSO

WATCH FOR FIRE!
¡VELAR POR EL FUEGO!

IN CASE OF EMERGENCY	
CALL	
AT	
EN CASO DE EMERGENCIA	
LLAME	
AL	

WARNING!

¡ADVERTENCIA!

THIS NEEDS TO BE PRINTED ON YELLOW PAPER BEFORE POSTING
SE DEBE IMPRIMIR ESTO EN PAPEL AMARILLO ANTES DE PONER A LA



Oxy-Acetylene Cutting & Welding Torch Inspection Checklist

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Name:		Date:	
Project:		Project Number	

	Items	Yes	No	N/A
1.	Have employees been trained?	☐	☐	☐
2.	Are the cylinders secured to a cart or structure?	☐	☐	☐
3.	Are the regulators and gauges in good condition?	☐	☐	☐
4.	Are the hoses in good condition?	☐	☐	☐
5.	Is the torch assembly in good condition?	☐	☐	☐
6.	Are flashback arrestors in use (in line or built in)?	☐	☐	☐
7.	Is proper eye protection on hand and used?	☐	☐	☐
8.	Are welding gloves available?	☐	☐	☐
9.	Is the wrench/handle kept on the cylinder valve?	☐	☐	☐
10.	Is acetylene pressure maintained below 10 PSI?	☐	☐	☐
11.	Are the valves closed and lines bled after use?	☐	☐	☐
12.	Are spare cylinders secured to prevent tipping?	☐	☐	☐
13.	Are oxygen cylinders stored at least 20 feet from fuel gasses or petroleum products?	☐	☐	☐
14.	Are cylinders stored away from heat sources?	☐	☐	☐
15.	Is a fire extinguisher available when using a torch?	☐	☐	☐
16.	Is a fire watch used when needed?	☐	☐	☐

Section 12

Hazardous Energy

Sub-Section	Topic	Page #
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2.0	Lockout/Tagout	4
3.0	Energized Electrical Work	8
4.0	Overhead Power Lines	13
5.0	Systems Testing	14

The Control of Hazardous Energies

(Includes Lockout/Tagout and Energized Electrical Work)

1.0 Introduction

The Control of Hazardous Energies (commonly referred to as Lockout/Tagout) section has been developed to establish the procedures for the control of potentially hazardous energy sources. These sources include electrical, hydraulic (fluid under pressure), pneumatic (air under pressure), gravity, thermal (excessive heat/cold), chemical, mechanical (moving parts), and radiological energies. The need to ensure that machines, equipment or fixtures are isolated from all potentially hazardous energy sources, before service, maintenance, demolition or installation activities commence is necessary for employee safety and the integrity of affected tools and equipment. The lockout/tagout protocol should apply to all permanently wired machines and equipment. *Cord and plug connected equipment are exempt from these requirements, provided that the cord is unplugged and under the exclusive control of the employee performing the service or maintenance.*

It is DPR's intent to perform all work in a de-energized state. Energized electrical work should not be allowed unless it can be demonstrated that de-energizing introduces additional hazards or increased risk, or it can be demonstrated it is infeasible to perform the work in a de-energized state. Increased risk and/or additional hazards may be other than electrical in nature, such as healthcare equipment, environmental controls, or other safety devices that rely on continuous power. Therefore, the analysis to perform energized electrical work (EEW) should include participation from the owner/client, qualified electrician(s), DPR project management, and DPR EHS Manager, at a minimum. Furthermore, an **Energized Electrical Work Permit** should be completed fully prior to work commencement (*refer to permit in the Forms Section at the end of the EHSP*).

Note: *The term infeasible and the term inconvenient are not interchangeable.*

An example of infeasible would be the need to power down life support systems in an operating hospital. An example of inconvenient would be the need to wait until after operating hours, a product run, or the next scheduled system shut down.

Reference: NFPA 70E, Electrical Safety Requirements for Employee Workplace; National Electrical Code (NEC); CFR Subpart S-Electrical Standard--General Industry 1910.331 to 1910.339; CFR Subpart K-Electrical Standard—Construction Industry and ASME testing procedures. Other applicable Standards, guidelines, testing guidelines and regulations may apply.

Definitions

Affected employee: An employee whose job requires them to operate or use machines or equipment that may be periodically locked-out/tagged-out during service, maintenance activities, or installation, or a person whose job requires them to work in the area of equipment or machines that may be periodically locked-out/tagged-out during service, maintenance, or installation activities

Arc flash hazard: A dangerous condition associated with the possible release of potentially enormous energy caused by an electric arc

Arc flash hazard analysis: A study investigating a worker's potential exposure to arc flash energy, conducted for the purpose of injury prevention and the determination of work practices, arc flash boundary, and the appropriate personal protective equipment (PPE)

Authorized employee: A person who implements or supervises the implementation of the lockout/tagout procedure for the purpose of performing maintenance, installation or service on machines or equipment with a potential hazardous energy source



Boundary, Arc flash: When an arc flash hazard exists, an approach limit at a distance from a prospective arc source within which a person could receive a second degree burn if an electrical arc flash were to occur

Boundary, Limited approach: An approach limit at a distance from an exposed energized electrical conductor or circuit part within which a shock hazard exists

Boundary, Restricted approach: An approach limit at a distance from an exposed energized electrical conductor or circuit part within which there is an increased risk of shock, due to electrical arc over combined with inadvertent movement, for personnel working in close proximity to the energized electrical conductor or circuit part

Disruption & Access Request (DAR): A form intended to notify an operating facility regarding the potential for disruption when working on or accessing a utility or sensitive area. Intent is to communicate and receive owner approval.

Electrically safe working condition: A state in which an electric conductor or circuit part has been disconnected from energized parts, locked/tagged in accordance with established standards, tested to ensure the absence of voltage, and grounded if deemed necessary

Energized Electrical Work (EEW): The performance of any and all operations on or near an exposed energized circuit, system or equipment regardless of the voltage or volt-amps

Energy source: Any source of electrical, mechanical, hydraulic, pneumatic, chemical, thermal or other energy

Ground: The earth (NFPA 70E)

Ground: A conducting connection, whether intentional or accidental, between an electrical circuit or equipment and the earth, or to some conducting body that serves in place of the earth. (CFR 1926.400 Subpart K)

Hazardous energy: The unexpected energization or start-up of the machines or equipment, or release of stored energy, which could harm employees

Incident energy analysis: A component of an arc flash hazard analysis used to predict the incident energy of an arc flash for a specified set of conditions.

Infeasible: It has been demonstrated that de-energization of those circuits will create additional or increased hazards or is infeasible due to equipment design or operational limitations. Operational limitations could include critical care in hospitals, research laboratories, or where it could cause severe economic impact to the client. Other considerations are made for emergency situations.

Lockout: The placement of a lockout device on an energy isolating device, in accordance with an established procedure, ensuring that the energy isolating device and the equipment being controlled cannot be operated until the lockout device is removed

Other employees: All employees not otherwise classified (i.e., such as affected employees, authorized employees or qualified persons)

Qualified person: One who has demonstrated skills and knowledge related to the construction and operation of electrical equipment and installations and has received safety training to identify and avoid the hazards involved

Shock hazard: A dangerous condition associated with the possible release of energy caused by contact or approach to energized electrical conductors or circuit parts

2.0 Lockout/Tagout

2.1 Responsibilities

All subcontractors are required to ensure that their respective hazardous energy control programs are implemented. All subcontractors are also responsible for designating a competent person who will be responsible for overseeing the program on the site and ensuring all lockout/tagout procedures are followed. Failure to comply with the requirements detailed within may result in disciplinary action, including removal from site or breach of contract.

2.2 Pre-Task Planning

Work activities involving hazardous energy are deemed to be High Hazard and therefore require a detailed Job Hazard Analysis and/or subcontractor work plan (i.e., Method Of Procedure(MOP), Safe Plan of Action(SPA), Standard Operating Procedure(SOP), additional owner/ facility requirements etc.), as well as detailed daily pre-task planning. These work activities include, but are not limited to:

- Tool hook up
- Introduction of process chemicals into systems
- Utility tie-ins
- Lockout/tagout
- Rough-in work tied in at existing systems
- Work on energized equipment
- Startup/Shutdown and systems testing activities

Refer to Pre-Planning Section 17 for additional pre-task planning guidelines and the Pre-task Planning worksheet in the Forms Section at the end of the EHSP.

2.2.1 Owner Coordination

On any existing facilities, DPR project management should meet with owner representatives to establish clear concise written procedures for the control of hazardous energy on existing equipment.

- Define the site specific protocols
- Understand owner's expectation for notification of equipment to be affected
- Establish responsible parties for the shutting down of existing owner equipment
- Disruption and access request process
- Review One-line diagrams with owner for all affected equipment/ systems
- As-built drawings available

2.3 Lockout/Tagout Procedures

The following lockout/tagout procedures apply to all systems, including but not limited to, electrical, mechanical and plumbing. This list details the general requirements and each project is responsible for implementation of site specific protocols. A Lockout/Tagout Checklist is provided in the Forms Section at the end of the EHSP to assist in the planning of these operations.

2.3.1 Lockout/Tagout Pre-Planning

- Pre-planning should occur to identify all potential sources of hazardous energy for all equipment, systems, and services prior to initiating any work
- Systems to be removed during demolition should be clearly marked and differentiated from those systems which are to remain intact
- If a system cannot be made safe-off without impeding facility operations, the system should be scheduled for a shutdown through DPR and/or the Owner

- All pumps, valves, control devices and/or electrical systems, etc. upon which work is to be performed should be properly locked and tagged out; specific methods of isolating electrical components and other related procedures are detailed below

2.3.2 *Equipment, Electrical or System De-Energizing*

- At no time should DPR or any subcontractor take it upon themselves to shutdown operating facility equipment or systems, except in life-threatening situations. All existing operating processes, mechanical or electrical systems should only be shut down by the Owner, or under the Owner's supervision.
- Equipment should be shut down through normal stop procedures; service disconnects, branch circuits and switches to the equipment, and power feeds upon which work is to be performed should be opened i.e., switched off
- All areas to be demolished should have lines traced to the power source, marked and de-energized
- Means should be taken to coordinate lockout/tagout with subcontractors on applicable equipment
- A master list of all equipment-systems locked out should be maintained

2.3.3 *Equipment Isolation*

- Lockout of the hazardous source should occur as close as possible to the point of work and should be in sight of affected personnel while they are performing the work
- The devices used to de-energize the equipment, system, or service, should be physically secured in the "safe" position
- The devices used to de-energize the equipment should be able to withstand the conditions of the space, such as heat, corrosion, chemicals etc.
- Mechanical isolating devices should also be used on mechanical systems
- Valves should be placed in the "safe" position and slip blinds ("pancakes") may be required on systems without mechanical isolation

2.3.4 *Lockout/Tagout Device Application*

- Only appropriately trained personnel may apply or remove lockout/tagout devices
- Each worker who is performing work on any equipment or systems should have their own lock and tag affixed to isolating devices for all potential energy sources, or use similarly protective devices
- Multiple lockout devices may be required when more than one employee is working on a piece of equipment or system; for multiple locks, a manufactured device should be used (e.g. alligator clip, multiple hasp), or similarly protective devices
- Sufficient numbers of lock-out devices and padlocks should be available in the event of a repair, emergency or a reasonably foreseeable need
- DPR may affix their lock and tag to any piece of equipment or system that has been de-energized to further ensure a safe work process
- Locks or other protective devices should be singularly identified and meet the following requirements:
 - no combination locks are to be used
 - no break-a-way locks are to be used
 - each locking device will be individually keyed
 - the key must remain in the possession of the person locking the equipment/system out
 - where locks/keys are not used, each qualified individual must have the ability to affix/remove their protective device without the risk of tampering and/or removal by another individual
 - should not be used for any other purpose
- Tags should be singularly identified and meet the following requirements:
 - information contained on the tag should include:
 - *contractor name*

- *date*
- *name of employee performing the work*
- *contact phone number*
- *a description of the work being performed*
- *a statement prohibiting unauthorized operation of the disconnecting means and removal of the tag*
 - tags should not to be removed except by the competent, trained employee responsible for it
 - tags should be legible and understandable by all employees in the work area
 - tags and their means of attachment should be able to withstand the workplace environmental conditions and should be securely attached
 - should not be used for any other purpose

2.3.5 *Stored Energy Control*

- If energized power feeds cannot be locked-out and tagged-out, they should be clearly marked as “live” and not removed
- All site employees must be informed of the potential hazard and corresponding work plan
- Special work procedures must be developed for all work performed around these energized systems or equipment
- Process equipment, vessels and piping should be isolated and drained prior to penetration
- Equipment and/or systems, which have contained corrosive, toxic or flammable substances, must be flushed or purged prior to starting work

2.3.6 *Isolation Verification*

- Lines that must be left energized should be tagged and identified and clearly marked out on the project throughout their length
- After locking the equipment, system, or building service the employee must check it to verify a “zero energy state”; and should be checked by:
 - trying to restart equipment
 - testing power feeds with a volt meter or similar means at all junction boxes
 - slowly opening valves on fluid or gas systems, etc.

2.3.7 *Lockout/Tagout Work Practices*

- Personnel should be appropriately trained and familiar with proper use of special precautionary techniques, PPE, insulating and shielding materials, insulated tools
- All personnel should use work practices that protect employees against contact with energized parts directly with any part of their body or indirectly through some other conductive object
- Conductive articles of clothing (e.g., synthetic fiber jewelry, metal frame glasses) should not be worn unless they are rendered non-conductive by insulating them
- Employees may not perform housekeeping duties at close distances to energize parts unless adequate safeguards are provided; electrically conductive cleaning materials cannot be used unless procedures are used to prevent all electrical contact
- No equipment or system should be operated, energized or otherwise altered when any lock or warning tag is attached to the isolating device

2.3.8 *Release from Lockout/Tagout and Lock Removal*

- No system should be re-energized until all locks, tags or other protective devices are removed and the system has been inspected to ensure safe operation
- Only the employee placing their lock/protective device on the equipment should be the one to remove the lock/device

- When used, DPR's lock and tag should be the last to be removed
- It is recommended that an employee from each affected subcontractor be present at the time the locks/devices are removed and the power source is re-energized
- When facilities are active/occupied, all equipment, systems or building services should not be re-energized until:
 - all affected personnel are notified
 - all employees are cleared away from potential hazardous area
 - the system has been verified safe by a competent person
- Only after all locks/tags/devices are removed and above requirements are met, the owner will place the isolating device in the ON or operating position
- Equipment/system is now energized and operating

2.3.9 *Prohibition against Restart*

- No one is to attempt to energize, restart or otherwise operate any machine or equipment that has a lock/tag/device attached to the energy control device
- No one is to remove a lock/tag/device other than authorized employee that applied it, except under the express direction and supervision of DPR project management, and only after following the procedures outlined in 2.2.10 below

2.3.10 *Removal of Locks/Tags/Devices in Absence of the Competent Employee*

- When the competent employee who applied the lockout/tagout device is not available to remove it, the lock/tag/device may be removed only under the direction of the DPR project management
- Removal should follow the following procedure:
 - verify that the individual is not on the premises
 - make all reasonable efforts to contact the individual and inform him/her that their lock/tag/device needs to be removed
 - assure that the individual knows that the lock/tag/device have been removed before they resume work at the facility
- After completing the above, follow normal removal procedures

2.3.11 *Multiple Shifts or Unusual Conditions*

- In the case of shift or personnel changes or other type of work that makes it extremely impractical to adhere to all the provisions of this lockout/tagout procedure, a written plan will be developed and submitted to DPR for approval
- For shift changes, a change-over period will be established so that the competent employees may exchange their locks/tags; shift change documentation forms will be completed and signed by each competent employee relieving the previous crew

2.4 *Lockout/Tagout Inspections*

A competent employee from each subcontractor involved in work on energized systems should perform a periodic detailed safety inspection of the Hazardous Energy Control Program and ensure that all safety requirements are being met.

2.5 *Lockout/Tagout Training*

Each subcontractor is responsible for training personnel in the proper use of locks, tags, and other means of protection against hazardous energy. All employees who are authorized to lockout and tagout pieces of equipment and/or systems should receive training upon initial assignment. All training is to be documented and made available upon request. Periodically employees will be provided with refresher training in lockout and tagout procedures in order to maintain their proficiency and keep current on new technologies.

Employees involved in work in and around hazardous energy, fall into one of four categories and the type of training they need. The following definitions and table will detail what training requirements are necessary:

Lockout/Tagout Training Requirements	Authorized Employees	Affected Employees	Other Employees
Description of all set procedures involving hazardous energy.	X	X	X
Prohibitions against restarting/reenergizing equipment or system.	X	X	X
Purpose and application of all procedures.	X	X	
Recognition of hazardous energy sources.	X		
Type and magnitude of hazardous energies.	X		
Methods and means for hazard isolation and control.	X		

3.0 Energized Electrical Work – (EEW)

It is DPR's position, and as defined in the governing standard NFPA 70e, that all electrical work is to be done in a de-energized state. Work on energized circuits presents two basic life threatening hazards, electrocution and burns by energy released during arc-flash.

In the event that de-energizing a circuit is determined infeasible, as mentioned in sub-section 1.0 of this section, an Energized Electrical Work (EEW) Permit should be completed including a written, detailed step-by-step Pre-task Plan (PTP) on how the work is to be completed. This EEW permit and pre-task plan should be submitted to DPR for review no less than 48 hours prior to the anticipated work taking place.

In addition to the EEW and detailed PTP, a detailed Disruption & Access Request (DAR) should be created, submitted, and then approved by the appropriate owner/facility representatives. The intent of the DAR is to establish and convey the potential consequences should a short circuit or arc-flash occur. Approval and implementation of an Energized Electrical Work Permit does not completely alleviate the potential for the equipment to be adversely affected and potentially shut down anyway.

Note: *Only qualified persons are allowed to be within the restricted boundary or the arc-flash boundary, whichever is the furthest distance from the hazard source.*

3.1 General EEW Guidelines

In the event that work cannot be performed in a de-energized state, the following general guidelines, at a minimum, apply:

- All power sources should be de-energized as feasible with lockout/tagout procedures followed
- A copy of the current NFPA 70E standard should be readily available during EEW activities.
- All covers, doors or similar protective means should be in place prior to performing EEW
- All circuits and systems should be tested prior to work for voltage using a meter that is properly rated, tested and calibrated
- Non-metallic pulling socks should be used when pulling wire into energized panels or where the potential exists for contacting energized systems; metal fish tape should never be used
- All warning tags should be applied at the nearest level upstream from the energized power source
- No EEW should be performed without approved insulated tools
- The hand tools should be manufactured to meet the requirements of the work to be done
- Areas around exposed/energized equipment should be properly barricaded and/or secured to prevent accidental contact and maintain a safe work environment
- Workers should not employ practices that provide a current path through any part of their body
- Every effort should be made to practice the "one-hand rule" when the task allows. *Note: the "one handed rule" requires personnel to work with one hand in their pocket or equivalent to prevent contact with grounded work surfaces, thus preventing electric current from flowing through the heart.*
- Adequate lighting should be provided in all areas of EEW
- Special tools should be used to install/remove fuses under load

- Panel doors should be closed prior to energizing circuits in which fuses have been replaced
- Tripped circuit breakers should not be reset or fuses replaced until the system which they service has been verified safe
- Employee clothing, jewelry, equipment and work materials should be removed, rendered non-conductive or used in a manner to prevent contact with energized electrical conductors
- Fabric materials containing nylon, rayon, polyester, acetate or their blends should not be worn
- All electrical and protective equipment should be inspected for damage prior to use; damaged items should be tagged and removed from service
- Temporary cords should be of extra heavy use type and should be protected from damage
- Work should not be done in damp or wet areas until these areas have been safely abated

3.2 **Limits of Approach**

Below are the distance boundaries of immediate concern when energized electrical work is performed:

- The electrical approach boundaries are: *Restricted and Limited*.
- The incident energy boundary is called: *Arc-Flash Boundary*.

Only qualified persons, wearing the appropriate PPE, are allowed within these boundaries. (Reference the graphic on page 13 of this Section)

A shock hazard analysis should be performed to determine the voltage to which an employee may become exposed. Electrical approach boundaries can be determined by referencing the tables within NFPA 70e. See reference Table 130.4 (C) (a) below.

NFPA 70e 2015 Table 130.4(D)(a) Approach boundaries to energized electrical conductors or circuit parts for shock protection for alternating-current systems *(All dimensions are distance from energized electrical conductor or circuit part to employee)*

Nominal System Voltage Range, Phase to Phase	Limited Approach Boundary		Restricted Approach Boundary: Includes Inadvertent Movement Adder
	Exposed Movable Conductor	Exposed Fixed Circuit Part	
<50V	Not specified	Not specified	Not specified
50 V – 150 V	3.0 m (10 ft 0 in)	1.0 m (3 ft 6 in)	Avoid contact
151 V – 750 V	3.0 m (10 ft 0 in)	1.0 m (3 ft 6 in)	0.3 m (1 ft 0 in)
751 V – 15 kV	3.0 m (10 ft 0 in)	1.5 m (5 ft 0 in)	0.7 m (2 ft 2 in)
15.1 kV – 36 kV	3.0 m (10 ft 0 in)	1.8 m (6 ft 0 in)	0.8 m (2 ft 7 in)
36.1 kV – 46 kV	3.0 m (10 ft 0 in)	2.5 m (8 ft 0 in)	0.8 m (2 ft 9 in)
46.1 kV – 72.5 kV	3.0 m (10 ft 0 in)	2.5 m (8 ft 0 in)	1.0 m (3 ft 3 in)
72.6 kV – 121 kV	3.3 m (10 ft 8 in)	2.5 m (8 ft 0 in)	1.0 m (3 ft 4 in)
138 kV – 145 kV	3.4 m (11 ft 0 in)	3.0 m (10 ft 0 in)	1.2 m (3 ft 10 in)
161 kV – 169 kV	3.6 m (11 ft 8 in)	3.6 m (11 ft 8 in)	1.3 m (4 ft 3 in)
230 kV – 242 kV	4.0 m (13 ft 0 in)	4.0 m (13 ft 0 in)	1.7 m (5 ft 8 in)
345 kV – 362 kV	4.7 m (15 ft 4 in)	4.7 m (15 ft 4 in)	2.8 m (9 ft 2 in)
500 kV – 550 kV	5.8 m (19 ft 0 in)	5.8 m (19 ft 0 in)	3.6 m (11 ft 10 in)
765 kV – 800 kV	7.2 m (23 ft 9 in)	7.2 m (23 ft 9 in)	4.9 m (15 ft 11 in)

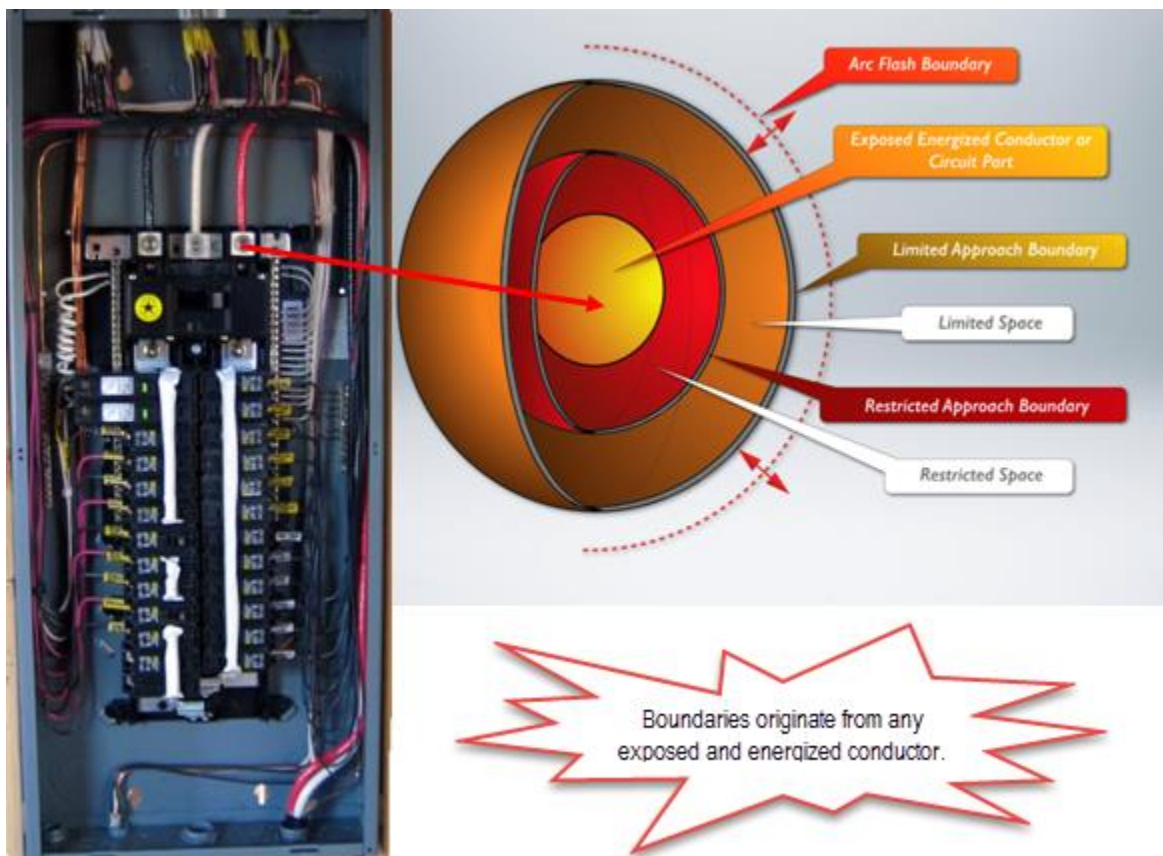
The above table is for alternating current only, the direct current reference is Table 130.4 (D)(b).

An arc flash hazard analysis should be performed to determine the arc flash boundary, the incident energy at the working distance, and the appropriate level of PPE to be used. There are two basic means to determine the arc flash boundary and the level of PPE necessary.

Incident energy calculation – There are several equations available for calculating the incident energy. These calculations should be carried out by an electrical engineer familiar with the system.

Use of NFPA 70e Tables 130.7(C)(15)(A)(b) and 130.7(C)(15)(B), when the requirements of these tables apply. Not all circuits or configurations are covered within these tables.

- Equipment labeling based on a previous incident energy calculation. The label should include:
 - Nominal system voltage.
 - Arc Flash boundary
 - At least one of the following:
 - Available incident energy and the corresponding working distance, or the arc flash PPE category in the tables listed above.
 - Minimum arc rating of clothing.
 - Site specific level of PPE.
- The label should contain “up to date” information with backup documentation.



3.3 **EEW Responsibilities**

Only qualified persons should work on energized systems. Each person will have the following general responsibilities:

- EEW Qualified Person Responsibilities:
 - ensure complete understanding of the scope of work and communicate this with each qualified person
 - fully participate in the generation of the detailed, written pre-task plan
 - comply with all electrical safe work procedures and requirements as described in this document and as part of the site specific/activity specific safe work practices
 - ensure that the buddy system is used
 - ensure that the EEW permit is followed

- have the necessary experience and training to work on the respective equipment or system and fully understand the hazards associated
- complete the applicable training to include basic, intermediate and advanced electrical safety as well as a current CPR/First Aid and AED certification (annually)
- EEW Buddy Responsibilities:
 - ensure complete understanding of the scope of work and communicate this with each qualified person
 - fully participate in the generation of the detailed, written pre-task plan
 - know the locations of disconnects and how to de-energize the system, equipment or circuit
 - should observe work without interfering with qualified person and without participating in the work
 - should determine the best emergency procedures prior to work beginning
 - review and sign the EEW permit
 - complete the applicable training to include the appropriate level of electrical safety training and a current CPR/First Aid and AED certification

3.4 **Equipment Labeling**

Electrical equipment such as switchboards, panel boards, industrial control panels, meter socket enclosures and motor control centers that are likely to require examination, adjustment, servicing or maintenance while energized, should be field marked with a label containing all of the following information:

- Nominal system voltage.
- Arc Flash boundary
- At least one of the following:
 - Available incident energy and the corresponding working distance, or the arc flash PPE category in the tables listed above.
 - Minimum arc rating of clothing.
 - Site specific level of PPE.
- The label should contain “up to date” information with backup documentation.

	WARNING
Arc Flash and Shock Hazard Appropriate PPE Required	
25.1 9 ft 7 in	Cal/cm² at 18 inches Flash Hazard Boundary
208 VAC 42 in 0 in	Nominal System Voltage Limited Approach Boundary Restricted Approach Boundary
Bus: MSB Prot.: Max Trip Time @ 2 sec.	
Equipment Name:	

All equipment to be installed should include signage. In the event electrical equipment is found to be unlabeled, no work should occur until the owner and/or facilities manager or equivalent ensures the equipment's rating is determined and the proper label is affixed.

3.5 EEW Personal Protection Equipment

PPE specific to the work activity should be clearly outlined via Pre-Task Planning and the Energized Electrical Work Permit and utilized for the duration of work. The proper level of personal protective equipment should be provided for the task. Some general requirements that relate to PPE include:

- Prior to use, each employee should review the use and limitations of their PPE
- PPE should be maintained in safe reliable condition
- Review insulation value of equipment and PPE
- Review specific types of equipment available and its use
- PPE used for EEW work should meet ASTM requirements

Types of PPE that may be used for EEW work include, but are not limited to:

- Hardhat
- Safety glasses with non-conductive frames
- Rubber insulated mats or boots
- Properly rated and tested insulated rubber gloves
- Properly rated and tested insulated rubber sleeves
- Body hook
- Properly rated and insulated tools
- Insulated blankets
- Non-conductive work platforms
- Face shield
- Flame retardant clothing (nomex or equivalent outer clothing)
- Properly rated and tested blast suit
- Cotton clothing
- Etc.

• Table 130.7(c)(7)(c) Rubber insulating Equipment Maximum Test Intervals		
Rubber Insulating Equipment	When to Test	Governing Standard for Test Voltage
Blanket	Before first issue; every 12 months thereafter†	ASTM F 479
Covers	If the insulating value is suspect	ASTM F 478
Gloves	Before first issue; every 6 months thereafter†	ASTM F 496
Line hose	If the insulating value is suspect	ASTM 478
Sleeves	Before first issue; every 12 months thereafter†	ASTM 496

Note: *If the insulating equipment has been electrically tested but not issued for service, it is not permitted to be placed into service unless it has been electrically tested within the previous 12 months*

The categories for Arc-Rated clothing have changed in the 2015 edition of NFPA 70e. Table 130.7(C)(16) only deals with arc flash protection only. The hazard/risk categories (HRC) are no longer represented. The "0" category has been eliminated.

Hazard/Risk Category
4 cal/cm²

1

Arc-rated long-sleeve shirt
Arc-rated pants or overall
Arc-rated face shield with hard hat
Safety glasses
Hearing protection
Leather & voltage rated gloves (as needed)
Leather work shoes



Hazard/Risk Category
8 cal/cm²

2

Arc-rated long-sleeve shirt
Arc-rated pants or overall
Arc-rated face shield & balaclava or
Arc flash suit with hard hat
Safety glasses, Hearing protection
Leather & voltage rated gloves (as needed)
Leather work shoes



Hazard/Risk Category
25 cal/cm²

3

Arc-rated long-sleeve jacket
Arc-rated pants
Arc-rated flash hood with hard hat
Safety glasses, Hearing protection
Leather & voltage rated gloves (as needed)
Leather work shoes



Hazard/Risk Category
40 cal/cm²

4

Arc-rated long-sleeve jacket
Arc-rated pants
Arc-rated flash hood with hard hat
Safety glasses, Hearing protection
Leather & voltage rated gloves (as needed)
Leather work shoes



4.0 Overhead Power Lines

Prior to beginning any work on a project, all locations where overhead power lines exist should be identified. Whenever possible, lines must be de-energized and grounded or other protective measures must be provided before work is started. In the event that it is infeasible to de-energize or ground the power line, the qualified person should determine if the overhead electric lines are insulated for the lines operating voltage. Refer to the most current NFPA 70E guidelines.

Table A – Minimum Clearance Distances

Voltage (nominal, kV, alternating current)	Minimum clearance distance (feet)
up to 50	10
over 50 to 200	15
over 200 to 350	20
over 350 to 500	25
over 500 to 750	35
over 750 to 1000	45
over 1000	(as established by the power line owner/operator or registered professional engineer who is a qualified person with respect to electrical power transmission and distribution)

Work activities can only be closer than indicated in the above chart if:

- The energized part is insulated both from all other conductive objects at a different potential and from the person
- The person is insulated from all conductive objects at a potential different from that of the energized part
- The person is insulated from all conductive objects at a different potential from that of the energized part

Vehicular or mechanical equipment that could have any part near overhead lines must keep a clearance of a minimum of 10 feet. (For voltages higher than 50 kV, a distance of 4 inches is to be added for every 10 kV over the voltage.) Exceptions to this rule are as follows:

- Vehicles in transit with structures lowered may keep a distance of 4 feet (for voltages higher than 50 kV, clearance to be raised 4 inches for every 10 additional kV)
- If using insulating barriers rated for the voltage involved, clearance can be reduced to that within the working dimensions of the barrier
- Aerial lifts insulated for voltage involved can use approach distances given earlier
- Employees on ground may not contact vehicle or mechanical equipment without using work practices for energized equipment
- Barriers to be used to protect employees from hazardous ground potentials
- Crane lifts within 20 feet of energized power lines are defined as critical lift. Refer to the Crane Section 05 and Material Handling Section 09 for specific requirements.

5.0 Systems Testing

While the hydrostatic test method is the preferred method of pressure testing, the selected testing method should be appropriate for the ultimate service of the system. All testing should be performed not only per component manufacturer's recommendations, but also per the design engineer's criteria and in full compliance with the Project Specifications.

Hydrostatic testing presents a much lower safety hazard than pneumatic testing. Other forms of testing are not to be used unless they specifically apply to the ultimate service of the piping system and all other methods have been considered and determined to be infeasible. Approval for alternative methods should be granted by DPR project management prior to start of testing. The designated DPR MEP coordinator as well as Regional EHS should also sign off on the testing procedure. Precautions should be taken to ensure that adequate protection is provided to prevent injury to personnel and damage to property from missile fragments, shock waves, or other consequences of a rupture or leakage during a pneumatic test.

The guidelines below are intended to provide a basis for project specific procedures that must be written by each subcontractor and submitted to DPR for review prior to the start of any pneumatic testing of piping systems and vessels. Notify DPR if these guidelines conflict with any project specifications or codes. The project specific test procedure should include, but not be limited to:

- Training required for the testing crew, and verification that all workers participating in the process have been trained
- DPR, and where applicable the owner, are notified prior to starting test
- Written confirmation of system walk down to identify all valves and proper valve position during test and drawing showing exact routing and size of pipe to be tested. All system walk downs must include DPR, and where applicable, the owner
- Test duration and maximum pressure of test must be included and defined
- Piping materials and joining methods, per manufacturer's and engineer's requirements
- Written confirmation that all supports and restraints are installed and have been visually verified to be in place and properly installed
- Hazard warning signs, i.e., "testing in process" or similar, placed in test area

- Bleed valves, relief valves, etc. are installed per engineering documentation and verified and properly working per manufacturer's and engineer's requirements
- Safe distance for testing crew and any other personnel in area of pipe under test must be clearly defined (Minimum safe distance should be calculated based on ASME Article 5.1 Mandatory Appendix II, PCC-2-2008)
- Testing equipment safety precautions must be established and clearly defined
- At conclusion of the test, written confirmation by the testing crew supervision, of completed piping depressurization must be completed and provided to DPR project management and EHS

Hazardous Energy Forms

Forms

Energized Electrical Work Permit
Lockout/Tagout Checklist – English
Lockout/Tagout Checklist – Spanish
Hazardous Energy/Pre-Start-up Meeting Checklist



Energized Electrical Work Permit

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Project Information			
Project Name:		DPR Project #:	
EEW Permit Reference #:		DAR Reference #:	
Electrical Contractor Name:		Expected Date of Work:	
Contractor Contact #:		EEW Proposed Start Time:	
Location of Activity:		EEW Expected End Time:	
Equipment #:		Circuit #:	
Compelling reason for performing work in an energized state:			
General description of activity:			

Shock Risk Assessment Information			
Name of person performing the shock risk assessment:	Date:		
Is there more than one voltage source?	☐	Yes	☐ No
What voltage will the qualified worker(s) be exposed to?	V1		V2
Describe the energized parts workers will be exposed to:			
Limited approach boundary distance:			
Restricted approach boundary distance:			

Arc Flash Risk Assessment Information			
Name of person performing the arc flash risk assessment:	Date:		
Is there more than one arc flash source?	☐	Yes	☐ No
Is there an existing label on the equipment?	☐	Yes	☐ No
If so, is the backup documentation available and current?	☐	Yes	☐ No
Available incident energy at the working distance:			
Or – Arc flash PPE category from NFPA tables (as appropriate):			
Arc flash boundary distance:			
List of necessary PPE required within the arc flash boundary:			

Specific Information			
Has a detailed, step by step pre-task plan and work procedure been created? If yes, attach to EEW Permit. If no, complete detailed pre-task plan before proceeding.	☐	Yes	☐ No
Has the owner been notified and all potential impacts to operations reviewed?	☐	Yes	☐ No
Can the equipment be shut down? If no, why?	☐	Yes	☐ No
Does the equipment have a disconnect?	☐	Yes	☐ No
If yes, is it close enough to the work that it can be turned off in case of emergency?	☐	Yes	☐ No
If no, what are the emergency procedures to remove a worker the energized circuit?	☐	Yes	☐ No
Compelling Reason:			
☐	Supports Life Sustaining Equipment		
☐	Severe Economic Impact (shutting down major operational systems, disrupting R&D prototypes etc.)		
☐	Emergency Event		
Detailed explanation of compelling reason:			



Energized Electrical Work Permit

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Required Signatures					
Description	Name (print)	Signature	Contact #	Trained?	Date
DPR Supervisor				Yes / No	
Subcontractor Supervisor				Yes / No	
DPR EHS Manager				Yes / No	
Subcontractor EHS Manager				Yes / No	
Qualified EEW Buddy				Yes / No*	
Qualified Person 1				Yes / No*	
Qualified Person 2				Yes / No*	
Qualified Person 3				Yes / No*	

* If no, this person is not eligible for EEW work

Specific Methods for Abating Hazards (check that devices are in place, as required)					
De-energize equipment / system	☐	Cords are extra hard use type	☐	Rubber mats	☐
Lockout /Tagout	☐	Testing equipment checked and calibrated	☐	Hard hat (Class E)	☐
Covers in place	☐	Proper tools and equipment inspected prior to use	☐	Non-conductive safety glasses	☐
Barricade / Secure area	☐	Approved / rated insulated tools	☐	Face shield	☐
Warning tags	☐	Cotton clothing	☐	Rubber boots	☐
Adequate lighting provided	☐	Arc Rated clothing (Nomex or equivalent)	☐	Rated and tested rubber gloves Last Test Date:	☐
Blankets Last Test Date:	☐	Removal of conductive clothing, jewelry etc.	☐	Rated and tested rubber sleeves Last Test Date:	☐
Ventilation	☐	HEARING PROTECTION REQUIRED	☐	Body hook	☐
Warning signage	☐	Rated blast suit	☐	Fire extinguisher	☐
Non-metallic pulling socks	☐	Area is dry		AED REQUIRED	☐

Additional Permits Required					
Hot Work	☐	Confined Space	☐	Other	☐

Does at least one crew member have current CPR and First Aid Training?	Yes	☐	No	☐
Location of AED:				

Emergency Contacts								
Name	CPR /First Aid Training**				AED Training**			
	Yes	☐	No	☐	Yes	☐	No	☐
	Yes	☐	No	☐	Yes	☐	No	☐
	Yes	☐	No	☐	Yes	☐	No	☐
	Yes	☐	No	☐	Yes	☐	No	☐
	Yes	☐	No	☐	Yes	☐	No	☐
	Yes	☐	No	☐	Yes	☐	No	☐

**Note: all personnel responsible for emergency response must be trained in CPR/ First Aid & AED training



Energized Electrical Work Permit

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Emergency Responder Numbers	
Emergency:	
Fire Department:	
Hospital:	
EHS:	

Other Considerations:

Description of Activity:
--

Owner Signature _____

By signing above, you verify that the above work is infeasible and have been adequately informed as to the risks and benefits of work on an energized system. Further, you verify that the work is authorized as per the procedures and precautions outlined below.



Lockout/Tagout Checklist

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As per DPR's Hazardous Energy Control Program any work on any powered equipment, machinery or system should not occur without appropriate Lockout/Tagout procedures followed. Lockout/Tagout procedures are required for any equipment, machinery or system that exposes the worker to the following energy sources/hazards:

Electrical	Chemical	Thermal	Pneumatic	Hydraulic	Mechanical	Potential (Gravity)	Kinetic
------------	----------	---------	-----------	-----------	------------	---------------------	---------

General Information			
Project/Project#		Date	
Contractor Performing Work			
Competent Person		Start Date/Time	
DPR Supervisor		End Date/Time	
Owner Work Area Supervisor		Building Location	
Nature of Work			

Lockout/Tagout Checklist			
Complete the following checklist for all steps taken / needed to protect the work area.			
Pre-Planning Stage		Procedures	
☐	All locations requiring lockout/tagout are identified?	☐	Machine, system or equipment is shut down, turned off or unplugged as per normal shut down procedures?
☐	Identify all primary and secondary sources via audit	☐	Circuits, breakers or disconnects opened / shut off?
☐	Consult drawings, prints, or cut sheets	☐	Valves are closed, blocked and/or blinded?
☐	All potential hazards identified?	☐	Lines are bled, purged or de-energized?
☐	All individuals performing work have appropriate training?	☐	Equipment or machinery has come to a stop and prevented from further motion?
☐	All affected individuals have been notified?	☐	Hazards due to gravity blocked?
☐	All affected individuals have appropriate training?	☐	Systems are allowed to cool / warm?
☐	Is any additional training required?	☐	Systems are decontaminated prior to work?
☐	Special permits required?	☐	Isolation devices applied at all energy sources?
☐	Confined Space	☐	Isolation devices sturdy and able to withstand all conditions?
☐	Hot Work	☐	Isolation devices inspected for defects and replaced as needed?
☐	Energized Electrical Work	☐	Isolation and lockout/tagout devices are clearly identifiable?
Locks		☐	Appropriate signage placed as needed?
☐	Standard individually keyed locks are utilized?	System Check	
☐	Lock is durable enough to meet the conditions where used?	☐	Equipment, machinery or system is checked for compliance?
☐	Lock identifies by name employee who owns it?	☐	Equipment, machinery or system tested to ensure isolation devices and lockout/tagout are working?
☐	Each employee working on system or equipment has their own lock applied?	Tags	
☐	Keys for locks are secured and maintained by the person performing the work?	☐	Tags name person who is performing work?
Release from Lock Out / Tag Out		☐	Tags list Contact Number?
☐	Prior to restoring energy an inspection of work area performed to ensure all guards in place, equipment or system is safe and area is safe?	☐	Tags indicate Date(s) of work?
☐	All affected personnel notified and accounted for and safely removed from area?	☐	Tags name purpose for doing work?
☐	Each isolation device, lock and tag removed by the authorized employee?	☐	Tags affixed securely to locking device?
		☐	Tags are legible?

Contractor Supervisor Signature

By signing above, you verify that the above location has been examined and that the above precautions have been taken.

DPR Supervisor Signature

By signing above, you verify that the above work has been completed, all locks have been removed and work area has been restored to a safe condition.

Work Area Supervisor Signature

By signing above, you verify that the above work has been completed, all locks have been removed and work area has been restored to a safe condition.

Lockout/Tagout Sign Off

Contractor Supervisor Signature

By signing above, you verify that the above work has been completed, all locks have been removed and work area has been restored to a safe condition.



Lockout/Tagout Checklist

Archive Document – DO NOT DISCARD

De acuerdo con el Programa de Control de Energía Peligrosa de DPR, no se puede realizar ningún trabajo con equipos, maquinarias o sistemas motrices sin acatar los procedimientos de interrupción de energía con candado y etiqueta de bloqueo (cierre y rotulación). Los procedimientos de interrupción de energía con candado y etiqueta de bloqueo se requerirán para cualquier equipo, maquinaria o sistema que exponga al trabajador a los/las siguientes peligros/fuentes de energía:

Eléctrico Químico Termal Neumático Hidráulico Mecánico Potencial (Gravedad) Cinético

Información general			
Contratista que realiza el trabajo		Fecha actual	
Persona competente		Fecha/Hora de comienzo	
Supervisor de DPR		Fecha/Hora de finalización	
Supervisor del área de trabajo del dueño		Ubicación de Edificio	
Naturaleza del trabajo			

Lista de comprobación de cierre y rotulación			
Complete la siguiente lista de comprobación para todos los pasos tomados/ necesarios para proteger el área de trabajo.			
Etapa previa a la planificación		Procedimiento	
☐	¿Se identificaron todas las áreas que requieren cierre y rotulación?	☐	¿Se cerró, desenchufó o se apagó la máquina, sistema o equipo según los procedimientos normales de apagado?
☐	Identifique las fuentes primarias y secundarias mediante auditoria	☐	¿Se abrieron / apagaron los circuitos, interruptores o desconectores?
☐	Consulte dibujos, impresiones u hojas aserradas	☐	¿Se cerraron, bloquearon y/o blindaron las válvulas?
☐	¿Se identificaron todos los peligros posibles?	☐	¿Se purgaron o desenergizaron las líneas?
☐	¿Todas las personas que realizan el trabajo tienen la capacitación adecuada?	☐	¿Se detuvo el equipo o maquinaria y se evitó que siguiera funcionando?
☐	¿Se ha notificado a todas las personas involucradas?	☐	¿Se bloquearon los peligros por gravedad?
☐	¿Todas las personas involucradas tienen la capacitación?	☐	¿Se permitió que los sistemas se enfriaran /calentaran?
☐	¿Se requiere capacitación adicional?	☐	¿Se descontaminaron los sistemas antes del trabajo?
☐	¿Se requieren permisos especiales?	☐	¿Se aplicaron dispositivos de aislamiento en todas las fuentes de energía?
☐	Espacios confinados	☐	¿Los dispositivos de aislamiento son resistentes y pueden soportar todas las condiciones?
☐	Trabajos peligrosos	☐	¿Se inspeccionaron los dispositivos de aislamiento para detectar defectos y reemplazarlos según fuera necesario?
☐	Trabajo eléctrico energizado	☐	¿Se identificaron claramente los dispositivos de aislamiento y de cierre y rotulación?
		☐	¿Se colocaron los avisos apropiados, según fuera necesario?
Cerrojos		Revisión del sistema	
☐	¿Se usan cerrojos estándar identificados individualmente?	☐	Se revisó el equipo, maquinaria o sistema para verificar su cumplimiento con las normas?
☐	¿El cerrojo es lo suficientemente resistente para hacer frente a las condiciones donde se usa?	☐	¿Se probó el equipo, maquinaria o sistema para verificar que los dispositivos de aislamiento y de cierre/rotulación funcionen?
☐	¿El cerrojo identifica por nombre al empleado al que le pertenece?		
☐	¿Cada empleado que trabaja en sistemas o equipos tiene su propio cerrojo?	Rótulos	
☐	¿La persona que realiza el trabajo guarda en un lugar seguro las llaves del cerrojo?	☐	¿Los rótulos indican el nombre de la persona que realiza el trabajo?
		☐	¿Los rótulos indican el número de contacto?
		☐	¿Los rótulos indican fecha(s) del trabajo?
		☐	¿Los rótulos mencionan el propósito del trabajo?
		☐	¿Los rótulos están adheridos firmemente al dispositivo de cierre?
		☐	¿Los rótulos son legibles?
Liberación del cierre y rotulación			
☐	Antes de restaurar la energía, ¿se realizó una inspección del área de trabajo para verificar que todas las protecciones estén en su lugar, que los equipos o sistemas estén seguros y que el área esté segura?		
☐	¿Se notificó y verificó la presencia de todo el personal y se alejó del área en forma segura?		
☐	¿El empleado autorizado quitó cada dispositivo de aislamiento, cierre y rotulación?		

Firma del supervisor del contratista

Al firmar arriba, usted confirma que se ha examinado el lugar mencionado y que se han tomado las precauciones indicadas arriba.

Firma del supervisor de DPR

Al firmar arriba, usted confirma que se autoriza el trabajo arriba indicado de acuerdo con los procedimientos y precauciones que se especificaron.

Firma del supervisor del área de trabajo

Al firmar arriba, usted confirma que se autoriza el trabajo arriba indicado de acuerdo con los procedimientos y precauciones que se especificaron.

Finalización del cierre y rotulación

Firma del supervisor del contratista

Al firmar arriba, usted confirma que se ha cumplido el trabajo indicado arriba, se han quitado todos los cerrojos y el área de trabajo se encuentra en condiciones seguras.



Hazardous Energy/Pre-Start-up Meeting Checklist

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Contractor Name:		Target Start Up Date:	
Scope of Work:			

Meeting Participants

Print Name	Company/Job Title	Contact Number(s)

Equipment Start-Up

☐ Temporary Power	☐ Chiller	☐ Fire Alarm	☐
☐ Permanent Power	☐ Air Handler Unit	☐ Fire Sprinkler	☐
☐ Auto Transfer Switch	☐ Boiler	☐ Equipment	☐
☐	☐ Chemical Line	☐	☐
☐ Emergency/Back-up	☐	☐	☐
☐ 2 nd Feed			
☐ Generator			
☐ UPS			
☐ Battery			

Hazardous Energy

☐ Electrical	☐ Gravity	☐ Mechanical
☐ Hydraulic	☐ Thermal	☐ Radiation
☐ Pneumatic	☐ Chemical	☐ Pressurized Piping

Power into the Building

☐ Low Voltage	☐ Medium Voltage	☐ High Voltage:
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Qualified Personnel

Name	Job Title	Date of Training	Training Record on File

Hazardous Energy

☐ One Line Diagram Available	☐ Tamper Lock Applied	☐ Owner Requirements
☐ Shut-off locations marked on drawings & communicated to qualified personnel	☐ LOTO devices identified	☐ Owner Notification
☐ LOTO Procedures Defined (attached)	☐ Individual Lock and Tag	☐ All sources Identified
☐ LO/TO Permit Provided	☐ EEW Permit Provided	☐ DPR Lock Applied
☐ MOP/SOP Reviewed by DPR		

Control Utilized	Yes/No/NA	Responsible Party	Due Date

Section 13

Incident Reporting

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Incident Reporting

1.0 Introduction

Incident reports are used to track, investigate and communicate information surrounding incidents or accidents to DPR management, insurance companies, attorneys and some state agencies. Incident reports help to provide a clear picture of an event, which will help depict opportunities for improvement on jobsites and the company's overall safety program. It is important, when filling out an incident report, that the report is complete (avoid leaving blanks) and reflects clear factual information. Information during the investigation process is used to determine the how and why an incident occurred as well as to determine the root cause and corrective actions necessary to prevent recurrence.

***Note:** Keep in mind that incident reports are prepared in anticipation of litigation and should be thoroughly reviewed prior to submission to the claims manager.*

Incident reports should be prepared and submitted to the claims manager within 24 hours of any event. The reports should document a clear picture of the incident, area where the incident occurred, parties involved and nature and extent of any injury or damage. **If you question whether an incident report is necessary, one should be completed.**

Along with an incident report, most events will require a Root Cause Analysis (RCA), which should be completed within 48 hours. A Root Cause Analysis should be completed with the injured or responsible party, his or her supervisor, the safety manager or safety coordinator and as many other team members possible. The Root Cause Analysis process is a fact finding mission, not a fault finding mission and should be completed in an open forum utilizing the RCA form in addition to the 5 Whys. The RCA should provide a detailed summary of events that occurred prior to the incident, all contributing factors and a corrective action plan. The lessons learned during the RCA process should be disseminated and shared with the region.

Reference: OSHA Recordkeeping CFR 1904. Other applicable Standards, guidelines, and regulations may apply.

Definitions

First Aid (FA) (Clinic): An injury that requires medical treatment provided by a licensed Emergency Medical Technician (EMT), physician or medical clinic where medication, physical therapy or restricted duty are not part of the treatment plan. This type of injury is classified as an OSHA First Aid.

Construction Defect (CD): A condition that results in a claim for property damage that is progressive in nature and arises out of the construction of buildings or structures and occurs sometime after construction operations are completed

Facility interruption: An incident that causes damage to services, systems or property resulting in interruption of normal business operations

Incident summary: An internal document used to communicate information to an owner or owner's representative

Lost Time (LT): An injury that requires medical treatment by a physician, medical clinic or hospital **and** the employee is unable to work for at least one full day **after** the injury. This type of injury is classified as an OSHA Recordable.

Motor Vehicle Accident (MVA): An automobile accident involving DPR fleet, regardless of fault or extent of damage

Near miss: An unplanned event that did not result in injury, property damage or interruption, but had the potential to do so

Non-industrial injury: An injury that occurred outside of the workplace, i.e., sports related or at home injury



On Site First Aid (OSFA): A minor injury that requires only on-site attention by the employer's personnel. This type of injury is classified as an On Site First Aid or Notice Only

Property Damage (PD): An incident that causes damage to property, service or equipment that results in any costs associated with repair

Recordable: An injury that requires medical treatment provided by a physician or medical clinic that requires in part; medication, stitches or sutures, physical therapy and/or restricted duty. This type of injury is classified as an OSHA Recordable. Reference the OSHA *First Aid vs. Recordable chart below*.

Root Cause Analysis (RCA): A fact finding process to determine cause(s), contributing factors and effective corrective actions to prevent recurrence

Service interruption: An incident that causes damage to services, systems, property, underground utilities, etc. and may result in an interruption of normal business operations

Theft: Property stolen from a job site, office, trailer or stolen vehicles

Water intrusion: An incident where intruding water, whether due to natural causes or an event, causes damage to existing property, equipment, utilities, tools or work being performed or incur costs associated with cleanup

First Aid Care	Recordable Medical Treatment
Over the counter medication at non-prescriptive strength	Any dosage of prescription medication
Any number of hot or cold treatments	Physical therapy
Wound coverings such as butterfly bandage or Steri-strips	Stitches or Staples
Foreign body removal using irrigation or swab	Removal of an embedded foreign body in eyes or skin
Loss of portion of day of injury	Days away from work – the inability to work at least one full day after day of injury
Non-rigid support such as ace bandages, back belts	Modified/ restricted duty or transfer to another job
Cleaning, flushing or soaking wounds	Debridement – removal of dead, damaged or infected tissue
Finger guards	Fracture
Drilling a fingernail or toenail	Overnight hospitalization
Tetanus shot	Chipped or broken tooth
Eye patches	Loss of consciousness – caused by work
Diagnostic procedures, including prescriptive medication for diagnostic purposes, x-rays, blood tests	Punctured ear drum – caused by work
Multiple visits to doctor for observation or counseling	Multiple visits to a medical professional for more than observation or counseling
	Cancer – caused by work
	Chronic irreversible disease – caused by work
	Asbestosis – caused by work
	Death – caused by work



2.0 Reporting

Contact your regional safety professional immediately. Once notified of an incident the Safety Coordinator, Safety Manager or other supervisor must contact the claims manager at 916.418.1910 or 916.826.9996 within 2 hours of any incident and determine what type of incident report to use to report the event. **A phone call is not a sufficient way to report an incident. It is considered notification.**

All incidents, regardless of size or severity, must be reported on an appropriate incident report. Most incidents also require a Root Cause Analysis (RCA) to be completed immediately following the investigation.

Note: Occasionally owners or owner's representatives will request copies of our incident reports to help them better understand the dynamics of an event. It is not acceptable to provide any internal documents to any outside sources, contacts or representatives. In the event a request is made for information regarding an incident, an Incident Summary Sheet can be completed and given to the owner or owner's representative; however, should be discussed with the claims manager or an attorney.

The following list provides types of incidents and the suggested report required to conduct an investigation.

Type of Incident	Definitions	Report Required
On Site First Aid (OSFA)	A minor injury that requires only on-site attention by the employer's personnel. This type of injury is classified as an On Site First Aid or Notice Only – NOTE: did not go to clinic, was not seen by ON-SITE HEALTH & SAFETY or 1st AID RESPONSE	On Site First Aid Grab & Go
Clinic First Aid (FA)	An injury that requires medical treatment provided by a licensed Emergency Medical Technician (EMT), physician or medical clinic where medication, physical therapy or restricted duty are not part of the treatment plan. This type of injury is classified as an OSHA First Aid. NOTE: was taken to CLINIC OR was seen by ON-SITE HEALTH & SAFETY or 1st AID RESPONSE	Off Site Medical Grab & Go Root Cause Analysis
Construction Defect (CD)	A condition that results in a claim for property damage that is progressive in nature and arises out of the construction of buildings or structures and occurs sometime after construction operations are completed. NOTE: not the same as property damage	General Liability Report – Construction Defects Root Cause Analysis
Recordable	An injury that requires medical treatment provided by a physician or medical clinic that requires in part; medication, stitches or sutures, physical therapy and/or restricted duty. This type of injury is classified as an OSHA Recordable. Reference the OSHA First Aid vs. Recordable chart.	Off Site Medical Grab & go Root Cause Analysis
Lost Time (LT)	An injury that requires medical treatment by a physician, medical clinic or hospital and the employee is unable to work for at least one full day after the injury. This type of injury is classified as an OSHA Recordable.	Off Site Medical Grab & go Root Cause Analysis
Non-Industrial Injury	An injury that occurred outside of the workplace, i.e., sports related or at home injury NOTE: requires a medical release in order to return to work.	Witness Statement
Near Miss	An unplanned event that did not result in injury, property damage or interruption, but had the potential to do so	Near Miss Report w/Root Cause Analysis
Motor Vehicle Accident (MVA)	An automobile accident involving DPR fleet, regardless of fault or extent of damage	Vehicle Accident Report
Facility Interruption	An incident that causes damage to services, systems or property resulting in interruption of normal business operations	General Liability Report – Property Damage, Theft or Bodily Injury Root Cause Analysis
Service Interruption	An incident that causes damage to services, systems, property, underground utilities, etc. and may result in an interruption of normal business operations	General Liability Report – Property Damage, Theft or Bodily Injury Root Cause Analysis
Property Damage (PD)	An incident that causes damage to property, service or equipment that results in any costs associated with repair	General Liability Report – Property Damage, Theft or Bodily Injury Root Cause Analysis
Theft	Property stolen from a job site, office, trailer or stolen vehicles	General Liability Report – Property Damage, Theft or Bodily Injury
Water Intrusion	An incident where intruding water, whether due to natural causes or an event, causes damage to existing property, equipment, utilities, tools or work being performed or incur costs associated with cleanup	Water Intrusion Report Root Cause Analysis

3.0 Investigation

A thorough investigation of any incident must be conducted immediately after an event occurs. The employee(s) controlling the investigation must ensure the safety and security of the individual(s) that were injured or involved, other people on site, the public, and the project.

3.1 Photographs

Photographs are a key component to an investigation and should be taken liberally. Photographs should be taken immediately and should encompass the entire area from a panoramic perspective, including but not limited to:

- the accident scene
- any and all surrounding areas
- the project site
- any contributing factors such as
 - machinery
 - tools
 - equipment
 - vehicles
 - associated environmental conditions

3.2 Witness Statements

Witness statements are a critical part of an accident investigation and should be obtained from any employee or subcontractor employee in the vicinity of the incident, ***whether they saw anything or not.***

- witnesses must be separated prior to providing any statement
- witness statements must be handwritten in the individual's native language (translations should be provided)
- witness statements must be signed by the employee providing the statement

Note: No one except emergency response personnel, owner or OSHA representatives or those associated with or retained by DPR should be allowed on the project site following an incident. Should other individuals require access, prior approval from DPR project management is required. Those individuals require DPR escort at all times.

The entire incident report must be completed thoroughly and submitted to the claims manager within 24 hours.

4.0 OSHA Reporting

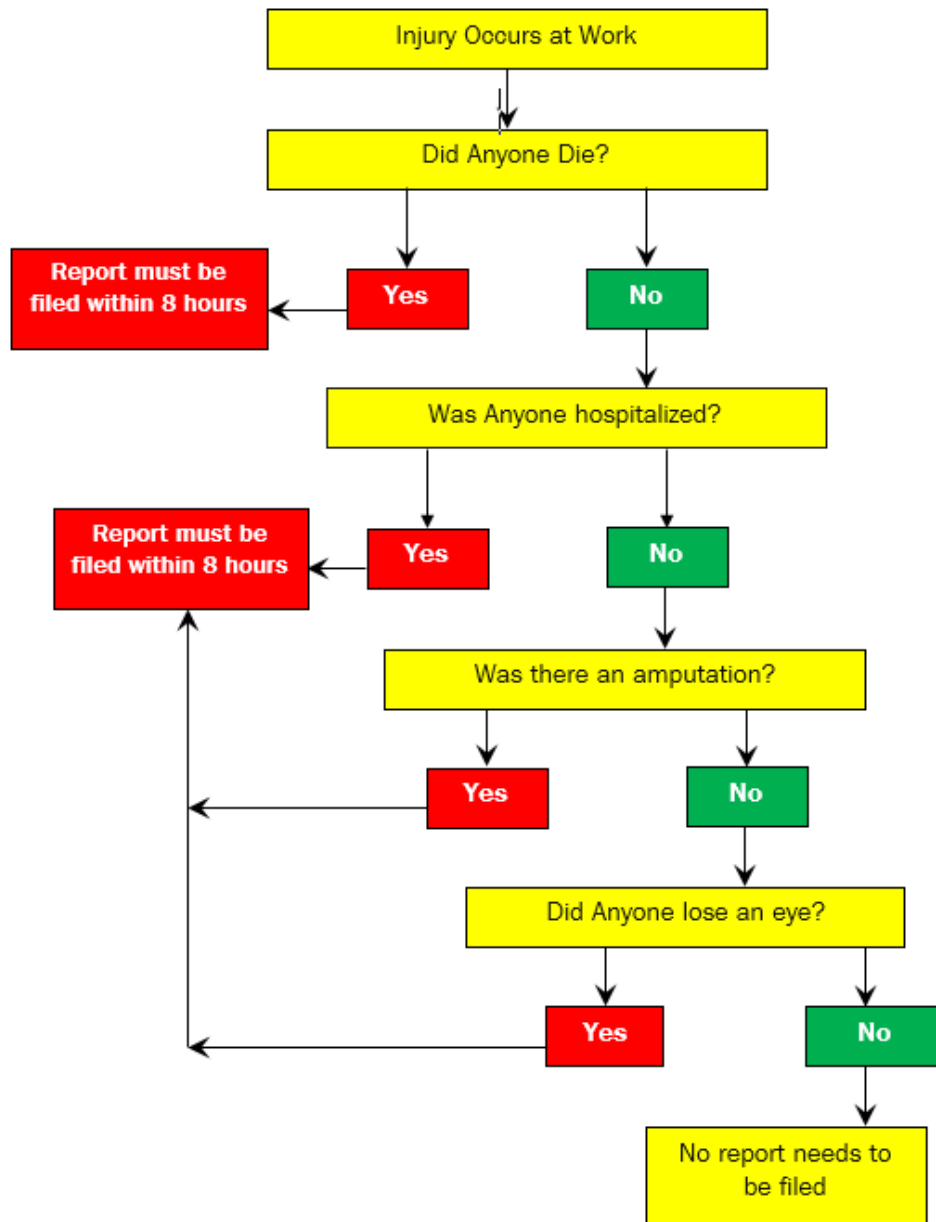
Occupational Safety and Health Administration (OSHA) requires serious injuries and fatalities to be reported within eight (8) hours of the incident or event, regardless if it is a weekend or holiday. Incidents that require direct reporting to OSHA within 24 hours from the time the employer learns of the incident include:

- FedOSHA – (OSHA 1904.39(a))
 - inpatient hospitalization
 - death
 - amputations
- loss of an eye
- CalOSHA – (Title 8, Calif. Code of Regulations, sec. 342a.)
 - hospitalization for more than 24 hours for anything other than observation
 - loss of any member of the body
 - permanent disfigurement
 - death

Reporting can be done by calling OSHA's free and confidential number 800-321-OSHA (6742) or by calling or visiting the closest OSHA area office during normal business hours.

Note: Failure to report a serious workplace injury or death will result in serious fines, which increase for repeat offenses and may result in an OSHA serious and willful citation.

Step-by-Step on the New OSHA Standard for Injury Reporting



5.0 Root Cause Analysis Process

A Root Cause Analysis (RCA) should be completed after every incident (except OSFA injuries) – regardless if it is a safety or quality issue. This process is a fact finding mission, not a fault finding mission. The process should be conducted in such a way that all involved do not feel threatened, which will aid in the information gathering process. Any person involved or witness to an incident should attend the process. Utilization of the 5 Why's practice will help to delve further into the actual root cause. The purpose behind the process is to identify the “one thing, if removed, would have prevented the incident”.

The RCA meeting should be conducted within 48 hours of an incident to ensure information is fresh in peoples' minds and that the parties involved are still present at the job site. Information learned through the Root Cause Analysis process will help provide lessons learned to others to aid in avoiding similar issues in the future.

The analysis portion is to engage all parties and ensure that input is gathered from everyone (injured worker/involved party, superintendents, foremen, project manager, project engineer, witnesses). The contributing factors should be documented as elements that may have contributed to the incident. Utilization of a timeline will help to determine what occurred prior to an event and help to develop contributing factors. The process should be a brainstorm of the group to work to an end goal of finding the Root Cause of an incident.

Note: *This timeline should not be limited to the day of the incident as it could start as early as the start of the job.*

This is an opportunity to utilize one another's knowledge and experience to determine what was the ultimate cause of an incident. The focus should not be what an individual “did”, but what could have been done to prevent an incident. Corrective actions should be determined to aid in preventing a similar incident in the future. The analysis should be followed up with tracing and verifying corrective actions and performing a plus/delta.

Lessons learned during this process should be shared with the project team and region.

Note: *If your last answer is something you cannot control, go back up to the previous answer.*

5 Why's Worksheet

Define the Problem:

1.

Why is that?
↓

2.

Why is that?
↓

3.

Why is that?
↓

4.

Why is that?
↓

5.

Why is that?
↓

Caution: If your last answer is something you cannot control, go back up to previous answer.

Incident Reporting Forms

Forms

- On-Site First Aid Grab & Go
- Off-Site Medical Treatment Grab & Go
- General Liability Report – Property Damage, Theft or Bodily Injury
- General Liability Checklist
- General Liability Report – Construction Defects
- Motor Vehicle Accident Report
- Water Intrusion Report
- Near Miss Incident Report
- Root Cause Analysis
- Incident Summary

On-Site First Aid Grab & Go

Incident Reporting Instructions

1. Ensure the safety and security of the individual(s) that were injured or involved, other people on site, the public, and the project. Ensure the injury is thoroughly cleaned and properly treated.
2. Contact your regional safety manager ***immediately***
3. Contact Heidi Marsac in the Sacramento office at 916.418.1910, cellular 916. 826.9996 **within 2 hours of any incident.**
4. Complete the entire GRAB & GO packet thoroughly. **The entire packet should be emailed to Heidi Marsac at heidir@dpr.com & jennap@dpr.com or faxed to 916-848-0269 within 24 hours.**

REQUIRED DOCUMENTS
Completed Grab & Go
Pre-task plan
Photographs, as applicable



ON SITE FIRST AID GRAB & GO

Incident Investigation Report

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

Date of Report: _____	Report Prepared by: _____	Phone Number: _____
☐ Contacted Heidi Marsac 916.826.9996	☐ Copy emailed to heidir@dpr.com & jennap@dpr.com	

#1 – PROJECT INFORMATION

Jobsite Name: _____	Region: _____
Address: _____	Job Number: _____
Subcontractor: _____	DPR Foreman: _____
Sub Foreman: _____	DPR Superintendent: _____
Sub Superintendent: _____	DPR Safety Coordinator: _____

#2 – EMPLOYEE/INCIDENT INFORMATION

Employee Name: _____	SS#: _____	DOB: _____
Address: _____	Phone: _____	
Date of Hire: _____	Job Title: _____	Wages: _____
Total Years in Occupation: _____	Start Time: _____	End Time: _____
Exact Location of Incident: _____		
General Activity at the time of Incident (i.e., concrete): _____		
Specific Task at Time of Incident (i.e., finishing): _____		

#3 – INJURY/ILLNESS INFORMATION

Date of Incident: _____	Day of Week: _____	Time of Incident: _____
Date reported: _____	Reported to: _____	Time Reported: _____
Type of Injury: _____	Body Part Injured: _____	
Was On-Site First Aid Given: ☐ Yes ☐ No	If Yes, by Whom: _____	
Describe First Aid Administered: _____		
Employee's Supervisor: _____	Number of Workers Managed by Supervisor: _____	
Working on a Crew: ☐ Yes ☐ No	Crew Size: _____	
PTP Required for Work Performed: ☐ Yes ☐ No	PTP Attached: ☐ Yes ☐ No	

#4 – DESCRIPTION OF THE INCIDENT

#5 – ADDITIONAL INFORMATION

Name of witness(es)/those working with injured worker: _____
Object, substance, equipment Involved in incident: _____
List PPE Worn at Time of Accident _____
Were standard work procedures followed ☐ Yes ☐ No If not, why? _____
Was a safety rule or specific instructions violated? ☐ Yes ☐ No If yes, what? _____

#6 – ACTIONS TO PREVENT RECURRENCE

#7 – INJURED WORKERS STATEMENT

In your own words, tell us what happened, tell a story (En sus propias palabras, díganos lo que sucedimos):

In addition to your statement, please answer the following questions (Además)

Where were you when the incident took place/Dónde estaba usted cuando ocurrió el incidente?

What activity were you performing prior to the event/Que actividad estaba realizando usted antes del incidente o evento?

Were others in the immediate area/Había otros trabajadores en las cercanías inmediatas en el momento del incidente o evento?

☐ Yes/Si ☐ No If yes, who/quienes eran? _____

What did you do immediately after the incident/event/Qué hizo usted inmediatamente después del incidente o evento?

Have you ever injured this body part before/ha sufrido alguna vez antes una lesión en esta parte del cuerpo?

☐ Yes/Si ☐ No

Employee Signature _____ Date _____

Note: If this statement is written in Spanish, it must be translated into English, attach a separate sheet if necessary.

#8 – SIGNATURES (all appropriate signatures should be included):

Foreman:

Print Name

Signature

Superintendent:

Print Name

Signature

DPR Safety:

Print Name

Signature

Project Manager:

Print Name

Signature

Off-Site Medical Treatment Grab & Go

Incident Reporting Instructions

1. Ensure the safety and security of the individual(s) that were injured or involved, other people on site, the public, and the project.

Note: No one except emergency response personnel, owner or OSHA representatives or those associated with or retained by DPR should be allowed on the project site following an incident. Should other individuals require access, prior approval from DPR project management is required. Those individuals require DPR escort at all times.

2. Contact your regional safety professional **immediately**.
3. Take photos of the accident scene and the surrounding area immediately.
4. Contact Heidi Marsac in the Sacramento office at 916.418.1910, cellular 916.826.9996 **within 2 hours of any incident**.
5. Complete the entire GRAB & GO packet thoroughly. **The entire packet should be emailed to Heidi Marsac at heidir@dpr.com & jennap@dpr.com or faxed to 916-848-0269 within 24 hours.**

REQUIRED DOCUMENTS
Completed Grab & Go and all witness statements
Any treatment documentation
Pre-task plan
Photographs
Root Cause Analysis
Recent Safety Meeting Minutes
Recent Safety Audits

Note: Root Cause Analysis must be completed and sent to Heidi Marsac no more than 48 hours after the incident.

6. Ask the treating physician to offer over the counter medications instead of prescriptions for minor injuries such as small lacerations or puncture wounds.
7. Before leaving the doctor's office, obtain the PHYSICIAN'S RELEASE/WORK STATUS or the JOB ANALYSIS/WORK RECOMMENDATIONS REPORT from the clinic/hospital doctor after each doctor's visit.
8. If necessary, identify an alternative work that meets the physician's requirements



OFF SITE MEDICAL GRAB & GO

Incident Investigation Report

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

Date of Report: _____	Report Prepared by: _____
Phone Number: _____	☐ Completed Substance Abuse Screen
☐ Contacted Regional Safety Manager	☐ Contacted Heidi Marsac – Cell: 916.826.9996 heidir@dpr.com & jennap@dpr.com

INCIDENT DESIGNATION

☐ On Site First Aid	☐ Clinic First Aid	☐ Recordable	☐ Lost Time	☐ Non-Industrial
☐ DPR Employee	☐ Subcontractor	☐ Valid Claim	☐ Suspicious claim	
☐ Unknown Claim (completely unaware of the incident)				

#1 – PROJECT INFORMATION

Jobsite Name: _____	Region: _____
Address: _____	Job Number: _____
Subcontractor: _____	DPR Foreman: _____
Sub Foreman: _____	DPR Superintendent: _____
Sub Superintendent: _____	DPR Safety Coordinator: _____

#2 – EMPLOYEE/INCIDENT INFORMATION

Employee Name: _____	SS#: _____	DOB: _____
Address: _____		Phone: _____
Date of Hire: _____	Job Title: _____	Wages: _____
Total Years in Occupation: _____	Start Time: _____	End Time: _____
Exact Location of Incident: _____		
General Activity at the time of Incident (i.e., concrete): _____		
Specific Task at Time of Incident (i.e., finishing): _____		

#3 – INJURY/ILLNESS INFORMATION

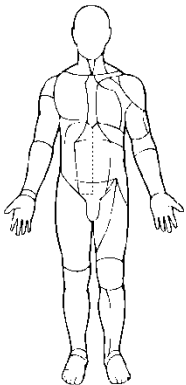
Date of Incident: _____	Day of Week: _____	Time of Incident: _____
Date reported: _____	Reported to: _____	Time Reported: _____
Type of Injury: _____	Body Part Injured: _____	
Was On Site First Aid Given: ☐ Yes ☐ No	If Yes, by Whom: _____	
Was Employee Taken to a Medical Clinic: ☐ Yes ☐ No	If so, When: _____	
Clinic Address/Phone Number: _____		
Transported by: ☐ Ambulance ☐ Company Vehicle ☐ Private Vehicle	Name of Driver: _____	
Employee Returned to Work: ☐ Regular Duty ☐ Alternative Duty ☐ Did not Return to Work		
Expected Return to Work Date: _____		
Number of Workers Managed by Foreman/Supervisor: _____	Working on a Crew: ☐ Yes ☐ No	Crew Size: _____
PTP Required for Work Performed: ☐ Yes ☐ No	PTP Attached: ☐ Yes ☐ No	
If no PTP attached, explain why: _____		

#4 – DESCRIPTION OF THE INCIDENT (not to be completed by the injured worker):

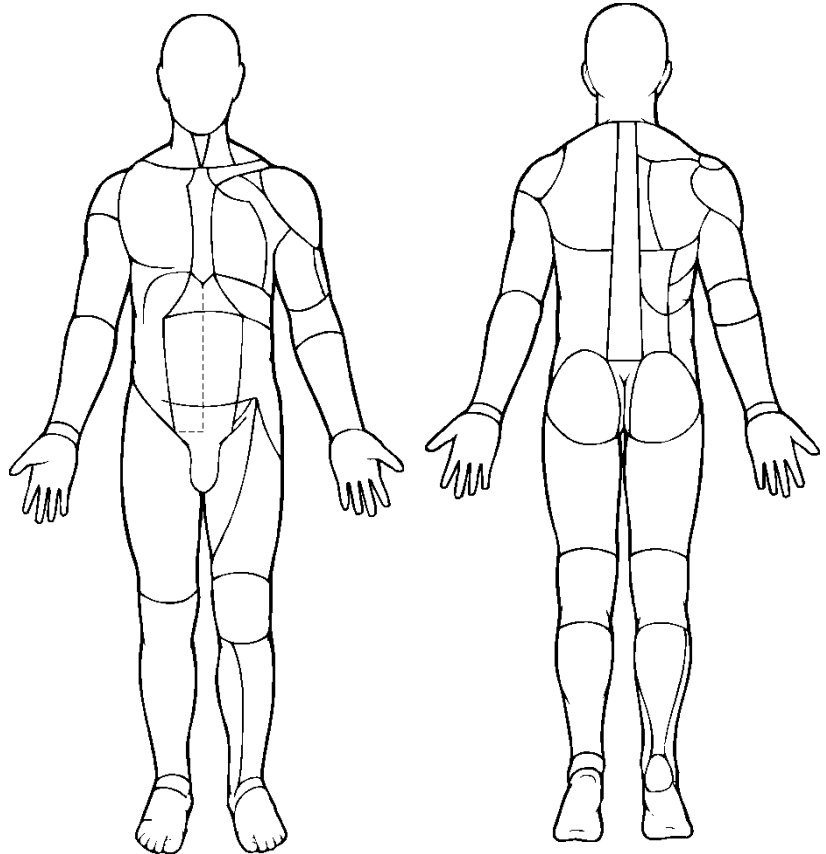
NOTE: This does not take the place of a witness statement. Describe in detail the circumstances of the incident (attach diagrams, drawings and/or photos of the accident scene.) Give a chronological sequence of events. If materials and/or equipment were involved, start before the materials/equipment were brought to the incident scene describing who, what, where, when, how:

Please indicate the location of all incurred injuries and describe the type of injury. For example, for a laceration to the left palm – shade the left hand palm and write laceration next to it connected by a line.

Example:



Laceration



#5 – ADDITIONAL INFORMATION

Name of witnesses and others working with injured worker (attach statement for each witness):

Object, substance, equipment involved in incident (description/model/serial #):

List PPE worn at time of incident:

Was PPE available at the time of incident? ☐ Yes ☐ No If no, explain _____

Does employee normally operate this equipment? ☐ Yes ☐ No ☐ N/A

Was employee instructed in the safe use of this equipment? ☐ Yes ☐ No ☐ N/A When/How (describe in detail/attach certification/training documentation): _____

Was any problem/defect with the equipment noted or reported prior to accident/incident? ☐ Yes ☐ No ☐ N/A

Was any recent maintenance/service performed on this equipment? ☐ Yes ☐ No ☐ N/A If yes, when/what (Describe in detail, attach a copy of the invoices/work order): _____

Were standard work procedures followed? ☐ Yes ☐ No If no, why not? (Describe in detail, attach additional sheets if necessary and attach a copy of the standard site procedures): _____

Was a safety rule or specific instruction violated? ☐ Yes ☐ No If yes, describe in detail:

When/How was this rule, regulation or specific instruction communicated to the injured worker(s)?

Date of last safety meeting conducted? _____ Date of last jobsite audit conducted? _____

Date **Root Cause Analysis** scheduled to be conducted on: _____

Attach copies of last safety meeting agenda w/sign-in sheet & last jobsite audit including corrective actions.

#6 – SIGNATURES

DPR Foreman:

Print Name

Signature

DPR Superintendent:

Print Name

Signature

DPR Safety:

Print Name

Signature

DPR Project Manager:

Print Name

Signature



Injured Worker's Statement

I _____ am submitting this statement made on _____
Employee Name Date

To _____ for DPR Construction. I am submitting this statement of my own
Name

free will. I have not been coerced or threatened in any way to submit this statement.

Date of Injury: _____ Time of Injury: _____

In your own word, tell us what happened, tell a story:

Additionally,

Where were you when the incident took place?

Were any other workers in the immediate area at the time of the incident/event? ☐ Yes ☐ No
If yes, who?

What did you do immediately following the incident/event?

If you were injured in the incident, have you ever injured this body part before? ☐ Yes ☐ No

I _____ *Permit DPR Construction and its agents to perform investigations,*
Employee Name

interviews, and any other activities necessary to process my claim. This authorization includes, but is not limited to: reviewing, requesting and copying past and present medical and employment records to process my current Worker's Compensation claim. I understand that any and all information gathered in connection with this claim will be handled in a confidential manner.

Employee Signature _____ I have received a copy of this statement ☐ Yes ☐ No

Today's Date: _____ Employer: _____

Home Address: _____

Home Phone: _____ Cell Phone: _____



Supervisor Statement

I _____ am submitting this statement made on _____
Employee Name Date

To _____ for DPR Construction. I am submitting this statement of my own
Name
free will. I have not been coerced or threatened in any way to submit this statement.

In your own word, tell us what happened, tell a story:

Additionally,
Where were you when the incident took place?

Did you witness the incident/event? ☐ Yes ☐ No

What activity were you performing prior to the incident/event?

To your knowledge, were any other workers in the immediate area at the time of the incident? ☐ Yes ☐ No
If yes, who?

What did you do immediately following the incident/event?

Supervisor Signature _____ I have received a copy of this statement ☐ Yes ☐ No

Today's Date: _____ Employer: _____

Home Address: _____

Home Phone: _____ Cell Phone: _____



Witness Statement

I _____ am submitting this statement made on _____
Employee Name Date

To _____ for DPR Construction. I am submitting this statement of my own
Name
free will. I have not been coerced or threatened in any way to submit this statement.

In your own word, tell us what happened, tell a story:

Additionally,
Where were you when the incident took place?

Did you witness the incident/event? ☐ Yes ☐ No

What activity were you performing prior to the incident/event?

To your knowledge, were any other workers in the immediate area at the time of the incident? ☐ Yes ☐ No
If yes, who?

What did you do immediately following the incident/event?

Witness Signature _____ I have received a copy of this statement ☐ Yes ☐ No

Today's Date: _____ Employer: _____

Home Address: _____

Home Phone: _____ Cell Phone: _____



Witness Statement

I _____ am submitting this statement made on _____
Employee Name Date

To _____ for DPR Construction. I am submitting this statement of my own
Name

free will. I have not been coerced or threatened in any way to submit this statement.

In your own word, tell us what happened, tell a story:

Additionally,
Where were you when the incident took place?

Did you witness the incident/event? ☐ Yes ☐ No

What activity were you performing prior to the incident/event?

To your knowledge, were any other workers in the immediate area at the time of the incident? ☐ Yes ☐ No
If yes, who?

What did you do immediately following the incident/event?

Supervisor Signature _____ I have received a copy of this statement ☐ Yes ☐ No

Today's Date: _____ Employer: _____

Home Address: _____

Home Phone: _____ Cell Phone: _____



Declaración del Accidentado

Yo _____ presento esta declaración hecha el _____
Nombre del Empleado/Testigo Fecha

A _____ para DPR Construction. Presento esta declaración
Name
por propia voluntad. No he sido objeto de coacción o amenaza alguna para presentar esta declaración.

En sus propias palabras, díganos qué sucedió, cuéntenos la historia:

Además,

¿Dónde estaba usted cuando ocurrió el incidente?

¿Que actividad estaba realizando usted antes del incidente o evento?

¿Había otros trabajadores en las cercanías inmediatas en el momento del incidente o evento? ☐ Sí ☐ No

Si contestó afirmativamente, ¿quiénes eran?

¿Qué hizo usted inmediatamente después del incidente o evento?

Si usted se lesionó en este incidente, ¿ha sufrido alguna vez antes una lesión en esta parte del cuerpo? ☐ Sí ☐ No

Yo _____ autorizo a DPR Construction y a sus agentes a llevar a cabo
Employee Name

investigaciones, entrevistas y toda otra actividad necesaria para procesar mi demanda. Sin limitarse a éstos, esta autorización incluye la evaluación, solicitud y copia de mis anteriores y actuales antecedentes médicos y laborales, con el fin de tramitar mi actual demanda de indemnización por accidente de trabajo. Entiendo que toda información que se obtenga en conexión con esta demanda será tratada en forma confidencial.

Firma del empleado: _____ He recibido una copia de esta declaración ☐ Sí ☐ No

Fecha de hoy: _____ Empleador: _____

Dirección particular: _____

Teléfono particular: _____ Celular particular: _____



Declaración del supervisor

(por ejemplo, Jefe de cuadrilla, Capataz, Supervisor que controla la actividad)

Yo _____ presento esta declaración hecha el _____
Nombre del Empleado/Testigo Fecha

A _____ para DPR Construction. Presento esta declaración
Name

por propia voluntad. No he sido objeto de coacción o amenaza alguna para presentar esta declaración.
En sus propias palabras, díganos qué sucedió, cuéntenos la historia:

Además,

¿Dónde estaba usted cuando ocurrió el incidente?

¿Que actividad estaba realizando usted antes del incidente o evento?

¿Fue testigo usted del evento? ☐ Sí ☐ No

¿Qué hizo usted inmediatamente después del incidente o evento?

Según su conocimiento, ¿había otros trabajadores en las cercanías inmediatas en el momento del incidente o evento? ☐ Sí ☐ No

Si contestó afirmativamente, ¿quiénes eran?

Yo _____ autorizo a DPR Construction y a sus agentes a llevar a cabo
Employee Name

investigaciones, entrevistas y toda otra actividad necesaria para procesar mi demanda. Sin limitarse a éstos, esta autorización incluye la evaluación, solicitud y copia de mis anteriores y actuales antecedentes médicos y laborales, con el fin de tramitar mi actual demanda de indemnización por accidente de trabajo. Entiendo que toda información que se obtenga en conexión con esta demanda será tratada en forma confidencial.

Firma del supervisor: _____ He recibido una copia de esta declaración ☐ Sí ☐ No

Fecha de hoy: _____ Empleador: _____

Dirección particular: _____

Teléfono particular: _____ Celular particular: _____



Declaración del testigo

Yo _____ presento esta declaración hecha el _____
Nombre del Empleado/Testigo Fecha

A _____ para DPR Construction. Presento esta declaración
Name

por propia voluntad. No he sido objeto de coacción o amenaza alguna para presentar esta declaración.

En sus propias palabras, díganos qué sucedió, cuéntenos la historia:

Además,

¿Dónde estaba usted cuando ocurrió el incidente?

¿Que actividad estaba realizando usted antes del incidente o evento?

¿Había otros trabajadores en las cercanías inmediatas en el momento del incidente o evento?

☐ Sí ☐ No

Si contestó afirmativamente, ¿quiénes eran?

¿Qué hizo usted inmediatamente después del incidente o evento?

¿Fue testigo usted del evento?

☐ Sí ☐ No

Yo _____ autorizo a DPR Construction y a sus agentes a llevar a cabo

Employee Name

investigaciones, entrevistas y toda otra actividad necesaria para procesar mi demanda. Sin limitarse a éstos, esta autorización incluye la evaluación, solicitud y copia de mis anteriores y actuales antecedentes médicos y laborales, con el fin de tramitar mi actual demanda de indemnización por accidente de trabajo. Entiendo que toda información que se obtenga en conexión con esta demanda será tratada en forma confidencial.

Firma del testigo: _____

He recibido una copia de esta declaración

☐ Sí ☐ No

Fecha de hoy: _____

Empleador: _____

Dirección particular: _____

Teléfono particular: _____

Celular particular: _____



Declaración del testigo

Yo _____ presento esta declaración hecha el _____
Nombre del Empleado/Testigo Fecha

A _____ para DPR Construction. Presento esta declaración
Name

por propia voluntad. No he sido objeto de coacción o amenaza alguna para presentar esta declaración.

En sus propias palabras, díganos qué sucedió, cuéntenos la historia:

Además,

¿Dónde estaba usted cuando ocurrió el incidente?

¿Que actividad estaba realizando usted antes del incidente o evento?

¿Había otros trabajadores en las cercanías inmediatas en el momento del incidente o evento? ☐ Sí ☐ No
Si contestó afirmativamente, ¿quiénes eran?

¿Qué hizo usted inmediatamente después del incidente o evento?

¿Fue testigo usted del evento? ☐ Sí ☐ No

Yo _____ autorizo a DPR Construction y a sus agentes a llevar a cabo
Employee Name

investigaciones, entrevistas y toda otra actividad necesaria para procesar mi demanda. Sin limitarse a éstos, esta autorización incluye la evaluación, solicitud y copia de mis anteriores y actuales antecedentes médicos y laborales, con el fin de tramitar mi actual demanda de indemnización por accidente de trabajo. Entiendo que toda información que se obtenga en conexión con esta demanda será tratada en forma confidencial.

Firma del testigo: _____ He recibido una copia de esta declaración ☐ Sí ☐ No
Fecha de hoy: _____ Empleador: _____
Dirección particular: _____

Teléfono particular: _____ Celular particular: _____



Authorization for Medical Treatment

Contact Information:	
Claims Manager:	Heidi Marsac
Phone Number:	916.418.1910
Cell Phone Number:	916.826.9996
Fax Number:	916.848.0269
Email:	Heidir@dpr.com

Injured Employee Information:	
Name of Injured Employee:	
Employer:	
Craft (Trade):	
Date of Hire:	
Date/Time of Injury (am/pm):	
Body Part:	
Type of Injury:	

Employer Authorization/Signature:	
Project Manager:	Or
Project Superintendent:	Or
Project Safety Coordinator/Manager:	

NOTE TO THE TREATING FACILITY:

DPR Construction has a "Return to Work" program. All injured workers will be returned to work within the treating physician's guidelines. Please contact Heidi Marsac 916.418.1910 or 916.826.9996 for all modified work cases, and be prepared to indicate how long the injured worker is anticipated to be under the treating physician's prescribed limitations.

PAYMENT:

Send all First Aid incident invoices directly to Heidi Marsac at

DPR Construction
2480 Natomas Park Drive, Suite 100
Sacramento, CA 95833

To ensure rapid payment, please have the treating physician complete the following:

- 1. Sign a release regarding whether the injured employee is to perform regular work or modified work*
- 2. Complete the attached "Job Analysis / Work Recommendations" form for each visit*
- 3. Fax copies of both documents to Heidi Marsac at 916.848.0269*

Return to Work Program

DPR Construction has developed a program designed to assist workers who are **temporarily** unable to perform their usual work activities due to a work related illness or injury. This program is called the "Return to Work Program." This includes a team effort, including injured workers, their attending physician, the insurance carrier and project management.

Studies show that return to work programs are therapeutic and help speed the recovery process. In addition, injured employees stay "in touch" with the work environment and with fellow employees, which helps to facilitate a smooth and speedy transition back to their normal job. This also creates an opportunity for cross training and developing new skills.

If the doctor determines that the employee qualifies for our Return to Work Program, the doctor will complete the appropriate forms indicating the restrictions and conditions for transitional work. **We will then attempt to provide an alternative work position until the employee is able to resume full duties.** All alternative work is temporary in nature and is designed to facilitate a return to full duties as soon as possible. Alternative work positions may be offered at any project, and/or any shift. Alternative work positions may also be offered on a varied schedule.

This program is not designed as a substitute for reasonable accommodation under any applicable federal and/or state regulations, such as the Americans with Disabilities Act, The Rehabilitation Act 1973 or other applicable laws.

To preserve the ability to meet company needs under changing conditions, we reserve the right to revoke, change or supplement these guidelines at any time with or without written notice. No permanent employment for any term is intended or can be implied by this policy. But, while in effect, all doctor recommended restrictions will be followed. The site management and the claims manager, on a two-week basis or after a doctor's visit, will review with the worker the availability of continued alternative work.

The policies and procedures in this program are not intended to be contractual commitments and they shall not be construed as such. This policy is not intended as a guarantee of continuity of benefits or rights.

Employee Responsibilities

Any work related injury, no matter how minor or near miss incident must be reported immediately to DPR project management. Failure to do so could result in your claim being delayed or denied.

You must inform your physician that there are alternative-jobs available. Once your doctor releases you to return to work (temporary, part-time, alternative work or regular duties), you must report to work immediately if possible, or the next scheduled shift, and inform your supervisor/foreman in charge of any physical restrictions or conditions. You must provide a written update on your condition by the next business day. **You cannot return to work without a release from your doctor.**

Failure to report for work at the designated time and place will be regarded as a voluntary resignation and could affect your time loss compensation and/or reemployment/reinstatement rights. If you are unable to report for any kind of work, you must call in daily no later than 30 minutes after start of shift (unless otherwise arranged) to your supervisor to report your status. It is also your responsibility to supply your supervisor with your current telephone number (unlisted or not) and an address where you can be reached.

I have read and fully understand all of the above procedures, and know my responsibilities. I understand that failure to complete my responsibilities as stated above may result in disciplinary action up to and including termination from my job and/or loss of my right to re-employment or reinstatement following injury. I have received a copy of this information.

Employee Signature

Date

Programa para regresar al trabajo

DPR Construction ha elaborado un programa para ayudar a los trabajadores que **temporalmente** se encuentran incapaces de cumplir con sus tareas normales debido a una enfermedad o lesión relacionada con el trabajo. Este programa se denomina "Programa para regresar al trabajo". Es un esfuerzo en equipo que incorpora al trabajador lesionado, su médico, la compañía de seguros y la gerencia del proyecto.

Los estudios demuestran que los programas para regresar al trabajo son beneficiosos y ayudan a acelerar el proceso de recuperación. Además, los empleados lesionados pueden mantener contacto con el lugar de trabajo y con sus compañeros, lo que facilita una transición rápida y sin obstáculos a su trabajo regular. También crea una oportunidad para su capacitación en otras tareas y el desarrollo de nuevas capacidades.

Si el doctor establece que el empleado/a se encuentra calificado/a para el programa para regresar al trabajo, completará los formularios respectivos indicando las restricciones y condiciones para realizar el trabajo de transición. **Intentaremos entonces ofrecer una posición alternativa de trabajo hasta el momento que el empleado pueda reanudar todas sus funciones.** Todo trabajo alternativo es de naturaleza temporal y tiene como fin facilitar el retorno lo antes posible al cumplimiento pleno de las tareas. Las posiciones alternativas podrán ofrecerse en cualquier proyecto, y/o para cualquier turno. También podrán ofrecerse en un horario variado.

Este programa no ha sido diseñado para sustituir un arreglo razonable conforme a los reglamentos federales y/o estatales, tales como el Decreto para los Americanos con Discapacidades, el Decreto de Rehabilitación de 1973, y toda otra ley de aplicación.

Para conservar nuestra capacidad para hacer frente a las necesidades de la compañía en situaciones cambiantes, nos reservamos el derecho a revocar, cambiar o suplementar estas normas en cualquier momento, con o sin aviso por escrito. Esta política no tiene la intención de ofrecer, y de ninguna forma quiere dar a entender que se está ofreciendo un empleo permanente por un cierto período. Sin embargo, mientras permanezca en vigencia se cumplirá con todas las restricciones recomendadas por el médico. Cada dos semanas, o después de una visita al médico, la gerencia del sitio de la obra y el administrador de demandas evaluarán, junto con el empleado, la disponibilidad continua de una posición alternativa de trabajo.

Las políticas y procedimientos en este programa no tienen la intención de ofrecer garantías o contratos, y no serán interpretados de dicha forma. Esta política no tiene como fin garantizar el continuo aporte de derechos o beneficios.

Responsabilidades del empleado

Toda lesión relacionada con el trabajo, por leve que sea e inclusive en casos de incidentes escasamente evitados, deberá ser informada inmediatamente a la administración del proyecto de DPR. Si usted deja de presentar este aviso, podría resultar en el rechazo o demora de su demanda.

Usted deberá informar a su médico que existe disponibilidad de trabajos alternativos. Una vez que su médico lo/la dé de alta para regresar al trabajo (tareas temporales, de tiempo parcial, alternativas o regulares), debe presentarse inmediatamente al trabajo, si es posible, o al próximo turno disponible, e informar a su supervisor/capataz acerca de cualquier restricción o condición física impuesta. Al siguiente día laboral, debe presentar una evaluación escrita acerca de su condición. **Usted no puede regresar al trabajo sin permiso de su médico.**

Si deja de presentarse al trabajo a la hora y en el lugar indicado, se considerará que ha renunciado voluntariamente y esto podría afectar su compensación por tiempo perdido y/o sus derechos al re-empleo o reinstauración. Si no puede presentarse para ningún tipo de trabajo, deberá llamar diariamente a su supervisor a más tardar 30 minutos después del comienzo del turno (salvo que se acuerde algo diferente), informándole acerca de su situación. Usted también tiene la responsabilidad de darle a su supervisor su número de teléfono actual (si figura o no en la guía telefónica) y una dirección donde se le pueda ubicar.

He leído y comprendo plenamente todos los procedimientos que anteceden, y conozco mis responsabilidades. Entiendo que mi falta de cumplimiento con mis responsabilidades según se indican arriba puede resultar en medidas disciplinarias que pueden incluir mi despido del trabajo y/o la pérdida a mi derecho al re-empleo o reinstauración posterior a una lesión. He recibido copia de esta información.

Firma del empleado

Fecha



Notice / Offer of Employment

Date: _____
Date of Injury: _____

Dear _____

Your physician has released you for modified work. We have identified a position for you with duties that Dr. _____ feels that you will be able to perform until you are able to return to your normal work duties.

Your position will be as follows:

Job Title: _____ Duration of Job: Up to 14 days – subject to review

Job Description (Based on the attached – physician approved – job analysis):

You will be receiving \$ _____ / per hour, the workers' compensation insurance company may supplement your wages with workers' compensation benefits.

We ask that you report for work on:

Date:	_____	Hours Per Day:	_____
Time:	_____	Days Per Week:	_____
Location:	_____	Duration of Job:	Is subject to periodic review
Phone Number:	_____		and cannot exceed 14 days
Report To:	_____		

If you received this letter on or after the day you are to report to work, please contact our office immediately and report to work within 24 hours from the date you receive this letter. Failure to report to work could affect you time loss compensation, vocational eligibility and/or re-employment / reinstatement rights.

We look forward to seeing you and wish you a speedy recovery.

Employer's Representative Signature

Date

☐
☐

I have reviewed and accept this alternative work position as offered.
I decline this modified work position.

Employee's Signature

Date



Notificación / Oferta de Empleo

Fecha: _____
Fecha del accidente/lesión: _____

Estimado/a _____

Su médico le ha autorizado para realizar algún trabajo modificado. Hemos identificado un puesto para usted con funciones que el Dr. _____ cree que podrá desarrollar hasta que esté en condiciones de volver a sus tareas habituales.

Su puesto será el siguiente:

Cargo: _____ Duración: Hasta 14 días – con sujeción a evaluación

Descripción de las funciones laborales (sobre la base del análisis de funciones laborales aprobado por el médico y que se adjunta):

Recibirá \$ _____ / por hora, la compañía del seguro para indemnizaciones por accidentes de trabajo podrá complementar su salario con los beneficios correspondientes a tal seguro.

Le solicitamos que se presente a trabajar el:

Fecha:	_____	Horas por día:	_____
Horario:	_____	Días por semana:	_____
Lugar:	_____	Duración del trabajo:	Será objeto de una evaluación periódica y no puede exceder 14 días
Teléfono:	_____		
Report To:	_____		

Si recibió esta carta el día en que se tiene que presentar a trabajar, o después de ese día, sírvase comunicarse con nuestra oficina de inmediato y preséntese a trabajar en un plazo de 24 horas de la fecha en que reciba esta carta. El hecho de no presentarse a trabajar podría afectar su compensación por horas de trabajo perdidas, elegibilidad vocacional y/o reconstrucción o restitución de derecho.

Esperamos verle pronto y le deseamos que se recupere lo antes posible.

Firma del empleador

Fecha

☐
☐

He leído la descripción de este puesto de trabajo modificado y acepto la oferta.
No acepto este puesto de trabajo modificado.

Firma del empleado

Fecha

General Liability Report

Property Damage, Theft or Bodily Injury *Incident Reporting Instructions*

1. Ensure the safety and security of the individual(s) that were injured or involved, if applicable.

Note: No one except emergency response personnel, owner or OSHA representatives or those associated with or retained by DPR should be allowed on the project site following an incident. Should other individuals require access, prior approval from DPR project management is required. Those individuals require DPR escort at all times.

2. Contact regional safety professional ***immediately***.
3. From a safe location - take photos of the accident scene, from all sides of the area and the surrounding area immediately.
4. Contact Heidi Marsac at (916) 826-9996 ***within 2 hours of any incident***.
5. Obtain witness names, addresses, phone numbers, and statements.
6. Email the completed form to Heidi Marsac at heidir@dpr.com, dennism@dpr.com & jennap@dpr.com.

Note: The General Liability checklist is available for additional guidance in preparing an incident report in anticipation of litigation.

REQUIRED DOCUMENTS
Completed General Liability report
Witness statements
Photographs of entire area
Copy of master contract
Copy of subcontracts, if subcontractors are involved
Insurance certificates & additional insured endorsements
Post incident drug screen results
Root Cause Analysis



General Liability Report

PROPERTY DAMAGE, THEFT OR BODILY INJURY

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

Reported by: _____	Phone: _____	Date: _____
Report Prepared for Attorney: _____		☐ Contacted Regional Safety Manager
☐ Contacted Heidi Marsac – Cell: 916.826.9996 heidir@dpr.com or Dennis Morey – cell: 650-995-6845 dennism@dpr.com		
☐ Post Incident Substance Abuse Screens for all parties, if applicable		

INCIDENT DESIGNATION

☐ Incident Only	☐ Property Damage	☐ Facility/Service Interruption	☐ Suspicious claim
☐ Valid Claim	☐ Unknown Claim (completely unaware of the incident)	☐ Other	
☐ Bodily Injury	☐ Owner Employee	☐ Member of General Public	☐ Vendor

#1 – PROJECT INFORMATION

Jobsite Name: _____	Region/Job Number: _____
Address/City/State/Zip: _____	
Project Start Date: _____	Project End Date: _____
DPR Foreman: _____	DPR Project Manager: _____
DPR Superintendent: _____	DPR Safety: _____

#2 –INCIDENT INFORMATION

Date of Incident: _____	Day of Week: _____	Time of Incident: _____
Date Reported: _____	Reported to: _____	Reported By: _____
Name/Party Involved: _____		
Involved Address: _____		Phone: _____
Damage/Theft/Injury: _____		
Was Employee Taken to a Medical Clinic: ☐ Yes ☐ No If so, When: _____		
Clinic Address/Phone Number: _____		

List names of persons involved in incident (list additional names on separate piece of paper if necessary):

Name: _____	Employer: _____	Phone: _____
Name: _____	Employer: _____	Phone: _____
Name: _____	Employer: _____	Phone: _____

#3 – ADDITIONAL INFORMATION

Exact Location of Incident: _____
Architect Name: _____
Architect Address/Phone: _____
Subcontractors & Trades Involved: _____
Name of person/company responsible for maintaining area: _____
Have any repairs been made: ☐ Yes ☐ No If yes, what repairs: _____
Estimated DPR exposure (costs to repair): _____



#4 – INCIDENT DESCRIPTION

NOTE: This does not take the place of any witness statements. Describe in detail the circumstances of the incident (attach diagrams, drawings and/or photos of accident scene). Give a chronological sequence of events. If materials and/or equipment were involved, start before the materials/equipment were brought to the incident scene describing who, what, where, when, how: (PRINT CLEARLY)

NOTE: ATTACH DIAGRAMS, DRAWINGS, PICTURES, ETC. TO CLEARLY DESCRIBE INCIDENT

#5 – SIGNATURES

DPR Superintendent:

Print Name

Signature

DPR Project Manager:

Print Name

Signature

DPR Safety:

Print Name

Signature



General Liability Checklist

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

PROJECT INFORMATION

Project/Project#:	_____	Date:	_____
Claim/Lawsuit:	_____	Incident Date:	_____

Contracts & Insurance (w/additional insured endorsement)	Date Provided	Due Date/By Whom
Owner – DPR Contract		
DPR – Subcontractor contract		
Subcontractor – Tiered Subcontractor contract		
Subcontractor naming Owner insurance certificate		
Tiered Subcontractors' naming DPR and/or Sub insurance certificate		
Scope of Work	Date Provided	Due Date/By Whom
DPR (if self-performed)		
Subcontractor		
Tiered Subcontractor(s)		
Subcontractor Documents	Date Provided	Due Date/By Whom
Subcontractor IIPP/Safety Plan		
Subcontractor Training Records		
Subcontractor Investigation		
Project Specific Data	Date Provided	Due Date/By Whom
Photographs of the area, if these are digital, please e-mail.		
DPR site specific EHSP		
Site Inspection for day of incident		
Site Inspections 30 days prior/30 days after		
Safety Meeting minutes		
Job schedule		
Blueprint drawings/schematic of incident area		
Pre-Construction (Safety) Meeting minutes		
Sub Coordination meeting minutes 30 days prior/30 days after		
Additional Documentation	Date Provided	Due Date/By Whom
Completed General Liability form		
Photographs – any pre incident which may exist		
DPR Grab & Go		
Witness Statements		
Medical Records/Release		
Pre-Task Plans – for incident, and 30 days prior/30 days after		
On-site safety orientation of injured party(s)		
On-site safety orientation of employees working in/responsible for the area		
Root Cause Analysis – worksheet and final report		
Corrective Actions identified with completion dates		
Lessons Learned shared with the project, region and global		
Any other information gathered during investigation/process		

General Liability Report

Construction Defect

Incident Reporting Instructions

1. Ensure the safety and security of the job site.

Note: No one except emergency response personnel, owner or OSHA representatives or those associated with or retained by DPR should be allowed on the project site following an incident. Should other individuals require access, prior approval from DPR project management is required. Those individuals require DPR escort at all times.

2. Take photos of the accident scene and the surrounding area immediately.
3. Contact your regional safety professional ***immediately***.
4. Contact Dennis Morey (650) 995-6845 ***within 2 hours of any incident***.
5. Complete the entire GRAB & GO packet thoroughly. **The entire packet should be emailed to Dennis Morey at dennism@dpr.com within 24 hours.**

REQUIRED DOCUMENTS
Completed General Liability report
Witness statements
Photographs of any/all property damaged
Copy of master contract
Copy of sub contracts, if sub-contractors involved
Insurance certificates & additional insured endorsements
Root Cause Analysis

Note: Root Cause Analysis must be completed and faxed to Dennis Morey no more than 48 hours after the incident.



General Liability Report

Construction Defect Incident Report

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

Reported by: _____	Phone: _____	Date: _____
Report Prepared for Attorney: _____		
☐ Contacted Regional Safety Manager		
☐ Contacted Dennis Morey Cell: 650.995.6845		
☐ Post Incident Substance Abuse Screens for all parties, if applicable		

#1 – PROJECT INFORMATION

Jobsite Name: _____	Project Start Date: _____
Address/City/State: _____	Project End Date: _____
Subcontractor: _____	DPR Project Manager: _____
Sub Contact: _____	DPR Superintendent: _____
Region/Job Number: _____	DPR Safety Coordinator: _____

#2 – INCIDENT FACTS

Date of Incident: _____	Day of Week: _____	Time of Incident: _____
Date Reported: _____	Reported to: _____	Time Reported: _____
Person Reporting Incident: _____		Job Title: _____
Weather Conditions: ☐ Raining ☐ Wet ☐ Dry ☐ Fog ☐ Snow ☐ Wind ☐ Other		

#3 – INCIDENT INFORMATION

List names of persons involved in incident (list additional names on separate piece of paper if necessary):		
Name: _____	Employer: _____	Phone: _____
Name: _____	Employer: _____	Phone: _____
Name: _____	Employer: _____	Phone: _____
Name: _____	Employer: _____	Phone: _____
Exact Location of Incident: _____		
General Activity at the time of Incident (i.e., concrete): _____		
Specific Task at Time of Incident (i.e., finishing): _____		

#4 – INCIDENT DESIGNATION

☐ Facility Interruption	☐ Property Damage	☐ Service Interruption	☐ Other: _____
<i>Note: If applicable, complete the General Liability Report</i>			
☐ Valid Claim	☐ Suspicious claim	☐ Unknown Claim (completely unaware of the incident)	
Reason for suspicions: _____			



General Liability Report

Construction Defect Incident Report

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

#5 – INCIDENT DESCRIPTION

NOTE: *This does not take the place of any witness statements.* Describe in detail the circumstances/facts of the incident (Attach: diagrams, drawings and/or photos of accident scene). Give a chronological sequence of events. If materials and/or equipment were involved, start before the materials/equipment were brought to the incident scene describing who, what, where, when, how: (Please Type)

NOTE: ATTACH DIAGRAMS, DRAWINGS, PICTURES, ETC. TO CLEARLY DESCRIBE INCIDENT

6 ADDITIONAL INFORMATION

Was the building occupied at the time of incident? ☐ Yes ☐ No ☐ N/A

Did occupants need to be relocated? ☐ Yes ☐ No ☐ N/A

Was there any business interruption? ☐ Yes ☐ No ☐ N/A

Was the customer/owner contacted & direction requested? ☐ Yes ☐ No ☐ N/A

Was there any damage to **NEW** work? ☐ Yes ☐ No ☐ N/A

If so, describe:

Was there any damage to **EXISTING** property? ☐ Yes ☐ No ☐ N/A

If so, describe:

What is the ROM to remedy all damage?

Was the customer/owner contacted & direction requested? ☐ Yes ☐ No ☐ N/A

Was the responsible subcontractor put on notice? ☐ Yes ☐ No ☐ N/A

Describe follow-up precautions to be implemented:

Was this a design/build project? ☐ Yes ☐ No ☐ N/A

Who is the Architect?

Were the Subs & Trades notified? ☐ Yes ☐ No ☐ N/A

Who were the Subs & Trades involved?

Separate job number used for remediation effort:

LMEO tracking mechanism:

#7 – SIGNATURES

DPR Superintendent:

Print Name

Signature

DPR Project Manager:

Print Name

Signature

DPR Safety:

Print Name

Signature

Motor Vehicle Accident Report

Incident Reporting Instructions

1. Ensure the safety and security of the individual(s) that were injured or involved.
2. Contact regional safety professional ***immediately***.
3. From a safe location - take photos of the accident scene, all sides of the vehicle(s) and the surrounding area immediately.
4. **Obtain other driver's information – including name, address, phone number, insurance company, policy number, vehicle plate number and VIN.**
5. Contact Heidi Marsac at (916) 826-9996 ***within 2 hours of any incident***.
6. Obtain witness names, addresses, phone numbers, and statements.
7. **Complete the entire GRAB & GO packet thoroughly. The entire packet should be emailed to Heidi Marsac at heidir@dpr.com, milverr@dpr.com & jennap@dpr.com or faxed to 916-848-0269 within 24 hours.**

REQUIRED DOCUMENTS
Completed Vehicle Accident Report
Witness statements
Copies of driver's license(s), insurance card(s), license plate(s)
Photographs of any/all damaged vehicles
Photographs of any/all property damage



MOTOR VEHICLE ACCIDENT REPORT

Incident Investigation Report

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

Date of Report: _____	DPR Region: _____
Reported by: _____	Phone Number: _____
☐ Contacted Regional Safety Manager ☐ Contacted Heidi Marsac 916.826.9996 & Milver Sweet 407.717.0747	

#1 - INSURED INFORMATION

Reported by: _____	Telephone Number: _____
Job Name: _____	Job Number: _____

#2 - VEHICLE A (DPR VEHICLE)

Driver's License Number: _____	DOB: _____	Telephone Number: _____
Address/City/State/Zip: _____		
Year/Make/Model: _____	Plate Number: _____	
VIN: _____	Damage: _____	

#3 - VEHICLE B (OTHER VEHICLE)

Other Driver/Owner: _____	Driver's License Number: _____	
Address/City/State/Zip: _____		
Telephone Number: _____	DOB: _____	
Year/Make/Model: _____	Plate Number: _____	
VIN: _____	Damage: _____	
Insurance Co.: _____	Policy #: _____	Phone #: _____

#4 - INJURY INFORMATION

Were You Injured: ☐ Yes ☐ No	Was Anyone Injured: ☐ Yes ☐ No	If yes, answer the following:
Name: _____		
Address/Phone: _____		
Medical Treatment Required: ☐ Yes ☐ No	Injuries: _____	
Name: _____		
Address/Phone: _____		
Medical Treatment Required: ☐ Yes ☐ No	Injuries: _____	

#5 - FACTS OF THE ACCIDENT

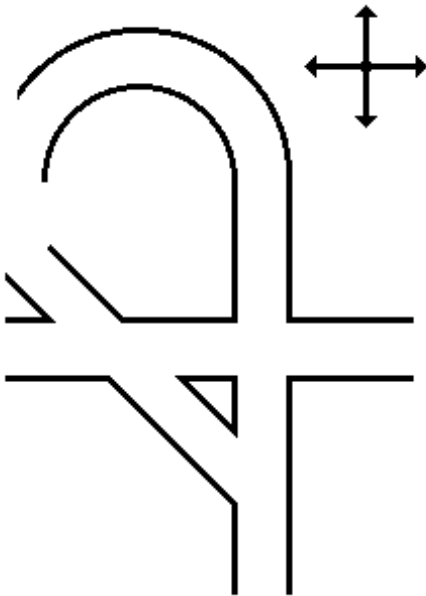
Date of Accident: _____	Time: _____	☐ a.m. ☐ p.m.	Day of Week: _____
Location: _____	City/State/Zip: _____		
Direction Car A (your car) was traveling: _____	What Street: _____	Speed: _____	
Direction Car B (other car) was traveling: _____	What Street: _____	Speed: _____	
Authorities Contacted: ☐ Yes ☐ No	What Station/Department: _____		
Driver Cited: ☐ A ☐ B	Reason: _____		
Indication of Intoxication: ☐ Yes ☐ No	Which driver: ☐ A ☐ B	Report #: _____	
Weather Conditions: ☐ Raining ☐ Wet ☐ Dry ☐ Fog ☐ Snow Other: _____			

#6 – WITNESS INFORMATION

Name: _____	Address: _____	Phone: _____
Name: _____	Address: _____	Phone: _____
Name: _____	Address: _____	Phone: _____
Name: _____	Address: _____	Phone: _____

#7 – DRIVER'S DESCRIPTION OF ACCIDENT

#8 – PLEASE ILLUSTRATE ON THIS DIAGRAM HOW ACCIDENT OCCURRED. PROVIDE STREET NAMES, ADDITIONAL VEHICLES, WITNESSES, OCCUPANTS AND/OR ADDITIONAL REMARKS:



SHOW: Car A (your car) as:

A	⇨
B	⇨

Car B (other car) as:

Reported by: _____ Date: _____

Signature: _____

Water Intrusion Report

Incident Reporting Instructions

1. Ensure the safety and security of the job site.

Note: No one except emergency response personnel, owner or OSHA representatives or those associated with or retained by DPR should be allowed on the project site following an incident. Should other individuals require access, prior approval from DPR project management is required. Those individuals require DPR escort at all times.

2. Take photos of the accident scene and the surrounding area immediately.
3. Contact Dennis Morey (650) 995-6845 **within 2 hours of any incident.**
4. Complete the entire GRAB & GO packet thoroughly. **The entire packet should be emailed to Dennis Morey at dennism@dpr.com within 24 hours.**

REQUIRED DOCUMENTS
Completed Water Intrusion report
Witness statements
Photographs of any/all property damaged
Copy of master contract
Copy of sub contracts, if sub-contractors involved
Insurance certificates & additional insured endorsements
Root Cause Analysis

Note: Root Cause Analysis must be completed and emailed to Dennis Morey no more than 48 hours after the incident.



WATER INTRUSION INCIDENT REPORT

Incident Investigation Report

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

Reported by: _____ Phone: _____ Date: _____

Report Prepared for Attorney: _____

- ☐ Contacted Dennis Morey 650.995.6845 dennism@dpr.com
- ☐ Contact Regional Safety Manager
- ☐ Post Incident Substance Abuse Screens for all parties, if applicable

#1 – PROJECT INFORMATION

Jobsite Name: _____	Region/Job Number: _____
Address: _____	Project End Date: _____
Subcontractor: _____	DPR Foreman: _____
Sub Foreman: _____	DPR Superintendent: _____
Sub Superintendent: _____	DPR Safety: _____

#2 – INCIDENT INFORMATION

List names of persons involved in incident (list additional names on separate piece of paper if necessary):

Name: _____	Employer: _____	Phone: _____
Name: _____	Employer: _____	Phone: _____
Name: _____	Employer: _____	Phone: _____
Name: _____	Employer: _____	Phone: _____

Exact Location of Incident: _____

General Activity at the time of Incident (i.e., concrete): _____

Specific Task at Time of Incident (i.e., finishing): _____

#3 – ADDITIONAL INFORMATION

Date of Incident: _____ Day of Week: _____ Time of Incident: _____

Date Reported: _____ Reported to: _____ Time Reported: _____

Person Reporting Incident: _____ Job Title: _____

Was Pre-task Planning Required for the work performed at the time of the incident: ☐ Yes ☐ No If PTP required, attach.

Weather Conditions: ☐ Raining ☐ Wet ☐ Dry ☐ Fog ☐ Snow ☐ Wind ☐ Other

#4 – INCIDENT DESIGNATION

☐ Facility Interruption ☐ Property Damage ☐ Service Interruption ☐ Other: _____

Note: If applicable, complete the General Liability Report

☐ Valid Claim ☐ Suspicious claim ☐ Unknown Claim (completely unaware of the incident)

Reason for suspicions: _____



WATER INTRUSION INCIDENT REPORT

Incident Investigation Report

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

#5 – INCIDENT DESCRIPTION

NOTE: This does not take the place of any witness statements. Describe in detail the circumstances of the incident (attach diagrams, drawings and/or photos of accident scene). Give a chronological sequence of events. If materials and/or equipment were involved, start before the materials/equipment were brought to the incident scene describing who, what, where, when, how:

(PRINT CLEARLY)

NOTE: ATTACH DIAGRAMS, DRAWINGS, PICTURES, ETC. TO CLEARLY DESCRIBE INCIDENT

6 ADDITIONAL INFORMATION

Was the building occupied at the time of incident? ☐ Yes ☐ No ☐ N/A

Did occupants need to be relocated? ☐ Yes ☐ No ☐ N/A

Were any of the following observed or identified?

☐ Musty Odors ☐ Visible Mold ☐ Sewage or Grey Water
☐ Water Damaged Building Materials ☐ Impact to the HVAC System ☐ Water Stained Building Material

Was there any damage to **NEW** work? ☐ Yes ☐ No ☐ N/A

If so, describe: _____

Was there any damage to **EXISTING** property? ☐ Yes ☐ No ☐ N/A

If so, describe: _____

What is the ROM to remedy all damage? _____

Was the customer/owner contacted & direction requested? ☐ Yes ☐ No ☐ N/A

Was the responsible subcontractor put on notice? ☐ Yes ☐ No ☐ N/A

Describe follow-up precautions to be implemented: _____

Were the services of an industrial hygienist retained? ☐ Yes ☐ No ☐ N/A

If yes, with whom: _____

Scope of their services: _____

Was clearance sampling performed: ☐ Yes ☐ No ☐ N/A (Attach laboratory results, if applicable)

If not, explain why: _____

What remediation methods were instituted: _____

Separate job number used for clean-up: _____

LMEO tracking mechanism: _____

#7 – SIGNATURES

DPR Superintendent:

Print Name

Signature

DPR Project Manager:

Print Name

Signature

DPR Safety:

Print Name

Signature

Near Miss Incident Report

Incident Reporting Instructions

1. Ensure the safety and security of the job site.
2. Substance abuse screening should be completed for each individual involved and anyone who could have contributed to the near miss incident.
3. Contact your regional Safety Manager **immediately**.
4. Complete the Near Miss Report packet thoroughly. Fax the completed documents to Heidi Marsac at 916.848.0269 heidir@dpr.com by end of business the day of the incident.

REQUIRED DOCUMENTS
Near Miss Incident Report
Pre-task Plan
Corrective Actions
Verification of Substance Abuse Screen(s)
Complete Root Cause Analysis portion, if applicable



Near Miss Incident Report

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

Project Information:

Date:	_____	Completed by:	_____
Project:	_____	Region:	_____
Project Type:	_____	Job Number:	_____

Incident Information:

Date of incident:	_____	Time of incident:	_____
Trade:	_____	Project phase:	_____
Incident key word(s):	_____		
General activity at time of incident:	_____		
Incident Severity:	_____		
Company:	_____		
Involved worker:	_____		
Drug screen complete?	_____		

Near Miss Information:

Facilitator Name:	_____	Date:	_____
Superintendent	_____	Safety Professional	_____
Foreman	_____	Involved Worker	_____
Project Manager	_____	Sub Foreman	_____
Safety Coordinator	_____	Sub Superintendent	_____
Other	_____	Other	_____
Other	_____	Other	_____
Other	_____	Other	_____

Description of Incident:

Severity Potential:

Complete worksheet below to determine potential impact of incident.

		Score – 1	Score – 2	Score – 3	Score – 4	Score – 5
A.	Injury Potential	None	No medical assistance	First aid	OSHA recordable	Lost time injury
B.	Potential for area of incident to be occupied	Unlikely	Occasional	Moderate	Likely	Definite
C.	Radius that injury can be expected	1-9 feet	10-25 feet	25-49 feet	50-99 feet	100+ feet
D.	Number of people in area of radius	None	1	2 to 3	4 to 8	9 or more
E.	Chance of escape	100% (Certain)	75% (Likely)	50% (Possible)	25% (Unlikely)	0% (None)

Incident potential = A + B + C + D + E = Total Score =

Near Miss Incident Rating Scale

1 to 7	MINIMAL	Incident is not a near miss but had no potential for injury – depending on incident, identify contributing factors and implement corrective action
8 to 12	MODERATE	Complete attached witness statements, review incident and identify contributing factors and record corrective actions in tables below
13 to 25	SIGNIFICANT	Investigate incident, complete entire Root Cause Analysis, complete witness statements and record corrective actions

Events/Timeline: *(complete this section in detail for Root Cause Analysis, otherwise, skip to Contributing Factors)*

1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	

Contributing Factors:

1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Root Cause:

Corrective Actions:

		Responsibility	Due Date
1			
2			
3			
4			
5			
6			
7			
8			

Comments:



ROOT CAUSE ANALYSIS

(This report and related documentation is confidential and was prepared in anticipation of future litigation)

Project Information:

Date:		Completed by:	
Project:		Region:	
Project Type:		Job Number:	

Incident Information:

Date of incident:		Time of incident:	
Trade:		Project phase:	
Incident key word(s):			
General activity at time of incident:			
Company:			
Is this company working under a CAP? If so, provide copy of CAP and review during corrective actions discussion. Attach to RCA.			

Root Cause Analysis Participants:

Facilitator Name:		Date:	
Superintendent		Safety Professional	
Foreman		Involved Worker	
Project Manager		Sub Foreman	
Safety Coordinator		Sub Superintendent	
Other		Other	
Other		Other	
Other		Other	

Description of Incident:

Events/Timeline:	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	
23	
24	
25	
26	
27	
28	
29	
30	
31	
32	
33	

Contributing Factors:

1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Root Cause:

Corrective Actions:

		Responsibility	Due Date
1			
2			
3			
4			
5			
6			
7			
8			

Comments:

INCIDENT SUMMARY

Project Information:

Date:		Completed by:	
Project:		Region:	
Company:		Job Number:	
Time of incident:		Date of incident:	
Injured worker:		Job title	
Alleged injuries/incident:			

Property damage/Incident:

--

Witnesses:

Incident site organizational structure:

--

Post incident findings:

--

Scope and limitations:

This incident fact sheet is based on observations and information available at the time of its preparation and is not intended nor should it be construed as providing all available evidence or cause of the incident.

Section 14

Infection & Contamination Control

Sub-Section	Topic	Page #
1.0	Introduction	2
2.0	Pre-Task Planning	2
3.0	Quality Control	2
4.0	General Construction Procedures	2
5.0	Mold Prevention	3
6.0	Healthcare (ICRA)	4
7.0	CleanBuild Technologies	7

Infection & Contamination Control

1.0 Introduction

It is DPR's intent to integrate, and where necessary enhance our EHS requirements with that of our owners to protect DPR and subcontractor workers, facilities employees and the general public from contaminants that could create health and environmental issues. Our core markets; the i.e., hospital populations, medical facilities, bio-technology and bio-pharmaceutical facilities, research laboratories, etc. require us to operate at a higher level of collaboration with facilities and our subcontractor management teams.

It is our intent to minimize the potential for exposure through the generation of construction debris, dust, biohazards, or anything that may cause adverse effects on people and product through various controls and collaboration. This plan will outline some of the most basic steps and guidelines for planning work in sensitive environments. Requirements specific to the owner or process should always be reviewed and integrated with these basic steps and guidelines.

Construction activity types are defined by the amount of dust that is generated, the duration of the activity, and the amount of shared HVAC systems. Prior to work commencement in an area, operations should be assessed to determine the type of construction activities that will occur. At a minimum, the entire project team (i.e., owner's representative(s), DPR, subcontractors, suppliers, etc. should plan and review any activity where the potential for contamination exists.

2.0 Pre-Task Planning

Work activities involving sensitive facilities and infection and contamination control are defined by DPR to be High Hazard and therefore require daily detailed, written pre-task planning. *Refer to the Pre-Planning Section 17 for additional pre-task planning guidelines.*

3.0 Quality Control

DPR/Owner's ICRA representative reserves the right to monitor air quality in the vicinity of construction work on an as needed basis. Whenever safe levels are exceeded, corrective measures should be implemented immediately, up to and including work stoppage. These corrective measures should be documented.

4.0 General Construction Procedures

- Maintain manpower and equipment including dust mops, wet mops, broom, and buckets for cleaning dust from surfaces adjacent to occupied areas
- Tacky mats should be maintained and placed appropriately, including at entrances/exits to worksite
- Contain work areas outside the construction barrier, including spaces above the ceilings, with full height visqueen sheet barrier, at least 6 mm thick, tightly taped and sealed
- Verify that all negative pressure rooms still have negative pressure and positive pressure rooms have positive pressure using a manometer, magnehelic or similar measuring device; general limits for pressure differential should read 0.01 inch w.g. (2.5 Pascals), at minimum
- DPR should coordinate with the hospital to ensure HEPA equipped air filtration machines run continuously for the duration of the project; at all times backup power support supply should be designated
- Only circuits and/or outlets designated for construction use should be utilized; *all workers should be trained to recognize designated circuits and/or outlets*
- Negative pressure filtration equipment should be vented/exhausted to the exterior of the building whenever possible
- Sufficient waste receptacles should be placed within the project area; cleaning and disposal should be ongoing with a final clean at the end of the day
- Areas of construction should be evaluated for necessary pest control, prior to any demolition activities

4.1 Barriers

Close unused doors with low tack adhesive tape applied over the frame to prevent particulate migration. Any taped off doors must be posted with a warning sign. Construction or demolition that cannot be contained in a single room should have the following barriers erected:

- Plastic or like material barriers that extend from the floor to ceiling and are terminated and sealed
- Seams should be sealed with low tack adhesive tape or equivalent to prevent dust and debris from escaping; ensure emergency or egress paths remain passable
- Barriers erected with joints should be covered or sealed to prevent dust or debris from escaping
- Penetrations in any existing barriers should be sealed
- Barriers at penetration of ceiling envelopes, chases, and ceiling spaces should be sealed to sufficiently stop movement of air and debris
- Barriers must be installed at elevator shafts and/or stairways within the area of operation
- An overlapping flap with a minimum of a 2' x 2' visqueen enclosure for personnel access and egress, including the removal of coveralls and shoe covers

4.2 Outdoor Construction

- Exhaust from equipment should be directed away from building air intakes
- Dirt excavation or building demolition should not occur unless building air intake filters are properly installed and operational
- Maximum HVAC re-circulation of building air during external construction should minimize construction dust penetration
- Seal windows adjacent to outside construction activities with low tack adhesive tape
- Cracks around windows and or frames should be caulked or weather stripped to prevent infiltration
- External construction debris should not be transported through hospital buildings at any time

5.0 Mold Prevention

Below are basic guidelines to be incorporated throughout construction phases to reduce the risk of contamination due to mold. Ensure every effort should be made to collaborate with owners to ensure any site specific requirements are also met.

- Ensure the building is protected and dried in during inclement weather seasons as soon as practicable
- When the building is not yet dried in, building materials should be protected from exposure to the elements
- Repair sources of moisture problems as soon as possible
- Repair leaky plumbing and leaks within and around the building as soon as possible
- Watch for condensation and wet spots
- Clean and dry wet or damp spots within 24 hours
- Prevent moisture due to condensation by increasing surface temperature or reducing the indoor humidity
 - to increase surface temperature, insulate or increase air circulation
 - to reduce the moisture level in air, repair leaks, increase ventilation (if outside air is cold and dry), or dehumidify (if outdoor air is warm and humid)
- Keep HVAC drip pans clean, flowing properly, and unobstructed
- Vent moisture-generating appliances, such as dryers, to the outside where possible
- Maintain low indoor humidity, below 60 percent relative humidity, ideally 30-50, if possible
- Perform regular building/HVAC inspections and maintenance as scheduled
- Do not let foundations stay wet; provide drainage and slope the ground away from the foundation
- Protect building materials from moisture
- Test, treat building materials that have become mold infested
- Discard building materials that cannot be adequately decontaminated

- Install drywall 5/8" off floor; maintain drywall mud/tape 5/8" off floor
- Clean drywall tract prior to closing up walls

6.0 Healthcare (ICRA)

The following sub-section addresses infection control measures applicable to health care environments. Infection control permitting may be required by the owner. All permitting requirements take precedence over this document where conflicts occur.

6.1 Construction Types

Construction activity types are defined by the amount of dust that is generated, the duration of the activity, and the amount of shared HVAC systems. Prior to starting work in an area the operation should be assessed to determine the type of construction activity that will be occurring. When in doubt always choose the category based on the type of work that has the greatest potential impact, even if the type of work is of a short duration or minimal scope.

Type "A" Inspection and non-invasive activities. For example, the removal of ceiling tile for inspection when limited to less than 1 tile per 50 square feet, painting (but no sanding) of wall covering, minor plumbing, electrical trim work, and activities which do not generate dust or require cutting walls or access ceilings other than for visual inspection.

Type "B" Small scale, short duration activities that create minimal dust. For example, installation of data and/or telecom lines, access to chase spaces, cutting of walls or ceilings where dust migration can be controlled.

Type "C" Any work that generates a moderate to high amount of dust or requires demolition or removal of any fixed building components or assemblies. For example, sanding walls for painting or wall covering, removal of floor coverings, ceiling tiles and casework, new wall construction, minor ductwork or electrical work above ceilings, major cabling activities, and any activity that cannot be completed during one shift.

Type "D" Major demolition and construction projects. Including, but not limited to, work requiring multiple consecutive work shifts, require heavy demolition and/or removal of the complete ceiling, floor or wall system, new construction and any activity creating a large breach in the existing containment system for a space.

6.2 Infection Control Risk Groups

After determining the type of construction activity that is to occur, the next step in the risk assessment is to determine the areas that will be impacted by the construction activity. When in doubt always choose the risk group based on the greatest potential impact, even if the exposure is of a short duration or minimal scope. Communicate with the owner to ensure that all areas of risk are appropriately categorized. The following general guidelines can be used to help you determine the risk groups:

Group 1 Lowest Risk	Group 2 Moderate Risk	Group 3 Moderate to High Risk	Group 4 High Risk
Office Areas Engineering Hospital Admitting Environmental Services Mail Room Warehousing & Storage Areas Shipping & Receiving Public Areas – lobbies, waiting rooms, etc. Equipment & Electrical Rooms	Patient Care Units Cardiac Rehab Respiratory Therapy Linen Rooms Outpatient Waiting Areas Physical Therapy Echocardiography Exercise Labs Non-invasive Lab Mechanical Rooms	Medical & Cardiac Telemetry Emergency Room Emergency Room Trauma Centers Post-Partum Newborn Nurseries Labor & Delivery Pediatrics Medical & Surgical Units Laboratories (specimen/research) Physical Therapy (tank areas) Radiology/MRI Nuclear Medicine Infusion Center Cafeteria/Kitchen Packaging Areas Fermentation Areas	Transplant Unit Operating Rooms, CVOR & Outpatient Surgery Sterile Processing ICU Burn Units Labor & Delivery Operating Rooms Cardiac Catheterization & Angiography Areas Pulmonary Function Dialysis Area Oncology Unit Radiation Oncology Cardiac Unit Pharmacy Admixture All Endoscopy Areas Anesthesia & Pump Area Post Anesthesia Care Pre-Induction Areas Negative Pressure Isolation Areas Production Areas Vialing Areas Weigh & Dispense Areas

6.3 Construction Activity and Infection Control Matrix

To determine the level of control measure needed, match the infection control risk group with the construction type in the matrix below and implement the appropriate controls.

Risk Level	Construction Type			
	Type A	Type B	Type C	Type D
Group 1	I	II	II	III/IV
Group 2	I	II	III	IV
Group 3	I	III	III/IV	IV
Group 4	II	III/IV	III/IV	IV

6.4 Control Measures

6.4.1 Class I

- Execute work by methods to minimize raising dust from construction operations
- Immediately replace any ceiling tiles displaced for visual inspection
- Cleanup and dispose debris immediately

6.4.2 *Class II*

- Provide active means to prevent air-born dust from dispersing into atmosphere (e.g. negative air machines, HEPA vacuums, dust collectors on tools, etc.)
- Water mist work surfaces to control dust while cutting
- Seal unused doors with low tack adhesive tape, visqueen or other similar product
- Block off and seal air vents
- Wipe surfaces with disinfectant
- Contain construction waste before transport in tightly covered containers
- Wet mop and/or vacuum with HEPA filtered vacuum before leaving work area
- Place tacky mats at entrances and exits of work area
- Remove or isolate HVAC system in areas where work is being performed
- Cleanup and dispose debris immediately

6.4.3 *Class III*

- Isolate HVAC system in area where work is being done to prevent contamination of the duct system
- Complete all critical barriers or implement control cube method before construction begins
- Maintain negative air pressure within the work site utilizing HEPA equipped air filtration units
- Air should be vented directly outside or to appropriate scrubber
- Do not remove barriers until the entire work area has been thoroughly cleaned and disinfected
- Vacuum work area with HEPA filtered vacuums
- Wet mop with disinfectant
- Remove barrier materials carefully to minimize spreading of dirt and debris associated with construction
- Contain construction waste before transport in tightly covered containers
- Cover transport receptacles or carts. Tapes covering if lid or covers are not tight
- Cleanup and dispose debris immediately

6.4.4 *Class IV*

- Isolate HVAC system in area where work is being done to prevent contamination of the duct system
- Complete all critical barriers or implement control cube method before construction begins
- Maintain negative air pressure within the work site utilizing HEPA equipped air filtration units
- Negative air should be vented directly outside or to appropriate scrubber
- Seal holes, pipes, conduits, and punctures appropriately
- Construct anteroom and require all personnel to pass through this room so they can be vacuumed using a HEPA vacuum cleaner before leaving work site or they can wear cloth or paper coveralls that are removed each time they leave the work site
- All personnel entering the work site are required to wear shoe covers
- Do not remove barriers until the entire work area has been thoroughly cleaned and disinfected
- Vacuum work with HEPA filtered vacuums
- Wet mop with disinfectant
- Remove barrier materials carefully to minimize spreading of dirt and debris associated with construction
- Contain construction waste before transport in tightly covered containers
- Cover transport receptacles or carts. Tape cover if lid or covers are not tight
- Cleanup and dispose debris immediately

7.0 CleanBuild Technologies

7.1 General guidelines

Every effort should be made to prevent contamination of construction materials by contact with dirt, debris, rodents, insects and other harbingers of disease or illness. Collaboration between DPR, owner and subcontractors is vital to ensure owner requirements are understood, implemented and maintained throughout all phases of construction. To that end, the following guidelines should be considered, as well as any owner requirement over and above those listed below:

- Permission from DPR project management should be obtained prior to mobilizing storage containers or construction of any structures
- Yard and work areas should be kept free of debris, rubbish, and wood scraps
- Containers, gang boxes, and equipment brought on site should be in good condition and have structural integrity to discourage harborage of rodents and insects
- Tools and materials should be stored on pallets or dunnage at least 4" from the ground to discourage harborage of rodents and insects
- Open ends of pipe, conduits, etc. should be capped or securely covered to discourage harborage of rodents and insects
- Piles of dirt, sand, sod, etc. should be delivered to the job and distributed as soon as possible to discourage harborage of rodents and insects
- Construction trailers should have skirting that closes off the bottom of the structure
- Daily cleanup of all work and storage areas
- Rodent traps throughout the project should not be disturbed. Notify DPR project management prior disturbing rodent traps
- No animals or domestic pets are permitted onsite at any time
- Do not feed wild animals
- No food, drink, chewing gum, or tobacco products use is allowed on site, except in designated areas

7.2 Soil contamination

- All vehicles and equipment should be inspected before entering the job site; those with leaking fluids or in poor condition should not be allowed on site
- Plastic diaper type containment should be utilized when feasible or another form of plastic leak-proof containment should be placed under any equipment
- Secondary containment should be provided for all stored liquids greater than one gallon and should be capable of handling 110% of the liquid capacity of the container
- Spill kits should be made available and accessible where the potential for spilled liquids exist, and all spills should be immediately cleaned up and reported to DPR project management
- Vehicles and equipment that are in need of maintenance or repair should be towed off site for repair
- Concrete washout containment areas or washout bins should be utilized; vehicles and equipment should not be washed on site
- Storage areas should be sloped and graded to prevent fluids from reaching storm drains; rain water is the only fluid allowed to enter storm drains
- In the event a spill occurs, coordinate cleanup and disposal with local waste disposal facilities to ensure compliance with federal, state and local requirements

Infection & Contamination Control Forms

Forms

ICRA Pre-Planning Checklist

ICRA Inspection – English

ICRA Inspection – Spanish

ICRA Pre-Planning Checklist

Archive Document – DO NOT DISCARD

General Information			
Name		Date	
Job Site		Project #	

Risk Assessment				
1.	Construction Activity Type	☐	Type A	Inspection / non-invasive activities
		☐	Type B	Small scale / short duration w/minimal dust
		☐	Type C	Moderate to high dust generation, i.e., demo, sanding
		☐	Type D	Major demo, new construction, multiple / consecutive shifts
2.	Risk Group			
		☐	Group 1	Low Risk
		☐	Group 2	Moderate Risk
		☐	Group 3	Moderate to High Risk
		☐	Group 4	High Risk
3.	Refer to Construction Activity Infection Control Matrix, define classification of control measures to be utilized, follow guidelines outlined under section 3.0 Control Measures			
		☐	Class 1	
		☐	Class 2	
		☐	Class 3	
		☐	Class 4	

Note: Any items answered “no” below should be immediately addressed, with corrective measures tracked to completion.

Pre-Construction		Yes	No
1.	Risk assessment completed?	☐	☐
2.	Methods for dust containment are established?	☐	☐
3.	Means for providing negative pressure in construction zone determined?	☐	☐
4.	Locations of fresh air intakes and filters are identified, and existing air ducts have been disconnected, as necessary, i.e., fully understand the HVAC system in the area.	☐	☐
5.	Traffic patterns for construction workers are planned and avoid sensitive areas? Attempt to establish one path of travel for entry, exit and materials.	☐	☐
6.	Route for debris removal is planned and avoids sensitive areas?	☐	☐
7.	Essential services with potential for disruption are identified, with potential impacts fully understood?	☐	☐
8.	Ventilation, distribution and potable water systems are identified?	☐	☐
9.	Faucet aerators and other stagnating features (i.e., plumbing dead ends) have been removed, where possible?	☐	☐
10.	High risk patients and operations have been relocated as practicable?	☐	☐
11.	Site orientation reflects project specific infection control measures?	☐	☐
12.	Air ducts and ceiling voids have been vacuumed prior to start of work?	☐	☐
13.	Baseline air quality assessments have been established?	☐	☐
14.	Document any and all pre-existing conditions where evidence of prior moisture damage and/or mold is present.	☐	☐

ICRA Pre-Planning Checklist

Archive Document – DO NOT DISCARD

During Construction		Yes	No
1.	Construction signs posted for the area?	☐	☐
	Doors and windows properly closed and sealed?	☐	☐
	Critical barrier intact and true sealing?	☐	☐
	All surfaces are clean, with no visible dust?	☐	☐
	Holes, pipes, conduits or other penetrations are sealed?	☐	☐
Air Handling		Yes	No
	All windows are closed behind barriers?	☐	☐
	Negative air pressure is maintained at barrier entrance?	☐	☐
	Negative air machine is running at all times?	☐	☐
	Alarms are in place and functioning properly for immediate notification of pressure loss?	☐	☐
	Air is vented directly to the outside or to an appropriate scrubber?	☐	☐
	Periodic monitoring of air quality is being performed, if determined to be necessary during pre-construction assessment?	☐	☐
Project Area		Yes	No
	Debris is removed immediately and continuously?	☐	☐
	Trash is disposed of in appropriate, covered containers?	☐	☐
	Areas are vacuumed with HEPA filters at least daily?	☐	☐
	Areas are wet mopped at least daily with hospital-grade disinfectant or similar?	☐	☐
	Tacky mats are placed at all entrance(s) and exit(s) and replaced at least daily?	☐	☐
Traffic Control		Yes	No
	Traffic is restricted to construction workers and essential staff only?	☐	☐
	All doorways and exits are free of debris?	☐	☐
	Route for debris removal avoids direct care areas?	☐	☐
Protective Clothing		Yes	No
	Protective wear has been evaluated and defined for workers in the project area?	☐	☐
	Protective clothing is removed prior to leaving construction area?	☐	☐
	Workers remove protective clothing and vacuum themselves with HEPA filtered vacuum if exiting construction area via patient areas?	☐	☐
Post Construction		Yes	No
	Entire construction area has been thoroughly vacuumed and disinfected prior to removal of dust barriers?	☐	☐
	Barrier materials have been wet wiped, HEPA vacuumed or water misted prior to removal?	☐	☐
	Construction area has been thoroughly vacuumed and disinfected again after removal of dust barriers?	☐	☐
	All blockages from air systems have been removed?	☐	☐
	Ventilation system has been re-balanced to design specifications?	☐	☐
	Construction area has been inspected by owner representative and approved for use and occupancy?	☐	☐
Post Construction		Responsible Party	Due Date



ICRA Inspection

Archive Document – DO NOT DISCARD

Project Information		General Information				Date									
Project Name		Construction Activity Type		A	B	C	D	Week Starting							
Contractor/Subcontractor		Risk Group		1	2	3	4	Week Ending							
Inspector's Name		Control Measure Level		I	II	III	IV								
Partial List of Safety Items – Identify & Correct All Unsafe Conditions															
A	Inspect twice daily, mark a "+" for items being performed well or in good condition.								MON	TUES	WED	THURS	FRI	SAT	SUN
B	For items that need improvement or corrective action, mark an "A" and comment on the back any action needed/taken.														
1	Construction Barricade														
	Barricades sealed, no penetrations														
	Tacky mats clean and adequate to contain construction dust														
	Doors closed and sealed properly														
	Signs posted cautioning about spread of dust														
	Immediate and adjacent ceiling tiles intact														
	Ceiling tiles replaced when space above ceiling not being accessed														
	Adjacent floor areas clean, no dust tracked														
2	Air Quality Management														
	All windows and doors closed behind barricade where appropriate														
	Return air vents taped/blocked														
	Exhaust fans functioning														
	No excess dust, fumes, vapor (e.g., project area dampened or vacuumed frequently to reduce dust														
	Personnel wearing cloth/paper coveralls and shoe covers, which are removed when exiting construction area, when appropriate														
	Negative pressure at entrance to barricade and results recorded														
	Negative air machine running														
3	Water Quality Management														
	Leaks contained/sealed and moisture build up stopped														
	Project area clean, debris removed daily														
	Debris removed in covered containers														
	Established traffic pattern identified and used														
	Traffic routes cleaned with wet mop or HEPA filtered vacuum routinely														
	Other														
5	Occupied Areas														
	Barricades(s) in place and properly sealed														
	Owner equipment and items removed from construction area														
	Door adjacent to sensitive areas are closed and sealed														
	Surrounding area clean														
	Other														



Specifics & Comments of Infection Control Inspection

Monday	
Tuesday	
Wednesday	
Thursday	
Friday	
Saturday	
Sunday	

Datos específicos y comentarios sobre la inspección del control de infecciones

Información del proyecto		Información general				Fecha	
Nombre del proyecto		Actividad de construcción tipo	A	B	C	D	Fecha comienzo de la semana
Contratista/subcontratista		Grupo de riesgo	1	2	3	4	Fecha fin de la semana
Nombre del inspector		Nivel de medida de control	I	II	III	IV	Localidad(es)

Lista parcial de artículos de seguridad–Identificar y corregir toda situación peligrosa													
			LUN	MAR	MEIRC	JUEV	VIERN	SAB	DOMI				
A	Inspeccionar dos veces al día, marque una “+” en todos los artículos que se hacen bien o que están en buenas condiciones.												
B	Marque una “Δ” en todos los artículos que necesitan mejorarse o corregirse, y comente al dorso sobre la acción tomada o necesaria.												
1	Barricadas de construcción												
	Las barricadas están cerradas, sin penetraciones												
	Las esterillas Tacky Mats están limpias y pueden adecuadamente captar el polvo de la construcción												
	Las puertas están bien cerradas y selladas												
	Se han colocado carteles advirtiendo sobre el esparcimiento de polvo												
	Los paneles del cielo raso inmediato y adyacente están intactos												
	Se volvieron a colocar los paneles del cielo raso cuando ya no hubo necesidad de entrar al espacio sobre el cielo raso												
	Los pisos de las zonas adyacentes están limpios y sin polvo												
2	Manejo de la calidad del aire												
	Cuando es apropiado, todas las puertas y ventanas están cerradas detrás de la barricada												
	Los conductos de aire de retomo han sido bloqueados o sellados con cinta adhesiva												
	Los extractores de aire están funcionando												
	No existe exceso de polvo, emanaciones, vapores (es decir, la zona del proyecto ha sido humedecida o aspirada frecuentemente para disminuir el polvo)												
	El personal usa overoles de tela o papel y forros para el calzado, que se quitan al salir de la zona de construcción, según sea apropiado												
	Instalación de presión negativa a la entrada de la barricada, y medición de resultados												
	Máquina de presión negativa de aire, en funcionamiento												
3	Manejo de la calidad de agua												
	Filtraciones contenidas o selladas y eliminación de acumulación de humedad												
4	Sitio de la obra												
	La zona del proyecto se encuentra limpia, con eliminación diaria de escombros												
	Escombros eliminados en contenedores con tapa												
	Se ha establecido y se usa una ruta de tránsito												
	Las rutas de tránsito se limpian de rutina con trapeador mojado o aspiradora de filtro HEPA												
	Otros												
5	Zonas ocupadas												
	La(s) barricada(s) se encuentra(n) colocada(s) y sellada(s) adecuadamente												
	El equipo o artículos del dueño han sido retirados de la zona de construcción												
	Las puertas adyacentes a zonas precarias están cerradas y selladas												
	La zona circundante está limpia												



Datos específicos y comentarios sobre la inspección del control de infecciones

Lunes	
Martes	
Miércoles	
Jueves	
Viernes	
Sábado	
Domingo	

Section 15

Motorized & Site Equipment

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Motorized Material Handling Devices & Motorized Equipment

1.0 Introduction

This section covers safe operation and maintenance of all vehicles that operate on off-highway workplaces as well as vehicles that operate on highways and roadways regulated by the Interstate Commerce Commission or US Department of Transportation.

Examples of off-highway vehicles include (but are not limited to): Concrete Pump Trucks, Drill Rigs, Earthmoving Equipment, Helicopters, Forklifts, and other Material Handling Devices.

Examples of highway vehicles covered include (but are not limited to): Company-owned-or-leased passenger vehicles, pickup trucks, light trucks and vans that do not require a commercial driver's license for operation.

Reference: CFR 1926 Subpart O (Motor Vehicles, Mechanized Equipment); CFR 1926 Subpart N (Helicopters, Hoists, Elevators, and Conveyors). Other applicable Standards, guidelines, and regulations may apply.

1.1 Pre-Task Planning

All motorized equipment activities, as well as material handling, are activities defined by DPR to be potentially high hazard and therefore require detailed, written pre-task planning. Refer to the Pre-Task Planning Section for additional guidelines and PTP form.

2.0 Earthmoving Equipment

Earthmoving equipment (i.e., scrapers, loaders, crawler or wheel tractors, bulldozers, off-highway trucks, graders, industrial tractors and similar equipment) should be operated in accordance with manufacturer's recommendations and limitations, and operators should be trained in the safe operation of the equipment they will be utilizing. Earthmoving equipment should be inspected each day prior to use to determine that they are in safe working condition. Operator should walk area and/or review as-builts and site plan to ensure understanding of locations of utilities, excavations, footings, obstructions, etc. *An operator training Verification and Daily inspection form in the Forms Section at the end of the EHSP.* These forms are provided as a tool for use on the project, and may become mandatory at the discretion of the DPR project team.

2.1 Seat Belts and Rollover Protective Structures (ROPS)

ROPS and seat belts should be installed and used on all equipment specified in this sub-section. Seat belts should be adequate for the intended service and in good repair. Seat belts need not be provided for equipment which is designed only for standup operation. ROPS should provide operator protection against the hazard of falling objects.

2.2 Brakes

All earthmoving equipment mentioned in this sub-section should have a service braking system capable of stopping and holding the equipment that is fully loaded.

2.3 Fenders

Pneumatic-tired earth-moving haulage equipment (trucks, scrapers, tractors, and trailing units) whose maximum speed exceeds 15 miles per hour should be equipped with fenders on all wheels.

3.0 Haulage Equipment & Vehicles

It is DPR's intent that all haulage equipment & vehicles (i.e., dump trucks, transfer trucks, water trucks, etc.) are inspected at the beginning of each shift and maintained per manufacturer's requirements. DPR project management reserves the right to verify documentation and maintenance has been performed. Should haulage equipment be found out of compliance, the respective owner/operator should remove equipment

from site immediately. Reference the Daily Inspection form in the Forms Section at the end of the EHSP (DVIR - Driver Vehicle Inspection Report).

3.1 General Haulage Equipment Operation

Haulage Equipment Operation which should be reviewed by the crew competent person, operators and all other workers associated with the project, should include, but not be limited to:

- Operators and drivers of all vehicles are required to do a walk around to check tires, connections, adjacent equipment and surrounding areas prior to operation
- On single-lane private roads with two-way traffic, turnouts should be provided. Where turnouts are not practicable, a control system should be provided to prevent vehicles from meeting on such single-lane roads
- On private roads used for two-way traffic, vehicles should travel on the right side as much as possible. Signs should be posted to clearly indicate variations.
- Separate haulage roads should be provided between loaded and empty units, where practicable
- Haulage roads should be wide enough to allow for safe passage. Safe distances between moving units should be maintained
- Private roads should be maintained free from holes and ruts that affect the safe control of the vehicle
- Flaggers or other controls measures should be used where hazards exist to employees and/or the public due to traffic or haulage conditions
- Workers exposed to vehicular traffic should wear high visibility safety apparel in accordance with the ANSI/ISEA 107-2004, High Visibility Safety Apparel and Headwear (i.e., Class 2 or Class 3 vests with retro-reflective stripes)
- Dust Control measures should be provided in accordance with federal, state and local requirements. Actions should be taken to control and prevent dust in work areas and prevent its migration off project boundaries.
- Operators should maintain control of their equipment at all times. Posted speeds limits should be followed, with due regard for weather, traffic, intersections, width and character of the roadway, type of motor vehicle, and any other existing conditions.
- Exhaust gases should be directed away from the operator's breathing zone
- Vehicle engines should not be allowed to run in closed garages or other enclosed places, unless provisions for effectively removing exhaust gases from the building are utilized
- Loads on vehicles should be secured against displacement
- Except for emergency field repairs, a safety tire rack, cage, or equivalent protection should be used when inflating truck or equipment tires after mounting on a rim, if such tires depend upon a locking ring or similar device to hold them on the rim
- Parking brakes should be set whenever equipment is parked. Chocks or other means should be used to secure equipment on inclines.
- Operators should be protected from all pinch points on the equipment
- Operators should maintain visibility and stability of the equipment at all times
- Loading device should not be left unattended, unless the load or bucket is lowered to the ground, or blocking or other precautions are taken to prevent accidental lowering
- Loading buckets, scoops, blades or similar attachments on haulage vehicles should not be used as work platforms or to elevate or transport employees
- Additional provisions are found in the Public Protection Section 16

4.0 Struck-By/Back-over Incidents

The purpose of this sub-section is to provide comprehensive protection to employees and employers exposed to struck-by and reverse operation traffic of vehicles, machinery and equipment in construction.

- Before operating the vehicle in reverse, the driver should visually determine that no employee is in the path of the vehicle
- Every vehicle used to haul dirt, rock, concrete, or other construction material should be equipped with a warning device that operates automatically while the vehicle is backing
 - The warning should sound an alarm above the surrounding noise level (200 feet in California) and will sound immediately on backing
 - In congested areas or areas with high ambient noise which obscures the audible alarm, a signaler or spotter, in clear view of the operator, should direct the backing operation
- Other safeguards that may be used with vehicles, machinery and equipment with an “obstructed view to the rear” includes, but is not limited to:
 - An automatic braking device at the rear of the vehicle that will apply the service brake immediately on contact with any obstruction to the rear
 - Operable video or similar technological capability providing the driver with a full view behind the vehicle

4.1 Traffic Control Plan

A traffic control plan should be developed that organizes work being done onsite in a manner that reduces the need for backing vehicles. At a minimum, the plan should include:

- Provisions to ensure operators and drivers of all vehicles perform a walk around to check tires, connections, adjacent equipment and surrounding areas prior to operation
- Worker-free zones to keep on-foot employees clear of operating equipment
- Details that identify hazards and traffic flow
- Communicate to all persons onsite during worker orientations, contractor meetings, OAC meetings and safety meetings
- Provisions for making changes to the plan when applicable, and means for communication to all affected crew members, including drivers and equipment operators

4.2 Training

All persons associated with material handling and haulage equipment should be trained to prevent struck-by/back-over incidents. Training should focus on the importance of safe backing operations. Operators and workers on-foot should receive training that includes hazard awareness and recognition. Training for Supervisors should focus on procedures such as how to develop a traffic control plan. Refer to the Struck By/Back Over Training Verification in the Forms Section at the end of the EHSP.

4.3 General Requirements for Off-Road Diesel Fueled Fleets (California only)

All self-propelled off-road diesel vehicles over 25 Horsepower and most two engine vehicles driven on or off-road are subject to the regulation. The overall purpose of the Off-Road regulation is to reduce emissions of oxides of nitrogen (NOx) and particulate matter (PM) from off-road diesel vehicles. Below are specific requirements of the regulation:

- Impose limits on idling and requires a written idling policy
- Requires all vehicles to be reported to the Air Resources Board
- Equipment Identification Number (EIN) labels are required to be affixed to both sides of the vehicle
- Restricts the adding of older vehicles into fleets
- Requires fleets to reduce their emissions by retiring, replacing, or repowering older engines, or installing Verified Diesel Emission Control Strategies, (i.e., exhaust retrofits)

5.0 Forklifts (Powered Industrial Trucks)

Forklifts should be inspected each day prior to use to determine that they are in safe working condition (refer to Forklift Inspection form in the Forms Section at the end of the EHSP or use manufacturer’s inspection guidelines) Forklifts should be operated in accordance with manufacturer’s recommendations and limitations.

5.1 Forklift Training Requirements

- Employees operating forklifts must be trained by a person qualified to recognize the hazards associated with aerial lifts and to understand the procedures to control or minimize those hazards
- The training must be in compliance with OSHA guidelines
- Operators must be trained on the equipment they will be operating or competency verified by a qualified person prior to vehicle operation (refer to Forklift Operator Proficiency Evaluation in the Forms Section at the end of the EHSP).
- Operators and drivers of all vehicles are required to do a walk around to check tires, connections, adjacent equipment and surrounding areas prior to operation
- Current certification must be maintained and presented to DPR project management upon request
- Certification will be revoked and refresher training provided and /or operator proficiency verified before commencement of operation when:
 - the operator is observed to operate the vehicle in an unsafe manner
 - the operator is involved in an accident or near miss incident
 - the operator is assigned to drive a different type vehicle
 - conditions in the workplace change that could affect safe operation

Note: A vehicle that is in need of repair, defective or in any way unsafe should be removed from service. The problem should be recorded on a log and reported to a supervisor immediately. This section discusses pre-operation and operational inspections that operators should perform to ensure that forklifts will operate safely. Only operators who have been trained and evaluated in accordance with [29 CFR 1910.178\(l\)](#) can operate forklifts.

5.2 Forklift Operation

Safely operating a forklift requires preparation, anticipation and careful attention in order to maintain control of the vehicle at all times. The operator shall be familiar with and follow manufacturer's operating procedures for the model of equipment they are operating. The following will identify recommended practices associated with operating a forklift:

- Operators shall be certified to operate the equipment they are using, certification must be current and carried on their person during operation.
- The operator is responsible for maintaining a safe and secure load at all times
- When loading or unloading, flag off back side of truck, and/or provide spotter to create a no access zone
- Operators and drivers of all vehicles are required to do a walk around to check tires, connections, adjacent equipment and surrounding areas prior to operation
- Loads should not be free-rigged, manufacturer approved forklift attachments should be used and taglines utilized to control loads, as needed
- Trucks should not be driven up to anyone standing in front of a bench or other fixed object
- No person should be allowed to stand or pass under the elevated portion of any truck, whether loaded or empty
- Seat belts should be provided and used on all equipment where rollover protection is installed
- Unauthorized personnel are not permitted to ride on equipment
- At no time should body parts be placed between the uprights of the mast or outside the running lines of the truck
- Prior to leaving a forklift unattended, the operator must fully lower the load, controls neutralized, power shut off, and brakes set. Wheels should be blocked if the truck is parked on an incline

Note: "Unattended" is when the operator is 25 ft. or more away from the vehicle, but within view; or whenever the operator leaves the vehicle and it is not in view.

- Brakes should be set and wheel blocks in place to prevent movement of trucks, trailers, or railroad cars while loading or unloading. Fixed jacks may be necessary to support a semitrailer during loading or unloading when the trailer is not coupled to a tractor
- The operators should slow down and sound the horn at intersections, blind spots and other locations where vision is obstructed. If the load being carried obstructs forward view, the driver should be required to travel with the load trailing
- Railroad tracks should be crossed diagonally wherever possible. Parking closer than 8 feet from the center of railroad tracks is prohibited
- When ascending or descending steep grades, loaded trucks should be driven with the load upgrade with forks tilted back and raised only as far as necessary to clear the road surface
- Dockboard or bridgeplates, should be properly secured before they are driven over and their rated capacity never exceeded
- Fuel tanks should not be filled while the engine is running and spills avoided
- Forklifts should be maintained per manufacturer's requirements and removed from service if found defective
- All repairs shall be made by authorized personnel

6.0 Concrete Pump Trucks

Concrete pump trucks and conveyor systems should be inspected each day prior to use to determine that they are in safe working condition (*refer to Concrete Pump Truck Inspection form in the Forms Section at the end of the EHSP or use manufacturer's inspection guidelines*). Concrete pump trucks should be operated in accordance with manufacturer's recommendations and limitations. Notice to DPR project team is required one week prior to concrete pump truck's arrival on site. Coordination between the DPR project team and concrete pump truck operators should occur to ensure proper conditions are maintained during setup, use and breakdown of equipment. The boom of the pump should not be used for hoisting at any time. Additionally, the following guidelines, at a minimum, should be followed:

- Overhead power lines
 - Power should be shut off to overhead power lines whenever possible, and verified de-energized prior to start
 - Where units must operator in proximity to energized overhead power lines, units should be set up a minimum of 20 feet away
- Setup
 - Work should not commence until the operator, concrete crew and DPR representative coordinate work and complete the pre-pump checklist and pre-task plan
 - Periodic review of setup conditions should occur throughout the pour
- Hose Whipping
 - No added metal fitting or couplings should be used at the end of the delivery hose
 - Hoses should not be allowed to become kinked
 - Personnel should maintain a safe distance from the discharge upon initial startup, restart or when air is in the hose. The safe distance is equivalent to the length of the discharge hose.
- Rock/Concrete Spray
 - Only trained personnel should be allowed in the area near the hopper
 - Barricading or other means of limiting access should be provided to ensure non-authorized personnel are kept away
- Environmental Hazards
 - All washout activities should be completed in their designated washout area
 - Concrete activities should be performed so as to contain and/or minimize residue or waste water; all such wastes should be properly disposed. Refer to the Chemical & Environmental Section 02 for additional environmental protection requirements

- Other Considerations
 - Chemicals and/or additives may cause chemical and/or thermal burns
 - Ensure work procedures include product specific requirements, i.e., additives may require additional PPE

7.0 Passenger Vehicles

This sub-section refers to vehicles designed to transport passengers, in addition to the operator. i.e., pickup trucks, automobiles, ATVs, golf carts, etc. Operators & passengers should be physically seated when the vehicle is moving.

- Drivers should operate in a manner as to eliminate or reduce all distractions, i.e., use of cell phones, etc.
- Operators and drivers of vehicles should determine that no person or object are in the path of the vehicle prior to operation or movement, where applicable, a spotter may be needed
- Posted speed limits should be observed at all times, with speeds being adjusted for adverse conditions, e.g., inclement weather, heavy traffic, poor road conditions, etc.
- Seat belts should be provided and used by the driver and any passengers at all times while the vehicle is in motion
- Vehicle maintenance should be performed at the designated intervals as outlined by manufacturer's recommendations
- Tire pressure should be checked regularly, maintained and tires visually inspected
- Windshields should be kept clean, clear of cracks and distortions; windshield wiper fluids should be maintained
- Passengers should not be allowed to ride in the back of a moving pickup
- Any accidents or incidents involving passenger vehicles on the project or in the course of work should be reported immediately to DPR project management. *Refer to the Incident Reporting Section 13 for further guidelines on accident and incident reporting.*
- Operators of DPR vehicles should also refer to the DPR Employee Handbook for additional DPR specific vehicle requirements

7.1 Golf Carts, Gators, Utility Vehicle

- Operator should review and understand owner's manuals
- Equipment Checklist should be completed prior to use
- When leaving unattended, keys should be removed and Parking brake applied
- When parking on an incline wheels should be chocked
- Windshields should be kept clean, clear of cracks, scratches and distortions; windshield wiper fluids should be maintained
- Should be equipped with minimum 2lb. fire extinguisher
- Whip mounted vehicle safety flags (min 8' off the ground) or strobe lights should be used
- Requires an audible back up alarm

8.0 Drill Rigs

For the purpose of this sub-section, a drill rig is an earth-excavating machine designed to excavate a circular shaft into which concrete or aggregate fill is typically placed. When a drill rig is set up to drill a shaft, the base of the unit remains stationary, allowing the shaft to be drilled without moving the base unit, and facilitates the discharge of spoils away from the shaft.

8.1 Drill Rig Pre-Mobilization

Documents required to be provided to DPR prior to mobilization on site include:

- An independent third party annual certification provided

- Drill rig contractor's certificate of insurance
- Drill rig operator's written verification of hands on/written competency testing with specific make and model of drill rig
- Written verification from drill rig owner that maintenance has been performed pursuant to manufacturer's recommendations

8.2 Inspection

- Annual inspections should be performed by an independent third party and must be received and reviewed by DPR project management prior to the arrival of equipment
- Inspections should be performed each day prior to operation and documentation should be maintained on the unit

8.3 Set Up and Operation

- The drill rig should be selected to ensure sufficient capacity and appropriate design features are suitable for the intended lift
- The operator should comply with the manufacturer's specifications and limitations applicable to the operation of the rig
- Fall protection should be provided for workers working within 6 feet any open holes. The following should be considered:
 - Guardrail system suitable to withstand 200lbs of force. (i.e. hay bale rings, movable panels, etc.)
 - Suitable anchorage points (i.e. deadman, with stamped, engineered drawing), full body harness and retractable lanyard should be provided for each affected worker
 - Ensure the number and placement of anchorage points are appropriate for the number of affected workers
 - Rope/flag off area around the perimeter of drilling operations, to include swing radius protection
- Written plan should be submitted and reviewed by DPR prior to start of work and should include at a minimum:
 - Number and sequencing of holes
 - Provisions for protection of open holes:
 - Covers
 - Back filling (back fill holes if concrete not poured the same day)
 - Guardrails or flagging
 - Identify vehicle traffic routes near and around drilling operations
 - Emergency Procedures in the event of a cave in or other mishap
- In lieu of having a worker enter a caisson to inspect a bell-bottom, alternative means of inspection must be utilized

Motorized & Site Equipment Forms

Forms

Driver Vehicle Inspection Report
Forklift Pre-Use Inspection – English
Forklift Pre-Use Inspection – Spanish
Forklift Operator Proficiency Evaluation – English
Forklift Operator Proficiency Evaluation – Spanish
Fork Lift Hand Signals
Heavy Equipment Training Verification Form – English
Heavy Equipment Training Verification Form – Spanish
Concrete Pump Subcontractor Pre-Planning Package
Concrete Truck Pre-Pump Checklist
Concrete Pump Hand Signals
Concrete Pump Truck Training Verification – English
Concrete Pump Truck Training Verification - Spanish
Water Truck Training Verification Form – English
Water Truck Training Verification Form - Spanish



Driver Vehicle Inspection Report

Project/Project#:		Date:	
Model:		Equipment Number:	
Company:		Hour Meter:	

**** If any item(s) are found to be missing or defective, the equipment is to be repaired/serviced prior to use**

To: _____ Date: _____

Vehicle Taken Out of Service ☐ Yes ☐ No

Date							
Initials of person performing inspection:							
Has the driver been instructed in the safe operation of this type of vehicle and are they Licensed?		Y/N	Y/N	Y/N	Y/N	Y/N	Y/N
Inspection Item & Description Pass Fail Status							
Inspection Item & Description Pass Fail Status		P/F	P/F	P/F	P/F	P/F	P/F
1.	Service Brakes/Parking/Trailer						
2.	Steering						
3.	Lighting and Reflectors						
4.	Tires						
5.	Wheels/Rims						
6.	Windshield Wipers						
7.	Rear View Mirrors						
8.	Horn						
9.	Engine Compartment						
10.	All Fluid Levels						
11.	Belts						
12.	Wiring						
13.	Leaks/Hoses/Air Lines						
14.	Seat Belt						
15.	All Gauges						
16.	Fuel Tanks						
17.	Oil Pressure						
18.	Heater/Defroster						
19.	Lights/Backup Alarm						
20.	Oil Pressure						
21.	Emergency Equipment						
22.	Body						
23.	Parking Brake						
24.	Hydraulic						
25.	Coupling Devices						

Repairs Made

By::		Date:	
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Forklift Pre-Use Inspection

Project/Project#:		Date:	
Model:		Equipment Number:	
Company:		Hour Meter:	

**** If any item(s) are found to be missing or defective, the equipment is to be repaired/serviced prior to use**

To: _____ Date: _____
Vehicle Taken Out of Service ☐ Yes ☐ No

Date							
Initials of person performing inspection:							
Has the operator been instructed in the safe operation of this type of lift and are they certified?	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N
Inspection Item & Description Pass Fail Status							
Inspection Item & Description Pass Fail Status	P/F	P/F	P/F	P/F	P/F	P/F	P/F
Pre-Start							
1 Fork Assembly Structure							
2 Mast/Boom							
3 Hydraulic Lines**							
4 Wheels/Tires**							
5 Axles/Under Carriage							
6 Placards (legible)							
7 Roll Cage – Structural							
8 Operator's Manual							
9 Engine Compartment							
10 All Fluid Levels							
11 Belts							
12 Wiring							
13 Leaks/Hoses**							
Start Up							
1 Seat Belt**							
2 All Gauges							
3 Warning Signs/Placards							
4 Steering (loose)**							
5 Levers/Switches							
6 Lights/Backup Alarm**							
7 Horn**							
Function Check							
1 Lift up/down**							
2 Telescope in/out**							
3 Tilt in/out **							
4 Outriggers							
5 Brakes, Parking Brake**							

Repairs Made

By:		Date:	
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Inspección previa al uso del elevador de horquilla

Proyecto/Proyecto#:		Fecha:	
Modelo:		Número del equipo:	
Compañía:		Medidor de horas:	

**** Si se encuentra cualquier artículo faltante o defectuoso, el equipo deberá ser reparado o recibir servicio de mantención antes de su uso.**

A: _____ Fecha _____

Elevador retirado de servicio ☐ Si ☐ No

FECHA							
Iniciales del inspector							
¿Ha sido instruido el operador acerca de la operación segura de este tipo de elevador, y tiene su certificación?	Sí/ No	Sí/ No	Sí/ No	Sí/ No	Sí/ No	Sí/ No	Sí/ No
Artículo inspeccionado y descripción (aprobado/desaprobado)							
Artículo inspeccionado y descripción (aprobado/desaprobado)	A/D	A/D	A/D	A/D	A/D	A/D	A/D
PRE PUESTA EN MARCHA							
1	Estructura del ensamblaje de la horquilla						
2	Mástil/pluma						
3	Líneas hidráulicas **						
4	Ruedas/neumáticos**						
5	Ejes/chassis inferior						
6	Carteles (legibles)						
7	Jaula antivuelcos – estructural						
8	Manual del operador						
9	Compartimiento del motor						
10	Niveles de todos los fluidos						
11	Correas						
12	Cableado						
13	Filtraciones/mangueras**						
PUESTA EN MARCHA							
1	Cinto de seguridad**						
2	Todos los indicadores						
3	Señales de advertencia/carteles						
4	Dirección (suelta)**						
5	Palancas/interruptores						
6	Luces/alarma de refuerzo**						
7	Bocina**						
CHEQUEO DE FUNCIONES							
1	Elevador sube/baja**						
2	Telescopio entra/sale**						
3	Inclinación dentro/fuera **						
4	Estabilizadores						
5	Frenos, freno de mano**						

Reparaciones hechas

Por:		Fecha	
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Forklift Operator Proficiency Evaluation

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Employee Name:		Equipment Manufacturer:	
Model:		Capacity:	

Pre-Start Inspection		Pass	Fail
1	Assure that no repairs are in progress	☐	☐
2	Check all fluid levels	☐	☐
3	Check tires/wheels	☐	☐
4	Check for oil leaks	☐	☐
5	Check the brakes	☐	☐
6	Check all hoses	☐	☐
7	Check battery	☐	☐
8	Check lights	☐	☐
9	Check horn	☐	☐
10	Check back-up alarm	☐	☐
11	Check fire extinguisher	☐	☐
12	Operator knows the capacity of forklift	☐	☐
13	Check operation of controls	☐	☐
14	Capacity chart posted	☐	☐
15	Operator's manual in cab	☐	☐

Operating Skills Evaluation		Pass	Fail
1	Does operator drive at a safe rate of speed?	☐	☐
2	Did operator slow down at stops or intersections?	☐	☐
3	Did the operator approach the load cautiously?	☐	☐
4	Does the operator know the weight of the load?	☐	☐
5	Did the operator place the forks under the load properly?	☐	☐
6	Did the operator raise and position the load properly?	☐	☐
7	Did the operator strike anything while moving the load?	☐	☐
8	Is the seat belt worn while operating?	☐	☐

Name/Signature of Operator

Date

Name/Signature of Examiner

Date



Evaluación del desempeño del operador de un elevador de horquilla

Documento de archivo – NO DESECHAR

Proyecto/Proyecto#		Fecha del examen	
Nombre del empleado		Equipo Fabricante	
Modelo		Capacity	

Inspección antes de la puesta en marcha		Aprobado	Desaprobado
1	Chequear que no exista una reparación en curso	☐	☐
2	Chequear todos los niveles de fluidos	☐	☐
3	Chequear ruedas y neumáticos	☐	☐
4	Chequear filtraciones de aceite	☐	☐
5	Chequear los frenos	☐	☐
6	Chequear todas las mangueras	☐	☐
7	Chequear la batería	☐	☐
8	Chequear luces	☐	☐
9	Chequear bocina	☐	☐
10	Chequear la alarma de refuerzo	☐	☐
11	Chequear el extinguidor de incendios	☐	☐
12	Operador conoce la capacidad del elevador de horquilla	☐	☐
13	Chequear la operación de los controles	☐	☐
14	Tabla de capacidades a la vista	☐	☐
15	Manual del operador en la cabina	☐	☐

Evaluación de capacidad de operación		Aprobado	Desaprobado
1	¿Maneja el operador a una velocidad prudente?	☐	☐
2	¿Redució el operador la velocidad en los signos "Pare" o en las intersecciones?	☐	☐
3	¿Se acercó el operador con cuidado a la carga?	☐	☐
4	¿Conoce el operador el peso de la carga?	☐	☐
5	¿El operador colocó correctamente las horquillas debajo de la carga?	☐	☐
6	¿El operador levantó y colocó la carga adecuadamente?	☐	☐
7	¿Al mover la carga, ¿chocó con algo el operador?	☐	☐
8	¿Usó el cinto de seguridad durante la operación?	☐	☐

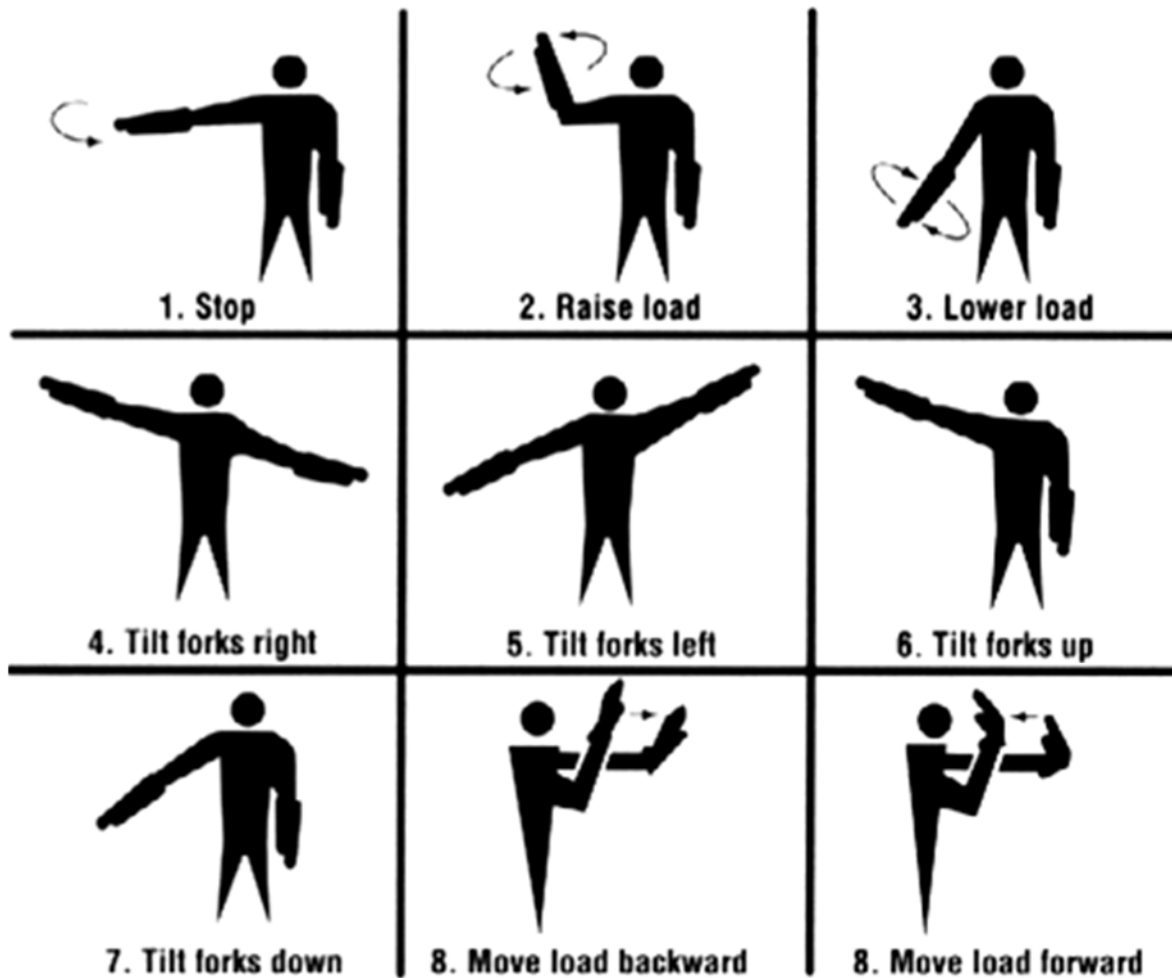
Nombre/Firma del Operador

Fecha

Nombre/Firma del Evaluador

Fecha

Fork Lift Hand Signals





Heavy Equipment Training Verification Form

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Name:		Company:	

Pre-planning

- Operators should be trained in the safe operation of the equipment they will be utilizing, and understand the equipment's limitations
- Utilities have been located and marked prior to start of work; critical systems/utilities have been pot holed (potholing requirements per the DPR EHSP have been reviewed and understood)
- Where applicable, the operator should reference blue prints and as-bults pertinent to the work area
- Operator's manual is readily available on the equipment

Inspection

- Perform equipment inspection per operator's manual. At a minimum, ensure the following are inspected:
 - o Ensure cab is free of loose materials and debris
 - o Ensure operator's manual is in the cab and in legible condition
 - o Check tires for cuts, bulges, chunking and proper inflation
 - o Check for proper level of fluids and ensure no leaks are present
 - o Check that belts, air pressure, air filter, etc are in place and in good working condition,
 - o All fittings should be adequately greased
 - o Look for broken welds, cracks in the paint/metal or obvious damage
 - o Inspect roll over protection for tightness of bolts/quality of welds, seat belt is in place and operational

Operation

- Provisions to ensure operators and drivers of all vehicles perform a walk around to check tires, connections, adjacent equipment and surrounding areas prior to operation
- When mounting or dismounting, maintain three points of contact with the machine at all times
- Use proper start procedures, consult the operator's manual for equipment specific requirements
- Perform equipment inspection
- Allow the equipment to warm up prior to operating the machine
- If applicable, the slope angle should be kept to a minimum
- If applicable, set stabilizers and raise the machine slightly off the ground, ensure the machine is level
- Make sure ALL personnel and other equipment are clear of the work area, and be alert to other people & equipment
- Match ground speed with conditions of terrain
- Allow no riders
- If applicable, keep the bucket low for stability and visibility

Name/signature of employee

Date

Name/signature of person conducting verification review

Date

A copy of this form should be faxed to the regional safety professional PRIOR TO START OF WORK.



Formulario de Verificación de Equipos Pesados

Documento para archivo – NO DESECHAR

Proyecto/Proyecto#:		Fecha:	
Nombre:		Compania:	

Planificación previa

- Los operadores deben recibir capacitación sobre el funcionamiento seguro del equipo que utilizarán, y entender las limitaciones de los equipos.
- Se deben ubicar y marcar los servicios públicos antes de comenzar el trabajo y hacer los pozos necesarios en los sistemas/servicios públicos de primera necesidad (se deben revisar y entender los requisitos para hacer pozos de acuerdo con la política EHSP de DPR).
- El operador debe consultar los planos y los dibujos pertenecientes al área de trabajo, si corresponde.
- El manual del operador debe estar disponible en el equipo.

Inspección

- Inspeccione el equipo según el manual del operador. Debe inspeccionar, como mínimo, lo siguiente:
 - o Verifique que no haya materiales sueltos ni escombros en la cabina.
 - o Verifique que el manual del operador se encuentre en la cabina y que se pueda leer bien.
 - o Revise los neumáticos para ver si están combados, si tienen cortes, si emiten ruidos y si tienen el nivel de inflado apropiado.
 - o Revise que el nivel de fluidos sea el correcto y que no haya ninguna pérdida.
 - o Verifique que los cinturones de seguridad, la presión de aire, el filtro de aire, etc., estén en su lugar correcto y en buenas condiciones.
 - o Todos los accesorios deben estar adecuadamente engrasados.
 - o Verifique que no hayan soldaduras rotas, grietas en la pintura/metal o daños visibles.
 - o Para inspeccionar la protección contra vuelcos verifique que los pernos estén ajustados correctamente, revise la calidad de las soldaduras y que los cinturones de seguridad estén en su lugar y en buen funcionamiento.

Operación

- Provisions to ensure operators and drivers of all vehicles perform a walk around to check tires, connections, adjacent equipment and surrounding areas prior to operation.
- Cuando deba subirse o bajarse de una máquina, mantenga tres puntos de contacto con la máquina en todo momento.
- Inicie la puesta en marcha del equipo de la manera correcta, consultando los requisitos específicos de los equipos en el manual del operador.
- Realice una inspección de los equipos.
- Deje que el equipo se caliente antes de operar la máquina.
- El ángulo de inclinación debe mantenerse al mínimo, si corresponde.
- Fije los estabilizadores y eleve la máquina suavemente del piso; verifique que la máquina se encuentre nivelada, si corresponde.
- Verifique que el área de trabajo se encuentre TOTALMENTE despejada de personas y equipos y esté alerta frente a las demás personas y equipos.
- Fije la velocidad de acuerdo con las condiciones del terreno.
- No acepte pasajeros en la máquina.
- Mantenga la cubeta baja, para mayor estabilidad y visibilidad, si corresponde.

Nombre/Firma del empleado

Fecha

Nombre/Firma de la persona que realiza la revision de verification

Fecha

Se debe enviar por fax este formulario a regional safety professional ANTES DE EMPEZAR EL TRABAJO.



Concrete Pump Subcontractor Pre-Planning Package

Project/Project#:		Date:	
Company:		Specific Location of Work:	

Dear Subcontractor

The following package is provided to assist you and your workers in maintaining an Injury Free Environment. As you know, concrete pump truck operations are potentially hazardous and there are many steps necessary that need to be planned prior to your arrival on the project. Though concrete pump truck accidents rarely occur, when they do they are potentially serious. Please review and complete the following documentation. This package should be filled out as completely as possible and submitted to DPR 1 week prior to mobilizing.

Required documentation:

- 1) Concrete Truck Pre-Pump Checklist – fill in as completely as possible
- 2) Provide copies of the annual boom certification
 - o Certification must match the pump truck that shows up on site
 - o Certification must be current
 - o Pump truck must have passed certification; including verification of completed repairs identified at time of certification
- 3) Provide copy of valid, current insurance
- 4) Written verification of operator's qualifications, specific to make and model of pump truck
- 5) Written verification from pump truck owner that maintenance has been performed pursuant to manufacturer's recommendations
- 6) Complete a pre-task/pre-lift plan, detailing likely sequencing and what safety precautions will be taken to address any potential hazards. A pre-task plan checklist and worksheet have been provided. Safety items to consider:

- | | |
|---------------------------------|-------------------------------|
| a. Fall protection | b. Guarding |
| c. Public and worker protection | d. Concrete Pump truck set-up |
| e. Weather | f. Communication |
| g. Control of pinch points | |

Additionally:

- 1) A safety meeting will be held prior to pour with operator and applicable crew members to review all of the above listed items
- 2) A daily, documented inspection must be performed on all pump trucks prior to operation. A sample inspection form is provided

This package is designed to assist you in running a smooth and safe operation while on our projects. If you have any questions regarding any information contained within, please contact the DPR Project Team. Thank you for your assistance in helping us to create an Injury Free Environment.

DPR Construction



Concrete Truck Pre-Pump Checklist

Archive Document – DO NOT DISCARD

Project/Project#: _____ Date: _____
Contractor Requesting Pump Truck: _____ Date of Pour: _____
Contractor Providing Pump Truck: _____ Location of Pour: _____

Documentation

☐ Valid pump truck certification with all repairs completed	
☐ Written verification from pump truck owner that maintenance has been performed pursuant to manufacturer's recommendations	
☐ Written verification of operator's qualifications specific to make and model of pump truck.	
☐ Verification of current, valid insurance	
☐ Complete detailed, written pre-task planning with operator, crew and foreman involved in pour. Items to be considered on the pre-task plan:	
☐ Start & finish times for all pours.	☐ Communication plan
☐ Protective measures for workers, the environment, and the public	☐ Weather considerations
☐ Pump truck set-up locations, with swing path, travel plan and distance	

Pre-Mobilization

☐ DPR Superintendent and requesting pump truck contractor perform site walk. Review, at minimum:				
☐ Condition of asphalt, concrete, soil, etc. where pump truck will be set-up & operating				
☐ Location of overhead electrical lines				
☐	Rated line voltage:	Kv		
☐	Distance of lift to lines:	ft. (must be >17')		
☐	Lines Energized?	☐ Yes	☐ No	If yes, contact DPR EHS. If no, contact owner and ground lines.
☐ Person designated to observe line clearance for operation				
☐ Pump truck set-up locations, with swing path, travel plan and distance				
☐ Distance to neighboring buildings, roadways, walkways, trenches, variations in soil stability, etc.				
☐ Pump truck contractor to provide:				
☐ Gross capacity of pump truck & verification of concrete mix design to be used				
Maximum Radius:		☐ Minimum Boom Angle:		
Maximum Boom Length:		☐ Building height:		
☐ Location of washout				

Mobilization – Continuous throughout Operation

☐ Site specific safety orientation completed for pump operator and/or pump crew	
☐ Documentation of pump truck Pre-task plan and Boom Inspection	
☐ A copy of the manufacturers Operating Manual is on the pump truck.	
☐ Conduct a safety meeting, covering at least	
☐ Pre-task plan	☐ Weather Conditions
☐ Evacuation Plan	☐ Priming Procedures
☐ Verify & review means of communication	☐ Safety precautions working around the pump truck & hose operations
☐ Location of emergency stop switch on pump truck	☐ Public Protection
☐ Outriggers must be fully extended, pads level, on solid bearing, and pinned. <u>Retracting outriggers without a signal man is prohibited.</u>	
☐ Barricading in place to prevent access to hopper area of pump truck	
☐ Should lifting or hoisting of material(s) will be needed, additional equipment must be available (the boom of the pump cannot be used for hoisting materials)	
☐ Traffic control measure in place, as applicable	
☐ Hoses are in good condition & additional hose is available, specific to pour requirements.	
☐ Signal Man (if applicable): _____	
☐ Ensure these are in place:	
☐ Hard hats	☐ Fall Protection (harnesses, lanyards as applicable)
☐ PPE (safety glasses, gloves, vests, dust masks, protective boots, face shields, knee pads, etc as applicable for scope of work)	

DPR Superintendent/EHS

Date

Contractor Supervisor

Date

Pump Truck Operator

Date



Concrete Pump Truck Pre-Task Plan

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Project/Project#:		Date:	
Company:		Specific Location of Work:	

Answer the following when evaluating your work YES, NO or N/A:		Are any of the following Required?							
1.	Have you walked your work area to identify live systems, overhead lines, utilities, underground conditions, public exposures, lighting, etc?	☐	YES	☐	NO	☐	N/A	PPE	
2.	If overhead powerlines are energized, is the rig set up a minimum of 17 feet away?	☐	YES	☐	NO	☐	N/A	☐	Hard Hat
3.	Are there any hazards created by any other workers in your area or does your work create hazards for others?	☐	YES	☐	NO	☐	N/A	☐	Fall Protection
4.	Have all equipment and tools been inspected prior to use? No steel hose ends, "S-Bend" device on hose, etc?	☐	YES	☐	NO	☐	N/A	☐	Eye / Face PPE
5.	Have you reviewed applicable SDSs to ensure proper work procedures are being utilized and to determine if additives or chemicals require additional PPE?	☐	YES	☐	NO	☐	N/A	☐	Hand / Arm PPE
6.	Are materials, tools and manpower adequate to perform the job safely?	☐	YES	☐	NO	☐	N/A	☐	Hearing PPE
7.	Have you addressed any barricading, warning system or signage requirements appropriate to the task?	☐	YES	☐	NO	☐	N/A	☐	Foot PPE
8.	Have you established a communication system for signaling and/or communication?	☐	YES	☐	NO	☐	N/A	☐	Dust Mask/Respirator
9.	Are outriggers fully extended with adequate pads?	☐	YES	☐	NO	☐	N/A	LOCATE	
10.	Have you addressed lifting, bending, twisting or pinch point hazards?	☐	YES	☐	NO	☐	N/A	☐	Emergency Telephones
11.	Has a method/location been established for workers to clean up in case of concrete over exposure?	☐	YES	☐	NO	☐	N/A	☐	Fire Extinguisher
12.	Have you completed Stretch & Flex today?	☐	YES	☐	NO	☐	N/A	☐	Emergency Exit Routes
13.	Are enough people assigned to safely complete the task? (lifting, spotting, etc.)	☐	YES	☐	NO	☐	N/A	☐	First Aid Equipment
14.	Is there a new hire, or new team member on the project who will need support?	☐	YES	☐	NO	☐	N/A	☐	Other
15.	Do you have an injury to report or were you injured the prior working day?	☐	YES	☐	NO	☐	N/A	☐	Other

Any safety item identified in the above question list must be addressed on the back page.

Tasks, Tools & Potential Hazards

Steps for Work	Tools	Hazards	Steps Taken to Address Hazards

*If you need more space, attach another PRE-TASK PLAN SHEET

Lift Supervisor		Foreman		Safety Coordinator	
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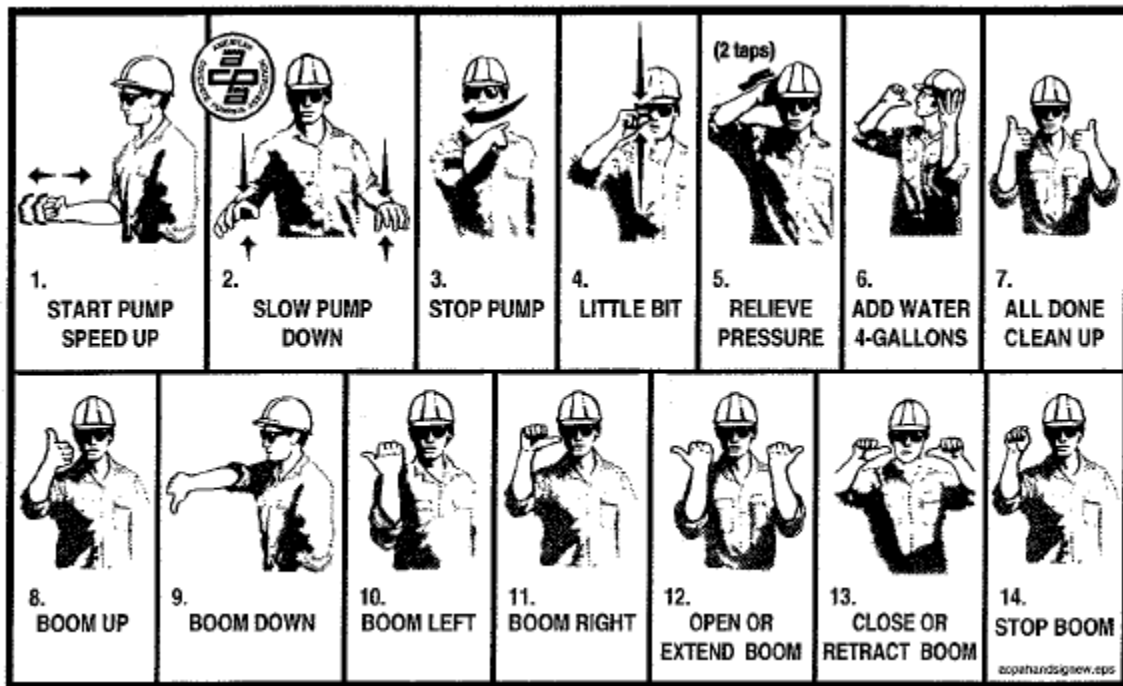
Crew Members – list additional crew members on back or a separate page if necessary:



Pump Plan

Please draw a map to detail set up points, sequencing of picks, protection etc.

Concrete Pump Hand Signals





Concrete Pump Truck Training Verification

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Project/Project#	Date
Name	Company

Overhead Power Lines

- o If a concrete pump hits/touches an overhead power line and becomes energized, everything that the pump is touching will also become energized
- o The minimum safety distance for energized power lines is 17 feet (even if it is a wrapped or insulated line); if working closer than 17 feet, power lines must be de-energized and verified dead prior to set up
- o Power lines have the potential to stretch up to 7 feet in very warm weather -- the distance a powerline is relation to the truck may change throughout the day and lines could come into contact with a pump/crane boom as temperatures rise
- o Power lines can create a magnetic field when moisture is present in the atmosphere and can “arc” and jump several feet

Hose Whippings

- o Keep personnel away from the end of the discharge hose when first starting, restarting or anytime there is air in the delivery line
- o Never kink a hose as the pump may straighten the hose by force causing it to whip
- o Devices like the “S-Bend” or metal fittings at the end of the hose are heavy and can become dangerous weapons if a pump hose whips -- all hose ends should be free of any steel fittings, etc

Concrete/Rock Spray Dangers

- o Stay clear of the rear end of the concrete pump, especially the hopper during a concrete pour --blow back can occur if the concrete pump “sucks air” causing material to be discharged from the hopper with great force
- o Set flagging at rear of pump by the hopper in such a way that concrete trucks can get to the hopper of the pump but keep unauthorized personal away from the hopper
- o Prior to opening or modifying any line, always wait until the operator reverses the pump and tells you that it is safe to open the line -- a plugged line may excessive pressure inside

Concrete Pump Tip – Over

- o Outriggers must be setup on firm, level ground -- outrigger pads and cribbing must be used to ensure proper weight distribtion
- o Upon arrival, coordination between the pump truck operator and the DPR project team should occur to ensure that all variables of a good setup location have been evaluated, i.e., proximity to work site, soil conditions, overhead obstructions such as power lines, flatness of area, etc.

Environmental Dangers

- o Concrete should not be discharged onto the ground, always ensure a washout area is designated and used

Other considerations

- o The boom of the pump truck should never be used for hoisting or lifting of materials
- o If blind pumping, i.e., the pump operator is not able to see pour, provide a suitable means of communication between the pump operator and the crew such as a relay person, siganler, radios, etc.
- o Only the operator should stand under the boom and only when it cannot be avoided and for as little time as possible
- o Ensure work procedures include product specific requirements and SDS are referenced, i.e., chemicals and/or additives may cause chemical and/or thermal burns; additional PPE may be required, etc.
- o Ensure the hose man never walks backwards and never hugs the hose. He should hold with both hands to allow the hose to move freely
- o Ensure the hose man never positions himself between the boom or boom hose and any fixed object like a wall or column
- o Ensure no one tries to support the tip hose with back or shoulders
- o Ensure no one positions their hands or any other body part between the end of the delivery system and a fixed object

Employee Signature	Date
--------------------	------

Person Conducting Verification Review Signature	Date
---	------

A copy of this form should be faxed to the DPR regional office PRIOR TO START OF WORK.



Concrete Pump Truck Training Verification

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Nombre	Fecha
Obra	Compañía

Tendido eléctrico

- o Si un camión de hormigón golpea o hace contacto con una línea de tendido eléctrico y se conecta a la energía eléctrica, todo lo que esté en contacto con la bomba también se conecta
- o La distancia de seguridad mínima para una línea eléctrica conectada es de 17 pies (incluso si se trata de una línea envuelta o aislada); si trabaja a menos de 17 pies, las líneas eléctricas deben desconectarse y verificar que estén desconectadas antes de la preparación
- o Las líneas eléctricas tienen el potencial de dilatarse hasta 7 pies en tiempo muy cálido – la distancia de una línea eléctrica en relación con el camión puede cambiar durante el día y las líneas pueden entrar en contacto con una bomba/pluma de grúa a medida que suben las temperaturas
- o Las líneas eléctricas pueden crear un campo magnético cuando haya humedad presente en la atmósfera y pueda formarse un “arco” y saltar una distancia de varios pies

Latigazos de las mangueras

- o Mantenga al personal apartado del extremo de la manguera de descarga cuando inicie o vuelva a iniciar el bombeo o en cualquier momento en que haya aire en la línea de suministro
- o No retuerza nunca una manguera, ya que la bomba puede enderezar la manguera a la fuerza y dar latigazos
- o Los dispositivos como “codos” o conexiones de metal en el extremo de la manguera son pesados y pueden ser armas peligrosas si la manguera de una bomba empieza a dar latigazos – ningún extremo de manguera debe tener conexiones de acero, etc.

Peligros de rociado de hormigón/rocas

- o No se acerque al extremo trasero de la bomba de hormigón, especialmente la tolva durante una colada de hormigón –se puede producir un retroceso si la bomba de hormigón “aspira aire” haciendo que el material se descargue de la tolva con gran fuerza
- o Fije los indicadores en la parte trasera de la bomba junto a la tolva de manera que los camiones de hormigón puedan llegar a la tolva de la bomba pero mantengan al personal no autorizado lejos de la tolva
- o Antes de abrir o modificar cualquier línea, espere siempre a que el operador invierta el sentido de la bomba y le indique que es seguro abrir la línea -- una línea obstruida puede causar una presión excesiva en el interior

Vuelco de la bomba de hormigón

- o Se deben configurar estabilizadores sobre un terreno firme horizontal – se deben usar apoyos de estabilizadores y asientos para asegurar una distribución apropiada del peso
- o Después de la llegada, debe haber una coordinación entre el operador del camión de la bomba y el equipo del proyecto DPR para asegurarse de que se hayan evaluado todas las variables de un buen lugar de configuración, es decir, proximidad a la obra, condiciones del suelo, obstrucciones superiores como tendido eléctrico, planeidad del área, etc.

Peligros medioambientales

- o El hormigón no debe descargarse sobre el terreno, asegúrese siempre de designar y usar un área de lavado

Otras consideraciones

- o La pluma del camión de la bomba no debe usarse nunca para elevar o levantar materiales
- o Si se bombea a ciegas, es decir, el operador de la bomba no puede ver la colada, proporcione un medio adecuado de comunicación entre el operador de la bomba y la cuadrilla como una persona mensajera, un señalizador, radios, etc.
- o Solamente el operador debe estar debajo de la pluma y sólo cuando no se pueda evitar y durante un tiempo mínimo posible
- o Asegúrese de que los procedimientos del trabajo incluyan requisitos específicos del producto y que haga referencia al SDS, es decir, productos químicos y/o aditivos químicos que pueden causar quemaduras; y Equipo de Protección Personal adicional que puede ser requerido, etc.
- o Asegúrese de que la persona con la manguera no camine nunca hacia atrás ni abraze la manguera. Debe sujetarla con las dos manos para permitir que la manguera se mueva libremente
- o Asegúrese de que la persona de la manguera no se coloque nunca entre la pluma o la manguera de la pluma y cualquier objeto fijo como una pared o columna
- o Asegúrese de que nadie trate de sujetar la punta de la manguera con la espalda o los hombros
- o Asegúrese de que nadie ponga las manos ni ninguna otra parte del cuerpo entre el extremo del sistema de suministro y un objeto fijo

Firma del empleado	Fecha
Firma de la persona que efectúa la revisión de erificación	Fecha

ANTES DE EMPEZAR EL TRABAJO se debe enviar una copia de este formulario por favor



Water Truck Training Verification Form

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Name:		Company:	

Equipment:			
Manufacturer:			
Model(s):		Capacity:	

Pre-planning

- Operators should be trained in the safe operation of the equipment they will be utilizing, and understand the equipment's limitations.
- Operator should walk area and/or review as-builts and site plan to ensure understanding of locations of utilities, excavations, footings, obstructions, etc.
- Operator's manual should be readily available on the equipment, and the operator has reviewed and understands the requirements outlined within.
- Area around water valve should be designated for water truck usage only.
- Means for protecting water valve should be in place; quick disconnect or breakaway device is recommended
- Sleeve should be provided when using water tank with overhead water valve to ensure overspills are minimize keeping ground around water tank dry
- Operator should be trained and hold a current dust control card
- Operator should understand the site water schedule, areas of priority and assist with maintaining the project water log.

Inspection

Perform equipment inspection per operator's manual. At a minimum, ensure the following are inspected:

- Perform vehicle walk around to check for new exterior frame and/or body damage.
- Ensure cab is free of loose materials and debris.
- Ensure operator's manual is in the cab and in legible condition.
- Check tires for cuts, bulges, chunking and proper inflation
- Check for proper level of fluids and ensure no leaks are present
- Check that belts, air pressure, air filter, etc. are in place and in good working condition,
- All fittings should be adequately greased
- Look for broken welds, cracks in the paint/metal or obvious damage to water spray equipment.
- Mirrors, lights, turn signals, horn, (manual) handbrake and backup alarm should be operable
- Fire extinguisher should be on the truck, in good working order and current on monthly and annual inspections.

Operation

- When mounting or dismounting, maintain three points of contact with the machine at all times.
- Use proper start procedures; consult the operator's manual for equipment specific requirements.
- Perform equipment inspection, *using the equipment inspection form*.
- Allow the equipment to warm up prior to operating the machine
- Imposed loads and how they affect adjacent excavations and / or other structures should be clearly understood
- Make sure ALL personnel and other equipment are clear of the work area, and be alert to other people & equipment
- Match ground speed with conditions of terrain
- Seatbelt should be worn at all times when the vehicle is in motion
- Do not leave the engine running if more than 15 feet from the vehicle
- Ensure visibility when backing up to the water filling station, or use a spotter
- Do not use tools when attaching/unhooking water valve, fittings are hand tightened/loosened only
- Verify the equipment is properly disconnected from water valve after filling, before pulling away.
- All occupants must wear their seat belt, if seat belt is not available or not in working order, no passengers should be in the vehicle.
- Operator should understand the project's oil/fluid cleanup procedures are, in case of leaks/spills.

Employee's Signature

A copy of this form should be faxed to the DPR regional office PRIOR TO START OF WORK.

Signature of person conducting verification review



Forma de Verificación de Entrenamiento De Trocas para Regar con Agua

Archivar este Documento – NO TIRAR

Proyecto/#Proyecto:		Fecha:	
Nombre:		Compañía:	
Equipo:			
Fabricante:			
Modelo(s):		Capacidad:	

Planificación Previa

- Los operadores deben ser entrenados en la operación segura del equipo que estén utilizando, y deben entender las limitaciones del equipo.
- El operador debe caminar el área y/o revisar planos de construcción y el plan del sitio para asegurar el entendimiento de las áreas de utilidades, excavaciones, cimientos, obstrucciones, etc.
- El manual del operador debe estar libremente disponible en el equipo, y el operador debe revisar y entender los requerimientos que estén explicados en los términos generales.
- El área alrededor de la válvula del agua debe ser diseñado para usuario del camión de agua solamente.
- Debe haber medidas para proteger las válvulas de agua; desconexión rápida o dispositivo de ruptura es recomendable.
- Fundas deberían ser proporcionadas durante el uso del tanque de agua con la válvula de agua en el techo para minimizar y asegurar que el suelo permanezca lo más seco posible alrededor del tanque.
- Operador debería ser entrenado y poseer una tarjeta de entrenamiento del control de polvo.
- Operador deberá entender el horario, áreas de prioridad y asistir en mantener y consignar en el cuaderno del agua del proyecto.

Inspecciones

- Se debe hacer inspección del equipo conforme los reglamentos del manual del operador. Por lo mínimo, asegúrese que lo siguiente sea inspeccionado:
 - o Caminar alrededor del vehículo para revisar nuevos daños hechos a la carrocería o cuerpo exterior.
 - o Asegúrese que la cabina este limpia y libre de materiales y escombros.
 - o Asegúrese que el manual de operador este en la cabina y esté en condiciones legibles.
 - o Revise las llantas por cortaduras, bolas, fragmentos, y que estén infladas apropiadamente.
 - o Revise que los fluidos estén al nivel apropiado y que no haya goteos presentes.
 - o Revise que los cintos, presión de aire, filtros de aire, etc. Estén en buenas condiciones.
 - o Todas las guarniciones deben estar engrasados adecuadamente.
 - o Revise que no haya soldaduras quebradas, quebraduras en el metal/pintura o daños obvios al equipo de pulverización de agua.
 - o Espejos, luces, direccionales, claxon, (manual) breca de mano y alarma de marcha en reversa deberán ser operables.
 - o Extintores deberán estar en el troque, en buenas condiciones y al corriente con las inspecciones mensuales y anuales.

Operación

- Cuando se esté subiendo o bajando, se debe mantener los tres puntos de contacto con la maquinaria en todo tiempo.
- Use los procedimientos apropiados para prender la maquinaria, consulte el manual del operador para requerimientos específicos del equipo.
- Haga la inspección del equipo, *usando la forma de inspección del equipo*.
- Permita que el equipo se caliente antes de operar la maquinaria.
- Se debe entender claramente el peso impuesto y como afectaran las excavaciones adyacentes y/o otras estructuras.
- Asegúrese que TODO el personal y otros equipos estén libres del área de trabajo, y este alerta a otras personas y equipo cercano.
- Tenga en cuenta la velocidad de la maquina con las condiciones del terreno.
- Cinturones deben ser usados en todo tiempo cuando el vehículo este andando.
- No deje la maquinaria encendida si está retirado más de 15 pies del vehículo.
- Asegúrese que tenga buena visibilidad cuando este moviendo de reversa para conectarse con la estación de agua, o use una persona que lo esté guiando.
- No use herramientas para poner o quitar las válvulas de agua, se deben cerrar o abrir a mano solamente.
- Verifique que el equipo este apropiadamente desconectado de la válvula de agua después de llenar y ANTES de moverse hacia el frente.
- Todos los ocupantes deberán usar los cinturones de seguridad. Si el cinturón de seguridad no está disponible o no en condiciones de funcionamiento, ningún pasajero estará en el vehículo.
- Operador deberá entender los procedimientos del proyecto para la limpieza de aceite y líquidos en caso de fugas o derrames.

Firma de Empleado

Firma de persona hacienda la verificación

Una copia de esta forma debe ser mandada por fax a la oficina regional de DPR ANTES DEL EMPIEZO DE TRABAJO

Section 16

Personal Protective Equipment

Sub-Section	Topic	Page #
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6.0	Respiratory Protection (<i>Reference Section 19 – Respiratory Protection</i>)	5
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Personal Protective Equipment

1.0 Introduction

Engineering controls should be the primary methods used to eliminate or minimize hazard exposure in the workplace. When such controls are not practical or applicable, personal protective equipment should be employed to reduce or eliminate personnel exposure to hazards. Personal protective equipment (PPE) will be provided, used, and maintained when it has been determined that its use is required and that such use will lessen the likelihood of occupational injuries and/or illnesses.

This plan is designed to aid in the evaluation and determination of proper personal protective equipment. The following requirements and guidelines apply to all DPR and contractor employees, tiered subcontractors, vendors, suppliers and visitors. Any personal protective equipment not considered to be everyday (street) clothing should be provided to all employees and purchased by their employer. All associated costs are the respective employer's responsibility.

Reference: Personal Protective and Life Saving Equipment CFR 1926 Subpart E; CFR 1910 Subpart I. Other applicable Standards, guidelines, and regulations may apply.

2.0 General Guidelines

Engineering controls or administrative procedures should be utilized whenever possible to eliminate or minimize the exposure to workers. When hazards cannot adequately be controlled by means of engineering controls or administrative procedures, then the use of PPE is required. Personal protective equipment should be worn as required by federal, state, or local rules, as well as facility policy, DPR's EHSP, manufacturer's recommendations and Safety Data Sheets (SDSs), whichever is more stringent.

Failure to comply with the various PPE requirements may result in disciplinary procedures, including immediate termination of work activities, removal from the work site and/or termination of employment / contract. Under no circumstance should work continue without the use of proper PPE required for the job.

- The expectation is that a hazard assessment should be performed, as part of a JHA or at a minimum during pre-task planning, to determine the level and type of PPE required. The choice of PPE should at minimum conform to the DPR EHSP. The PPE should provide the appropriate level of protection for the task and be included within the written JHA and PTP.
- All DPR employees, subcontractors, tiered subcontractors, vendors, suppliers or visitors, during the course of employment, contract involvement, or visit to the project should use appropriate work-related PPE
- The employer should provide to its workers all PPE required to perform their work
- Workers have the responsibility to obtain and wear personal protective equipment as required
- Appropriate attire should be worn while on DPR projects. This includes:
 - shirts with sleeves that fully extend 4" past the shoulder
 - durable, long pants that reach from the waist to the top of the foot (sweat pants are not permitted)
 - suitable work boots as defined below
- Types of PPE that may be worn include, but are not limited to:
 - head protection (hard hats)
 - hearing protection (ear plugs, muffs etc.)
 - respiratory protection (respirators, nuisance masks etc. – refer to Respiratory Protection Section 19)
 - eye and face protection (glasses, goggles, face shields, shades etc.)
 - hand and arm protection (gloves, Kevlar sleeves, gauntlets etc.)
 - fall protection (refer to Elevated Work Section 07)
 - protective clothing (Tyvek type coveralls, welding bibs, chemical suits etc.)

- high visibility apparel (safety vests or clothing, etc.)
- knee and leg protection (chaps, knee pads etc.)
- foot protection (toe caps, neoprene, metatarsal guards, over-boot protection, etc.)

3.0 Head Protection

Hard hats are to be worn at all times on DPR projects, for the duration of the project. The hard hat should be adjusted to fit properly at all times and be worn in accordance with manufacturer's recommendations; no alterations are allowed. Hard hats should meet ANSI requirements.

The following general guidelines apply:

- Hard hats are to be worn with the bill facing forward and the suspension worn in manner approved by the manufacturer
- Hard hat stickers should be ¼" away from the outer edge of the hard hat
- To improve communication on a jobsite – names on hard hats are recommended
- Hard hats not meeting this standard should be immediately removed from service

Hard hats come in three classes, and should be worn as applicable to the scope of work being performed:

- Class G – General (formerly Class A) hard hats are made from insulating material for protection from falling objects and electric shock by voltages of up to 2,200 volts
- Class E – Electrical (formerly Class B) hard hats are made from insulating material for protection from falling objects and electric shock by voltages of up to 20,000 volts
- Class C – conductive hard hats are designed for protection from falling objects, but are not designed for use around live electrical wires or where corrosive substances are present

4.0 Hearing Protection

Hearing protection is required whenever noise levels exceed OSHA guidelines (see table below) or it is determined that the hazard is significant enough to warrant hearing protection.

- Use earplugs or earmuffs when noise prevents conversation in a normal voice at a distance of 3 feet; this is a "rule of thumb" that indicates noise levels are exceeding 85 decibels
- Double protection (e.g., earplugs and muffs) may be needed for excessively high noise level activities, i.e., sandblasting, jackhammering, pile driving, etc.
- Signage should be posted for any area where the possibility of sound levels exceeding 85 decibels exists
- Noise assessment surveys should be conducted as appropriate

Examples of work that require hearing protection include, but are not limited to:

- Grinding operations
- Heavy equipment operation
- Powder actuated tools
- Chop saws/ cutoff saw
- Roto-hammering
- Manufacturing areas
- Other hazardous noise creating activities and environments

Note: Workers in areas adjacent to the above listed activities may also require the use of hearing protection.

<i>Duration – Per Day Hours</i>	<i>Sound Level dBA Slow Response</i>
8	90
6	92
4	95
3	97
2	100
1 ½	102
1	105
½	110
¼ or less	115

5.0 Eye and Face Protection

- Eye protection should be worn when required by federal, state and local regulations, owner, DPR's EHSP, a manufacturer, the SDS or when hazardous conditions exist.
- Visitors should abide by the same requirements for protective eyewear as those that apply to site personnel.
- FDA-approved, impact hardened "street glasses" should not be used in place of ANSI 87.1-1968 approved industrial safety glasses.
- Personnel who wear prescriptive lenses and are also required to wear safety glasses should be provided with goggles or over-the glass safety glasses.
- Contact lenses should not be worn in working environments where there is a potential for exposure to hazardous substances, flying dusts or light flashes.
- Face shields, in addition to other eye protection, should be used to provide added protection to the face and neck from flying particles, sprays of hazardous liquids, splashes of molten metal, extremely hot/cold solutions, use of chop saws or other hazardous operations identified by the project management team and DPR EHS or SDS.
- Face shields are not a primary form of eye protection; they are useful only as additional protection over basic eye protection, therefore, if face shields are utilized, safety glasses or goggles should also be worn.
- Face shields may also be used to provide anti-glare protection.
- Where workers may be exposed to splashes of acids, alkalis or other hazardous liquids/chemicals, chemical resistant hoods and/or goggles should be used; for severe exposures, a face shield should be worn in combination with the chemical-resistant goggles or hood.
- When performing overhead work such as drilling, cutting or other particulate generating tasks, face shields, goggles or a combination of the two should be worn in lieu of basic eye protection.
- Refer to Hazardous Energies Section 12 and the most current NFPA 70E edition, for PPE requirements related to energized electrical work.
- Employees involved in welding operations should wear filter lenses or plates of the proper shade number.
- Eye protection with the proper tint as outlined by OSHA should be used during all welding, torching and other operations involving extreme heat; refer to the table below:

Hot Work Operation	Shade Number
Shielded metal-arc welding 1/16-, 3/32-, 1/8-, 5/32 - inch diameter electrodes	10
Gas-shielded arc welding (nonferrous) 1/16-, 3/32-, 1/8-, 5/32 - inch diameter electrodes	11
Gas-shielded arc welding (ferrous) 1/16-, 3/32-, 1/8-, 5/32 - inch diameter electrodes	12
Shielded metal-arc welding 3/16-, 7/32-, 1/4 - inch diameter electrodes	12
5/16-, 3/8-inch diameter electrodes	14
Atomic hydrogen welding	10-14
Carbon-arc welding	14
Soldering	2
Torch brazing	3 or 4
Light cutting, up to 1 inch	3 or 4
Medium cutting, 1 inch to 6 inches	4 or 5
Heavy cutting, over 6 inches	5 or 6
Gas welding (light), up to 1/8-inch	4 or 5
Gas welding (medium), 1/8-inch to 1/2-inch	5 or 6
Gas welding (heavy), over 1/2-inch	6 or 8

6.0 Respiratory Protection

Refer to the Respiratory Protection Section 19 for requirements.

7.0 Hand and Arm Protection

Hand protection should be worn any time the nature of the work has the potential to cause a hand and/or finger injury. All gloves should be inspected, worn and stored in accordance with manufacturer's recommendations. Store gloves and sleeves/gauntlets in a designated location when not in use to protect from damage; such areas should be cool, dark and dry. Refer to the following general guidelines:

- The correct glove for the job should be used.
- Gloves should be appropriately sized to fit the individual's hand.
- Fabric gloves should be used for mild heat or cold.
- Appropriately rated rubber gloves with insulated liners should be worn for electrical work; these gloves are color coded for their resistance to different types of voltage. (Refer to Hazardous Energies Section 12 and the most current NFPA 70E edition, for PPE requirements related to energized electrical work.)
- Chemical resistant gloves should be worn when handling chemicals or corrosives, e.g., rubber, neoprene, vinyl, etc. depending upon the chemical.
- Leather gloves should be worn for work with sparks, rough or abrasive materials, scraping, debris removal, material handling, general site cleanup, etc.
- Metal mesh or kevlar gloves should be worn when working with saws, knives, axes and materials with sharp edges e.g., metal stud, glass, sheet metal, spiral duct, cleanup of such items, etc.
- Flame-retardant gloves should be worn around open flames.
- Reflective gloves, such as aluminized gloves, should be used for intense, radiant heat.
- Anti-vibration gloves should be worn when appropriate, e.g., jackhammering, roto-hammering, screw guns, scabbling, etc.
- Kevlar sleeves should be worn for activities that expose the arms to cuts, punctures, etc., including but not limited to; above ceiling work, demolition activities, metal stud framing, etc.
- Full leather gauntlets and bibs should be worn for welding activities.
- Latex or surgical gloves should be worn for protection from blood borne pathogens, infectious materials, etc.

- Gloves should not be worn around moving equipment or machine parts, e.g., fan motors, rotating axles, belts, drill presses, augers, etc.

8.0 Fall Protection

Refer to the Elevated Work Section 07 for requirements on fall protection equipment.

9.0 Foot Protection

Substantial leather work boots that cover the ankle should be worn by all individuals on the project.

- Work boots soles should be:
 - puncture resistant
 - slip resistant
 - in good condition
- Sneaker type work boots are not permitted.
- Full foot protection, i.e., toe caps that provide metatarsal protection should be worn for operating tampers, jackhammers, etc or in exposure to heavy objects or potential crush injuries.
- Chemical resistant boots should be worn to prevent skin contact with irritant and/or hazardous chemicals.
- The worker should wear footwear as required by owner, facility or SDS.

10.0 Miscellaneous PPE

- Workers exposed to vehicular traffic should wear high visibility safety apparel in accordance with the ANSI/ISEA 107-2004, High Visibility Safety Apparel and Headwear (i.e., Class 2 or Class 3 vests with retro-reflective stripes).
- Employees working over or near water where danger of drowning exists should wear U.S. Coast Guard approved life jackets or buoyant work vests.
- The following general guidelines apply to protective clothing:
 - full or partial chemical suits should be worn for exposure to irritants or hazardous chemicals
 - Nomex, Tyvek, blast suits and other similar types of suits should be worn as appropriate e.g. energized electrical work, hazardous substances, infectious materials, bloodborne pathogens, etc.
- Knee pads should be worn when kneeling or crawling for extended periods of time.
- Chaps, aprons or other similar protective clothing should be worn as appropriate e.g. material handling, chainsaw use, glass work, reinforcing steel, pressure washing, compressed air use, cutting operations, etc.
- Refer to the Hazardous Energies Section 12 for PPE requirements related to energized electrical work.

Personal Protective Equipment Forms

Forms

There are currently no forms in this section



Section 17

Pre-Planning

(DPR & Subcontractors)

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7.0	Disruption and Access Request (DAR) Process	13

Pre-Planning

1.0 Introduction

DPR's intent is to eliminate incidents, accidents and business interruptions by utilizing and standardizing best known methods. Through planning, DPR strives to ensure all tasks are performed incident free and meet project specific quality standards. The following should, at a minimum, be considered in the planning process during pre-construction safety meetings, subcontractor coordination meetings, etc.:

1. Work physical requirements
2. Work environmental conditions
3. Personal protective equipment required
4. Sequence of basic job steps
5. Potential accident or hazards associated with each step
6. Safe work practice for each step, in order to mitigate potential risks and hazards

2.0 DPR Clinic Selection & Introduction

Our mission is to have a Safe and Injury Free Environment; however, there are occasion when employees need medical attention outside of what DPR can provide. In this instance, the project team, along with the safety and claims managers, need to meet the physician establish a relationship with an industrial/occupational medical clinic within close proximity to the jobsite (the closest clinic to the jobsite does not always mean that is the right fit for DPR). The clinic selection and introduction process is imperative to our OSHA sensitivity and overall outcome of claims. Creating relationships with physicians and their staff allow for better, stronger communication.

Note: This process should be done prior to mobilization to ensure the proper information is disseminated to the project team.

2.1 Medical Provider Shared Vision

Sometimes we fall short of our mission and when injuries do occur our number one priority then becomes providing the best medical treatment for our workers. To accomplish this, we must have an established relationship and a mutual understanding, what we like to call a "Shared Vision," with our designated medical facilities. Together we must be innovative, creative, and aggressive. Communicating our expectations is essential in developing this Shared Vision. We believe the stronger the relationship with our clinics, the better results we will achieve.



 DPR
CONSTRUCTION

Medical Clinics and DPR Construction

Heidi Marsac, Claims Manager - Desk: 916-418-1910 Fax: 916-848-0289 Cell: 916-826-9986

Our mission is to have a Safe and Injury Free Environment.

Sometimes we fall short of our mission and when injuries do occur our number one priority then becomes providing the best medical treatment for our workers. To accomplish this, we must have an established relationship and mutual understanding, what we like to call a "Shared Vision," with our designated medical facilities. Together we must be innovative, creative, and aggressive. Communicating our expectations is essential in developing this Shared Vision. We believe the stronger the relationship with our clinics, the better results we will achieve.

Here are some areas when a team approach can make the difference:

- For minor strains, recommend conservative care such as: ice, home therapy and over-the-counter medications versus prescribing physical therapy, prescription medications, and multiple follow up appointments. We would prefer that follow up and discharge appointments are one and the same, thus allowing a first-aid case. We understand that this may not be possible in every case. If the worker has increased pain or symptoms, we will follow up and interest, this may not be a firm



3.0 Safety Start Up Checklist

It's DPR's intent that all projects are planned to ensure successful completion. The project team members should meet to establish roles and responsibilities for specific safety related items. The Safety Start Up checklist is intended to be used as a tool to help guide the team as they set up their job at mobilization.

4.0 Subcontractors

This (Sub) Contractor Safety Program is designed to protect company and contractor employees, equipment and facilities from injury, accident or loss. Contractors are persons not directly employed by the company who provide specific labor or services.

DPR Construction's expectation is that every subcontractor and tiered-subcontractor participates in a preconstruction safety/EHS meeting to ensure consistent understanding of the project's EHS requirements, and Insurance requirements prior to mobilization. The following attendees should be present for this meeting; Subcontractor Project Manager, Subcontractor Superintendent, Subcontractor Safety Manager, Subcontractor onsite full time lead person, DPR Project Manager, DPR Superintendent, DPR Foreman, and DPR EHS representative.

Subcontractor participation should not end at the preconstruction meeting. DPR is looking for full collaborative efforts with our subcontractors as they are an integral part of DPR's projects and culture. Our subcontractors should be actively engaged at all times. It is required that all employees attend a job site orientation to ensure that employees working on site are up to speed with DPR Construction's requirements and various jobsite nuances. Our subcontractors are expected to participate in DPR's daily Stretch and Flex program as well as participate in Injury Free Environment (IFE) training when performed.

5.0 Pre-Task Planning

5.1 Scope

Daily, written, pre-task planning is required for all work activities. All activities should be pre-task planned and reviewed and/or updated as tasks and/or conditions change (PDCA). Additionally, a Job Hazard Analysis (JHA – *reference 3.0 in this Section*) and /or written plans with greater detail should be prepared for any high hazard activity. These activities include, but are not limited to:

- Chemicals: hazardous & irritant
- Concrete: pre-cast, tilt up, vertical, form work
- Confined Space
- Crystalline Silica (*re: site specific Silica Exposure Control Plan – reference Section 2.0*)
- Hoisting/rigging activities: including cranes, derricks, forklifts, straddle buggies, etc.
- Demolition activities & hazardous materials assessment: asbestos, lead, biohazards or other chemicals in the workplace, as well as general demolition hazard assessment
- Framing activities (All, including drywall)
- Excavation & trenching
- Fall hazards: exposures 6+ feet, overhead work
- Material handling
- Public exposure: phased occupancy, partial demolition, traffic control, etc.
- Scaffolding
- Steel erection
- Start Up/Shut Down/System Testing activities: tool hook up, introduction of process chemicals into systems, utility tie ins, Lockout/Tagout, work on energized equipment
- Non-routine activities: activities not performed in the previous six months

5.2 Pre-Task Planning Process

Prior to start of work, the crew/individual assigned to perform the task prepares the pre-task planning worksheet. The crew or individual should physically walk the area the same day the work is to be performed to identify potential hazards.

The top of the form should be completed with the date (daily), identify the name of the project, the company performing the work, and the specific location the work will be performed. If work is to be performed in multiple locations, individual pre-task plan forms may be needed. The box at the top of the page is for the DPR project team to audit the pre-task plan and provide coaching and input to the foreman & crew performing the work.



Pre-Task Plan

Archive Document – DO NOT DISCARD

Audited by: _____
 Date: _____
 Time: _____

Project	ABC Project	Date	xx - xx - 2017
Project Number	D2-12345-00	Company	Company D
Specific Location of Work	Penthouse Area, Section B		
Answer the following when evaluating your work:			Circle YES, NO or N/A
Prior to start, have you walked your work area to address lighting, housekeeping, slip/trip issues, etc.?			

The questions at the top of the form should be discussed and agreed upon by all members of the crew. If a question does not apply, mark “N/A”.

Answer the following when evaluating your work:	Circle YES, NO or N/A		
Prior to start, have you walked your work area to address lighting, housekeeping, slip/trip issues, etc.?	YES	NO	N/A
Has a Hazardous Material Survey been conducted on the project/clearance records? (asbestos, lead, PBC's, etc.)	YES	NO	N/A
Is there a new hire, or new team member on the project who will need support?	YES	NO	N/A
Are enough people assigned to safely complete the task? (lifting, repetition, spotters etc.)	YES	NO	N/A
Are there any hazards created by any other workers in your area or does your work create hazards for others?	YES	NO	N/A
Does your task require the use of a personal fall arrest system? Has a rescue plan been developed and communicated to all crew members?	YES	NO	N/A
Are you working around live systems or energized equipment? Will you need to use Lockout/Tagout procedures?	YES	NO	N/A
Does your work require you to be exposed to pinch points, cave-ins, articulating equipment (caught in-between); falling or flying materials or debris, vehicular traffic, moving equipment (struck by).	YES	NO	N/A
Are operators certified/trained/authorized for the equipment they are operating? (Scissor lift, powder actuated tools, forklift, mobile equipment, rigging, etc.)	YES	NO	N/A

Question: Prior to start, have you walked your work area to address lighting, housekeeping, slip & trip issues, etc.?

Example: Walk entire area involved the same day as work is to be performed, including areas above and below, identifying all potential hazards.

Question: Has as a Hazardous Material Survey been conducted on the project/clearance records? (asbestos, lead, PBC's, etc.)

Example: If hazardous materials were in the work area, a hazardous materials survey should be performed, with results shared with the applicable project team members.

Question: Is there a new hire or new team member on the project who will need support?



Example: Are there newly hired employees within the crew? Have they completed DPR site specific safety orientation? Are all crew members aware of the level of experience of these people?

Question: *Are enough people assigned to safely complete the task?*

Example: Have sufficient personnel been scheduled to safely perform the task? Is a buddy system, spotter, flag personnel, point communications, etc. needed?

Question: *Are there any hazards created by any other workers in your area or does your work create hazards for others?*

Example: Verify the work you will be performing is compatible with work being performed by other crews, e.g., hot work next to PVC gluing operations, chemical incompatibilities, etc. Have all crews in the immediate area been notified of work to be performed? Has work been coordinated to ensure no adverse impacts to the work of others? Is the work area congested with other crafts? If yes, report to supervision. Would scheduling alternative times for work to be performed resolve over-congestion issues?

Question: *Does your task require the use of a personal fall arrest system? Has a rescue plan been developed and communicated to all crew members?*

Example: Are sufficient anchorage points available for the work to be performed? Has the fall systems to be used been inspected and have they been installed and reviewed by competent person? Should a fall occur, has a plan been developed to safely and efficiently retrieve the worker from his fall system?

Question: *Are you working around live systems or energized equipment? Will you need to use Lockout/Tagout procedures?*

Example: Is there the potential to strike or bump valves, contact electrical shock hazards, disturb or break chemical lines, etc.? If the hazard is electrical in nature, an Energized Electrical Work permit will be required. If Lockout/Tagout is required, ensure a qualified electrician performs the work, and complete the Lockout/Tagout checklist prior to start of work.

Question: *Does your work require you to be exposed to pinch points, cave-ins, articulating equipment (caught in-between); falling or flying materials or debris, vehicular traffic, moving equipment (struck-by)*

Question: *Are operators certified/trained/authorized for the equipment they are operating? (Scissor lift, powder actuated tools, forklift, mobile equipment, rigging etc.)?*

Have training requirements been reviewed for work to be performed? If training requirements exist, are all affected crew members appropriately trained? Is training current, and are affected members able to demonstrate proficiency?

Question: *Does this task require any special permits, procedures or inspection forms? (Confined Space, Hot Work, Excavation, Elevated Work, Energized Electrical Work, Scaffold/Scissor/Boom/Forklift inspections, etc).*

Example: Identify required permits and procedures and ensure all are in place prior to start of work. Have training requirements been reviewed for work to be performed? If training requirements exist, are affected crew members appropriately trained? Is training current, and are affected members able to demonstrate proficiency?

- *Confined Space Permit* – Is work to be performed in a confined space? Are all team members trained in confined space entry? Refer to DPR EHSP for additional details.
- *Hot Work Permit* – Will work being performed produce flames or sparks?
- *Excavation/Trenching* – Has the initial excavation/trenching assessment been completed? Are daily inspections being performed and documented?
- *Energized Electrical Work* -- Will work to be performed require systems to remain live? Has the work been approved by DPR project management?



- *Scaffolding/Scissor/Boom/Forklift Inspection Forms* - Has all equipment been inspected by a qualified person? Inspections should be documented prior to start of work.

Question: *Do you have the right type of work platform or equipment to reach your work? Have you been trained to use this equipment?*

Example: Do all crew members understand safe ladder procedures? Do they understand our elevated work policy and fiberglass ladder policy? Have all crew members completed ladder training? Is the ladder or scaffold to be used of sufficient capacity to hold any/all anticipated loads? (only type 1, 1A or 1AA fiberglass ladders are allowed) Is the ladder the correct height to reach the area of work? Is the scaffolding of sufficient weight and construction to hold any/all anticipated loads? Are affected crew members appropriately trained in scaffold use? Is training current, and are affected members able to demonstrate proficiency?

Question: *Do you need to review SDSs (safety data sheets) to proceed with this work?*

Example: Do all crew members understand the potential impacts of chemical exposure? Are chemicals to be used new to any crew members? Have all crew members received HAZCOM training and chemical specific information? Are SDSs readily available for reference?

Question: *Have you addressed any barricading, warning system or signage requirements appropriate to the task?*

Example: Are other crews working in the area? Have you notified all who may potentially be impacted by the work? If barricading is needed, have you identified all areas, which should be barricaded to provide complete protection? Does the work involve traffic control? Should red or yellow barricade tape be used? Have you prepared ownership signage to be attached to all sides of the barricading? Is the signage in both English and Spanish?

Question: *Have all the tools, equipment and materials been inspected prior to use and are they adequate to perform the work safely?*

Example: Have all tools been inspected and have all defective or damaged items been corrected or removed from service? Are GFCIs available, or is assured grounding program in compliance? Are inspection checklists being used for forklifts, scissor lifts, etc. If tools/equipment are to be used within the cleanroom, have all protocol requirements been met?

Has a thorough inventory of needed tools been completed? Will all tools be readily available to complete work? If not, notify supervisor to ensure appropriate tools are available for use.

Question: *Will this task require any lifting, bending or twisting?*

Example: Is the object too big or bulky to do a squat lift (bent legs, straight back)? Will you have to carry an object on only one side of your body? Will the task be repetitive? Do you have a co-worker to help with heavy lifting? Can your work or your body be re-positioned to reduce the amount of bending or twisting?

Question: *Have you completed Stretch & Flex today?*

Example: If you are a DPR employee, have you been trained in Stretch & Flex? Have you completed the exercises that address the parts of your body affected by the work to be performed?

Question: *Do you have an injury to report or were you injured the prior working day?*

Example: Have you been injured on the job? Report any injuries immediately to your supervisor, no matter how minor.

Quality Questions

Question: Identify the drawing you are working from today, is it the current version?

Example: Has the crew been provided with the correct drawings to perform their work that day?

Question: Have you reviewed all construction details associated with our work?

Example: Have the crew members been walked through the construction details that will ensure work is installed right the first time? What are those details that each crew member should understand? Does any special training or review need to occur to ensure work is installed correctly?

Question: Who on the crew is responsible for quality control today?

Example: How will the crew ensure quality control occurs? Has one person on the crew been identified to verify details, install, etc?

Question: What is the quality item you will be focusing on today? What will you do today that will prevent rework tomorrow?

Example: What is the most critical portion of the work for the day that, when installed correctly, will ensure rework is not required later to correct? What can the crew focus on to ensure work is installed correctly the first time?

Identify the PPE to be used for the task being performed; mark the applicable PPE to be used for the *specific* hazards

Are any of the following PPE required?	
☒	Hard Hat/Safety Glasses
☐	Fall Protection/Rescue Plan
☒	Goggles/Faceshield
☒	Hand / Arm PPE
☐	Hearing PPE
☐	Foot PPE
☐	Respirator

Identify emergency equipment and exit routes; identify where equipment is located in the work area and the best route out of the work area. Note: if project specific

Locate and identify:	
☒	Emergency Telephones house phone - 7333
☒	Fire Extinguisher
☒	Emergency Exit Routes
☒	First Aid Equipment in DPR office
☐	Other
☐	Other
☐	Other

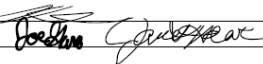
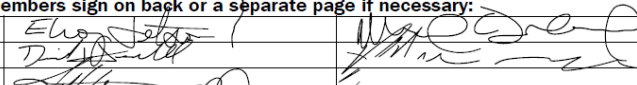
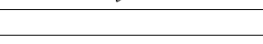
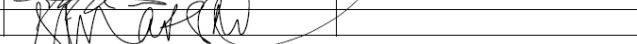
On the back side of the form:

Describe task to be performed, i.e. framing priority wall.

Steps for Work	Tools	Hazards	Steps Taken to Address Hazards
Stretch & flex, complete PTP, gather tools and materials	drill, tie wire, rebar, saws, nails, lumber, hand tools, extension cords, saw horses	Pulled muscle, cuts, improper lifting, pushing/pulling,	Do not overstretch muscles, use proper PPE, use proper lifting mechanics (lift with knees not back), don't force pushing/pulling, cut materials on elevated surface, route cords to prevent tripping
Layout concrete curbs in penthouse area, identify where holes will be drilled, barricade area	tape measure, chalk line, caution tape	Slip/trip hazards, hazards from other trades in area	Walk area to i.d. slip/trip or other hazards in the area, caution tape off work area to warn others of potential hazards
Drills holes and cut rebar, install rebar with epoxy, tie all special cut rebar in place, cap/protect impalement hazards	concrete drill, chop saw, epoxy, tie wire, rebar caps, extension cords	Cuts, strained muscle tying rebar, trip hazards, loose material caught in drill bit, flying debris, impalement hazards	Take frequent breaks from bending over tying rebar, ensure guard is on saw, do not wear loose fitting gloves using drill, cap rebar as it is placed, wear proper eye protection when drilling

- Fill in the boxes at the bottom of the form:
 - Identify steps for work, in order. Use as many line items as needed to adequately plan the work. The back of the form or a separate sheet of paper may be used if necessary, i.e., *an example for framing*:
 1. attach top & bottom track, *ensuring door rough openings are accurate on bottom track*
 2. cut studs to length
 3. install studs
 4. . . .
 5. . . .
 6. Cleanup (i.e., last step/ leave the area ready to hand over to the next contractor)
 - For each of the steps of work, identify tools that will be used for each step – remember that PPE is addressed in the boxes at the side of the form, this sub-section is specific to tools & equipment (using the example above):
 1. Gas or powder actuated tool, rolling scaffold, etc.
 2. Chop saw, fire extinguisher, cut station, etc.
 3. Screw gun, drill, c-clamps, etc.
 4. . . .
 5. . . .
 6. Gondola, brooms, shovels, cut-off boxes or debris baskets, floor sweep, etc.
 - Identify actual or potential hazards associated for each step of work
 1. Working at heights/overhead, existing MEP systems, concrete spauling, etc.
 2. Noise, flying metal shavings, sharp edges, fire, etc.
 3. Sharp edges, fall/unsecured studs, studs properly staged (i.e., no leaning) etc.
 4. . . .
 5. . . .
 6. Dust, sharp objects overhanging gondola, lifting, bending, etc.
 - For each hazard identified, indicate the steps that will be taken to address the hazard:
 1. Fall protection, live systems marked, faceshield, hearing protection,
 2. Properly functioning guard, support for long material, spark shield/backstop, etc.
 3. Cut resistant gloves/sleeves, secure materials with C-clamp, studs
 4. . . .
 5. . . .
 6. Floor sweep, hand & arm protection, materials straight to bin/debris basket, materials cut/placed inside gondola, etc.

*If you need more space, attach another PRE-TASK PLAN SHEET

Work Planner	Mike Black	Foreman	Joe White	Safety Coordinator	Frank Fellow
Crew Members – additional crew members sign on back or a separate page if necessary:					
					
					

- Supervisor, safety coordinator, superintendent, project engineer, etc., may help to identify potential hazards and effects on systems
- Once the form is complete, the crew and supervisor jointly discuss the work sequencing to be performed and review hazards identified to verify completeness of the plan, and assign individuals to perform the given tasks identified; supervisors are responsible for communication and coordination activities
- Supervisor and crew members sign the form to indicate understanding of the task to be performed and potential hazards involved with task completion
- The completed form is posted in a prominent location within the work area, and remains posted for the duration of the task

Note 1: *If performance of the task requires deviation from the posted plan, the crew should immediately cease work and notify supervision. Supervisor and crew should reassess the plan, make necessary modifications, re-communicate any additional changes to all crew members, initial the worksheet and post the modified plan before resuming work.*

Audited by: 

Date: xx-xx-2017

Time: 8:17am

- DPR project management/ DPR EHS should routinely audit, coach and provide feedback to the crews on the pre-task planning process in the work area(s). Utilize the audit box provided at the top of form.

Note 2: *Should an incident/accident occur during the course of work, the pre-task planning form is expected to be utilized during the investigation and root cause analysis process.*

- Upon completion of the work, the PTP form should be submitted to DPR. Project generated pre-task plans are archived with project records

5.3 Pre-Task Planning Responsibilities

- Each crew member performing each task should:
 - Conduct walk down of area and complete the pre-task plan worksheet the same day work is to be performed
 - Discuss contents of the plan and make change as needed
 - Understand, sign and follow pre-task planning
 - Post the pre-task plan in the work area for duration of task
 - Remove signage and barricading upon completion of work
 - Submit pre-task plan to DPR at work completion
- Supervision:
 - Review the information on pre-task plan form with crew the same day work is to be performed

- Periodically audit the work to ensure compliance with the PTP and quality items, coach the crews to update/amend the plan should work or environment change (PDCA)
- Sign Pre-Task Planning worksheet (utilizing the audit box)
- Project Management:
 - Conduct periodic field audits to ensure that the pre-task planning is conducted properly
 - Periodically evaluate the program for effectiveness and continuous improvement opportunities
 - Update documents and/or process changes
 - Ensure all Pre-Task plans are archived at end of project

6.0 Job Hazard Analysis

6.1 Purpose

Job Hazard Analysis (JHA) is a tool that can be used by project teams and/or subcontractors to preplan high hazard tasks and procedures. Whereas a pre-task plan is task specific, the JHA is intended to be used as a high-level planning tool that will serve to breakdown an entire scope of work into tasks and define the potential hazards and controls for each task. The goal to eliminate hazards and/or re-work through successful planning and such methods as engineering controls, training, site and craft coordination, etc.

6.2 Job Hazard Analysis Scope

As required, written, detailed JHA's should be conducted for all high hazard activities for all scopes of work on DPR projects. Site specific requirements may require JHA's to be completed more frequently. Reference the example of a completed JHA below:

 Soft TI Demolition Plan Archive Document – DO NOT DISCARD		
DPR Construction: XYZ project Demo interior walls, cabinets		Date: xx/xx/2015
Sequence Of Basic Job Steps	Potential Hazards	Recommended Action Or Procedure
Review current as-builts, verify with electrician, and plumbing personnel if any live systems are within wall.	Electrocution, unnecessary water damage.	Walk the demolition area before the crew starts to verify safe conditions. Identify marking system to establish universal understanding of live vs dead systems, communicate to all crews. Verify crew members understand the marking system. Verify spill kits are available in the work area and that all crew members have been trained to use contents. Verify shutoff locations for all systems and communicate to crews.
Electricians complete safe-off for electrical in divider wall	Electrical contact, cuts, etc.	Prepare detailed safe-off plan, communicate to crews and follow.
Fire alarm personnel will move and/or protect sprinkler heads, but they will be live during demo.	Water damage, electrical, unnecessary alarms	Fire detectors will be covered during work hours and uncovered at finish of day. Reference spills kits/shutoff, locations, and training above.
Flag off area with barricade tape & signage	Fall, slips, trips Hazards	Flag off area of work to ensure pedestrian traffic is rerouted around area for demo, install fencing panels if possible, post signage.
Inspect tools & equipment prior to start. Verify that all tools are working properly. Inspect ICPA for	Unnecessary damage to tools	Verify GFCI in use for all tools when plugging into house power, inspect tools for proper function, inspect for damage, inspect ladder

6.3 Job Hazard Analysis Process

6.3.1 Sequence of Basic Job Steps

- Break the job down into individual basic steps. Each step should accomplish some major task. The step should begin with a verb or action work such as “hammer”, “turn” or “move”. The wording in each step should be concise and to the point.
- If a step cannot be written in one simple sentence, it may be more than one task item. Reference example below:

Sequence Of Basic Job Steps
Review current as-builts, verify with electrician, and plumbing personnel if any live systems are within wall.
Electricians complete safe-off for electrical in divider wall
Fire alarm personnel will move and/or protect sprinkler heads, but they will be live during demo.
Flag off area with barricade tape & signage
Inspect tools & equipment prior to start. Verify that all tools are working properly. Inspect ICRA for any deficiencies.

6.3.2 **Potential Hazards**

Potential Hazards are events which could occur during the individual step that may result in personal injury, equipment damage, or an environmental incident. For each work step identified in the first column, list the potential hazards. Items to identify may include or more of the following per step:

1. Caught in, under or between
2. Fall to same level
3. Fall to different level
4. Struck by flying object
5. Struck by falling object
6. Strike against a stationary object
7. Strains, sprains, or pulls from pushing, pulling, bending, lifting or twisting
8. Cuts, lacerations or contusions
9. Struck by moving objects
10. Overexposure to dust, fumes, mists, vapors or gases
11. Skin irritation from oils, solvents, soaps
12. Exposure to chemicals
13. High noise levels
14. Ionizing radiation (X or gamma rays, alpha or beta particles)
15. Non-ionizing radiation (illumination, infra-red ultra violet lasers)
16. Exposure to environmental extremes (hot or cold)
17. Ergonomic stresses
18. Fire or explosion hazard

Potential Hazards	
Electrocution, unnecessary water damage.	W
Electrical contact, cuts, etc.	F
Water damage, electrical, unnecessary alarms	F
Fall, slips, trips Hazards	F
Unnecessary damage to tools and injuries to crewmembers Infectious dust particulates and other harmful debris.	W F S W S

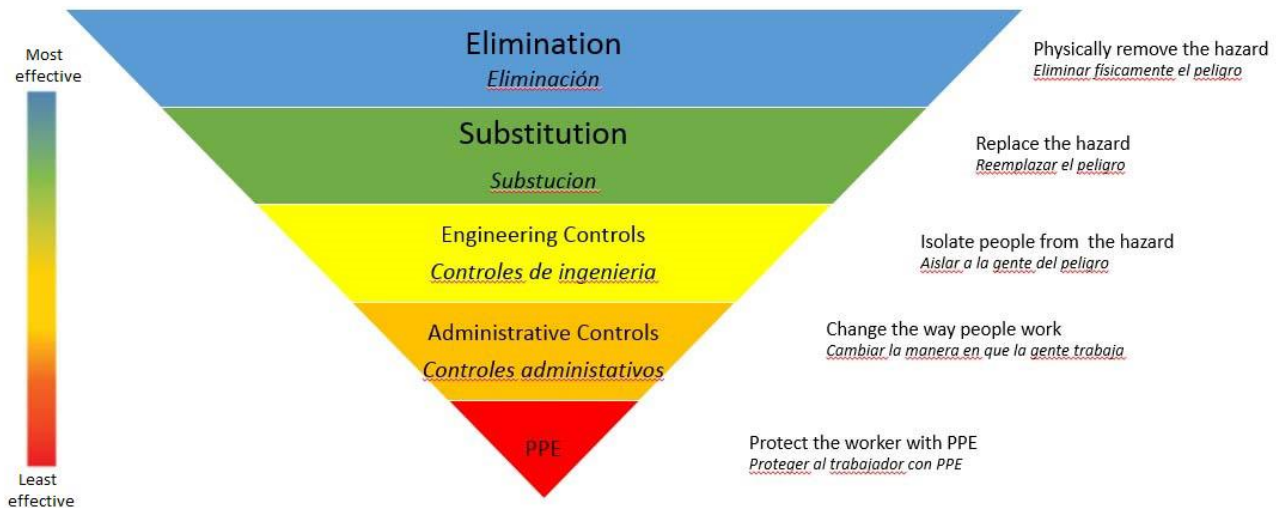
6.3.3 Recommended Procedures or Controls

For each potential hazard listed in the previous column list the actions that can be taken to reduce or eliminate the specific hazard. When completing this sub-section, the solutions must be specific and complete. “Stay alert” and “pay attention” are not acceptable measures to prevent injury.

Recommended Action Or Procedure
Walk the demolition area before the crew starts to verify safe conditions. Identify marking system to establish universal understanding of live vs dead systems, communicate to all crews. Verify crew members understand the marking system. Verify spill kits are available in the work area and that all crew members have been trained to use contents. Verify shutoff locations for all systems and communicate to crews.
Prepare detailed safe-off plan, communicate to crews and follow.
Fire detectors will be covered during work hours and uncovered at finish of day. Reference spills kits/shutoff, locations, and training above.
Flag off area of work to ensure pedestrian traffic is rerouted around area for demo, install fencing panels if possible, post signage.
Verify GFCI in use for all tools when plugging into house power, inspect tools for proper function, inspect cords for damage, inspect ladders, scissor lifts, scaffolding for damaged parts. Remove any equipment, tools, and cords from service if not in good condition. Ensure that the whole crew knows where eye wash unit and first aid kits are located. Ensure both are filled and contents are fresh/current. Walk the ICRA system once an hour or as necessary looking for air leaks as well as and damages that could compromise integrity.

For each step with an associated hazard, ask “is this step necessary to complete the job?” If it can be eliminated, it would eliminate the exposure to the hazard as well. If it cannot be eliminated, then consider the following by order of effectiveness; Substitution, Engineering, Administrative, PPE. This is shown in the hierarchy below.

Hierarchy of Controls



Elimination and Substitution

The elimination and substitution of hazards is most effective during the design stage.

Engineering Controls

Engineering controls are favored over administrative and personal protective equipment (PPE) for controlling existing worker exposures in the workplace because they are designed to remove the hazard at the source. Examples include, but are not limited to:

- Exhaust ventilation to capture/remove airborne silica dust
- Machine guards on cut off saws to shield workers
- Hoists for lifting or moving heavy items

Administrative Controls

While safe work practices can be considered forms of administrative controls, OSHA uses the term administrative controls to mean other measures aimed at reducing employee exposure to hazards. These types of controls are normally used in conjunction with other controls that more directly prevent or control exposure to the hazard, some include, but are not limited to:

- Additional relief workers
- Hydration and/or rest breaks
- Rotation of workers

Personal Protective Equipment (PPE)

When exposure to hazards cannot be engineered out of normal operations or maintenance work, PPE should be used to control hazards.

7.0 Disruption and Access Request (DAR) Process

7.1 Purpose

The purpose of the Disruption and Access Request Process (DAR) is to aid in the coordination between DPR, owner and subcontractors when areas or systems controlled by the owner may be affected and/or interrupted

during construction activities. This process allows the identification and planning of high hazard activities at start of project to identify those items requiring focused project team planning, review & monitoring to ensure safe execution of work and owner systems representative review and approval.

7.2 *DAR Pre-Planning Process*

During the project planning phase, DPR should identify work that may require interface with the owner and incorporate those activities into the project schedule. For each DAR activity identified, the subcontractor planning the work will develop a detailed, written job hazard analysis (JHA). The JHA will detail the steps of work to be performed, potential hazards and the subcontractor's plan for addressing each hazard. The JHA should include all required DPR and/or owner permits, checklists, inspections, procedures, etc. as well as a logistics plan for the area of work.

Prior to start of work, the JHA submitted, reviewed and accepted by DPR project management should be presented to designated owner's representative(s) for final review and acceptance prior to start of work. Work should be monitored by DPR project team member(s) prior to start of work and throughout the work being performed, and log for tracking and closing out DARs.

7.3 *Schedule Updates of DAR Activities*

7.3.1 *Identification of DAR Activities:*

- At start of project or initiation of new project phases, DPR project team and subcontractors should jointly identify those activities that trigger the DAR process.
- DAR activities should be clearly identified on the project schedule and shared with subcontractors.

7.3.2 *Periodic Review and Update of DAR Activities:*

- Periodic review of DAR activities should be completed by DPR project team using updated project schedule. Periodic review should be updated as deemed necessary by the DPR project team to ensure a safe, incident free project. Periodic review of DAR activities should include, at a minimum:
 - Any date changes or updates to the project schedule or activities
 - Review of recent add-ons and change orders
 - Prior safety performance by subcontractor and/or tiered subcontractor personnel
- Any change or addition of DAR activities should be analyzed for potential impact to work and/or DARs currently in progress or in the planning stages
- Updated DARs should be clearly identified on the project schedule and communicated to all applicable parties

7.4 *Development of DAR*

7.4.1 *Subcontractor DAR*

Subcontractors and/or their tiered subcontractors should develop a detailed, written plan of the identified DAR activity. The DAR should utilize the JHA format (reference sub-section 3.0) and should be submitted for review to the DPR project team a minimum of two weeks prior to the scheduled activity, or per owner requirements. In addition to the JHA requirements the DAR should include, at a minimum:

- Specific services, facilities and/or utilities
- Shut off points
- Timeline with key interface points of impact to critical systems, i.e., closing valves, opening/closing circuit breakers, etc.
- Coordination and utility provider integration, if applicable
- When applicable, all-party (i.e., owner, utilities, DPR, subcontractor, etc.) validation/sign-off
- Contingency plan

7.5 *DAR Checklist*

Upon completion of the detailed, written DAR, the subcontractor performing the work should attach a DAR checklist, JHA and any associated documents for approval and routing purposes to the DPR project team and owner representatives. Associated documents could include, but are not limited to:

- Concrete Pump Truck Pre-Planning Checklist
- Crane Pre-Planning Checklist
- Demolition Pre-Planning Checklist
- Excavation & Trenching Pre-Planning Checklist
- Hot Work Permit
- ICRA Pre-Planning Checklist and/or Inspection
- Lockout/Tagout Checklist
- Energized Electrical Work Permit
- Confined Space Entry Permit
- Scaffold Inspection
- Owner specific forms or documents

7.6 *DAR Review and Approval Process*

Subcontractor submits their completed detailed, written DAR, DAR checklist and associated documents (DAR package) to DPR project management for review. Should additional information be required, the subcontractor should be notified for needed updates.

Upon review and acceptance of the subcontractor's DAR Package, the DPR Project Team will review the entire package with the applicable Owner Representative and building/systems owners. Upon review and approval by the Owner Representatives and applicable building/system owners, the DAR package will be returned to the subcontractor lead assigned to perform the work. Upon receipt of the approved DAR Package, work may proceed.

7.7 *Field Work*

A copy of the approved DAR package must be reviewed with the entire crew who will be performing the work. Any questions or concerns must be answered and understood by the crew. Daily Pre-Task planning will be completed using the DAR package as reference. The Plan must be followed without deviation.

Should a change of plan occur, DPR project management should be notified and changes/updates made to the plan and approved by Owner Representative and building/systems owners before work may proceed.

7.8 *Lessons Learned*

At the close of each permit, a plus/delta session should be scheduled with the crew performing the work, DPR and all parties, if applicable. Plus/delta comments are gathered and lessons learned developed from each comment. Lessons learned are shared with applicable crews during subsequent planning sessions.

Pre-Planning / Subcontractor Forms

Forms

- Pre-Task Plan – English
- Pre-Task Plan – Spanish
- Job Hazard Analysis – English
- Job Hazard Analysis – Spanish
- Disruption & Access Request
- Safety Start-Up Checklist
- Safety/EHS Pre-Construction Meeting Checklist
- OSHA Training & Competent Person Requirements - Federal OSHA 1910 & 1926
- Scope Specific Safety Expectations



Pre-Task Plan

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Audited by: _____
Date: _____
Time: _____

Project		Date	
Project Number		Company	
Specific Location of Work			

Answer the following when evaluating your work:	Circle YES, NO or N/A		
Prior to start, have you walked your work area to address lighting, housekeeping, slip/trip issues etc.?	YES	NO	N/A
Has a Hazardous Material Survey been conducted on the project/clearance records? (asbestos, lead, PBC's, etc.)	YES	NO	N/A
Is there a new hire, or new team member on the project who will need support?	YES	NO	N/A
Are enough people assigned to safely complete the task? (lifting, repetition, spotters etc.)	YES	NO	N/A
Are there any hazards created by any other workers in your area or does your work create hazards for others?	YES	NO	N/A
Does your task require the use of a personal fall arrest system? Has a rescue plan been developed and communicated to all crew members?	YES	NO	N/A
Are you working around live systems or energized equipment? Will you need to use Lockout/Tagout procedures?	YES	NO	N/A
Does your work require you to be exposed to pinch points, cave-ins, articulating equipment (caught in-between); falling or flying materials or debris, vehicular traffic, moving equipment (struck by).	YES	NO	N/A
Are operators certified/trained /authorized for the equipment they are operating? (Scissor lift, powder actuated tools, forklift, mobile equipment, rigging, etc.)	YES	NO	N/A
Does this task require any special permits, procedures or inspection forms? (Confined Space, Hot Work, Excavation, Elevated Work, Energized Electrical Work, Scaffold/Scissor/Boom/Forklift Inspection, etc.)	YES	NO	N/A
Do you have the right type of work platform or equipment to reach your work? Have you been trained to use this equipment?	YES	NO	N/A
Do you need to review SDS's (safety data sheets) to proceed with this work?	YES	NO	N/A
Have you addressed any barricading, warning system or signage requirements appropriate to the task?	YES	NO	N/A
Have all tools, equipment and materials been inspected prior to use and are they adequate to perform work safely?	YES	NO	N/A
Will this task require any lifting, bending or twisting?	YES	NO	N/A
Have you completed Stretch & Flex today?	YES	NO	N/A
Do you have an injury to report or were you injured the prior working day?	YES	NO	N/A
Quality:			
Identify the drawing you are working from today, is it the current version?	YES	NO	N/A
Have you reviewed all construction details associated with our work?	YES	NO	N/A
Who on the crew is responsible for quality control today?			
What is the quality item you will be focusing on today? What will you do today that will prevent rework tomorrow?			

Are any of the following PPE required?	Locate and identify:
☐ Hard Hat/Safety Glasses	☐ Emergency Telephones
☐ Fall Protection/Rescue Plan	☐ Fire Extinguisher
☐ Goggles/Faceshield	☐ Emergency Exit Routes
☐ Hand / Arm PPE	☐ First Aid Equipment
☐ Hearing PPE	☐ Other
☐ Foot PPE	☐ Other
☐ Respirator	☐ Other



Pre-Task Plan

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Audited by: _____
Date: _____
Time: _____

Safety items identified on the front side question list must be addressed on the table below:

Steps for Work	Tools	Hazards	Steps Taken to Address Hazards

***If you need more space, attach another PRE-TASK PLAN SHEET**

Work Planner		Foreman		Safety Coordinator	
--------------	--	---------	--	--------------------	--

Crew Members – additional crew members sign on back or a separate page if necessary:



Planificación previa a las tareas

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Revisado por: _____
Fecha: _____
Hora: _____

Proyecto		Fecha	
Proyecto #		Compania	
Ubicación específica del trabajo			

Responda a lo siguiente cuando evalúe su trabaj	Haga un círculo alrededor de SI, NO o N/A		
¿Caminó por su zona de trabajo para verificar la iluminación, el aseo, causas de tropiezos y resbalones, etc.?	SI	NO	N/A
¿Se ha realizado una encuesta de material peligroso en el proyecto/registros de compensación? (asbestos, plomo, PCB, etc)	SI	NO	N/A
¿Hay algún empleado nuevo, o un nuevo miembro del equipo que esté trabajando en el proyecto y que necesite ayuda?	SI	NO	N/A
¿Se asignaron suficientes personas para completar la tarea sin peligro? (personas que vigilen, levanten pesos, repitan tareas, etc.)	SI	NO	N/A
¿Existen peligros creados por los demás trabajadores en su zona, o su propio trabajo crea un peligro para los demás?	SI	NO	N/A
¿Su trabajo requiere al uso de un sistema personal de detención de caídas? Tiene un plan de rescate? Se lo a comunicado a los miembros de su equipo?	SI	NO	N/A
¿Está usted trabajando alrededor de sistemas eléctricos conectados o equipo energizado? ¿Necesitará usar el procedimiento de bloqueo y rotulación?	SI	NO	N/A
Su trabajo requiere que usted este expuesto a machucones, derrumbes, equipo en movimiento (vehículos) que lo atrapen en medio, materiales que caen o vuelan o escombros. ¿Los operadores están certificados/capacitados? (elevador tipo tijera, herramientas a pólvora, elevador de horquilla, etc.)	SI	NO	N/A
¿Exige esta tarea algún permiso o procedimiento especial, o formularios de inspección? (espacios confinados, trabajos peligrosos, excavaciones, trabajo en alturas, trabajo eléctrico energizado, inspecciones de andamios, elevador de tijera, cabrias, elevador de horquilla, etc.)	SI	NO	N/A
¿Cuenta usted con el tipo y tamaño adecuado de andamio y de escalera portátil? ¿Ha sido capacitado para usar este equipo?	SI	NO	N/A
¿Necesita revisar las SDS antes de proceder con este trabajo?	SI	NO	N/A
¿Contempló la necesidad de instalar barricadas, sistemas de alerta o carteles apropiados para la tarea?	SI	NO	N/A
¿Se inspeccionaron todos los equipos y herramientas antes de su uso?	SI	NO	N/A
¿Deberá levantar peso, agacharse o torcerse durante esta tarea?	SI	NO	N/A
¿Hizo hoy sus ejercicios de estiramiento y flexión?	SI	NO	N/A
¿Tiene una lesión que deba informar o se lesionó el día laboral anterior?	SI	NO	N/A
¿Caminó por su zona de trabajo para verificar la iluminación, el aseo, causas de tropiezos y resbalones, etc.?	SI	NO	N/A
Calidad:			
¿Identificar el plano de trabajo en que se está trabajando hoy, es la versión actual?	SI	NO	N/A
¿Ha revisado todos los detalles de construcción relacionados con su trabajo?	SI	NO	N/A
¿Quién en su equipo de trabajo es responsable del control de calidad hoy?	SI	NO	N/A
¿En qué punto referente a calidad se enfocara el día de hoy? ¿Qué va a hacer hoy para evitar reelaborar mañana?			

Equipo de protección personal:	Ubique e identificar:
☐ Casco	☐ Teléfonos de emergencia
☐ Protección contra caídas /Plan de rescate	☐ Extintidor de incendio
☐ PPE de ojos/cara	☐ Rutas hacia salida de emergencia
☐ PPE de manos/brazos	☐ Equipo de primeros auxilios
☐ PPE auditiva	☐ Ducha/lavado de ojos
☐ PPE para pies	☐ Other
☐ Respirador	☐ Teléfonos de emergencia



Planificación previa a las tareas

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Todo punto relativo a la seguridad identificado en la lista de arriba debe ser abordado en la tabla siguiente.

Pasos del trabajo	Herramienta	Peligros	Medidas tomadas para resolver los peligros

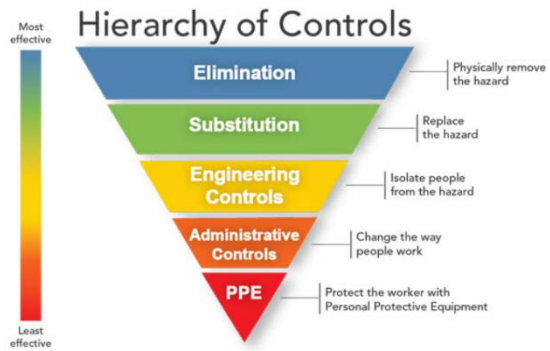
*Si necesita más espacio, adjunte otra hoja de planificación previa a las tareas **Si necesita un diagrama del trabajo, adjúntelo en una hoja separada.

Planificador del trabajo		Nombre del capataz		Coordinador de seguridad	
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Cuadrilla

Job Hazard Analysis (BACK)

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Sequence of Basic Job Steps	Potential Hazards	Recommended Procedures or Controls
Break the job down into individual basic steps. Each step should accomplish some major task. The step should begin with a verb or action work such as “hammer”, “turn” or “move”. The wording in each step should be concise and to the point. If it cannot be written in one simple sentence, it may be more than one task item.	Potential Hazards are events which could occur during the individual step that may result in personal injury, equipment damage, or an environmental incident. Items to identify may include one or more of the following per step: 1. Caught in, under or between 2. Fall to same level 3. Fall to different level 4. Struck by flying object 5. Struck by falling object 6. Strike against a stationary object 7. Strains, sprains, or pulls from pushing, pulling, bending, lifting or twisting 8. Cuts, lacerations or contusions 9. Struck by moving objects 10. Overexposure to dust, fumes, mists, vapors or gases 11. Skin irritation from oils, solvents, soaps 12. Exposure to chemicals 13. High noise levels 14. Ionizing radiation (X or gamma rays, alpha or beta particles) 15. Non-ionizing radiation (illumination, infra-red, ultraviolet lasers) 16. Exposure to environmental extremes (hot or cold) 17. Ergonomic stresses 18. Fire or explosion hazard	List the actions that can be taken to reduce or eliminate the hazard. When completing this section, the solutions must be specific and complete. “Stay alert” and “pay attention” are not acceptable measures to prevent injury. For each step with an associated hazard, ask “is this step necessary to complete the job?” If it can be eliminated, it would eliminate the exposure to the hazard as well. If it cannot be eliminated, then consider mechanical, administrative or environmental changes. Examples of Hazard Controls:  The diagram illustrates the Hierarchy of Controls as an inverted pyramid with five levels. From top to bottom, the levels are: Elimination (blue), Substitution (green), Engineering Controls (yellow), Administrative Controls (orange), and PPE (red). To the left of the pyramid is a vertical color bar with a gradient from blue at the top to red at the bottom, labeled 'Most effective' at the top and 'Least effective' at the bottom. To the right of the pyramid, each level is associated with a specific control measure: Elimination (Physically remove the hazard), Substitution (Replace the hazard), Engineering Controls (Isolate people from the hazard), Administrative Controls (Change the way people work), and PPE (Protect the worker with Personal Protective Equipment).

Job Hazard Analysis (BACK)

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Sample Job Hazard Analysis Step		
Sequence of Job Steps	Potential Hazards	Recommended Procedures or Controls
1. Remove access panel	Pinch point when lifting panel away from front	Install handles on panel front for use in removing panel from unit.
2. Overhead concrete drilling	Dust falling in eyes, Falling concrete chunks	Wear face shield in addition to safety glasses, hard hat and gloves



Job Hazard Analysis

Operation Type: Deck Pour

SEQUENCE OF BASIC JOB STEPS	POTENTIAL HAZARDS	RECOMMENDED ACTION OR PROCEDURE
Morning safety meeting	None	Review pre-task plan, stretch 'n flex, review hazard analysis, review scheduled work.
General cleanup	Cuts, abrasions, lacerations Trips	Utilize gloves when handling sharp or awkward material. Utilize caution when walking, place material and debris out of the work/walk area.
Load wire mesh/rebar to work area	Back injuries Cuts, abrasions, lacerations Falls	Utilize proper lifting techniques. Utilize gloves when handling sharp or awkward material. Maintain clean, orderly work area.
Install wire mesh/rebar on deck	Cuts, abrasions, lacerations Trips Back injuries	Utilize gloves when handling sharp or awkward material. Utilize caution when walking, place material and debris out of the work/walk area. Utilize proper lifting techniques.
Install chairs	Trips, falls Back strains	Work front to back, secure protruding mesh or rebar. Alternate tasks.
Tie chairs, mesh or rebar	Trips Knee and back injuries Cuts, lacerations	Verify footing while working. Alternate tasks, take frequent breaks to stretch. Bend wire and mesh over.
Set up system and lay out piping/ connection	Back injuries	Use proper lifting techniques, verify system layout to ensure mobility.
Set pump truck	Overhead hazards` Impact to facility or other craft personnel	Verify visibility of operator and utilize flagmen if necessary. Barricade, secure area as needed. Utilize spotter in



EHSP Rev 9
Revised August 2017

Análisis de peligros en el trabajo - (DORSO)

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Secuencia de pasos básicos del trabajo	Posibles peligros	Procedimientos o controles recomendados
Dividir el trabajo en pasos individuales básicos. Cada paso deberá lograr una tarea importante. El paso debe comenzar con un verbo o palabra que denote una acción, como por ejemplo “martillar”, “gírar” o “mover”. Use palabras breves y concisas para explicar cada paso. Si no lo puede expresar con una sola frase sencilla, quizás sea más de una tarea.	Los posibles peligros son acontecimientos que pueden ocurrir durante el paso individual, resultando en una lesión personal, daño al equipo, o un incidente de contaminación ambiental. Las situaciones a identificar pueden incluir una o más de las siguientes en cada paso. La persona: 1. Se encuentra trabada, debajo, o entre algo 2. Se cae en un mismo nivel 3. Se cae a otro nivel 4. Recibe un golpe de un objeto volador 5. Recibe un golpe de un objeto que cae 6. Se golpea contra de un objeto fijo 7. Sufre distensión muscular, dislocación o torcedura como resultado de empujar, jalar, agacharse, levantar peso o torcerse 8. Sufre cortes, laceraciones o contusiones 9. Recibe golpes de objetos en movimiento 10. Se sobre expone al polvo, emanaciones, vahos, vapores o gases 11. Sufre irritación a la piel provocada por aceites, solventes o jabones 12. Está expuesta a sustancias químicas 13. Está expuesta a niveles altos de ruidos 14. Está expuesta a la radiación ionizadora (rayos X o gama, partículas alfa o beta) 15. Está expuesta a radiación no-ionizadora (iluminación, láser infrarrojos o ultra violeta) 16. Está expuesta a temperaturas ambientales extremas (calor o frío) 17. Está sometida a estrés ergonómico 18. Está expuesta a peligro de incendio o explosión	Haga una lista de las acciones a tomar para disminuir o eliminar el peligro. Al completar esta sección, las soluciones deberán ser específicas y completas. “Manténgase alerta” y “preste atención” no son recomendaciones aceptables para prevenir las lesiones. Para cada caso vinculado con un peligro, pregúntese, ¿es necesario este paso para completar el trabajo? Si se puede eliminar, eliminaría a su vez la posibilidad del peligro. Si no se puede eliminar, considere entonces diferentes cambios mecánicos, administrativos o ambientales. Ejemplos de control de peligros: Controles mecánicos: barreras protectoras, uso de herramientas, elevadores de horquilla, etc. Controles administrativos: rotación de tareas, cambios en procedimientos de capacitación, PPE, etc. Controles ambientales: ventilación, iluminación, aislamiento de procedimientos, etc.

Ejemplo: análisis del peligro de un paso

Secuencia de pasos de la tarea	Posibles peligros	Procedimientos o controles recomendados
1. Quitar el panel de acceso	Punto de posible pellizco al levantar y quitar el panel frontal	Instalar manijas en el panel frontal para usar al quitar el panel de la unidad.



Disruption & Access Request

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Project/Project#:		Date:	
Project Manager:		Phone:	
Reference Number:		Change Control #:	
Signature:			

Disruption System	Yes	No	N/A
Electrical circuit (One-line Diagram/As-Builts) review required?	☐	☐	☐
Electrical circuit review completed?	☐	☐	☐
Lockout/tagout or facility support required?	☐	☐	☐
Underground work required?	☐	☐	☐
MOP/SOP submitted for review and Approved by all Stakeholders?	☐	☐	☐
Project manager and EHS have copies of required SDS (DPR Trailer)?	☐	☐	☐

By signing below, you acknowledge that the protection procedures outlined in the construction protection plan are acceptable and that the work can be executed accordingly.

Name	Job Title	Signature	Date

Comments

Access Section - Description of Work/Reason for Access

System/ Area Impacted							
☐	Electrical	☐	Air Handler Unit	☐	Fire Alarm	☐	Public Space
☐	Water	☐	Chiller	☐	Fire Sprinkler	☐	General Space
☐	Chemical Line	☐	ICRA	☐	Emergency/Back-up	☐	Loading Dock
☐	Equipment	☐	Boiler	☐	Emergency Exit	☐	Admin. Space
☐	Sewer/ Waste Water	☐	Operating Room/ ER/ ICU	☐	Entrance/ Exit	☐	Other:

Date	Start	End	Weekend				Area
			☐	Y	☐	N	
			☐	Y	☐	N	
			☐	Y	☐	N	
			☐	Y	☐	N	
			☐	Y	☐	N	

Contractors – List all companies and representatives performing work

Name	Company	Phone



Disruption & Access Request

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Project/Project#:				Date:				
Tracking No:		Area:		ICRA Required	Yes	☐	NO	☐
Originator:		Company:		Phone:				
Owner Contact:								

Facility Response

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Approved By	
-------------	--

Safety Start-Up Checklist

Please use the following checklist to assist you in ensuring that all the basic safety information is set-up on your job during mobilization. If you have any questions on the checklist below, please do not hesitate to contact your local safety professional.

Regulatory Binders	
	Current version of State OSHA binder (if applicable)
	Current version of Federal OSHA Standards for the Construction Industry
	Current version of Federal OSHA Standards for General Industry
	Copy of DPR's current revision of the Environmental Health & Safety Plan (EHSP) – to be made site specific and kept in the Right To Know/Understand Area (see below).

Right To Know Area	
	SDS Binder obtained and all SDS's for chemicals used on site stored in this binder Other sets of binders may be needed for field copies of SDS's
	Chemical Inventory Checklist given to all subcontractors for completion and submittal to DPR. Once submitted, check list to ensure we have all SDS's for each subcontractor and then file completed list in SDS binder.
	Signage for the Right To Know/Understand Area that is clearly distinctive.
	Right To Know Area is located where all workers have easy access to the information. The areas should not be set up in an intimidating place such as over the superintendent's desk and all safety documentation can be easily stored here. Ideally all safety documents will be stored and filed in the Right To Know Area.
	A copy of the OSHA 300 logs should be posted between February 1 and April 30 . After April 30th store a copy on file.

Signage	
	IFE Signage – miscellaneous
	DPR Construction jobsite signage - Project Trailer - Road Signs - Project Information Sign
	Site Visitor Check-in Sign
	Stretch & Flex Sign
	Current Labor Postings - English and Spanish (California only - add current annual Excavation permit and Activity Notification Form) ensure we fill in sign with appropriate information as needed
	Egress/Exit Signs
	Parking Signage
	Fire Extinguisher/First Aid Signs
	Any other site specific signage needed

Safety Documentation	
	Emergency Response Plan Completed - Clinic location and map obtained from local safety professional. - Emergency Response Plan Template highlighted areas reviewed and completed as per project staffing, emergency numbers, emergency response responsibilities and other site specific safety requirements - Completed plan copied and posted around trailer, in Right to Know Area and shared with subcontractors. - Any site specific changes added as per project needs. - Emergency Response Plan shared with all workers during orientation and at all hands safety meetings.
	Orientation - Site Specific pages of the Orientation is filled out and included into the Orientation package, with the same information made to Spanish Orientation, as applicable - The entire orientation is reviewed and edited to match site specific requirements – discuss any changes you might make with your local safety professional. - Copies of the Orientation are made and kept for easy use during site specific safety orientations - Binder or File System created to file completed orientations by subcontractor - During Orientation, training of workers should be verified (use Spot Check Training Questionnaire to check knowledge or by making copies of applicable training cards, equipment operation cards, etc.). - If used, Orientation stickers/badging should be acquired and used per site requirements
	Preconstruction Meeting Minutes - First nine pages of Preconstruction Meeting Template copied and ready to be used for precon meetings - Any site specific changes should be added/included in this template. - Trade specific addendum pages in same location to be quickly grabbed and used w/template as per scope - Completed copies should be filed in a binder or file system by subcontractor

Safety Start-Up Checklist

	o Contract log should also be updated to show precon meeting with each sub has been completed when done and precon log completed and filed at front of folder/binder that precon meeting minutes filed in.
	Pretask Plans - o Current copy of pretask plan obtained, updated to make site specific (e.g., Add questions in checklist, fill in project name etc.) and copies made for subs/DPR to pick up - o Binder or File System created to file completed pretask plans in an orderly fashion - o Binder or File System should be organized by subcontractor and by date
	Safety Audits - o Electronic auditing system and database set-up and people trained on use of system. - o For paper Safety Audits, modify to make site specific, print and make copies for everyone. Site Specific changes usually include completing Project Name section and/or pre-completing the boxes under Supervisor Name. - o Binder or File System created for orderly filing of completed (and fully closed/corrected) paper safety audits.
	Safety Meeting Minutes - o Copies of Safety Meeting Sign-in/Agenda made, modified with site specific information and available for use (most project teams file blank forms in same location as completed forms) - o Completed safety meeting minutes filed in binder or file system. - o Subcontractor toolbox meeting minutes should be gathered and filed in the same location.
	Grab & Go/Root Cause Analysis o Current copies of the Grab & Go and Root Cause Analysis forms are copied, modified if needed to include any site specific notification procedures needed on the front checklist and posted on a clipboard in the Right To Know Area.
	Crane Information - o Copies of Crane Subcontractor Pre-Planning Package copied and made available for subcontractor pick-up and use - o Completed Crane Documentation to be stored in separate binder or file system
	Permits As per job needs, Hot Work, Confined Space, Excavation, Height, Energized Electrical Work, , Crane, etc. permits copied, posted for pick-up and use by workers and a binder or file system used to store completed copies
	Miscellaneous safety forms copied, posted for pick-up and use by workers and a binder or file system used to store completed copies - preferably filed by subcontractor by date of inspection, to include, but not limited to: - o Equipment Inspection forms (forklift, scissorlift/boomlift, scaffolding, rolling scaffolding, etc.) - o Trench & Excavation Pre-Planning Checklist & Inspection - o Scaffold Inspection - o Barricade Signage - o Chemical Inventory and Checklist - o SWPPP Checklist - o Demolition Checklist - o Steel Erection Checklist
	Visitor Sign In – form located in the Public Protection Section - o Clip board with several copies of the Visitor sign-in/sign-out log put out – completed sheets are typically filed on bottom of stack in clipboard. - o Complete project information on the form to make site specific, e.g., building entrance/exits, worker access, location of Right to Know Area.
	Daily Reports o Edit Daily Report to make it site specific (i.e. add project name and address) and make copies for distribution to all subcontractors. All completed Daily Reports should be filed, preferably by subcontractor, by date.
	Safety Training Verification - o Make copies of safety training verification and have available to update DPR personnel training as needed: ✓ Bloodborne pathogens ✓ Excavation awareness ✓ Fall protection ✓ Fire Watch ✓ Flagging ✓ Forklift ✓ Hazard Communication ✓ Heat Illness ✓ Heavy equipment ✓ Ladders ✓ Lasers ✓ Powder actuated tools ✓ Scaffolding (rolling, frame, swingstage) ✓ Scissor/boomlift

Safety Start-Up Checklist

	o If training is provided, ensure the form is completely filled out, by both the worker and the trainer prior to filing, i.e., signatures, date, checks in all boxes, etc.
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Site Specific EHSP (Region and Project Specific)

The standard EHSP binder you receive is not a project specific document and thus you will need to update it with regional specific and site specific information. The following checklist will help you make specific changes to this global document and aide in the establishment of a site specific EHSP binder. Each item corresponds to a specific section within the EHSP binder.

	Contents
	o EHSP Core – you may need to review this section to make any site specific changes to basic safety language. Examples of common changes are with PPE requirements, owner required Drug and Alcohol policies, Stretch and Flex, Orientations etc.
	o Bloodborne Pathogens – if your project may occupationally expose workers to bloodborne pathogens and thus enact the standard for bloodborne pathogens, a Bloodborne Pathogen Plan will need to be written and included within this plan.
	o Chemical & Environmental Management – as applicable, an Environmental & Chemical Management Plan may need to be written and included within this plan.
	o Concrete & Masonry – any site specific requirements as they relate to concrete, masonry or precast
	o Confined Spaces – any site specific changes to the protocol should be added. Generally, these are owner specific requirements.
	o Cranes – typically, no site specific changes need to be made to this section, unless owner specified requirements are needed.
	o Demolition – If demolition activities are present on the project, a site specific demolition plan will need to be developed, written and added to the EHSP, and posted for site information
	o Elevated Work – Any site specific fall protection guidelines should be made to this plan, to include site specific rescue plans.
	o Emergency Procedures – ensure the site specific crisis management plan is included in this section and the boilerplate template within is replaced. Attach logistics plan. Complete site posting with applicable information and post.
	o Ergonomics, Material Handling & Mechanical Handling Aids typically there are no regional or site specific changes in this section
	o Excavation and Trenching – insert any site specific changes. Generally these are owner specific protocols or requirements.
	o Fire Protection & Prevention – insert any site specific changes (i.e., many owners have their own hot work permits or hot work procedures).
	o Hazardous Energy – ensure any site specific changes to the protocols are added. Some owners have their own EEW permit or Lockout/Tagout Procedures.
	o Incident Reporting – update Grab & Go with regional specific contact information and names on front page. Include any CCIP or OCIP required additional information on applicable projects.
	o Infection & Contamination Control -- ensure any site specific changes to the protocols are added. Some owners have their own Infection Control Procedures which should be included.
	o Motorized & Site Equipment – insert any site specific changes, i.e., owner required equipment identification systems, site specific fueling requirements, etc.
	o Personal Protective Equipment – ensure site specific changes are included here, i.e., typical changes include vest use, gowning requirements, 100% glove programs, etc.
	o Pre-Planning –if applicable, insert a regional specific PTP. You may also have some site specific amendments to add to the pre-task. Ensure preconstruction template matches regional expectations and that site specific section of the template is complete.
	o Public Protection –any site specific changes as it relates to badging or security should be included in this section.
	o Respiratory Protection – ensure site specific requirements as it relates to the Respiratory Protection Plan are included.
	o Safety Training – ensure the site specific page(s) of the Orientation is completed for the project. Also, ensure that any site specific items are included within the orientation to all workers.
	o Sanitation & Housekeeping – add site specific housekeeping protocols, if applicable, to the EHSP.
	o Steel Erection – As applicable, a copy of the erection plan may also be included within. Ensure written documentation of concrete strength to erection company and verification of any anchor bolt pattern changes are also documented, and at a minimum include in the project files.
	o Tools and Equipment – typically there are no regional or site specific changes in this section.



Safety Start-Up Checklist

Site Specific EHSP (Region and Project Specific)	
	o Water Intrusion Prevention & Response – include any site specific water intrusion prevention measures, owner required water intrusion kit contents, etc. if used; update the water intrusion spill response kit label with applicable emergency contact information & post.
	o Work Environment & Employee Health - typically there are no regional or site specific changes in this section.
	o Forms Section of the EHSP – Add the regional “Consistency Survey Template” and “Consistency Survey Expectations List” to this section.
NOTE: Each site specific EHSP binder is an important safety document. Ensure all changes are made, the binder is maintained and kept current and that a copy goes into the archives at the completion of the project. As such, each project will require their own site specific EHSP.	



Safety/EHS Pre-Construction Meeting Checklist

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Project/Project#:		Date:	
Contractor:		Mobilization Date:	
Scope of work being pre-planned:			

Meeting Participants:

Print Name	Company/Job Title	Contact Number(s)

****Items that should be in place/submitted to DPR prior to start of work.**

Item	
Notes	DPR Philosophy
	- Injury Free Environment - Behavior Based Safety – DPR Culture - Project Goals – Zero - Zero Accidents - Zero Incidents - Zero Near Misses - Zero Unplanned Business Interruptions
	Subcontractor Pre-Qualification
	- Subcontractor has completed the DPR Pre-Qualification process and been conditionally approved - Should Safety Corrective Action Plan (CAP) be required, the CAP must be submitted and approved prior to mobilization - If a CAP is required, ensure CAP is incorporated into site specific safety plan
	Insurance Requirements**
	- Insurance certificates - Additional endorsement - Point of contact
	Tiered Subcontractors
	Contractor field leads may be required to be on-site, actively managing any tiered subcontractor. Will any portion of your scope of work be subcontracted? If so, identify: Tiered Subcontractors: _____ Scope of work: _____
	Temporary Labor

Safety/EHS Pre-Construction Meeting Checklist

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	Will you be using temporary labor? If so, DPR must be notified, and subcontractors / staffing agencies jointly ensure: • verify temporary workers have been trained and are competent to perform assigned work. • temporary workers will be treated as company's own employees in case of injury/emergency.
	Site Specific Safety Plan**
	Contractor safety plan submitted & reviewed by DPR prior to mobilization or letter of acceptance of DPR safety plan • Include any site-specific requirements pertinent to the project • If a CAP is required, CAP should be included as part of the site specific requirements. - o If a CAP is required, the CAP will be reviewed during any subsequent incident reviews, instances of non-safety compliance and/or during all Root Cause Analysis discussions – should any portion of the CAP not be in compliance, corrective measures must be immediately implemented to address gaps.
	Site Specific Quality Plan**
	Contractor written quality control plan submitted & reviewed by DPR. At a minimum, quality plan should include the following: • A schedule of all scope specific submittals required by specification. • A Quality Control Team Organizational Chart • The process for managing document control • Describe process for identification of project specific Distinguishing Features of Work (DFOW) - o Owner expectations - o Designer expectations - o DPR & Subcontractor expectations ▪ A list of all Mock-ups and “first installed work” required ▪ Dates for completion of all mockups and first Install work inspections should be loaded into Pull Plans or schedules as quality milestones • Describe process for development of measurable acceptance criteria for DFOW - o For MEPFP and Structural trade partners: List the process ensuring construction per the BIM model • Describe a process for tracking deficiencies from initial identification, identifying the root cause and implementing preventative actions • Describe process for material transportation, storage and handling - o (How are you going to protect the materials) - o A materials and equipment procurement schedule • Describe personnel training and certification - o Describe your company's quality training program that ensure each employee is committed and understands the company's quality control plan as well as the quality expectations required of them. ▪ A list of all individuals and their required certification and licenses.) • An outline of all Start-up Schedules, Commissioning Requirements, Spare Parts Requirements, Punch List Procedures, As-built Drawings and Warranty Provisions

Safety/EHS Pre-Construction Meeting Checklist

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	Testing and Inspection Plan, Procedures and schedule.
	Safety Representative**
	- Identify on-site safety representative, approved by DPR, with contact number - When full-time safety responsibilities are required, cannot perform other work - Submit resume for DPR review/approval prior to mobilization - Minimum OSHA 30-hour training & applicable competent person designations - Identify company safety representative, with contact number
	Safety Orientation**
	- Site specific – all project personnel should complete prior to start of work. - Contractors to provide other task specific orientation as needed. - Notify DPR of all new hires immediately - Workers off-site for 3 months or more repeat the orientation process
	Weekly Toolbox Safety Meetings
	- All personnel attend daily DPR mass safety meetings (i.e., safety briefings) - Submit contractor toolbox safety meeting documentation - Content sufficient to provide on-going task specific OSHA safety training - Copies of detailed agenda, sign-in sheets submitted to DPR weekly
	Jobsite Safety Audits
	Daily audits performed by the Subcontractor and Tiered Subcontractor Field Lead(s), documented a minimum of once/week
	English Language/Non-English
	If all employees are not fluent in English, contractor should provide alternative means/methods to ensure communication of all safety training and safety related issues, at a minimum: - Conduct all safety training in the alternate language - Ensure each crew has a foreman or designated individual who is fluent in English - Post signs in alternate language, as necessary
	Chemical Management Plan
	- SDS submitted for any chemical brought on site, prior to arrival - DPR Right-to-Know Area - Flammable material storage: “Just-in-time” fueling methods preferred - Fuel tanks: prior approval from DPR project management required - Written plan for spill response and hazardous materials storage & disposal
	Pre-Planning (PTPs and JHAs)
	Pre-Task Planning (PTP) - Daily, written, pre-task planning is required, at a minimum, for all work activities. All activities should be pre-task planned and reviewed and/or updated as tasks and/or conditions change. Job Hazard Analysis (JHA) - JHAs are high-level planning tools that will break down an entire scope of work into tasks and define the potential hazards and controls for each task. Written, detailed JHAs should be prepared for any high hazard activity, which includes, but are not limited to: - Chemicals: hazardous & irritant

Safety/EHS Pre-Construction Meeting Checklist

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	• Concrete: precast, tilt up, post tension, vertical, form work • Confined Space • Crystalline Silica (site specific Silica Exposure Control Plan - SEC) • Hoisting/Rigging Activities: including cranes, derricks, forklifts, straddle buggies, etc. • Demolition Activities & Hazardous Materials Assessment: asbestos, lead, biohazards or other chemicals in the workplace, as well as general demolition hazard assessment • All Framing Activities (including drywall) • Excavation and Trenching • Fall Hazards: exposures 6+ feet, overhead work • Material Handling • Public Exposure: phased occupancy, partial demolition, traffic control, etc. • Scaffolding • Steel Erection • Startup/Shutdown/System Testing Activities: tool hookup, introduction of process chemicals into systems, utility tie-ins, lockout/tagout, work on energized equipment • Non-Routine Activities: activities not performed in last 6 months						
	Safety Training Verification**						
	Training should be documented at the time of site specific orientation, per scope of work. Documentation of applicable training should be provided upon request to DPR. Training required, includes, but is not limited to (refer to Construction Industry Training Matrix at the end of this table): Safety Training: <table border="0"> <tr> <td> • Electrical • Qualified Operators • Qualified Riggers • First Aid/CPR </td> <td> • Hazard Communication • Powder Actuated Tools • Aerial Lifts • Scaffolding – workers and erectors </td> <td> • Forklifts • Ladders • Confined Space • Bloodborne Pathogens • Lock-Out/Tag-Out • Lasers </td> </tr> </table> Competent Person identified: <table border="0"> <tr> <td> • Rigging • Scaffolding • Fall Protection </td> <td> • Fall Protection • Steel Erection • Confined Space </td> <td> • Crystalline Silica • Excavation • Demolition </td> </tr> </table>	• Electrical • Qualified Operators • Qualified Riggers • First Aid/CPR	• Hazard Communication • Powder Actuated Tools • Aerial Lifts • Scaffolding – workers and erectors	• Forklifts • Ladders • Confined Space • Bloodborne Pathogens • Lock-Out/Tag-Out • Lasers	• Rigging • Scaffolding • Fall Protection	• Fall Protection • Steel Erection • Confined Space	• Crystalline Silica • Excavation • Demolition
• Electrical • Qualified Operators • Qualified Riggers • First Aid/CPR	• Hazard Communication • Powder Actuated Tools • Aerial Lifts • Scaffolding – workers and erectors	• Forklifts • Ladders • Confined Space • Bloodborne Pathogens • Lock-Out/Tag-Out • Lasers					
• Rigging • Scaffolding • Fall Protection	• Fall Protection • Steel Erection • Confined Space	• Crystalline Silica • Excavation • Demolition					
	Medical Management						
	• On-site first aid kit • Injured individuals should be accompanied to the medical facility, regardless of severity, by their foreman or company designee • Identify designated medical provider • Return to Work program • CPR/First Aid trained workers (one per crew)						
	Stretch & Flex						
	• DPR required • Contractors – encouraged to participate						

Safety/EHS Pre-Construction Meeting Checklist

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	• Project teams reserve the right to make stretch and flex a requirement for all subcontractors			
	Substance Abuse Testing			
	• Post accident/post incident, including near miss • For cause			
	PPE (personal protective equipment)			
	Minimum: • Hardhat • Safety glasses/approved eye protection • Substantial leather work boots (no tennis shoes look-alikes) • Clothing -- shirts w/sleeves, pants w/o holes As required for work activities: <table><tr><td> • Gloves • Leather -- material handling • Kevlar -- sharps, potential for cuts • Rubber -- chemicals, concrete • Insulated – per electrical activities • Anti-vibration – jack/roto hammering, etc. </td><td> • Kevlar sleeves – demolition • Foot protection - o Metatarsal guards -- jackhammering, tamping - o Rubber boots -- chemicals, concrete • Hearing protection • Faceshields/goggles • Respiratory protection -- as required by hazard </td><td> • Welding hoods, shields, protective wear • High visibility safety attire –Class 2 or 3 vests for vehicular traffic, dayglow t-shirts • Other as needed or required by OSHA, SDS, manufacturer or DPR </td></tr></table>	• Gloves • Leather -- material handling • Kevlar -- sharps, potential for cuts • Rubber -- chemicals, concrete • Insulated – per electrical activities • Anti-vibration – jack/roto hammering, etc.	• Kevlar sleeves – demolition • Foot protection - o Metatarsal guards -- jackhammering, tamping - o Rubber boots -- chemicals, concrete • Hearing protection • Faceshields/goggles • Respiratory protection -- as required by hazard	• Welding hoods, shields, protective wear • High visibility safety attire –Class 2 or 3 vests for vehicular traffic, dayglow t-shirts • Other as needed or required by OSHA, SDS, manufacturer or DPR
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	Accident Reporting, Investigation and Root Cause Analysis			
	• Immediate reporting of any injury to DPR • Reporting of all incidents & near misses immediately to DPR • Completion of DPR Grab ‘n Go • Root Cause Analysis (RCA) – - o Performed for all incidents (safety or quality) except OSFA injuries - o If CAP is in place, review of CAP is completed as part of the RCA process. Corrective actions developed and tracked through to completion. Intent of injury reporting - DPR Culture			
	Emergency Response Plan			
	• Explain DPR emergency response plan			
	Hot Work Permits			
	• Daily/per shift Hot work permits required in occupied facilities • Bottle cart storage – ½ hour fire wall on cart or separate 20’ • Fire extinguisher within 10’ of hot work • Welding shields and blankets as applicable • Welding machines set-up as per exhaust and prevailing wind • Chop saw tables should have protective shields placed behind them			
	Progressive Discipline Program			
	o Violations may result disciplinary action, up to and including removal from the project			

Safety/EHS Pre-Construction Meeting Checklist

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	- o Zero tolerance issues may include these areas: - o Fall protection - o Lockout/Tagout - o Confined space entry - o Horseplay - o Substance Abuse - o Other site specific zero tolerance items
	Hazardous Energies Management
	- o Coordination of all activities with owner - o Detailed JHA, PTP and/or work procedures required - o Designated subcontractor's competent person for LO/TO & EEW operations; all affected workers trained per their scope of work - o Work on energized circuits allowed <i>only if no other feasible means</i> of de-energizing can be demonstrated - o Systems testing - o Performed per specifications and/or manufacturer requirements - o Hydro testing preferred or compelling reason for use of pneumatic testing
	ICRA Requirements
	• Maintain negative air pressure: ante-chamber doors maintained closed at all times, no work in ante-chambers, filters checked daily/replaced as needed, immediately address/correct holes in flex tubing or gaps in temp ICRA walls • Clean Work Areas: maintain/use tacky mats, booties inside work spaces, clean body/equipment/tools before exiting workspace, clean work at all times
	Fall Protection
	- o Competent person identified - o Documented pre-task planning required - o Fall protection measures required at 6' - o All workers trained, documentation made available to DPR upon request - o Fall protection systems inspected daily by competent person, including anchorage points and equipment - o Rescue plan developed by competent person, included on PTP
	Excavation & Trenching (exterior and interior)
	• 811/OneCall/Bluestake/USA current & verified • Private utility locator • "Soft Dig" potholing to visually locate existing utilities REQUIRED on existing facilities or where impact to facilities, public health and/or safety has the potential to be compromised (<i>if "soft dig" cannot be used, a compelling reason must be submitted in writing and approved by the DPR project team prior to start of work.</i>) • Pre-Excavation Checklist & Pre-Task Plan completed by competent person; reviewed by DPR prior to start of excavation; utilities location map & 811/OneCall/Bluestake/USA tracking board maintained current • Subcontractor must have excavation competent person at the/each location of excavation at all times while work is in progress; DPR reserves the right to request and verify credentials • All workers trained to recognize hazards associated with the work to be performed (documentation made available to DPR upon request) • Equipment operators trained (documented) • Daily, documented equipment inspection • Fall protection provided for work within 6' of excavation 6' or more in depth • Walkways/bridges with handrails provided at crossover areas

Safety/EHS Pre-Construction Meeting Checklist

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	- Contingency plan in place on existing facilities prior to start of work <i>DPR reserves the right to require, at the subcontractor's expense, protective systems to be installed at 4', at no cost to the owner, regardless of subcontractor's competent person findings</i>
	Confined Space Entry
	- Spaces should be evaluated prior to entry for permit / non permit entry - Host Employer responsibilities - Controlling Contractor - Entry Employers - Worker training - Rescue and emergency services
	Crystalline Silica Exposure Control Plan (SEC/JHA)
	- If required, the following should be submitted to the DPR project team, in writing, two weeks prior to start of work. The Silica Exposure Control Plan (SEC/JHA) should include, at a minimum, the items listed below and be submitted on the DPR SEC Plan template (reference Section 2 Chemical & Environmental Management Forms section for the blank template): - Description of tasks to be performed - Housekeeping measures to be used - Engineering controls (such as water or ventilation) to be used to limit worker exposure to airborne silica - Provisions for providing respiratory protection when engineering controls cannot adequately limit exposure to airborne silica, including their medical surveillance program; - Provisions to limit worker access to high exposure areas; - Training workers on silica risks and how to limit exposures. - Identify company's competent person
	Cranes
	- Documentation submitted & approved prior to crane arrival on-site: - Crane/lift annual certification - Crane/lift contractor's certificate of insurance - Operator's certification - Copy of load charts pertinent to the specific crane (in the configuration to be used) being brought on site - Written verification from crane/lift owner that maintenance has been performed pursuant to manufacturer's recommendations. - Documented verification of training for riggers, signalers and assembly/disassembly director and/or qualified lift supervisor - Written verification of assembly/disassembly procedures - On-site documented daily inspections should be completed Integrated anti-two block system - Work around power lines - Tower Cranes: - Weekly inspection of OSHA retrieval basket - Trained riggers/signalers (documented) - Colored vest, "X" on top of hardhat - Whistles or air horns for load movement - Personnel on receiving end prior to load flying

Safety/EHS Pre-Construction Meeting Checklist

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	• Taglines on all loads
	Water/Spill Management
	• Water spill response kits required to be provided by the contractor in areas where work is being conducted with the potential for water spills, releases or leaks • Multiple kits may be required, depending upon scope of work (minimum is one kit per floor, per contractor as applicable) • Shut off locations, accessibility, facility contact if required
	Tools & Equipment
	• Daily inspection (documentation required for forklifts, scissor lifts, boom lifts, heavy equipment and cranes) • Maintenance per manufacturer's specifications • Employee training • Spotters as required for congestion/hazardous areas • Ladders: - o only fiberglass ladders are allowed - o extension ladders should have a minimum rating - o labels must be legible • Electrical: - o Minimum 12 gauge electrical cords - o GFCI protection preferred, or Assured Grounding Program - o No patches on cords - o Elevate cords and protect from damage
	Core Drilling / Saw Cutting*
	• The subcontractor competent person should be on the job site at all times to monitor coring/cutting • Daily, detailed, written pre-task plan required • Before all cores/cuts: - o Verify as-builts - o Consult with facilities personnel - o Utility locating - o Additionally, one of the following must be performed: ▪ Ground Penetrating Radar (GPR) ▪ X-ray • Coring/cutting above occupied areas must be coordinated with DPR • Coring/cutting operators should be equipped with appropriately rated rubber gloves and boots • All coring/cutting requires full cleanup and water containment as work is being performed • Materials must be removed from the work area and disposed of properly *Site specific Silica Exposure Control Plan (if applicable)
	Forklifts
	• Daily Inspections • Training Certifications for all operators (current, within 3 years, carried on person) • Backup alarm • Seatbelts • Engines turned off when leaving unit unattended

Safety/EHS Pre-Construction Meeting Checklist

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	• Rigging directly to forks not allowed, manufacturer's approved attachment device required • Secure loads on lifts • Material unloading: - o Flag off back side of truck, and/or - o Provide spotter to create a no access zone
	Material Handling & Storage
	• Sufficient, appropriate manpower • Plan material flow -- minimize the need for double handling of materials • Material carts for movement • Spotters • Pinch Points • Utilize elevated cut station, i.e., saw horses, debris containers, etc. • Use of mechanized equipment whenever possible, or proper body mechanics when not possible • Protect materials from moisture/water • Placement of materials - o access/egress paths maintained clear at all times - o do not store within 6 feet of hoistway or inside floor openings - o do not store within 10 feet of solid exterior wall high than materials • Use racks and/or carts whenever possible • Consolidate material when possible • Protect from trade damage • Secure material from displacement • If lifting/hoisting system is needed, all components are labeled with capacity • Material unloading: - o Flag off back side of truck, and/or - o Provide spotter to create a no access zone - o Maintain eye contact between operator/driver/spotter
	Unmanned Aerial Vehicles (Drones)
	• The use must be pre-approved by DPR project team • Detailed pre-plan must be developed and reviewed by DPR project to flight. • Only Skycatch and KESPRY drones are approved for flight • Flights must stay below 400 ft • The person controlling the drone must have a valid FAA pilot license • A second contact needs to be present during the flight (next to the pilot) to act as a spotter • On-site personnel must be informed of the flight • The owner must approve the flight
	Meetings
	• Site safety orientations (daily, as needed) • Mass safety meetings (start of the week, beginning of shift) • Daily huddles • Subcontractor/foreman coordination meetings • Scheduling/weekly work plan (WWP) meetings for interior and exterior areas • Map sessions, as needed

Safety/EHS Pre-Construction Meeting Checklist

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	Project Logistics
	• Evacuation Plan & Meeting Area • Vehicle Access and worker parking • Trailer complex • Site & Logistics Plan • Project hours • Laydown areas • Visitor and Driver/Delivery personnel PPE guidelines • Lunch/eating area location • Lunch truck • Worker Access • Restricted areas • Badging requirements
	Site Cleanup
	• Performed daily • Site specific clean-up plan • Dumpsters, cart and trash receptacle availability • Materials consolidated and secured at end of shift • Compressed gas cylinders properly secured/stored at the end of shift
	Site Communications
	• Radios or other communication devices • Emergency channels • Contact list of all supervisory personnel • No iPods, Bluetooth technologies or other items to be used in the field •
	Public/Owner Relations Issues
	• Noise • Indoor air quality • Dust • Restroom usage • Access
	Daily Reports
	• Completed daily, turned in weekly at a minimum • Hours worked
	RFIs
	• Process and procedures
	Overtime
	• DPR approval required, hours verified prior to work
	Schedule
	• Permit status • Start date/end date • Weekly work plans • Subcontractor input • Constraint removal • Substantial completion • Critical milestones
	Drawings
	• Review drawings • Site Logistics • As-builts



Safety/EHS Pre-Construction Meeting Checklist

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	Critical Success Factors
	• If you're not keeping score, it doesn't count: - o Preconstruction Excellence - o Schedule - o Safety - o Change Management - o Zero Defect - o Project Close Out • Subcontractor's role in CSFs • Benefits
	Rolling Completion List
	• Zero Defect • Finish as you go • Reviewed weekly
	Site Walk

Action Plan		
Description	Person Responsible	Due Date

Comments:

--

Signature of Subcontractor Representative

Date

(Representative Responsible for Coordinating Outstanding Corrective Actions)



Safety/EHS Pre-Construction Meeting Checklist

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It is DPR's expectation that all subcontractors' office and field based personnel shall actively participate in creating an injury free work environment on all DPR projects. As such, I recognize that our company has the overall responsibility for accident prevention and implementation of DPR's Safety Program for our personnel on this project site, including our vendors and suppliers. I acknowledge that I have been provided with the following information and have been given the opportunity to ask questions regarding the DPR Environmental Health and Safety Program (EHSP).

Subcontractor Company Name:	_____	_____
	Print Name	Date of Receipt
Project Manager:	_____	_____
	Print Name	Signature
Superintendent:	_____	_____
	Print Name	Signature
Foreman:	_____	_____
	Print Name	Signature



OSHA Training & Competent Person Requirements - Federal OSHA 1910 & 1926

Note: the information provided below is Federal requirements, state and/or local requirements may vary -- **California specific requirements are noted as CCR standards.**

Training Requirements	OSHA Reference	Certification Required	Minimum Training Frequency
Access to Employee Exposure & Medical Records	CFR-1926.33 (see 1910.1020(g)(1))		Upon employment, annually thereafter
Accident Prevention: Qualified Operators	CFR-1926.20 (b) (4)		
Accident Prevention: Supervisors/Accident Investigators	CCR 3203 (a)(7)(F)		Upon initial assignment, Supervisors/Accident Investigators
Acetylene & Fuel Gas Safety	CCR 1740 (k)(1)		Initial, Users
Asbestos	CFR-1926.1101(k)(9), (o)(4)(i-ii)		Upon initial assignment, annually thereafter
Benzene	CFR-1926.1128(j)(3), 1910.1028 (j)(3)(i)		Upon initial assignment, annually thereafter
Bloodborne Pathogens	CFR-1910.1030(g)(2)		Upon initial assignment, annually thereafter
Boatswain Chair	CCR 1662 (a)		Upon initial assignment, users
Cadmium	CCR 5207 (m)(4)		Upon initial assignment, annually thereafter for Exposed workers and Supervisors
Confined Space	CFR-1926.1207(a)	yes	Prior to use, thereafter with change of duty/process sufficient to maintain proficiency
Control of Employee Exposure from Dust- Generating Operations Conducted on Concrete and Masonry Materials	CCR 1530.1		Upon initial assignment, annually thereafter for Exposed workers and Supervisors
Cranes: Assembly/Disassembly Director	CFR-1926.1430(d)		Upon initial assignment, re-train as needed to maintain proficiency
Cranes: Operator	CFR-1926.1430(c)	yes	Prior to use, CCO card, re-train if failure to demonstrate competency occurs, every 5 years
Cranes: Rigger (Qualified)	CFR-1926.1404(r)(1), 753(c)(2)		Prior to use, re-train if failure to demonstrate proficiency occurs
Cranes: Signal Person (Qualified)	CFR-1926.1430(b), 1428(c)		Prior to use, re-train if failure to demonstrate proficiency occurs, company provided training is not portable
Employee Alarm Systems	CFR-1910.165(b)(4)		
Employee Emergency Action Plans	CFR-1926.35(e)		With initial plan, and as responsibilities/plan changes
Fall Protection	CFR-1926.503	yes	Upon initial assignment, then as/if equipment changes; as needed to maintain proficiency
Fire Protection: Fire Brigades	CFR-1926.150 (a)(5)		
First Aid	CFR-1926.50(c)	yes	Consistent with First Aid program requirements
Flaggers (Traffic)	CCR 1599 (f & (g)		Upon initial assignment, Assigned Employees



OSHA Training & Competent Person Requirements - Federal OSHA 1910 & 1926

Note: the information provided below is Federal requirements, state and/or local requirements may vary -- **California specific requirements are noted as CCR standards.**

Training Requirements	OSHA Reference	Certification Required	Minimum Training Frequency
Hazard Communication	CFR-1926.59 (see 1910.1200(h))		Initial assignment, then as/if new hazards are introduced
Hazardous Waste Operations/Emergency Response	CFR-1926.65(b)(1)(ii), (d), (e), (q)	yes	Upon initial assignment, annually thereafter
Heat Illness (CA only)	3395		Upon initial assignment
Injury & Illness Prevention Program	CCR 1509 (e)		All Employees through Supervisor Tailgates
Ionizing Radiation	CFR-1926.53(b)		
Job Hazards	CCR 1510 (a & c)		Before job assignment and with new hazards
Laser Equipment	CCR 1801		Operators and Qualified Person
Lead	CFR-1926.62(l)		Prior to initial assignment, then annually
Lock-out/Tag-out of Hazardous Energy	CFR-1910.147(c)(7)	yes	Upon initial assignment, with change in job/hazards for authorized & affected employees, with sufficient frequency to maintain proficiency
Machinery and Equipment	CCR 1510 (b)		Before use, Qualified Person
Non-ionizing Radiation	CFR-1926.54 (a)	yes	
Portable Fire Extinguishers	CFR-1910.157(g)		Upon initial assignment, annually thereafter
Powder Actuated Tools	CFR-1926.302 (e) (1)	yes	
Power Transmission and Distribution: Emergency Procedures	CFR-1926.950 (e), (e)(1)		
Powered Industrial Trucks (Forklifts)	CFR-1910.178(l)	yes	Prior to use, after accident/incident or unsafe operation; with new models; every 3 years thereafter
Process Safety Management	CFR-1926.64(g) & (h)	yes	Upon initial assignment, then every 3 years
Reinforcing Steel/Similar Projections	CCR 1712 (f)(A)		Qualified Person
Respirable Crystalline Silica	CFR-1926.1153(i)(2); CCR 1532.3(i)(2)		Upon initial assignment
Respiratory Protection	CFR-1926.103, 1910.134 (k)		Prior to use and then at least annually
Safety Training and Requirements	CFR-1926.21 (b)		
Scaffolding: Aerial lifts	CFR-1926.453 (b)(2)(ii)		
Scaffolding: Employees Working on Scaffolding	CFR-1926.454 (a), (c)	Yes in CA	Upon initial assignment, then as/if equipment changes; as needed to maintain proficiency
Scaffolding: Erection Crews	CFR-1926.454 (b), (c); App D	Yes in CA	Upon initial assignment, then as/if equipment changes; as needed to maintain proficiency
Stairways & Ladders	CFR-1926.1060(a)		Prior to use, thereafter sufficient to maintain proficiency
Steel Erection	CFR-1926.761		



OSHA Training & Competent Person Requirements - Federal OSHA 1910 & 1926

Note: the information provided below is Federal requirements, state and/or local requirements may vary -- **California specific requirements are noted as CCR standards.**

Underground Construction	CFR-1926.800(d), (g)(5)(iii & v)		Annually
Ventilation: Open Surface Tanks	CFR-1926.57(i)(9) (i)		
Welding: Fuel Gas	CFR-1926.350 (d)		
Welding: Arc Welding and Cutting	CFR-1926.351 (d)		
Welding: Fire Watch	CFR-1926.352 (e)		

Competent Person Requirements	OSHA Reference	Verification Required
Accident Prevention	CFR-1926.20 (b) (2)	
Asbestos	CFR-1926-1101(o)(1-3)	
Confined Spaces in Construction	CFR-1926.1203(a)	
Cranes: Assembly/Disassembly Director	CFR-1926.1404(a)(1);	
Demolition	CFR-1926.850(a), .852(c), .859(g)	
Electrical: Assured Grounding Program	CFR-1926.404 (b)(1)(iii)(B)	
Excavations	CFR-1926.651 (c)(1)(i), (h)(2-3), (k)(1); .652(a)(1)(ii), (d)(3)	
Fall Protection	CFR-1926.502 (c)(4)(ii), (d)(19), (h)(1)(i), (k)(4) and .503(a)(2)	Competent and Qualified in CA
Hearing Protection	CFR-1926.101(b); .1404(r)(1)	
Ionizing Radiation	CFR-1926.53 (b)	
Ladders	CFR-1926.1053(b)(15); .1060(a)(1)	
Respirable Crystalline Silica	CFR-1926.1153(g)(4); CCR 1532.3(g)(4)	
Rigging Equipment	CFR-1926.251(a)(6); .753(c)(2)	
Roofing – Initial and Qualified	CCR 1509 (a) 1730 (b)(8)-(9)	
Scaffolding	CFR-1926.451 (b)(10&11), (d) 10&18),(e)(9)(i), (f)(3,7,12), (g)(2)	
Slings	CFR-1910.184(d), (e)(3)(i-iii)	
Steel Erection	CFR-1926.753(c)(1,2), .754(d)(1), .755(a)(4), .756(a)(2), .757(a)(2,4), .761(a)	
Underground Construction	CFR-1926.803(a)	
Welding: Preservative Coatings	CFR-1926.354 (a)	

Carpentry: Rough and Formwork

Skillsaws

- Proper guarding
- Utilize elevated cut station, i.e., saw horses, debris containers, etc.
- Inspection of tools and cords

Stripping

- Bending nails, clean-up, limited access etc.
- Prybars – body position
- Prybars, cats paws, crow bar, etc. in place of hammers for stripping

Controlled Access Zones

- Signage, barricading etc.

Rebar Protection

- Caps, shields, covers etc. for vertical and horizontal rebar
- NOTE: hook and 90-bars are not considered adequate impalement protection

Material Storage

- Consolidate material when possible
- Secured/stabilized
- Safe access, egress

Pressure treated wood

- PPE required per SDS (e.g., leather gloves)

Ceiling: T-Bar, drop-in, etc.

PPE

- Hard hats
- Gloves & kevlar sleeves
- Hearing protection – e.g., powder actuated tools, etc.
- Respiratory protection, as indicated by SDS, e.g., cutting of tiles if necessary
- Faceshield or goggles for overhead drilling/shooting

Scissor Lifts

- Daily inspections
- Operator training
- 2 hand or 1 hand/1foot controlled lifts only if possible

Rolling Scaffolds

- Fall protection required at 6' and above
- Locking wheels required, no self-propelling of scaffolds
- Worker training: user for workers

Powder Actuated Tools*

- Operator training
- Signage
- Cannot be left unattended, secure when not in use
- Disposal of shells; *soak spent shells in water for 24 hours prior to disposal*

*Site specific Silica Exposure Control Plan (if applicable)

Guarding

- Ensure all holes in work area are covered prior to scaffold/scissor lift use
- Fall Protection
- Wires turned up/protected to ensure worker protection

Laser

- Set-up – avoid eye height
- Signage

Knives

- Ensure all knives are kept sharp and have appropriate sheaths
- Kevlar gloves

Clean-up

- Daily cleaning (dry sweeping is prohibited)
- Appropriate manpower

Concrete Finishers, Cutters, Core Drillers

PPE

- Appropriate, e.g., glasses, hardhats, gloves and respiratory protection
- Rubber boots over workboots
- Knee pads as necessary
- Mesh faceshields, e.g., discharge end of pump
- Hearing protection as required by activity, i.e., cutting, coring, mechanical trowels, etc.

Protection Plan

- Protection of floor and area at and below coring operation
- Water containment plan
- Clean-up plan
- Handwash provisions: water, non- alkaline soap, drying materials
- Impalement protection plan (note: hook and 90 bars are not considered adequate impalement protection)

Respiratory Protection*

- As appropriate, e.g., wet saw on expansion joints, bushing operations, etc.

*Site specific Silica Exposure Control Plan

High Voltage Vacuums

- Appropriate grounding and protection against static charge

Material Handling

- Rigging and removal plan

Concrete Pump Trucks

- Insurance, maintenance verification, operator qualification submitted to DPR
- Operator's manual on truck
- Utilize pre-pump checklist
- Hoses
 - o Adequate personnel for weight of hoses/
 - o Use hooks, taglines, etc. to drag/move hoses
 - o Keeper pins and safety chains on pump connections

Concrete Cutting / Coring*

- Competent person on jobsite at all times to monitor core drilling
- Pre-Planning:
 - o Verify as-builts
 - o Consult with facilities personnel
 - o Perform utility locating
 - o In addition, one f the two below should be used:
 - Ground Penetrating Radar (GPR)
 - X-ray
- Slurry cleanup & disposal
- Protection of holes after
- Side handles on drill motors (and gloves)
- Coordination and protection below cuts

*Site specific Silica Exposure Control Plan (if applicable)

Troweling Machines

- Appropriately maintained
- Refueled with approved gas containers

Demolition

Engineering Survey

- Completed by competent person (qualified in CA) for structural demo (framing, floors, ceilings, walls)

Written Demolition Plan*

Written Plan for safe demolition activities, submitted & reviewed by DPR, to include at a minimum:

- Safe work procedures
- PPE
- Dust/crystalline silica control & prevention measures (*Site specific Silica Exposure Control Plan, if applicable)
- Employee training
- Plan for communication of hazards

Hazardous Materials

- Disturbance, demolition and/or removal of any hazardous material (asbestos, lead, PCBs, etc.) should be performed by qualified personnel under the guidance of third party oversight.

Utility Identification & Protection

- All existing utilities i.d.'d & protected (underground, overhead, under floor, under slab, in/on walls, etc.)
- Areas of demo clearly marked w/standardized system, communicated to all workers
- Electrical demolition:
 - Live circuits clearly marked
 - Safe systems will have wires pulled & appropriate flagging
 - Systems that cannot have wires pulled protected with Lockout/Tagout measures & flagged or marked

Concrete Cutting / Coring*

- Competent person on jobsite at all times to monitor core drilling
- Pre-Planning:
 - Verify as-builts
 - Consult with facilities personnel
 - Perform utility locating
 - In addition, one of the two below should be used:
 - Ground Penetrating Radar (GPR)
 - X-ray
- Slurry cleanup & disposal
- Protection of holes after
- Side handles on drill motors (and gloves)
- Coordination and protection below cuts

*Site specific Silica Exposure Control Plan (if applicable)

PPE

- Fall protection required at 6' – rescue plan required
- Respirators as appropriate to hazard
- Kevlar gloves/sleeves as applicable
- Hearing protection
- High visibility apparel

Protection of Public and Workers

- Plan for protection – lockout tagout, barricades, signage, nets, catch platforms, spotters, exit/evacuation routes clearly marked, etc.

Clean-up / Material Handling

- Clean-up plan
- Dust Control
- Removal of debris from building – chutes, carts, etc.



Scope Specific Safety Expectations

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Hot Work

- Hot work permits
- Bottle cart storage – ½ hour fire wall on cart or separate 20'
- Fire extinguisher within 10' of hot work
- Welding shields as applicable
- Welding leads inspected daily and repaired as necessary
- Welding machines set-up as per exhaust and prevailing wind

Bobcats

- Operator training (documented)
- Daily inspections
- Backup alarms
- Scrubbers for exhaust

Scissor Lifts

- Daily inspections
- Training Certifications for all operators
- 2 hand or 1 hand/1foot controlled lifts only if possible

Jackhammering*

- Removal of debris
- Dust Control – respiratory protection
- Appropriate PPE
 - *Anti-vibration gloves*
 - *Hearing protection*
 - *Metatarsal guards*
 - *Hearing protection*
 - *Faceshield*

*Site specific Silica Exposure Control Plan (if applicable)

Controlled Access Zones

- Signage
- Barricading

Drywall and Taping

PPE

- Gloves as appropriate - e.g., Fingerless – screwing, hanging
- Full Fingert – framing, handling
- Kevlar sleeves – framing
- Hearing protection – e.g., chop saws, powder actuated tools, etc.
- Respiratory protection, as indicated by SDS, e.g., Sanding
- Faceshield -- choppersaws

Scissor Lifts

- Daily inspections
- Operator training
- 2 hand or 1 hand/1foot controlled lifts only if possible

Scaffolds

- Fall protection at > 6' high
- Locking wheels
- Worker training: user for workers, additional for erection crews

Cut Stations

- Utilize elevated cut station, i.e., saw horses, etc.
- Debris containers
- Fire extinguisher & hot work permit
- Spark protection
- Guards in place on chop saws

Powder Actuated Tools

- Operator training
- Daily inspection of tools
- Cleaning of tools per manufacturer's requirements
- Signage
- Tools cannot be left unattended, secure when not in use
- Loads kept in separate, secure containers
- Disposal of spent loads per manufacturer specific requirements

Guarding

- Protect all holes > 2"
- Shaft work
- Fall Protection

Laser

- Set-up – avoid eye height
- Signage

Knives

- Ensure all knives are kept sharp and have appropriate sheaths
- Kevlar gloves

Clean-up*

- Dust control plan (dry sweeping is prohibited)
- Daily cleaning of drywall debris
- Appropriate manpower
- Taping mud washout

*Site specific Silica Exposure Control Plan (if applicable)

Earthwork, Paving and Grading

PPE

- As appropriate, e.g., hardhats, glasses, vests etc.

Site Speed Limit

Heavy Equipment

- Daily inspections
- Training of all operators (documented)
- Backup alarms
- Seatbelts
- Spotters
- Engine turned off when leaving unit unattended

Protection of Holes and Drops in Elevation

Flagging, barricades, signage, delineation etc.

Trenching and Excavations

- Appropriate protection e.g., sloping and benching, shoring, etc.
- Trench plates
- Pot holing
- Daily trenching and excavation inspections
- Also review “Excavation & Trenching” section of the preconstruction meeting checklist form

Protection of Material

- Proper delineation, barricading etc. to protect against trade damage

Traffic Control

- Traffic Control Plan e.g., trained flaggers, lane delineation, signage, vehicle blocking, lane closures, city notification etc.

Dust Control*

- Dust Control plan

*Site specific Silica Exposure Control Plan (if applicable)

Protection of Public Roadways

- Tracking of mud, dirt and debris
- Vehicle wash out

SWPP

- Site specific plan

Site specific conditions

- Utilities locations
- Environmental concerns

Electricians

Electrical Systems

- Lockout/Tagout Plan; PPE plan for testing or diagnostics (until systems are verified dead)
- All hot panels and live electrical rooms need to be secured/locked at all times unless a supervisor is present at all times; with appropriate signage
- All exposed circuits protected
- EEW work needs prior approval from DPR project management, detailed, written PTP in addition to EEW permit

Temp Lighting - Maintenance

- Who is responsible?
- Elevate with appropriate materials
- Protection of nomex

Conduit

- Develop plan to address and prevent slip/trip hazards with rollers/cut offs on floor, daily cleanup
- Cut material on benches, carts or table
- Secure materials when cutting
- Storage on racks or carts if possible

Scissor Lifts

- Daily Inspections
- Operator training
- 2 hand or 1 hand/1foot controlled lifts only if possible

Rotohammer or Overhead Drilling*

- Safety goggles or faceshield
- Side handle

*Site specific Silica Exposure Control Plan (if applicable)

Concrete Cutting / Coring*

- Competent person on jobsite at all times to monitor core drilling
- Pre-Planning:
 - o Verify as-builts
 - o Consult with facilities personnel
 - o Perform utility locating
 - o In addition, one of the two below should be used:
 - Ground Penetrating Radar (GPR)
 - X-ray
- Slurry cleanup & disposal
- Protection of holes after
- Side handles on drill motors (and gloves)
- Coordination and protection below cuts
- Handwash provisions: water, non- alkaline soap, drying materials

*Site specific Silica Exposure Control Plan (if applicable)

Guarding

- Protect all holes $\geq 2"$
- Shaft work
- Fall protection, rescue plan required

Powder Actuated Tools

- Operator training
- Hearing protection, as required
- Daily inspection of tools
- Cleaning of tools per manufacturer's requirements
- Signage



Scope Specific Safety Expectations

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- Tools cannot be left unattended, secure when not in use
- Loads kept in separate, secure containers
- Disposal of spent loads per manufacturer specific requirements

Fireproofing / Insulators

Cleanup

- Cleanup plan
- Containment of overspray
- Dust control
- Washout areas

Respiratory Protection

- According to SDSs
- Fans, ventilation etc.
- Off Hours work coordination

PPE

- Hardhat, glasses, workboots, fall protection if required, etc.
- Gloves when cutting

Preparation/Mixing of Materials*

- Adequate ventilation –performed outside whenever possible

*Site specific Silica Exposure Control Plan (if applicable)

Equipment

- Properly maintained
- Hoses w/locking claims and whip checks
- Scaffolding, scissor lifts – maintained per manufacturer's requirements, cleaned of material buildup

Flooring – carpet, tile, base, laminate, etc.

Respiratory Protection

- According to SDSs
- Fans, ventilation etc.
- Off Hours work coordination

PPE

- Hardhat, glasses, workboots, fall protection if required, etc.
- Knee pads
- Gloves, e.g., kevlar for cutting operations

Knives

- Maintain in sharp condition
- Gloves, i.e., Kevlar or cut resistant
- Sheaths when not in use
- Proper body positioning

Protection of Work from Trade Damage

- Signage, barricading, paper, masonite, etc.

Hot Work

- Permits
- Fire extinguishers within 10' of flame

Preparation/Mixing of Materials*

- Adequate Ventilation – performed outside whenever possible
- Appropriate manpower
- Utilize elevated cut station, i.e., saw horses, etc.
- Handwash provisions: water, non- alkaline soap, drying materials

*Site specific Silica Exposure Control Plan (if applicable)



Scope Specific Safety Expectations

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Glazers

Scissor Lifts

- Daily inspections
- Training Certifications for all operators
- 2 hand or 1 hand/1foot controlled lifts only if possible

PPE

- As appropriate PPE – e.g., gauntlets and/or aprons when handling, cut resistant gloves, etc.
- Fingerless gloves with screwing, installation etc.
- Hearing protection as necessary

Controlled Access Zones

- Exterior installation plan
- Signage, barricading etc.

HVAC

Material Handling

- Wear gloves
- Appropriate manpower
- Secure loads on lifts
- Placement of materials (access/egress paths maintained)
- Approved, inspected and maintained material hoists; all parts of system labeled & rated for capacity
- Utilize elevated cut station, i.e., saw horses, benches, etc.

PPE

- Cut resistant gloves
- Fall protection as required
- Respiratory protection as required, e.g., duct sealant, etc.

Cleaning/Flushing/Draining Operations

- Develop Plan
- Proper disposal and equipment for materials/fluids
- SDSs
- Signage & communication prior to start

Scissor Lifts

- Daily inspections
- Training Certifications for all operators
- 2 hand or 1 hand/1foot controlled lifts only if possible

Rotahammer or Overhead Drilling*

- Safety goggles or faceshield
- Hearing protection
- Side handle on tool

*Site specific Silica Exposure Control Plan (if applicable)

Hot Work

- Hot work permits
- Bottle cart storage – ½ hour fire wall on cart or separate 20'
- Fire extinguisher within 10' of hot work
- Welding shields as applicable
- Welding leads inspected daily and repaired as necessary
- Welding machines set-up as per exhaust and prevailing wind

Concrete Cutting / Coring*

- Competent person on jobsite at all times to monitor core drilling
- Pre-Planning:
 - o Verify as-builts
 - o Consult with facilities personnel
 - o Perform utility locating
 - o In addition, one of the two below should be used:
 - Ground Penetrating Radar (GPR)
 - X-ray
- Slurry cleanup & disposal
- Protection of holes after
- Side handles on drill motors (and gloves)
- Coordination and protection below cuts
- Handwash provisions: water, non- alkaline soap, drying materials

*Site specific Silica Exposure Control Plan (if applicable)



Scope Specific Safety Expectations

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Guarding

- Protect all holes > 2"
- Shaft work
- Fall protection

Powder Actuated Tools

- Operator training
- Daily inspection of tools
- Cleaning of tools per manufacturer's requirements
- Signage
- Tools cannot be left unattended, secure when not in use
- Loads kept in separate, secure containers
- Disposal of spent loads per manufacturer specific requirements

Iron Workers – Erectors, Deckers, Misc. Steel

Fall Protection

Fall protection systems

- 100% fall protection required for exposures of 6' or more
- Rescue plan

Perimeter Cables

- Mid-cables installed as deck is installed
- Kickers installed at corners if necessary to ensure proper strength
- Stamped engineered drawings if used as part of a fall arrest system
- When guardrail systems are used around holes which are used as points of access (such as ladder ways), they should be provided with a gate, or be so offset that a person cannot walk directly into the hole

Crane Pre-Op Planning

Erection Plan

- Submitted to DPR for approval

Hoisting & Rigging

- Single pick hoisting & rigging preferred
- Single pick only for decking bundles, no nylon chokers
- Taglines on all members being picked

If Multiple Lifts are approved:

- o Manufactured rigging unit
- o Trained riggers
- o Minimum 7' between members
- o Maximum 3 members per lift

Documentation

- Concrete strength verification for anchor bolts
- Notice on changes to drawing/patterns

Hot Work

- Hot work permits
- Bottle cart storage – ½ hour fire wall on cart or separate 20'
- Fire extinguisher within 10' of hot work
- Welding shields as applicable
- Welding leads inspected daily and repaired as necessary
- Welding machines set-up as per exhaust and prevailing wind

Scissor Lifts

- Daily inspections
- Training for all operators
- 2 hand or 1 hand/1foot controlled lifts only if possible

Taglines

Required on all loads

Connections

- 2 – bolts minimum for all connections – hand tight
- After torquing, cleanup of load indicator tips

Guarding

- Protect all holes > 2"
- Shaft work
- Fall protection



Scope Specific Safety Expectations

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Iron Workers – Erectors, Deckers, Misc. Steel

Controlled Access Zones

- Signage, barricading etc.
- Release of floors coordinated with DPR project management

Rebar Protection

- Protection in the form of caps, shields, covers etc. maintained below any elevated work for vertical rebar
- NOTE: hook and 90-bars are not considered adequate impalement protection

Iron Workers – Reinforcing Bar

Personal Protective Equipment

- 100% tie off above 6' – rescue plan required
- Full body harness only (body belts only for positioning, in addition to full body harness)
- Arm & hand protection required

Hot Work

- Hot work permits
- Bottle cart storage – ½ hour fire wall on cart or separate 20'
- Fire extinguisher within 10' of hot work
- Welding shields as applicable
- Welding leads inspected daily and repaired as necessary
- Welding machines set-up as per exhaust and prevailing wind

Rebar Protection

Caps, shields, covers etc. for vertical and horizontal rebar

NOTE: hook and 90-bars are not considered adequate impalement protection

Drilling/Doweling*

- Appropriate PPE – respirators, earplugs, knee pads, etc.
- Dust control
- Side handles on Rotohammers
- Safe ergonomic positioning
- SDS for epoxy's, fuels etc.

*Site specific Silica Exposure Control Plan (if applicable)

Walkways

Whenever possible provide walkways across rebar grid/web

Guarding

- Protect all holes > 2"
- Shaft work
- Fall protection

Crane Pre – Op Planning

See attached crane checklist

Controlled Access Zones

- Signage, barricading etc.
- Release of floors

Rebar Rollers/Cut-offs on slab/floor

Develop Plan to address

Ergonomics

- Utilize tie stations to elevate work when possible

Masons, Plasterers, Lathers and Exterior Skin

Fall Protection

- Fall protection required with exposures of 6' or more and needed when working near leading edge or through guardrails
- When guardrail systems are used around holes which are used as points of access (such as ladder ways), they should be provided with a gate, or be so offset that a person cannot walk directly into the hole.

Rescue plan

Scaffolds

- Scaffolds should meet OSHA standards
- E.g., fully planked, guardrails, toeboards, etc.
- Inspected daily
- Safe access and egress:
- Scaffolds greater than 24 feet in height:
 - o access options: internal access systems, a stair tower or other similar access means
 - o at no time will an employee climb on the exterior of a scaffold greater than 24 feet in height without some other means of personal fall protection
- Load limits verified & communicated

Rebar Protection

Caps, shields, covers etc. for vertical and horizontal rebar

NOTE: hook and 90-bars are not considered adequate impalement protection

Wet Saw/Demo Saw*

- Appropriate PPE – respirators, earplugs, faceshield etc.
- Dust Control
- Material stockpile
- Handwash provisions: water, non- alkaline soap, drying materials

*Site specific Silica Exposure Control Plan

PPE

- As appropriate e.g., glasses, hardhats, gloves etc.

Respiratory Protection*

- According to SDSs
- Mixing operations, mortars, grouts, applications

*Site specific Silica Exposure Control Plan

Controlled Access Zones

- Signage, barricading etc.

Material Storage

- Protection of material from trade damage
- Overloading of scaffold (load limits verified)
- Locations should be away from storm sewers (hazardous materials)

Material Handling

- Plan to get material to work area

Material Disposal

- Plan for disposing of waste, waste water or excess material



Scope Specific Safety Expectations

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Millwork

Tools & Equipment

- Proper guarding on all tools, i.e., per manufacturer's requirements
- Use of elevated work stations: saw horses or work benches
- Inspection of tools and cords

Painters and Epoxy Flooring

Respiratory Protection*

- According to SDSs
- Fans, ventilation etc.
- Off Hours work coordination

*Site specific Silica Exposure Control Plan (if applicable)

Paint Clean-up/Washout

- Proper disposal of paints, etc.

Material Storage/Building Loading

- Protection of material from trade damage
- Storage inside building – fire protection, no more than 25 gallons stored in any one location, appropriate signage, etc.
- Locations should be away from storm sewers or floor drains
- Double containment

PPE

- As appropriate, e.g. hardhat, glasses, work boots, fall protection if required, etc.

Protection of Work from Trade Damage

- Signage and barricading

Preparation/Mixing of Materials

- Adequate Ventilation – performed outside whenever possible

Pile Driving

Hearing Protection

- Protection per OSHA requirements; provide dosimeter testing documentation upon request

Forklifts and Heavy Equipment

- Operator Certifications for forklifts (current within the past 3 years)
- Trained operators for heavy equipment (documented, specific to the company)
- Daily inspections
- Backup alarms
- Seatbelts
- Plan to avoid carrying piles over material or workers
- Engines turned off when leaving unit unattended

Pile, Pile Cap and Hole Protection

- Hole guarding, rebar protection, marking of piles and endcaps

Cleanup

- Removal and cleanup of pilecap material, bend nails, disposal plan

Jackhammering*

- Appropriate PPE
- Dust Control – respiratory protection
- Removal of debris

*Site specific Silica Exposure Control Plan

Controlled Access Zones

- Signage, barricading etc.

Hot Work

- Hot work permits
- Bottle cart storage – ½ hour fire wall on cart or separate 20'
- Fire extinguisher within 10' of hot work
- Welding shields as applicable
- Welding leads inspected daily and repaired as necessary
- Welding machines set-up as per exhaust and prevailing wind

Body Positioning

- Maintain body location clear of operation

Plumbers/Pipefitters/Sprinkler Fitters

Pipe

- Develop plan to address and prevent slip/trip hazards with rollers/cut offs on floor, daily cleanup
- Cut material on benches, carts or table
- Secure materials when cutting
- Storage on racks or carts if possible

Hot Work

- Hot work permits
- Bottle cart storage – ½ hour fire wall on cart or separate 20'
- Fire extinguisher within 10' of hot work
- Welding shields as applicable
- Welding leads inspected daily and repaired as necessary
- Welding machines set-up as per exhaust and prevailing wind

Scissor Lifts

- Daily inspections
- Training Certifications for all operators
- 2 hand or 1 hand/1foot controlled lifts only if possible

Rotohammer or Overhead Drilling*

- Safety goggles or faceshield
- Hearing protection
- Side handle
- Cleanup immediately of any debris

*Site specific Silica Exposure Control Plan

Material Storage

- Pipes stored on racks or carts if possible, maintaining clear access/egress paths
- Fittings consolidated
- Combustibles removed from project

Concrete Cutting / Coring*

- Competent person on jobsite at all times to monitor core drilling
- Pre-Planning:
 - o Verify as-builts
 - o Consult with facilities personnel
 - o Perform utility locating
 - o In addition, one of the two below should be used:
 - Ground Penetrating Radar (GPR)
 - X-ray
- Slurry cleanup & disposal
- Protection of holes after
- Side handles on drill motors (and gloves)
- Coordination and protection below cuts
- Handwash provisions: water, non- alkaline soap, drying materials

*Site specific Silica Exposure Control Plan

Guarding

- Protect all holes > 2"
- Shaft work
- Fall Protection

Cleaning/Flushing/Draining Operations

- Develop Plan, to include lockout/tagout requirements, as applicable
- Proper disposal and equipment for materials/fluids



Scope Specific Safety Expectations

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- SDSs
- Signage & communication prior to start

Powder Actuated Tools

- Operator training
- Daily inspection of tools
- Cleaning of tools per manufacturer's requirements
- Signage
- Tools cannot be left unattended, secure when not in use
- Loads kept in separate, secure containers
- Disposal of spent loads per manufacturer specific requirements

Roofers

Fall Protection

- Fall protection required with exposures of 6' or more and needed when working near leading edge or through guardrails
- When guardrail systems are used around holes which are used as points of access (such as ladder ways), they should be provided with a gate, or be so offset that a person cannot walk directly into the hole
- Safety monitors – only with prior approval from DPR
- Protection around skylights
- Rescue plan

Hot Work

- Hot work permits
- Bottle cart storage – ½ hour fire wall on cart or separate 20'
- Fire extinguisher within 10' of hot work
- Welding shields as applicable
- Welding leads inspected daily and repaired as necessary
- Welding machines set-up as per exhaust and prevailing wind Position kettle with consideration to prevailing winds

Controlled Access Zones

- Signage, barricading etc.

PPE

- Faceshields with tar carts and kettle
- Hardhats, glasses, boots etc.

Material Loading

- Secure Loads while hoisting
- Verify roof capacity prior to loading
- All materials secured in storage on roof
- Materials stored away from roof edge; secured from displacement

Protection of Roof Edge from pipes/hoses

- Protect roof edge, parapet wall from trade damage

Crane Pre – Op Planning

- See attached crane checklist

Scaffold Erectors

Scaffold Set-up

- Proper structural components provided
- Capable of supporting 4x maximum intended load
- Toeboards on all working levels
- Headache bars on scaffolds or stair towers
- Fully planked and secured
- Load limits verified (posted for areas designed for material loading/storage)
- Scaffolds greater than 24 feet in height:
 - o access options: internal access systems, a stair tower or other similar access means
 - o at no time will an employee climb on the exterior of a scaffold greater than 24 feet in height without some other means of personal fall protection
- Plan for material storage locations/limitations on scaffolding

Inspection

- Daily, document inspection (who will perform?)

Scaffold User Training

- Is there a training program that they offer?

Material Storage

- Protection of material from trade damage
- Consolidate all excess material or remove from site when possible

Fall Protection

- Fall Protection required
- Where tie-off is infeasible, Fall Protection Plan for erection and dismantling reviewed by DPR prior to start of work
- Competent person to verify anchorage and fall system set up prior to work, inspected daily
- Rescue plan

PPE

- Hardhat, gloves, glasses, etc.

Sheet Metal

Fall Protection

- Fall protection required with exposures of 6' or more and needed when working near leading edge or through guardrails
- When guardrail systems are used around holes which are used as points of access (such as ladder ways), they should be provided with a gate, or be so offset that a person cannot walk directly into the hole.
- Rescue plan

Scissor Lifts

- Daily inspections
- Operator training
- 2 hand or 1 hand/1foot controlled lifts only if possible

PPE

- As appropriate – e.g., Fingerless gloves when screwing
- Full fingered when handling

Controlled Access Zones

- Signage, barricading etc.

Guarding

- Protect all holes > 2"
- Roof work
- Fall Protection

Powder Actuated Tools

- Operator training
- Daily inspection of tools
- Cleaning of tools per manufacturer's requirements
- Signage
- Tools cannot be left unattended, secure when not in use
- Loads kept in separate, secure containers
- Disposal of spent loads per manufacturer specific requirements



Scope Specific Safety Expectations

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Wood Framing (floor joist installation & decking)

Fall Protection

- Fall protection required with exposures of 6' or more (measured from the work platform to the bottom of the sole of the foot)
- Worker training
- Walking top plates of framed walls is strictly prohibited unless 100% fall protection is in place
- Rescue plan
- Reference OSHA Guidance Document: Fall Protection in Residential Construction

Section 18

Public Protection

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Public Protection

1.0 Introduction

Many hazards have the potential to injure not only on site crews but members of the public and visitors as well. This section outlines processes to minimize or eliminate exposure to hazardous work activities to the public and visitors. All work to be performed in areas occupied by the public requires considerable pre-planning and careful review by DPR Project Management.

It is pertinent that work activities do not commence until all aspects of the scope-of-work are reviewed and appropriate measures are in place in order to maximize employee and public safety and minimize both exposure to injuries and general liability. The following details some general requirements as it relates to public protection and public protection measures.

Definitions

Activity area: the area where the construction work takes place. It is comprised of the work space, traffic space and the buffer space

Advance warning area: the section of the roadway where road users are informed about the upcoming work zone. In general, the placement of warning signs should be at a distance of 4-8 times the speed limit of the affected urban roadways. Table 6C-1 below lists suggested distances for road types:

Table 6C-1. Recommended Advance Warning Sign Minimum Spacing

Road Type	Distance Between Signs**		
	A	B	C
Urban (low speed)*	100 feet	100 feet	100 feet
Urban (high speed)*	350 feet	350 feet	350 feet
Rural	500 feet	500 feet	500 feet
Expressway / Freeway	1,000 feet	1,500 feet	2,640 feet

* Speed category to be determined by the highway agency

** The column headings A, B, and C are the dimensions shown in Figures 6H-1 through 6H-46. The A dimension is the distance from the transition or point of restriction to the first sign. The B dimension is the distance between the first and second signs. The C dimension is the distance between the second and third signs. (The "first sign" is the sign in a three-sign series that is closest to the TTC zone. The "third sign" is the sign that is furthest upstream from the TTC zone.)

Buffer space: The lateral and longitudinal area that separates the workers from traffic flow

Termination area: The area used to return traffic to their normal path. It extends as far as the last traffic control device such as an "End of Construction Zone" sign

Dual gate, Reserve gate, or Two gate systems: A procedure under the National Labor Relations Act which isolates a disputing union and employer, thereby allowing other neutral contractors to continue working

Gate contamination: A gate "contaminated" by employees or suppliers entering or exiting the wrong gate

Neutral gate: The entrance used by employers (and their suppliers) who do not have a dispute with the union. The Neutral Gate is designated as Gate A. All unaffected employers, their employees and suppliers should enter this gate.

Reserved (Primary) gate: The entrance used by the primary employer (and their suppliers) who has a dispute with the union. The Reserved or Primary Gate is designated as Gate B. All non-union employers, their employees and suppliers will enter the site through this gate.

Traffic space: The portion of the roadway where traffic is routed through

Transition area: The area where road users are redirected out of their normal path. These areas usually involve use of tapers. Tables 6C-3 and 6C4 below describes typical taper lengths, and formulas for determining the appropriate taper for the situation.

Table 6C-3. Taper Length Criteria for Temporary Traffic Control Zones

Type of Taper	Taper Length
Merging Taper	at least L
Shifting Taper	at least 0.5 L
Shoulder Taper	at least 0.33 L
One-Lane, Two-Way Traffic Taper	50 feet minimum, 100 feet maximum
Downstream Taper	50 feet minimum, 100 feet maximum

Note: Use Table 6C-4 to calculate L

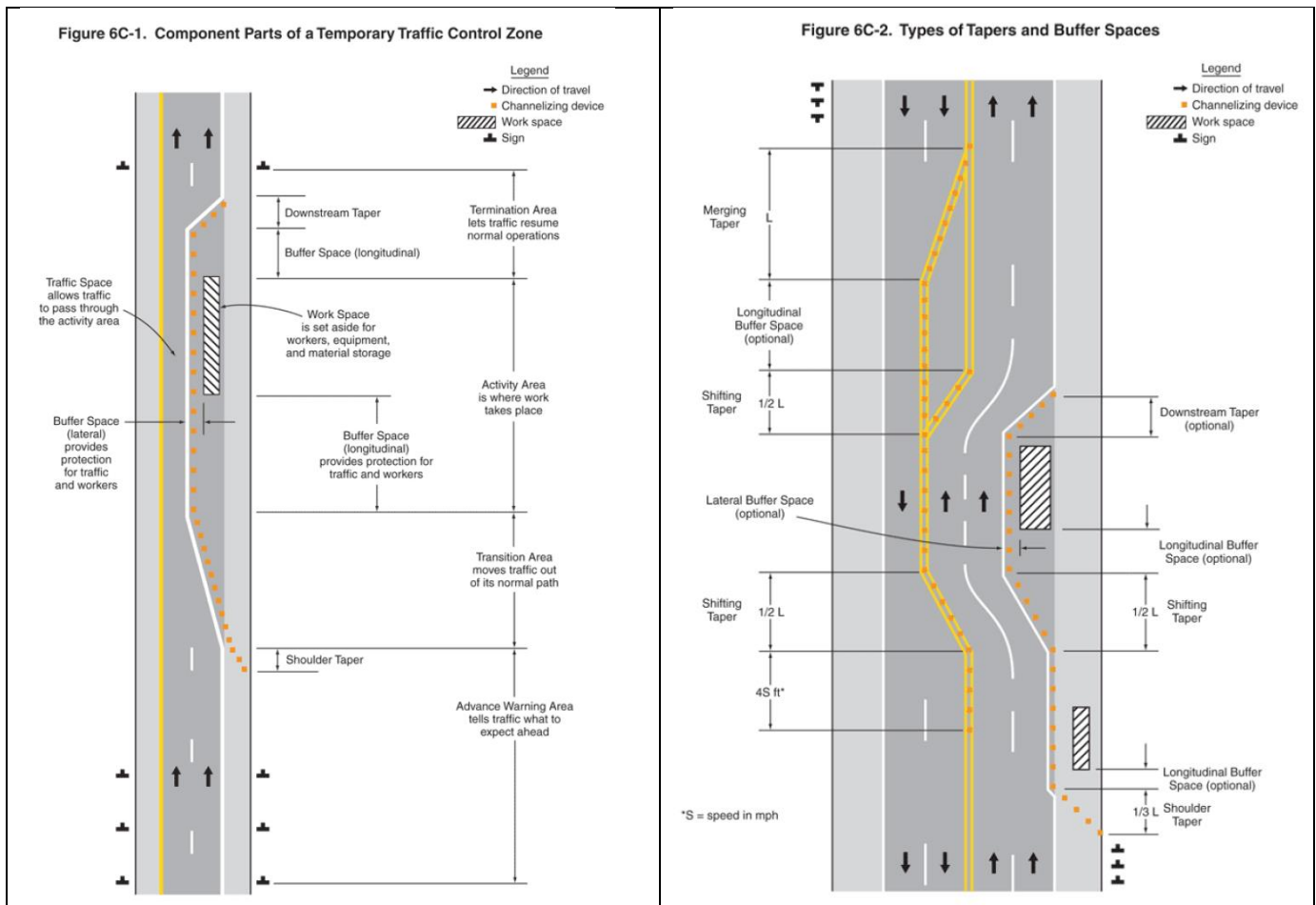
Table 6C-4. Formulas for Determining Taper Length

Speed (S)	Taper Length (L) in feet
40 mph or less	$L = \frac{WS^2}{60}$
45 mph or more	$L = WS$

Where: L = taper length in feet
W = width of offset in feet
S = posted speed limit, or off-peak 85th-percentile speed prior to work starting, or the anticipated operating speed in mph

Work space: The area that is closed to road users and set aside for workers, equipment and material and a shadow vehicle if necessary. Work spaces are usually delineated with temporary barriers or channeling devices (cones, delineators etc.)

Temporary Traffic Control Zone (TTC): An area of a highway where road user conditions are changed as a result of construction activities through the use of traffic control devices. The TTC extends from the first warning sign to the last traffic control device. See Figure 6C-1 below.



2.0 Pre-Task Planning

Work activities with the potential for public exposure (e.g., traffic control, barricading, work with potential exposures to the general public, etc.) are defined as High Hazard Activities and therefore require detailed, daily written pre-task planning. Refer to the Pre-Planning Section 17 for additional guidelines.

3.0 General Requirements

At a minimum, the following precautions should be taken when working around public or owner occupied facilities:

- All workers should take such necessary action as is needed to protect and maintain the public's use of sidewalks, entrances to buildings, lobbies, corridors, aisles, doors, exits and vehicular roadways; such protective measures may include:
 - sidewalk sheds
 - canopies
 - catch platforms
 - fences
 - guardrails
 - barricades
 - shields
 - adequate visibility

- signage
- warning lights and signals
- Public protection measures should guard against noise, flying materials, falling or moving materials and equipment, hot or poisonous materials, flammable or toxic liquids and gases, open flames, smoke, dust, fumes, mist, and vapors, energized electric conductors or circuits or other harmful exposures
- A trained signal person should be posted to control the movement of motorized equipment in areas where there exists the potential for public exposure
- Temporary sidewalks, ramps, stairs or elevated areas should be provided with guardrails on both sides whenever permanent sidewalks, ramps or stairs are obstructed by the work. See ADA requirements prior to removal of any existing structure.
 - DPR project management may authorize barricades, secured against accidental displacement and meeting the requirements of local authorities where fences, sheds, walkways and/or guardrails are impractical
 - during the period when any barricade, fence, shed, walkway or guardrail is removed for the purpose of work, a watchman should be placed at all openings
 - guardrails should be made of rigid materials complying with the requirements for standard guardrails as defined by OSHA and the DPR EHSP (refer to the Elevated Work Section 07)
- Appropriate warnings, signs and instructional safety signs should be conspicuously posted where necessary, and the expected alternative path should be clearly marked
- Adequate provisions should be taken in consideration of those members of the public with disabilities when designing and implementing public protection measures
- Warning lights, including lanterns, torches, flares and electric lights, meeting the requirements of governing authorities, should be provided and maintained from dusk to sunrise along guardrails, barricades, and temporary sidewalks and at every obstruction to the public; these warning signs and lights should be placed at both ends of such protection or obstruction to include adequate lighting to provide both worker and public protection
- When signs, signals and barricades do not provide necessary protection on or adjacent to a highway or street, flag personnel or other appropriate traffic controls should be used; flag personnel should wear high visibility safety apparel
- All Department of Transportation and/or municipality requirements for public safety should be strictly observed on operations being performed on public roadways, including permitting
- Verification of permitting should be provided to DPR project management and DPR EHS
- Fire hydrants and all designated fire lanes should remain clear at all times for the use of emergency vehicles
- At no time are subcontractor personnel or vehicles to obstruct traffic on public streets or at entry driveways
- All material deliveries should be scheduled in advance with DPR project management and should be completed within the time segment allocated for a specific delivery
- A temporary fence or similar means of protection in compliance with laws and regulations of governing authorities should be provided, secured and maintained, as practicable, around the perimeter of all operations on the job site to control access to the work area(s) by employees, to protect the public, and to restrict access by unauthorized individuals. Special attention should be paid to ensuring base plates or other protruding members are painted, delineated or otherwise labeled for identification and visibility.
- Access to the site should be limited to the entrance designated for construction traffic as indicated on the site plans issued with the construction documents
- Special attention should be given to the emergency evacuation of buildings, structures and jobsites and at no time should construction activities affect this evacuation

3.1 Additional Precautions

Whenever possible, pedestrian and vehicular traffic should be prohibited from any area closer than one-fourth of the height of the structure where an exposure to falling objects, flying materials, moving materials and equipment, sparks etc. exists. In the event it is infeasible to move pedestrian and vehicular traffic greater than one-fourth the height of the structure away, additional precautions such as sidewalk sheds, catch platforms and other protective measures should be taken.

3.1.1 Sidewalk Sheds

A sidewalk shed is a protective structure located along routes used for pedestrian or vehicular traffic, with the potential for debris and falling object hazards to be present, as a result of operations being performed. The following guidelines, at a minimum, should be followed:

- The sidewalk shed should run the entire length of the walkway and be sufficient in width to accommodate pedestrian traffic without causing congestion
- Adequate provisions should be taken in consideration of those members of the public with disabilities when designing and constructing sidewalk sheds
- Sidewalk sheds should be lighted by natural or artificial light at all times to a degree sufficient to ensure the public's safety
- The entire structure of the sidewalk shed should be crafted of sound and sturdy materials
- The roof of a sidewalk shed must consist of sound planking not less than two inch (2") nominal thickness, closely and evenly laid and made watertight as necessary
- The outside edge and ends of the roof of the sidewalk shed should be provided with a substantial enclosure projecting to a height of at least 42 inches above the roof of the shed
- The enclosure on top of the shed may be vertical or inclined outward, but should not be more than a 45 degree angle
- Roofs of sidewalk sheds should be capable of safely sustaining a load of 150 pounds per square foot, at a minimum
- If material is stored on the sidewalk shed, it must be able to withstand the weight of the material plus an additional 150 pounds
- Sidewalk sheds should be at least eight feet in height from the base of the walkway
- Unless the inside face of the walkway lays directly against the structure, a solid fence should be erected on the inside face for the full height of the walkway

3.1.2 Catch Platforms

A catch platform is a temporary structure erected around or attached to a building being demolished to safeguard and protect employees, and where applicable, the public. The purpose of the structure is to catch and retain falling objects or debris. The following guidelines, at a minimum, should be followed:

- Catch platforms should be used when pedestrian or vehicular traffic are within the distance of one-fourth the height of the structure being demolished and the distance from the ground to the area being demolished is greater than 55 feet
- Catch platforms should be able to withstand at least 125 pounds per square foot
- Catch platforms should be maintained and inspected daily
- Catch platforms should meet the following minimal requirements:
 - outriggers should be constructed of 3 by 10 inch planks, or their equivalent, set on edge no more than 8 feet apart
 - the platform should be at least 7 feet wide, and pitched so that the outer edge is at least 6 inches higher than the inner edge
 - the floor should be constructed of solid 2 inch nominal planks and safely secured to the outriggers

- the outer edge of the platform should consist of an apron that lies against the floor of the platform and extends at least 48 inches at a 60-degree angle
- if a tubular scaffold is being used instead of a catch platform, it should be tied off to the structure at least every 26 feet vertically and 30 feet horizontally

3.1.3 Other Walkways

Where there exists a means of rerouting pedestrian traffic to a distance greater than one-fourth the height of the structure being worked upon, a cleared temporary pedestrian walkway should be used. The temporary pedestrian walkway should be clearly marked and consist of a continual fence on the inward side and suitable railing on the outer side. Where possible, site perimeter fencing posts should be buried. When buried posts are not possible, special attention should be paid to ensure base plates or other protruding members are painted, delineated or otherwise labeled for identification and visibility to prevent potential tripping hazards.

4.0 Project Signage and Barricading

Signs should be placed in positions where they will convey their messages most effectively and placement must therefore be accommodated to enhance design and alignment. Safety signage is important to keep constant visual reminders in front of workers, public and owners. Key points of interest should be clearly marked with signage (i.e. entrance, construction office, emergency gathering location, etc.) The following guide lines should be considered when selecting project signage:

- Safety signage should follow any applicable ANSI requirements
- Flammable storage areas should be clearly signed
- Fire extinguisher locations should be clearly signed and visible
- Authorized Personnel Only signs placed in areas where a hard barricade cannot be placed to delineate construction areas from areas accessible to the public
- Construction Areas

4.1 General Signs

General signage includes, but is not limited to the following signs:

- DPR Project sign*
- IFE Signage
- DPR Construction office *
- Fire Extinguisher*
- Visitors Must Check In
- Subcontractor office trailers w/ contractor logo*
- Park at Your Own Risk
- Speed limit “#” mph*
- Construction area: Hardhats, safety glasses, work boots required*
- Construction area: Authorized Personnel Only
- First - Aid*
- Emergency Gathering (Evacuation) Area*

4.2 Optional/ site specific Signs

Parking signage includes, but is not limited to the following signs:

- DPR safety office
- Safety first, be alert, be alive
- Safety does not stop here - buckle up
- Safety first - zero accidents is our goal
- Jobsite injury statistics & communications board
- Staging/ lay down area

- No construction parking
- Owner worker parking only
- Owner parking lot
- No parking - fire lane
- Keep clear - fire lane
- Visitor parking
- Stop
- Not a construction entrance*
- Construction material deliveries*
- Water Shut-off
- Electrical Shut-off
- Danger – Electrical
- Flammable Storage/Refueling Area – No Smoking
- Flammable & Combustible Storage – No Smoking
- Compressed Gas Storage
- Danger – Underground Utilities
- Gate identification signs; Gate #1: Gate #2 : Gate #3 : Gate #4*
- No truck deliveries
- Right (left) turn only
- Caution – construction traffic
- Caution – construction entrance
- Laydown area (zone 1, zone 2, zone 3, ...)
- Security (Security Station #1, #2, ...)
- Controlled Entry
- Badges Required

Signage Examples (additional signage is available via OES and/or the DPR Toolbox)



4.3 Barricading

Barricades, stanchions, delineators, safety cones, or safety warning tape or ribbon should be provided as required to isolate and protect unsafe work areas from workers, pedestrians or vehicle traffic. Substantial hard barricades should be utilized to protect personnel from exposure to vehicular traffic – safety warning tape, cones and delineators should not be utilized for public protection purposes.

Barricades should be appropriately marked or tagged with the name, phone number, reason for barricading, and duration of hazards. Barricades should be maintained at all times and removed when the hazard no longer exists. The following barricade general designations and definitions apply:

Hazard Type	Color	Type of Barricade	Action
Occupational Hazard	Yellow	Caution: Physical barricade required, i.e., caution tape or sign: - Rope, chain or tape may be used in the majority of these applications YELLOW TAPE SHOULD BE USED FOR THIS CLASS.	Do not cross until hazard is identified and safe passage an access is assured. Use to identify the presence of a possible hazard. <i>Examples: tripping hazards, overhead work, open fab floor grates</i>
High Hazard Immediate Hazard Imminent Danger	Red	Danger: Physical barricade required; i.e., danger tape or sign: - PVC, chains, screens, metal, wood, visqueen or tape RED TAPE SHOULD BE USED FOR THIS CLASS.	DO NOT CROSS -- if access is required, coordinate with contact person on barricade signage. <i>Examples: energized electrical work in progress, overhead suspended load, critical high-pressure tests, chemical introduction and/or fall exposure.</i>

5.0 Traffic Control and Flagging

Temporary traffic control procedures should be implemented any time that road users and vehicular traffic are impacted by construction and construction related activities. All state, federal and local requirements should be followed and at a minimum the system of traffic controls and requirements should be in conformance with those found in the current Manual on Uniform Traffic Control Devices (MUTCD – *current copy available on DPR Toolbox*). For more information on these requirements please refer to the U.S. Department of Transportation, Federal Highway Division, Manual on Uniform Traffic Control Devices for Streets and Highways.

5.1 General Traffic Control & Flagging Guidelines

For the protection of workers, motorists, bicyclists and pedestrians, the following general guidelines apply:

- A written temporary traffic control plan (TTC) should be completed and submitted to DPR project management for approval prior to any work
- The TTC plan should be completed by a competent person and should detail the scope of the work, safety measures implemented, placement of traffic control devices, any lane or road closures and appropriate means to protect workers and the public
- All Department of Transportation and/or municipality requirements for public safety should be strictly observed on operations being performed on public roadways, including permitting
- Verification of permitting should be provided to DPR project management
- All workers within a designated Temporary Traffic Control Zone should wear high visibility safety apparel (*reference the Personal Protective Equipment Section 16 for additional requirements*)
- Temporary Traffic Control Zones should be designed to minimize the impact to traffic flow
- Adequate warning, delineation, signage and channeling should be provided and maintained to assist road users in advance of the Temporary Traffic Control Zone
- Frequent and abrupt changes in geometrics such as lane narrowing, dropped lanes, or main roadway transitions that require rapid maneuvers should be avoided
- Traffic control devices that are inconsistent with the intended travel path and Temporary Traffic Control Zone should be covered or removed as practical
- Road users should be encouraged to use alternate routes that do not include Temporary Traffic Control Zones

- Roadway occupancy should be scheduled during off-peak hours whenever possible and if necessary night work should be performed
- All appropriate traffic control devices should be in place prior to any construction activities taking place and should be maintained through the duration of the activity
- All traffic control devices should be removed as soon as practical when they are no longer needed. When work is suspended for a short duration, traffic control devices should be covered or removed.

5.2 Traffic Flagging

- Traffic flaggers should be used at locations where barricades and other warning signs cannot practicably control the moving traffic
- Only in emergency situations should flagger stations not be protected by advance warning signs
- When flaggers are required, they should be placed appropriately so that adequate warning is given to vehicular traffic and in such a manner as to ensure the flagger's safety
- Flaggers should wear highly visible safety/reflectorized vests or garments at all times
- Flaggers should be trained in the fundamentals of flagging before being assigned to duty as a flagger; such training should at a minimum cover the following:
 - flagger equipment that should be used
 - layout of the control zone and flagging station(s)
 - methods to signal traffic to stop, proceed or slow down
 - methods of one-way traffic control
 - demonstration of proper flagging protocols
 - emergency vehicles traveling through the work zone
 - handling emergency situations
 - methods of dealing with hostile drivers
- Traffic signaling directions should be in conformance with the DOT and the MUTCD
- A STOP/SLOW paddle should be used, be octagonal in shape and no less than 18' wide
 - the STOP paddle should be red with white letters and border
 - the SLOW paddle orange with black letters and border

5.2.1 Flagger Procedures

The following general guidelines should be used for signaling traffic:

STOP - face traffic and aim the STOP paddle at oncoming traffic with the arm fully extended to the side and stationary. The free hand is bent and above the shoulder with the palm facing forward.

SLOW/PROCEED – face traffic and aim the SLOW paddle at oncoming traffic with the arm fully extended to the side and stationary. Use the free hand to motion traffic to proceed.

ALERT/SLOW – face traffic and aim the SLOW paddle at oncoming traffic with the arm fully extended to the side and stationary. Use the free hand to motion for traffic to slow (raising it up and down).

See figure 6E-1 below for further clarification of traffic signaling for flaggers.



Stop traffic

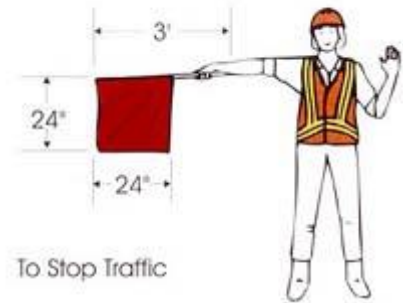
- ☐ Stand on the edge of the road facing traffic
- ☐ Hold up your left hand palm outward, like a Police Officer
- ☐ Hold the paddle out away from your body, the STOP side toward traffic
- ☐ If you are using a flag, hold it out horizontally. Do not wave the flag! It confuses drivers. Make eye contact with the driver
- ☐ After the first car has stopped, you may go to the center of the road to stop succeeding cars.

Traffic Proceed

- ☐ Step back to the shoulder of the road, turn your paddle to SLOW (or drop flag by your side).
- ☐ With your free hand, motion traffic to proceed.
- ☐ Do not wave the flag! It confuses drivers.

Alert/Slow Traffic

- ☐ Face traffic and aim SLOW paddle towards traffic, use free hand to way traffic forward
- ☐ With your free hand, motion traffic to proceed, raising it up and down.
- ☐ Using flag, slowly move flag from directly in front of body towards direction of travel.
- ☐ Do not wave the flag! It confuses drivers.



5.3 Inspections

Daily inspections of the Temporary Traffic Control Zone should be performed by a competent person for compliance and conformance of the TTC with all local, state, federal requirements in addition to those requirements detailed within DPR's EHSP. These inspections should be documented and filed with DPR.

6.0 Site Security

The principal purposes of the jobsite security program are:

- To promote security and security awareness among jobsite personnel
- To help protect the health and welfare of jobsite personnel
- To reduce the risk of property damage or loss
- To protect the public from all construction activities

6.1 Authorized Visitors

- All visitors to the site should register with DPR project management or designee upon arrival. Visitors should sign in and sign out of the Visitor Sign-In Sheet (refer to the Visitor Sign-In Sheet in the Forms Section at the end of the EHSP).
- Subcontractors should regulate their visitors accordingly
- On projects that require the use of visitor passes, all visitor passes expire upon departure from the site and should be surrendered prior to leaving the site to the gate security guard or member of DPR project management

6.2 Employee Identification (Project Specific)

- All job site employees will be issued an identification badge and/or hardhat sticker upon completion of their initial safety orientation and after having completed any and all other site specific requirements (e.g. passed their alcohol and drug test, completed background check, completed owner orientation etc. as required)
- All persons without a hardhat identification sticker or badge should report to DPR project management for verification of employment status, attendance at an orientation session or issuance of a single day visitor pass, as applicable

6.3 Badging (Project Specific)

6.3.1 Permanent Photo Badge Procedure (Sample Procedure)

The purpose of the photo badge is to allow applicable personnel access and the ability to work on-site on a long-term basis. Only approved DPR workers and subcontractor workers can be badged. The procedure is as follows:

- Permanent photo badges or other similar type are for personnel assigned to this jobsite and are numbered; these badges are NOT to be issued to a visitor (refer to visitor badge guidelines below)
- The DPR project management or designee will issue badges
- Badges are only issued following site specific safety orientation and have completed all other site specific requirements such as drug screen with a negative result, background check and/or owner specific orientation/training, as applicable
- Badges have an expiration date and will only be issued for set period of time
- Identification badges will remain the property of DPR and the owner
- The identification badge should be maintained in good condition and on the person to whom it is issued
- The identification badge should be returned to DPR or designee when employment on the project is terminated or when requested by DPR project management or designee
- All lost or stolen identification cards should be immediately reported to DPR project management

- Identification badges should be worn in a manner to be visible at all times, i.e., between the beltline & neck; they should not be placed in pockets or hardhat liners
- Any worker with an expired photo badge will not be allowed on site; the expired badge should be confiscated and the worker escorted to the badging office for issuance of a new badge
- Tracking:
 - a log should be maintained for all badges issued
 - all subcontractors terminating workers from the site, for any reason, are responsible for the immediate return of badges along with termination form; a fee may be assessed to each subcontractor for each badge not returned

6.3.2 Visitor Badge Procedure (Sample Procedure)

Principle purpose of the visitor badge procedure is to allow visitors and vendors access to the construction office area; the following procedures apply:

- Visitor badges will be issued on a daily basis only
- The visitor badge will be worn in plain sight at all times
- All badges are valid for one day only, and must be returned to DPR project management or designee when exiting the site
- Visitors should be escorted at all times by a permanent badged individual
- Tracking:
 - all visitor badges issued and returned should be maintained on a log daily

6.4 Project Tours

All persons touring the job site should be escorted and protected. The personnel responsible for the organization, direction and safe conduct of the tours should follow the following guidelines:

- All group tours should be approved in advance by the owner's representative and/or DPR project management, allowing for maximum advanced notice
- All tours should be coordinated by DPR project management to accommodate the project schedule, to make necessary preparations and to assure safety precautions are observed
- DPR project management should review the following items with the person requesting the tour:
 - individual tour groups in non-hazardous areas may be limited in size depending on the environment
 - tour groups should wear appropriate clothing and personal protective equipment in accordance with site standards including hard hats, safety glasses, proper footwear, full-length pants (are preferred), etc., unless other precautions have been made
 - children under the age of 12 should not be permitted on tours
 - children ages 12 to 18 should be accompanied continuously by an adult
 - hard hats, boots, raincoats, earplugs, etc., may be supplied or provided at the discretion of DPR project management
 - each visitor should read and sign the Visitor Sign-In sheet prior to the start of the tour (in the Forms Section at the end of the EHSP)
- Immediately prior to entering the job site, all visitors should be briefed about the need for careful and orderly conduct, including mention of any special hazards that may be encountered
- Technical and official visitor tours should be conducted in accordance with the above safety precautions; since technical tours are often conducted through areas of more hazardous work, it is recommended that the number of people on such tours be proportionate to the degree of hazard involved

6.5 Unmanned Aerial Systems (UAS), i.e., Drones

The use of UAS (drones) on DPR projects has increased. We have begun to use them for: daily logistics planning, existing conditions documentation, underground utility as-built drawings, change documentation,

exterior skin production tracking, as supplements to owner billings, and are beginning to develop map systems for strategic planning purposes. We have developed an understanding of drone best practices. DPR filed for a 333 exemption (FAA exemption) to help clear up the grey area around the use of drones in commercial applications. At the time, DPR was one of three construction companies that were granted a 333 exemption for commercial use of drones in construction. For additional information contact the DPR UASCoregroup@DPR.com .

- The following are the most notable items from this exemption:
- Only use of authorized aerial systems approved for flight
- Flights must stay below 400 ft.
- The person controlling the drone must have a pilot license
- A second contact needs to be present during the flight (next to the pilot) to act as a spotter
- On-site personnel must be informed of the flight
- The owner must approve the flight
- UAS specific pre-task safety survey: [LINK](#)

6.6 Bomb Threats

Bomb threat procedures should be updated and included in the project Crisis Management Plan (*refer to Emergency Procedures Section 08 for additional information and Bomb Threat Information Sheet*). Bomb threats may originate from pranksters, political terrorists, criminal extortionists, disgruntled personnel or an employee looking for a few hours off work, to name a few. The threat can arrive over the telephone, in the mail, by written message left in a public place, scrawled on the wall, in person, etc. **ALL THREATS SHOULD BE TAKEN SERIOUSLY!**

6.6.1 Jobsite Evacuations

When DPR Project Management has called an evacuation, site security also becomes involved in the event. DPR Project Management and site security (where applicable) should provide a continual flow of communication and assist in the evacuation of people in a calm and orderly fashion and assist in any related search.

In the event of a major jobsite evacuation, DPR Project Management and site security will be required to block off entry into parking lots and direct emergency vehicles to the incident scene.

Employees should be directed to search their private belongs; i.e., purses, lunchboxes, and similar personal property taken with them during the evacuation, and leave all other property behind until clearance has been given to return to the site.

Jobsite evacuation will be initiated by a pre-arranged signal. When this signal is sounded, all employees should evacuate the site or building using the nearest exit and proceed to the designated gathering area immediately.

Refer to the Emergency Procedures Section 08 for additional guidelines and project specific information.

7.0 Union and Labor Disputes

7.1 Establishment of a Dual Gate System

When picketing occurs on the jobsite, the following steps should be taken by DPR personnel (and/or security personnel) to assure the contractors and subcontractors not directly involved in the dispute may continue working:

7.1.1 Day 1 of Picket

- Determine from picketers reason for the picket
- Inform DPR project management immediately; DPR should inform the owner immediately

- All conversations and events with respect to the picketing should be documented, i.e., what is written on each picket sign is important to record, including taking pictures of the picket sign if necessary
- Contact the Building Trade Council
 - indicate a dual gate system will be established immediately
 - indicate the gates will be posted with signage and established prior to beginning of shift the following day
 - identify the location of the gates
- Letters should be sent via courier service with verification of delivery to picketing union, union contractors and subcontractors, as well as the Building Trades Council informing them of a dual gate system
 - it is extremely important the documents be delivered, even if the party to which it is delivered does not wish to accept
 - courier should be instructed to leave document, and keep a written record of each delivery
 - copies of written record of deliveries should be maintained in the DPR project files
- letters should also be sent to all non-union contractors, subcontractors and DPR suppliers to notify them of the dual gate system
- Install signage for Neutral Gate A and Primary Gate B (picketed)
- Assure names of current contractors and subcontractors are listed on the appropriate gate signs:

Gate A, Neutral (union)	Gate B, Reserved (non-union)
[list all Neutral union contractors/subcontractors]	[list all non-union and primary contractors/subcontractors]

- Prior to starting work the second day, the gate entrances should be established
- Clearly mark each entrance with signage and take pictures of the gates prior to starting time
- Prepare a site diagram of the project gates
- Security officers should be notified of date and time of dual gate implementation
 - emphasize the importance of security officers' responsibilities for assuring gate contamination does not occur
 - arrange for additional manpower to perform traffic flagging duties, if required
 - security officers should be responsible for verifying the identity of each person/vehicle/delivery at each gate
 - anyone at the wrong gate entrance should be instructed to enter the correct gate only
 - security and flagmen will assist misdirected persons/vehicles/deliveries in moving to the correct gate
- Each person/vehicle/delivery should be reminded they must exit from the same gate they enter
- Assure passes designating gate entry are distributed to all incoming persons/vehicles/deliveries not in possession of a valid site ID badge

7.1.2 Day 2 – Picketing the Neutral Gate

If the neutral gate is being picketed on the second day, the following steps should be taken:

- Determine why the union(s) is/are picketing the neutral gate
- Document conversations/action and take pictures of pickets in front on the neutral gate
- Alert DPR project management and owner about neutral gate picketing
- If the picketing is illegal, inform the picket in charge that they are illegally picketing in front of the neutral gate and that picketing should be restricted to the primary gate
- Contact the Building Trades Council by phone to request pickets be removed and follow with written documentation

7.1.3 Contamination of Gates

- DPR and/or security personnel should be assigned to monitor the integrity of the gates and ensure that gates do not become contaminated
- Gate may be “contaminated” by employees, suppliers, visitors entering or exiting the wrong gate
- Should a gate become “contaminated”, DPR project management should:
 - contact the Union/Building Trades Council and find out what caused the “contamination”
 - re-establish dual gate system by re-sending notification(s) to respective parties

7.1.4 Establishing a Reserve Time

On occasion, it may be more convenient to establish a Reserve Time for the picketed subcontractor. This may be done in such a manner that the picketed subcontractor is working during some hours or days when none of the other neutral subcontractors are performing work on the jobsite. For instance, that contractor may work on Saturday and Sunday provided none of the neutral contractors are working on those days or the picketed contractor may work from 6:00 p.m. on; again, provided none of the neutral contractors are performing work or are receiving materials during those hours.

- Determine the appropriate Reserve Time
- Inform the picketed subcontractor that he is to perform work and that his suppliers are to deliver materials only during that Reserve Time
- Inform the picketing union/Building Trades Council of the Reserve Time
- Inform all other neutral contractors of the Reserve Time
- Inform the Building Trades Council of the Reserve Time

Public Protection Forms

Forms

Visitor Sign-In – English
Visitor Sign-In - Spanish
Barricade Signage
Temporary Traffic Control Zone Inspection
Flagging Training Verification – English
Flagging Training Verification – Spanish



Visitor Sign-In

Archive Document – DO NOT DISCARD

Note: Individuals whose visit involves high hazard activities, i.e., confined space entry, fall protection, excavation, etc. (reference Pre-Planning Section 17 of EHSP) require site specific safety orientation. **Visitors must be escorted by DPR or subcontractor designee at all times.**

Entry & Exit - All subcontractors, suppliers, and visitors are required to enter the building using	
Right To Know & Understand Area – is located	
Personal Protective Equipment – Meets job requirements and ANSI standards and includes but not limited to: substantial leather work boot (above ankle), long pants, shirts with sleeves, safety glasses. Gloves should be worn if the site requires them.	
Drinks Allowed – Only drinking water is allowed inside the building. (May be job specific)	
Smoking, eCigarettes, Vaporized Cigarettes/Tobacco Products – Smoking and/or tobacco product use is only permitted in designated areas.	
Zero Tolerance Items – Harassment, horseplay, theft, drug and/or alcohol use are violations of the safety rules and may result in immediate disciplinary action, including termination from the project.	
Incident Reporting - All incidents, accidents, injuries, and close calls/near misses must be reported to any DPR personnel immediately.	
Emergencies – In the case of an emergency contact any DPR personnel immediately. If evacuation is necessary, ensure you check in with DPR before leaving the site. No one is permitted back in the building until DPR gives the all-clear.	
Ladder Safety – Should ladders be needed, only heavy duty type 1A fiberglass ladders are allowed. Maintain three points of contact at all times. Your mid-line (Belt Buckle) should never cross outside the outer rail. Do not stand on the cap or top rung. All extension ladders used for access must extend 3 ft. past the landing and be properly secured.	

These Basic Site Safety Rules are in addition to any rules, regulations or requirements required by any public agency with appropriate jurisdiction. I understand that it is my responsibility to learn and follow all of the applicable safety rules and regulations that pertain to my scope of work and visit to this Project. I have had the opportunity to ask questions and understand that violation of the site safety rules may result in revocation of visitor privileges. I have also asked about any site specific hazards and conditions and am aware of where to get further information about DPR's safety rules.

In consideration for granting permission to the undersigned to enter such premises, the undersigned does hereby RELEASE AND FOREVER DISCHARGE DPR Construction, their agents and servants, and all other persons, firms and corporations connected therewith of and from any and all liability, actions, claims, demands, or suits whatsoever for personal injuries or death suffered or resulting from any acts or omissions of said Company whether from known or unknown, apparent or unapparent hazards on the jobsite. Further, the undersigned hereby states that before executing this Release (signing this form), that he or she has read it, understands the same and executes it with full knowledge of its content and meaning.

Print Name	Signature	Date	Time In	Time Out	No high hazard work	Read & understand requirements
					Y/N	Y/N
					Y/N	Y/N
					Y/N	Y/N
					Y/N	Y/N
					Y/N	Y/N
					Y/N	Y/N
					Y/N	Y/N



Visitor Sign-In

Archive Document – DO NOT DISCARD

Nota: Las personas quienes su visita involucra actividades de alto riesgo, ej.: espacios confinados, protección contra caídas, excavaciones, etc. (Referente a sección planificación previa EHSP) requieren de orientación de seguridad específica. **Los visitantes deben ser acompañados por DPR o subcontratista designado en todo tiempo.**

Entradas y salidas – Todo subcontratista, proveedor y visitante tiene la obligación de entrar al edificio usando	
Zona de derecho a información – Está ubicada en	
Equipo personal de protección (PPE) – debe cumplir con las normas ANSI (Instituto Americano para Normas Nacionales) e incluye: botas pesadas de cuero (sobre el tobillo), pantalones largos de trabajo, camisa con mangas, gafas de protección, guantes de trabajo, y/o protección para las manos. La protección para las manos debe ser llevada por la persona en todo momento y usada cuando existe la posibilidad de lesión a las manos.	
Agua Potable Para Tomar – Solo se permite agua potable para tomar. (Pueda que sea según cada trabajo)	
Productos de Tabaco/Fumar, eCigarettes o Cigarrillos en forma de vapor - Uso de productos de Tabaco o el acto de fumar es permitido solamente en áreas designadas para ello.	
Cero Tolerancia- El hostigar, jugar pesado, robo, uso de drogas/alcohol, son violaciones de las reglas de seguridad y puede que resulte en acción disciplinaria inmediata, incluyendo ser despedido del proyecto.	
Reportes de incidentes – Todo accidente, lesión o accidentes escasamente evitados o salvados deben ser reportados inmediatamente a DPR.	
Procedimientos de emergencia – En caso de emergencia, comuníquese inmediatamente con DPR. Si es necesaria una evacuación, asegúrese de haber verificado con DPR antes de dejar el sitio. Está prohibido el regreso de toda persona al edificio hasta no recibir el visto bueno de DPR.	
Seguridad de las escaleras portátiles – En el sitio de trabajo, sólo se permite el uso de escaleras portátiles de gran resistencia, fabricadas de fibra de vidrio 1A. Usted debe mantener tres puntos de contacto en todo momento. Nunca cargue materiales al subir o bajar por la escalera. Use un cinturón de herramientas o pida que se los pasen. Su punto medio (la hebilla de su cinturón) jamás debe pasar por fuera del riel externo. Jamás debe pararse en el descansillo. Todas las escaleras de extensión deberán prolongarse 3 pies pasado el descanso cuando se usan para dar acceso y deben estar bien firmes.	

Estas reglas de sitio básicas son en conjunto a cualquier otras reglas o requerimientos requeridos por cualquier agencia pública con poder de jurisdicción. Entiendo que tengo la responsabilidad de aprender y cumplir todas las reglas y normas de seguridad que tienen que ver con el alcance de mi trabajo y mis visitas a este Proyecto. También entiendo que soy el último responsable de tomar medidas para que mi zona de trabajo se encuentre libre de peligros antes de entrar y realizar mi trabajo. He tenido la oportunidad de hacer preguntas y entiendo que la violación de las reglas de seguridad del sitio podría resultar en medidas disciplinarias, incluyendo el despido del sitio del proyecto. También he preguntado acerca de riesgos específicos o condiciones peligrosas y estoy informado acerca del lugar donde puedo conseguir mayor información acerca de las reglas de seguridad de DPR.

En consideración del permiso que se otorga al que suscribe para entrar en dichos recintos, el que suscribe por la presente DEJA LIBRE PARA SIEMPRE a DPR Construcción, Inc., a sus agentes y empleados, y a toda otra persona, firma y corporación vinculada, de toda obligación, queja, demanda o pleito de cualquier tipo vinculado con lesiones personales o muerte ocurrida o resultante de cualquier acto u omisión de dicha Compañía que surja de riesgos conocidos o desconocidos, evidentes o no evidentes, en el sitio de la obra. Además, el que suscribe por la presente declara que ha leído y entendido esta Declaración antes de firmarla, y que la firma en pleno conocimiento de su contenido y significado.

Nombre en letra de imprenta	Firma	Fecha	Hora de entrada	Hora de salida	No high hazard work	Read & understand requirements
					Y/N	Y/N
					Y/N	Y/N
					Y/N	Y/N
					Y/N	Y/N
					Y/N	Y/N
					Y/N	Y/N

Barricade Signage

Letreros de las barricadas

Contact Name: _____
(Nombre del contacto)

Contact Number: _____ Company: _____
(Número del contacto) (Compañía)

Start Date: _____ End Date: _____
(Fecha de comienzo) (Fecha de terminación)

Hazard: _____
(Peligro)



Temporary Traffic Control Zone Inspection

Archive Document – DO NOT DISCARD

Project Information		General Information				Date			
Project Name		Purpose of Work Zone		Week Starting					
Project Number		Location		Week Ending					
Contractor/Subcontractor		Contact Number							
Inspectors Name		Approximate Size (sf)							
Weather (check all that apply)		Hot	Dry	Clear	Windy	Cool	Wet	Overcast	Raining
Partial List of Safety Items – Identify & Correct All Unsafe Conditions									
A	Inspect at a minimum daily, mark a “+” for items being performed well or in good condition								
B	For items that need improvement or corrective action, mark a “Δ” and comment on the back any action needed/taken								
1	General Requirements								
	A written temporary traffic control plan (TTC) has been completed and DPR has approved								
	Permitting has been obtained and DPR has been provided with verification								
	All workers within the TTC are wearing high visibility reflectorized vests or apparel								
	Permanent traffic control devices inconsistent with intended travel path have been removed or covered								
	Road users have been encouraged to use alternative routes where possible								
	Warning control devices are in place and being maintained								
	Warning control devices no longer needed have been covered or removed								
	Other								
2	Speed Reduction Areas								
	Continuity of traffic has been planned without undue impact: transit buses, railroad services, business, etc.								
	Frequent changes in speed limits have been avoided								
	Traffic flow is designed to allow vehicles to travel through TTC w/minimal reduction in speed, <10mph								
	Only where restrictive features of the work zone require it should speed be reduced more than 10mph,								
	Additional driver notification is in place for speed reductions more than 10mph								
	Other								
3	Advance Warning Area								
	Distances between warning signs should be in accordance with Temporary Traffic Control Zone Plan & permitting								
	Adequate warning, delineation, signage and channeling are in place and maintained								
	Other								
4	Transition Area								
	Frequent/abrupt changes in geometrics (land narrowing, dropping lanes, etc) have been avoided where possible								
5	Activity Area								
	Buffer space is provided prior to work area -- no work activity, storage of equipment, materials, or non-safety related vehicles								
	Emergency vehicle storage area is not extended into buffer space								
	Each work space is adequately signed /barricaded / protected from traffic and clearly delineated from traffic								
	Other								
6	Termination Area								
	Road users have been directed back to the normal travel path								
	END OF WORK signs have been posted, if required								
	Other								



Specifics & Comments of Construction Work Zone Inspection

Ensure all open items are closed/addressed in a timely manner.

Monday	
Tuesday	
Wednesday	
Thursday	
Friday	
Saturday	
Sunday	



Flagging Training Verification

Archive Document – DO NOT DISCARD

Print Name

Date

Jobsite

Company

Only personnel trained in safe flagging procedures should be permitted to flag traffic in construction work zones. Traffic control is required to ensure the protection of workers, motorists, bicyclists and pedestrians. Traffic control and associated flagging activities are considered High Hazard activities by DPR Construction. Therefore, a written, detailed pre-task plan is required to be completed by crew members, reviewed and approved by supervision prior to start of work.

High-Visibility Clothing

For daytime work, the flagger's vest, shirt, or jacket should be orange, yellow, strong yellow green or fluorescent versions of these colors. For nighttime work, similar outside garments should be retro-reflective. The retro-reflective material should be orange, yellow, white, silver, strong yellow green, or a fluorescent version of one of these colors and should be visible at a minimum distance of 1,000 feet. The retro-reflective clothing must be designed to identify clearly the wearer as a person and be visible through the full range of body motions.

Hand-Signaling Devices

Hand-signaling devices, such as STOP/SLOW paddles, lights, and red flags are used to control traffic through construction work zones. The STOP/SLOW paddle, should be the primary hand-signaling device. The standard STOP/SLOW sign paddle should be 18 inches wide and octagonal in shape with letters at least 6 inches high. A rigid handle should be provided. Flashing lights can also be used on the STOP face. When used at night, the STOP/SLOW paddle should be retro-reflectorized. Flag use should be limited to emergency situations and a low-speed and/or low-volume locations which can best be controlled by a single flagger. Flags used for signaling should be a minimum of 24 inches square, made of a good grade of red material, and securely fastened to a staff about 3 feet long. When used at night, flags should be retro-reflective red.

Other Personal Protective Equipment:

- Hard hat
- Suitable clothes/footwear
- Air horn or whistle to communicate between flaggers
- "FLAGGER AHEAD" sign or similar device and other work zone traffic control devices

Proper Training

The first rule of flagging is: **do not stand in the travel lane!** The proper place to stand in order to stop traffic is at the right shoulder of the road. That way, if the driver does not see the signal to stop or slow down, the flagger should not get hit. Flaggers should always have an escape route. It is preferable for drivers to hit traffic cones and stop in the buffer area than to hit the flagger. Make sure that the sight distance is long enough to allow drivers time to see and react (500 feet on faster roads, 200 feet in slower areas). It is important to train all flaggers to adhere to standard signals. Non-standard actions will confuse drivers. To maintain attention level and alertness, it is important to relieve flaggers regularly.

Flagging Do's	Flagging Don'ts
DO use stop/slow paddles whenever possible	DON'T become distracted
DO be alert	DON'T stand in the travel lane
DO face traffic	DON'T wave the flag, it confuses drivers
DO hold up your left hand to stop traffic	DON'T assume traffic will stop
DO hold the flag or paddle away from your body, horizontally	DON'T use headphones, radio's, etc.
DO set up proper advanced warning signal for approaching traffic at 500 feet on higher speed roads; 200 feet in lower speed areas	DON'T start traffic until communication with other end of work zone is complete
DO step to the road shoulder to release traffic; drop the flag or turn the paddle to "SLOW"	DON'T turn your back on traffic unless it is absolutely necessary after approaching vehicles have stopped



To Stop Traffic



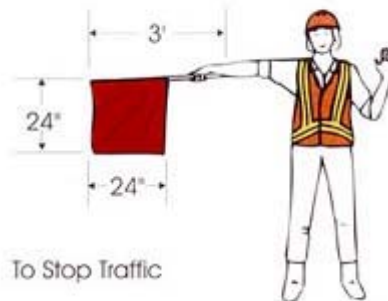
Traffic Proceed



To Alert and Slow Traffic

To Stop traffic

- ☐ Stand on the edge of the road facing traffic
- ☐ Hold up your left hand palm outward, like a Police Officer
- ☐ Hold the paddle out away from your body, the STOP side toward traffic
- ☐ If you are using a flag, hold it out horizontally. Do not wave the flag! It confuses drivers. Make eye contact with the driver
- ☐ After the first car has stopped, you may go to the center of the road to stop succeeding cars.



To Stop Traffic



Traffic Proceed

Traffic Proceed

- ☐ Step back to the shoulder of the road, turn your paddle to SLOW (or drop flag by your side).
- ☐ With your free hand, motion traffic to proceed.
- ☐ Do not wave the flag! It confuses drivers.

Alert/Slow Traffic

- ☐ Face traffic and aim SLOW paddle towards traffic, use free hand to way traffic forward
- ☐ With your free hand, motion traffic to proceed, raising it up and down.
- ☐ Using flag, slowly move flag from directly in front of body towards direction of travel.
- ☐ Do not wave the flag! It confuses drivers.



To Alert and Slow Traffic

This document hereby certifies that I have been instructed on the items listed above and I have had the opportunity to ask questions. I fully understand the DPR requirements for flagging

Employee Signature

Date

Person Conducting Verification Review Signature

Date

A copy of this form must be faxed to the DPR regional office PRIOR TO START OF WORK.



Verificación de capacitación en señalización

Documento de archivo – NO DESECHAR

Nombre:		Fecha:	
Lugar de la obra:		Compañía:	

En las zonas de obras en construcción, sólo se debe permitir al personal entrenado en los procedimientos de señalización segura que se ocupe de la señalización con bandera al tráfico. Se debe controlar el tránsito para garantizar la protección de los trabajadores, motoristas, ciclistas y peatones. La dirección del tránsito y las actividades de señalización relacionadas son consideradas actividades de Alto Riesgo por DPR Construction. Por lo tanto, se requiere una planificación previa a las tareas en forma detallada y por escrito, que deben completar los miembros de la cuadrilla y que debe ser revisada y aprobada por la supervisión antes de comenzar el trabajo.

Vestimenta de alta visibilidad

Para el trabajo diurno, el chaleco, camisa o chaqueta del banderillero debe ser de color naranja, amarillo o verde amarillento fuerte o las versiones fluorescentes de estos colores. Para el trabajo nocturno, la misma vestimenta exterior debe ser retroreflectiva. El material retroreflectivo debe ser de color naranja, amarillo, blanco, plateado, verde amarillento fuerte o las versiones fluorescentes de uno de estos colores y debe ser visible a una distancia mínima de 1.000 pies. La ropa retroreflectiva debe estar diseñada para identificar claramente a la persona que la usa y ser visible a través de todo el rango de movimientos corporales.

Dispositivos manuales de señalización

Los dispositivos manuales de señalización, como las paletas con mensaje PARE/DESPACIO (STOP/SLOW), luces, y banderas rojas se usan para controlar el tránsito que circula en las áreas de obras en construcción. Las paletas con mensaje PARE/DESPACIO (STOP/SLOW) deben ser el principal dispositivo de señalización manual. La paletas estándar con mensaje PARE/DESPACIO (STOP/SLOW) deben tener 18 pulgadas de ancho y forma octogonal con letras de al menos 6 pulgadas de altura. Debe tener un mango rígido. También se pueden usar luces destellantes en la faz del mensaje PARE (STOP). Cuando se use de noche, la paleta con mensaje PARE/DESPACIO (STOP/SLOW) debe ser retroreflectante. El uso de la bandera se debe limitar a situaciones de emergencia y en áreas de baja velocidad y/o de poco volumen, las cuales se pueden controlar mejor mediante un solo banderillero. Las banderas que se usan para señalar deben tener un mínimo de 24 pulgadas cuadradas, estar confeccionadas en tela de buena calidad, de color rojo y fijadas a un asta de aproximadamente 3 pies de largo. Cuando se usen de noche, las banderas deben ser de color rojo retroreflectante.

Otros equipos de protección personal:

- Casco
- Ropa/ calzado adecuados
- Silbato o bocina neumática para comunicarse entre banderilleros
- Cartel "BANDERILLERO ADELANTE" (FLAGGER AHEAD) o dispositivo similar y otros dispositivos de control del tránsito en zonas de construcción

Capacitación adecuada

La primera regla de la señalización con bandera es **¡no pararse en el carril de circulación!** El lugar correcto para pararse a fin de detener el tránsito es en la banquina derecha del camino. De esa manera, si el conductor no ve la señal para detenerse o bajar la velocidad, no atropellará al banderillero. Los banderilleros deben siempre tener una ruta de escape. Es preferible que los conductores golpeen contra los conos de tránsito y que se detengan en el área de amortiguación y no que atropellen al banderillero. Verifique que la distancia de visión sea lo suficientemente grande como para dar tiempo a los conductores a ver y reaccionar (500 pies en los caminos más rápidos y 200 pies en las áreas más lentas). Es importante entrenar a los banderilleros para que se adhieran a las señales estándar. Las acciones no estándar confundirán a los conductores. Para mantener el nivel de atención y alerta es importante relevar a los banderilleros periódicamente.

Lo que el banderillero debe	no debe hacer
SIEMPRE que sea posible use las paletas con mensaje de PARE/DESPACIO (STOP/SLOW)	NO se distraiga
SIEMPRE esté alerta	NO se pare en el carril de circulación
SIEMPRE levante la mano izquierda para parar el tránsito	NO asuma que el tránsito se detendrá
SIEMPRE mantenga la bandera o paleta alejada del cuerpo, en forma horizontal	NO flamee la bandera pues confunde a los conductores
SIEMPRE coloque señales de advertencia con la debida anticipación para el tránsito que se aproxima, a 500 pies en los caminos de más alta velocidad y a 200 pies en las áreas de velocidad más lenta	NO use auriculares, radios, etc.
SIEMPRE párese de cara al tránsito	NO inicie la circulación del tránsito hasta que se haya comunicado con el otro extremo de la zona de trabajo
SIEMPRE hágase a un lado en la banquina del camino para dar paso al tránsito. Baje la bandera o gire la paleta a "DESPACIO" (SLOW)	NO dé la espalda al tránsito, salvo que sea absolutamente necesario y después que los vehículos que se aproximan se hayan detenido.



To Stop Traffic



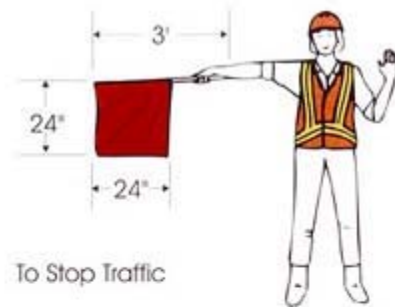
Traffic Proceed



To Alert and Slow Traffic

Para parar el tránsito

- ☐ Párese al borde del camino de cara al tránsito
- ☐ Levante la mano izquierda con la palma hacia fuera, como un oficial de policía
- ☐ Mantenga la paleta alejada del cuerpo, con el cartel de PARE (STOP) hacia el tránsito
- ☐ Si usa una bandera, manténgala en posición horizontal. No flamee la bandera. Ello confunde a los conductores. Mire al conductor a los ojos
- ☐ Después que el primer automóvil se haya detenido, puede dirigirse al centro del camino para detener a los automóviles que le siguen.



To Stop Traffic



Traffic Proceed



To Alert and Slow Traffic

Dar paso al tránsito

- ☐ Hágase a un lado del camino en la banquina, gire la paleta a LENTO (SLOW) (o deje caer la bandera a su lado).
- ☐ Con su mano libre, indíquelo al tránsito que proceda.
- ☐ ¡No flamee la bandera!
- ☐ Ello confunde a los conductores.

Señal de Alerta/despacio al tránsito

- ☐ Párese de cara al tránsito y apunte la paleta con el mensaje DESPACIO (SLOW) hacia el tránsito, use su mano libre para indicar al tránsito que se mueva hacia adelante
- ☐ Con su mano libre, indique al tránsito que proceda, levantándola y bajándola.
- ☐ Usando la bandera, lentamente mueva la bandera directamente en frente del cuerpo hacia el sentido de circulación.
- ☐ ¡No flamee la bandera! Ello confunde a los conductores.

Este documento por el presente certifica que he recibido instrucción sobre los puntos listados anteriormente y que he tenido oportunidad de formular preguntas. Entiendo en forma cabal los requisitos de señalización de DPR.

Firma del empleado

Firma de la persona que realiza la verificación del repaso

Se debe enviar por fax una copia de este formulario a la oficina regional de DPR ANTES DE COMENZAR EL TRABAJO.

Section 19

Respiratory Protection

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Respiratory Protection

1.0 Introduction

In the Respiratory Protection Program, hazard assessment and selection of proper respiratory protective equipment is conducted in the same manner as for other types of personal protective equipment (PPE).

In the control of occupational diseases caused by breathing air contaminated with harmful dusts, fogs, fumes, mists, gases, smokes, sprays, or vapors, the primary objective should be to prevent atmospheric contamination. This may be accomplished as far as feasible by accepted engineering control measures; i.e., enclosure or confinement of the operation or of employees, general and local ventilation, substitution of less toxic materials, or change of the work process such as utilizing wet methods, vacuum sanding, dust collectors, etc.

When effective engineering controls are not feasible, or while they are being instituted, appropriate respirators shall be used.

Reference: OSHA Standards Respiratory Protection 29 CFR 1910.134, Respirable Crystalline Silica 29 CFR 1926.1153 and Chemical and Environmental Section 2 or any other applicable Standards, guidelines, and regulations may apply.

Definitions

Air-purifying respirator: A respirator with an air-purifying filter, cartridge, or canister that removes specific air contaminants by passing ambient air through the air-purifying element.

Atmosphere-supplying respirator: A respirator that supplies the user with breathing air from a source independent of the ambient atmospheres. This includes supplied-air respirators (SARs) and self-contained breathing apparatus (SCBA) units.

Canister or cartridge: A container with a filter, sorbent, or catalyst, or combination of these items, which removes specific contaminants from the air passed through the container.

Crystalline silica: A basic component of soil, sand, granite, and many other minerals. Quartz is the most common form of crystalline silica. Cristobalite and tridymite are two other forms of crystalline silica. All three forms may become respirable size particles when workers chip, cut, drill or grind objects that contain crystalline silica.

Demand respirator: An atmosphere-supplying respirator that admits breathing air to the facepiece only when a negative pressure is created inside the facepiece by inhalation.

Emergency situation: Any occurrence such as, but not limited to, equipment failure, rupture of containers, or failure of control equipment that may or does result in an uncontrolled significant release of an airborne contaminant.

End-of-Service-Life Indicator (ESLI): A system that warns the respirator user of the approach of the end of adequate respiratory protection, for example, that the sorbent is approaching saturation or is no longer effective.

Escape-only respirator: A respirator intended to be used only for emergency evacuation.

Filter or air purifying element: A component used in respirators to remove solid or liquid aerosols from the inspired air.

Filtering facepiece (dust mask): A negative pressure particulate respirator with a filter as an integral part of the facepiece or with the entire facepiece composed of the filtering medium.

Fit test: The use of a protocol to qualitatively or quantitatively evaluate the fit of a respirator on an individual. (See also Qualitative fit test QLFT and Quantitative fit test QNFT)

High Efficiency Particulate Air (HEPA) Filter: A filter that is at least 99.97% efficient in removing monodisperse particles of 0.3 micrometers in diameter. The equivalent NIOSH 42 CFR 84 particulate filters are the N100, R100, and P100 filters.

Immediately Dangerous to Life or Health (IDLH): An atmosphere that poses an immediate threat to life, would cause irreversible adverse health effects, or would impair an individual's ability to escape from a dangerous atmosphere.

Loose-fitting facepiece: A respiratory inlet covering that is designed to form a partial seal with the face.

Negative pressure respirator (tight fitting): A respirator in which the air pressure inside the facepiece is negative during inhalation with respect to the ambient air pressure outside the respirator.

Positive pressure respirator: A respirator in which the pressure inside the respiratory inlet covering exceeds the ambient air pressure outside the respirator.

Powered Air-Purifying Respirator (PAPR): An air-purifying respirator that uses a blower to force the ambient air through air-purifying elements to the inlet covering.

Pressure demand respirator: A positive pressure atmosphere-supplying respirator that admits breathing air to the facepiece when the positive pressure is reduced inside the facepiece by inhalation.

Qualitative Fit Test (QLFT): A pass/fail fit test to assess the adequacy of respirator fit that relies on the individual's response to the test agent.

Quantitative Fit Test (QNFT): An assessment of the adequacy of respirator fit by numerically measuring the amount of leakage into the respirator.

Self-Contained Breathing Apparatus (SCBA): An atmosphere-supplying respirator for which the breathing air source is designed to be carried by the user

Supplied-Air Respirator (SAR) or Airline Respirator: An atmosphere-supplying respirator for which the source of breathing air is not designed to be carried by the user.

Tight-fitting facepiece: A respiratory inlet covering that forms a complete seal with the face.

2.0 Pre-Task Planning

Work activities that expose or have the potential to expose workers to respiratory hazards are activities defined by DPR to be High Hazard and therefore require daily detailed, written pre-task planning and may require a Job Hazard Analysis (JHA) if required by the DPR project team. *Refer to the Pre-Planning-Subcontractor Section 17 for additional pre-task planning guidelines.* A management level JHA and daily, detailed, written pre-task plan should be required, at a minimum, for the following activities:

- Asbestos
- Lead
- Grinding, sanding, drilling, chipping or cutting activities that create crystalline silica dust from materials such as concrete, drywall mud, mortar, brick, stone
- Biohazards
- Work with hazardous or irritant substances
- Oxygen deficient environments
- Welding on or with potentially hazardous materials and metals
- Environmental hazards such as mold
- Others as determined through a hazard assessment

3.0 Responsibilities

The contractor's designated Respiratory Protection Program Administrator should coordinate the company's Respiratory Protection Program and ensure compliance with DPR EHSP and OSHA requirements. The Administrator's responsibilities should include:

- Selection and purchase of approved respirators
- Fit testing
- Organization of training
- Administration of the program to include medical clearance, use, maintenance, storage and sanitary care

Program Evaluation: The Respiratory Protection Program should be evaluated at least annually to ensure it complies with regulatory requirements and adequately meets the needs of the contractor's written safety program and the DPR EHSP.

4.0 Hazard Assessment and Control

Hazard assessment and the need for respiratory protection should be addressed during the preparation of the contractor's written Respiratory Protection Program. The company's written program and operating procedures should be made available to workers who may have the need to wear respiratory protection.

Appropriate selection criteria should be established so that the selected respirators are applicable and suitable for the purpose intended. Selection of respirators should be in accordance with the guidelines defined in the Respiratory Protection Program and should be limited to NIOSH/MSHA approved equipment. The selection of a respirator for any given situation should also consider the following factors:

- Nature of respiratory hazard; consideration should be given to the specific nature of the respiratory hazard, including:
 - *presence or absence of oxygen* – atmosphere supplying respirators should be used in oxygen deficient environments
 - *contaminant concentration* - air purifying respirators are only approved for use in environments with known airborne contaminant concentrations, i.e. not to exceed the permissible exposure limit (PEL), short term exposure limit (STEL) and/or time weighted average (TWA)
 - *physical and chemical properties of the contaminant* – if the contaminant or mixture exists in different physical states (gas, vapor, mist, dust, fume), the respirator should provide protection for all forms
 - an air purifying device may require a chemical sorbent and particulate filter for adequate protection
 - the presence of two or more incompatible contaminants that might react in the cartridges or canisters would exclude the use of air purifying devices
- Workplace and User Factors
 - *size and configuration of the workspace*
 - *ease of worker communication*
 - *ease or difficulty of the work or rate of activity*
 - *workplace conditions such as temperature and humidity*
 - *workplace conditions such as movement of equipment, personnel and material*

When the nature of the hazards is unknown, it should be assumed that the environment is Immediately Dangerous to Life and Health (IDLH) and all reasonable and necessary precautions should be taken. Monitoring, sampling or other method of determining exposures should be done to accurately estimate worker exposures and to assist in selecting respiratory protection.

The following activities have been identified as typical activities where respirator use may be required by DPR employees:

- Concrete work such as drilling, jackhammering, cutting, chipping, grinding, scabbling, drilling, finishing, etc.
 - *Reference Demolition Section 06 for additional information.*
- Mixing of dry concrete, grout, sack & patch compounds, etc.
- Sanding of drywall joint compounds
- Demolition
- Work with pressure treated lumber e.g., cutting, sawing, drilling, etc.
- Sandblasting or bead blasting operations
- insulation, etc.
- Application and removal of fireproofing materials – monokote, wool/fiber, insulation, etc.
- Sweeping – general and heavy
- Caulking and sealing operations
- Welding operations

Note: welding operations on stainless steel should be pre-planned and the work plan should address potential chromium exposure and associated protective measures.

- Clean-up operations using chemical or hazardous substances such as solvents and cleaners e.g., alcohol, acetone, acids, etc.
- Drywall and/or lead board installation
- Work with hazardous chemicals and substances (various)
- Work in and around other operations where respiratory protection is required

Refer to the Recommended Use Guidelines below for assistance in selecting the appropriate respirators.

4.1 Voluntary Respirator Use

When the use of respirators will not present a hazard in itself due to misuse, other hazards or conditions in the workplace that do not exceed any regulated PEL's or employee medical condition, a worker may use respirators voluntarily. For the sake of this sub-section, "voluntary use" will be defined as when a worker chooses to wear a respirator, even though the use of a respirator is not required by DPR, OSHA or other. Voluntary use does not apply to elastomer or rubber face piece type masks (half or full). Voluntary users should be trained on respirator use, maintenance, etc. and should be provided with the Voluntary Use Respirator Advisory Form.

5.0 Medical Evaluation

A medical evaluation should be performed to determine a worker's ability to use a respirator before the worker is fit tested, before the respirator is used for the first time and at a minimum annually thereafter, while the worker continues to use a respirator. A medical questionnaire is completed by the worker and evaluated by a trained and licensed health care professional. Further evaluation may be required, at the discretion of the health care professional, in addition to routine follow-up evaluations.

Information obtained via the medical evaluation process should be kept strictly confidential. Upon completion of the medical evaluation, the health care professional should complete a written determination of the employee's ability to use a respirator under the particular circumstances they will be exposed to using the Medical Clearance Report or equivalent.

Medical evaluations are not required for voluntary use of respirators.

6.0 Fit Testing

Workers should be fit tested before they are permitted to use any type of respiratory protection (with the exception of nuisance dust masks and positive pressure air hoods). The fit testing should be performed using the same make, model, style and size of respirator that will be used by the worker. Fit testing of respirators should be conducted prior to assignment, at least annually thereafter and any time there are significant

changes in the employee's physical condition that could affect fit (e.g., significant weight loss or gain, dental surgery), (unless otherwise specified. Fit testing should be conducted and be performed by the Program Administrator or designee using an approved method per 29 CFR 1926.103/1910.134, Appendix A. Upon completion of a successful fit test, the type of fit test methodology used and the size, make and model of the respirator should be documented using the Respirator Fit Test Record form or equivalent.

7.0 Respiratory Training

Following medical approval and successful fit testing, training should be completed prior to the worker using a respirator. Re-training should be completed on an annual basis or more frequently if required. Each respirator wearer should be provided formal training, which should include the following:

- Requirements of the OSHA respiratory protection standard (29 CFR 1926.103/1910.134)
- Instruction in the nature of the respiratory hazards and what may happen if the respirator is not used properly
- An explanation of engineering and administrative control measures being used and why respirators are needed to provide protection
- Instruction in the selection, use, sanitary care, maintenance, proper storage and limitations of each applicable respirator type
- Demonstration and practice in proper fitting, wearing, adjusting, inspecting, maintaining and assuring of the face-to-face-piece seal of each applicable respirator type (e.g., positive and negative pressure seal checks)
- An opportunity to handle the respirator and to wear it in a safe atmosphere for an adequate period of time so that the wearer is familiar with the operational characteristics of the respirator
- An explanation of the medical surveillance program as it relates to the use of respiratory protective equipment
- An explanation of the requirements for maintaining the respirator face to facepiece seal, including beard and facial hair policies

8.0 Respirator Use, Care and Maintenance

- Supervisors are responsible for ensuring the safe use, care and maintenance of respirators by their workers
- Workers who use respirators should have no conditions as to affect the seal of the respirator e.g., facial hair, corrective glasses, or PPE
- An inspection of the respirator should be performed prior to each use at minimum
- User seal checks should be performed each time a respirator is worn
- Appropriate cartridges should be utilized at all times e.g.; dust filters provide no protection against organic vapors
- Respirators should be maintained at all times and all defective, missing or damaged parts replaced or repaired prior to use
- Filtering face pieces and air purifying respirators should have their filters and cartridges changed when the worker first experiences difficulty (resistance) breathing while wearing the mask; chemical cartridges should be maintained and regularly changed as per the exposure
- Respirators should be regularly cleaned and disinfected using appropriate cleaning solutions and disinfecting agents; refer to the manufacturers suggestions for further guidelines
- In general, warm soapy water and a disinfectant spray or wipe containing 70% isopropyl alcohol may be used
- In those cases, where respirators are assigned to individual workers, those respirators should be marked with the worker's name
- When respirators are not in use, they should be stored in such a manner as to provide protection from mechanical damage, dust, chemicals, sunlight, extreme heat, cold or moisture

- If a respirator is utilized in a regulated area, proper doffing procedures shall be followed to prevent the spread of potential hazards outside of the regulated area.

9.0 Assigned Protection Factors (APFs)

Employers must use the assigned protection factors listed in Table 1 to select a respirator that meets or exceeds the required level of employee protection. When using a combination respirator (e.g., airline respirators with an air-purifying filter), employers must ensure that the assigned protection factor is appropriate to the mode of operation in which the respirator is being used.

Table 1. -- Assigned Protection Factors⁵

Type of respirator ^{1, 2}	Quarter mask	Half mask	Full facepiece	Helmet/hood	Loose-fitting facepiece
1. Air-Purifying Respirator	5	³ 10	50		
2. Powered Air-Purifying Respirator (PAPR)		50	1,000	⁴ 25/1,000	25
3. Supplied-Air Respirator (SAR) or Airline Respirator					
• Demand mode		10	50		
• Continuous flow mode		50	1,000	⁴ 25/1,000	25
• Pressure-demand or other positive-pressure mode		50	1,000		
4. Self-Contained Breathing Apparatus (SCBA)					
• Demand mode		10	50	50	
• Pressure-demand or other positive-pressure mode (e.g., open/closed circuit)			10,000	10,000	

Notes:

¹Employers may select respirators assigned for use in higher workplace concentrations of a hazardous substance for use at lower concentrations of that substance, or when required respirator use is independent of concentration.

²The assigned protection factors in Table 1 are only effective when the employer implements a continuing, effective respirator program as required by this section (29 CFR 1910.134), including training, fit testing, maintenance, and use requirements.

³This APF category includes filtering facepieces, and half masks with elastomeric facepieces.

⁴The employer must have evidence provided by the respirator manufacturer that testing of these respirators demonstrates performance at a level of protection of 1,000 or greater to receive an APF of 1,000. This level of performance can best be demonstrated by performing a WPF or SWPF study or equivalent testing. Absent such testing, all other PAPRs and SARs with helmets/hoods are to be treated as loose-fitting facepiece respirators, and receive an APF of 25.

⁵These APFs do not apply to respirators used solely for escape. For escape respirators used in association with specific substances covered by 29 CFR 1910 subpart Z, employers must refer to the appropriate substance-specific standards in that subpart. Escape respirators for other IDLH atmospheres are specified by 29 CFR 1910.134 (d)(2)(ii).



10.0 Recordkeeping

All medical records pertinent to workers wearing respiratory protection should be maintained for the duration of the worker's employment, plus 30 years. Each Respiratory Fit Testing Record should be kept for respirator users until the next fit test is administered.

The Respirator Use Guidelines details DPR's recommended use guidelines for typical operations. This table is to be used as a guide. Specific hazard assessments should be performed to determine the exact exposure levels, nature of the hazard and type of protection appropriate.

Type of Work	Duration of Activity	Recommended Respirator	General Hazard Controls (Ventilation and Isolation to be used for all)
Any activity with the potential for occupational exposure to respirable crystalline silica, e.g. concrete chipping, grinding, drilling, jackhammering, sawing, milling, crushing, demolition, etc.	Any duration	Refer to CFR 1926.1153 Respirable Crystalline Silica; CCR Section 1532.3 Occupational Exposure to Respirable Crystalline Silica; DPR EHSP Chemical and Environmental Management Section 02; and/or manufacturers SDS	
Beadblasting	Any duration	Refer to the manufacturer's SDS	
Sandblasting	Any duration	Atmosphere supplying respirator	Alternative abrasive materials
Drywall sanding (Other than pole sanding)	Any duration	Refer to the manufacturer's SDS for the specific joint compound used	- Wet methods - Dust collectors - Vacuum sanders
Drywall Sanding (Pole Sanding) Open Spaces > 12' x 12'	Any duration	None or Nuisance dust mask (no medical exam or fit testing required)	- Wet Methods - Dust collectors - Vacuum sanders
Drywall Sanding (Pole Sanding) Spaces < 12' x12'	< 2 hours/day, Per person	None or Nuisance dust mask (no medical exam or fit testing required)	- Wet Methods - Dust collectors - Vacuum sanders
Drywall Sanding (Pole Sanding) Spaces < 12' x12'	> 2 hours/day, Per person	Half-face purifying respirator with N-95 or higher particulate cartridges	- Wet Methods - Dust collectors - Vacuum sanders
Drywall Installation	Any duration	None or Nuisance dust mask (no medical exam or fit testing needed)	
Leadboard Installation	Any duration	Task specific - ensure a specific hazard assessment is performed based on the individual exposures.	Hand cutting when possible
Chopsaw Use	Any duration	None or Nuisance dust mask (no medical exam or fit testing needed)	Smoke hogs – vacuum devices
Demolition Activities	Any duration	Task specific – perform hazard assessment based on the individual exposures.	Various
Sweeping or brushing (dry sweeping or brushing of any duration is prohibited)	Any duration	None or Nuisance dust mask (no medical exam or fit testing required)	- Wet methods - Sweep compounds - Vacuuming (HEPA preferred)
Clean-up operations using chemical or hazardous substances such as	Any durations	Refer to the manufacturer's SDS	- Substitution - Dispensing containers

Type of Work	Duration of Activity	Recommended Respirator	General Hazard Controls (Ventilation and Isolation to be used for all)
solvents, cleaners etc. (e.g. alcohol, acids, acetone etc.)			• Proper handling & disposal methods
Caulking, Sealing, Epoxy, Adhesives etc. application	Any duration	Refer to the manufacturer's SDS	• Substitution • Mechanical fastening
Work with hazardous chemicals and materials (various)	Any duration	Refer to the manufacturer's SDS Task specific - ensure a specific hazard assessment is performed based on the individual exposures.	• Substitution • Floor fans • Neutralizing compounds • Various • Proper handling & disposal methods
Work with and around contaminated soils	Any duration	Task specific - ensure a specific hazard assessment is performed based on the individual exposures.	• Containment/Encapsulation • Neutralizing compounds • Removal
Sanding, cutting, drilling of treated or specialty wood products	Any duration	Refer to manufacturers recommended guidelines. In general a half-face particulate respirator with N-95 or higher designation.	• Substitution • Prefabrication • Hand of low speed tools
Cutting, drilling, sanding, grinding on stone, ceramic, fiberglass, resin products etc.	Any duration	Task specific - ensure a specific hazard assessment is performed based on the individual exposures.	• Substitution • Vacuum systems • Wet Methods
Application or removal of fireproofing materials e.g. monokote, wool/fiber insulation	Any duration	Refer to the manufacturer's SDS	• Wet methods • Vacuum systems • Extension tools
Cutting of metal studs	Any duration	None or Nuisance dust mask (no medical exam or fit testing needed)	• Smoke eaters – vacuum systems • Redesign – alternate fastening, strengthening
Welding, cutting, grinding – general	Any duration	Task specific - ensure a specific hazard assessment is performed based on the individual exposures.	• Smoke eaters – vacuum systems

Note: Silica concentrations can vary from region to region and from concrete mixture to mixture, therefore it is advisable to conduct project and task specific industrial hygiene sampling if there are questions regarding the exact exposure. **For specific information on hazard evaluations, sampling and exposure data for the specific activities listed above, please refer to DPR Archives**

Respiratory Protection Forms

Forms

Respirator Medical Evaluation Questionnaire – English
Respirator Medical Evaluation Questionnaire – Spanish
Medical Clearance Report
Respirator Fit Test Record
Voluntary Use Respirator Advisory – English
Voluntary Use Respirator Advisory – Spanish



RESPIRATOR MEDICAL EVALUATION QUESTIONNAIRE

The following information must be provided by every employee who has been selected to use any type of respirator.
(Please print)

DATE		Time		# of Pages (including cover)	
ATTENTION				FAX #	
COMPANY				REFERENCE	Respiratory Medical Evaluation
FROM	DPR Construction			Questionnaire	

Confidential Fax

Note to person sending fax:

The attached document is confidential. The cover sheet is to remain on top of document, once the document has been faxed, return the original to the worker, retain the fax confirmation sheet verifying the fax was successfully sent.

Note to fax recipient:

The information contained within this document is intended solely for the recipient and is of a confidential nature. If this fax is received in error, please notify DPR Construction at 916-418-1910, return the fax to the sender and destroy all copies RECEIVED.



RESPIRATOR MEDICAL EVALUATION QUESTIONNAIRE

The following information must be provided by every employee who has been selected to use any type of respirator.
(Please print)

Project/Project#:		Date:	
Name:			
Job Title:		SS#:	
Height:		Weight:	
Age:		Sex:	Male/Female

Phone number where you can be reached by the health care professional reviewing this questionnaire			
Phone:		Best time to call:	am/pm

Has your employer told you how to contact the health care professional who will review this questionnaire?			
Yes	☐	No	☐

Check the type of respirator(s) you will use:					
☐	Disposable respirator (N95/P95) filter mask, non – cartridge type only.				
☐	Other type: (for example, half or full face piece type, powered-air purifying, supplied-air, self-contained breathing apparatus)				
Have you ever worn a respirator?					
☐	Yes	☐	No	If yes, what type(s)	

Questions 1-9 must be answered by every employee who has been selected to use any type of respirator.

YES	NO	SECTION 1			
		1. Do you currently smoke tobacco, or have smoked tobacco in the last month?			
		2. Have you ever had any of the following conditions?			
		- Seizures (fits) - Diabetes (sugar disease) - Allergic reactions that interfere with your breathing - Claustrophobia (fear of closed-in or confined space) - Trouble smelling odors			
		3. Have you ever had any of the following pulmonary or lung problems?			
YES	NO		YES	NO	
		Asbestosis			Asthma
		Chronic Bronchitis			Emphysema
		Pneumonia			Tuberculosis
		Silicosis			Pneumothorax (collapsed lung)
		Lung cancer			Broken ribs
		Any chest injuries/surgeries			Any other lung problems
		Pleurisy			Allergies
		4. Do you currently have any of the following symptoms of pulmonary or lung illness?			
		- Shortness of breath - Shortness of breath when walking fast on level ground or walking up a slight incline - Shortness of breath when walking with others at an ordinary pace on level ground - Have to stop for breath when walking at your own pace on level ground - Shortness of breath when washing or dressing yourself - Shortness of breath that interferes with your job - Coughing that produces phlegm (thick sputum) - Coughing that occurs mostly when you are lying down - Coughing that wakes you early in the morning - Coughing up blood in the last month.			

RESPIRATOR MEDICAL EVALUATION QUESTIONNAIRE

The following information must be provided by every employee who has been selected to use any type of respirator.
(Please print)

		• Wheezing • Wheezing that interferes with your job • Chest pain when you breathe deeply. • Any other symptoms that you think may be related to lung problem?										
YES	NO	5. Have you ever had any of the following cardiovascular or heart problems??										
		• Heart attack • Stroke • Angina • Heart failure • Swelling in your legs or feet (not due to walking) • Irregular heart beat • High blood pressure • Any other heart problems?										
YES	NO	6. Have you ever had any of the following cardiovascular or heart symptoms?										
		• Frequent pain or tightness in your chest • Pain or tightness in your chest during physical activity • Pain or tightness in your chest that interferes with your job • In the past two years, have you noticed your heart skipping or missing a beat • Heartburn or indigestion that is not related to eating • Any other symptoms that you think may be related to heart or circulation problems										
YES	NO	7. Do you take medication for the following problems?										
		<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 50%; background-color: #d3d3d3;">PROBLEM</th> <th style="width: 50%; background-color: #d3d3d3;">MEDICATION (please list)</th> </tr> </thead> <tbody> <tr> <td>• Breathing or lung problems</td> <td></td> </tr> <tr> <td>• Heart problem</td> <td></td> </tr> <tr> <td>• Blood pressure</td> <td></td> </tr> <tr> <td>• Seizures</td> <td></td> </tr> </tbody> </table>	PROBLEM	MEDICATION (please list)	• Breathing or lung problems		• Heart problem		• Blood pressure		• Seizures	
PROBLEM	MEDICATION (please list)											
• Breathing or lung problems												
• Heart problem												
• Blood pressure												
• Seizures												
YES	NO	8. If you have used a respirator, have you ever had any of the following problems?										
		• Eye irritation • Skin allergies or rashes • Anxiety • General weakness or fatigue • Any other problems that interfere with your use of a respirator?										
YES	NO											
		9. Would you like to talk to the health professional who will review this questionnaire about your answers?										
Questions 10 to 15 below must be answered by every employee who has been selected to use either a full-facepiece respirator or a self-contained breathing apparatus (SCBA). For employees who have been selected to use other types of respirators answering these questions is voluntary.												
YES	NO											
		10. Have you ever lost vision in either eye temporarily or permanently?										
		11. Do you currently have any of the following vision problems?										
		• Wear contact lens • Wear glasses • Color blind • Other vision/eye problems?										
YES	NO											

RESPIRATOR MEDICAL EVALUATION QUESTIONNAIRE

The following information must be provided by every employee who has been selected to use any type of respirator.
(Please print)

		12. Do you currently have any of the following hearing problems?
		• <i>Difficulty hearing</i>
		• <i>Wear a hearing aid</i>
		• <i>Other hearing/ear problems?</i>
YES	NO	
		13. Have you ever had a back injury?
YES	NO	
		14. Do you currently have any of the following musculoskeletal problems?
		• <i>Weakness in any of your arms, hands, legs or feet?</i>
		• <i>Back pain</i>
		• <i>Difficulty fully moving your arms and legs</i>
		• <i>Pain or stiffness when you lean forward or backward at the waist</i>
		• <i>Difficulty fully moving your head up and down</i>
		• <i>Difficulty fully moving your head side to side</i>
		• <i>Difficulty bending at your knees</i>
		• <i>Difficulty squatting to the ground</i>
		• <i>Climbing a flight of stairs or a ladder carrying more than 25 lbs.</i>
		• <i>Any other muscle or skeletal problem that interferes with using a respirator?</i>

YES	NO	SECTION 2
		15. At work or at home, have you ever been exposed to hazardous solvents, airborne chemicals (e.g., gases, fumes or dust), or have you come into contact with hazardous chemicals?
YES	NO	
		16. Have you ever worked with any of the materials, or under any of the conditions listed below?
		• <i>Asbestos</i>
		• <i>Silica</i>
		• <i>Tungsten/cobalt (e.g. grinding/welding this material)</i>
		• <i>Beryllium</i>
		• <i>Aluminum</i>
		• <i>Coal</i>
		• <i>Iron</i>
		• <i>Tin</i>
		• <i>Dusty environments</i>
		• <i>Any other hazardous exposures?</i>
17. List your previous occupations:		
18. List your current and previous hobbies:		
YES	NO	
		19. Have you ever worked on a HAZMAT team?
20. Other than medications for breathing and lung problems, heart trouble, blood pressure and seizures mentioned earlier in this questionnaire, are you taking any other medications for any reason (including over-the-counter medications)? Please list them.		
YES	NO	



RESPIRATOR MEDICAL EVALUATION QUESTIONNAIRE

The following information must be provided by every employee who has been selected to use any type of respirator.
(Please print)

		21. How often are you expected to use the respirator(s)? Check yes for all answers that apply to you otherwise, check no.
		• <i>Escape only (no rescue)</i>
		• <i>Emergency rescue only</i>
		• <i>Less than 5 hours per week</i>
		• <i>Less than 2 hours per day</i>
		• <i>2 to 4 hours per day</i>
		• <i>Over 4 hours per day</i>
YES	NO	
		22. During the period you are using the respirator(s), is your work effort:
		• <i>Light (less than 200kcal per hour) e.g. sitting, typing standing while operating drill</i>
		<i>If yes, how long does this period last during the average shift?</i> <i>Hrs/mins.</i>
		• <i>Moderate (200kcal per hour) e.g. driving a truck in traffic; standing while drilling; performing assembly work or transferring a moderate load (35 lbs.) or pushing a heavy load (100 lbs.)</i>
		<i>If yes, how long does this period last during the average shift?</i> <i>Hrs/mins.</i>
		• <i>Heavy (above 350 kcal per hour) e.g. lifting a heavy load (50 lbs.) from the floor to your waist or shoulder; working on a loading dock; shoveling, climbing stairs with heavy load</i>
		<i>If yes, how long does this period last during the average shift?</i> <i>Hrs/mins.</i>
		23. Will you be working under hot conditions (temperatures exceeding 77 degrees)?
24. Describe any special or hazardous conditions you might encounter when you're using your respirator(s). For example: confined spaces, life-threatening gases):		

I have answered all questions to the best of my knowledge.

Employee Signature	Date
--------------------	------

Tympanic Membranes				
Intact:		Right		Left
Cleared for Respirator Use		Yes		No
Restricted from Respirator Use				
Restriction is:		Permanent		Temporary
Review Date:				

Provider Signature	Date
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CUESTIONARIO DE EVALUACIÓN MÉDICA PARA PODER USAR RESPIRADOR

Cada empleado seleccionado para usar cualquier tipo de respirador deberá completar la información que sigue.

(Por favor use letra de imprenta)

Proyecto/Proyecto#:		Fecha:	
Nombre:			
Título del cargo:		Número de Seguro Social	
Estatura:		Peso:	
Edad:		Sexo:	Male/Female

Número telefónico donde el profesional de cuidados de la salud que evalúa este cuestionario puede comunicarse con usted

Código de área:		La mejor hora para llamarlo/la a este número	am/pm
-----------------	--	--	-------

¿Le ha informado su empleador sobre la forma de comunicarse con el profesional de cuidados de la salud que evaluará este cuestionario?

Si	☐	No	☐
----	--------------------------	----	--------------------------

Marque el tipo de respirador o respiradores que usted usará:

☐	Respirador desechable (N95/P95) máscara de filtro, tipo sin cartucho exclusivamente.
☐	Otros tipos: (por ejemplo, respirador con máscara facial parcial o completa, respirador purificador de aire a batería, respirador de manguera exterior, respirador autónomo).

¿Alguna vez ha usado un respirador?

☐	Yes	☐	No	Si contestó afirmativamente, ¿de qué tipo(s)?	
--------------------------	-----	--------------------------	----	---	--

Todo empleado seleccionado para usar cualquier tipo de respirador deberá responder a las preguntas 1-9.

Si	NO	SECCION 1			
		1. ¿Fuma tabaco actualmente, o ha fumado tabaco durante el último mes?			
		2. Have you ever had any of the following conditions?			
		• Convulsiones • Diabetes (enfermedad del azúcar) • Reacciones alérgicas que le dificultan la respiración • Claustrofobia (temor a los espacios encerrados o confinados) • Dificultad en percibir olores (problemas del olfato)			
		3. Have you ever had any of the following pulmonary or lung problems?			
Si	NO		Si	NO	
		Asbestosis			Asma
		Bronquitis crónica			Enfisema
		Pulmonía			Tuberculosis
		Silicosis			Neumotórax (colapso del pulmón)
		Cáncer del pulmón			Costillas fracturadas
		Cualquier tipo de lesión o cirugía al tórax			Cualquier otro problema pulmonar
		Pleuresía			Alergias
Si	NO	4. ¿Actualmente tiene algunos de los siguientes síntomas de enfermedad al pulmón?			
		• Falta de aire • Falta de aire al caminar rápidamente en una superficie plana o al subir una pendiente leve • Falta de aire al caminar con otras personas a paso normal en superficie plana • Necesita parar y recuperar el aliento cuando camina a su propia velocidad en superficie plana • Falta de aire al lavarse o vestirse • Falta de aire que interfiere con su trabajo • Tos que produce flema (expectoraciones gruesas) • Tos que ocurre principalmente cuando usted está acostado/a • Tos que le despierta temprano en la mañana • Tos con sangre durante el último mes • Resuellos al respirar • Resuellos que interfieren con su trabajo • Dolor al pecho cuando respira profundo			

CUESTIONARIO DE EVALUACIÓN MÉDICA PARA PODER USAR RESPIRADOR

Cada empleado seleccionado para usar cualquier tipo de respirador deberá completar la información que sigue.

(Por favor use letra de imprenta)

		<i>¿Tiene otros síntomas que piensa que pueden estar relacionados con problemas al pulmón?</i>	
Si	NO	5. ¿Alguna vez ha tenido alguno de los siguientes problemas cardíacos o cardiovasculares?	
		• Ataque al corazón • Ataque apopléjico (derrame cerebral) • Angina • Insuficiencia cardíaca • Hinchazón de piernas o pies (no producidos al caminar) • Arritmia del corazón • Presión alta	
		<i>¿Algún otro problema cardíaco?</i>	
Si	NO	6. ¿Alguna vez ha tenido alguno de los siguientes síntomas cardíacos o cardiovasculares?	
		• Frecuente dolor o sensación de presión en el pecho • Dolor o sensación de presión en el pecho al realizar una actividad física • Dolor o sensación de presión en el pecho que interfiere con su trabajo • Durante los últimos dos años, ¿ha notado un cambio en el ritmo del latido de su corazón? • Acidez o indigestión no ocasionada al comer	
		<i>¿Tiene otros síntomas que piensa que pueden estar relacionados con problemas cardíacos o circulatorios?</i>	
Si	NO	7. ¿Toma medicamento para los siguientes problemas?	
		PROBLEMA	MEDICAMENTO (hacer lista)
		• Problemas pulmonares o de respiración • Problema cardíaco • Presión arterial • Convulsiones	
Si	NO	8. Si usted ha usado un respirador, ¿ha tenido alguna vez alguno de los siguientes problemas?	
		• Irritación a los ojos • Alergias a la piel o sarpullidos • Ansiedad • Fatiga o debilidad general	
		<i>¿Tiene otros problemas que interfieren con su uso de un respirador?</i>	
Si	NO		
		9. ¿Desea hablar sobre sus respuestas con el/la profesional de la salud a cargo de evaluar este cuestionario?	
Todo empleado seleccionado para usar un respirador de máscara facial completa o un aparato respirador autónomo (SCBA) deberá responder a las preguntas 10 a 15 que siguen. Las respuestas de aquellos empleados que han sido seleccionados para usar otros tipos de respiradores son voluntarias.			
Si	NO		
		10. ¿Alguna vez ha perdido visión en un ojo, en forma temporal o permanente?	
		11. ¿Actualmente tiene alguno de los siguientes problemas a la vista?	
		• Usa lentes de contacto • Usa anteojos • Sufre de daltonismo (dificultad en distinguir colores)	
		<i>¿Tiene otros problemas de los ojos o de la vista?</i>	
Si	NO		
		12. ¿Actualmente tiene alguno de los siguientes problemas auditivos?	
		• Dificultades auditivas • Usa un aparato auditivo	
		<i>¿Tiene otros problemas auditivos o del oído?</i>	
Si	NO		
		13. ¿Alguna vez ha tenido una lesión a la columna?	

CUESTIONARIO DE EVALUACIÓN MÉDICA PARA PODER USAR RESPIRADOR

Cada empleado seleccionado para usar cualquier tipo de respirador deberá completar la información que sigue.

(Por favor use letra de imprenta)

SI	NO	
		14. ¿Tiene actualmente alguno de los siguientes problemas músculo esqueléticos?
		- Debilidad en uno o ambos brazos, manos, piernas o pies - Dolor de espalda - Dificultad para mover sus brazos y piernas - Dolor o rigidez al doblar la cintura hacia adelante o hacia atrás - Dificultad para mover la cabeza hacia arriba o hacia abajo - Dificultad para mover la cabeza completamente de un lado a otro - Dificultad para doblar las rodillas - Dificultad para ponerse de cuclillas - Dificultad para subir gradas o una escalera portátil cargando más de 25 libras
		¿Tiene algún otro problema muscular o del esqueleto que interfiere con el uso de un respirador?
SI	NO	SECCION 2
		15. En casa o en el trabajo, ¿alguna vez ha estado expuesto/a a solventes peligrosos, sustancias químicas en el aire (es decir, gases, vapores o polvo) o ha tenido contacto con sustancias químicas peligrosas?
SI	NO	
		16. ¿Alguna vez ha trabajado con cualquiera de los materiales, o bajo cualquiera de las condiciones que se indican a continuación?
		- Asbesto/amianto - Sílice - Tungsteno/cobalto (por ejemplo, al pulir o soldar este material) - Berilio - Aluminio - Carbón - Hierro - Estaño - Lugares de mucho polvo
		¿Ha estado expuesto/a a otros peligros?
17. Indique sus anteriores ocupaciones:		
18. Indique sus pasatiempos actuales o anteriores:		
YES	NO	
		19. ¿Ha trabajado alguna vez en un equipo de emergencia para el manejo de materiales peligrosos (equipo HAZMAT)?
20. Aparte de los medicamentos que pueda estar tomando para los problemas que hemos mencionado anteriormente en este cuestionario, es decir, convulsiones, problemas pulmonares o de respiración, problemas cardíacos o de la presión arterial, ¿está usted tomando algún otro medicamento por cualquier otro motivo (incluyendo los medicamentos no recetados)? Por favor indíquelos.		
YES	NO	
		21. ¿Con qué frecuencia se anticipa que usted usará el respirador o respiradores? Indique "Sí" en todas las respuestas que le correspondan. Indique "No" para las que no le corresponden.
		- Escape solamente (sin rescate) - Rescate de emergencia solamente - Menos de 5 horas por semana - Menos de 2 horas por día 2 a 4 horas por día



CUESTIONARIO DE EVALUACIÓN MÉDICA PARA PODER USAR RESPIRADOR

Cada empleado seleccionado para usar cualquier tipo de respirador deberá completar la información que sigue.

(Por favor use letra de imprenta)

		• Más de 4 horas por día	
YES	NO		
		22. Durante el período en que usted está usando el respirador o respiradores, ¿cómo describiría la intensidad de su trabajo?:	
		<i>Liviano (menos de 200kcal por hora), por ejemplo, estar sentado/a, escribir a máquina, estar de pie al operar un taladro</i>	
		Si contestó afirmativamente, ¿cuánto dura este período en un turno promedio?	Horas/minutos
		• Moderado (200kcal por hora), por ejemplo manejar un camión en tráfico; estar de pie al taladrar; hacer trabajo de ensamblaje; correr una carga mediana (35 libras); o empujar una carga pesada (100 libras)	
		Si contestó afirmativamente, ¿cuánto dura este período en un turno promedio?	Horas/minutos
		• Pesado (más de 350 Kcal. por hora), por ejemplo levantar una carga pesada (50 libras) del piso hasta el hombro o la cintura; trabajar en un muelle de carga; trabajar con pala; subir gradas con cargas pesadas	
		Si contestó afirmativamente, ¿cuánto dura este período en un turno promedio?	Horas/minutos
		23. ¿Estará trabajando en condiciones de calor (temperaturas sobre los 77 grados)?	
24. Describa toda condición especial o peligrosa que pueda usted enfrentar al usar el respirador o respiradores. (Por ejemplo, espacios confinados, gases mortales):			

He contestado todas las preguntas a mi mejor saber y entender.

Firma del empleado

Fecha de hoy

Membrana timpánica:				
Intacta:		Derecha		Izquierda
No Intacta:		Derecha		Izquierda
Autorizado/a para uso de respirador		Yes		No
Restringido del uso de respirador				
Restricción es:		Permanente		Temporal
Fecha de evaluación:				

Firma del proveedor

Fecha



Medical Clearance Report

for

DPR Construction

Project/Project#:		Date:	
-------------------	--	-------	--

Employee Name:	
Employee SS#:	
Examination Date:	
Review Date:	

☐	Passed the OSHA Respiratory Certification Requirements	
☐	Did Not Pass the OSHA Respiratory Certification Requirements	
☐	Further Medical Evaluation Recommended:	

Examination Site:	
Address/City/State/Zip	
Phone Number:	

Print Provider Name

Provider Signature



Respirator Fit Test Record

for

DPR Construction

Project/Project#:		Date:	
-------------------	--	-------	--

Employee Name:			
Employee SS#:			
Fit Test Date:			
Respirator Type:			
Manufacturer:			
Model:		Size:	

Conditions that could affect respirator fit:

☐	Clean shaven	☐	1 – 2 days beard growth (do not test)
☐	Moustache	☐	2+ days beard growth (do not test)
☐	Glasses	☐	Dentures absent (do not test)
☐	None	☐	Other:

Fit Checks:

Negative Pressure	☐	Pass	☐	Fail	☐	Not Tested
Positive Pressure	☐	Pass	☐	Fail	☐	Not Tested

Irritant Smoke Fit Test:

☐	Normal breathing	☐	Deep breathing
☐	Side to side	☐	Up and down
☐	Rainbow passage	☐	Jogging in place
☐	Normal breathing	☐	Employee reacts to irritant smoke

Employee Acknowledgement of test results:

Employee Signature	Date
Test Conducted by	Date



Voluntary Use Respirator Advisory

Project/Project#:		Date:	
Name:			
Job Title:		SS#:	

Respirators are an effective method of protection against designated hazards when properly selected and worn. Respirator use is encouraged, even when exposures are below the exposure limit, to provide an additional level of comfort and protection for workers. However, if a respirator is used improperly or not kept clean, the respirator itself can become a hazard to the worker. Sometimes, workers may wear respirators to avoid exposures to hazards, even if the amount of hazardous substance does not exceed the limits set by OSHA standards. If DPR provides respirators for your voluntary use, or if you provide your own respirator, you need to take certain precautions to be sure that the respirator itself does not present a hazard.

You should do the following:

1. Read and heed all instructions provided by the manufacturer on use, maintenance, cleaning and care, and warnings regarding the respirators limitations.
2. Choose respirators certified for use to protect against the contaminant of concern. NIOSH, the National Institute for Occupational Safety and Health of the U.S. Department of Health and Human Services, certifies respirators. A label or statement of certification should appear on the respirator or respirator packaging. It will tell you what the respirator is designed for and how much it will protect you.
3. Do not wear your respirator into atmospheres containing contaminants for which your respirator is not designed to protect against. For example, a respirator designed to filter dust particles will not protect you against gases, vapors, or very small solid particles of fumes or smoke.
4. If you feel that you may require a respirator to perform work you should notify your supervisor and ensure that appropriate protective measures are taken to protect you.
5. Keep track of your respirator so that you do not mistakenly use someone else's respirator.
6. Ensure that you properly dispose of your respirator after use.

I acknowledge that I have read and understand the above and have had the opportunity to ask questions.

Print Name

Signature/Date



Notificación sobre respiradores

Información para los usuarios voluntarios de respiradores

Project/Project#:		Date:	
Name:			
Job Title:		SS#:	

Los respiradores constituyen un método eficaz de protección contra ciertos peligros cuando se seleccionan y usan correctamente. Se recomienda el uso de respirador, aún cuando las exposiciones sean inferiores al límite de exposición, para brindar un nivel adicional de confort y protección para los trabajadores. Sin embargo, si el respirador no se usa correctamente o no se mantiene limpio, el respirador en sí mismo puede convertirse en un peligro para el trabajador. En algunas ocasiones, los trabajadores pueden usar respiradores para evitar la exposición a peligros, aún cuando la cantidad de sustancia peligrosa no supere los límites establecidos por las normas OSHA. Si DPR suministra respiradores para su uso voluntario, o si usted trae su propio respirador, deberá tomar ciertas precauciones para estar seguro de que el respirador no represente en sí mismo un peligro.

Deberá hacer lo siguiente:

1. Lea y acate las instrucciones del fabricante sobre el uso, mantenimiento, limpieza y cuidado y las advertencias con respecto a las limitaciones de los respiradores.
2. Escoja respiradores cuyo uso esté certificado para brindar protección contra el contaminante en cuestión. El Instituto Nacional de Salud y Seguridad Ocupacional del Departamento de Salud y Servicios Humanos de EE.UU. (*National Institute for Occupational Safety and Health of the U.S. Department of Health and Human Services*, o *NIOSH por su sigla en inglés*) se ocupa de certificar los respiradores. El respirador o el envoltorio del respirador deberán contener una etiqueta o declaración de certificación, que indique para qué fue diseñado el respirador y cuánto le protegerá.
3. No use el respirador en atmósferas que contengan contaminantes para los cuales el respirador no fue diseñado. Por ejemplo, un respirador diseñado para filtrar partículas de polvo no lo protegerá contra gases, vapores, o partículas sólidas muy pequeñas de gases o humo.
4. Si siente que podría necesitar un respirador para realizar un trabajo, deberá comunicarlo a su supervisor y asegurarse de que se tomen las medidas necesarias para protegerlo.
5. Sepa cuál es su respirador para no usar el de otra persona inadvertidamente.
6. Asegúrese de disponer correctamente del respirador después del uso.

Por el presente reconozco que he leído y entiendo lo precedente y que he tenido oportunidad de formular preguntas.

Nombre y apellido en letra de imprenta

Firma/Fecha

Section 20

Rigging & Signaling

Sub-Section	Topic	Page #
1.0	Introduction	2
2.0	Signal Person Qualification	4
3.0	Pre-Planning	4

Rigging and Signaling

1.0 Introduction

Many types of rigging and signaling methods are used on jobsites for lifting and moving materials. It is DPR's intent to maintain a safe workplace for all employees and contractors; therefore, it cannot be overemphasized that only qualified individuals should direct and perform these operations. The safety rules and guidance in this section apply to all operations that involve the use of rigging and/or signaling of suspended loads. This applies to all personnel involved in such activities.

DPR reserves the right to require independent third party evaluation, re-certification or re-training of cranes or personnel.

Reference: Cranes & Derricks in Construction CFR 1926.1400 (Subpart CC); 1926.1428; 1926.1401

Definitions

Qualified Person: a person, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training and experience, successfully demonstrated the ability to solve/resolve problems relating to the subject matter, the work, or the project

Rigging: lines, cables, slings, and other equipment used for lifting loads

Signaling: to do something by means of gestures or signs rather than explicit orders

Shackle: a metal link, typically U-shaped, closed by a bolt, used to secure a chain or rope to something

Sling: a loop of material, which connects the load to the lifting device. Slings can be made of chain, wire, metal mesh, natural, and synthetic materials

1.1 Rigger Qualification

All rigging operations must be performed by a qualified rigger. Documentation of rigger's qualification should be on the employer's company letterhead or documentation provided by an evaluating third party and submitted to the DPR project team prior to start of work. Copies of these qualifications will be kept on file at the jobsite.

1.2 Qualified Rigger Responsibilities

The person designated as the qualified rigger must have the ability to properly rig the load for a particular job, including inspection of all rigging to be used. It does not mean that a rigger must be qualified to do every type of rigging job. Each load that requires rigging has unique properties that can range from the simple to the complex. Riggers must ensure all loads are rigged with a "tag-line", and have the ability to stop any lift deemed unsafe.

Once it has been verified that a person is qualified to rig, that person needs to be outfitted, at the expense of their employer, with a differentiating colored reflective vest, a whistle or air horn, and a "X" on the top of their hardhat made of a reflective tape or sticker. Whistle or air horn is to be blown during picks to make other workers aware that a load is being lifted. Alternative means should be reviewed and signed off by EHS.

1.2.1 Rigging Inspections

Rigging equipment for material handling shall be inspected prior to use on each shift and as necessary during its use to ensure that it is safe. Defective rigging equipment shall be removed from service. The qualified rigger and the employer of the qualified rigger are responsible for ensuring that all rigging:

- Has permanently affixed and legible identification markings as prescribed by the manufacturer that indicate the recommended safe working load
- Is not loaded in excess of its recommended safe working load as prescribed on the identification markings by the manufacturer
- When not in use, shall be removed from the immediate work area so as not to present a hazard to employees

1.3 Rigging Equipment

The hardware, tools, and machines used for moving loads.

1.3.1 Wire rope sling

A sling composed of wire rope that is either, braided, bridled, cable laid, or strand laid. All wire rope slings should be:

- Equipped with permanently affixed and legible identification markings that shows the name or trademark of the manufacturer, rated load capacity for the type of hitches, and the angle upon which it is based, and the diameter or size
- Should be inspected per the manufactures recommendations, and removed from service when found to be damaged or showing signs of wear
- Slings shall not be shortened with knots or bolts or other makeshift devices
- Slings shall be padded or protected from the sharp edges of their loads

1.3.2 Synthetic Rope Slings

A sling made of conventional three strand construction synthetic materials such as nylon, polyester, and polypropylene. All synthetic rope slings should:

- Synthetic web slings (nylon/polyester) shall be marked or coded to show the name or trademark of the manufacturer, the rated capacities, and the type of material
- Should be inspected per the manufactures recommendations, and removed from service when found to be damaged or showing signs of wear
- Slings shall be padded or protected from the sharp edges of their loads
- Web slings shall not be used where fumes, vapors, sprays, mists or liquids of acids or phenolics are present

1.3.3 Chain Sling Rigging

Other forms of rigging are preferred over the use of chain slings, and allowed only where manufacturer requirements apply. Subcontractors planning to use chain slings must submit a written request, including the compelling reason, to DPR project management prior to use. Upon review and approval, the following requirements should be met:

- Chain has been inspected, and signed-off on by a competent 3rd party entity. These documents need to be submitted to DPR prior to sling being used on site, and chains need to be re-certified quarterly while chain has been in use
- Chain is made of, at minimum, grade 80 alloy steel
- Welded alloy steel chain slings shall have permanently affixed durable identification stating size, grade, rated capacity, and sling manufacturer
- Prior to use, Chain Sling is inspected link by link prior to the start of work by a competent person, and the Chain Sling Inspection form is completed and signed off

1.4 Shackles

All shackles must be, at minimum, a grade B, and made of forged North American alloy steel. Shackles should also:

- Have a known load rating legibly marked on them
- Be used and inspected per the manufacturers recommendations
- Be properly closed and secured prior to lifting
- Manufacture provided components should be used. Bolts or pins that are not rated or designed for use in the shackle should not be used.

2.0 Signal Person Qualification

All signaling must be performed by a qualified signal person. Verification of signal person(s) qualification should be submitted on the employer's company letterhead with qualified evaluator identified **OR** documentation from third party qualified evaluator. Hand signal charts must be either posted on the equipment or conspicuously posted in the vicinity of the hoisting operations.

2.1 Qualified Signal Person Responsibilities

Individuals deemed as a qualified signal person:

- Responsible for knowing and understanding the type of signals used at the worksite and is competent in those signals
- Knowing and understanding the relevant signal person qualification requirements specified in the crane standard
- Understanding the operation and limitations of the equipment, including the crane dynamics involved in swinging, raising, lowering and stopping loads and in boom deflection from hoisting loads
- Has taken and passed an oral or written test, and a practical test
- Is responsible for ensuring no loads are left suspended in the air, and has the authority to stop any lift deemed unsafe

Once it has been verified that a person is qualified to signal, that person needs to be outfitted, at the expense of their employer, with a differentiating colored reflective vest, a whistle or air horn, and an "X" on the top of their hardhat made of a reflective tape or sticker. Whistle or airhorn is to be sounded during picks to make other workers aware that a load is being lifted. A building "map" should be distributed to the Crane Operator as well as to all signalers so that clear verbal direction can be given as to where the load is heading once it is rigged and lifted. Alternative means should be reviewed and signed off by EHS.

3.0 Pre-Planning

Work involving rigging activities is defined by DPR to be High Hazard and require detailed, written pre-task planning, and, if requested by the DPR project team, a management level Job Hazard Analysis. Refer to the Pre- Planning section 17 for additional JHA and PTP guidelines and the Forms Section at the end of the EHSP.

Rigging & Signaling Forms

Forms

Chain Sling Daily Inspection

Crane Rigging/Signaling Training Verification



Chain Sling Daily Inspection

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Competent Person:		Company:	

	Mon		Tues		Wed		Thurs		Fri		Sat		Sun	
Item	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No

DATE:									
Initials of person performing inspection:									
Has the inspector been trained in the safe inspection of this type of rigging?		Y/N	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N
Daily Inspection Items (To be performed on a daily basis rigging is in operation) (Yes / No)									
Item Description		Y/N	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N
1.	Does any part of rigging show signs of being stretched or overloaded?								
2.	Is any part of rigging bent or twisted?								
3.	Is any part of rigging gouged or pitted?								
4.	Does any part of the rigging show signs of broken welds or cracking?								
5.	Does any part of the rigging show signs of corrosion or heat damage?								
6.	Has the riggings identification markings or tag been removed?								
7.	Is there any paint or covering material on any part of rigging?								
8.	Is there any excessive wear anywhere on the rigging?								
9.	Have the safety latches been removed from the hooks?								
10.	Is there any other obvious damage to any portion of rigging?								
Inspection Results (Ok / Repair / Scrap)									
Additional Comments:									

****If any item(s) are found to be missing or marked Yes, the equipment is to be repaired/serviced prior to use.****

Defects Reported To:		Date:	
Rigging Taken Out of Service?	☐ YES	☐ No	

Signature of Competent Person

Date



Crane Rigging/Signaling Training Verification

Archive Document – DO NOT DISCARD

Name:		Date:	
Project/Project#:		Company	

Workers should be trained in proper rigging and signaling methods prior to performing such a task. A hazard evaluation should be completed to determine that use of a crane is the safest means available to perform the task. Use of crane is considered a high hazard activity by DPRH and therefore requires daily, detailed, written pre-task planning prior to use. Below is a review of DPRH guidelines:

Qualified Person: A person who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training and experience, successfully demonstrated the ability to solve/resolve problems relating to the subject matter, the work, or the project.

All rigging and signaling must be performed by a qualified employee. Documentation of employee's qualifications should be on the employer's company letterhead or documentation provided by an evaluating third party and submitted to the DPRH project team prior to start of work.

Riggers:

Qualified Rigger:

Is a person who meets the criteria for a qualified person. Employers must determine whether a person is qualified to perform specific rigging tasks. Each qualified rigger may have different credentials or experience. A qualified rigger is a person that:

- possesses a recognized degree, certificate, or professional standing, or
- has extensive knowledge, training, and experience, and
- can successfully demonstrate the ability to solve problems related to rigging loads

Rigger Responsibilities:

The person designated as the qualified rigger must have the ability to properly rig the load for a particular job, including inspection of all rigging to be used. It does not mean that a rigger must be qualified to do every type of rigging job. Each load that requires rigging has unique properties that can range from the simple to the complex. Riggers have the ability to stop any lift deemed unsafe.

Signalers:

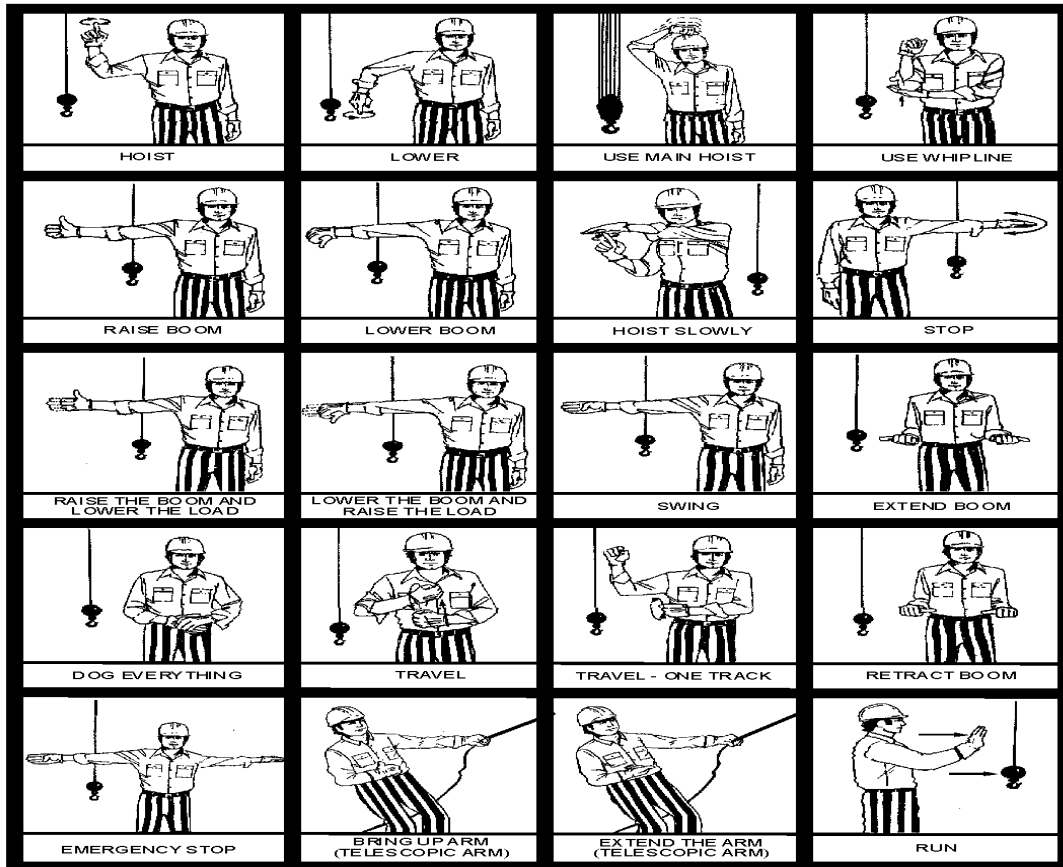
Qualified Signal Person:

All signaling must be performed by a qualified signal person. Verification of signal person(s) qualification should be submitted on the employer's company letterhead with qualified evaluator identified **OR** documentation from third party qualified evaluator. Hand signal charts must be either posted on the equipment or conspicuously posted in the vicinity of the hoisting operations.

Signal Person Responsibilities:

The signal person is responsible for:

- Knowing and understanding the type of signals used at the worksite and is competent in those signals
- Understanding the operation and limitations of the equipment, including the crane dynamics involved in swinging, raising, lowering and stopping loads and in boom deflection from hoisting loads
- Knowing and understanding the relevant signal person qualification requirements specified in the crane standard
- Having passed an oral or written test and a practical test
- Having the authority to stop any lift deemed unsafe



Tower Crane Signalers/Riggers:

The following additional provisions are required for signalers/riggers of tower cranes:

- Red vest
- High-visibility gloves
- "X" on top of hard hat
- Whistles to be used for load movement
- Personnel on receiving end **prior** to load flying

This document hereby certifies that I have been instructed on the items listed above and I have had an opportunity to ask questions. I fully understand the DPR requirements for work involving the use of signaling or rigging loads for cranes.

Name/signature of employee

Date

Name/signature of person conducting verification

Date

A copy of this form should be faxed to the regional safety professional

PRIOR TO START OF WORK.

Section 21

Safety Training

Sub-Section	Topic	Page #
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2.0	Site Specific Safety Orientation	2
3.0	Worker Information Sheet	2
4.0	Spot Check Training Questionnaire	2
5.0	OSHA Training & Competent Person Requirements	3
6.0	Safety Training Document	3
7.0	Safety Meeting Agenda	3

Safety Training

1.0 Introduction

Training is one of the most important elements of any safety and health program. Training is designed to enable employees to learn their jobs properly, reinforce safety policies, procedures, and safe work practices. Safety training also provides an opportunity to communicate safety principles, expectations and the commitment of DPR Construction to a safe and healthful work place.

2.0 Site Specific Safety Orientation

It is DPR's expectation that every worker on a jobsite, including DPR and subcontractor/tiered subcontractor project management, complete a site specific safety orientation prior to start of work. Documentation of this orientation is to be kept on site for the duration of the project, and then archived with project files. The generic template is available on the DPR Toolbox Safety Documents section in both English & Spanish. This template should be used to make the project's Orientations site specific.

3.0 Worker Information Sheet

Documentation should be kept for each worker who completes the site specific safety orientation. The Worker Information Sheet is used to document their attendance, and includes the following:

- *Employee Information/Emergency Medical Information* – this information is used in the case of an emergency where contacting the individual's emergency contact person may be need or a medical condition occurs during their hours of employment. This information is kept confidential and only used in the case of an emergency, or should a medical incident occur (*note: medical information is voluntary, but highly recommended*).
- *Training Information* – the worker is to use this section to document the training they have received, prior to arriving at the project. The worker should check the box marked "yes" if they have received this training (training indicated with a "*" indicates training that requires proof of training – a copy of the card of training record should be attached to their individual Worker Information Sheet). If the worker has never received training on a particular topic, they should mark the box marked "no". This section should be spot checked by a member of the DPR project team to ensure the worker has been trained by their employer pertinent to their scope of work. (Reference the Spot Check Training Questionnaire document in this section for sample questions that could be asked to check training competency). Should training be deficient, the training should occur prior to the worker engaging in activities where specific training is required, and be provided by their employer. A number of training verification forms are provided throughout the EHSP to aid in documenting training, i.e., Fire Watch in the Fire Protection & Prevention Section 11, Fall Protection in the Elevated Work Section 07, etc.). Ensure the worker signs the bottom of this section.
- *Acknowledgement of DPR Drug & Alcohol Restriction Policy* – the worker is to read this section, understand DPRs Drug & Alcohol Policy and sign off. Ensure the worker signs the bottom of this section.
- *Acknowledgement of Hazard Communication Training* - the worker is to read this section, understand the elements of the Hazard Communication Training received by their company and sign off. Ensure the worker signs the bottom of this section.
- *Site Agreement* – the worker reads and understands the intent of DPRs site specific orientation, and the information therein, and signs off that each of the identified items have been explained and any questions addressed during the orientation. Ensure the worker signs off on the bottom of this section.

4.0 Spot Check Training Questionnaire

The Spot Check Training Questionnaire is a tool that can be used by the project team to assist in the Spot Check Training process of applicable, required training. Should training be found to be deficient, the worker



should be trained/re-trained prior to the worker engaging in activities where specific training is required. This training/re-training should be provided by their employer. Should training/re-training be necessary, an option for documenting this training is to use one of the DPR Training Verification forms location throughout the DPR EHSP. A list of these training verification forms is provided at the end of the Spot Check Training Questionnaire, i.e., Fire Watch in the Fire Protection & Prevention Section 11, Fall Protection in the Elevated Work Section 07, etc.

Note: The Spot Check Training Questionnaire is intended as a tool to gauge an individual's level of understanding; it is not intended to be used as a definitive measurement of competency.

5.0 OSHA Training & Competent Person Requirements

This sub-section includes a matrix of 29 CFR Part 1910 & 1926 Training and Competent Person Requirements that includes all Federal requirements, as well as California specific CCR requirements. State & local requirements may vary, and should be verified as needed. This matrix is a tool intended for use in the verification of training & competent person requirements for personnel (DPR, subcontractor and tiered subcontractor) pertinent to their scope of work. It is DPR's expectation, as well as the law, that personnel are trained and competent persons' knowledge/level of competency is verified prior to start of work.

6.0 Safety Training Document

The Safety Training Document provides a Sign-in Sheet for use in the event a training class/session is performed. The use of this document ensures all attendees are accounted for who attended the training class/session. When this sheet is used, additional information should be attached to document the information presented during the class/session.

7.0 Safety Meeting Agenda

It is DPR's intent that mass safety meetings and DPR Toolbox Meetings (DPR personnel only) are held weekly with all project personnel in attendance. Meetings are primarily led by DPR project management team; however, subcontractor input should be encouraged and welcomed depending on the topic. Lessons learned, hazard identification for upcoming activities, review of the prior week's safety deficiencies, etc. should be discussed. All personnel present should sign the Safety Meeting sign-in sheet. Detailed agendas should be developed and outline the topics discussed. If any items outside of the prepared meeting minutes are raised, these are captured on the meeting minutes.

Safety Training Forms

Forms

Worker Information Sheet – English
Worker Information Sheet – Spanish
Spot Check Training Questionnaire
OSHA Training & Competent Person Requirements - Federal OSHA 1910 & 1926
Safety Training Attendance Sign-In Sheet
Safety Meeting Minutes
Safety Meeting Sign In Sheet

The English & Spanish versions of the site specific safety orientation program are located on-site or refer to the DPR Toolbox for the template files.

New Worker's Site Specific Safety Orientation **(project name)**



Worker Information Sheet

Archive Document – DO NOT DISCARD

Facilitator: _____
Date: _____
*initial training spot checked per
scope of work

Project/Project Number:				
Employee Information				
Employee Name		Company Name		
Date of Birth		Position		Years of Experience
Emergency Contact Name		Phone Number		
Emergency Contact Address		Relationship		

Emergency Medical Information				
<i>This information is strictly voluntary. Any information you choose to disclose in this area will help to assist the medical facility to provide the best possible care that you require in case of an emergency. This is not a requirement of employment and is optional.</i>				
Do you have any allergies?	☐	Yes	☐	No
If so, please describe:				
Do you have any allergic reaction to particular medications?	☐	Yes	☐	No
If so, please describe:				
Do you have any medical condition that requires medical attention; for example, heart disease, thyroid disease, asthma, high blood pressure, diabetes, etc.?	☐	Yes	☐	No
If so, please describe:				

Training Information							
<i>It is your employer's responsibility to provide any training that you may need to perform your job. If you are asked to perform something that you are not trained for, you must contact your employer or DPR Project Management and ensure that you are adequately trained prior to performing the task. OSHA requires that all workers are suitably trained and DPR therefore reserves the right to verify training records at any time. Please check the training that you have received below:</i>							
Required Training	Yes	No	DPR Spot Check	Required Training	Yes	No	DPR Spot Check
Asbestos Abatement	☐	☐	☐	Heat Illness	☐	☐	☐
Bloodborne Pathogen Awareness	☐	☐	☐	Ladders	☐	☐	☐
Boom/Aerial Lifts	☐	☐	☐	Lasers/Non-ionizing Radiation	☐	☐	☐
Concrete Dust (CA Only)	☐	☐	☐	Lead Abatement	☐	☐	☐
Confined Space Entrant/Attendant	☐	☐	☐	Lockout/Tagout	☐	☐	☐
Confined Space - Supervisor	☐	☐	☐	Powder Actuated Tools	☐	☐	☐
Confined Space Rescuer	☐	☐	☐	Respiratory Protection	☐	☐	☐
CPR and First Aid	☐	☐	☐	Respiratory Fit Test	☐	☐	☐
Excavation and Trenching	☐	☐	☐	Rigger, Qualified	☐	☐	☐
Electrical Awareness Training	☐	☐	☐	Scaffold User	☐	☐	☐
Electrical (Authorized)	☐	☐	☐	Scaffold Erector/Dismantler	☐	☐	☐
Fall Protection	☐	☐	☐	Scissor Lifts	☐	☐	☐
Fire Extinguisher Use/ Inspection	☐	☐	☐	Steel Erection	☐	☐	☐
Fire Watch	☐	☐	☐	Other Project Specific Training Requirements:			
Forklift-certification card required	☐	☐	☐	Dust Control	☐	☐	☐
Flagging Training	☐	☐	☐	Infection Control (ICRA)	☐	☐	☐
Hazard Communication Training/GHS	☐	☐	☐	Silica Awareness	☐	☐	☐
Hazardous Materials Training	☐	☐	☐	Other:	☐	☐	☐

By signing below, you acknowledge that the above information provided regarding training is correct

Employee Signature	Date
--------------------	------



Acknowledgement of DPR Drug and Alcohol Restriction Policy	Acknowledgement of Hazard Communication Training
I acknowledge that I have been informed of DPR's Drug and Alcohol Restriction policy in this project's site-specific DPR Environmental Health and Safety Program. I also acknowledge that the sale, possession, or use of illegal drugs, drug paraphernalia, and/or alcohol during the working day, or reporting to work under the influence of illegal drugs or alcohol, is strictly prohibited and is a dischargeable offense. I also understand that substance abuse screening may include pre-employment, (project specific), post-accident/post-incident, near miss, and probable cause. I further understand, however, that this policy does not alter DPR's policy that employment is terminable at will at the option of DPR or any employee. Notwithstanding various state laws regarding licensed use of medical marijuana, no worker is permitted to use, be impaired by or possess marijuana on DPR jobsites at any time.	The potential hazards I may be exposed to as a part of my job on this site have been communicated to me by my employer. The topics discussed include, but are not restricted to: • The existence and availability of OSHA Hazard Communication Standard - 29CFR 1926.59 and DPR's EHS Program • The existence and availability of a complete list of hazardous chemicals that I, as an employee, may be exposed to on this site • An explanation of the hazard warnings and labeling system for all hazardous chemicals used on this site • A description and the availability of personal safety equipment, emergency procedures, and safe operation practices My responsibility includes a review of SDSs for all materials I must use in my duties, and to inquire about any areas of these SDSs that I do not fully understand. By signing below, I acknowledge that I understand the above and that this information has been given to me.

Print Name	Date
Signature	Signature

This document hereby certifies that I have been oriented by DPR Construction, and that I understand the extreme importance of creating and maintaining an injury free workplace. In addition, I also certify that I will comply with all medical procedures as outlined in this orientation and as established, written or otherwise, on this jobsite. I also agree to comply fully with any and all safety and environmental laws, rules, regulations, procedures, and/or posted markings in effect on the site. I also understand that any violation or breach of this agreement may result in removal from the site, fines, and civil and/or criminal prosecution and penalties under any applicable state and/or federal laws.

The items listed below were discussed with me today and I had an opportunity to ask questions. I fully understand the project's policy and position on these items. The location of the DPR Environmental Health and Safety Plan has been identified, and is available for my review at any reasonable time.

Topic	Yes	No	Topic	Yes	No
DPR's Philosophy	☐	☐	Project Disciplinary Procedures	☐	☐
DPR's Dress Code of Safe Work Practices	☐	☐	Zero Tolerance Items	☐	☐
Personal Protective Equipment	☐	☐	Drug and Alcohol Restriction Policy	☐	☐
Hazard Communication	☐	☐	Site Emergency Procedures	☐	☐
Equipment Qualification and Training	☐	☐	Miscellaneous Site Information	☐	☐
Pre-task Planning	☐	☐	Site Specific Information	☐	☐
Correcting Unsafe Conditions	☐	☐		☐	☐

Print Name	Date of Orientation
Signature	Company Name – Name of Foreman & Supervisor

This Team is committed to an Injury Free Environment. Join the Team!

The English & Spanish versions of the site specific safety orientation program are located on-site or refer to the DPR Toolbox for the template files.

Orientación de seguridad para el empleado nuevo – Lugar específico de trabajo (Project name)

Oja de informacion del trabajador

Documento archivos - NO SE DESCARTEN POR FAVOR

Información del empleado				
Nombre del empleado			Nombre de la compañía:	
Fecha de nacimiento		Posición	Años de experiencia	
Nombre de contacto de emergencia			Número de teléfono	
Dirección de contacto de emergencia			Relación/parentesco	

Información médica de emergencia				
<i>Esta información es estrictamente voluntaria. Toda información que usted opte por revelar en esta sección ayudará a asistir a la institución de salud a proveer la mejor atención médica posible que requiera en caso de una emergencia. Esta información no es un requisito del empleo y es opcional.</i>				
¿Padece de alguna alergia?	☐	<i>Sí</i>	☐	<i>No</i>
En caso afirmativo, describa:				
¿Tiene alguna reacción alérgica a algún medicamento en particular?	☐	<i>Sí</i>	☐	<i>No</i>
En caso afirmativo, describa:				
¿Tiene alguna patología que requiera de atención médica: por ejemplo, enfermedad cardíaca, de tiroides, asma, hipertensión arterial, diabetes, etc.?	☐	<i>Sí</i>	☐	<i>No</i>
En caso afirmativo, describa:				

Información de formación							
Es responsabilidad de su empleador brindar la capacitación que usted pueda necesitar para realizar su trabajo. En caso que se le solicite que realice una tarea para la cual no ha sido capacitado, deberá comunicarse con su empleador o con la Administración de proyectos de DPR y asegurarse de recibir la capacitación adecuada antes de realizar dicha tarea. OSHA requiere que todos los trabajadores estén adecuadamente capacitados y DPR por lo tanto se reserva el derecho de verificar los antecedentes de capacitación en cualquier momento. Sírvase marcar la capacitación que ha recibido en el cuadro a continuación:							
Capacitación requerido	Sí	No	Repasado por DPR*	Capacitación requerido	Sí	No	Repasado por DPR*
Asbesto	☐	☐	☐	Enfermedad de Calor	☐	☐	☐
Patógenos transmitidos por la sangre	☐	☐	☐	HazMat (materiales peligrosos)	☐	☐	☐
Polvo de Concreto (CA only)	☐	☐	☐	Escaleras	☐	☐	☐
Elevadores aéreos / telescópicos	☐	☐	☐	Láser/Radiación no ionizada	☐	☐	☐
Espacios confinados-principiante/ asistente	☐	☐	☐	Plomo	☐	☐	☐
Espacios confinados-supervisor	☐	☐	☐	Candado y etiqueta de bloqueo	☐	☐	☐
Rescate en espacios confinados	☐	☐	☐	Herramientas detonantes por polvo	☐	☐	☐
RCP (CPR en inglés) y Primeros auxilios	☐	☐	☐	Protección respiratoria	☐	☐	☐
Excavación y excavación de zanjas	☐	☐	☐	Prueba de ajuste de equipos respiradores	☐	☐	☐
Capacitación (Conocimiento) eléctrica	☐	☐	☐	Aparejos (quolidado)	☐	☐	☐
Eléctrica- (Autorizado)	☐	☐	☐	Uso de andamios	☐	☐	☐
Protección contra caídas	☐	☐	☐	Armado/ Desarmado de andamios	☐	☐	☐
Prevención y protección contra incendios	☐	☐	☐	Elevadores tipo tijera	☐	☐	☐
Brigada de bomberos	☐	☐	☐	Otro proyecta requisitos de entrenamiento específicos:			
Montacargas- requiere tarjeta	☐	☐	☐	Control de polvo	☐	☐	☐
Capacitación de flagger	☐	☐	☐	Control de la infección	☐	☐	☐
Capacitación en HazCom (norma de comunicación en caso de peligro)/GHS	☐	☐	☐	La conciencia de sílice	☐	☐	☐

Al firmar a continuación reconoce que la información suministrada precedentemente sobre capacitación es verdadera:

Firma	Fecha
-------	-------

Reconocimiento de la política de Restricción del consumo de alcohol y drogas de DPR	Reconocimiento de Capacitación de comunicación de peligros
Reconozco que he sido informado sobre la política de Restricción del consumo de alcohol y drogas de DPR de este Programa de Seguridad y Salud Ambiental de DPR, específico para este sitio del proyecto. Asimismo reconozco que la venta, posesión o consumo de drogas ilícitas, parafernalia de drogas y/o alcohol durante el horario laboral o la asistencia al trabajo bajo los efectos del consumo de drogas ilícitas o alcohol, está estrictamente prohibido y constituye causal de despido. También entiendo que la prueba de detección de consumo de sustancias tóxicas puede obedecer a una causa previa al empleo, (para un proyecto en particular), pos-accidente/pos-incidente, cuasi-accidente y causa probable. Asimismo entiendo que esta política no altera la política de DPR con respecto a que el empleo puede darse por terminado a criterio y opción de DPR o cualquier empleado. No obstante las diversas leyes estatales relacionadas con el uso autorizado de la marihuana medicinal, ningún trabajador está permitido utilizar, ser perjudicado por tener marihuana o en lugares de trabajo DPR en cualquier momento.	Mi empleador me ha comunicado los peligros potenciales a los cuales puedo estar expuesto como parte de mi trabajo en este sitio. Entre los temas que se trataron se incluyen, en forma no limitativa: - Existencia y disponibilidad de la Norma OSHA de Comunicación de peligros - 29CFR 1926.59 y el Programa EHS de DPR. - Existencia y disponibilidad de una lista completa de sustancias químicas peligrosas a las que yo, como empleado, puedo estar expuesto en este sitio. - Explicación de las advertencias del peligro y el sistema de etiquetado de todas las sustancias químicas peligrosas que se utilizan en este sitio. - Existencia, disponibilidad y uso de las Hojas de datos de seguridad de los materiales (SDS por sus siglas en inglés). - Descripción y disponibilidad del equipo de protección personal, procedimientos de emergencia y prácticas de operación segura. Es mi responsabilidad examinar las Hojas de datos de seguridad de los materiales que debo usar en el ejercicio de mis tareas y formular preguntas si tengo dudas sobre alguna sección de estas Hojas de seguridad de los materiales. Al firmar a continuación, reconozco que entiendo lo precedente y que se me ha suministrado la siguiente información:

Nombre (letra de imprenta)	Fecha
Firma	Firma

Esta documento certifica que he sido orientado por DPR Construction, y que comprendo la enorme importancia de crear y mantener un lugar de trabajo sin lesiones. Asimismo, certifico que cumpliré con todos los procedimientos médicos tal como se describieron en la orientación y tal como se establecieron, escribieron o de otra manera, en este lugar de trabajo. También acepto cumplir con todas y cada una de las leyes sobre la seguridad y el medio ambiente, las reglas, los reglamentos, los procedimientos y/o carteles exhibidos en el lugar de trabajo. Además comprendo que cualquier violación o incumplimiento contractual podrá causar la remoción del sitio, multas y sanciones penales y/o civiles y penalidades de acuerdo con la legislación federal y/o estatal aplicable.

Hoy se han tratado los temas listados a continuación conmigo y tuve oportunidad de formular preguntas. Entiendo en forma cabal la política y la posición del proyecto en relación a estos temas. Se me ha informado la ubicación del Plan de seguridad y salud ambiental de DPR, el cual se encuentra a mi disposición para revisión en cualquier momento.

Topic	Si	No	Topic	Si	No
Filosofía de DPR	☐	☐	Procedimientos disciplinarios del proyecto	☐	☐
Código de prácticas laborales seguras de DPR	☐	☐	Puntos de tolerancia cero	☐	☐
Equipo de protección personal	☐	☐	Política de restricción del consumo de drogas y alcohol	☐	☐
Protección contra caídas	☐	☐	Notificación de lesiones/incidentes/enfermedades	☐	☐
Comunicación de peligros	☐	☐	Procedimientos de emergencia en el sitio	☐	☐
Calificación y capacitación en equipos	☐	☐	Información adicional sobre el sitio	☐	☐
Planificación previa a las tareas	☐	☐	Información específica sobre el sitio	☐	☐

En letra de imprenta	Fecha de orientación
Primo	Nombre de la empresa - Nombre del capataz y Supervisor

Este equipo se ha comprometido a un ambiente libre de lesiones. ¡Únete al equipo!

Spot Check Training Questionnaire

NOTE: ** indicates Training Verification documentation form available in EHSP, see additional notes at end of document.

Topic	Sample Questions to ask	Sample Answers
Asbestos Abatement	What is "Class 1" asbestos work?	Activities involving the removal of thermal system insulation and servicing asbestos containing material, and material presumed to contain asbestos
Bloodborne Pathogens	Name some bodily fluids that could potential contain bloodborne pathogens	Any bodily fluid, i.e., blood, urine, saliva, mucus, etc.
Boomlifts/Aerial Lift**	When should you be tied off in a boomlift?	As soon as your feet touch the platform – before moving, raising, etc.
Cleanroom Protocol	When should you put booties on?	Prior to entry into the clean room space.
Confined Space – Entrant	As an entrant, if the attendant sounds the alarm that a hazardous condition exists, what are you required to do?	Immediately exit the confined space.
Confined Space – Attendant	As an attendant, can you have other duties?	No, the attendant must not have any duties that interfere with his primary responsibility, which to to monitor the worker in the confined space and no leave unless replaced by another attendant.
Confined Space – Supervisor	As a supervisor, limited or restricted access is one condition of a confined space, name at least one other condition that determines a confined space.	For a confined space: • Not designed for human occupancy or continuous work • Large enough and so configured that a worker can bodily enter and perform assigned work For a permit-required confined space: • Contains or has the potential to contain a hazardous atmosphere • Contains material that has the potential to engulf an entrant • Has an internal configuration such that an entrant could become trapped or asphyxiated by inwardly converging wall or a floor that slopes downward that tapers to a small cross section • Contains another recognized serious safety or health hazard
Confined Space – Rescuer	When can you enter a confined space to rescue a downed worker?	Only after the hazard has been abated or appropriate protective measures are in place to protect your health or life
Control of Employee Exposure from Dust-Generating Operations Conducted on Concrete and Masonry Material (CA Only)	Workers who bush and grind concrete as part of their job assignment need only wear a respirator and Protective Suit to protect themselves.	No! In California, during operations in which powered tools or equipment are used to cut, grind, core, or drill, concrete or masonry materials, a dust reduction system shall be applied to effectively reduce airborne particulate.
CPR/First Aid	If someone on the project is down and not breathing and you detect a pulse, should you perform chest compressions?	NO, if a worker has a pulse but is not breathing, you should only perform rescue breaths.
Excavation & Trenching – Competent Person	As the competent person, name two instances when an excavation should be inspected?	• Prior to worker entry or start of work, and • Periodically throughout the day if workers are in the excavation • After a rainstorm • If water accumulates in the trench/excavation

Spot Check Training Questionnaire

NOTE: ** indicates Training Verification documentation form available in EHSP, see additional notes at end of document.

Topic	Sample Questions to ask	Sample Answers
		If the integrity of the excavation is potentially compromised, such as fissures or cracks appear in the wall of the excavation
Excavation & Trenching – worker**	Is it necessary for your company's competent person to inspect the excavation prior to you entering it?	Yes
Electrical -- Awareness Level (non-electricians)	Is it okay for you to repair damaged cords on your tools?	No, needs to be done by a qualified electrician
Electrical – Authorized or Qualified Electricians	At what voltage do you need to be protected from exposure to live electrical equipment?	50 volts or higher
Fall Protection – worker**	How much weight does your anchorage point need to be good for?	5,000 lbs (a structural member of the building, needs to be strong enough to hang a pickup truck off of)
Fall Protection – Competent Person	Can a harness and/or lanyard be reused after a fall?	Yes, but only after it's been inspected by a competent person and found to be in good working order. NOTE: for DPR personnel, the equipment MUST BE REMOVED FROM SERVICE.
Fire Extinguisher -- Use	What is a class A fire?	Ordinary combustibles, i.e., wood, paper, plastic, cardboard, etc.
Fire Extinguisher - Inspection	What often should a fire extinguisher be inspected?	Monthly
Fire Watch**	Should you sign off on a hot work permit?	Yes, as a fire watch, they are required to review and sign off on the hot work permit.
Flaggers	Where would you find the answer to this question? How many flaggers should be utilized on a construction site where barricades and warning signs cannot control the moving traffic?	It would be found in the Manual on Uniform Traffic Control Devices for Streets and Highways (MUTCD)
Forklift- certification card required ***	How long is a forklift certification good for?	Three years Or the operator has to be trained more often if: - They are involved in an accident or incident - They no longer demonstrate proficiency - Or the equipment changes
Gowning	Can you re-use your gown after exiting a clean space?	No
Hazard Communication **	Where on this project would you find an SDS sheet?	In the DPR office, DPR trailer or DPR designated location (<i>where ever you just told them during orientation</i>)
Hazardous Materials	On the NFPA triangle, what is more hazardous, a number 1 or a number 4	Number 4, the higher the number, the more hazardous the material
Heat Illness (CA Only)	Name the important components of the Heat Illness standard?	Cal/OSHA 3395 requires these items to be followed: Monitoring The Weather, Water, Shade, Work Severity, Rest Periods, Acclimatization, High Heat Procedures, Work Clothing/PPE, Emergency Response and Training
Ladders**	Can you stand on the top step of a ladder?	No
Lasers/Non-ionizing radiation**	Should a laser be left unattended and operating?	No
Lead Abatement	When should a blood lead level test be performed?	Prior to working with lead

Spot Check Training Questionnaire

NOTE: ** indicates Training Verification documentation form available in EHSP, see additional notes at end of document.

Topic	Sample Questions to ask	Sample Answers
Lockout / Tagout	Is it okay to remove someone else's lock or tag?	Generally – no (unless the individual cannot be located and management has jumped through hoops to verify all procedures have been followed)
Infection Control	What is a negative pressure space?	When the pressure within the dirty space is less than the pressure in the “clean” spaces – causing the dirt, debris, etc. to remain inside the “dirty” space and not contaminate the “clean” space
Powder Actuated Tools**	What should be done with spent loads or cartridges after use?	Follow the manufacturer's recommendation for disposal
Reinforcing Steel	How should employees who work above protruding reinforcing steel be protected?	They should be protected by the use of a guardrail or other fall protection systems and protective covers
Respiratory Protection	Can any type of filtering media be used with any type of contaminate?	No, the filter has to be specifically designed for the contaminate present in the atmosphere.
Respiratory Fit Test	How often should fit testing be performed?	Any of the following are acceptable answers: • Annually for most situations, • If the type of respirator changes, • if the individual gains or loses a significant amount of weight
Riggers (qualified)	Which sling hitch is stronger, a single leg, basket or choker	Basket
Scaffold User**	Do you need to be trained prior to working on a scaffold?	Yes
Scaffold Erector/Dismantler	Do you need to be trained in the specific type of scaffold you will be erecting?	Yes
Scaffold – competent person	Do you need to be onsite and present during scaffold erection, dismantling or reconfiguring?	Yes
Scissor Lifts**	Should you be tied off in a scissor lift?	Only if your feet leave the platform, i.e., standing on toeboard, standing on handrails, etc. (for additional information only: you ONLY tie off to the lift IF it's recommended to do so by the manufacturer OR use a different piece of equipment that provides adequate fall protection)
Silica	Name one health hazard associated with working with silica?	Lung cancer, silicosis, kidney disease, other respiratory disease
Steel Erection – connectors, erectors and/or deckers	As a member of a steel erection crew, are you required to be fall protection trained?	Yes
Steel Erection – competent person	Is it okay to allow non-authorized personnel to enter a restricted access zone?	No



Spot Check Training Questionnaire

*NOTE: ** indicates Training Verification documentation form available in EHSP, see additional notes at end of document.*

DPR Training Verification Documentation:

** DPR Training verification sheets available in the DPR EHSP.

*** Forklift Operator Proficiency Evaluation sheet available in DPR EHSP, however, this is ONLY for previously trained operators that need to be re-trained and/or re-qualified. This is not to be used for first time operators without prior training (requires extended classroom, observation & hands on training).

Other training verification documents available in the DPR EHSP:

- Bloodborne Pathogens
- Hepatitis B Vaccine
- Concrete Pump Truck
- Ergonomics
- Excavation & Trenching Awareness
- Fall Protection
- Fire Watch
- Flagging
- Hazard Communication
- Heat Illness
- Heavy Equipment
- Ladders
- Lasers
- Powder Actuated Tools
- Rigging & Signaling
- Scaffold User
- Silica Awareness
- Stilts
- Suspended Scaffold User
- Water Truck



OSHA Training & Competent Person Requirements - Federal OSHA 1910 & 1926

Note: the information provided below is Federal requirements, state and/or local requirements may vary -- **California specific requirement are noted as CCR standards.**

Training Requirements	OSHA Reference	Certification Required	Minimum Training Frequency
Access to Employee Exposure & Medical Records	CFR-1926.33 (see 1910.1020(g)(1))		Upon employment, annually thereafter
Accident Prevention: Qualified Operators	CFR-1926.20 (b) (4)		
Accident Prevention: Supervisors/Accident Investigators	CCR 3203 (a)(7)(F)		Upon initial assignment, Supervisors/Accident Investigators
Acetylene & Fuel Gas Safety	CCR 1740 (k)(1)		Initial, Users
Asbestos	CFR-1926.1101(k)(9), (o)(4)(i-ii)		Upon initial assignment, annually thereafter
Benzene	CFR-1926.1128(j)(3), 1910.1028 (j)(3)(i)		Upon initial assignment, annually thereafter
Bloodborne Pathogens	CFR-1910.1030(g)(2)		Upon initial assignment, annually thereafter
Boatswain Chair	CCR 1662 (a)		Upon initial assignment, users
Cadmium	CCR 5207 (m)(4)		Upon initial assignment, annually thereafter for Exposed workers and Supervisors
Confined Space	CFR-1926.1207(a)	yes	Prior to use, thereafter with change of duty/process sufficient to maintain proficiency
Confined Space	CFR-1910.146(g), (h), (l), (j)	yes	Prior to use, thereafter with change of duty/process sufficient to maintain proficiency
Control of Employee Exposure from Dust- Generating Operations Conducted on Concrete and Masonry Materials	CCR 1530.1		Upon initial assignment, annually thereafter for Exposed workers and Supervisors
Cranes: Assembly/Disassembly Director	CFR-1926.1430(d)		Upon initial assignment, re-train as needed to maintain proficiency
Cranes: Operator	CFR-1926.1430(c)	yes	Prior to use, CCO card, re-train if failure to demonstrate competency occurs, every 5 years
Cranes: Rigger (Qualified)	CFR-1926.1404(r)(1), 753(c)(2)		Prior to use, re-train if failure to demonstrate proficiency occurs
Cranes: Signal Person (Qualified)	CFR-1926.1430(b), 1428(c)		Prior to use, re-train if failure to demonstrate proficiency occurs, company provided training is not portable
Employee Alarm Systems	CFR-1910.165(b)(4)		
Employee Emergency Action Plans	CFR-1926.35(e)		With initial plan, and as responsibilities/plan changes
Fall Protection	CFR-1926.503	yes	Upon initial assignment, then as/if equipment changes; as needed to maintain proficiency
Fire Protection: Fire Brigades	CFR-1926.150 (a)(5)		
First Aid	CFR-1926.50(c)	yes	Consistent with First Aid program requirements
Flaggers (Traffic)	CCR 1599 (f & (g)		Upon initial assignment, Assigned Employees



OSHA Training & Competent Person Requirements - Federal OSHA 1910 & 1926

Note: the information provided below is Federal requirements, state and/or local requirements may vary -- **California specific requirement are noted as CCR standards.**

Hazard Communication	CFR-1926.59 (see 1910.1200(h))		Initial assignment, then as/if new hazards are introduced
Training Requirements	OSHA Reference	Certification Required	Minimum Training Frequency
Hazardous Waste Operations/Emergency Response	CFR-1926.65(b)(1)(ii), (d), (e), (q)	yes	Upon initial assignment, annually thereafter
Heat Illness (CA only)	3395		Upon initial assignment
Injury & Illness Prevention Program	CCR 1509 (e)		All Employees through Supervisor Tailgates
Ionizing Radiation	CFR-1926.53(b)		
Job Hazards	CCR 1510 (a & (c)		Before job assignment and with new hazards
Laser Equipment	CCR 1801		Operators and Qualified Person
Lead	CFR-1926.62(l)		Prior to initial assignment, then annually
Lock-out/Tag-out of Hazardous Energy	CFR-1910.147(c)(7)	yes	Upon initial assignment, with change in job/hazards for authorized & affected employees, with sufficient frequency to maintain proficiency
Machinery and Equipment	CCR 1510 (b)		Before use, Qualified Person
Non-ionizing Radiation	CFR-1926.54 (a)	yes	
Portable Fire Extinguishers	CFR-1910.157(g)		Upon initial assignment, annually thereafter
Powder Actuated Tools	CFR-1926.302 (e) (1)	yes	
Power Transmission and Distribution: Emergency Procedures	CFR-1926.950 (e), (e)(1)		
Powered Industrial Trucks (Forklifts)	CFR-1910.178(l)	yes	Prior to use, after accident/incident or unsafe operation; with new models; every 3 years thereafter
Process Safety Management	CFR-1926.64(g) & (h)	yes	Upon initial assignment, then every 3 years
Reinforcing Steel/Similar Projections	CCR 1712 (f)(A)		Qualified Person
Respirable Crystalline Silica	CFR-1926.1153(i)(2); CCR 1532.3(i)(2)		Upon initial assignment
Respiratory Protection	CFR-1926.103, 1910.134 (k)		Prior to use and then at least annually
Safety Training and Requirements	CFR-1926.21 (b)		
Scaffolding: Aerial lifts	CFR-1926.453 (b)(2)(ii)		
Scaffolding: Employees Working on Scaffolding	CFR-1926.454 (a), (c)	Yes in CA	Upon initial assignment, then as/if equipment changes; as needed to maintain proficiency
Scaffolding: Erection Crews	CFR-1926.454 (b), (c); App D	Yes in CA	Upon initial assignment, then as/if equipment changes; as needed to maintain proficiency
Stairways & Ladders	CFR-1926.1060(a)		Prior to use, thereafter sufficient to maintain proficiency
Steel Erection	CFR-1926.761		
Underground Construction	CFR-1926.800(d), (g)(5)(iii & v)		Annually



OSHA Training & Competent Person Requirements - Federal OSHA 1910 & 1926

Note: the information provided below is Federal requirements, state and/or local requirements may vary -- **California specific requirement are noted as CCR standards.**

Ventilation: Open Surface Tanks	CFR-1926.57(i)(9) (i)		
Welding: Fuel Gas	CFR-1926.350 (d)		
Welding: Arc Welding and Cutting	CFR-1926.351 (d)		
Welding: Fire Watch	CFR-1926.352 (e)		

Competent Person Requirements	OSHA Reference	Verification Required
Accident Prevention	CFR-1926.20 (b) (2)	
Asbestos	CFR-1926-1101(o)(1-3)	
Confined Spaces in Construction	CFR-1926.1203(a)	
Cranes: Assembly/Disassembly Director	CFR-1926.1404(a)(1);	
Demolition	CFR-1926.850(a), .852(c), .859(g)	
Electrical: Assured Grounding Program	CFR-1926.404 (b)(1)(iii)(B)	
Excavations	CFR-1926.651 (c)(1)(i), (h)(2-3), (k)(1); .652(a)(1)(ii), (d)(3)	
Fall Protection	CFR-1926.502 (c)(4)(ii), (d)(19), (h)(1)(i), (k)(4) and .503(a)(2)	Competent and Qualified in CA
Hearing Protection	CFR-1926.101(b); .1404(r)(1)	
Ionizing Radiation	CFR-1926.53 (b)	
Ladders	CFR-1926.1053(b)(15); .1060(a)(1)	
Respirable Crystalline Silica	CFR-1926.1153(g)(4); CCR 1532.3(g)(4)	
Rigging Equipment	CFR-1926.251(a)(6); .753(c)(2)	
Roofing – Initial and Qualified	CCR 1509 (a) 1730 (b)(8)-(9)	
Scaffolding	CFR-1926.451 (b)(10&11), (d) 10&18),(e)(9)(i), (f)(3,7,12), (g)(2)	
Slings	CFR-1910.184(d), (e)(3)(i-iii)	
Steel Erection	CFR-1926.753(c)(1,2), .754(d)(1), .755(a)(4), .756(a)(2), .757(a)(2,4), .761(a)	
Underground Construction	CFR-1926.803(a)	
Welding: Preservative Coatings	CFR-1926.354 (a)	



Safety Training Attendance Sign-In Sheet

Archive Document – DO NOT DISCARD

Course Name		Date	
Jobsite/Region		Instructor	
Class Duration		Start time	
Class Location			

	Print Name	Signature	Jobsite/Office	Company	Last 4 of SS# of Employee ID
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Section 22

Sanitation & Housekeeping

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Sanitation & Housekeeping

1.0 Introduction

All employees share the responsibility for maintaining good housekeeping practices and following the established housekeeping procedures contained in this plan. Further, this section details sanitation and quality of life measures to ensure a safe and healthful workplace. Attention to general cleanliness, storage and housekeeping can prevent numerous accidents, including fire, and illness.

Reference: Sanitation “Potable Water” CFR 1926.51; Sanitation CFR 1926.27; Housekeeping CFR 1926.25. Other applicable Standards, guidelines, and regulations may apply.

2.0 Sanitation Facilities

- At a minimum, one separate toilet should be provided for each 20 workers or fraction thereof, *for each gender*
 - Where there are 4 or less workers, one toilet may be provided for both genders, provided doors are lockable from the inside.
- Toilets should be maintained in a clean and sanitary condition, with adequate supplies of toilet paper
- Toilet facilities should be designed and maintained in a manner which ensures privacy
- When half toilets are used, a greater level of privacy should be provided (e.g., privacy screens, enclosures, etc. may be needed)
- Workers should make all attempts to maintain the cleanliness of toilet and handwashing facilities
- Graffiti on toilet walls should not be permitted; workers found defacing toilet facilities may be subject to disciplinary action, up to and including removal from the project
- DPR project management should designate the locations of toilet and handwashing facilities, with consideration for worker accessibility; where practical, toilets and handwashing facilities should be no further than 200 feet from work areas and no more than one floor to floor flight of stairs from working areas
- Hand wash facilities should be maintained in close proximity to toilet facilities and break areas, and/or locations where workers are engaged in activities where they may come into contact with hazardous materials (including, but not limited to: painting, coatings application, hazardous material handling, demolition activities, lead materials handling, placement of concrete, etc.) Where applicable, the manufacturer’s SDS should be reviewed to ensure specific requirements are met. At a minimum, these facilities should:
 - be provided and maintained in a clean and sanitary condition
 - be provided with adequate supplies of water, cleansing and drying materials
 - located in close proximity to those activities
 - equipped with cleansing and drying supplies applicable to the potential exposure
 - Hand wash facilities must be provided at the same ratio as toilets (CA Required)
- Owner controlled bathroom facilities should not be utilized unless authorized by DPR project management

3.0 Potable Water

- An adequate supply of potable water should be supplied at every project
- Each subcontractor will be responsible to provide potable water at the work site for their employees
- Containers for drinking water should not be filled with a hose and should be regularly cleaned and maintained in a sanitary condition
- Containers should be capable of being tightly closed and marked as to contents

- Water should not be dipped from containers and containers should be secured to prevent contamination
- Where single service or one-time use cups are used and supplied, a sanitary container for unused cups should be provided
- A receptacle for disposing of used cups should be provided and regularly emptied
- The use of a shared, common drinking cup is prohibited
- Refer to Heat Illness Prevention Section 27 for additional drinking water requirements.

3.1 Procedures for disinfecting and cleaning water containers

- Wash your hands thoroughly and use impervious gloves when possible
- Wash the container thoroughly with soap and water
- Pour a solution of 1 tablespoon of liquid household bleach to one gallon of water
- Let the solution remain in the container for 10 minutes, then pour out the solution
- Thoroughly rinse out the container with clean water; ensure the container and spigot are thoroughly rinsed
- Fill the container with clean drinking water and tape the lid shut. On the tape write down the name of the person who cleaned the container and the date it was cleaned
- Provide single service cups in a sanitary container, and a receptacle for disposing of the used cups

4.0 Break Areas

- DPR project management should designate the locations of lunch/break areas; eating and drinking should only occur in those areas
- An adequate number of garbage containers should be present for the disposal of food refuse and garbage and regularly emptied
- All personnel should ensure that food debris is properly disposed of into trash containers and is not left in the building for possible infestation by insects and rodents
- In general, the use of sunflower seeds, chewing tobacco, etc. should only occur in the designated lunch/break areas

5.0 Smoking, eCigarettes & Vaporized Cigarettes

- Smoking, of any kind, on projects should comply with owner requirements
- Smoking, of any kind, should be prohibited inside any building and in any public or eating areas
- A smoking area should be designated on the project and all smoking should only occur within this area
- All cigarette butts should be disposed of into appropriate receptacles
- Any worker found smoking, of any kind, in the building may be subject to disciplinary action, up to and including removal from the project
- As necessary, appropriate “No Smoking” signage should be posted

6.0 Housekeeping

All areas of the project should be maintained in a clean, healthy and sanitary condition at all times. Prior to moving to a new work area, the area being vacated should be broom swept and cleared of tools, materials and debris. Good housekeeping on the job is mandatory and every worker should do their part **daily** to minimize dust and to clean up their work area to keep the job clean for safety and efficiency. Owner requirements for housekeeping and material storage should be followed. The following general guidelines apply:

- Each subcontractor should be solely responsible for the storage, handling and timely removal of all debris, materials and equipment associated with their scope of work
- Subcontractors are responsible for supplying an adequate number of debris containers and buggies; these containers should be in close proximity to the work being performed
- All work areas, passageways, means of access and egress and areas surrounding buildings and/or structures should remain free of debris, trip hazards, obstructions and depressions
- Materials and equipment should not be stored under stairways or in stair passageways
- Combustible materials should be removed from the building as soon as practicable
- Flammable liquids should not be stored inside buildings overnight unless in a flammable storage container
- Construction equipment should be stored and/or placed in an orderly manner
- All stored materials, equipment and debris should be stored in a manner that will minimize or eliminate sliding, shifting, falling or collapsing; special consideration should be given to local and environmental factors including, but not limited to, wind, earthquake, vibration etc.
- Cords should be rolled up after use and stored neatly
- Protruding nails or other impalement or puncture hazards should be completely removed, or at a minimum, bent sufficient to prevent injury, prior to discard.
- Materials stored during the construction process should be coordinated with DPR, with the understanding that the construction process is dynamic and multiple moves of the same material may be required in order to promote the safety of the project.
- Only material that can readily be used and safely stored should be stocked in work areas. In general, no more than one-week of material should be stocked, unless prior approval has been given by DPR project management.
- Controls should be observed to minimize tracking of dirt into owner occupied areas and public roadways, i.e., wash out areas, rumble strips, tacky mats, crushed rock, first layer of asphalt, etc.
- Dust control measures should be utilized to prevent distribution of dust into the air, i.e., watering trucks, screens, etc.

7.0 Contractual Obligations for Clean-up and Housekeeping

Daily cleanup is part of the contract documents and failure to abide by the provisions of the contract may result in sanctions being taken against a subcontractor. Failure to maintain adequate housekeeping and to perform daily cleanup will result in notification to the subcontractor to clean the identified areas. Notification includes, but is not limited to the following actions:

- DPR reserves the right to cease work activities to address housekeeping issues determined to present an immediate hazard
- Verbal Notice will be given to subcontractors who fail to maintain their work areas
- Written Notice may be issued and upon receipt, the subcontractor will have 48 hours, depending on the notice given, to take immediate action to abate any identified housekeeping and cleanup deficiencies
- If the subcontractor fails to clean up after having been given notice, the cleanup work may be performed by others and the errant subcontractor may be back charged for all related costs
- Continued failure to complete required levels of cleanup may result in additional action, up to and including, the removal of a contractor from site or the removal of scopes of work at the subcontractor's expense not to exceed 150% of the actual cost, with no cost to DPR and the owner

Sanitation & Housekeeping Forms

Forms

Clean Up Notice



Clean Up Notice
VIA CERTIFIED MAIL AND FACSIMILE

[Date]

[Addressee]

ATTN: _____

RE: [Project]
[Project Location]
DPR Contract Number [Project Number]

Subject: Clean-Up Notice

Dear _____:

THIS IS A NOTICE THAT YOU MAY BE IN DEFAULT OF YOUR SUBCONTRACT. THE CONDITIONS CITED BELOW MUST BE CURED AS DESCRIBED HEREIN OR SERIOUS CONSEQUENCES TO YOUR COMPANY, FINANCIAL AND OTHERWISE, MAY RESULT. YOUR IMMEDIATE ATTENTION AND ACTION IS REQUIRED.

This Clean-up Notice is provided pursuant to our Subcontract section 14, Recourse by Contractor and section 23, Clean-up. DPR believes that [subcontractor], herein referred to as Subcontractor, is not performing their work in a manner so as to maintain the site in a clean, safe and orderly condition. The following conditions are jeopardizing the successful performance of your obligations under our Subcontract, are affecting the ability of other trades to perform their work, and must therefore be cured:

- 1.
- 2.
- 3.

If within 48 hours from receipt of this written notice to cure, should Subcontractor fail to commence and continue satisfactory correction of such default with diligence and promptness, then DPR Construction Inc., without prejudice to any rights or remedies, shall have the right to any or all of the following remedies:

- a) Supplying such numbers of workers, and quantity of materials, equipment and other facilities as DPR Construction, Inc. deems necessary for the completion of Subcontractor's work or any part thereof which Subcontractor has failed to perform, and charge the cost thereof to subcontractor, who shall be liable for the payment of same including reasonable overhead, profit and actual attorney's fees incurred as a result of Subcontractor's failure to perform;
- b) Contract with one or more additional contractors to perform such part of Subcontractor's work as DPR Construction, Inc. shall determine will provide the most expeditious completion of the total work and charge the cost thereof to Subcontractor; and
- c) Withhold payment of any monies due Subcontractor pending corrective action to the extent required by and to the satisfaction of DPR Construction.

In the event of an emergency immediately affecting the safety of persons or property, DPR Construction, Inc. may proceed as above without notice.

Should these conditions continue uncured, Subcontractor will be in default under the terms of our Subcontract, and DPR Construction, Inc. may terminate your Subcontract for default in accordance with section 14.1.2. Absent said cure, you will be subject to any of the aforementioned actions, up to and including termination of your subcontract for default, withholding all further payments pending completion of your scope of work by other means and resources. Subcontractor will be responsible for payment for its work not performed, in addition to any costs or damages arising from its default, to the extent that such costs exceed the unpaid balance of the Contract Price. All such financial settlements shall be in accordance with the terms of the Subcontract.

If you have any questions or concerns with the above, please contact me at [phone].

Sincerely,

DPR Construction

cc:

Section 23

Steel Erection

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Steel Erection

1.0 Introduction

These guidelines are in place to access, plan, minimize or eliminate risks and hazards associated with steel erection activities. To prevent incidents resulting from steel erection activities there are several considerations. What are the general requirements for steel erection? Where will these activities take place on the job site? What is the level of training for workers, working under loads, hoisting, landing and placing decking, fall protection, ladders, the use of aerial and scissor lifts, and controlled access zones?

Reference: Steel Erection 29 CFR 1926 Subpart R. Other applicable Standards, guidelines, and regulations may apply.

All steel erection activities, including, but not limited to, are covered by this section:

- Laying out
- Hoisting
- Placing
- Erecting
- Connecting
- Bolting
- Guying and bracing
- Plumbing
- Welding
- Installation of decking
- Installation of sheet metal panels
- Installation of curtain walls or window wall systems
- Miscellaneous metal work

2.0 Pre-task Planning

Steel erection activities are defined as high hazard activities and therefore require detailed, written pre-task planning by all crews. *Refer to the Pre-Planning Section 17 for additional pre-task planning guidelines and the Pre-Task Planning Worksheet.*

3.0 Steel Erection Pre-Planning and Documentation

The following general requirements should be completed prior to steel erection activities commencing on the project:

- DPR should ensure that the steel erector is provided with documentation verifying that the concrete in the footings, piers and walls and the mortar in the masonry piers and walls have attained, either 75% of the intended minimum compressive design strength or sufficient strength to support the intended loads. This should be documented by the Structural Engineer of Record and a copy provided to the steel erection company prior to start of erection activities.
- DPR should ensure that the steel erector is provided with documentation of any repairs, replacements or modifications to the anchor bolts; such changes should be approved by the Structural Engineer of Record
- DPR should ensure that there are adequate access roads into and on the site for safe delivery and movement of cranes, trucks, derricks and other essential equipment
- The steel erector will be responsible for maintaining pedestrian and vehicle control in their work area and around their work activities
- DPR should ensure a safe and adequate laydown and staging area, to include:
 - Firm, properly graded (this area must be compacted for CA)

- Drained
 - Readily accessible from access roads
- In-depth pre-task planning of all work activities should be performed by the steel erector and submitted to DPR for review. All hoisting activities should be coordinated as outlined in the Crane Section 05.
- The erector should be responsible for providing a written and detailed site specific erection plan
 - the site-specific erection plan should be developed by a qualified person
 - the plan should comply with this sub-section and all applicable federal, state and local rules and regulations

4.0 Site Specific Steel Erection Plan

In developing a site-specific erection plan, the following item, at a minimum, should be addressed:

- The sequence of erection activity, developed in coordination with DPR project management, to include the following:
 - material deliveries
 - material staging and storage
 - coordination with other trades and construction activities
- A description of the crane selection and placement procedures, to include the following:
 - site preparation
 - swing path for overhead loads
 - critical lifts, including rigging supplies and equipment
- A description of steel erection activities and procedures, to include the following:
 - stability considerations requiring temporary bracing and guying
 - erection bridging terminus point
 - anchor bolt notifications regarding repair, replacement and modifications
 - columns and beams (including joists and purlins)
 - connections
 - decking
 - ornamental and miscellaneous iron
- A description of the fall protection procedures to be utilized
- A description of procedures that will be used to protect workers from hazards of falling objects
- A description of the special procedures required for hazardous non-routine tasks
- A certification that each employee has received training for performing steel erection activities
- A list of all qualified and competent persons
- A description of the procedures that will be utilized in the event of rescue or emergency response; these procedures should include the identification of the site and project, be signed and dated by the qualified person responsible for its preparation, implementation and modification
- A chemical inventory and all applicable SDSs

5.0 Hoisting and Rigging

Refer to DPR Cranes Section 05 of this EHSP for additional hoisting and rigging requirements. In addition to these requirements, it is DPR's intent that single lift rigging and hoisting will be utilized.

5.1 General Hoisting and Rigging Guidelines

- A qualified rigger should inspect the rigging at minimum daily
- All hooks shall have safety latches. Safety latches on hooks should not be de-activated or made inoperable
- All suspended loads should be pre-planned to ensure that no employees are working below except for those involved in landing and connecting loads
- All loads should be rigged by a qualified rigger

- Tag lines or other suitable means of controlling a load are to be used at all times
- Bundles of decking should only be single picked; nylon chokers should not be used for picking bundles of decking
- Rigging should be selected based on known working load weights and type of material or items being hoisted

5.2 Multiple Lift Rigging and Hoisting

DPR recognizes that at certain times multiple lift rigging and hoisting may be necessary for tie-ins and other similar situations where single member lifting and rigging may create a greater hazard. All multiple lift hoisting and rigging operations should be in strict conformance with 29 CFR 1926.753 with the exception that at no time should more than **three** members be hoisted per lift. Should multiple lift hoisting be utilized, the following applies:

- Components of the multiple lift rigging assembly should be specifically designed and assembled with a maximum capacity for total assembly and for each individual attachment point
- The capacity of the rigging assembly should be certified by the manufacturer and have a 5 to 1 safety factor for all components
- The total load should not exceed:
 - the rated capacity of the hoisting equipment specified in the hoisting equipment load charts
 - the rigging capacity specified in the rigging rating chart
- The multiple lift rigging assembly should be rigged with members:
 - attached at their center of gravity and maintained reasonably level
 - rigged from top down
 - rigged a minimum of 7 feet apart
 - the members on the multiple lift rigging assembly should be set from the bottom up
 - controlled load lowering should be used whenever the load is over the connectors
 - each member should be equipped with a tag line

6.0 Structural Steel Assembly

All steel erection operations should be performed in firm accordance with all federal, state and local regulations, including but not limited to, the following:

- An appropriate railing system should be established at the earliest opportunity
- A fully planked deck or decked floor should be maintained at all times within two stories or 30 feet, whichever is less, directly under all erection work being performed
- When placing structure, structural members should not be cut loose from hoisting equipment until at least 4 bolts are in each connection and are wrench tight, except in the case of cantilevered sections where all connections are to be bolted and wrench tight prior to releasing the load and diagonal bracing which only require one bolt per connection wrench tight
- Turnbuckles should be secured from unwinding under stress
- Guy wires should be installed and removed only under the supervision of a competent person
- All openings in the floor and perimeter should be appropriately covered and protected in accordance with federal, state and local requirements
- All columns should be anchored by a minimum of 4 anchor bolts

7.0 Falling Object Protection

All materials, equipment and tools which are not in use while aloft, should be secured against accidental displacement. In addition:

- All fasteners should be housed in a secured container
- Any and all necessary means to prevent falling objects should be utilized
- No work should be permitted below steel erection activities unless overhead protection for the employees below is provided

8.0 Fall Protection

It is DPR's intent that all works involved in steel erection activities be protected from falling due to the high incidents of falls and fatalities among iron workers.

8.1 Perimeter Cables

On multi-story structures, perimeter safety cables should follow code requirements with mid-cables installed at the final interior and exterior perimeters of the floors as soon as the metal decking has been installed or as early as practicable. Where perimeter cables are to be used as part of a fall arrest system, they must be designed to meet all applicable requirements for fall protection. Stamped engineered drawings of perimeter systems will be required.

When perimeter cables are turned over for general project usage, complete handrail systems must be in place. Top cables must be flagged and both mid cables and toeboards must be in place prior to releasing the areas for general use. *Reference the Elevated Work Section 07 for specific requirements.*

8.2 Fall Protection Systems

The following general requirements apply to all steel erection and decking activities except as otherwise noted. Any deviation from these requirements requires approval from DPR project management and should be included in the site-specific erection plan.

- 100% fall protection is required for all steel erection activities, including connectors.
- Each employee engaged in a steel erection activity on a walking/working surface with an unprotected side or edge more than 6 feet above a lower level should be protected from fall hazards by guardrail systems, safety net systems, personal fall arrest systems, positioning device systems or fall restraint systems.

8.3 Controlled Decking Zone

DPR does not recognize the use of controlled decking zones as an effective means of ensuring the safety of persons working in and around decking installation activities. Therefore, **the use of a controlled decking zone is only allowed when all other means have been demonstrated to be infeasible.**

A written fall protection plan must be developed to address the actual work to be performed. The written plan must be submitted and reviewed by DPR project management and DPR EHS prior to start of work. A controlled decking zone is an area in which certain steel erection and decking activities may take place where there is a leading edge, and portions of the decking may not be fully installed and/or secured.

Should a controlled decking zone be necessary, the following guidelines will apply:

- Access to the controlled decking zone will be limited to only those engaged in leading edge work
- All workers within the area have completed controlled decking zone training
- Area should not exceed 90 feet wide by 90 feet deep from any leading edge
- Must be marked by the use of control lines
- Unsecured decking should not exceed 3,000 square feet

9.0 Training

Steel erection activities require several types of training. Each subcontractor is required to provide their workers with the necessary training. Training should be provided by a qualified person. The following requirements detail the safety training instruction required:

9.1 General Fall Protection Training

All employees exposed to fall hazards should be trained. This program should include training and instruction in the following:

- Recognition and identification of fall hazards in the work area
- The use and operation of guardrail systems, personal fall arrest systems, positioning device systems, fall restraint systems, safety net systems, and other protection to be used
- Correct procedures for erecting, maintaining, disassembling, and inspecting the fall protection systems to be used
- Procedures to be followed to prevent falls to lower levels, through or into holes and openings in walking/working surfaces and walls
- Fall protection requirements of this part

9.2 Multiple Lift Training

The employer should provide special training to employees engaged in the following activities:

- Proper rigging procedures and equipment for rigging of multiple lifts
- The hazards associated with multiple lifts
- The hazards associated with connecting activities
- The establishment, access, proper connecting techniques and work practices for safe connecting activities
- Decking procedures, including access to, and proper installation techniques and work practices for safe decking procedures

9.3 Connectors

Each connector should be trained in the nature of all hazards associated with connecting, to include, at a minimum:

- The hazards associated with steel erection activities
- The hazards associated with connecting activities
- Access
- Proper connecting techniques
- Work practices
- Controlled access zones, if applicable

Steel Erection Forms

Forms

Steel Erection Checklist
Deck Release Form



Steel Erection Checklist

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Note: any item marked “NO” should be addressed prior to start of work.

Approval to Begin Steel Erection	Yes	No	N/A
Before authorizing the commencement of steel erection, DPR should provide written notification to the steel erector that:			
Concrete footings, piers and walls have been cured to a level that will provide adequate strength to support any forces imposed during steel erection (75%)	☐	☐	☐
Anchor bolt repairs, replacements and modifications have been performed with the approval of the project Structural Engineer of Record	☐	☐	☐

Site Layout	Yes	No	N/A
DPR should ensure that the following is provided and maintained:			
Adequate access roads into and through the site for the safe delivery of derricks, cranes, trucks and materials to be erected	☐	☐	☐
Means and methods for pedestrian and vehicular control (this does not apply to roads outside of the construction site)	☐	☐	☐
Adequate space for the safe storage of materials and the safe operation of the erectors equipment. The space must be firm, properly graded, drained and readily accessible to the work.	☐	☐	☐

Column Anchorage	Yes	No	N/A
Before the erection of a column, DPR should provide written notification to the steel erector if there has been any repair, replacement or modification of the anchor bolts of that column	☐	☐	☐

Falling Object Projection (non-hoist)	Yes	No	N/A
The steel erector should ensure that no other trades or work is allowed below steel erection unless overhead protection for workers below is provided	☐	☐	☐

Fall Protection	Yes	No	N/A
DPR requires 100% fall protection for all steel erection activities, including connectors, at 6 feet.	☐	☐	☐
Perimeter Cables:			
Perimeter safety cables have mid-cable installed at the final interior & exterior perimeters of floor as soon as the metal decking has been installed, or as early as practicable	☐	☐	☐
Where perimeter cables are to be used as part of a personal fall arrest system, they should be designed to meet all applicable requirements for fall protection (e.g., 5000 lbs per person to be attached)	☐	☐	☐
Stamped, engineered drawings of perimeter systems are required	☐	☐	☐
Top cables must be flagged and both mid cables and toeboards must be in place prior to releasing the areas for general use	☐	☐	☐

Site Specific Erection Plan	Yes	No	N/A
The steel erector must consider the following elements when developing a site specific erection plan. Also, the plan should be signed and dated by the qualified person(s) responsible for its preparation/modification.	☐	☐	☐
- The sequence of erection activity, developed in coordinator with DPR should include the following: - Material deliveries - Material staging and storage - Coordination with other trades and construction activities	☐	☐	☐
- A description of the crane/derrick selection and placement procedures should include: - Site preparation - Path for overhead loads - Critical lifts, including rigging supplies and equipment	☐	☐	☐
- A description of steel erection activities and procedures, including the following: - Stability considerations requiring temporary bracing and guying - Erection bridging terminus point - Notification regarding repair, replacement and modifications of anchor bolts - Columns and beams (including joists and purlins) - Connections - Decking	☐	☐	☐

Steel Erection Checklist

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Site Specific Erection Plan	Yes	No	N/A
o Ornamental and miscellaneous iron			
▪ A description of fall protection procedures that will be used	☐	☐	☐
▪ A description of the procedures that will be used to prevent falling object hazards	☐	☐	☐
▪ A description of the special procedures required for hazardous non-routine tasks	☐	☐	☐
▪ Certification for each worker who has received training for steel erection activities	☐	☐	☐
▪ A list of the qualified and competent persons	☐	☐	☐
▪ A description of the procedures that will be used in the event of rescue or emergency response	☐	☐	☐

Cranes and Rigging	Yes	No	N/A
Cranes:			
Prior to the crane's arrival on site, the responsible subcontractor should provide the following information to DPR:	☐	☐	☐
o Certificate of insurance	☐	☐	☐
o Verification of certified operator	☐	☐	☐
o Most recent annual inspection on the crane	☐	☐	☐
o Load charts pertinent to specific boom configuration to be used	☐	☐	☐
o Written verification that maintenance has been performed pursuant to manufacturer's recommendations	☐	☐	☐
o Written verification of rigger's training	☐	☐	☐
Prior to the first pick, the crane operator should complete the following:	☐	☐	☐
o Documented, daily inspection (should be performed daily thereafter)	☐	☐	☐
o Complete the crane lift package, including pre-lift checklist, crane pre-task plan, lift plan, and FAA notification if applicable	☐	☐	☐
Crane operators are responsible for operations under their direct control. The operator has the authority to stop any/all hoisting activities and refuse to handle loads until safety is assured	☐	☐	☐
▪ Crane should be equipped with integrated anti-two block system	☐	☐	☐
▪ Safety latches should be in place and operable	☐	☐	☐
Rigging:	☐	☐	☐
▪ Single lift rigging preferred, maximum of 3 members	☐	☐	☐
▪ Multi lift rigging assembly should be specifically designed & assembled to accommodate maximum capacity of lifts with 5 to 1 safety factor	☐	☐	☐
▪ Single unit picking & rigging only for decking bundles, no nylon chokers allowed	☐	☐	☐
▪ All materials hoisted should be rigged by a qualified rigger	☐	☐	☐

Metal Buildings	Yes	No	N/A
Purlins and girts are prohibited from being used as an anchorage point for a fall arrest system, unless written approval is obtained from the subcontractor's qualified person, and approved by DPR project management (project team should defer to the Structural Engineer of Record)	☐	☐	☐

Structural Steel Assembly	Yes	No	N/A
Plumbing up	☐	☐	☐
Plumbing up equipment should be installed during the steel erection process, when deemed necessary by a competent person to ensure the stability of the structure	☐	☐	☐
When plumbing up equipment is used, it should be in place and properly installed before the structure is loaded with construction materials such as loads of joists, bundles of decking or bundles of bridging	☐	☐	☐
Plumbing up equipment may be removed only with the approval of a competent person	☐	☐	☐
Columns: All columns should be evaluated by a competent person to determine whether guying or bracing is needed. If needed, it should be installed	☐	☐	☐
Anchor Bolts/Rods:	☐	☐	☐
Approval by the project structural engineer of record is required before anchor bolts/rods can be repaired, replaced or field modified	☐	☐	☐
Before the erection of a column, DPR must ensure written notification is presented to the steel erector if there has been any repair, replacement or modification of the anchor bolts/rods of that column	☐	☐	☐

Steel Erection Checklist

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Structural Steel Assembly	Yes	No	N/A
Beams and Columns: During the final placing of solid web structural members, the load should not be released from the hoisting line until: - The members are secured with at least four bolts per connection, of the same size and strength shown in the erection drawings; - The bolts are drawn up wrench-tight; or - The equivalent, as specified by the project structural engineer of record	☐	☐	☐
A competent person must determine if more than two bolts are necessary to ensure the stability of cantilevered members; if additional bolts are needed, they must be installed	☐	☐	☐
Diagonal Bracing: Solid web structural members used as diagonal bracing should be secured by: - At least four bolts per connection drawn up wrench tight; or - The equivalent, as specified by the project structural engineer of record	☐	☐	☐

Open Web Steel Joists	Yes	No	N/A
Where constructability does not allow a steel joist to be installed at the column:			
- An alternate means of stabilizing joists should be installed on both sides near the column and it must: - Provide equivalent stability to a steel joist field-bolted at the column - Be designed by a qualified person - Be shop installed - Be included in the erection drawings	☐	☐	☐
- Where steel joists at or near columns span more than 60 feet: - The joists should be set in tandem with all bridging installed, or - An alternative method of erection should be used, provided it: - Provides equivalent stability to the steel joist, - Is designed by a qualified person, and - Is included in the site specific erection plan	☐	☐	☐
Any modification that affects the strength of a steel joist or steel-joist girder should be made with the approval of the project structural Engineer of Record	☐	☐	☐
Steel joists and steel joist girders should not be used as anchorage points for fall arrest systems unless written approval to do so is obtained from a qualified person.	☐	☐	☐
When bolted diagonal erection bridging is required the following should apply: - The bridging should be indicated on the erection drawing - The erection drawing should be the exclusive indicator of the proper placement of this bridging	☐	☐	☐

Landing and Placing Loads	Yes	No	N/A
During the construction period, the contractor placing a load on steel joists should ensure that the load is distributed so as to not exceed the carrying capacity of any steel joist.	☐	☐	☐
No bundle of decking should be placed on steel joists until: - All bridging has been installed and anchored - All joist bearing ends are attached, <i>unless all of the following conditions are met:</i> - The contractor has first determined that the structure or portion of the structure is capable of supporting the load. This should be: - Determined by a qualified person, and - Documented in the site specific erection plan - The bundle of decking is placed on a minimum of three structural joists - The joists supporting the bundle of decking are attached at both ends - At least one row of bridging is installed and anchored - The total weight of the bundle of decking does not exceed 4,000 lbs - Placement of the bundle of decking should be within 1 foot of the bearing surface of the joist end	☐	☐	☐

Training	Yes	No	N/A
Fall Hazards			
Worker training should be provided by a qualified person.	☐	☐	☐
All workers exposed to fall hazards should be trained in the following areas: The recognition and identification of fall hazards in the work area	☐	☐	☐



Steel Erection Checklist

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▪ The use and operation of protective systems such as guardrail systems, personal fall arrest systems, positioning device systems, fall restraint systems, safety net systems and other protection to be used ▪ The correct procedures for erection, maintaining, disassembling and inspecting and inspecting the fall protection systems to be used ▪ Procedures for protection from falls to lower levels and into/through holes and openings			
Special Training			
The subcontractor should also provide special training to workers involved in the following activities:	☐	☐	☐
▪ Multiple lift rigging operations (including multiple lift hazards and proper procedures/equipment required to perform multiple lifts) ▪ Connecting operations (including connecting hazards and the establishment, access, proper connecting techniques and required work practices) ▪ Controlled decking zones, where allowed (including the hazards associated with working in controlled decking zones, the establishment, access, proper connecting techniques and required work practices)	☐	☐	☐



Deck Release Form

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Steel Erection Contractor:		Date:	
Project:			
Steel Erector Representative:			
DPR Representative:			
Floor Level/Area:			
Sequence:			
Yes	No		
☐	☐	Are the wire rope perimeter guardrails in place and at the proper height?	
☐	☐	Are all wire rope cables tight, with a maximum of 3" of deflection?	
☐	☐	Are cable clamps properly installed, i.e., 3 clamps minimum, clamps installed in the correct direction, and clamps are of drop forged material?	
☐	☐	Are all wire rope cables tight, with a maximum of 3" of deflection?	
☐	☐	Are uprights of sufficient strength to provide fall protection as required (i.e., designed and installed to meet 200 lb guardrail system or 5,000 lbs. per person tie-off requirements?)	
☐	☐	Do guardrails or permanent guard railings protect all stair openings?	
☐	☐	Is all decking secured?	
☐	☐	Is the entire area clean, with all excess materials removed?	
☐	☐	Is the area ready to be turned over?	
☐	☐	Any other issues?	
Sign off			
<i>This form should be completed jointly by DPR Construction and the steel erection company's authorized representative prior to DPR accepting responsibility for the care, custody and control of the specific area of work outlined above.</i>			

Steel Erector Print Name / Signature

Date

By signing above, you verify that the above described area meets turn over requirements, and all requested changes/corrections have been made.

DPR Supervisor Print Name / Signature

Date

By signing above, you verify that the above described area meets turn over requirements, and all requested changes/corrections have been made.

Section 24

Tools & Equipment

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Tools and Equipment

(Includes hand, electrical, pneumatic, powder driven, hydraulic tools, saws, grinders, grounding, and lasers)

1.0 Introduction

Use of tools makes many tasks easier. However, the same tools that assist us, if improperly used or maintained, can create significant hazards in our work areas. DPR's intent is that all employees who use tools must be properly trained to use, adjust, store and maintain tools properly.

Retraining is required when proficiency can no longer be demonstrated and/or following any accident or incident involving referenced tools and equipment. All tools and equipment should be brought onto the jobsite in good working order free from defects. Defective tools are to be removed from service, repaired or removed from site immediately.

Reference: Hand and Power Tools CFR 1926 Subpart I; Grounding 1926.404; Pneumatic and Portable Tools 1926.951. Other applicable Standards, guidelines, and regulations may apply.

2.0 Electrical Tools & Equipment, Portable

2.1 General Guidelines

The following general requirements apply to all portable and/or plug-connected tools and equipment:

- Tools and equipment should be GFCI protected (including double insulated tools)
- The non-current carrying metal parts of portable and/or plug-connected equipment should be grounded
- Portable tools and appliances protected by an approved system of double insulation should be distinctly marked, with polarized plugs
- Each tool, cord and plug, and any equipment connected by cord and plug should be visually inspected before each day's use for external defects and indication of possible internal damage
- Tools and equipment found damaged or defective should be removed from service and not used until properly repaired
- Tools should not be carried by their power cord
- Electrical tools and equipment should not be used in hazardous locations without prior approval and issuance of the appropriate work permit (i.e. Confined Space, Energized Electrical Work, Hot Work, etc.) by DPR EHS and project management, additionally; tools used in hazardous environments should be intrinsically safe
- All tools and equipment should be unplugged and/or Lockout/Tagout (LOTO) prior to performing any alteration, maintenance or servicing operation such as cleaning and changing blades, bits or accessories
- Appropriate guarding of all tools and equipment should be provided, maintained and used as per manufacturer's guidelines
- Where tools are used which present a "flying debris" hazard (grinding operations, chop saws, drilling overhead, chain saws, etc.), operators should use double eye protection, i.e., goggles and a face shield or a face shield with safety glasses; *for additional guidelines on Personal Protective Equipment refer to the Personal Protective Equipment Section 16*

2.2 Drills, Screw Guns and Impact Guns

- Ensure bits and sockets are inspected and replaced routinely
- Ensure tools are not forced beyond capacity
- Ensure bits are sharp and the appropriate bit is selected for the task

- Appropriate protective measures should be taken to protect against flying debris for overhead operations
- Ensure tools and material are adequately secured
- Body parts should be clear of the tool's line of action

2.3 *Extension Cords*

- Only heavy duty, three-wire ground extension cords should be used with portable electric tools and appliances. All cords should conform to the type and configuration required by OSHA standards
- **Minimum acceptable size of electrical cord is 12 gage, 3 wire, heavy duty**
- Cords should be protected from damage by doors, equipment, traffic, chemicals etc.
- Cords should not be run down the middle of passageways, in stairwells and other paths of travel and a cord management program should be established on all projects
- Whenever possible, all cords and cables should be elevated above the floor/ground a minimum of seven feet (7'); when elevating cords and cables, non-conductive fasteners should be used
- Cords should be inspected daily before use for external defects such as deformed or missing pins or insulation damage and potential internal damage
- Damaged or defective equipment should be immediately removed from service, repaired and tested prior to being returned to service
- "Daisy-chaining" of extension cords should not occur unless manufacturer's recommendations allow for such use.

2.4 *Grinders*

- Operators of equipment should be trained in the safe operation of the tool prior to use
- Use and inspection of wheel and equipment should follow manufacturer's guidelines
- Ensure that the wheel is properly rated and sized for the tool and material being cut or ground
- Appropriate guards of all grinders should be provided and used
- Guards and other protective devices should not be removed or rendered ineffective or inoperable
- Power should be disconnected and/or LOTO during all cleaning, maintenance or servicing of the grinder, e.g., changing the wheel, cleaning, etc.
- Ensure that material is adequately secured prior to grinding or cutting
- Ensure that two hands are utilized and the tool be kept under control at all times, including but not limited to, overhead work

Note: Saws should not be used for any grinding type operation

2.5 *Saws (Band Saw, Chain Saw, Chop Saw, Circular Saw, Table Saw, Portable Cutoff Saw, Reciprocating Saw)*

- Follow all manufacturers guidelines
- Areas around cut stations should be kept free and clear of material and debris
- Appropriate guarding of all saws should be provided and used
- Guards and other protective devices should not be removed or rendered ineffective or inoperable
- Power should be disconnected and/or LOTO during all cleaning, maintenance or servicing of the saw, e.g., changing the blade, cleaning, oiling etc.
- Ensure that the blade is properly rated and sized for the material being cut
- Ensure that the blade is sharp
- Ensure that two hands are utilized when the tool is designed for two handed use
- Ensure push sticks are used Note: Two people may be required when cutting large and/or bulky material

- Ensure that appropriate height work benches, saw horses or cut stations are used for cutting operations
- Material to be cut should not be supported off any body part
- Ensure that material is adequately secured to prevent displacement prior to cutting
- Ensure that appropriate guarding or barriers are used to guard against flying debris
- Cut resistant trousers or chaps should be worn when using a chain saw
- Chain saws should not be used without kickback protection on the chain

Note: Saws should not be used for any grinding type operation

3.0 Hand Tools

3.1 General Requirements

- Individuals should inspect hand tools prior to use
- Faulty or damaged tools should be removed from service and/or the site immediately
- Only appropriate tools should be used for the job
- Impact tools should be maintained free of mushroomed heads
- Proper body positioning and ergonomics should be maintained at all times
- Wooden tool handles should be kept free of splinters or cracks
- Tools should be routinely inspected to assure a tight connection between the tool head and the handle
- Wheelbarrows should not be pushed with handles in an upright position
- Wheelbarrows should have manageable loads and tires properly inflated at all times
- Pointed tools, such as chisels and screwdrivers, should never be carried point-up in any pocket, nor should they be carried point-down in a front pocket; they should be hand carried with the sharp edge or point away from the body, in a toolbox, bag, pouch or special tool belt
- Tools should never be carried in a way that interferes with a worker's ability to use both hands while climbing a ladder or structure; tools should be raised or lowered by rope, using a bucket if necessary
- Tools should not be dropped, thrown or tossed to one another; they should be handed off carefully
- Tools on scaffolds, ladders or elevated work platforms should not be left or stored on elevated work surfaces

Refer to the Excavation & Trenching Section 10 for further guidelines regarding hand tools and hand digging around utilities and services

3.2 Axes/Mattock

- Axes should be carried with sheaths or guards in place, or at one's side with the handle up and shank in the palm of the hand
- Ensure that the head is securely fastened to the handle
- Ensure the handle is free of cracks and splinters
- Ensure that the blade is sharp
- Ensure the axe is the proper weight and size for the task
- Aim downswing of axe/mattock in a safe manner so as not to impact body

3.3 Chisels

- Use the right size chisel for the job and a hammer of the appropriate weight
- Ensure that chisels are sharp
- Chisels that have begun to mushroom should be ground to avoid the danger of flying chips or removed from service

3.4 Crowbars

- Ensure that the appropriate size of crowbar is used
- Do not use substitutes such as pipes, iron bars or lumber to pry or fulcrum material
- Ensure that all body parts are out of pinch points and impact zones

3.5 Drywall Finishing Tools

- Ensure tools are cleaned and maintained at all times, and adequate measures are taken to ensure floors and walkways remain clear of spills and slippery conditions
- Ensure vacuums on dust collection systems are appropriately grounded
- When not in use, store taping blades down to prevent potential cut hazards.
- Ensure bazooka tubes and taping boxes are not overloaded and the appropriate size is used for the task
- Set bazookas horizontally on folding scaffold when not in use – do not leave on floor.
- All mud containing taping tools (10" box, 12" box, nail spotters, etc.) should be set with the heaviest part of the tool down and leaned against a corner where they cannot slide down splattering mud or damage the tool

3.6 Files

- Files should be equipped with handles
- Files should not be used to punch or pry
- A file should never be cleaned by striking it against a metal object

3.7 Hammers and Sledges

- Inspect all handles for cracks, splinters and looseness
- Hammers should be replaced when the hammerhead begins to mushroom
- Hammers should only be swung only as hard as necessary and safe
- Hit the target straight on, never at an angle
- A hammer should never be used to strike another hammer; a soft mallet should be used instead to prevent chipping
- Claw-hammers whose handles are not designed for continuous pulling should not be used for prying. Use a nail puller, 'cats claw' or pry bar for this work
- Hammers and sledges should be sized for the user

3.8 Knives

- When not in use, fixed blade knives should be sheathed
- Sheathed knives should be carried above the hip or in back, never in front
- Knives should be kept sharp to avoid the use of excessive force
- Knives should always be carried with the blade away from the body
- Knives are not to be used as screw drivers, pry bars or picks
- The use of pocket knives, Leatherman knives or other multi-tools is prohibited

3.9 Shovels

- Ensure the right type of shovel is used for the task
- Inspect shovels for cracked, splintered or otherwise damaged handles
- Ensure shovels are sharp to avoid excessive force being applied
- Proper lifting techniques should be utilized when using a shovel

- Twisting the spine should be avoided; the legs, rather than the arms, shoulders and back should be used
- Use the ball of the foot instead of the arch to push the shovel

3.10 Wire and Bolt Cutters

- Use the proper cutter for the job and ensure that the cutter's capacity rating is not exceeded
- Do not "rock" cutters
- Cutters should never be used as pry bars or nail pullers
- Cutters should have insulated grips and no wiring or conduit should be cut unless indicated safe to do so by a competent person
- In the event where EEW is required, all established procedures must be in place prior to start of work. EEW tools and equipment must be insulated. Refer to the Hazardous Energy Section 12

3.11 Wrenches

- Wrenches should not be used when jaws are sprung to the point that slippage occurs
- Wrenches should not be altered by the addition of handle-extensions or "cheaters"
- Pipe or Stillson wrenches should not be used as a substitute for other wrenches

4.0 Electrical Equipment, Fixed

- Exposed, non-current carrying, metal parts of fixed electrical equipment, including motors, generators, frames and tracks of electrically operated cranes, electrically driven machinery, etc., should be grounded at all times
- All portable and/or fixed, power driven woodworking tools should be provided with a disconnect switch that can be either locked or tagged in the off position.
- The path from circuits, equipment, structures, and conduits or enclosures to ground should:
 - be permanent and continuous
 - have ample carrying capacity to safely conduct the current likely to be imposed on it

5.0 Illumination

Illumination should meet the following minimum requirements:

Foot-Candles	Areas of Operation
5	General construction area lighting.
5	General low activity construction areas, waste areas, inactive storage areas, parking and field maintenance areas.
5	Corridors, hallways, and exit ways.
10	General construction plant and shops (e.g. mechanical rooms, electrical equipment rooms, fab shops, active store rooms, indoor toilets and workrooms.)
30	First aid stations and offices.

5.1 Temporary Lighting

- Follow manufacturer's recommendations for installation, maintenance and repair of temporary lighting
- Temporary lighting should be fed on a separate circuit from construction use power
- All construction lighting must be protected from accidental contact or breakage; cages must be provided

- Electrical lights cannot be hung from their cords unless specifically designed to do so; hang only with non-conductive material
- All open sockets must be secured against accidental contact
- All temporary lighting devices should be disposed of following federal, state and local guidelines
- Use additional task lighting where necessary to provide adequate light to work safely
- Head lamps and flashlights are not typically adequate to provide task lighting
- Additional lights should be used in work locations

6.0 Lasers

The following general requirements apply to all lasers:

- Only trained workers are allowed to operate lasers
- Signs should be posted in areas where lasers are in use
- Workers should wear proper eye protection where there is a potential exposure to laser light greater than 0.005 watts (5 milliwatts)
- Beam shutters or caps should be utilized, or the laser turned off when laser transmission is not actually required
- Lasers should be turned off if they are to be left unattended for a substantial period of time
- All attempts should be made to set up lasers at a height sufficient to reduce exposure to other works in the area

7.0 Pneumatic Tools

The following general requirements apply to all pneumatic or air powered tools on a DPR project:

- All hoses and pneumatic equipment should be inspected before each use
- All damaged hoses and pneumatic equipment should be removed from service and repaired to prevent failure
- All air tools should be disconnected prior to altering, performing maintenance, cleaning, servicing, changing bits or accessories, etc.
- All pneumatic tools should be secured to the hose in a positive manner to prevent accidental disconnection: all other hose connections should be secured to prevent accidental disconnection
- Install and maintain safety clips, retainers and hose whip restraints on all pneumatic tools to prevent attachments from being accidentally disconnected
- All hoses exceeding 1/2 inch inside diameter require safety devices at the source of supply to reduce pressure in case of hose failure
- Hoses should be protected from damage by doors, equipment, traffic, chemicals etc.
- Hoses should not be run down the middle of passageways, in stairwells and other paths of travel and a hose control program should be established on all projects
- Compressed air used for cleaning purposes should not exceed 30 psi, and then only in conjunction with effective chip guarding and personal protective equipment
- Exceptions to 30 psi are allowed only for concrete form, mill scale and similar cleaning operations
- When using compressed air (or water) in cleaning operations or other operations which present "flying debris" hazards (blowing off pan decks, power washing etc.), operators should use double eye protection, e.g., goggles or a face shield with safety glasses
- The use of compressed air to clean yourself or other workers is not permitted
- Prevent the tip from coming into contact with the operator

7.1 Nailers and Staplers

- All pneumatically-driven nailers and staplers should be operated and maintained in accordance with the manufacturer's operating and safety instructions and appropriate PPE should be worn
- All tools should have a safety device on the muzzle to prevent the tool from operating unless the muzzle is in contact with the work surface, to prevent accidental discharge. This trigger type will only fire a nail when the controls are activated in a certain order.
- Never disable or bypass nail gun safety features.
- Operating controls and safety devices should not be removed, tampered with, altered or otherwise disabled
- Tools should be connected to the air supply with a safety disconnect that consists of a spring loaded shut-off valve and a positive locking mechanism to prevent the tool from becoming accidentally disconnected
- Tools should be equipped with a fitting that will discharge all compressed air in the tool at the time the fitting or hose coupling is disconnected
- Tools should be disconnected from the air supply prior to performing any maintenance or repair or clearing a jam
- When performing roof or elevated activities air hose shall be secured in such a manner as to provide ample, but not excessive amounts of hose
- The employer's written Code of Safe Practices shall include provisions for the use of pneumatically-driven nailers and staplers where applicable (CA required)

8.0 Powder Actuated Tools

The following general requirements apply to all powder actuated tools:

- Only trained and qualified workers should operate powder-actuated tools
- Operators of powder-actuated tools should be trained in safe operation, specific to the tool being used
- Tools should be held in a secure manner during use
- All powder-actuated tools should be inspected daily before use; damaged or defective equipment should be removed from service and repaired prior to use
- Signs should be posted in areas where powder-actuated tools are in use, within 50 feet
- Tools should not be loaded until immediately before use
- Tools should not be left unattended
- Loads should be kept in separate and secure containers (CA 1687 requirement is a lockable, properly labeled container)
- Spent loads and misfires should be disposed of per manufacturer and any Federal, State and Local requirements
- In the event of misfires, the tool should be held against the work surface for 30 seconds before attempting another shot
- Appropriate eye and hearing protection should be worn within 25' of the operation

9.0 Temporary Power & Wiring

- The electrical subcontractor is responsible for the installation, maintenance and testing of all electrical system components supplying power for the project, including all grounding references
- All electrical work, installation and wire capacities should be in accordance with the pertinent provisions of the National Electrical Code, unless otherwise provided by federal, state and local jurisdictions, whichever is more stringent
- All temporary wiring should be effectively grounded in accordance with the National Electrical Code
- All switches should be enclosed and grounded

- Panel boards should have provisions for closing and locking the main switch and fuse box compartment; panels should be secured from entry by unauthorized individuals at all times
- Temporary power panels should be elevated off the ground sufficient to protect from moisture damage.
- Temporary power panels, j-boxes, etc. should be fully enclosed with no missing or damaged openings or knockouts
- Only manufactured electrical temporary power cords/cables, temporary adapters, j-boxes, etc. are allowed. No improvised assemblies are allowed
- Precautions should be taken to ensure that all open wiring is covered, secured and otherwise protected at all times; effective LO/TO programs are required
- Temporary power conductors should not rest on floors and should not be installed so as to be subject to physical damage; conductors should be elevated where possible using non-conductive means
- All 120 volt 15- and 20-amp receptacle outlets on the site that are not part of the permanent wiring of the building should use ground fault circuit interrupters
- Cables passing through work areas should be covered or elevated to protect them from damage and to eliminate hazards to workers
- Suitable means should be provided for identifying all electrical equipment and circuits, especially when two or more voltages are used on the same job
- All circuits should be marked for the voltage and the area of service they provide
- The only electrical circuits and outlets that should be utilized for construction power are those designated by DPR project management
- When permanent power is used to supply construction power, red outlets are never to be used

9.1 *Ground Fault Circuit Protection*

Ground fault circuit protection is required for all temporary power throughout the length of the construction project. Ground fault circuit interrupters (GFCIs) should be inspected and tested prior to use and at minimum monthly for continuity and proper functioning. Where construction tools and/or equipment will be plugged directly into “house” power, a GFCI adaptor should be used between the construction tool/equipment and “house” power.

9.2 *Assured Equipment Grounding*

In some cases, owners may require an Assured Equipment Grounding Conductor Program for the construction site. This program will not negate DPR’s GFCI policy. Equipment Grounding Conductor Program will meet all federal, state and local requirements. This program will cover all cord sets, receptacles that are not part of the permanent wiring of the building or structure and equipment connected by cord and plug, which are available for use or used by all workers. This program should be implemented by competent person(s).

Tools & Equipment Forms

Forms

Powder Actuated Tools Training Verification – English

Powder Actuated Tools Training Verification - Spanish

Laser Training Verification – English

Laser Training Verification – Spanish



Powder Actuated Tools Training Verification

Archive Document – DO NOT DISCARD

Name:		Date:	
Project/Project #:		Company:	

Equipment

Manufacturer(s):			
Model(s):		Capacity:	

Training

☐	Only personnel trained in the safe use of powder actuated tools should be permitted to utilize this equipment
☐	Training should be completed prior to using the equipment, and be specific to the equipment to be used

PPE & Signage

☐	Operators and bystanders should wear approved eye protection; some manufacturers recommend the use of safety goggles to protect eyes from flying particles. (Refer to specific Manufacturer's guidelines)
☐	Operators and bystanders should wear hearing protection when using or in the general vicinity of powder actuated tools
☐	Anti-vibration gloves should be worn with prolonged use and consideration should be given to tight confines and tool recoil as it relates to hand protection
☐	Warning signage should be used in the work area to notify others of powder actuated tools in use, within 50 feet

Tool Inspection

☐	Tools should be inspected prior to use
☐	Tools should be used with the correct shield, guard or attachment recommended by the manufacturer
☐	Damaged or defective equipment should be removed from service and repaired prior to use
☐	

General Use Guidelines

☐	Tools should not be loaded until immediately before use
☐	Tools should not be altered or changed.
☐	Only manufacturer's compatible fasteners, loads and tool parts should be used
☐	Verify the tool is not fully or partially loaded with a load or fastener prior to handling.
☐	Do not fire a tool without a fastener
☐	Never place hand or other body part in front of the muzzle, as the piston and/or fastener may pass through the operator's body in the event of accident discharge
☐	Point tool in a safe direction at all times.
☐	Do not close tool against work surface. Tool should be manually closed, with hand away from muzzle end to prevent unintentional discharge.
☐	Always hold tool perpendicular to work surface to limit the possibility of a fastener bouncing off the surface and striking operator or bystander
☐	In the event you decide not to make a fastening after the tool has been loaded, always remove the load first, then the fastener
☐	Never use powder actuated tools in an explosive or flammable atmosphere
☐	Loaded tools and shots should not be left unattended, and must remain locked in a tool case or toolbox when not in use
☐	Loads should be kept in separate and secure containers
☐	Spent loads and misfires should be disposed of per manufacturer and any Federal, State and Local requirements

Test Fastenings & Material Compatibility

☐	Test fastenings should be made with the lightest load level designed for use in the tool. If the lightest load does not set the fastener, try the next highest load level until the proper level is attained. This will eliminate the possibility of an overpowered fastener fully penetrating your work, thereby injury someone.
☐	Perform the center punch test if you are unsure of your fastening material. This will ensure the work can contain your fastening.



Powder Actuated Tools Training Verification

Archive Document – DO NOT DISCARD

☐	Tool should not be fired within three inches (3") of the edge of a concrete base material or within one-half inch (1/2") in steel. Do not attempt a fastening application closer to another previously inserted fastener, i.e., 3" in concrete, 1/2" in steel. (Refer to operator's manual for guidance.)
☐	Do not fasten into a concrete base material less than 3 times as thick as the fastener penetration unless otherwise permitted and never fasten into steel base material thinner than the fastener shank.
☐	Do not fasten into cast iron, tile, glass or other brittle materials
☐	Do not fasten through pre-drilled holes
☐	Do not fasten into a spalled area as fastener may bend and bounce off the spall and strike operator or bystander. Place fastener at least 3 inches away from spalled areas.

Miss Fires

☐	In the event a load fails to discharge, the tool should be depressed for at least 30 seconds to ensure protection to operator or bystander in the event of a delayed load discharge
☐	Spent loads and misfires should be disposed of per manufacturer and any Federal, State and Local requirements
☐	Jammed, stuck or broken tools should not be unloaded or disassembled; improper handling may cause it to discharge, striking operator or bystander
☐	Point the jammed tool in a safe direction; tag and lock up the tool and call your equipment vendor/supplier for assistance

This document hereby certifies that I have been instructed on the items listed above and I have had an opportunity to ask questions. I fully understand the DPR requirements for the use of powder actuated tools.

Name/signature of employee	Date
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Name/signature of person conducting verification review	Date
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A copy of this form must be faxed to the DPR regional office PRIOR TO START OF WORK.



Verificación de la capacitación en herramientas accionadas por cartucho de pólvora

Documento de archivo – NO DESECHAR

Nombre:		Fecha:	
Proyecto/Proyecto#:		Compañía:	

Equipo

Fabricante(s)			
Modelo(s)		Capacidad(es)	

Entrenamiento

☐	Solo se debe permitir que el personal entrenado en el uso seguro de las herramientas accionadas por cartucho de pólvora utilice este equipo
☐	Debe cumplirse la capacitación antes de usar el equipo y ser específico al equipo usado

Equipo de protección personal y señalización

☐	Los operadores y los espectadores deben usar protección para los ojos aprobada. Algunos fabricantes recomiendan el uso de gafas de seguridad para proteger los ojos contra las partículas en el aire. (Refiera a las guías de consulta del fabricante específico.)
☐	Los operadores y los espectadores deben usar protección para los oídos cuando usen o se encuentren cerca de herramientas accionadas por cartucho de pólvora
☐	Se deben usar guantes contra la vibración en caso de uso prolongado y se debe prestar especial consideración a los recintos cerrados y el efecto del retroceso de la herramienta en lo que se refiere a la protección de las manos
☐	Se deben usar carteles de advertencia en el área de trabajo para notificar a los demás que se están usando herramientas accionadas por cartucho de pólvora, dentro de 50 pies

Inspección de las herramientas

☐	Se deben inspeccionar las herramientas antes de su uso
☐	Las herramientas se deben usar con la protección, barrera o accesorio adecuados recomendados por el fabricante
☐	Se debe quitar de servicio el equipo dañado o defectuoso y repararlo antes de usarlo

Guías generales de uso

☐	Se deben inspeccionar las herramientas antes de su uso
☐	No se deben cargar las armas hasta justo antes de su uso
☐	No se deben alterar o cambiar las herramientas
☐	Solo se deben usar sujetadores, cargas y partes de las herramientas compatibles con las del fabricante
☐	Verifique que la herramienta no esté cargada total o parcialmente con una carga o con un sujetador antes de manejarla
☐	No dispare una herramienta sin sujetadores
☐	Nunca coloque la mano u otra parte del cuerpo en frente del orificio, ya que el pistón y/o un sujetador puede atravesar el cuerpo del operador en el caso de un disparo accidental
☐	Apunte la herramienta en sentido seguro en todo momento
☐	No cierre la herramienta contra la superficie de trabajo. La herramienta debe cerrarse con la mano lejos del orificio para evitar que se dispare por accidente.
☐	Siempre mantenga la herramienta perpendicular a la superficie de trabajo para limitar la posibilidad de que un sujetador rebote en la superficie y golpee a un operador o espectador
☐	Si usted decide no disparar un sujetador después de haber cargado la herramienta, primero siempre quite la carga, luego el sujetador
☐	Nunca use herramientas accionadas por cartucho de pólvora en una atmósfera explosiva o inflamable
☐	No se deben dejar las herramientas cargadas sin supervisión
☐	Los instrumentos y los disparos cargados no deben ser dejados desatendidos, y deben quedarse encerrados en un toolcase o caja de herramientas cuando no en uso
☐	Las cargas se deben guardar en recipientes separados y seguros rotulados
☐	Las cargas usadas y fallos de disparos deben ser arrojadas según los reglamentos del fabricante y requerimientos federales, estatales y locales



Verificación de la capacitación en herramientas accionadas por cartucho de pólvora

Documento de archivo – NO DESECHAR

Prueba de sujetadores y compatibilidad del material

☐	Los sujetadores se deben probar con el menor nivel de carga diseñado para el uso en la herramienta. Si la carga más liviana no fija el sujetador, pruebe con la carga que le sigue en nivel hasta que se obtenga el nivel correcto. Esto eliminará la posibilidad de que un sujetador sobrecargado penetre por completo en la obra y lastime a alguien.
☐	Realice la prueba del punzón de perforar si no está seguro del material del sujetador. De esta manera se asegurará que la obra pueda contener el sujetador
☐	No se debe accionar la herramienta dentro de las tres pulgadas (3") del borde de un material de concreto o dentro de la pulgada y media (1/2") de material de acero. No intente fijar un sujetador cerca de otro sujetador insertado previamente, es decir, 3" en concreto, 1/2" en acero. (Consulte el manual del operador si desea una guía).
☐	No fije el sujetador en un material de concreto de menos de 3 veces el grosor de la penetración del sujetador, salvo que se permita lo contrario y nunca fije el sujetador en material de base de acero que sea más delgado que la espiga del sujetador
☐	No fije en hierro fundido, losas, vidrio u otros materiales quebradizos
☐	No fije a través de orificios pre-taladrados
☐	No fije en un área fragmentada ya que el sujetador puede doblarse y rebotar del fragmento y golpear a un operador o espectador. Coloque el sujetador al menos a 3 pulgadas de las áreas fragmentadas

Cuasi disparos

☐	Si el disparo de la carga falla, se debe mantener la herramienta presionada durante al menos 30 segundos para protección del operador o espectador en el caso de una descarga demorada, luego quite la carga y colóquela en agua durante 24 horas antes de desecharla
☐	Las cargas usadas y fallos de disparos deben ser arrojadas según Los reglamentos del fabricante y requerimientos federales, estatales y locales
☐	Apunte la herramienta atascada en un sentido seguro, rotule y guarde la herramienta bajo llave y pida ayuda al proveedor/distribuidor del equipo

Este documento por el presente certifica que he recibido instrucción sobre los puntos listados anteriormente y que he tenido oportunidad de formular preguntas. Entiendo en forma cabal los requisitos de DPR con respecto a las herramientas accionadas por cartucho de pólvora.

Nombre/Firma del empleado

Fecha

Nombre/Firma de la persona que lleva a cabo la verificación

Fecha

Enviar por fax una copia de este formulario a la oficina regional de DPR ANTES DE COMENZAR A TRABAJAR.



Laser Training Verification

Archive Document – DO NOT DISCARD

Name:		Date:	
Project/Project #:		Company:	

Equipment

Manufacturer(s):			
Model(s):		Capacity:	

Training

☐	Only personnel trained in the safe use of lasers should be permitted to utilize this equipment
☐	Training should be completed prior to using the equipment, and be specific to the equipment to be used

PPE & Signage

☐	Operators and bystanders should wear approved eye protection; avoid direct eye exposure
☐	Warning signage should be used in the work area to notify others that a laser is in use

Tool Inspection

☐	Tools should be inspected prior to use
☐	Tools should be used with the correct shield, guard or attachment recommended by the manufacturer
☐	Damaged or defective equipment should be removed from service and repaired prior to use

General Use Guidelines:

☐	Laser should not be altered or changed
☐	Never use a laser in an explosive or flammable atmosphere
☐	Lasers should not be left unattended
☐	Use only the specifically designed batteries
☐	Store idle lasers out of reach from untrained persons
☐	Do not use optical tools i.e. telescopes, transits to view the laser beam
☐	Do not place the laser in a position which may allow someone to intentionally or unintentionally stare at the laser beam
☐	Turn the laser off when it is not being used
☐	Do not disassemble or attempt to repair the laser

This document hereby verifies that I have been instructed on the items listed above and I have had an opportunity to ask questions. I fully understand the DPR requirements for the use of lasers.

Name/signature of employee

Date

Name/signature of person conducting verification review

Date

A copy of this form must be faxed to the DPR regional office PRIOR TO START OF WORK.



Verificación de Capacitación en el uso de Láser

Documento para archivo – NO DESECHAR

Nombre:		Fecha:	
Proyecto/Proyecto #:		Compañía:	

Equipo

Fabricante(s)			
Modelo(s)		Capacidad(es)	

Capacitación

☐	Sólo el personal capacitado en el uso seguro del láser estará autorizado para utilizar este equipo
☐	Se debe cumplir la capacitación antes de usar el equipo, la que deberá referirse específicamente al equipo que se utilizará

Equipo de protección personal y señalización

☐	Los operadores y transeúntes deben usar protección para los ojos y evitar la exposición directa a los ojos
☐	Se debe usar señalización de advertencia en el área de trabajo para notificar a los demás que se está usando un láser

Inspección de Herramientas

☐	Se deben inspeccionar las herramientas antes de usarlas
☐	Se deben usar las herramientas con la protección, barrera o accesorio correcto, según las recomendaciones del fabricante
☐	Se deben retirar de servicio los equipos dañados o defectuosos y repararlos antes de usarlos

Pautas Generales de Uso

☐	No altere ni cambie un láser
☐	Nunca use un láser en una atmósfera explosiva o inflamable
☐	No deje el láser sin supervisión
☐	Use solamente las baterías diseñadas específicamente
☐	Mantenga el láser que no esté en uso fuera del alcance de las personas no capacitadas en su uso
☐	No use herramientas ópticas, como telescopios, tránsitos para ver el rayo de láser
☐	No coloque el láser en una posición que pueda permitir que alguien, intencional o inadvertidamente, mire el rayo de láser
☐	Apague el láser cuando no esté en uso
☐	No desarme ni intente reparar un láser

Este documento verifica que he recibido instrucción sobre los puntos enumerados arriba y he tenido oportunidad de realizar preguntas. Entiendo completamente los requisitos de DPR para el uso del láser.

Nombre/Firma del empleado	Fecha
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Nombre/Firma de la persona que lleva a cabo la verificación	Fecha
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Enviar por fax una copia de este formulario a la oficina regional de DPR ANTES DE COMENZAR A TRABAJAR

Section 25

Warehousing & Delivery

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Warehousing & Delivery

1.0 Introduction

The purpose of this section is to establish a set of best practices for the warehousing and delivery of materials and equipment. This section is intended to supplement the overall Environmental Health and Safety Plan (EHSP) without diminishing the importance and relevance of the other sections as they apply.

Understanding the complexities of securement (tying down a load), Department of Transportation (DOT) regulations, SDS, load restrictions, and local traffic & transportation laws requires training. No employee is permitted to perform any work for which they have not been properly equipped and trained or which they feel is unsafe. If an employee is asked to perform tasks which he/she feels they are not properly trained for, that employee is expected to bring it to the attention of the Branch Manager or the Regional EHS Manager.

2.0 Responsibilities for Safety and Loss Control

2.1 Branch Manager

The Branch Manager is responsible for oversight and daily operations management. Their EHSP responsibilities include:

- Oversee that work is performed in accordance with the DPR EHSP.
- Foster a proactive injury free environment.
- Participate in incident investigations and ensure reports are accurate and timely.
- Evaluate employees and ensure all training needs are addressed.
- Maintain open lines of communication with the Regional EHS Manager.

2.2 Driver / Warehouse Personnel

Drivers and warehouse personnel are potentially interchangeable positions. They are responsible for the loading, transporting, and delivery of materials and equipment. They are responsible for the safe operation of company owned vehicles. In addition:

- Understand and obey traffic laws.
- Secure loads adequately using proper tie-downs and techniques.
- Understand the requirements associated with chemical transport.
- Daily inspection of vehicle(s).
- Perform daily stretch & flex.
- The employee should ensure the correct number of workers have been assigned to perform the work safely, and that they have the correct skill level and training along with the correct tools and equipment to perform the job.

2.3 Mechanic

A Mechanic uses electric, pneumatic, as well as hand tools to repair equipment. They may come into contact with oils, gases, solvents, and other chemicals. They are expected to understand the safe use of these components as well as:

- Be aware of stored and potential energy sources.
- Be familiar with the site specific chemical disposal plan.
- Do not use tools, equipment, compressed air, etc., beyond their intended use.
- Understand and wear the appropriate PPE for the task.
- Daily stretch and flex.

2.4 **Inside Sales / Counter Person**

Some branches include sales to the public which means there is an element of public risk within our facilities. Inside Sales and Counter Persons also includes work within the warehouse environment. In addition to compliance with our EHSP, inside sales should:

- Assist in keeping fire exits and walkways clear.
- Use mechanical means for lifting if trained, get help when necessary.
- Comply with DPR incident reporting requirements, especially when a member of the public is involved.
- Do not perform operations for which they are not trained.

2.5 **Outside Sales / Copier Technician**

When performing tasks within the public domain or around construction sites, the DPR EHSP is applicable. EHSP duties include:

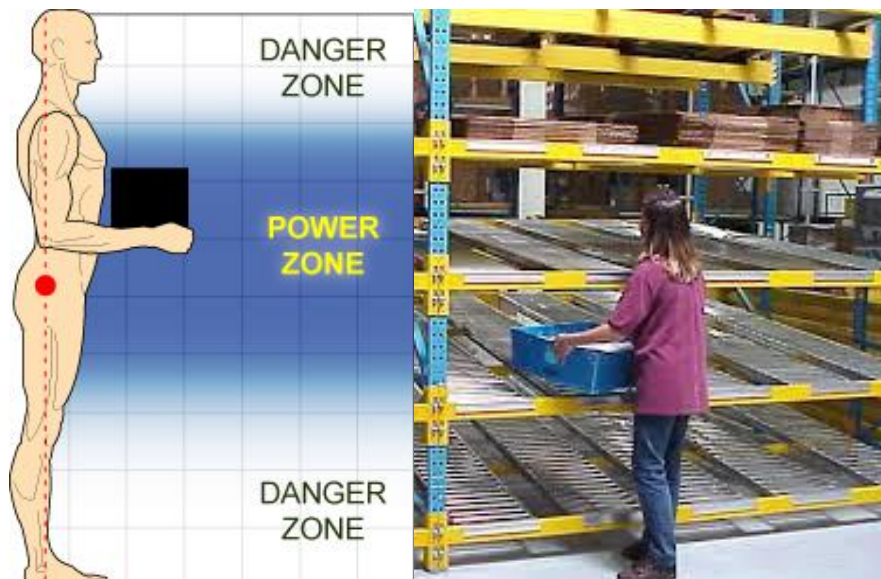
- Understand and obey traffic laws.
- Wear PPE as appropriate for the location.
- Follow site specific safety requirements.
- Do not perform operations for which they are not trained.

3.0 **Warehousing and Storage**

3.1 **Racking**

The use of racking is a common condition found in warehouses and other high density storage facilities. The design, installation, and method for stocking and retrieval of material should be planned. Hazards include:

- Collapse from overloading – Understand the rack system's capacity and the weight of the materials being stored. Rack capacity should be clearly marked.
- Tip over – Racking should be anchored to the floor per manufacturer's recommendations. Heavier objects that are loaded with forklifts should be placed lower on the system. Potential seismic activity should be considered. Forklift operators should proceed slowly and with additional caution when driving near racks.
- Product retrieval – Materials that are to be handled by hand, should be stored with weight, awkwardness, and access in mind.



- Catwalks – Racking systems with integrated catwalks fall into the elevated work category. Guardrails are the preferred method of fall protection, including toe boards. Open ended catwalks and mezzanine areas where material is loaded require some sort of fall protection. There are several “moveable” handrail devices on the market that prevent employee fall exposure. Working on a ladder next to a guardrail poses a threat as well. A personal fall arrest system (PFAS) may be needed. Use of a PFAS requires specific training. Please consult your local safety professional for assistance in designing adequate protection and providing the necessary training for employees.
- Falling objects – Smaller objects should be contained in closed containers. Opened containers with partially used materials should be secured to prevent contents from spilling out.
- ONLY TRAINED, QUALIFIED, AND CERTIFIED FORKLIFT OPERATORS should operate forklift equipment. Training can be scheduled through your local safety professional.

3.2 Chemical Management

Reference: DPR EHSP Section 02 - Chemical Management & Environmental Management

It is the responsibility of all OES employees to maintain the SDS's for material brought into the warehouse and kept in a known location. They shall provide SDS's for materials when being delivered.

3.2.1 Flammable / Combustible Materials

References: DPR EHSP Section 11 – Fire Protection & Prevention
29CFR1910.126 – Flammable Liquids
29CFR1910.157 – Portable Fire Extinguishers

The storage, handling, and transportation of flammable and combustible materials are regulated by multiple agencies, which includes but is not be limited to OSHA, DOT, NFPA, and local jurisdictions. The intent of this sub-section is to address common questions and paradigms. The expectation is that if quantities of flammable liquids or gases are to be stored in a warehouse environment, an individual plan will be put together to address and verify compliance with governing jurisdictions.

Flammable liquids are typically classified into Categories 1 through 4, based on flashpoint and boiling point. Handling requirements vary depending on the category designation. A flammable liquid fire (such as gasoline) is considered a class B fire.

- Only OSHA approved metal safety cans are allowed. Plastic cans should not be used.
- Safety cans are considered to hold a maximum of 5 gallons.
- No more than 25 gallons of gasoline may be stored outside of an approved storage cabinet.
- No more than 60 gallons may be stored inside of an approved cabinet.
- Cabinets should be properly labeled.
- A portable fire extinguisher should be placed within 50' [ref. 1910.157(d)(4)] of stored fuel cans and/or cabinets.

Should the need arise to store greater quantities of flammable liquids or gases, please consult with the local regional safety manager.

When storing and/or handling these types of materials the manufactures recommendations need to be followed. Reference: DPR EHSP Section 11 - Fire Protection and Prevention for additional information

3.3 Material Handling

Reference: DPR EHSP Section 09 - Ergonomics and Manual Material Handling
DPR EHSP Section 15 - Motorized and Site Equipment – Forklifts

The primary function of warehousing and distribution is moving material and equipment. Back strains, sprains, soft tissue injuries, can result from improper lifting and twisting. Other serious injuries can result from improper use of material handling equipment.

3.3.1 *Lifting and Ergonomics*

The most common injuries occur from improper lifting -- see pictures below for examples of some of the proper techniques.

Some of the most common and preventable injuries are strains and sprains. Injuries caused by putting too much tension or pressure on our muscles and other body parts. DPR EHSP Section 09 - Ergonomics & Manual Material Handling addresses subjects such as, lifting techniques, awkward body positions, effects of twisting, unequal lifting, etc., just to name a few, which are intended to help eliminate undue strain. Workers should become familiar with Section 09.

The depictions below are meant to illustrate some of the content within Section 09.

Good Lifting Practices

Keep load close to your body	Lift comfortably with/without a straight back & use your legs
	
For heavy or awkward loads use a team lift	Avoid lifting by using equipment
	

What Not to Do

Don't bend or reach out of your power zone



Keep center of gravity and avoid twisting or jerking motions with a load



3.4 **Housekeeping**

3.4.1 **Access, Egress, and Fire Exits**

Aisles, walkways, stairways, and doorways should be kept free of clutter and other obstructions. Temporary storage of boxes or other items should be kept out of walk paths. Fire exits should be clearly marked. Stairways should not be used for storage of materials, temporary or otherwise.

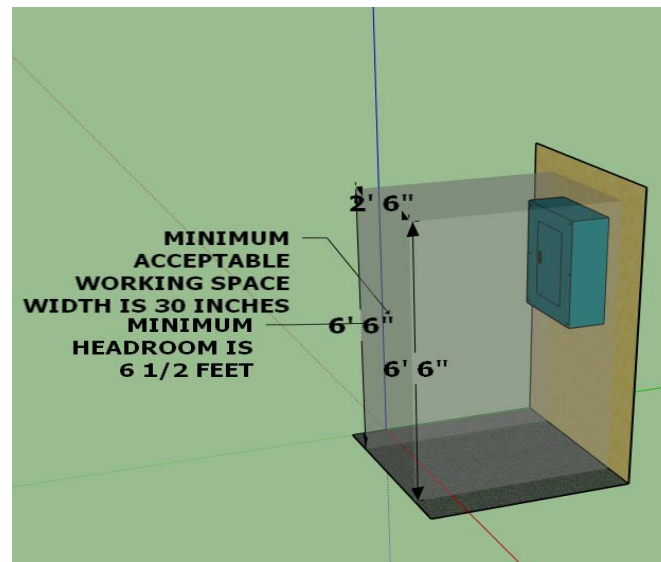
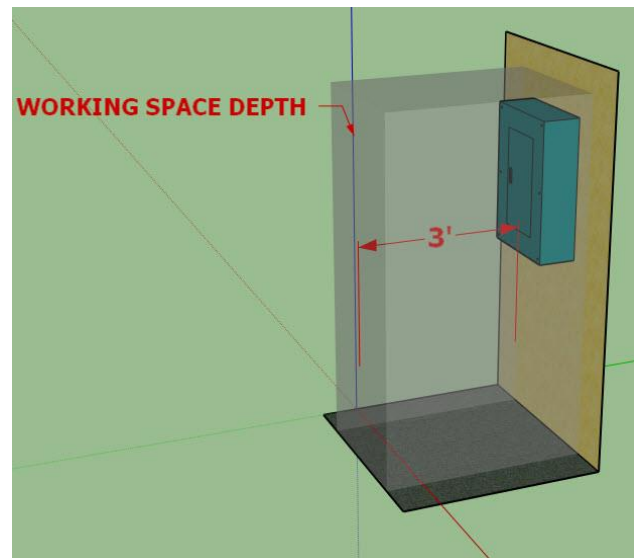
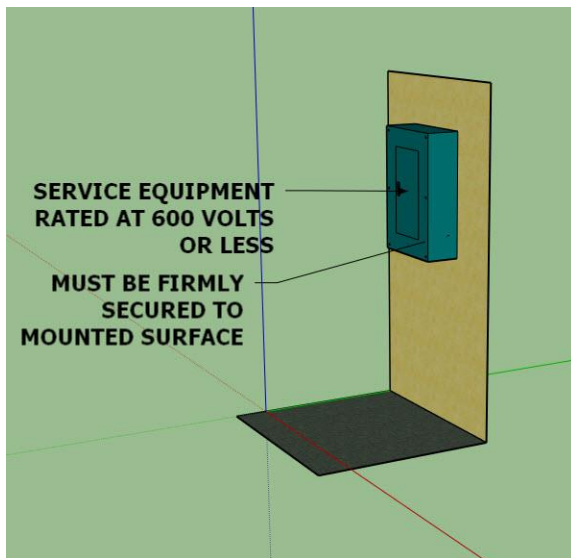
Exit doors and panic hardware should be kept in good working order. Required exits should be kept unlocked during business hours. Except as specifically allowed by local codes, two exit routes should be provided, sized, and designed per local building codes. Routes should be clearly marked.

Reference: 29 CFR 1910.36 & 37 – Maintenance, safeguards, and operational features for exit routes.

3.4.2 **Electrical Panel Clearance**

Materials should not be stored inside of dedicated electrical rooms. In addition, there are clearance requirements around exposed electrical panels per the National Electric Code (ref. Table 110.26). Best practice would be to clearly identify the space to be kept clear with paint on the floor. Electrical panels should be clearly marked with signage depicting the hazard, i.e. **DANGER – ELECTRICAL SHOCK HAZARD**.

Electrical Panel Considerations



3.4.3 Public Access and Common Areas

Regarding areas where the public has the right of entry, including outdoor entry areas, counter and display areas, restrooms, and paths of travel between these areas, vigilance should be given to maintaining unobstructed, properly sized, clean, clear, and dry walk paths and surfaces. Consideration should be given to the American with Disabilities Act standards, as applicable.

4.0 Deliveries

4.1 Motor Vehicles and Trailers

Reference: 49CFR390.5 – Federal Motor Carrier Safety Regulations; Title 49 Code of federal regulations – Transportation – Part 392 – Driving of commercial motor vehicles.

A commercial motor vehicle (CMV) is defined as:

- Has a gross combination vehicle weight of 10,001 pounds or more.
- Is designed or used to transport more than 8 passengers (including the driver) for compensation, or

- Is designed or used to transport more than 15 passengers (including the driver) without compensation, or
- Is used to transport hazardous material as defined in 49CFR5103 and in sufficient quantity as to require placarding.

A commercial driver's license is required to operate a commercial vehicle on the highway.

Whether operating a commercial vehicle, making a delivery in a pickup truck, driving, loading, unloading, accessing construction sites, or even climbing into a truck bed, there are hazards and risks. It is important to recognize these hazards in order to address them.

- Drivers should inspect their vehicle(s) daily, at the beginning of their shift, including pick-up truck deliveries. If deficiencies are found, they should be noted and submitted to the Branch Manager. Unsafe vehicles should not be driven.
- Trucks and trailers should be kept free of loose debris (rocks, dirt, sand, etc.). A broom should be available when delivering products that may spill.
- Reflective tape should be intact and kept clean.
- When parking in the public right of way for a delivery, cones should be strategically placed to alert other drivers of your presence and activities.
- Know the weight capacity of the vehicle and the weight of the cargo being hauled. Do not exceed the capacity of the vehicle or trailer.
- Know the capacity of the truck bed and do not exceed the capacity with the combined weight of equipment and/or material
- Get help lifting heavy or awkward objects.
- Be aware of, and avoid, the pinch points associated with lift gates.
- Gloves should be worn while loading and unloading.
- Be aware of slip hazards on wet surfaces.
- When (un)loading on a public right of way, additional traffic control measures may be necessary.

When accessing a project site, it is important to adhere to the site-specific safety rules. Be aware of designated drop off areas.

4.2 Weight & Vehicle Capacity

The gross vehicle weight rating (GVWR) is the total of the curbside weight of the vehicle plus the cargo, people, and potential tongue weight of a trailer. You can think of it as the overall weight limit for the vehicle, including the vehicle itself. It is important to be familiar with the GVWR, and the weight of the vehicle to determine the maximum load carrying capacity.

Note: A vehicle carrying a load handles differently than one that is empty. Steering and maneuverability may feel sluggish, stopping distances may be longer. Do not load a vehicle beyond its' capacity. Know the weight of your cargo.

5.0 Load Securement

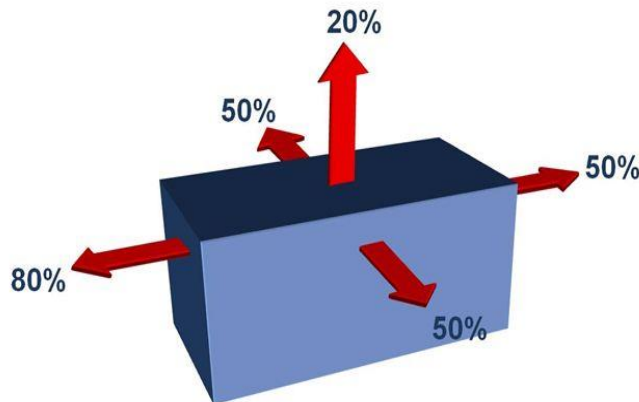
Prior to making any delivery, OES employees should first be trained on load securement

It is a sinking feeling when making a delivery and discovering the merchandise is no longer in the truck. An improperly secured load can produce catastrophic results. It is the responsibility of the driver to verify and double check that equipment, material, and other goods are properly tied down and secured, and that the vehicle is not overloaded.

5.1 Forces & Load Shift Prevention

Unexpected events occur on the road that may require evasive maneuvers. While the expectation is to drive at an appropriate speed for traffic conditions, allow enough space to stop, and look down the road for changing conditions, unforeseen conditions can come up. In this event, if a load is not secured properly, a shift may occur which can result in damage, load loss, or worse.

The forces that may be imparted on a load are shown below. Securement systems should be chosen and implemented to accommodate these reactions.



- Forward force: 80% of cargo weight (braking and sudden stops)
- Rearward force: 50% of cargo weight (accelerating, climbing hills, or shifting)
- Sideways force: 50% of cargo weight (turning or changing lanes)
- Upward force: 20% of cargo weight (bumps and cresting hills)

If the vehicle or trailer is equipped with a header board and it is intended to provide the securement in the forward direction, it should be adequately sized to block the entire load and be of sufficient strength to do so.

5.2 Loose Objects

It is all too common to throw smaller objects into the bed of a stake-bed truck or in the cab of the vehicle for convenience. In a highway/vehicle accident, these objects may become projectiles and cause serious injury to the driver or others. Care and consideration should be given to containing objects which cannot be tied down independently.

5.3 Securement System(s)

A securement system can be broken down into three basic elements, vehicle structure, securing devices, and blocking & bracing equipment. Each of these elements should be considered independently and as a working assembly that provides the necessary strength, durability, and flexibility for the cargo.

For example, chains and binders might be the right choice for tying down a tractor, but may not have the flexibility needed to hold down a stack of drywall without damage to the product.

5.4 Working Load Limit

The (WLL) is the maximum load which should ever be applied to a securing device, in a straight line pull. Side loading should be avoided and reduces the actual WLL of the system. WLL is NOT the proof load or breaking strength. When combining securing devices, the WLL is determined by the device with the lowest working load limit, including the anchor points. Remember the saying, a chain is only as strong as its' weakest link.

Individual securing devices should be labeled with their WLL. In the event the label is missing or is illegible, tables may be utilized to determine the working load limit, see below for an example.

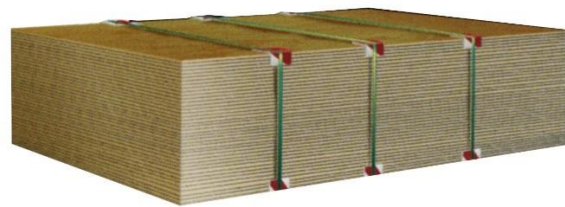
Reference: 49CFR393.108 – Working load limit tables

[Working Load Limits (WLL), Chain]

Size mm (inches)	WLL in kg (pounds)				
	Grade 30 proof coil	Grade 43 high test	Grade 70 transport	Grade 80 alloy	Grade 100 alloy
1. 7 (1/4)	580 (1,300)	1,180 (2,600)	1,430 (3,150)	1,570 (3,500)	1,950 (4,300)
2. 8 (5/16)	860 (1,900)	1,770 (3,900)	2,130 (4,700)	2,000 (4,500)	2,600 (5,700)
3. 10 (3/8)	1,200 (2,650)	2,450 (5,400)	2,990 (6,600)	3,200 (7,100)	4,000 (8,800)
4. 11 (7/16)	1,680 (3,700)	3,270 (7,200)	3,970 (8,750)		
5. 13 (1/2)	2,030 (4,500)	4,170 (9,200)	5,130 (11,300)	5,400 (12,000)	6,800 (15,000)
6. 16 (5/8)	3,130 (6,900)	5,910 (13,000)	7,170 (15,800)	8,200 (18,100)	10,300 (22,600)
Chain Mark Examples:					
Example 1	3	4	7	8	10
Example 2	30	43	70	80	100
Example 3	300	430	700	800	1000

Chains, straps, binders, ratchets, anchor points, etc., should be inspected daily and prior to use.

Edge protectors should be used to protect straps when securing loads that may cause damage.



Palletized material and stacks of material are typically secured with web straps, placed over the load, and tightened down. Verify there is an adequate number of straps for both the weight and the size of the load being carried. A single twist in a web strap is acceptable to prevent wind whipping, which gives the appearance of a loose strap.

5.4 Guidelines for number of straps to use:

When the cargo can't be blocked or positioned to prevent movement in the forward direction; the following guidelines at a minimum should be followed:

- Use one tie-down for articles five feet long or less and 1100 pounds or less
- Use two tie-downs if the cargo is five feet long or less and more than 1100 pounds
- Use two tie-downs if the article is longer than five feet but no longer than ten feet regardless of weight
- Use one tie-down for every ten linear feet or fraction thereof for cargo greater than ten feet (49 CFR part 393.100)

5.5 Anchor Points

Anchor points are typically part of the truck/trailer structure and may be overlooked when analyzing the securement system. Bolt on D-rings for example may not have the strength needed to tie down a heavy or large piece of equipment. Be familiar with the load ratings of the anchor points that are chosen and the potential forces that may be imparted to them.

When anchoring to a piece of equipment directly, it is best to use the manufacturers intended anchor points. If you are unsure of the capacity of a particular anchor point, verify with the manufacturers recommendations.



5.6 Weatherization

When protecting cargo from the elements, with tarps, shrink wrap, or other coverings, these components require consideration for tie down(s) themselves. Covers can become wind whipped and may need multiple points of securement.

Note: Shrink wrapped cargo is not considered a secured load.

Section 27

Work Environment & Employee Health

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Work Environment and Employee Health

1.0 Introduction

This program is to provide guidance for protecting employees from hazards of high heat and cold conditions; venomous snakes, bees, other insects and animals as well as prevention and response for Infectious Disease/Pandemic outbreak. The intent is to provide information on engineering, administrative, safe work practices and PPE controls to eliminate and or reduce the risk.

Being uncomfortable is not the major problem with working in high and low temperatures and humidity. Workers who are suddenly exposed to working in a hot or cold environment face additional and generally avoidable hazards to their safety and health.

This program applies to all outdoor and indoor places of employment at those times when the environmental risk factors including heat or cold illness are present. It also applies to preventing bites and stings from insects and animals.

Reference: Heat Illness Prevention Standard Title 8 California Code of Regulations 3395, Sanitation CFR 1926.51; National Institute for Occupational Safety and Health (NIOSH). Other applicable Standards, guidelines, and regulations may apply.

Note: DPR has recognized Heat Illness as a serious and preventable condition and has selected the Cal/OSHA standards as a means of best practice method for controlling for heat illness risks.

Definitions

Acclimatization: Temporary adaptation of the body to work in the heat, that occurs gradually when a person is exposed. Acclimatization peaks in most people within four to fourteen days of regular work for at least two hours per day in heat.

Heat Illness: A serious medical condition resulting from the body's inability to cope with a particular heat load, and includes heat cramps, heat exhaustion, heat syncope and heat stroke.

Environmental Risk Factors for heat illness: Working conditions that create the possibility that heat illness could occur, may include: air temperature, relative humidity, radiant heat from the sun and other sources, conductive heat sources, air movement, workload (severity and duration), protective clothing and personal protective equipment worn by employees.

Landscaping: Providing landscape care and maintenance services and/or installing trees, shrubs, plants, lawns, or gardens, or providing these services in conjunction with the design of landscape plans and/or the construction (i.e., installation) of walkways, retaining walls, decks, fences, ponds, and similar structures, except for employment by an employer who operates a fixed establishment where the work is to be performed and where drinking water is plumbed.

Pandemic: An outbreak of disease that occurs over a wide geographic area and affects an exceptionally high proportion of the population.

Personal Risk Factors for heat illness: Factors such as an individual's age, degree of acclimatization, health, water consumption, alcohol consumption, caffeine consumption, and use of prescription medications that affect the body's water retention or other physiological response to heat.

Shade: Blockage of direct sunlight. One indicator that blockage is sufficient is when objects do not cast a shadow in the area of blocked sunlight. Shade is not adequate when heat in the area of the shade defeats the purpose of shade, which is to allow the body to cool. For example, a car sitting in the sun does not provide acceptable shade to a person inside it, unless the car is running with air conditioning. Shade may be provided by any natural or artificial means that does not expose employees to unsafe or unhealthy conditions and that does not deter or discourage access or use.

Temperature: The dry bulb temperature, in degrees Fahrenheit, is obtainable by using a thermometer to measure the outdoor temperature in an area where there is no shade. While the temperature measurement

must be taken in an area with full sunlight, the bulb or sensor of the thermometer should be shielded while taking the measurement, e.g., with the hand or some other object, from direct contact by sunlight.

Note: All contractors should identify their designated competent person who, prior to each workday, will be responsible for monitoring the weather using www.nws.noaa.gov or with the aid of a simple thermometer at the worksite.

2.0 Heat Illness

All contractors with any outdoor places of work must comply with the Heat Illness Prevention Standard Title 8 California Code of Regulations 3395. This sub-section applies to the control of risk of occurrence of heat illness and other environmental factors. This sub-section applies to all outdoor places of employment at those times when the environmental risk factors for heat illness are present. This sub-section also applies to transportation or delivery of construction materials or other heavy materials (e.g. furniture, lumber, freight, cargo, cabinets, industrial or commercial materials), except for employment that consists of operating an air-conditioned vehicle and does not include loading or unloading.

2.1 Provisions of Water

Employees should have access to safe drinking water that includes but not limited; that it be fresh, pure, suitably cool and provided to employees free of charge. The water should be located as close as practicable to areas where employees are working. The following items must be addressed:

- An adequate supply of potable water should be provided in all places of employment
- Portable containers used to dispense drinking water should be:
 - equipped with a faucet or drinking fountain
 - capable of being tightly closed
 - otherwise designed, constructed and serviced so that sanitary conditions are maintained
- Outlets for non-potable water, such as water for industrial or firefighting purposes only, shall be identified by signs meeting regulatory-requirements, to indicate clearly that the water is unsafe and is not to be used for drinking, washing, or cooking purposes
- There shall be no cross-connection, open or potential, between a system furnishing potable water and a system furnishing non-potable water
- Any container used to store or dispense drinking water should be clearly marked as to the nature of its contents and should not be used for any other purpose
- Water should not be dipped from containers
- Where drinking fountains are not provided, single service cups (to be used but once) should be supplied
- Where single service cups are supplied, a sanitary container for the unused cups and a receptacle for disposing of the used cups should be provide
- Where re-usable containers for individual use are relied upon for compliance with this section, the employer shall ensure the containers are marked to identify the user and maintained in a sanitary condition
- Re-usable containers for individual use and drinking cups shall not be shared or used in common
- Drinking water should be provided in sufficient quantity at the beginning of the work shift to provide one quart per employee per hour for drinking for the entire shift. Employers may begin the shift with smaller quantities of water if they have effective procedures for replenishment during the shift as needed to allow employees to drink one quart or more per hour
- The frequent drinking of water should be encouraged
- Ice is recommended to cool water in high heat
- All contractors should identify the designated person(s) that has been assigned the task of providing drinking water containers to the site

- As part of effective replenishment procedures, the designated person will check the water level as needed to ensure a constant fresh supply is available at all times. When the temperature exceeds 80 degrees F the water levels should be checked more frequently

2.2 Shade

In California, shade should be present when the temperature exceeds 80 degrees Fahrenheit. When the outdoor temperature in the work area exceeds 80 degrees Fahrenheit, the employer should have and maintain one or more areas with shade at all times while employees are present that is either open to the air or provided with ventilation or cooling. The amount of shade present should be at least enough to accommodate the number of employees on recovery or rest periods, so that they can sit in a normal posture fully in the shade without having to be in physical contact with each other. The shaded area should be located as close as practicable to the areas where employees are working. Subject to the same specifications, the amount of shade present during meal periods should be at least enough to accommodate the number of employees on the lunch or break period who remain onsite. Shade should be available when the temperature does not exceed 80 degrees Fahrenheit. When the outdoor temperature in the work area does not exceed 80 degrees Fahrenheit employers should either provide shade at all times or provide timely access to shade upon an employee's request.

In all other states, shade should be provided as temperatures rise. The shaded area should be located as close as practicable to the areas where employees are working, and workers should be encouraged to use rest/hydration breaks as necessary to prevent signs/symptoms of heat illness.

Employees should be allowed and encouraged to take a preventative cool-down rest in the shade when they feel the need to do so to protect themselves from overheating. Such access to shade should be permitted at all times.

An individual employee who takes a preventative cool-down rest should be:

- Monitored and asked if he or she is experiencing symptoms of heat illness
- Encouraged to remain in the shade
- *Should not* be ordered back to work until any signs or symptoms of heat illness have been abated, but in no event less than 5 minutes in addition to the time needed to access the shade
- If an employee exhibits signs or reports symptoms of heat illness while taking a preventative cool-down rest or during a preventative cool-down rest period, the employer shall provide appropriate first aid or emergency response according to subsection 2.4 of this sub-section.
- Exceptions to the shade requirements above:
- Where the employer can demonstrate that it is infeasible or unsafe to have a shade structure, or otherwise to have shade present on a continuous basis, the employer may utilize alternative procedures for providing access to shade if the alternative procedures provide equivalent protection.
- Except for employers in the agricultural industry, cooling measures other than shade (e.g., use of misting machines) may be provided in lieu of shade if the employer can demonstrate that these measures are at least as effective as shade in allowing employees to cool.
- All contractors should identify the designated person(s) who will ensure that shade structures are opened and placed as close as practical to the workers when the temperature equals or exceeds 80 degrees F.

2.3 High Heat Procedures

In California, the employer should implement high-heat procedures when the temperature equals or exceeds 95 degrees Fahrenheit (all other states are encouraged to implement High Heat Procedures as temperatures and/or humidity rise). These procedures should include the following to the extent practicable:

- Ensuring that effective communication by voice, observation, or electronic means is maintained so that employees at the work site can contact a supervisor when necessary. An electronic device, such

as a cell phone or text messaging device, may be used for this purpose only if reception in the area is reliable.

- Observing employees for alertness and signs or symptoms of heat illness. The employer should ensure effective employee observation/monitoring by implementing one or more of the following;
 - Supervisor or designee observation of 20 or fewer employees, or
 - Mandatory buddy system, or
 - Regular communication with sole employee such as by radio or cellular phone
 - Other effective means of observation
- Designating one or more employees on each worksite as authorized to call for emergency medical services, and allowing other employees to call for emergency services when no designated employee is available
- Reminding employees throughout the work shift to drink plenty of water.
- Pre-shift meetings before the commencement of work to review the high heat procedures, encourage employees to drink plenty of water and remind employees of their right to take a cool-down rest when necessary.
- All contractors should identify the designated person(s) who will ensure High Heat Procedures are put into place when the temperature equals or exceeds 95 degrees F.

2.4 *Emergency Response Procedures*

The employer should implement effective emergency response procedures including:

- Ensure that effective communication by voice, observation or electronic means is maintained so that employees at the work site can contact a supervisor or emergency medical services when necessary. An electronic device, such as a cell phone or text messaging device, may be used for this purpose only if reception in the area is reliable. If an electronic device will not furnish reliable communication in the work area, the employer will ensure a means of summoning emergency medical services
- Respond to signs and symptoms of possible heat illness including but not limited to first aid measures and how emergency medical services will be provided
 - If a supervisor observes or any employee reports any signs or symptoms of heat illness in any employee, the supervisor should take immediate action and commensurate with the severity of the illness
 - If the signs or symptoms are indicators of severe heat illness (such as, but not limited to, flushed skin, high body temperature, decreased level of consciousness, staggering, vomiting, disorientation, irrational behavior or convulsions), the employer must implement emergency response procedures.
 - An employee exhibiting signs or symptoms of heat illness should be monitored and should not be left alone or sent home without being offered onsite first aid and /or being provided with emergency medical services in accordance with the employer's procedures
- Contact emergency medical services and, if necessary transporting employees to a place where they can be reached by emergency medical provider.
- Ensure that, in the event of an emergency, clear and precise directions to the work site can and will be provided as needed to emergency responders.

2.5 *Acclimatization*

- All employees should be closely observed by a supervisor or designee during a heat wave. For this sub-section only, "heat wave" means any day in which the predicted high temperature for the day will be at least 80-degrees Fahrenheit and at least ten degrees Fahrenheit higher than the average high daily temperature in the preceding five days.
- An employee who has been newly assigned to a high heat area should be closely observed by a supervisor or designee for the first 14 days of the employee's employment.

2.6 Training

Employee training. Effective training in the following topics should be provided to each supervisory and non-supervisory employees before the employee begins work. Training should be provided to any employees whose work could reasonably be anticipated to result in exposure to the risk of heat illness. The following are training topics to covered:

- The environmental and personal risk factors for heat illness, as well as the added burden of heat load on the body caused by exertion, clothing, and personal protective equipment.
- The employer's procedures for complying with the requirements of this standard, including but not limited to, the employer's responsibility to provide water, shade, cool-down rests, and access to first aid as well as the employees' under this standard without retaliation.
- The importance of frequent consumption of small quantities of water, up to 4 cups per hour, when the work environment is hot and employees are likely to be sweating more than usual in the performance of their duties.
- The concept, methods and importance, of acclimatization.
- The common signs and symptoms of heat illness; appropriate first aid and/or emergency responses to the different types of heat illness, and in addition, that heat illness may progress quickly from mild symptoms and signs to serious and life threatening illness
- The importance to employees of immediately reporting to the employer, directly or through the employee's supervisor, symptoms or signs of heat illness in themselves, or in co-workers.
- The employer's procedures for responding to signs or symptoms of possible heat illness, including how emergency medical services will be provided should they become necessary.
- The employer's procedures for contacting emergency medical services, and if necessary, for transporting employees to a point where they can be reached by an emergency medical service provider.
- The employer's procedures for ensuring that, in the event of an emergency, clear and precise directions to the work site can and will be provided as needed to emergency responders. These procedures should include designating a person to be available to ensure that emergency procedures are invoked when appropriate.

Supervisor training. Prior to assignment to supervision of supervising employees performing work that should reasonably be anticipated to result in exposure to the risk of heat illness working in the heat, effective training on the following topics should be provided to the supervisor:

- The information required to be provided by sub-section 2.7 above.
- The procedures the supervisor is to follow to implement the applicable provisions in this sub-section.
- The procedures the supervisor is to follow when an employee exhibits signs or reports symptoms consistent with possible heat illness, including emergency response procedures.
- How to monitor weather reports and how to respond to hot weather advisories.

2.7 Heat Illness Prevention Plan

All contractors shall establish, implement, and maintain, an effective heat illness prevention plan. The plan shall be in writing in both English and the language understood by the majority of the employees and shall be made available at the worksite to employees and to representatives of the Division upon request. The Heat Illness Prevention Plan may be included as part of the employer's Illness and Injury Prevention Program, and shall, at a minimum, contain:

- Procedures for the provision of water and access to shade.
- The high heat procedures referred to in subsection.
- Emergency Response Procedures in accordance with subsection.
- Acclimatization methods and procedures in accordance with subsection.

2.7.1 Heat Related Illness

Heat Stroke

Heat stroke is the most serious heat-related illness. It occurs when the body becomes unable to control its temperature: the body's temperature rises rapidly, the sweating mechanism fails, and the body is unable to cool down. When heat stroke occurs, the body temperature can rise to 106°F or higher within 10 to 15 minutes. Heat stroke can cause death or permanent disability if emergency treatment is not given.

Symptoms of heat stroke:

- Confusion, altered mental status, slurred speech
- Loss of consciousness (coma)
- Hot, dry skin or profuse sweating
- Seizures
- Very high body temperature
- Fatal if treatment delayed

First Aid steps to be taken to treat a worker with heat stroke:

- Call 911 for emergency medical care.
- Stay with worker until emergency medical services arrive.
- Move the worker to a shaded, cool area and remove outer clothing.
- Cool the worker quickly with a cold water or ice bath if possible; wet the skin, place cold wet-cloths on skin, or soak clothing with cool water.
- Circulate the air around the worker to speed cooling.
- Place cold wet-cloths or ice on head, neck, armpits, and groin; or soak the clothing with cool water.

Heat Exhaustion

Heat exhaustion is the body's response to an excessive loss of the water and salt, usually through excessive sweating. Workers most prone to heat exhaustion are those that are elderly, have high blood pressure, and those working in a hot environment.

Symptoms of heat exhaustion:

- Headache
- Nausea
- Dizziness
- Weakness
- Irritability
- Thirst
- Heavy sweating
- Elevated body temperature
- Decreased urine output

First Aid steps to be taken to treat a worker with heat exhaustion:

- Take worker to a clinic or emergency room for medical evaluation and treatment.
- If medical care is unavailable, call 911.
- Someone should stay with worker until help arrives.
- Remove worker from hot area and give liquids to drink.
- Remove unnecessary clothing, including shoes and socks.
- Cool the worker with cold compresses or have the worker wash head, face, and neck with cold water.
- Encourage frequent sips of cool water.

Rhabdomyolysis

Rhabdomyolysis is a medical condition associated with heat stress and prolonged physical exertion, resulting in the rapid breakdown, rupture, and death of muscle. When muscle tissue dies, electrolytes and large proteins are released into the bloodstream that can cause irregular heart rhythms and seizures, and damage the kidneys.

Symptoms of rhabdomyolysis include:

- Muscle cramps/pain
- Abnormally dark (tea or cola colored) urine
- Weakness
- Exercise intolerance
- Asymptomatic

First Aid steps to be taken to treat a worker with rhabdomyolysis:

- Stop activity.
- Increase hydration (water preferred).
- Seek immediate care at the nearest medical facility.
- Ask to be checked for rhabdomyolysis (i.e., blood sample analyzed for creatine kinase).

Heat Syncope

- Heat syncope is a fainting (syncope) episode or dizziness that usually occurs with prolonged standing or sudden rising from a sitting or lying position. Factors that may contribute to heat syncope include dehydration and lack of acclimatization.

Symptoms of heat syncope include:

- Fainting (short duration)
- Dizziness
- Light-headedness during prolonged standing or suddenly rising from a sitting or lying position

First Aid steps to be taken to treat a worker with heat syncope:

- Sit or lie down in a cool place.
- Slowly drink water, clear juice, or a sports drink.

Heat Cramps

Heat cramps usually affect workers who sweat a lot during strenuous activity. This sweating depletes the body's salt and moisture levels. Low salt levels in muscles causes painful cramps. Heat cramps may also be a symptom of heat exhaustion.

Symptoms of heat cramps:

- Muscle cramps, pain, or spasms in the abdomen, arms, or legs

First Aid steps to be taken to treat a worker with heat cramps:

- Drink water and have a snack and/or carbohydrate-electrolyte replacement liquid (e.g., sports drinks) every 15 to 20 minutes.
- Avoid salt tablets.
- Get medical help if the worker has heart problems, is on a low sodium diet, or if cramps do not subside within 1 hour.

Heat Rash

Heat rash is a skin irritation caused by excessive sweating during hot, humid weather.

Symptoms heat rash include:

- Looks like red cluster of pimples or small blisters
- Usually appears on the neck, upper chest, groin, under the breasts, and in elbow creases

First Aid steps to be taken to treat a worker with heat rash:

- Keep rash area dry.
- Powder may be applied to increase comfort.
- Ointments and creams should not be used.

In the event that persons exhibit any of these symptoms notify DPR's project management immediately. In an emergency employees are allowed to call emergency services without contacting a supervisor first. Even early symptoms may indicate serious heat exposure. If first aid trained personnel are not immediately available on-site to make an assessment and workers show of the signs and symptoms of heat illness, you should notify DPR Project management and refer to and follow site specific emergency response procedures. Regardless of the worker's protests, no employee with any of the symptoms of possible serious heat illness should be sent home or left unattended without medical assessment and authorization. For Emergency information please review Emergency Procedures Section 08.

2.8 Site Specific Procedures

1. Identify the designated person(s) that has been assigned the applicable task(s) (e.g. supervisor, foreman, safety coordinator, crew leader). Please review and complete Site Specific Procedures Checklist in the Forms Section at the end of the EHSP.
2. Provide specific details required to carry out the task and ensure that the task is accomplished successfully (e.g. how many water containers/shade structures, of what size, distance to placement, frequency of water-level replenishment/weather-tracking/water breaks/reminders, etc.). Specify how these procedures will be communicated to your employees and in particular to the persons assigned these responsibilities (e.g. via training, meeting), and how it will be ascertained that these company instructions and procedures are followed.
3. Each project should designate persons to implement the provisions in this Heat Illness Prevention Plan and may use the Site Specific Procedures Checklist in the Forms Section at the end of the EHSP.

2.9 Cleaning a Water Storage Container Before Use

Water containers should be cleaned daily and sanitized weekly, at a minimum. Before filling with safe water, use these steps to clean and sanitize water storage containers:

Daily cleaning:

- Wash the storage container with a solution of dishwashing soap and water or baking soda and water.
- Rinse completely with clean water.

Weekly sanitization:

- Sanitize the container with a solution made by mixing 1 teaspoon of unscented liquid household chlorine bleach in one quart of water.
- Cover the container and shake it well so that the sanitizing bleach solution touches all inside surfaces of the container.
- Wait at least 30 seconds and then pour the sanitizing solution out of the container.
- Let the empty sanitized container air-dry before use OR rinse the empty container with clean, safe water that is available already.

3.0 Cold Emergencies

3.1 Hypothermia

Hypothermia is the lowering of the entire body's core temperature. Hypothermia is most likely at very cold temperatures, but it can occur even at cool temperatures (above 40°F), if a person becomes chilled from rain, sweat, or submersion in cold water.

- Symptoms: shivering, confusion, irrationality, lethargy or drowsiness; skin may be pale and cool, breathing may slow and the affected individual may have changing levels of responsiveness
- Preventative measures:
 - Dress in layers that can be added and removed as work progresses
 - Choose a coat with wind and/or waterproof outer layer
 - Wear a hat; most body heat is lost from the head
 - Consume warm drinks and high-energy food
- Treatment:
 - Call 911 immediately in an emergency; otherwise seek medical assistance as soon as possible.
 - Move the person to a warm, dry area.
 - Remove wet clothes and replace with dry clothes, cover the body (including the head and neck) with layers of blankets; and with a vapor barrier (e.g. tarp, garbage bag). Do not cover the face.
 - If medical help is more than 30 minutes away:
 - Give warm sweetened drinks if alert (no alcohol), to help increase the body temperature. Never try to give a drink to an unconscious person.
 - Place warm bottles or hot packs in armpits, sides of chest, and groin. Call 911 for additional rewarming instructions.
 - If a person is not breathing or has no pulse:
 - Call 911 for emergency medical assistance immediately.
 - Treat the worker as per instructions for hypothermia, but be very careful and do not try to give an unconscious person fluids.
 - Check him/her for signs of breathing and for a pulse. Check for 60 seconds.
 - If after 60 seconds the affected worker is not breathing and does not have a pulse, trained workers may start rescue breaths for 3 minutes.
 - Recheck for breathing and pulse, check for 60 seconds.
 - If the worker is still not breathing and has no pulse, continue rescue breathing.
 - Only start chest compressions per the direction of the 911 operator or emergency medical services*
 - Reassess patient's physical status periodically.
 - *Chest compression are recommended only if the patient will not receive medical care within 3 hours.

3.2 Frostbite

Frostbite is an injury to the body that is caused by freezing of the skin and underlying tissues. The lower the temperature, the more quickly frostbite will occur. Frostbite typically affects the extremities, particularly the feet and hands.

- Symptoms: Reddened skin develops gray/white patches, numbness in the affected part, affected part may feel firm or hard, blisters may occur, in severe cases
- Preventative measures
 - Most common areas for frostbite are exposed skin on head, face, hands or feet. Keep cloths dry and dress in layers
 - Wind chill increases the risk of frostbite, prevent exposure to high winds whenever possible

- Treatment:
 - Move exposed person to a warm environment
 - Do not rub the affected area to warm it because this action can cause more damage.
 - Do not apply snow/water. Do not break blisters.
 - Loosely cover and protect the area from contact.
 - Do not try to rewarm the frostbitten area before getting medical help; for example, do not place in warm water. If a frostbitten area is rewarmed and gets frozen again, more tissue damage will occur. It is safer for the frostbitten area to be rewarmed by medical professionals.
 - Give warm sweetened drinks, if the person is alert.
 - Seek medical attention

3.3 Trench Foot

Trench Foot or immersion foot is caused by prolonged exposure to wet and cold temperatures. It can occur at temperatures as high as 60°F if the feet are constantly wet. Non-freezing injury occurs because wet feet lose heat 25-times faster than dry feet. To prevent heat loss, the body constricts the blood vessels to shut down circulation in the feet. The skin tissue begins to die because of a lack of oxygen and nutrients and due to the buildup of toxic products.

- Symptoms: Redness of the skin, swelling, numbness, blisters
- Preventative measures: Wear insulated and waterproof boots (or other footwear). Where possible, keep extra clothing handy in case you get wet and need to change.
- Treatment:
 - Call 911 immediately in an emergency; otherwise seek medical assistance as soon as possible.
 - Remove the shoes, or boots, and wet socks.
 - Dry the feet.

4.0 Portable Electronics

The use of iPods, MP3 players, Bluetooth technology, radios and other such portable electronic devices etc. are prohibited within the work area or areas where use of such devices have the potential to interfere with surrounding activities, awareness of surroundings and/or communication between individuals on the project. Cell phones and Bluetooth technology may be used for business related communication or in case of emergency, but not for entertainment purposes. At no time should cell phones or Bluetooth technology be used by the operator of mobile equipment when the equipment is being operated unless required by another section. See Section 5 Cranes (10.0)

5.0 Animals and Wildlife

5.1 Snake Bites

- Avoiding Snakebites:
 - Know the area you are working in and if snakes are likely.
 - When walking in areas known to have venomous snakes keep your eye out and make noise to warn the snakes of your presence.
 - Leave snakes alone. Many people are bitten because they try to kill a snake or get a closer look at it
 - Stay out of tall grass unless you wear thick leather boots, and remain on paths as much as possible
 - Keep hands and feet out of areas you can't see. Don't pick up rocks, lumber or firewood unless you are out of a snake's striking distance. (A snake can strike half its length, Hardy says.)
 - Be cautious and alert when climbing rocks
- Treatment:
 - If bitten seek medical attention immediately from a facility capable of rendering advanced care, including possible anti-venom administration and airway support.

- Briefly attempt to identify the snake (especially, note the color pattern). If possible, take a digital photo of the snake from a safe distance. Do not attempt to catch the snake, efforts to catch or kill the animal can result in wasted time and further bites.
- Rapidly apply a compressive bandage (e.g., elastic bandage, crepe bandage, torn clothing) to the bitten extremity, starting distally and progressing to encompass the entire limb. Wrap it as tightly as one would wrap a severe sprained ankle. Then, splint the extremity and, if possible, keep it at approximately heart level. Do not elevate the area. This technique may significantly delay systemic absorption of elapid venoms, including coral snake venom. Research suggests that, in a simulated snakebite scenario, people tend to underestimate how tight they can get the bandage for the technique to be effective.

5.2 Animal Bites & Stings

- Report all animal bites to local animal control or police; follow local procedures for rabies
- Suspect rabies in cases of unprovoked attacks of strange acting animals or wild animals
- Do not attempt to catch animals of unprovoked attacks
- Treatment:
 - Wash affected area thoroughly with soap and water
 - Run water over wound for at least five minutes
 - Control bleeding
 - Cover wound with sterile dressing and bandage
 - Seek immediate medical attention

5.3 Spider Bites

- Only the venom of black widow and brown recluse spider is serious or potentially fatal
 - Black widow spiders are shiny black and have red hourglass-shaped marking on underside of abdomen. If bitten, the following symptoms may occur:
 - Pain at bite site
 - Red skin at site
 - After 15 minutes to hours: sweating, nausea, stomach and muscle cramps, increased pain at site, dizziness or weakness, difficulty breathing
 - Brown recluse spiders are usually light brown in color with a violin-shaped marking on its back
 - Stinging sensation at site
 - Over 8 to 48 hours: increasing pain, blistering at site, fever, chills, nausea or vomiting, joint pain, open sore at site
- Treatment:
 - *If the victim has difficulty breathing, call 911 and be prepared to administer basic life support*
 - *Keep bite area below level of the heart*
 - *Wash affected area thoroughly with soap and water*
 - *Put ice or cold pack on bite area*
 - *Seek medical attention immediately if black widow or brown recluse bites are suspected*

5.4 Bee and Wasp Stings

- Insect stings are not poisonous but can cause life-threatening allergic reactions in affected workers with severe allergies
- Signs & symptoms:
 - *Pain, burning or itching at sting site*
 - *Redness or swelling*
 - *Difficulty breathing*
- Treatment:
 - *Remove stinger from skin by scraping it away gently with credit card or knife blade (call 911 for known allergy to stings)*

- Wash area with soap and water
- Put ice pack on sting site
- Watch victim for 30 minutes for any signs or symptoms of allergic reaction
- If symptoms occur, call 911 and treat for shock

5.5 Scorpion Stings

- Some types in the American Southwest are dangerous
- Antivenin may be available in some areas
- Signs & symptoms:
 - Scorpions sting with tail
 - Severe burning pain at sting site (later numbness, tingling)
 - Nausea or vomiting
 - Difficulty swallowing
 - Possible convulsions or coma
- Treatment:
 - Contact a qualified health care provider or poison control center for advice and medical instructions.
 - Ice may be applied directly to the sting site (never submerge the affected limb in ice water).
 - Remain relaxed and calm.
 - Do not take any sedatives.
 - Capture the scorpion for identification if it is possible to do so safely.

5.6 Fire Ants

Fire ants bite and sting. They are aggressive when stinging and inject venom, which causes a burning sensation. Red bumps form at the sting, and within a day or two they become white fluid-filled pustules.

- Employers should protect their workers from fire ants by training them about:
 - Their risk of exposure
 - How to identify fire ants and their nests
 - How to prevent exposure
 - What to do if they are bitten or stung
- Workers should take the following steps to prevent fire ant stings and bites:
 - Do not disturb or stand on or near ant mounds.
 - Be careful when lifting items (including animal carcasses) off the ground, as they may be covered in ants.
 - Fire ants may also be found on trees or in water, so always look over the area before starting to work.
- Treatment:
 - Workers with a history of severe allergic reactions to insect bites or stings should consider carrying an epinephrine auto injector (EpiPen) and should wear a medical identification bracelet or necklace stating their allergy.
- Workers should take the following steps if they are stung or bitten by fire ants:
 - Rub off ants briskly, as they will attach to the skin with their jaws.
- Antihistamines may help.
 - Follow directions on packaging.
 - Drowsiness may occur.
- Take the worker to an emergency medical facility immediately if a sting causes severe chest pain, nausea, severe sweating, loss of breath, serious swelling, or slurred speech.

5.7 Mosquito-Borne Diseases

Workers are at risk when they are working where mosquitoes are biting. Different species of mosquitoes are found in different geographic locations, are most active at different times, and spread different diseases. The risk to workers varies with:

- Where they are working in the United States or elsewhere
- Type of habitat at the work site
- Season
- Time of day.
 - Mosquitoes breed in standing water. Employers should decrease mosquito populations at worksites by
 - Removing, turning over, covering, or storing equipment
 - Removing debris from ditches
 - Filling in ruts and other areas that collect standing water
 - Removing tires, buckets, bottles, and barrels that collect water
 - Placing drain holes in containers that collect water and cannot be discarded.
- Employers can keep mosquitoes out of indoor worksites by ensuring that doors and windows have screens and are kept closed when possible.
- Employers should also protect workers by providing training about:
 - The risk of exposure to mosquitoes
 - How they can protect themselves from mosquito bites
 - Symptoms of diseases spread by mosquitoes
 - Protective clothing, including long-sleeved shirts and long pants
 - EPA-registered insect repellents to use on exposed skin and clothing
- Workers can reduce their risk of mosquito bites while working outdoors, by:
 - Helping to reduce sources of standing water
 - Wearing long-sleeved shirts and long pants
 - Using EPA-registered insect repellents on exposed skin and clothing, as directed on the product labels.
- Workers who develop symptoms of a mosquito-borne disease should report this promptly to their supervisor and get medical attention.

6.0 Poisonous Plants

Poison ivy, poison oak, and poison sumac release an oil, urushiol, when the leaf or other plant parts are bruised, damaged, or burned. When the oil gets on the skin an allergic reaction, referred to as contact dermatitis, occurs in most exposed people as an itchy red rash with bumps or blisters. Employers should protect their workers from poisonous plants by training them about:

- Their risk of exposure to poisonous plants
- How to identify poisonous plants
- How to prevent exposure to poisonous plants
- What they should do if they are exposed to poisonous plants
- Workers can prevent contact with poisonous plants by taking these steps:
 - Wear long sleeves, long pants, boots, and gloves.
 - Wash exposed clothing separately in hot water with detergent.
- Barrier skin creams, such as a lotion containing bentoquatam, may offer some protection before contact.
 - Barrier creams should be washed off and reapplied twice a day.
 - After use, clean tools with rubbing alcohol (isopropanol or isopropyl alcohol) or soap and lots of water. Urushiol can remain active on the surface of objects for up to 5 years.
 - Wear disposable gloves during this process.

- Do not burn plants that may be poison ivy, poison oak, or poison sumac. Inhaling smoke from burning plants can cause severe allergic respiratory problems.
- Treatment:
 - Immediately rinse skin with rubbing alcohol, specialized poison plant washes, degreasing soap (such as dishwashing soap) or detergent, and lots of water.
 - Rinse frequently so that wash solutions do not dry on the skin and further spread the urushiol.
 - Scrub under nails with a brush.
 - Apply wet compresses, calamine lotion, or hydrocortisone cream to the skin to reduce itching and blistering.
 - Follow the directions on any creams and lotions. Do not apply to broken skin, such as open blisters.
 - Oatmeal baths may relieve itching.
 - Drowsiness may occur.
 - In severe cases or if the rash is on the face or genitals, seek professional medical attention.
 - Call 911 or go to a hospital emergency room if the worker is suffering a severe allergic reaction, such as swelling or difficulty breathing, or has had a severe reaction in the past.

7.0 Infectious Disease/Pandemic Prevention, Preparedness and Response

DPR Construction will take proactive steps to protect the workplace (both offices and project worksites) in the event of an infectious disease/pandemic outbreak such as COVID-19. It is the goal of DPR Construction during any such time period to strive to operate effectively and ensure that all essential services are continuously provided and that employees, subcontractors, clients, and other stakeholders are safe within the workplace and areas occupied or controlled by DPR Construction.

DPR Construction is committed to identifying risk levels in the workplace and providing authoritative information about the nature and spread of infectious diseases, including symptoms and signs to watch for, as well as required steps to be taken in the event of an illness or outbreak. DPR Construction will monitor and implement guidance based upon recommendations of the U.S. Centers for Disease Control and Prevention (CDC) and/or the World Health Organization (WHO). DPR will implement engineering, administrative and work practice controls and personal protective equipment (PPE) as well as considerations for doing so as much as practicable and appropriate.

Unless otherwise notified, our normal attendance and leave policies will remain in place. Individuals who believe they may face particular challenges reporting to work during an infectious disease/pandemic outbreak should take steps to develop any necessary contingency plans. For example, employees might want to arrange for alternative sources of child-care should schools close and/or speak with supervisors about the potential to work from home temporarily or on an alternative work schedule.

For the latest detailed information and guidance on Infectious disease/pandemic prevention and response provided by DPR Construction see

<https://dprprod.sharepoint.com/sites/toolbox/communicationsbrand/pages/coronavirus.aspx>

(Hyperlink for DPR Employee Access Only)

Subcontractors or other Non-DPR stakeholders, please request the latest guidance from the DPR Project Team.

7.1 Preventing the Spread of Infection in the Workplace

DPR Construction will work to identify risks ensure a clean workplace, including the regular cleaning and disinfecting of objects and areas that are frequently used, such as bathrooms, break rooms, conference rooms, door handles, light switches and railings. Only EPA approved cleaning agents will be used for cleaning and disinfecting per the manufacturer's recommendations. A committee, working group or task force may

be designated to monitor and coordinate events around an infectious disease/pandemic outbreak, as well as to create work rules that could be implemented to promote safety through infection control.

For a list of EPA approved Disinfectants see: <https://www.epa.gov/pesticide-registration/list-n-disinfectants-use-against-sars-cov-2>

7.2 Implement Workplace Controls

DPR may implement to the extent practicable, workplace controls that follow a generally accepted hierarchy of controls in an attempt to systematically remove hazards associated with Infectious Disease/Pandemic outbreaks rather than relying solely on workers to reduce their exposure. During an Infectious Disease/Pandemic outbreak, when it may not be possible to eliminate the hazard, the most effective protection measures are (listed from most effective to least effective): engineering controls, administrative controls, safe work practices (a type of administrative control), and PPE. There are advantages and disadvantages to each type of control measure when considering the ease of implementation, effectiveness, and cost. In most cases, a combination of control measures will be necessary to protect workers from exposure to Infectious Disease/Pandemic outbreaks.

In addition to the types of workplace controls discussed below, CDC may offer additional guidance for infection prevention strategies. Employers should check the CDC website regularly for updates about workplace controls.

Engineering Controls

Engineering controls involve isolating employees from work related hazards. In workplaces where they are appropriate, these types of controls reduce exposure to hazards without relying on worker behavior and can be the most cost-effective solution to implement. Engineering controls that may be practicable and appropriate include:

- Installing high-efficiency air filters in offices and project trailers
- Increasing ventilation rates in the work environment.
- Installing physical barriers, such as clear plastic sneeze guards.

Administrative Controls

Administrative controls require action by the worker or employer. Typically, administrative controls are changes in work policy or procedures to reduce or minimize exposure to a hazard. Examples of administrative controls that may be practicable and appropriate include:

- Encouraging sick workers to stay at home.
- Minimizing contact among workers, clients, and customers by replacing face-to-face meetings with virtual communications and implementing telework if feasible.
- Establishing alternating days or extra shifts that reduce the total number of employees in a facility at a given time, allowing them to maintain distance from one another while maintaining a full onsite work week.
- Discontinuing nonessential travel to locations with ongoing Infectious Disease/Pandemic outbreaks. Employers should check the CDC website regularly for updates about travel advisories.
- Developing emergency communications plans, including a forum for answering workers' concerns and internet-based communications, if feasible.
- Providing workers with up-to-date education and training on Infectious Disease/Pandemic risk factors and protective behaviors (e.g., cough etiquette and care of PPE).
- Training workers who need to use protecting clothing and equipment how to put it on, use/wear it, and take it off correctly, including in the context of their current and potential duties. Training material should be easy to understand and available in the appropriate language and literacy level for all workers.

Safe Work Practices

Safe work practices are types of administrative controls that include procedures for safe and proper work used to reduce the duration, frequency, or intensity of exposure to a hazard. Examples of safe work practices for Infectious Disease/Pandemic outbreak include:

- Providing resources and a work environment that promotes personal hygiene. For example, no-touch trash cans, hand soap, alcohol-based hand rubs containing at least 60 percent alcohol, disinfectants, and disposable towels for workers to clean their work surfaces.
- Requiring regular hand washing or using of alcohol-based hand rubs. Workers should always wash hands when they are visibly soiled and after removing any PPE.
- Post handwashing and other hygiene and social distancing protocol signs in conspicuous locations including break areas, office lobbies, project trailers and restrooms.

Personal Protective Equipment (PPE)

While engineering and administrative controls are considered more effective in minimizing exposure to Infectious Disease/Pandemic outbreaks, PPE may also be needed to prevent certain exposures. While correctly using PPE can help prevent some exposures, it should not take the place of other prevention strategies. Examples of PPE include: gloves, goggles, face shields, face masks, and respiratory protection, when appropriate. During an outbreak of an Infectious Disease/Pandemic, recommendations for PPE specific to occupations or job tasks may change depending on geographic location, updated risk assessments for workers, and information on PPE effectiveness in preventing the spread of the Infectious Disease/Pandemic. Employers should check the OSHA and CDC websites regularly for updates about recommended PPE.

All types of PPE must be:

- Selected based upon the hazard to the worker.
- Properly fitted and periodically refitted, as applicable (e.g., respirators).
- Consistently and properly worn when required.
- Regularly inspected, maintained, and replaced, as necessary.
- Properly removed, cleaned, and stored or disposed of, as applicable, to avoid contamination of self, others, or the environment. Employers are obligated to provide their workers with PPE needed to keep them safe while performing their jobs. The types of PPE required during an Infectious Disease/Pandemic outbreak will be based on the risk of being infected while working and job tasks that may lead to exposure.

Follow Existing OSHA Standards

Existing OSHA standards may apply to protecting workers from exposure to and infection with an Infectious Disease/Pandemic. While there is no specific OSHA standard covering Infectious Disease/Pandemic exposure, some OSHA requirements may apply to preventing occupational exposure.

Among the most relevant are:

- OSHA's Personal Protective Equipment (PPE) standards (in general industry, 29 CFR 1910 Subpart I), which require using gloves, eye and face protection, and respiratory protection.
- When respirators are necessary to protect workers or where employers require respirator use, employers must implement a comprehensive respiratory protection program in accordance with the Respiratory Protection standard (29 CFR 1910.134).
- The General Duty Clause, Section 5(a)(1) of the Occupational Safety and Health (OSH) Act of 1970, 29 USC 654(a)(1), which requires employers to furnish to each worker "employment and a place of employment, which are free from recognized hazards that are causing or are likely to cause death or serious physical harm."
- OSHA's Bloodborne Pathogens standard (29 CFR 1910.1030) applies to occupational exposure to human blood and other potentially infectious materials that typically do not include respiratory

secretions that may transmit Infectious Disease/Pandemic. However, the provisions of the standard offer a framework that may help control some sources of the virus, including exposures to body fluids (e.g., respiratory secretions) not covered by the standard.

Engagement and Participation

We ask employees, subcontractors, clients, and other stakeholders to cooperate in taking steps to reduce the transmission of infectious disease/pandemic outbreak in the workplace. The best strategy remains the most obvious— social distancing, frequent hand washing with warm, soapy water; covering your mouth whenever you sneeze or cough; and discarding used tissues in wastebaskets. We will also provide alcohol-based hand sanitizers containing at least 60% alcohol throughout the workplace and in common areas. There may be additional strategies implemented based upon guidance from the U.S. Centers for Disease Control and Prevention (CDC) and/or the World Health Organization (WHO). Since those strategies may be rapidly changing based upon the most recent information available, DPR will publish and distribute separate guidance to address additional strategies.

7.3 Limiting Travel

All nonessential travel should be avoided during times of infectious disease/pandemic outbreaks. Employees who travel as an essential part of their job should consult with management on appropriate actions. International Business-related travel may not be authorized during times of infectious disease/pandemic outbreaks.

Employees should avoid crowded public transportation when possible. Alternative scheduling options, ride-share resources should be considered.

Employers should check the CDC website regularly for updates about travel advisories.

7.4 Telecommuting

Telework requests will be handled on a case-by-case basis. While not all positions will be eligible, all requests for temporary telecommuting should be submitted to your manager/supervisor for consideration.

7.5 Staying Home When Ill

Many times, with the best of intentions, employees report to work even though they feel ill.

During an infectious disease/pandemic outbreak, it is critical that employees do not report to work while they are ill and/or experiencing symptoms associated with the infectious disease/pandemic. Examples include fever, cough, sore throat, runny or stuffy nose, body aches, headache, chills and fatigue]. As of the date of this document, the Centers for Disease Control and Prevention recommends:

All employees should stay home if they are sick until at least 24 hours after their fever* (temperature of 100 degrees Fahrenheit or 37.8 degrees Celsius or higher) is gone. Temperature should be measured without the use of fever-reducing medicines (medicines that contains ibuprofen or acetaminophen). Not everyone with flu will have a fever. Individuals with suspected or confirmed flu, who do not have a fever, should stay home from work at least 4-5 days after the onset of symptoms. Persons with the flu are most contagious during the first 3 days of their illness.

See: <https://www.cdc.gov/flu/business/stay-home-when-sick.htm>

Employees who report to work ill may be sent home in accordance with these health guidelines.

7.6 Requests for Medical Information and/or Documentation

If you are out sick or show symptoms of being ill, it may become necessary to request information from you and/or your health care provider. In general, we would request medical information to confirm your need to be absent, to show whether and how an absence relates to the infection, and to know that it is appropriate



for you to return to work. As always, we expect and appreciate your cooperation if and when medical information is sought.

7.7 Confidentiality of Medical Information

Our policy is to treat any medical information as confidential. In furtherance of this policy, any disclosure of medical information is in limited circumstances with supervisors, managers, first aid and safety personnel, and government officials as required by law.

7.8 Social Distancing Guidelines for Workplace Infectious Disease/Pandemic Outbreaks

In the event of an infectious disease/pandemic outbreak, DPR Construction will follow the recommendations of the CDC and/or WHO and may implement these or other social distancing guidelines to minimize the spread of the disease among the employees, subcontractors, clients and other stakeholders.

During the workday, employees are requested to:

- Avoid meeting people face-to-face. Employees are encouraged to use the telephone, online conferencing, e-mail or instant messaging to conduct business as much as possible, even when participants are in the same building.
- If a face-to-face meeting is unavoidable, minimize the meeting time, choose a large meeting room and sit at least six feet from each other if possible; avoid person-to-person contact such as shaking hands.
- Avoid any unnecessary travel and cancel or postpone nonessential meetings, gatherings, workshops and training sessions.
- Do not congregate in work rooms, pantries, copier rooms or other areas where people socialize.
- Bring lunch and eat at your desk or away from others (avoid lunchrooms and crowded restaurants).
- Encourage members and others to request information and orders via phone and e-mail in order to minimize person-to-person contact. Have the orders, materials and information ready for fast pick-up or delivery.

7.9 Outside activities

- Employees might be encouraged to the extent possible to:
- Avoid public transportation (walk, cycle, drive a car) or go early or late to avoid rush-hour crowding on public transportation.
- Avoid recreational or other leisure classes, meetings, activities, etc., where employees might come into contact with contagious people.

End of Section 27

Warehousing & Delivery Forms

Forms

Vehicle Inspection Form



Vehicle Inspection Form

DPR Region:		Week of Date Range:	
Model/Type of Vehicle:		License Number:	
Gross Vehicle Weight:		Mileage:	

**** If any item(s) are found to be missing or defective, the equipment is to be repaired/serviced prior to use**

ReportedTo _____ Date: _____
Vehicle Taken Out of Service ☐ Yes ☐ No

Day of the Week:							
Date of Inspection:							

Clear Initials of person performing inspection:							
--	--	--	--	--	--	--	--

Has the driver been instructed in the safe operation of this type of vehicle and are they Licensed?	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N
---	-----	-----	-----	-----	-----	-----	-----

Inspection Item & Description Pass Fail Status							
--	--	--	--	--	--	--	--

Inspection Item & Description Pass Fail Status	P/F	P/F	P/F	P/F	P/F	P/F	P/F
1. Oil Level							
2. Coolant Level							
3. Fuel Level							
4. Other Fluid Levels							
5. Hoses & Belts							
6. Head lights, running lights							
7. Running Lights							
8. Foot Brakes							
9. Trailer Brake Actuator							
10. Parking Brake							
11. Brake Lights							
12. Back-up Alarm							
13. Gauges							
14. Windshield & Wipers							
15. Horn							
16. Seat Belts							
17. Air Tank Pressure (if one)							
18. Fluid Leaks							
19. Tire Pressure							
20. Bumpers							
21. Exterior Reflective Tape							
22. Cones on board (as needed)							
23. Fire extinguisher							
24. Trailer Hitch Secured							
25. Emergency Reflector Kit							
26. Other -							

Notes: (please record notable observations, with dates)
--

Repairs Made (Does this make sense to have here???)

By::		Date:	
------	--	-------	--

Section 26

Water Intrusion Prevention & Response

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Water Intrusion Prevention & Response

1.0 Introduction

To mitigate the risk of damages and litigation arising from water intrusion and/or potential mold contamination in buildings, all subcontractors involved in the project must proactively work with DPR to prevent water intrusions during construction that could lead to potential mold growth on the building materials (wood, drywall, etc.). The key to mold control is moisture control. It is important to dry water-damaged areas and items within 24-48 hours to prevent mold growth.

For this reason, opportunities to prevent water intrusion occur during all phases of construction: design, pre-installation, installation and maintenance or warranty. In addition to effective pre-planning, DPR and all subcontractors must respond immediately and appropriately when a water intrusion event occurs in a building during all phases of construction and/or during renovation.

2.0 Water Intrusion Prevention Planning

Planning to prevent water intrusion should start in the design and preconstruction stage. Water intrusion can be prevented by working closely with the owner, architect, third party consultant and subcontractors to review the building envelope components, product submittals, schedule with a weather projection plan and potential sources of water intrusion. A water intrusion pre-planning checklist has been developed and should be reviewed and discussed during the design stage and through all other phases of construction. The pre-planning checklist should be reviewed during OAC meetings, subcontractor pre-construction meetings, coordination meetings, worker orientations, safety meetings etc. (Reference the Water Intrusion Pre-Planning Checklist in the Forms Section of the EHSP)

2.1 Pre-Planning Checklist

The pre-planning checklist has been developed to identify potential water intrusions that could occur in the scheduled construction and focuses on phases below:

- Pre-Construction
- Foundation Systems
- Exterior Wall Systems
- Sealants
- Window Installation
- Roofing Systems
- Interior Wall Systems
- Plumbing and Fire Sprinklers
- Installation and Balancing of HVAC Systems
- Clean-up for Construction
- Water Intrusion Clean-up

3.0 Water Intrusion Procedures

DPR and subcontractors can minimize the risk of water intrusion and possible resulting mold contamination claims by effective planning for and implementing water intrusion procedures. It is the responsibility of the project management team to develop a project specific rapid response plan. Below are general guidelines to assist with the development of this plan:

3.1 Water Intrusion Pre-Planning Guidelines

A notification list should be developed that includes, but is not limited to:

- Owner contacts
- Third party contacts for mold cause & origin

- Environmental Consultant/Industrial Hygienist
- Water Damage Restoration Professional
- Licensed Abatement Contractor
- Equipment Material Supply Company
- Local Utility Companies
- DPR EHS and DPR Risk Management (Refer to site specific emergency response plan)
- Water shut off procedures

3.2 Spill Response Kit

Prepare a water containment/water remediation spill kit. Included on the kit should be:

- At a minimum, one spill response kit should be readily available; more if needed depending on area to be covered
- Spill response kits should be readily accessible, easily identifiable and in close proximity to appropriate work areas
- Applicable workers should be trained in the proper use of the spill response kit
- The kit should include, but not be limited to:
 - 55-gallon water tight container, on wheels
 - Fire sprinkler head shut-off tool
 - Various lengths of PVC pipe (in appropriate diameters to capture water from broken pipes) – determine if 180° or other bends are needed in pipe to reach affected areas
 - 50-100-foot length of hose to attach to barrel for discharging water
 - Roll(s) of visquene (6 mil or thicker recommended)
 - Roll(s) of duct tape
 - Squeegees and water vacuums (shop-vacuums or similar)
 - Extension cords and GFCI protection
 - Project specific shut-off procedures (location and any specific requirements for shutoff)
 - Affix a large sticker with Emergency Contact Information on spill response kit (reference the suggested label in the Forms Section at the end of the EHSP)

4.0 Initial Response

Identify and shutoff or repair all sources of water leaks or intrusions (if not already completed). This is essential for the prevention of mold growth.

- Be certain before allowing workers into any flooded area that there is no chance of a live electrical outlet in contact with the water
- Contact DPR Project Management, DPR Safety and DPR Risk Management
- Shut down the HVAC system only after receiving direction from the building owner and DPR project management
- Use circulating fans and area dehumidifiers to boost air circulation and remove water vapor from the air. (i.e., Dehumidify impacted areas). Do not push any air over visually moldy materials.
- Contact Environmental Consultant/Industrial Hygienist to investigate and develop a water remediation plan
- Contain visually impacted areas with 6-mil polyethylene sheeting to reduce chance of mold spores spreading to other areas of the building
- Take photographs: before, during and after remediation
- Document all remediation activities, including containment measures and moisture meter readings before, during and after remediation
- Complete the Water Intrusion Grab & Go and forward to the appropriate parties (reference the Forms Section at the of the EHSP)
- Develop checklists for inspection at critical points during re-construction

- Negative air machines (Air Filtering Devices) set up inside the contained area may also be employed to further reduce spread of contamination

5.0 Remediation

- **Do not** use biocides and/or anti-microbial chemicals as a substitute for microbial removal (don't just use Clorox).
- **Do not** impact any materials that may contain lead or asbestos without testing and assessment. These materials require special handling and disposal, by law.
- **Do not** impact mold contaminated materials or handle biocides without proper PPE
- **Do** cut (not tear) out wallboard
- **Do** remove and contain materials at least two (2) feet beyond visible water damage and/or visible mold growth
- Some specialty porous items (i.e., expensive rugs or clothing) must be wrapped in plastic prior to removal from the building for professional cleaning by qualified contractors
- **Do not** remove contaminated materials from the contained work area without securely wrapping in plastic
- **Do** ensure negative air machines and other equipment is decontaminated /cleaned prior to mobilization and demobilization
- **Do** wet-wipe clean and/or HEPA vacuum surfaces in the remediation area to remove excessive settled spores. These areas should be free of dust and debris before post-remediation monitoring.
- **Do** dispose of microbial waste in a regular landfill (it is currently non-regulated) unless asbestos or lead-containing materials are also present (special handling required)
- **Do** wrap microbial waste in plastic prior to disposal to prevent further release of spores
- **Do not** begin reconstruction prior to post-remediation sampling and receipt of successful sampling/visual inspection results
- **Do not** run 'post-remediation' sampling immediately following remediation; utilize negative air machines equipped with HEPA filtration for 24-48 hours to "scrub" the air before sampling

Remediation Guidelines	
Building Component	Actions to be Taken
Carpet	Remove impacted Carpet if recommended by Environmental Consultant/Industrial Hygienist
Ceiling Tiles	Remove if water stained or fungi- contaminated place in a plastic bag for removal
Ceramic Tile Flooring	If grout is compromised remove and wet wipe sub flooring
Concrete, cinder block or brick	Wire brush visible fungi within negative pressure containment. Wet wipe clean and HEPA vacuum. Apply biocide / sealant per third party consultant recommendations.
Drywall	Remove two (2) feet beyond any visibly impacted or water stained material, double bag in 6 mil polyethylene plastic and remove from containment or area
HVAC duct system components (hard components)	Wet wipe clean
Insulation	Remove two (2) feet beyond any visibly impacted or water stained material, double bag in 6 mil polyethylene plastic and remove from containment or area.
Padding	Remove impacted padding
Plywood Floor	If structurally unsound remove, otherwise wire brush or hand sand visible fungi within a negative pressure containment. Wet wipe clean and HEPA vac. Application of biocide / sealant may be necessary.
Unlined Flex Duct	Remove and dispose of properly
Wood Flooring	If flooring cannot be restored, remove. Wet wipe sub flooring clean.
Wood Studs/ braces/ rafters	If structurally unsound remove, otherwise wire brush and hand sand visible fungi within a negative pressure containment. Wet wipe clean and HEPA vac. Application of biocide/ sealant maybe necessary.

5.1 *Cleaning a Flooded Building*

Flood water is divided into three categories that require different cleanup measures. Clear water is tap or rain water. Gray water is from sinks, showers, tubs, and washers. Black water is contaminated with human and/or animal waste. Flood water from lakes or rivers is considered black water.

5.1.1 *Clear Water (tap or rain water)*

- Remove standing water as soon as possible
- Clean washable items in a washing machine with detergent
- Clean hard surfaces with soap and water
- Use a “wet-dry” shop vacuum to remove water from beds and couches. Dry within 24-48 hours
- Clean and dry carpets within 24-48 hours, or remove
- Plaster and drywall that have been saturated should be removed and replaced to at least (2) feet above the water line

5.1.2 *Gray Water (sink, shower, tub, and washer water)*

5.1.3 *Black Water (river, lake - contains human and/or animal waste)*

- Black water remediation should be performed by a qualified person
- Wear eye goggles, approved respirator, rubber gloves and rubber boots, at a minimum, when cleaning. Change work clothes and boots and wash hands thoroughly with soap and water after working in the contaminated area.
- Keep people and animals out of the contaminated area until clean-up has been completed
- Remove standing water as soon as possible. Ask your local health department for directions on removal and disposal
- Clean washable items in a washing machine with detergent and hot water. Wash contaminated items separately from those that were not contaminated
- Clean hard surfaces with soap and water
- Remove and dispose items that cannot be washed and disinfected (mattresses, carpet, carpet padding, upholstered furniture, pillows, books, and most foam and paper products)
- Remove and replace drywall that have been saturated at least two (2) feet above the water line
- Mops and brooms, and other items used in cleaning should be thoroughly washed and disinfected. Rinse thoroughly with water and then soak in an appropriate cleaning solution(s). Rinse thoroughly with water, wring out, and dry thoroughly outside in the open air.

5.1.3 *Sewage Cleanup*

- Sewage remediation should be performed by a qualified person
- Wear proper protective clothing such as rubber boots and waterproof gloves while cleaning up the spill or handling contaminated items
- Contain the spill immediately, and divert it to a sanitary sewer if possible
- Restrict access to contaminated areas
- Assume anything touched by sewage to be contaminated. Every surface touched by sewage will need to be cleaned, disinfected, and sanitized
- Ensure that all liquid and solid materials are removed prior to decontamination
- Discard porous building materials and personal protective equipment by enclosing them in sealed plastic bags and disposing of them into a solid waste container
- All drywall materials that have been contaminated by sewage water should be cut out and removed. Extend the area for removal approximately two (2) feet beyond the watermark or height of the floodwater
- Discard any ceiling tiles that have been contaminated by sewage waters

- All insulation materials, whether Styrofoam, fibrous glass, sponge rubber or cellulose wetted by black water should be discarded
- All wall coverings that are wetted should be removed and discarded
- Any upholstered furniture, fabrics or furnishings wetted by sewage-contaminated waters, and furniture made from pressed wood products, will be difficult to properly decontaminate and therefore should be disposed
- Any contaminated papers, cardboard boxes or files should be disposed
- Any foodstuffs, or opened drink containers should be disposed of
- For non-porous materials, apply an approved disinfectant and follow the directions printed on the label to ensure sufficient contact time
- Do not mix cleaning / disinfecting products or chemicals. Cleaning products can react with one another to produce toxic vapor or liquid substances.
- Thoroughly wash hands with soap and water immediately following cleanup efforts and before preparing food, eating, or drinking.

6.0 Post Water Intrusion

Complete Water Intrusion Report and forward completed report and photos to DPR EHS and DPR Risk Management.

- Collect reports from any third-party experts
- Alert the owner in writing; include steps being implemented to address the issue. In the event you uncover pre-existing mold, fully document findings and notify owner in writing
- Contact an environmental testing agency to visually inspect area and provide proposal for testing and estimate of abatement costs
- Conduct testing after obtaining approval from the owner
- Should test results indicate that remediation or abatement is required, hire a Licensed and certified remediation or abatement subcontractor, after obtaining owner approval

Water Intrusion Prevention & Response Forms

Forms

Water Intrusion Pre-Planning Checklist
Water Intrusion Emergency Contacts



Water Intrusion Pre-Planning Checklist

Archive Document – DO NOT DISCARD

Project/Project#:		Date:	
Planner:		Company:	
Scope of work being pre-planned:			
DPR Superintendent:		DPR Project Manager:	
Construction Start/End Date:		Pre-Plan Date:	

Pre-Construction	Yes	No	N/A
Has the schedule of work and sequencing plan been integrated with a weather projection plan to protect as-built components and various stored building materials?	☐	☐	☐
Have you obtained a third party expert review of the exterior envelope design details and specifications (walls, windows, roofing, flashing, expansion joints, waterproofing, drainage, etc.) to identify potential risk, areas for water intrusion, and to offer alternative or corrective design details?	☐	☐	☐
Has a photo and progress inspection log been implemented breaking down crucial points of construction? (i.e., pre-pour foundation photographs & documentation, pre-skin and pre-dry-in of roof elements, landscape, hardscape and drainage at completion, etc.)?	☐	☐	☐
Have water intrusion issues been added as a standing agenda topic to subcontractor coordination meetings?	☐	☐	☐
Are all subcontractor coordination meeting minutes accurately capturing attendees, specific issues addressed, questions and/or problems as they relate to specific water intrusion issues.	☐	☐	☐
Have areas been walked, with shutoff valve location(s) and any specific procedures identified to allow for rapid response in the event of spills.	☐	☐	☐
Have you obtained a third party expert review of the exterior envelope design details and specifications (walls, windows, roofing, flashing, expansion joints, waterproofing, drainage, etc.) to identify potential risk, areas for water intrusion, and to offer alternative or corrective design details?	☐	☐	☐

Foundation Systems	Yes	No	N/A
Has a peer review been performed as it relates to drainage at and around the footprint and foundation?	☐	☐	☐
Has a peer review been performed regarding the performance and installation of water proofing system?	☐	☐	☐
Have you taken into consideration proper curing and dry-time as recommended by various associations' in order to eliminate moisture transmission through capillary voids created through hydration through the concrete slab?	☐	☐	☐

Exterior Wall Systems	Yes	No	N/A
Has your designer/architect incorporated the appropriate vapor barrier and assembly in order to prevent thermal bridging relative to colder climates?	☐	☐	☐
Has your designer/architect incorporated the appropriate vapor retarder relative to the building assemblies in order to allow proper drying, due to potential humid air to the exterior?	☐	☐	☐
Have the on-site terminations of various vapor barrier locations been terminated and/or sealed properly to eliminate water intrusion?	☐	☐	☐



Water Intrusion Pre-Planning Checklist

Archive Document – DO NOT DISCARD

Sealants	Yes	No	N/A
Are the sealants, which are being used in the construction process, compatible with the materials in which they are being integrated?	☐	☐	☐
Do you have specific data relative to installation recommendations?	☐	☐	☐
Do the sealants meet the requirements of the manufacturer's, relative to exterior cladding (i.e., EIFS, etc.)?	☐	☐	☐
Have the appropriate contractor(s) and suppliers attended the preconstruction and construction meetings relative to the installation of the specified sealants?	☐	☐	☐

Window Installation	Yes	No	N/A
Do the windows and doors as specified, qualify for the area based on rate of failure?	☐	☐	☐
Is it clear how the windows and doors are to be integrated into exterior lathing for proper waterproofing?	☐	☐	☐
Have the window openings been accepted as-is by the window installer?	☐	☐	☐
Do all window sill details call for positive drainage slopes (away from the windows)?	☐	☐	☐
Do the window details require drip edges?	☐	☐	☐
Has a pre-construction or construction meeting taken place where manufacturer's installation instructions have been incorporated into the meeting which would include all parties related to the installation (i.e., architect, window and door installation subcontractor, exterior lathing subcontractor, general contractor, etc.)?	☐	☐	☐

Roofing Systems	Yes	No	N/A
Has a pre-roofing meeting been conducted with the roofer, manufacturer and other related subs to review the details, manufacturer requirements, code requirements, UL requirements, etc.?	☐	☐	☐
Has the roofer and manufacturer walked and signed off on the substrate, openings, joints and penetrations prior to installation of roofing? A copy of the sign off checklist must be provided to and kept by DPR.	☐	☐	☐

Interior Wall Systems	Yes	No	N/A
If a wood structure is being constructed, is the moisture content in the wood between 15-18%?	☐	☐	☐
The installation of gypsum board and various taping and topping compounds, including textures, should also be in a range of 15-18% before installation of any wall finishes or wall coverings. Have moisture meters been utilized in documenting conditions pre-wall finish?	☐	☐	☐

Plumbing & Fire Sprinklers	Yes	No	N/A
Have all water supply lines been thoroughly checked for proper installation, protection and insulation to ensure a leak-proof system?	☐	☐	☐

Installation and Balancing of HVAC Systems	Yes	No	N/A
Have you confirmed that the HVAC system is providing fresh air and proper distribution into the building cavity?	☐	☐	☐
Have unoccupied spaces, such as chase areas, crawlspaces, etc. been properly ventilated and detailed by the architect?	☐	☐	☐
If not done through preconstruction peer review, it is recommended this be done during construction. Identify areas of concern in writing and distribute RFIs to architect, mechanical subcontractor, owner, or other pertinent subcontractors and/or any parties involved for documentation purposes.	☐	☐	☐
Have you had the HVAC system cleaned and certified by a third party with certification capability?	☐	☐	☐
If so, copies of the test and balance reports should be given to the architect, mechanical subcontractor and any other subcontractors involved in the system for their records.	☐	☐	☐



Water Intrusion Pre-Planning Checklist

Archive Document – DO NOT DISCARD

Has a plan been developed to maintain a 30-55% relative humidity level within the building (de-humidifier, desiccant/gas, etc.)?	☐	☐	☐
Clean-up for Construction	Yes	No	N/A
Is your construction site maintained on a continuous clean-up basis throughout the project?	☐	☐	☐
If so, good maintenance habits are in place. It is important to understand that all construction debris and dust must be cleaned up in all areas prior to the finished wall systems being installed, which could entrap various trash, debris and dust.	☐	☐	☐
System Failure and/or Damage due to Water or Moisture Intrusion	Yes	No	N/A
Clean-up for Construction			
Have you analyzed the damaged areas thoroughly (such as below floor insulation) evaluated the exterior of damage from loss and documented these findings?	☐	☐	☐
After making the appropriate repairs, you must get the proper clearance certificates from the appropriate trades relative to the loss. Copies of these reports relative to specific repairs should be forwarded to the architect, appropriate subcontractor(s) and the owner and/or the owner's representative.	☐	☐	☐

Employee's Name/Signature

Date



Label for spill Response Kit

WATER INTRUSION **EMERGENCY CONTACTS**

DPR EHS - (enter name & contact info)

DPR Regional Office – (enter phone #)

ENVIRONMENTAL CONSULTANT
(enter name & contact info)

WATER REMEDIATION SPECIALISTS
(enter name & contact info)

ENVIRONMENTAL ABATEMENT
(enter name & contact info)

Section 27

Work Environment & Employee Health

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Work Environment and Employee Health

1.0 Introduction

This program is to provide guidance for protecting employees from hazards of high heat and cold conditions; venomous snakes, bees, other insects and animals as well as prevention and response for Infectious Disease/Pandemic outbreak. The intent is to provide information on engineering, administrative, safe work practices and PPE controls to eliminate and or reduce the risk.

Being uncomfortable is not the major problem with working in high and low temperatures and humidity. Workers who are suddenly exposed to working in a hot or cold environment face additional and generally avoidable hazards to their safety and health.

This program applies to all outdoor and indoor places of employment at those times when the environmental risk factors including heat or cold illness are present. It also applies to preventing bites and stings from insects and animals.

Reference: Heat Illness Prevention Standard Title 8 California Code of Regulations 3395, Sanitation CFR 1926.51; National Institute for Occupational Safety and Health (NIOSH). Other applicable Standards, guidelines, and regulations may apply.

Note: DPR has recognized Heat Illness as a serious and preventable condition and has selected the Cal/OSHA standards as a means of best practice method for controlling for heat illness risks.

Definitions

Acclimatization: Temporary adaptation of the body to work in the heat, that occurs gradually when a person is exposed. Acclimatization peaks in most people within four to fourteen days of regular work for at least two hours per day in heat.

Heat Illness: A serious medical condition resulting from the body's inability to cope with a particular heat load, and includes heat cramps, heat exhaustion, heat syncope and heat stroke.

Environmental Risk Factors for heat illness: Working conditions that create the possibility that heat illness could occur, may include: air temperature, relative humidity, radiant heat from the sun and other sources, conductive heat sources, air movement, workload (severity and duration), protective clothing and personal protective equipment worn by employees.

Landscaping: Providing landscape care and maintenance services and/or installing trees, shrubs, plants, lawns, or gardens, or providing these services in conjunction with the design of landscape plans and/or the construction (i.e., installation) of walkways, retaining walls, decks, fences, ponds, and similar structures, except for employment by an employer who operates a fixed establishment where the work is to be performed and where drinking water is plumbed.

Pandemic: An outbreak of disease that occurs over a wide geographic area and affects an exceptionally high proportion of the population.

Personal Risk Factors for heat illness: Factors such as an individual's age, degree of acclimatization, health, water consumption, alcohol consumption, caffeine consumption, and use of prescription medications that affect the body's water retention or other physiological response to heat.

Shade: Blockage of direct sunlight. One indicator that blockage is sufficient is when objects do not cast a shadow in the area of blocked sunlight. Shade is not adequate when heat in the area of the shade defeats the purpose of shade, which is to allow the body to cool. For example, a car sitting in the sun does not provide acceptable shade to a person inside it, unless the car is running with air conditioning. Shade may be provided by any natural or artificial means that does not expose employees to unsafe or unhealthy conditions and that does not deter or discourage access or use.

Temperature: The dry bulb temperature, in degrees Fahrenheit, is obtainable by using a thermometer to measure the outdoor temperature in an area where there is no shade. While the temperature measurement

must be taken in an area with full sunlight, the bulb or sensor of the thermometer should be shielded while taking the measurement, e.g., with the hand or some other object, from direct contact by sunlight.

Note: All contractors should identify their designated competent person who, prior to each workday, will be responsible for monitoring the weather using www.nws.noaa.gov or with the aid of a simple thermometer at the worksite.

2.0 Heat Illness

All contractors with any outdoor places of work must comply with the Heat Illness Prevention Standard Title 8 California Code of Regulations 3395. This sub-section applies to the control of risk of occurrence of heat illness and other environmental factors. This sub-section applies to all outdoor places of employment at those times when the environmental risk factors for heat illness are present. This sub-section also applies to transportation or delivery of construction materials or other heavy materials (e.g. furniture, lumber, freight, cargo, cabinets, industrial or commercial materials), except for employment that consists of operating an air-conditioned vehicle and does not include loading or unloading.

2.1 Provisions of Water

Employees should have access to safe drinking water that includes but not limited; that it be fresh, pure, suitably cool and provided to employees free of charge. The water should be located as close as practicable to areas where employees are working. The following items must be addressed:

- An adequate supply of potable water should be provided in all places of employment
- Portable containers used to dispense drinking water should be:
 - equipped with a faucet or drinking fountain
 - capable of being tightly closed
 - otherwise designed, constructed and serviced so that sanitary conditions are maintained
- Outlets for non-potable water, such as water for industrial or firefighting purposes only, shall be identified by signs meeting regulatory-requirements, to indicate clearly that the water is unsafe and is not to be used for drinking, washing, or cooking purposes
- There shall be no cross-connection, open or potential, between a system furnishing potable water and a system furnishing non-potable water
- Any container used to store or dispense drinking water should be clearly marked as to the nature of its contents and should not be used for any other purpose
- Water should not be dipped from containers
- Where drinking fountains are not provided, single service cups (to be used but once) should be supplied
- Where single service cups are supplied, a sanitary container for the unused cups and a receptacle for disposing of the used cups should be provide
- Where re-usable containers for individual use are relied upon for compliance with this section, the employer shall ensure the containers are marked to identify the user and maintained in a sanitary condition
- Re-usable containers for individual use and drinking cups shall not be shared or used in common
- Drinking water should be provided in sufficient quantity at the beginning of the work shift to provide one quart per employee per hour for drinking for the entire shift. Employers may begin the shift with smaller quantities of water if they have effective procedures for replenishment during the shift as needed to allow employees to drink one quart or more per hour
- The frequent drinking of water should be encouraged
- Ice is recommended to cool water in high heat
- All contractors should identify the designated person(s) that has been assigned the task of providing drinking water containers to the site

- As part of effective replenishment procedures, the designated person will check the water level as needed to ensure a constant fresh supply is available at all times. When the temperature exceeds 80 degrees F the water levels should be checked more frequently

2.2 Shade

In California, shade should be present when the temperature exceeds 80 degrees Fahrenheit. When the outdoor temperature in the work area exceeds 80 degrees Fahrenheit, the employer should have and maintain one or more areas with shade at all times while employees are present that is either open to the air or provided with ventilation or cooling. The amount of shade present should be at least enough to accommodate the number of employees on recovery or rest periods, so that they can sit in a normal posture fully in the shade without having to be in physical contact with each other. The shaded area should be located as close as practicable to the areas where employees are working. Subject to the same specifications, the amount of shade present during meal periods should be at least enough to accommodate the number of employees on the lunch or break period who remain onsite. Shade should be available when the temperature does not exceed 80 degrees Fahrenheit. When the outdoor temperature in the work area does not exceed 80 degrees Fahrenheit employers should either provide shade at all times or provide timely access to shade upon an employee's request.

In all other states, shade should be provided as temperatures rise. The shaded area should be located as close as practicable to the areas where employees are working, and workers should be encouraged to use rest/hydration breaks as necessary to prevent signs/symptoms of heat illness.

Employees should be allowed and encouraged to take a preventative cool-down rest in the shade when they feel the need to do so to protect themselves from overheating. Such access to shade should be permitted at all times.

An individual employee who takes a preventative cool-down rest should be:

- Monitored and asked if he or she is experiencing symptoms of heat illness
- Encouraged to remain in the shade
- *Should not* be ordered back to work until any signs or symptoms of heat illness have been abated, but in no event less than 5 minutes in addition to the time needed to access the shade
- If an employee exhibits signs or reports symptoms of heat illness while taking a preventative cool-down rest or during a preventative cool-down rest period, the employer shall provide appropriate first aid or emergency response according to subsection 2.4 of this sub-section.
- Exceptions to the shade requirements above:
- Where the employer can demonstrate that it is infeasible or unsafe to have a shade structure, or otherwise to have shade present on a continuous basis, the employer may utilize alternative procedures for providing access to shade if the alternative procedures provide equivalent protection.
- Except for employers in the agricultural industry, cooling measures other than shade (e.g., use of misting machines) may be provided in lieu of shade if the employer can demonstrate that these measures are at least as effective as shade in allowing employees to cool.
- All contractors should identify the designated person(s) who will ensure that shade structures are opened and placed as close as practical to the workers when the temperature equals or exceeds 80 degrees F.

2.3 High Heat Procedures

In California, the employer should implement high-heat procedures when the temperature equals or exceeds 95 degrees Fahrenheit (all other states are encouraged to implement High Heat Procedures as temperatures and/or humidity rise). These procedures should include the following to the extent practicable:

- Ensuring that effective communication by voice, observation, or electronic means is maintained so that employees at the work site can contact a supervisor when necessary. An electronic device, such

as a cell phone or text messaging device, may be used for this purpose only if reception in the area is reliable.

- Observing employees for alertness and signs or symptoms of heat illness. The employer should ensure effective employee observation/monitoring by implementing one or more of the following;
 - Supervisor or designee observation of 20 or fewer employees, or
 - Mandatory buddy system, or
 - Regular communication with sole employee such as by radio or cellular phone
 - Other effective means of observation
- Designating one or more employees on each worksite as authorized to call for emergency medical services, and allowing other employees to call for emergency services when no designated employee is available
- Reminding employees throughout the work shift to drink plenty of water.
- Pre-shift meetings before the commencement of work to review the high heat procedures, encourage employees to drink plenty of water and remind employees of their right to take a cool-down rest when necessary.
- All contractors should identify the designated person(s) who will ensure High Heat Procedures are put into place when the temperature equals or exceeds 95 degrees F.

2.4 *Emergency Response Procedures*

The employer should implement effective emergency response procedures including:

- Ensure that effective communication by voice, observation or electronic means is maintained so that employees at the work site can contact a supervisor or emergency medical services when necessary. An electronic device, such as a cell phone or text messaging device, may be used for this purpose only if reception in the area is reliable. If an electronic device will not furnish reliable communication in the work area, the employer will ensure a means of summoning emergency medical services
- Respond to signs and symptoms of possible heat illness including but not limited to first aid measures and how emergency medical services will be provided
 - If a supervisor observes or any employee reports any signs or symptoms of heat illness in any employee, the supervisor should take immediate action and commensurate with the severity of the illness
 - If the signs or symptoms are indicators of severe heat illness (such as, but not limited to, flushed skin, high body temperature, decreased level of consciousness, staggering, vomiting, disorientation, irrational behavior or convulsions), the employer must implement emergency response procedures.
 - An employee exhibiting signs or symptoms of heat illness should be monitored and should not be left alone or sent home without being offered onsite first aid and /or being provided with emergency medical services in accordance with the employer's procedures
- Contact emergency medical services and, if necessary transporting employees to a place where they can be reached by emergency medical provider.
- Ensure that, in the event of an emergency, clear and precise directions to the work site can and will be provided as needed to emergency responders.

2.5 *Acclimatization*

- All employees should be closely observed by a supervisor or designee during a heat wave. For this sub-section only, "heat wave" means any day in which the predicted high temperature for the day will be at least 80-degrees Fahrenheit and at least ten degrees Fahrenheit higher than the average high daily temperature in the preceding five days.
- An employee who has been newly assigned to a high heat area should be closely observed by a supervisor or designee for the first 14 days of the employee's employment.

2.6 *Training*

Employee training. Effective training in the following topics should be provided to each supervisory and non-supervisory employees before the employee begins work. Training should be provided to any employees whose work could reasonably be anticipated to result in exposure to the risk of heat illness. The following are training topics to covered:

- The environmental and personal risk factors for heat illness, as well as the added burden of heat load on the body caused by exertion, clothing, and personal protective equipment.
- The employer's procedures for complying with the requirements of this standard, including but not limited to, the employer's responsibility to provide water, shade, cool-down rests, and access to first aid as well as the employees' under this standard without retaliation.
- The importance of frequent consumption of small quantities of water, up to 4 cups per hour, when the work environment is hot and employees are likely to be sweating more than usual in the performance of their duties.
- The concept, methods and importance, of acclimatization.
- The common signs and symptoms of heat illness; appropriate first aid and/or emergency responses to the different types of heat illness, and in addition, that heat illness may progress quickly from mild symptoms and signs to serious and life threatening illness
- The importance to employees of immediately reporting to the employer, directly or through the employee's supervisor, symptoms or signs of heat illness in themselves, or in co-workers.
- The employer's procedures for responding to signs or symptoms of possible heat illness, including how emergency medical services will be provided should they become necessary.
- The employer's procedures for contacting emergency medical services, and if necessary, for transporting employees to a point where they can be reached by an emergency medical service provider.
- The employer's procedures for ensuring that, in the event of an emergency, clear and precise directions to the work site can and will be provided as needed to emergency responders. These procedures should include designating a person to be available to ensure that emergency procedures are invoked when appropriate.

Supervisor training. Prior to assignment to supervision of supervising employees performing work that should reasonably be anticipated to result in exposure to the risk of heat illness working in the heat, effective training on the following topics should be provided to the supervisor:

- The information required to be provided by sub-section 2.7 above.
- The procedures the supervisor is to follow to implement the applicable provisions in this sub-section.
- The procedures the supervisor is to follow when an employee exhibits signs or reports symptoms consistent with possible heat illness, including emergency response procedures.
- How to monitor weather reports and how to respond to hot weather advisories.

2.7 *Heat Illness Prevention Plan*

All contractors shall establish, implement, and maintain, an effective heat illness prevention plan. The plan shall be in writing in both English and the language understood by the majority of the employees and shall be made available at the worksite to employees and to representatives of the Division upon request. The Heat Illness Prevention Plan may be included as part of the employer's Illness and Injury Prevention Program, and shall, at a minimum, contain:

- Procedures for the provision of water and access to shade.
- The high heat procedures referred to in subsection.
- Emergency Response Procedures in accordance with subsection.
- Acclimatization methods and procedures in accordance with subsection.

2.7.1 Heat Related Illness

Heat Stroke

Heat stroke is the most serious heat-related illness. It occurs when the body becomes unable to control its temperature: the body's temperature rises rapidly, the sweating mechanism fails, and the body is unable to cool down. When heat stroke occurs, the body temperature can rise to 106°F or higher within 10 to 15 minutes. Heat stroke can cause death or permanent disability if emergency treatment is not given.

Symptoms of heat stroke:

- Confusion, altered mental status, slurred speech
- Loss of consciousness (coma)
- Hot, dry skin or profuse sweating
- Seizures
- Very high body temperature
- Fatal if treatment delayed

First Aid steps to be taken to treat a worker with heat stroke:

- Call 911 for emergency medical care.
- Stay with worker until emergency medical services arrive.
- Move the worker to a shaded, cool area and remove outer clothing.
- Cool the worker quickly with a cold water or ice bath if possible; wet the skin, place cold wet-cloths on skin, or soak clothing with cool water.
- Circulate the air around the worker to speed cooling.
- Place cold wet-cloths or ice on head, neck, armpits, and groin; or soak the clothing with cool water.

Heat Exhaustion

Heat exhaustion is the body's response to an excessive loss of the water and salt, usually through excessive sweating. Workers most prone to heat exhaustion are those that are elderly, have high blood pressure, and those working in a hot environment.

Symptoms of heat exhaustion:

- Headache
- Nausea
- Dizziness
- Weakness
- Irritability
- Thirst
- Heavy sweating
- Elevated body temperature
- Decreased urine output

First Aid steps to be taken to treat a worker with heat exhaustion:

- Take worker to a clinic or emergency room for medical evaluation and treatment.
- If medical care is unavailable, call 911.
- Someone should stay with worker until help arrives.
- Remove worker from hot area and give liquids to drink.
- Remove unnecessary clothing, including shoes and socks.
- Cool the worker with cold compresses or have the worker wash head, face, and neck with cold water.
- Encourage frequent sips of cool water.

Rhabdomyolysis

Rhabdomyolysis is a medical condition associated with heat stress and prolonged physical exertion, resulting in the rapid breakdown, rupture, and death of muscle. When muscle tissue dies, electrolytes and large proteins are released into the bloodstream that can cause irregular heart rhythms and seizures, and damage the kidneys.

Symptoms of rhabdomyolysis include:

- Muscle cramps/pain
- Abnormally dark (tea or cola colored) urine
- Weakness
- Exercise intolerance
- Asymptomatic

First Aid steps to be taken to treat a worker with rhabdomyolysis:

- Stop activity.
- Increase hydration (water preferred).
- Seek immediate care at the nearest medical facility.
- Ask to be checked for rhabdomyolysis (i.e., blood sample analyzed for creatine kinase).

Heat Syncope

- Heat syncope is a fainting (syncope) episode or dizziness that usually occurs with prolonged standing or sudden rising from a sitting or lying position. Factors that may contribute to heat syncope include dehydration and lack of acclimatization.

Symptoms of heat syncope include:

- Fainting (short duration)
- Dizziness
- Light-headedness during prolonged standing or suddenly rising from a sitting or lying position

First Aid steps to be taken to treat a worker with heat syncope:

- Sit or lie down in a cool place.
- Slowly drink water, clear juice, or a sports drink.

Heat Cramps

Heat cramps usually affect workers who sweat a lot during strenuous activity. This sweating depletes the body's salt and moisture levels. Low salt levels in muscles causes painful cramps. Heat cramps may also be a symptom of heat exhaustion.

Symptoms of heat cramps:

- Muscle cramps, pain, or spasms in the abdomen, arms, or legs

First Aid steps to be taken to treat a worker with heat cramps:

- Drink water and have a snack and/or carbohydrate-electrolyte replacement liquid (e.g., sports drinks) every 15 to 20 minutes.
- Avoid salt tablets.
- Get medical help if the worker has heart problems, is on a low sodium diet, or if cramps do not subside within 1 hour.

Heat Rash

Heat rash is a skin irritation caused by excessive sweating during hot, humid weather.

Symptoms heat rash include:

- Looks like red cluster of pimples or small blisters
- Usually appears on the neck, upper chest, groin, under the breasts, and in elbow creases

First Aid steps to be taken to treat a worker with heat rash:

- Keep rash area dry.
- Powder may be applied to increase comfort.
- Ointments and creams should not be used.

In the event that persons exhibit any of these symptoms notify DPR's project management immediately. In an emergency employees are allowed to call emergency services without contacting a supervisor first. Even early symptoms may indicate serious heat exposure. If first aid trained personnel are not immediately available on-site to make an assessment and workers show of the signs and symptoms of heat illness, you should notify DPR Project management and refer to and follow site specific emergency response procedures. Regardless of the worker's protests, no employee with any of the symptoms of possible serious heat illness should be sent home or left unattended without medical assessment and authorization. For Emergency information please review Emergency Procedures Section 08.

2.8 Site Specific Procedures

1. Identify the designated person(s) that has been assigned the applicable task(s) (e.g. supervisor, foreman, safety coordinator, crew leader). Please review and complete Site Specific Procedures Checklist in the Forms Section at the end of the EHSP.
2. Provide specific details required to carry out the task and ensure that the task is accomplished successfully (e.g. how many water containers/shade structures, of what size, distance to placement, frequency of water-level replenishment/weather-tracking/water breaks/reminders, etc.). Specify how these procedures will be communicated to your employees and in particular to the persons assigned these responsibilities (e.g. via training, meeting), and how it will be ascertained that these company instructions and procedures are followed.
3. Each project should designate persons to implement the provisions in this Heat Illness Prevention Plan and may use the Site Specific Procedures Checklist in the Forms Section at the end of the EHSP.

2.9 Cleaning a Water Storage Container Before Use

Water containers should be cleaned daily and sanitized weekly, at a minimum. Before filling with safe water, use these steps to clean and sanitize water storage containers:

Daily cleaning:

- Wash the storage container with a solution of dishwashing soap and water or baking soda and water.
- Rinse completely with clean water.

Weekly sanitization:

- Sanitize the container with a solution made by mixing 1 teaspoon of unscented liquid household chlorine bleach in one quart of water.
- Cover the container and shake it well so that the sanitizing bleach solution touches all inside surfaces of the container.
- Wait at least 30 seconds and then pour the sanitizing solution out of the container.
- Let the empty sanitized container air-dry before use OR rinse the empty container with clean, safe water that is available already.

3.0 Cold Emergencies

3.1 Hypothermia

Hypothermia is the lowering of the entire body's core temperature. Hypothermia is most likely at very cold temperatures, but it can occur even at cool temperatures (above 40°F), if a person becomes chilled from rain, sweat, or submersion in cold water.

- Symptoms: shivering, confusion, irrationality, lethargy or drowsiness; skin may be pale and cool, breathing may slow and the affected individual may have changing levels of responsiveness
- Preventative measures:
 - Dress in layers that can be added and removed as work progresses
 - Choose a coat with wind and/or waterproof outer layer
 - Wear a hat; most body heat is lost from the head
 - Consume warm drinks and high-energy food
- Treatment:
 - Call 911 immediately in an emergency; otherwise seek medical assistance as soon as possible.
 - Move the person to a warm, dry area.
 - Remove wet clothes and replace with dry clothes, cover the body (including the head and neck) with layers of blankets; and with a vapor barrier (e.g. tarp, garbage bag). Do not cover the face.
 - If medical help is more than 30 minutes away:
 - Give warm sweetened drinks if alert (no alcohol), to help increase the body temperature. Never try to give a drink to an unconscious person.
 - Place warm bottles or hot packs in armpits, sides of chest, and groin. Call 911 for additional rewarming instructions.
 - If a person is not breathing or has no pulse:
 - Call 911 for emergency medical assistance immediately.
 - Treat the worker as per instructions for hypothermia, but be very careful and do not try to give an unconscious person fluids.
 - Check him/her for signs of breathing and for a pulse. Check for 60 seconds.
 - If after 60 seconds the affected worker is not breathing and does not have a pulse, trained workers may start rescue breaths for 3 minutes.
 - Recheck for breathing and pulse, check for 60 seconds.
 - If the worker is still not breathing and has no pulse, continue rescue breathing.
 - Only start chest compressions per the direction of the 911 operator or emergency medical services*
 - Reassess patient's physical status periodically.
 - *Chest compression are recommended only if the patient will not receive medical care within 3 hours.

3.2 Frostbite

Frostbite is an injury to the body that is caused by freezing of the skin and underlying tissues. The lower the temperature, the more quickly frostbite will occur. Frostbite typically affects the extremities, particularly the feet and hands.

- Symptoms: Reddened skin develops gray/white patches, numbness in the affected part, affected part may feel firm or hard, blisters may occur, in severe cases
- Preventative measures
 - Most common areas for frostbite are exposed skin on head, face, hands or feet. Keep cloths dry and dress in layers
 - Wind chill increases the risk of frostbite, prevent exposure to high winds whenever possible

- Treatment:
 - Move exposed person to a warm environment
 - Do not rub the affected area to warm it because this action can cause more damage.
 - Do not apply snow/water. Do not break blisters.
 - Loosely cover and protect the area from contact.
 - Do not try to rewarm the frostbitten area before getting medical help; for example, do not place in warm water. If a frostbitten area is rewarmed and gets frozen again, more tissue damage will occur. It is safer for the frostbitten area to be rewarmed by medical professionals.
 - Give warm sweetened drinks, if the person is alert.
 - Seek medical attention

3.3 Trench Foot

Trench Foot or immersion foot is caused by prolonged exposure to wet and cold temperatures. It can occur at temperatures as high as 60°F if the feet are constantly wet. Non-freezing injury occurs because wet feet lose heat 25-times faster than dry feet. To prevent heat loss, the body constricts the blood vessels to shut down circulation in the feet. The skin tissue begins to die because of a lack of oxygen and nutrients and due to the buildup of toxic products.

- Symptoms: Redness of the skin, swelling, numbness, blisters
- Preventative measures: Wear insulated and waterproof boots (or other footwear). Where possible, keep extra clothing handy in case you get wet and need to change.
- Treatment:
 - Call 911 immediately in an emergency; otherwise seek medical assistance as soon as possible.
 - Remove the shoes, or boots, and wet socks.
 - Dry the feet.

4.0 Portable Electronics

The use of iPods, MP3 players, Bluetooth technology, radios and other such portable electronic devices etc. are prohibited within the work area or areas where use of such devices have the potential to interfere with surrounding activities, awareness of surroundings and/or communication between individuals on the project. Cell phones and Bluetooth technology may be used for business related communication or in case of emergency, but not for entertainment purposes. At no time should cell phones or Bluetooth technology be used by the operator of mobile equipment when the equipment is being operated unless required by another section. See Section 5 Cranes (10.0)

5.0 Animals and Wildlife

5.1 Snake Bites

- Avoiding Snakebites:
 - Know the area you are working in and if snakes are likely.
 - When walking in areas known to have venomous snakes keep your eye out and make noise to warn the snakes of your presence.
 - Leave snakes alone. Many people are bitten because they try to kill a snake or get a closer look at it
 - Stay out of tall grass unless you wear thick leather boots, and remain on paths as much as possible
 - Keep hands and feet out of areas you can't see. Don't pick up rocks, lumber or firewood unless you are out of a snake's striking distance. (A snake can strike half its length, Hardy says.)
 - Be cautious and alert when climbing rocks
- Treatment:
 - If bitten seek medical attention immediately from a facility capable of rendering advanced care, including possible anti-venom administration and airway support.

- Briefly attempt to identify the snake (especially, note the color pattern). If possible, take a digital photo of the snake from a safe distance. Do not attempt to catch the snake, efforts to catch or kill the animal can result in wasted time and further bites.
- Rapidly apply a compressive bandage (e.g., elastic bandage, crepe bandage, torn clothing) to the bitten extremity, starting distally and progressing to encompass the entire limb. Wrap it as tightly as one would wrap a severe sprained ankle. Then, splint the extremity and, if possible, keep it at approximately heart level. Do not elevate the area. This technique may significantly delay systemic absorption of elapid venoms, including coral snake venom. Research suggests that, in a simulated snakebite scenario, people tend to underestimate how tight they can get the bandage for the technique to be effective.

5.2 Animal Bites & Stings

- Report all animal bites to local animal control or police; follow local procedures for rabies
- Suspect rabies in cases of unprovoked attacks of strange acting animals or wild animals
- Do not attempt to catch animals of unprovoked attacks
- Treatment:
 - Wash affected area thoroughly with soap and water
 - Run water over wound for at least five minutes
 - Control bleeding
 - Cover wound with sterile dressing and bandage
 - Seek immediate medical attention

5.3 Spider Bites

- Only the venom of black widow and brown recluse spider is serious or potentially fatal
 - Black widow spiders are shiny black and have red hourglass-shaped marking on underside of abdomen. If bitten, the following symptoms may occur:
 - Pain at bite site
 - Red skin at site
 - After 15 minutes to hours: sweating, nausea, stomach and muscle cramps, increased pain at site, dizziness or weakness, difficulty breathing
 - Brown recluse spiders are usually light brown in color with a violin-shaped marking on its back
 - Stinging sensation at site
 - Over 8 to 48 hours: increasing pain, blistering at site, fever, chills, nausea or vomiting, joint pain, open sore at site
- Treatment:
 - *If the victim has difficulty breathing, call 911 and be prepared to administer basic life support*
 - *Keep bite area below level of the heart*
 - *Wash affected area thoroughly with soap and water*
 - *Put ice or cold pack on bite area*
 - *Seek medical attention immediately if black widow or brown recluse bites are suspected*

5.4 Bee and Wasp Stings

- Insect stings are not poisonous but can cause life-threatening allergic reactions in affected workers with severe allergies
- Signs & symptoms:
 - *Pain, burning or itching at sting site*
 - *Redness or swelling*
 - *Difficulty breathing*
- Treatment:
 - *Remove stinger from skin by scraping it away gently with credit card or knife blade (call 911 for known allergy to stings)*

- Wash area with soap and water
- Put ice pack on sting site
- Watch victim for 30 minutes for any signs or symptoms of allergic reaction
- If symptoms occur, call 911 and treat for shock

5.5 Scorpion Stings

- Some types in the American Southwest are dangerous
- Antivenin may be available in some areas
- Signs & symptoms:
 - Scorpions sting with tail
 - Severe burning pain at sting site (later numbness, tingling)
 - Nausea or vomiting
 - Difficulty swallowing
 - Possible convulsions or coma
- Treatment:
 - Contact a qualified health care provider or poison control center for advice and medical instructions.
 - Ice may be applied directly to the sting site (never submerge the affected limb in ice water).
 - Remain relaxed and calm.
 - Do not take any sedatives.
 - Capture the scorpion for identification if it is possible to do so safely.

5.6 Fire Ants

Fire ants bite and sting. They are aggressive when stinging and inject venom, which causes a burning sensation. Red bumps form at the sting, and within a day or two they become white fluid-filled pustules.

- Employers should protect their workers from fire ants by training them about:
 - Their risk of exposure
 - How to identify fire ants and their nests
 - How to prevent exposure
 - What to do if they are bitten or stung
- Workers should take the following steps to prevent fire ant stings and bites:
 - Do not disturb or stand on or near ant mounds.
 - Be careful when lifting items (including animal carcasses) off the ground, as they may be covered in ants.
 - Fire ants may also be found on trees or in water, so always look over the area before starting to work.
- Treatment:
 - Workers with a history of severe allergic reactions to insect bites or stings should consider carrying an epinephrine auto injector (EpiPen) and should wear a medical identification bracelet or necklace stating their allergy.
- Workers should take the following steps if they are stung or bitten by fire ants:
 - Rub off ants briskly, as they will attach to the skin with their jaws.
- Antihistamines may help.
 - Follow directions on packaging.
 - Drowsiness may occur.
- Take the worker to an emergency medical facility immediately if a sting causes severe chest pain, nausea, severe sweating, loss of breath, serious swelling, or slurred speech.

5.7 Mosquito-Borne Diseases

Workers are at risk when they are working where mosquitoes are biting. Different species of mosquitoes are found in different geographic locations, are most active at different times, and spread different diseases. The risk to workers varies with:

- Where they are working in the United States or elsewhere
- Type of habitat at the work site
- Season
- Time of day.
 - Mosquitoes breed in standing water. Employers should decrease mosquito populations at worksites by
 - Removing, turning over, covering, or storing equipment
 - Removing debris from ditches
 - Filling in ruts and other areas that collect standing water
 - Removing tires, buckets, bottles, and barrels that collect water
 - Placing drain holes in containers that collect water and cannot be discarded.
- Employers can keep mosquitoes out of indoor worksites by ensuring that doors and windows have screens and are kept closed when possible.
- Employers should also protect workers by providing training about:
 - The risk of exposure to mosquitoes
 - How they can protect themselves from mosquito bites
 - Symptoms of diseases spread by mosquitoes
 - Protective clothing, including long-sleeved shirts and long pants
 - EPA-registered insect repellents to use on exposed skin and clothing
- Workers can reduce their risk of mosquito bites while working outdoors, by:
 - Helping to reduce sources of standing water
 - Wearing long-sleeved shirts and long pants
 - Using EPA-registered insect repellents on exposed skin and clothing, as directed on the product labels.
- Workers who develop symptoms of a mosquito-borne disease should report this promptly to their supervisor and get medical attention.

6.0 Poisonous Plants

Poison ivy, poison oak, and poison sumac release an oil, urushiol, when the leaf or other plant parts are bruised, damaged, or burned. When the oil gets on the skin an allergic reaction, referred to as contact dermatitis, occurs in most exposed people as an itchy red rash with bumps or blisters. Employers should protect their workers from poisonous plants by training them about:

- Their risk of exposure to poisonous plants
- How to identify poisonous plants
- How to prevent exposure to poisonous plants
- What they should do if they are exposed to poisonous plants
- Workers can prevent contact with poisonous plants by taking these steps:
 - Wear long sleeves, long pants, boots, and gloves.
 - Wash exposed clothing separately in hot water with detergent.
- Barrier skin creams, such as a lotion containing bentoquatam, may offer some protection before contact.
 - Barrier creams should be washed off and reapplied twice a day.
 - After use, clean tools with rubbing alcohol (isopropanol or isopropyl alcohol) or soap and lots of water. Urushiol can remain active on the surface of objects for up to 5 years.
 - Wear disposable gloves during this process.

- Do not burn plants that may be poison ivy, poison oak, or poison sumac. Inhaling smoke from burning plants can cause severe allergic respiratory problems.
- Treatment:
 - Immediately rinse skin with rubbing alcohol, specialized poison plant washes, degreasing soap (such as dishwashing soap) or detergent, and lots of water.
 - Rinse frequently so that wash solutions do not dry on the skin and further spread the urushiol.
 - Scrub under nails with a brush.
 - Apply wet compresses, calamine lotion, or hydrocortisone cream to the skin to reduce itching and blistering.
 - Follow the directions on any creams and lotions. Do not apply to broken skin, such as open blisters.
 - Oatmeal baths may relieve itching.
 - Drowsiness may occur.
 - In severe cases or if the rash is on the face or genitals, seek professional medical attention.
 - Call 911 or go to a hospital emergency room if the worker is suffering a severe allergic reaction, such as swelling or difficulty breathing, or has had a severe reaction in the past.

7.0 Infectious Disease/Pandemic Prevention, Preparedness and Response

DPR Construction will take proactive steps to protect the workplace (both offices and project worksites) in the event of an infectious disease/pandemic outbreak such as COVID-19. It is the goal of DPR Construction during any such time period to strive to operate effectively and ensure that all essential services are continuously provided and that employees, subcontractors, clients, and other stakeholders are safe within the workplace and areas occupied or controlled by DPR Construction.

DPR Construction is committed to identifying risk levels in the workplace and providing authoritative information about the nature and spread of infectious diseases, including symptoms and signs to watch for, as well as required steps to be taken in the event of an illness or outbreak. DPR Construction will monitor and implement guidance based upon recommendations of the U.S. Centers for Disease Control and Prevention (CDC) and/or the World Health Organization (WHO). DPR will implement engineering, administrative and work practice controls and personal protective equipment (PPE) as well as considerations for doing so as much as practicable and appropriate.

Unless otherwise notified, our normal attendance and leave policies will remain in place. Individuals who believe they may face particular challenges reporting to work during an infectious disease/pandemic outbreak should take steps to develop any necessary contingency plans. For example, employees might want to arrange for alternative sources of child-care should schools close and/or speak with supervisors about the potential to work from home temporarily or on an alternative work schedule.

For the latest detailed information and guidance on Infectious disease/pandemic prevention and response provided by DPR Construction see

<https://dprprod.sharepoint.com/sites/toolbox/communicationsbrand/pages/coronavirus.aspx>

(Hyperlink for DPR Employee Access Only)

Subcontractors or other Non-DPR stakeholders, please request the latest guidance from the DPR Project Team.

7.1 Preventing the Spread of Infection in the Workplace

DPR Construction will work to identify risks ensure a clean workplace, including the regular cleaning and disinfecting of objects and areas that are frequently used, such as bathrooms, break rooms, conference rooms, door handles, light switches and railings. Only EPA approved cleaning agents will be used for cleaning and disinfecting per the manufacturer's recommendations. A committee, working group or task force may

be designated to monitor and coordinate events around an infectious disease/pandemic outbreak, as well as to create work rules that could be implemented to promote safety through infection control.

For a list of EPA approved Disinfectants see: <https://www.epa.gov/pesticide-registration/list-n-disinfectants-use-against-sars-cov-2>

7.2 Implement Workplace Controls

DPR may implement to the extent practicable, workplace controls that follow a generally accepted hierarchy of controls in an attempt to systematically remove hazards associated with Infectious Disease/Pandemic outbreaks rather than relying solely on workers to reduce their exposure. During an Infectious Disease/Pandemic outbreak, when it may not be possible to eliminate the hazard, the most effective protection measures are (listed from most effective to least effective): engineering controls, administrative controls, safe work practices (a type of administrative control), and PPE. There are advantages and disadvantages to each type of control measure when considering the ease of implementation, effectiveness, and cost. In most cases, a combination of control measures will be necessary to protect workers from exposure to Infectious Disease/Pandemic outbreaks.

In addition to the types of workplace controls discussed below, CDC may offer additional guidance for infection prevention strategies. Employers should check the CDC website regularly for updates about workplace controls.

Engineering Controls

Engineering controls involve isolating employees from work related hazards. In workplaces where they are appropriate, these types of controls reduce exposure to hazards without relying on worker behavior and can be the most cost-effective solution to implement. Engineering controls that may be practicable and appropriate include:

- Installing high-efficiency air filters in offices and project trailers
- Increasing ventilation rates in the work environment.
- Installing physical barriers, such as clear plastic sneeze guards.

Administrative Controls

Administrative controls require action by the worker or employer. Typically, administrative controls are changes in work policy or procedures to reduce or minimize exposure to a hazard. Examples of administrative controls that may be practicable and appropriate include:

- Encouraging sick workers to stay at home.
- Minimizing contact among workers, clients, and customers by replacing face-to-face meetings with virtual communications and implementing telework if feasible.
- Establishing alternating days or extra shifts that reduce the total number of employees in a facility at a given time, allowing them to maintain distance from one another while maintaining a full onsite work week.
- Discontinuing nonessential travel to locations with ongoing Infectious Disease/Pandemic outbreaks. Employers should check the CDC website regularly for updates about travel advisories.
- Developing emergency communications plans, including a forum for answering workers' concerns and internet-based communications, if feasible.
- Providing workers with up-to-date education and training on Infectious Disease/Pandemic risk factors and protective behaviors (e.g., cough etiquette and care of PPE).
- Training workers who need to use protecting clothing and equipment how to put it on, use/wear it, and take it off correctly, including in the context of their current and potential duties. Training material should be easy to understand and available in the appropriate language and literacy level for all workers.

Safe Work Practices

Safe work practices are types of administrative controls that include procedures for safe and proper work used to reduce the duration, frequency, or intensity of exposure to a hazard. Examples of safe work practices for Infectious Disease/Pandemic outbreak include:

- Providing resources and a work environment that promotes personal hygiene. For example, no-touch trash cans, hand soap, alcohol-based hand rubs containing at least 60 percent alcohol, disinfectants, and disposable towels for workers to clean their work surfaces.
- Requiring regular hand washing or using of alcohol-based hand rubs. Workers should always wash hands when they are visibly soiled and after removing any PPE.
- Post handwashing and other hygiene and social distancing protocol signs in conspicuous locations including break areas, office lobbies, project trailers and restrooms.

Personal Protective Equipment (PPE)

While engineering and administrative controls are considered more effective in minimizing exposure to Infectious Disease/Pandemic outbreaks, PPE may also be needed to prevent certain exposures. While correctly using PPE can help prevent some exposures, it should not take the place of other prevention strategies. Examples of PPE include: gloves, goggles, face shields, face masks, and respiratory protection, when appropriate. During an outbreak of an Infectious Disease/Pandemic, recommendations for PPE specific to occupations or job tasks may change depending on geographic location, updated risk assessments for workers, and information on PPE effectiveness in preventing the spread of the Infectious Disease/Pandemic. Employers should check the OSHA and CDC websites regularly for updates about recommended PPE.

All types of PPE must be:

- Selected based upon the hazard to the worker.
- Properly fitted and periodically refitted, as applicable (e.g., respirators).
- Consistently and properly worn when required.
- Regularly inspected, maintained, and replaced, as necessary.
- Properly removed, cleaned, and stored or disposed of, as applicable, to avoid contamination of self, others, or the environment. Employers are obligated to provide their workers with PPE needed to keep them safe while performing their jobs. The types of PPE required during an Infectious Disease/Pandemic outbreak will be based on the risk of being infected while working and job tasks that may lead to exposure.

Follow Existing OSHA Standards

Existing OSHA standards may apply to protecting workers from exposure to and infection with an Infectious Disease/Pandemic. While there is no specific OSHA standard covering Infectious Disease/Pandemic exposure, some OSHA requirements may apply to preventing occupational exposure.

Among the most relevant are:

- OSHA's Personal Protective Equipment (PPE) standards (in general industry, 29 CFR 1910 Subpart I), which require using gloves, eye and face protection, and respiratory protection.
- When respirators are necessary to protect workers or where employers require respirator use, employers must implement a comprehensive respiratory protection program in accordance with the Respiratory Protection standard (29 CFR 1910.134).
- The General Duty Clause, Section 5(a)(1) of the Occupational Safety and Health (OSH) Act of 1970, 29 USC 654(a)(1), which requires employers to furnish to each worker "employment and a place of employment, which are free from recognized hazards that are causing or are likely to cause death or serious physical harm."
- OSHA's Bloodborne Pathogens standard (29 CFR 1910.1030) applies to occupational exposure to human blood and other potentially infectious materials that typically do not include respiratory

secretions that may transmit Infectious Disease/Pandemic. However, the provisions of the standard offer a framework that may help control some sources of the virus, including exposures to body fluids (e.g., respiratory secretions) not covered by the standard.

Engagement and Participation

We ask employees, subcontractors, clients, and other stakeholders to cooperate in taking steps to reduce the transmission of infectious disease/pandemic outbreak in the workplace. The best strategy remains the most obvious— social distancing, frequent hand washing with warm, soapy water; covering your mouth whenever you sneeze or cough; and discarding used tissues in wastebaskets. We will also provide alcohol-based hand sanitizers containing at least 60% alcohol throughout the workplace and in common areas. There may be additional strategies implemented based upon guidance from the U.S. Centers for Disease Control and Prevention (CDC) and/or the World Health Organization (WHO). Since those strategies may be rapidly changing based upon the most recent information available, DPR will publish and distribute separate guidance to address additional strategies.

7.3 Limiting Travel

All nonessential travel should be avoided during times of infectious disease/pandemic outbreaks. Employees who travel as an essential part of their job should consult with management on appropriate actions. International Business-related travel may not be authorized during times of infectious disease/pandemic outbreaks.

Employees should avoid crowded public transportation when possible. Alternative scheduling options, ride-share resources should be considered.

Employers should check the CDC website regularly for updates about travel advisories.

7.4 Telecommuting

Telework requests will be handled on a case-by-case basis. While not all positions will be eligible, all requests for temporary telecommuting should be submitted to your manager/supervisor for consideration.

7.5 Staying Home When Ill

Many times, with the best of intentions, employees report to work even though they feel ill.

During an infectious disease/pandemic outbreak, it is critical that employees do not report to work while they are ill and/or experiencing symptoms associated with the infectious disease/pandemic. Examples include fever, cough, sore throat, runny or stuffy nose, body aches, headache, chills and fatigue]. As of the date of this document, the Centers for Disease Control and Prevention recommends:

All employees should stay home if they are sick until at least 24 hours after their fever* (temperature of 100 degrees Fahrenheit or 37.8 degrees Celsius or higher) is gone. Temperature should be measured without the use of fever-reducing medicines (medicines that contains ibuprofen or acetaminophen). Not everyone with flu will have a fever. Individuals with suspected or confirmed flu, who do not have a fever, should stay home from work at least 4-5 days after the onset of symptoms. Persons with the flu are most contagious during the first 3 days of their illness.

See: <https://www.cdc.gov/flu/business/stay-home-when-sick.htm>

Employees who report to work ill may be sent home in accordance with these health guidelines.

7.6 Requests for Medical Information and/or Documentation

If you are out sick or show symptoms of being ill, it may become necessary to request information from you and/or your health care provider. In general, we would request medical information to confirm your need to be absent, to show whether and how an absence relates to the infection, and to know that it is appropriate

for you to return to work. As always, we expect and appreciate your cooperation if and when medical information is sought.

7.7 Confidentiality of Medical Information

Our policy is to treat any medical information as confidential. In furtherance of this policy, any disclosure of medical information is in limited circumstances with supervisors, managers, first aid and safety personnel, and government officials as required by law.

7.8 Social Distancing Guidelines for Workplace Infectious Disease/Pandemic Outbreaks

In the event of an infectious disease/pandemic outbreak, DPR Construction will follow the recommendations of the CDC and/or WHO and may implement these or other social distancing guidelines to minimize the spread of the disease among the employees, subcontractors, clients and other stakeholders.

During the workday, employees are requested to:

- Avoid meeting people face-to-face. Employees are encouraged to use the telephone, online conferencing, e-mail or instant messaging to conduct business as much as possible, even when participants are in the same building.
- If a face-to-face meeting is unavoidable, minimize the meeting time, choose a large meeting room and sit at least six feet from each other if possible; avoid person-to-person contact such as shaking hands.
- Avoid any unnecessary travel and cancel or postpone nonessential meetings, gatherings, workshops and training sessions.
- Do not congregate in work rooms, pantries, copier rooms or other areas where people socialize.
- Bring lunch and eat at your desk or away from others (avoid lunchrooms and crowded restaurants).
- Encourage members and others to request information and orders via phone and e-mail in order to minimize person-to-person contact. Have the orders, materials and information ready for fast pick-up or delivery.

7.9 Outside activities

- Employees might be encouraged to the extent possible to:
- Avoid public transportation (walk, cycle, drive a car) or go early or late to avoid rush-hour crowding on public transportation.
- Avoid recreational or other leisure classes, meetings, activities, etc., where employees might come into contact with contagious people.

End of Section 27

Work Environment & Employee Health Forms

Forms

Heat Illness Prevention Training Verification – English
Heat Illness Prevention Training Verification - Spanish
Site Specific Procedures Checklist to Prevent Heat Illness
High Heat Procedures Pre-shift Meeting Checklist
High Heat Procedures Sign-In Sheet



Heat Illness Prevention Training Verification

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Print Name	Date
Project/Project Number	Company

Workers should be trained in procedures for preventing heat illness when weather conditions have the potential to impact worker health.

Provisions of Water

- Adequate supplies of potable water should be provided in all places of employment including but not limited to the requirements that it be fresh, pure, and suitably cool and provided to employees free of charge. The water should be located as close as practicable to areas where employees are working
- Portable containers used to dispense drinking water should have a facet or spigot, be tightly closed and labeled as to the nature of its contents; maintained in a sanitary condition and should not be used for any other purpose
- Water should not be dipped from containers
- Where drinking fountains are not provided, single use cups should be used
- Where single service cups are supplied, a sanitary container for the unused cups and a receptacle for disposing of the used cups should be provide
- Where re-usable containers for individual use are relied upon for compliance with this section, the employer shall ensure the containers are marked to identify the user and maintained in a sanitary condition (Required in CA, highly recommended in all other states)
- Re-usable containers for individual use and drinking cups shall not be shared or used in common
- The frequent drinking of water should be encouraged, one quart per hour is recommended in hot weather
- Non-potable water:
 - o Outlets for non-potable water, such as water for industrial or firefighting purposes only, shall be identified by signs to indicate clearly that the water is unsafe and is not to be used for drinking or washing
 - o There shall be no cross-connection, open or potential, between a system furnishing potable water and a system furnishing non-potable water

Shade

- In California, shade should be present when temperature exceeds 80 degrees Fahrenheit. The amount of shade present should be at least enough to accommodate the number of employees on recovery, break, or lunch periods, so that they can sit in a normal posture fully in the shade without having to be in physical contact with each other. The shaded area should be located as close as practicable to the areas where employees are working. Subject to the same specifications, the amount of shade present during meal periods should be at least enough to accommodate the number of employees on the meal period who remain onsite.
- In all other states, shade should be provided as temperatures rise. The shaded area should be located as close as practicable to the areas where employees are working, and workers should be encouraged to use rest/hydration breaks as necessary to prevent signs/symptoms of heat illness.
- When necessary, workers should remain in shaded areas until symptoms of heat illness subside
 - o If an employee exhibits signs or reports symptoms of heat illness while taking a preventative cool-down rest or during a preventative cool-down rest period, the employer shall provide appropriate first aid or emergency response.
- At no time should a worker with symptoms of possible serious heat illness be left unattended or sent home without medical assessment and authorization

Environmental & Personal Risk Factors

Workers need to be aware of the following potential risk factors:

- Environmental risk factors for heat illness means working conditions that create the possibility that heat illness could occur, including air temperature, relative humidity, radiant heat from the sun and other sources, conductive heat sources from the ground, air movement, workload severity and duration, protective clothing and personal protective equipment worn by employees.
- Personal risk factors for heat illness means factors such as an individual's age, degree of acclimatization, health, water consumption, alcohol consumption, caffeine consumption, and use of prescription medications that affect the body's water retention or other physiological response to heat.
- Pre-shift meetings before the commencement of work to review the high heat procedures, encourage employees to drink plenty of water and remind employees of their right to take a cool-down rest when necessary.



Heat Illness Prevention Training Verification

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Signs & Symptoms and First Aid Treatment of Heat Illnesses

Heat Stroke

Heat stroke is the most serious heat-related illness. It occurs when the body becomes unable to control its temperature: the body's temperature rises rapidly, the sweating mechanism fails, and the body is unable to cool down. When heat stroke occurs, the body temperature can rise to 106°F or higher within 10 to 15 minutes. Heat stroke can cause death or permanent disability if emergency treatment is not given.

Symptoms of heat stroke:

- Confusion, altered mental status, slurred speech
- Loss of consciousness (coma)
- Hot, dry skin or profuse sweating
- Seizures
- Very high body temperature
- Fatal if treatment delayed

First Aid steps to be taken to treat a worker with heat stroke:

- Call 911 for emergency medical care.
- Stay with worker until emergency medical services arrive.
- Move the worker to a shaded, cool area and remove outer clothing.
- Cool the worker quickly with a cold water or ice bath if possible; wet the skin, place cold wet cloths on skin, or soak clothing with cool water.
- Circulate the air around the worker to speed cooling.
- Place cold wet cloths or ice on head, neck, armpits, and groin; or soak the clothing with cool water.

Heat Exhaustion

Heat exhaustion is the body's response to an excessive loss of the water and salt, usually through excessive sweating. Workers most prone to heat exhaustion are those that are elderly, have high blood pressure, and those working in a hot environment.

Symptoms of heat exhaustion:

- Headache
- Nausea
- Dizziness
- Weakness
- Irritability
- Thirst
- Heavy sweating
- Elevated body temperature
- Decreased urine output

First Aid steps to be taken to treat a worker with heat exhaustion:

- Take worker to a clinic or emergency room for medical evaluation and treatment.
- If medical care is unavailable, call 911.
- Someone should stay with worker until help arrives.
- Remove worker from hot area and give liquids to drink.
- Remove unnecessary clothing, including shoes and socks.
- Cool the worker with cold compresses or have the worker wash head, face, and neck with cold water.
- Encourage frequent sips of cool water.

Rhabdomyolysis

Rhabdomyolysis is a medical condition associated with heat stress and prolonged physical exertion, resulting in the rapid breakdown, rupture, and death of muscle. When muscle tissue dies, electrolytes and large proteins are released into the bloodstream that can cause irregular heart rhythms and seizures, and damage the kidneys.

Symptoms of rhabdomyolysis include:

- Muscle cramps/pain
- Abnormally dark (tea or cola colored) urine
- Weakness



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- Exercise intolerance
- Asymptomatic

First Aid steps to be taken to treat a worker with rhabdomyolysis:

- Stop activity.
- Increase hydration (water preferred).
- Seek immediate care at the nearest medical facility.
- Ask to be checked for rhabdomyolysis (i.e., blood sample analyzed for creatine kinase).

Heat Syncope

- Heat syncope is a fainting (syncope) episode or dizziness that usually occurs with prolonged standing or sudden rising from a sitting or lying position. Factors that may contribute to heat syncope include dehydration and lack of acclimatization.

Symptoms of heat syncope include:

- Fainting (short duration)
- Dizziness
- Light-headedness during prolonged standing or suddenly rising from a sitting or lying position

First Aid steps to be taken to treat a worker with heat syncope:

- Sit or lie down in a cool place.
- Slowly drink water, clear juice, or a sports drink.

Heat Cramps

Heat cramps usually affect workers who sweat a lot during strenuous activity. This sweating depletes the body's salt and moisture levels. Low salt levels in muscles causes painful cramps. Heat cramps may also be a symptom of heat exhaustion.

Symptoms of heat cramps:

- Muscle cramps, pain, or spasms in the abdomen, arms, or legs

First Aid steps to be taken to treat a worker with heat cramps:

- Drink water and have a snack and/or carbohydrate-electrolyte replacement liquid (e.g., sports drinks) every 15 to 20 minutes.
- Avoid salt tablets.
- Get medical help if the worker has heart problems, is on a low sodium diet, or if cramps do not subside within 1 hour.

Heat Rash

Heat rash is a skin irritation caused by excessive sweating during hot, humid weather.

Symptoms heat rash include:

- Looks like red cluster of pimples or small blisters
- Usually appears on the neck, upper chest, groin, under the breasts, and in elbow creases

First Aid steps to be taken to treat a worker with heat rash:

- When possible, a cooler, less humid work environment is best treatment.
- Keep rash area dry.
- Powder may be applied to increase comfort.
- Ointments and creams should not be used.

Acclimatization

People need time for their bodies to adjust to working in the heat, for this reason, the following guidelines should be followed:

- Work up to exposure to the heat slowly; especially after illness, prolonged leave of absence; new arrival to area
- For heavy work under extremely hot conditions, it is advisable to progressively increase work time exposure each day, starting with approximately 2 hours
- An employee who has been newly assigned to a high heat area should be closely observed by a supervisor or designee for the first 14 days of the employee's employment.
- All employees should be closely observed by a supervisor or designee during a heat wave. For the purpose of this section only, "heat wave" means any day in which the predicted high temperature for the day will be at least 80



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degrees Fahrenheit and at least ten degrees Fahrenheit higher than the average high daily temperature in the preceding five days.

Cleaning a Water Storage Container Before Use

Water containers should be cleaned daily and sanitized weekly, at a minimum. Before filling with safe water, use these steps to clean and sanitize water storage containers:

Daily cleaning:

- Wash the storage container with a solution of dishwashing soap and water or baking soda and water.
- Rinse completely with clean water.

Weekly sanitization:

- Sanitize the container with a solution made by mixing 1 teaspoon of unscented liquid household chlorine bleach in one quart of water.
- Cover the container and shake it well so that the sanitizing bleach solution touches all inside surfaces of the container.
- Wait at least 30 seconds and then pour the sanitizing solution out of the container.
- Let the empty sanitized container air-dry before use OR rinse the empty container with clean, safe water that is available already.

Emergency Procedures and Reporting

- Any symptoms of heat illness, either in yourself or your co-workers, should be immediately reported to your supervisor
- In an emergency employees are allowed to call emergency services without contacting a supervisor first.
- Ensure you are familiar with any site-specific reporting requirements that may be necessary at your particular work location
- Ensure you are familiar with the emergency procedures in the event of an emergency
- In the event that persons exhibit any of these symptoms notify DPR's project management immediately. Even early symptoms may indicate serious heat exposure. If first aid trained personnel are not immediately available on-site to make an assessment and workers show an abnormal response to the heat, you should notify DPR Project Management and follow site specific emergency response procedures. Regardless of the workers' protests, no employee with any of the symptoms of possible serious heat illness noted above should be sent home or left unattended without medical assessment and authorization.

This document hereby certifies that I have been instructed on the items listed above and I had an opportunity to ask questions. I fully understand the DPR heat illness prevention requirements.

Employee Signature

Date

Person Conducting Verification Review Signature

Date

A copy of this form must be faxed to the DPR regional office PRIOR TO START OF WORK.

Verificación de Entrenamiento sobre la prevención de enfermedades Provocadas por el Calor

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Nombre	Fecha
Proyecto/Proyecto Numero	Compañía

Los trabajadores deben recibir entrenamiento sobre los procedimientos de prevención de enfermedades provocadas por el calor cuando las condiciones del clima tienen el potencial de afectar la salud del trabajador.

Suministro de agua

- Un suministro adecuado de agua potable debe ser proveído en todos los lugares de empleo
- Los contenedores portátiles utilizados para distribuir el agua potable deben tener un grifo o espita, estar bien cerrados y etiquetado hacia la naturaleza de sus contenidos; y mantenerse en una condición sanitaria y no debería ser utilizado para cualquier otro propósito
- No se deben sumergir vasos en los contenedores para sacar el agua
- Cuando no haya bebederos de agua potable, deben utilizarse vasos de un solo uso
- Donde se utilicen vasos de un solo uso, debe haber un contenedor higiénico para los vasos no utilizados y un recipiente para desechar los vasos usados
- Donde contenedores que se pueden volver a usar para uso individual son invocados para el cumplimiento de esta sección, el empleador deberá asegurar que los contenedores estén marcados para identificar usuario y mantener en condición sanitaria (Requerido en CA y muy recomendado en otros estados)
- Contenedores que se pueden volver a usar para uso individual y vasos para beber no serán compartidos o utilizados en común
- El consumo frecuente de agua deberá ser alentado. Un cuarto de agua por hora es recomendado en clima caloroso
- Agua no potable:
 - o Salidas para el agua no potable, como agua para uso industrial o propósitos contra incendios deberán ser identificadas con letreros para indicar claramente que el agua no es segura y no se deberá usar para tomar o lavar
 - o No deberá ver contra conexión, abierta o posibilidad entre un sistema que proporcione agua potable y un sistema que proporcione agua no potable

Sombra

- En California, sombra deberá estar presente cuando la temperatura exceda 80 grados Fahrenheit. La cantidad de sombra presente deberá ser lo suficiente para complacer el número de empleados en recuperación, descansos y horas de comida para que se puedan sentar en posturas normales en la sombra evitando tener que estar en contacto físico unos a otros. El área sombreada deberá ser ubicada lo más cerca posible a las áreas donde empleados están trabajando. Sujeto a las mismas especificaciones, la cantidad de sombra presente durante las horas de comida deberán ser lo suficiente para complacer el número de empleados en su hora de comida que permanezcan en el sitio.
- En todos los demás estados, sombra deberá ser proporcionada conforme como suba la temperatura. El área sombrada deberá ser colocada lo más cerca posible a las áreas donde empleados estén trabajando, e igual, los trabajadores deberán ser animados para usar descansos de hidratación tal como sea necesario para prevenir signos/síntomas de enfermedad de calor
- Cuando sea necesario, los trabajadores deben permanecer en la sombra hasta que desaparezcan los síntomas de la enfermedad provocada por el calor
- Si un empleado exhibe signos o reporta síntomas de enfermedad de calor mientras tome un descanso preventivo para refrescarse o durante un periodo preventivo de descanso para refrescarse, el empleador deberá proveer primeros auxilios apropiados o respuesta de emergencia
- En ningún momento se dejará sin vigilancia a un trabajador con síntomas graves de una enfermedad provocada por el calor; atención medica deberá de ser evaluada y autorizada

Factores de Riesgos Personales y del Ambiente

Los trabajadores deben estar atentos a los factores de riesgos potenciales:

- Riesgos del ambiente en cuanto a enfermedades por el calor quiere decir condiciones de trabajo que puedan crear la posibilidad de que ocurra una enfermedad por el calor, incluyendo la temperatura del aire, la humedad, calor radiante del sol, y también otras fuentes de calor como el calor que produce la tierra, movimiento del aire, dureza del trabajo, duración del trabajo, ropa protectora, y equipo de protección personal usada por el trabajador



Verificación de Entrenamiento sobre la prevención de enfermedades Provocadas por el Calor

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- Factores de riesgo personales para enfermedades del calor quiere decir factores como la edad del trabajador, aclimatación del individuo, salud, el consumo de agua, el consumo de alcohol, el consumo de cafeína, y el uso de medicinas recetadas que puedan afectar la retención de agua del cuerpo, y también como el cuerpo responde psicológicamente al calor

Signos y Síntomas y el Tratamiento de Primeros Auxilios de Enfermedades por el Calor

INSOLACION DE CALOR

La insolación de calor es la enfermedad más seria relacionada a enfermedades de calor. Ocurre cuando el cuerpo ya no puede controlar su temperatura: la temperatura del cuerpo se eleva rápidamente, el mecanismo de poder sudar falla y el cuerpo es incapaz de enfriarse. Cuando ocurre la insolación de calor, la temperatura del cuerpo puede elevarse a 106 grados Fahrenheit o más dentro de 10 a 15 minutos. Insolación de calor puede causar la muerte o discapacidad permanente si no se les da tratamiento de emergencias.

Síntomas de insolación de calor:

- Confusión, alteración del estado mental y pronunciación incomprensible
- Pérdida de conocimiento (coma)
- Piel seca, caliente o sudar profusamente
- Convulsión
- Alta temperatura
- Fatal si se dilata el tratamiento

Medidas de Primeros Auxilios que se deben tomar para el tratamiento de un trabajador con insolación de calor:

- Llamar al 911 para recibir atención médica de emergencia
- Permanezca con el trabajador hasta que lleguen los servicios médicos de emergencias
- Colocar al trabajador en área fresca con sombra y remover ropaje exterior
- Rápidamente refresque al trabajador con agua fría o baño de hielo si es posible, moje la piel, coloque paños húmedos y fríos en la piel o moje ropaje con agua fresca
- Circule el aire alrededor del trabajador para refrescar rápidamente
- Coloque paños húmedos y mojados o hielo en la cabeza, cuello, axilas, ingle o empape ropaje con agua fresca

POSTRACION CAUSADA POR EL CALOR EXCESIVO

Postración causada por el calor excesivo es la respuesta del cuerpo a una pérdida excesiva de agua y sal, por lo regular de sudor excesivo. Trabajadores más propensos a postración causada por el calor excesivo son esos ya mayores de edad, con alta presión y esos trabajadores trabajando en ambientes calurosos.

Síntomas de postración causada por el calor excesivo:

- Dolor de cabeza
- Náusea
- Mareos
- Debilidad
- Irritable
- Sed
- Sudor demasiado
- Temperatura del cuerpo elevada
- Disminución de orina

Medidas de Primeros Auxilios que se deben tomar para tratar a un trabajador con postración causada por el calor excesivo:

- Llevar al trabajador a la clínica o a la emergencia para una evaluación médica y tratamiento
- Si atención médica no es disponible, llamar al 911
- Alguien deberá permanecer con el trabajador hasta que llegue ayuda
- Remueva al trabajador de áreas calurosas y dar líquidos para tomar
- Remueva ropa innecesaria, incluyendo zapatos y calcetines
- Refresque al trabajador con compresas frías o haga que el trabajador se lave su cabeza y cuello con agua fría
- Animar beber a sorbos agua fresca



Verificación de Entrenamiento sobre la prevención de enfermedades Provocadas por el Calor

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RHABDOMYOLYSIS

Rhabdomyolysis es una condición médica asociada con estrés de calor y esfuerzo físico prolongado, dando como resultado la rápida descomposición, ruptura y la muerte del músculo. Cuando muere el tejido muscular, electrolitos y proteínas grandes se liberan en el torrente sanguíneo que puede causar ritmos irregulares del corazón, convulsiones y daño a los riñones.

Síntomas de Rhabdomyolysis incluyen:

- Calambres musculares/dolor
- Orín anormalmente oscuro (color de té o de cola)
- Debilidad
- Intolerancia al ejercicio
- Asintomática

Medidas de Primeros Auxilios que se deben tomar para tratar a un trabajador con Rhabdomyolysis:

- Parar actividad
- Aumento de hidratación (agua es preferida)
- Busque atención médica inmediata en la instalación medica más cercana
- Pregunte a comprobar la aparición de Rhabdomyolysis (es decir, la muestra de sangre analizada para la creatina quinasa)

SINCOPE de CALOR

- Síncope de calor es un episodio de desmayo (síncope) o mareos que ocurre generalmente cuando de pie por tiempo prolongado o de repente se levanta de una posición sentada o acostada. Factores que podrán contribuir a la síncope de calor incluyen deshidratación y la falta de aclimatación.

Síntomas de Síncope de Calor incluyen:

- Desmayo (corto plazo)
- Mareos
- Sensación de mareo durante mucho tiempo de pie o de repente se levanta de una posición sentada o acostada

Medidas de Primeros Auxilios que se deben tomar para tratar a un trabajador con Síncope de Calor:

- Sentarse o acostarse en un lugar fresco
- Beber agua lentamente, jugo claro o bebida deportiva

CALAMBRES POR CALOR

Los calambres por calor suelen afectar a los trabajadores que sudan mucho durante la actividad vigorosa. Sudar de esta manera reduce los niveles de sal y humedad del cuerpo. Los bajos niveles de sal en los músculos provocan calambres dolorosos. Los calambres por calor también pueden ser un síntoma de agotamiento por calor.

Síntomas de calambres por calor:

- Calambres musculares, dolor, o espasmos en el abdomen, brazos y piernas

Medidas de Primeros Auxilios que se deben tomar para tratar a un trabajador con calambres por calor

- Beba agua y tome un aperitivo y/o liquido de sustitución de electrolitos y carbohidratos (por ejemplo, bebidas deportivas) cada 15 a 20 minutos
- Evite tabletas de sal
- Busque atención medica si el trabajador tiene problemas del corazón, esta en una dieta baja en sodio, o si los calambres no desaparecen dentro de 1ª hora

SARPULLIDO

Sarpullido es irritación de la piel causado por el exceso de sudor durante clima caliente y húmedo.

Síntomas de sarpullido incluyen:

- Parecen conjuntos de granitos enrojecidos o pequeñas ampollas
- Por lo general, aparece en el cuello, parte superior del pecho, en la ingle, debajo de los senos y en los pliegues del codo

Medidas de Primeros Auxilios que se deben tomar para tratar a un trabajador con sarpullido

- Cuando sea posible, un ambiente de trabajo más fresco con menos humedad es mejor para tratamiento
- Mantener el área de sarpullido seco



Verificación de Entrenamiento sobre la prevención de enfermedades Provocadas por el Calor

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- Polvo higiénico puede ser aplicado para aumentar la comodidad
- Pomadas y cremas no deberán ser utilizadas

Aclimatación

Las personas necesitan tiempo para que su cuerpo se adapte al trabajo en lugares sometidos a temperaturas elevadas; por esta razón es necesario seguir estas lecciones:

- Trabajar gradualmente y lentamente su exposición al calor en particular si ha estado enfermo, ha disfrutado de una ausencia prolongada o es nuevo en la zona
- Para trabajos pesados bajo condiciones extremadamente calurosas, se recomienda aumentar el tiempo de exposición de forma progresiva cada día, comenzando con aproximadamente 2 horas

Limpieza de los Contenedores de Agua Antes de Usar

Contenedores de agua deben limpiarse y desinfectarse por lo mínimo cada semana. Antes de llenar con agua limpia y segura, utilice estos pasos para limpiar y desinfectar los contenedores almacenados de agua.

Limpieza Diaria:

- o Lave el recipiente de almacenamiento con jabón de lavaplatos y agua o con bicarbonato de sodio y agua
- o Enjuague completamente con agua limpia

Desinfección Semanal:

- o Desinfecte el contenedor con una solución preparada mezclando 1ª cucharadita de cloro de uso doméstico sin aroma en un litro de agua
- o Cubra el contenedor y agitar bien para que la solución de cloro desinfectante toque todas las superficies interiores del contenedor
- o Espere por lo mínimo 30ª segundos y luego vaciar la solución desinfectante fuera del contenedor
- o Deje el contendor desinfectado vacío que se seque al aire libre O enjuagar el contendor vacío con agua limpia y segura que ya este disponible

Procedimientos de Emergencia y la Notificación

- Cualquier síntoma de enfermedad causada por el calor, tanto si la sufre usted como otro compañero de trabajo, debe ser notificada inmediatamente a su supervisor
- En una emergencia empleados están autorizados a llamar a los servicios de emergencia sin tener que primero avisar a un supervisor
- Asegúrese de estar familiarizado con los requisitos de notificación específicos al sitio y particularmente en su lugar de trabajo
- Asegurase de que esté familiarizado do los procedimientos de emergencia en caso de una emergencia
- El el caso de que la persona presente cualquier tipo de estos síntomas notificar de inmediato la gerencia de DPR. Incluso los primeros síntomas pueden indicar una grave exposición al calor. Si la persona capacitada de primeros auxilios no está inmediatamente disponible para hacer una evaluación y el trabajador muestra una respuesta anormal al calor, debe de llamar al 911 inmediatamente o referirse al procedimiento de emergencia específico a su lugar de trabajo. Independientemente de las protestas de los trabajadores, ningún empleado con cualquiera de los síntomas de una posible grave enfermedad por el calor señalados anteriormente deberá ser mandado a casa o dejarse sin vigilancia sin ninguna evaluación médica sin autorización.

Este documento certifica que he recibido instrucción en lo referente a los elementos enumerados y he tenido la oportunidad de hacer preguntas. Entiendo plenamente los requisitos de DPR para la prevención de las enfermedades provocadas por el calor.

Firma del empleado

Firma de la persona que realiza la verificación del repaso

Se debe enviar por fax una copia de este formulario a la oficina regional de DPR ANTES DE COMENZAR EL TRABAJO. Una copia de esta forma deberá ser acompañada con la documentación de la orientación del trabajador



Site Specific Procedures Checklist to Prevent Heat Illness

The following designated person or persons have the authority and responsibility for implementing the provisions of this program at this worksite.

Name/Title/Phone Number	
1.	
2.	
3.	
4.	
5.	

Procedures for Provision of Water (include but are not limited to the following):	
☐	Drinking water will be brought to the site, so that at least two quarts per employee are available at the start of the shift. All workers whether working individually or in smaller crews, shall have access to drinking water. Drinking water shall be provided in sufficient quantity at the beginning of the work shift to provide one quart per employee per hour for drinking for the entire shift. Drinking water containers (of five to ten gallons each) may be used to supply water. On some projects drinking water bottles will be available in the DPR tool container, office trailer, or at the work station in the field. Drinking water may also available in the DPR office trailers where five-gallon chilled water dispensers are located with a reserve supply of water is available. Another source of drinking water may be supplied by tap water. There is a continuous supply of water so that at least two quarts are available at the start of the shift. All of the above mentioned water is fresh, pure, and suitably cool, and provided to employees free of charge.
	Designated Person: _____
☐	Paper cone rims or bags of disposable cups and the necessary cup dispensers will be made available to workers and will be kept clean until used.
	Designated Person: _____
☐	As part of the Effective Replenishment Procedures, the water levels and/or quantities will be checked periodically. Water will be replaced as needed.
	Designated Person: _____
☐	Water containers will be placed as close as possible to the workers (given the working conditions and layout of the worksite), to encourage the frequent drinking of water. If field terrain prevents the water from being placed as close as possible to the workers, bottled water or personal water containers will be made available, so that workers can have drinking water readily accessible.
	Designated Person: _____
☐	Placing water only in designated shade areas or where toilet facilities are located is not sufficient. When employees are working across large areas, water will be placed in multiple locations. For example, on a multi-story construction site, water should be placed in a safe accessible location on every floor where employees are working.
	Designated Person: _____
☐	All water containers will be kept in a clean and sanitary condition. Water from non-approved or non-tested water sources (e.g., untested wells) is not acceptable. If hoses or connections are used, they must be governmentally approved for potable drinking water systems, as shown on the manufactures label.
	Designated Person: _____
☐	Daily, workers will be reminded of the location of the water coolers and of the importance of drinking water frequently.
	Designated Person: _____

Site Specific Procedures Checklist to Prevent Heat Illness

☐	When the temperature equals or exceeds 95 degrees Fahrenheit or during a heat wave, pre-shift meetings will be conducted before the commencement of work to encourage employees to drink plenty of water, and remind employees of their right to take a cool-down rest when necessary (document the meeting using the High Heat Pre-Shift Meeting Checklist). The number of water breaks will be increased. Supervisors/foreman will lead by example and workers will be reminded throughout the work shift to drink water.
	Designated Person:
☐	Where individual water containers or bottled water is provided to workers, containers will be adequately identified to eliminate the possibility of drinking from a co-workers' container or bottle.
	Designated Person:
Procedures for Access to Shade (include but are not limited to the following):	
☐	Shade structures will be opened and placed as close as practical to the workers, when the temperature equals or exceeds 80 degrees Fahrenheit. When the temperature is below 80 degrees Fahrenheit, access to shade will be provided promptly, when requested by an employee. Note: The interior of a vehicle may not be used to provide shade unless the vehicle is air-conditioned and the air conditioner is on.
	Designated Person:
☐	The amount of shade present should be at least enough to accommodate the number of employees on recovery or rest periods, so that they can sit in a normal posture fully in the shade without having to be in physical contact with each other.
	Designated Person:
☐	Daily, workers will be informed of the location of the shade structures and will be encouraged to take a five minute cool-down rest in the shade. An employee who takes a preventative cool-down rest break will be monitored and asked if he/she is experiencing symptoms of heat illness and in no case will the employee be ordered back to work until signs or symptoms of heat illness have abated. (see also the section on Emergency Response for additional information)
	Designated Person:
☐	Shade structures will be relocated to follow along with the crew and they will be placed as close as practical to the employees, so that access to shade is provided at all times.
	Designated Person:
☐	In situations where trees or other vegetation are used to provide shade, the thickness and shape of the shaded area will be evaluated, before assuming that sufficient shadow is being cast to protect employee
	Designated Person:
☐	In situations where it is not safe or feasible to provide access to shade (e.g., during high winds), a note will be made of these unsafe or unfeasible conditions, and of the steps that will be taken to provide shade upon request.
	Designated Person:
☐	In situations where it is not safe or feasible to provide shade, a note will be made of these unsafe or unfeasible conditions, and of the steps that will be taken to provide alternative cooling measures but with equivalent protection as shade
	Designated Person:

Site Specific Procedures Checklist to Prevent Heat Illness

Procedures for Monitoring the Weather (include but are not limited to):

☐	The supervisor will be trained and instructed to check in advance the extended weather forecast. Weather forecasts can be checked with the aid of the internet (http://www.nws.noaa.gov/), or by calling the National Weather Service phone numbers or by checking the Weather Channel TV Network. The work schedule will be planned in advance, taking into consideration whether high temperatures or a heat wave is expected. This type of advance planning should take place all summer long.
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	Designated Person:	
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☐	A thermometer will be used at the jobsite to monitor for sudden increases in temperature, and to ensure that once the temperature exceeds 80 degrees Fahrenheit, shade structures will be opened and made available to the workers. In addition, when the temperature equals or exceeds 95 degrees Fahrenheit, additional preventive measures such as the High Heat Procedures will be implemented.
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	Designated Person:	
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Handling a Heat Wave:

☐	For purposes of this section only, "heat wave" means any day in which the predicted high temperature for the day will be at least 80 degrees Fahrenheit and at least ten degrees Fahrenheit higher than the average high daily temperature in the preceding five days.
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	Designated Person:	
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☐	During a heat wave or heat spike, the scheduled work day will be analyzed by the Project Superintendent. The Project Superintendent may determine that the work day will be cut short or rescheduled (example conducted at night, started earlier in the morning, or during cooler hours).
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	Designated Person:	
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☐	During a heat wave or heat spike, and before starting work, tailgate meetings will be held, to review the company heat illness prevention procedures, the weather forecast and emergency response. In addition, if schedule modifications are not possible, workers will be provided with an increased number of water and rest breaks and will be observed closely for signs and symptoms of heat illness.
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	Designated Person:	
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☐	Each employee will be assigned a "buddy" to be on the lookout for signs and symptoms of heat illness and to ensure that emergency procedures are initiated when someone displays possible signs or symptoms of heat illness.
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	Designated Person:	
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Sample High Heat Procedures (include, but are not limited to)

High Heat Procedures are additional preventive measures that this company will use when the temperature equals or exceeds 95 degrees Fahrenheit.

☐	Effective communication by voice, direct observation (applicable for work crews of 20 or fewer), mandatory buddy system, or electronic means will be maintained, so that employees at the worksite can contact a supervisor when necessary. If the supervisor is unable to be near the workers (to observe them or communicate with them), then an electronic device, such as a cell phone or text messaging device, may be used for this purpose if reception in the area is reliable.
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	Designated Person:	
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☐	Frequent communication will be maintained with employees working by themselves or in smaller groups (keep tabs on them via phone or two-way radio), to be on the lookout for possible symptoms of heat illness.
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	Designated Person:	
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☐	Effective communication and direct observation for alertness and/or signs and symptoms of heat illness will be conducted frequently. When the supervisor is not available, a designated alternate responsible person must be assigned, to look for signs and symptoms of heat illness. If a supervisor, designated observer, or any employee reports any signs or symptoms of heat illness in any employee, the supervisor or designated
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Site Specific Procedures Checklist to Prevent Heat Illness

	person will take immediate action commensurate with the severity of the illness (see Emergency Response Procedures).	
	Designated Person:	
☐	Employees will be reminded throughout the work shift to drink plenty of water and take preventative cool-down rest break when needed.	
	Designated Person:	

Procedures for Acclimatization (include, but are not limited to)

Acclimatization is the temporary and gradual physiological change in the body that occurs when the environmentally induced heat load to which the body is accustomed is significantly and suddenly exceeded by sudden environmental changes. In more common terms, the body needs time to adapt when temperatures rise suddenly, and an employee risks heat illness by not taking it easy when a heat wave strikes or when starting a new job that exposes the employee to heat to which the employee's body hasn't yet adjusted.

Inadequate acclimatization can be significantly more perilous in conditions of high heat and physical stress. Employers are responsible for the working conditions of their employees, and they must act effectively when conditions result in sudden exposure to heat their employees are not used to.

☐	The weather will be monitored daily. The supervisor will be on the lookout for sudden heat wave(s), or increases in temperatures to which employees haven't been exposed to for several weeks or longer.	
	Designated Person:	
☐	New employees, or those employees who have been newly assigned to a high heat area will be closely observed by the supervisor or designee for the first 14 days. The intensity of the work will be lessened during a two-week break-in period (such as scheduling slower paced, less physically demanding work during the hot parts of the day and the heaviest work activities during the cooler parts of the day (early-morning or evening). Steps taken to lessen the intensity of the workload for new employees will be documented.	
	Designated Person:	
☐	The supervisor will be extra-vigilant with new employees and stay alert to the presence of heat related symptoms.	
	Designated Person:	
☐	New employees will be assigned a "buddy" or experienced coworker to watch each other closely for discomfort or symptoms of heat illness.	
	Designated Person:	
☐	During a heat wave, all employees will be observed closely (or maintain frequent communication via phone or radio), to be on the look-out for possible symptoms of heat illness.	
	Designated Person:	
☐	Employees and supervisors will be trained on the importance of acclimatization, how it is developed and how these company procedures address it.	
	Designated Person:	



Site Specific Procedures Checklist to Prevent Heat Illness

Sample Procedures for Emergency Response (include but are not limited to):

☐	Prior to assigning a crew to a particular worksite, workers and the foreman will be provided a map of the site, along with clear and precise directions (such as streets or road names, distinguishing features and distances to major roads), to avoid a delay of emergency medical services.
	Designated Person:
☐	Prior to assigning a crew to a particular worksite, efforts will be made to ensure that a qualified and appropriately trained and equipped person is available at the site to render first aid if necessary.
	Designated Person:
☐	Prior to the start of the shift, a determination will be made of whether or not a language barrier is present at the site and steps will be taken (such as assigning the responsibility to call emergency medical services to the foreman or an English speaking worker) to ensure that emergency medical services can be immediately called in the event of an emergency.
	Designated Person:
☐	All foremen and supervisors will carry cell phones or other means of communication, to ensure that emergency medical services can be called. Checks will be made to ensure that these electronic devices are functional prior to each shift.
	Designated Person:
☐	When an employee is showing symptoms of possible heat illness, steps will be taken immediately to keep the stricken employee cool and comfortable once emergency service responders have been called (to reduce the progression to more serious illness). Under no circumstances will the affected employee be left unattended.
	Designated Person:
☐	At remote locations, the supervisor will designate an employee or employees to physically go to the nearest road or highway where emergency responders can see them. If daylight is diminished, the designated employee(s) should be given reflective vest or flashlights in order to direct emergency personnel to the location of the worksite, which may not be visible from the road or highway.
	Designated Person:
☐	During a heat wave or hot temperatures, workers will be reminded and encouraged to immediately report to their supervisor any signs or symptoms they are experiencing.
	Designated Person:
☐	Employees and supervisors training will include every detail of these written emergency procedures.
	Designated Person:

Handling a Sick Employee:

☐	When an employee displays possible signs or symptoms of heat illness, a trained first aid worker or supervisor will check the sick employee and determine whether resting in the shade and drinking cool water will suffice or if emergency service providers (911) will need to be called. A sick worker should not be left alone in the shade.
	Designated Person:
☐	When an employee displays possible signs or symptoms of heat illness and no trained first aid worker or supervisor is available at the site, call 911.
	Designated Person:



Site Specific Procedures Checklist to Prevent Heat Illness

☐	Emergency service providers (911) will be called immediately if an employee displays signs or symptoms of heat illness (decreased level of consciousness, staggering, vomiting, disorientation, irrational behavior, incoherent speech, convulsions, red and hot face), does not look OK or does not get better after drinking cool water and resting in the shade. While the ambulance is in route, first aid will be initiated (cool the worker: place the worker in the shade, remove excess layers of clothing, place ice pack in the armpits and groin area and fan the victim). Do not let a sick worker leave the site, as they can get lost or die before reaching a hospital.
	Designated Person:
☐	If an employee does not look OK and displays signs or symptoms of severe heat illness (decreased level of consciousness, staggering, vomiting, disorientation, irrational behavior, incoherent speech, convulsions, red and hot face), and the worksite is located more than 20 minutes away from a hospital, call emergency service providers (911), communicate the signs and symptoms of the victim and request Air Ambulance.
	Designated Person:

Sample Procedures for Employee and Supervisory Training (include but are not limited to):

☐	Supervisors will be trained prior to being assigned to supervise other workers. Training will include this company's written procedures and the steps supervisors will follow when employees' exhibit symptoms consistent with heat illness.
	Designated Person:
	Supervisors will be trained on their responsibility to provide water, shade, cool-down rests, and access to first aid as well as the employees' right to exercise their rights under this standard without retaliation.
	Designated Person:
	Supervisors will be trained in appropriate first aid and/or emergency responses to different types of heat illness, and in addition, that heat illness may progress quickly from mild symptoms and signs to serious and life threatening illness.
	Designated Person:
☐	Supervisors will be trained on how to track the weather at the job site (by monitoring predicted temperature highs and periodically using a thermometer). Supervisors will be instructed on, how weather information will be used to modify work schedules, to increase number of water and rest breaks or cease work early if necessary.
	Designated Person:
☐	All employees and supervisors will be trained prior to working outside. Training will include all aspects of implementing an effective Heat Illness Prevention Plan including but not limited to; providing sufficient water, providing access to shade, high-heat procedures, emergency response procedures and acclimatization contained in the company's written prevention procedures.
	Designated Person:
☐	Employees will be trained on the steps that will be followed for contacting emergency medical services, including how they are to proceed when there are non-English speaking workers, how clear and precise directions to the site will be provided and the importance of making visual contact with emergency responders at the nearest road or landmark to direct them to their worksite.
	Designated Person:
☐	New employees will be assigned a "buddy" or experienced coworker to ensure that they understand the training and follow company procedures.
	Designated Person:



High Heat Procedures Pre-shift Meeting Checklist

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REQUIRED IN CALIFORNIA: Conduct meeting prior to the work shift when the temperature will equal or exceed 95 degrees Fahrenheit.

Project/Project#:		Date:	
Conducted by:		Time:	
Signature:			

High Heat Pre-shift Topics	Yes	No	N/A
Remind workers to drinking plenty of water throughout the work shift	☐	☐	☐
Water is located:			
Shade is located			
Remind workers of their right to take cool-down rest periods when necessary	☐	☐	☐
Discuss with workers how effective communication is maintained so that employees at the work site can contact a supervisor when necessary	☐	☐	☐
Describe how frequent communication will be maintained with employees working by themselves or in smaller groups	☐	☐	☐
Remind workers to report to their supervisors immediately if they believe they require access to shade, and a preventative cool down period	☐	☐	☐
Supervisors authorized to call for emergency services: Names:			
Designated employees to call for emergency services (list below) Names:			
Remind workers that if heat illness is suspected (altered coordination and speech, mental confusion, unusual behavior, nausea, vomiting, hot dry skin, unusually profuse sweating, loss of consciousness, and seizures) emergency medical personnel should be contacted immediately.	☐	☐	☐
Anyone who detects heat illness in anyone onsite should summon emergency medical personnel immediately.	☐	☐	☐
DPR's emergency response plan with responsibilities is posted in the DPR office trailer.	☐	☐	☐

Additional Site Specific Considerations



High Heat Procedures Sign-In Sheet

Course Name:	High Heat Procedures	Date:	
Jobsite/Region:		Trainer:	

	Print Name	Signature	Jobsite/Office	DPR/Subcontractor
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